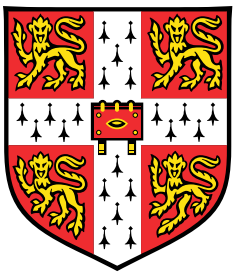
Combinatorial Problems with Geometric Flavour



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This dissertation is submitted for the degree of
Doctor of Philosophy

Trinity Hall May 2021

To my grandmother Georgeta-Paula Dimitriu

Declaration

I hereby declare that except where specific reference is made to the work of others, the contents of this dissertation are original and have not been submitted in whole or in part for consideration for any other degree or qualification in this, or any other university. This dissertation is my own work and contains nothing which is the outcome of work done in collaboration with others, except as specified in the text and Acknowledgements.

Marius Tiba
May 2021

Acknowledgements

I am tremendously grateful to my supervisor, Professor Béla Bollobás for taking me into an amazing journey and for all his care and support every step of the way. Béla, together with the incredible community of mathematicians gravitating towards him, has been a continuous source of inspiration and has taught me how to appreciate the beauty in mathematics as well as in many other aspects of life. I can't thank Béla and his delightful wife Gabriella enough for welcoming me into their academic family and for always being there for me.

I am extremely thankful to Professor Imre Leader for all his kindness and enthusiasm. The spectacular lectures, the gorgeous riddles and the numerous mathematical discussions with Imre meant so much to me.

I am very indebted to Professor Paul Balister, Professor Robert Morris and Dr Julian Sahasrabudhe, for taking me alongside them on numerous lovely projects. Being surrounded for the last several years by such superb mathematicians has had a profound influence on my development.

I need to thank my dear friend Peter van Hintum for all the wonderful moments we shared throughout our PhD. I have been extraordinarily fortunate to find a companion with whom I bonded on so many levels and, in particular, with whom I had a common understanding of the aesthetic of mathematics.

I am deeply grateful to Professor Alessio Figalli for an exciting ongoing collaboration across mathematical disciplines. I have benefited greatly from his enthusiasm to bridge different areas of mathematics and his eye for beautiful ideas.

I owe a great debt of gratitude to my contemporary academic siblings: Hunter Spink for numerous projects and valuable conversations; Ivailo Hartarsky and Vojtěch Dvořák for a couple of very enjoyable collaborations; Stefan David for many interesting discussions, explorations and projects gathered over the course of two decades.

I want to express my heartfelt appreciation to Professor Tadashi Tokieda and Professor Thomas Körner for their close guidance throughout my undergraduate studies and to Dan Schwarz, Adrian Zanoschi and Gabriel Popa for nurturing my passion for mathematics in school.

I want to thank my girlfriend Andreea-Cristina Stanciu for bringing so much joy into my life. I am truly lucky to have such an amazing partner by my side whose emotional support has no limits.

Finally, I need to thank my mother Cornelia-Livia Bejan and grandmother Georgeta-Paula Dimitriu for everything. Your infinite love and care meant to me more than words could ever describe. I will always love you and hope one day I can reciprocate the support.

Through Béla's generosity, I enjoyed numerous fantastic research visits in places I had never expected to see otherwise. Thus, I would like to thank the University of Memphis, IMT Institute for Advanced Studies in Lucca and Instituto de Matemática Pura e Aplicada in Rio for their hospitality. My PhD would have been impossible without the support from Trinity Hall and from DPMMS and EPSRC, for all of which I'm very grateful.

Abstract

This thesis consists of an introduction and nine chapters, each devoted to a different combinatorial problem. What gives these problems coherence is that although at first sight they look very different, the essential difficulties in most are geometric.

In Chapters 2 and 3, we investigate sumset inequalities. For sets A, B in an abelian group G, define the sumset $A+B = \{a+b : a \in A, b \in B\}$. We consider the ambient groups $\mathbb{Z}^k$ and $\mathbb{R}^k$, endowed with the measure $|\cdot|$ representing cardinality and outer Lebesgue measure, respectively. One of the central questions in additive combinatorics is the inverse sumset problem of characterizing the finite subsets A with small *doubling constant* $|A+A| \cdot |A|^{-1}$. Sets A in $\mathbb{R}^k$ have doubling constant at least 2^k; this is no longer true for sets A in $\mathbb{Z}^k$, unless some *non-degeneracy* condition is imposed. Our main result describes the structure of *non-degenerate* sets A in $\mathbb{Z}^k$ with doubling constant close to 2^k.

We prove a sharp stability result for the classical Freiman–Bilu 2^k-inequality (improved by Green–Tao) and a generalization of Freiman's $3k-4$ theorem to arbitrary dimension. For $\delta > 0$ sufficiently small and $A \subset \mathbb{Z}^k$ with $|A+A| \leqslant (2^k+\delta)|A|$, we show either A is covered by $m_k(\delta)$ parallel hyperplanes, or it satisfies $|\widehat{\mathrm{co}}(A) \setminus A| \leqslant c_k \delta |A|$, where $\widehat{\mathrm{co}}(A)$ is the smallest convex progression (convex set intersected with an affine sub-lattice) containing A.

In particular, we deduce the sharp stability result for the Brunn–Minkowski inequality for equal sets, conjectured by Figalli and Jerison. Given $\delta > 0$ sufficiently small and $A \subset \mathbb{R}^k$ with $|A+A| \leqslant (2^k+\delta)|A|$, we show $|\mathrm{co}(A) \setminus A| \leqslant c_k \delta |A|$, where $\mathrm{co}(A)$ is the smallest convex set containing A.

Finally, we prove a strengthen version of the aforementioned conjecture by Figalli and Jerison for a specific class of geometric objects. We find the optimal constants c_k, when $A \subset \mathbb{R}^k$ is the hypograph of a function defined on a convex domain in dimension $k-1 \leqslant 3$.

In Chapters 4, 5 and 6, we investigate covering systems. A covering system is a finite collection of arithmetic progressions $\{a_1 \pmod{m_1}, a_2 \pmod{m_2}, \ldots, a_k \pmod{m_k}\}$ that cover the integers, i.e., $\cup_i \{a_i + n m_i : n \in \mathbb{Z}\} = \mathbb{Z}$. Since their introduction by Erdős in 1950, covering systems have been extensively studied, and numerous questions and conjectures have been posed regarding the existence of covering systems with various properties. More than fifty years ago, Erdős asked if the moduli can be distinct and all arbitrarily large, Erdős and Selfridge asked if the moduli can be distinct and all odd, and Schinzel conjectured that in any covering system there exists a pair of moduli, one of which divides the other.

Another beautiful conjecture, proposed by Erdős and Graham in 1980, states that if the moduli are distinct elements of the interval $[n, Cn]$, and n is sufficiently large, then the density of integers

uncovered by the union is bounded below by a constant (depending only on C). This conjecture was confirmed (in a strong form) by Filaseta, Ford, Konyagin, Pomerance and Yu in 2007, who moreover asked whether the same conclusion holds if the moduli are distinct and sufficiently large, and $\sum_{i=1}^{k} \frac{1}{d_i} < C$. Although, as it turns out, this condition is not sufficiently strong to imply the desired conclusion, as one of the main results of this paper we give an essentially best possible condition which is sufficient. More precisely, we show that if all of the moduli are sufficiently large, then the union misses a set of density at least $e^{-4C}/2$, where

$$C = \sum_{i=1}^{k} \frac{\mu(d_i)}{d_i}$$

and μ is a multiplicative function defined by $\mu(p^i) = 1 + (\log p)^{3+\varepsilon}/p$ for some $\varepsilon > 0$. We also show that no such lower bound (i.e., depending only on C) on the density of the uncovered set holds, when $\mu(p^i)$ is replaced by any function of the form $1 + O(1/p)$.

Our method has a number of further applications. Most importantly, we prove the conjecture of Schinzel. In addition, we give an alternative (somewhat simpler) proof of a breakthrough result of Hough, who resolved Erdős' minimum modulus problem, with an improved bound on the smallest difference. Moreover, we make further progress on the problem of Erdős and Selfridge, which, in particular, we solve in the square free case.

Finally, we answer another natural question of Erdős, asked in 1952, on the *number* of minimal covering systems. Addressing this question requires very different methods. More precisely, we show that the number of minimal covering systems with exactly n elements is

$$\exp\left(\left(\frac{4\sqrt{\tau}}{3} + o(1)\right)\frac{n^{3/2}}{(\log n)^{1/2}}\right)$$

as $n \to \infty$, where

$$\tau = \sum_{t=1}^{\infty} \left(\log \frac{t+1}{t}\right)^2.$$

En route to this counting result, we obtain a structural description of all covering systems that are close to optimal in an appropriate sense.

Chapter 7 is devoted to Littlewood polynomials. Answering a question of Erdős from 1957 and confirming a conjecture of Littlewood from 1966, we show that there exist absolute constants $\Delta > \delta > 0$ such that, for all $n \geqslant 2$, there exists a polynomial P of degree n, with coefficients in $\{-1, 1\}$, such that

$$\delta\sqrt{n} \leqslant |P(z)| \leqslant \Delta\sqrt{n}$$

for all $z \in \mathbb{C}$ with $|z| = 1$. Over time, this problem attracted considerable attention and, in particular, Littlewood included it in his well-known monograph containing 30 of his favourite problems.

Chapter 8 is devoted to judicious partitions. Bollobás, Reed and Thomason proved that every 3-uniform hypergraph with m hyperedges has a vertex-partition into 3 parts such that each part meets at least $\frac{1}{3}(1-\frac{1}{e})m$ hyperedges. Halsegrave optimized the value to $0.6m$, confirming a special case of a conjecture of Bollobás and Thomason from 1993. For large values of m, Ma and Yu improved the bound asymptotically to $0.65m+o(m)$. We further improve this asymptotic bound to $\frac{19}{27}m+o(m)$, which is best possible up to the error term, resolving the next open case of a conjecture of Bollobás and Scott from 2000.

Chapter 9 is devoted to coloured structures in the Boolean lattice. Given a collection of coloured chain posets, we estimate the number of coloured subsets of the Boolean lattice which avoid all chains in this collection. Our proof relies on the recent powerful method of hypergraph containers developed independently by Balogh, Morris and Samotij as well as Saxton and Thomason, inspired by the previous work of Conlon and Gowers. In order to prove results about coloured chain posets we need to apply the hypergraph container lemma recursively, with different uniformities at each stage, using a balanced supersaturation result for a certain non-uniform hypergraph encoding forbidden configurations. Our work extends to a coloured setting the previous work of Collares and Morris as well as Balogh, Mycroft, and Treglown on the number of antichains in a random subset of the Boolean lattice.

Chapter 10 is devoted to positional games. For two graphs B and H the strong Ramsey game $\mathscr{R}(B,H)$ on the board B and with target H is played as follows. Two players alternately claim edges of B. The first player to build a copy of H wins. If none of the players win, the game is declared a draw. A notorious open question of Beck asks whether the first player has a winning strategy in $\mathscr{R}(K_n,K_k)$ in bounded time as $n\to\infty$. Surprisingly, in a recent paper Hefetz, Kusch, Narins, Pokrovskiy, Requilé and Sarid constructed a 5-uniform hypergraph $\mathscr{H}$ for which they proved that the first player does not have a winning strategy in $\mathscr{R}(K_n^{(5)},\mathscr{H})$ in bounded time. They naturally asked whether an analogous result holds for graphs. We make further progress towards this question.

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Chapter 1

Introduction

This thesis is organised into ten chapters of which this introduction is the first. Chapters 2 and 3 investigate the stability of sumset inequalities. Chapters 4, 5 and 6 investigate covering systems. Chapter 7 is devoted to Littlewood polynomials. Chapter 8 is devoted to judicious partitions. Chapter 9 is devoted to coloured structures in the Boolean lattice. Chapter 10 is devoted to positional games.

1.1 Stability of sumset inequalities

The results in Chapters 2 and 3 are all joint with Peter van Hintum and Hunter Spink [95–98].

The fundamental notion that we explore here is that of *sumset*. Given two subsets A and B of an abelian group G, we define their *sumset*

$$A+B=\{a+b : a\in A, b\in B\}.$$

The ambient groups G that we consider are $\mathbb{Z}^k$ and $\mathbb{R}^k$. In the former case $|\cdot|$ refers to cardinality and in the latter case $|\cdot|$ refers to the outer Lebesgue measure.

1.1.1 Discrete Setting

In the discrete setting, the most basic sumset inequality says that if A and B are non-empty finite subsets of $\mathbb{Z}$ then

$$|A+B|\geqslant |A|+|B|-1$$

and equality holds if and only if A and B are arithmetic progressions with the same common difference or one of them is a singleton. Cauchy [41] was the first to study sumsets in $\mathbb{Z}_p$ rather than $\mathbb{Z}$, establishing a corresponding statement there. Over one hundred years later, his result was rediscovered by Davenport [48, 49] and was further generalized by Kneser [106] to abelian groups.

One of the central questions in additive combinatorics is the inverse sumset problem of characterizing the finite subsets A of abelian groups with small *doubling constant* $|A+A|\cdot|A|^{-1}$.

We consider the inverse sumset problem in torsion-free abelian groups $\mathbb{Z}^k$, which has been studied from a variety of perspectives by Freiman [79, 80], Green and Tao [85], Chang [42], and Sanders [133] among others. However, we remark that the setting of arbitrary abelian groups has been studied by Green and Ruzsa [83]; the setting of non-abelian groups represents the scope of the recent literature on approximate groups which culminates with the work of Breuillard, Green and Tao [34].

In his seminal work [79], Freiman gave the first qualitative answer to the inverse sumset problem in $\mathbb{Z}^k$. There are two closely related formulations of Freiman's theorem of which one relies on the notion of *generalized arithmetic progressions* and the other relies on the notion of *convex progressions*.

Definition 1.1.1. An *r-dimensional generalized arithmetic progression* B in $\mathbb{Z}^k$ is a subset of the form $B = b + \phi(\mathbb{Z}^r \cap K)$, where $b \in \mathbb{Z}^k$, K is a axis-aligned box in $\mathbb{R}^r$ and $\phi : \mathbb{Z}^r \to \mathbb{Z}^k$ is a homomorphism. Equivalently, an r-dimensional generalized arithmetic progression B in $\mathbb{Z}^k$ is a subset of the form

$$B(n_1, \ldots, n_r; v_1, \ldots, v_r; b) := \left\{ b + \sum_{i=1}^{r} \ell_i v_i : 0 \leqslant \ell_i < n_i \right\},$$

where $n_1, \ldots, n_r \in \mathbb{N}$ and $b, v_1, \ldots, v_r \in \mathbb{Z}^k$.

More generally, an *r-dimensional convex progression* P in $\mathbb{Z}^k$ is a subset of the form $P = p + \phi(\mathbb{Z}^r \cap K)$, where $p \in \mathbb{Z}^k$, K is a *convex*[1] body in $\mathbb{R}^r$ and $\phi : \mathbb{Z}^r \to \mathbb{Z}^k$ is a homomorphism.

We also define a notion of degeneracy, which turns out to be central in our results.

Definition 1.1.2. We call B a *non-degenerate* r-dimensional generalized arithmetic progression and P a *non-degenerate* r-dimensional convex progression in $\mathbb{Z}^k$ if additionally $\phi : \mathbb{Z}^r \to \mathbb{Z}^k$ is an injective homomorphism.

Equivalently, a non-degenerate r-dimensional generalized arithmetic progression B in $\mathbb{Z}^k$ is a subset of the form $B(n_1, \ldots, n_r; v_1, \ldots, v_r; b) := \{b + \sum_{i=1}^{r} \ell_i v_i : 0 \leqslant \ell_i < n_i\}$, where $n_1, \ldots, n_r \in \mathbb{N}$ and $b, v_1, \ldots, v_r \in \mathbb{Z}^k$ and $v_1, \ldots, v_r$ are linearly independent.

A non-degenerate r-dimensional convex progression P in $\mathbb{Z}^k$ is a subset of the form $P = \Lambda \cap K$ where Λ is an r-dimensional *affine sub-lattice*[2] in $\mathbb{Z}^k$ and K is a convex body in $\mathbb{R}^k$.

We digress to justify the equivalence above. Say $P = p + \phi(\mathbb{Z}^r \cap K)$, where $p \in \mathbb{Z}^k$, K is a convex body in $\mathbb{R}^r$ and $\phi : \mathbb{Z}^r \to \mathbb{Z}^k$ is an injective homomorphism. We can extend ϕ to an injective linear map $\phi : \mathbb{R}^r \to \mathbb{R}^k$. Then $P = p + \phi(\mathbb{Z}^r \cap K) = (p + \phi(\mathbb{Z}^r)) \cap (p + \phi(K))$, where $p + \phi(\mathbb{Z}^r)$ is an affine sub-lattice and $p + \phi(K)$ is convex. Conversely, say $P = \Lambda \cap K$ where Λ is an r-dimensional affine sub-lattice and K is convex. By the fundamental theorem of finitely generated abelian groups, it follows that $\Lambda = p + \phi(\mathbb{Z}^r)$, where $\phi : \mathbb{Z}^r \to \mathbb{Z}^k$ is an injective homomorphism. We can extend ϕ to an injective linear map $\phi : \mathbb{R}^r \to \mathbb{R}^k$. We conclude $P = \Lambda \cap K = p + (-p + \Lambda) \cap (-p + K) = p + \phi(\mathbb{Z}^r) \cap (-p + K) = p + \phi(\mathbb{Z}^r \cap \phi^{-1}(-p + K))$, and $\phi^{-1}(-p + K)$ is a convex set in $\mathbb{R}^r$.

[1] Set $K \subset \mathbb{R}^r$ is *convex* if $pX + (1-p)X = X$ for all $p \in [0, 1]$.

[2] An *affine sub-lattice* in $\mathbb{Z}^k$ is a coset of the additive group $\mathbb{Z}^k$.

Note that any k-dimensional convex progression in $\mathbb{Z}^k$ that is not contained in a single hyperplane is non-degenerate. Conversely, any non-degenerate convex progression in $\mathbb{Z}^k$ is (at most) k-dimensional (and we can always view it as being exactly k-dimensional).

Freiman's theorem states that for any $\lambda > 0$ there exist $b_1(\lambda), b_2(\lambda), b_3(\lambda) > 0$ such that the following holds. Any finite subset $A \subset \mathbb{Z}^k$ with doubling constant less than λ can be covered by $b_1(\lambda)$ translates of a generalized arithmetic progression (or alternatively convex progression) of size at most $b_2(\lambda)|A|$, and of dimension at most $b_3(\lambda)$.

Freiman introduced the notion of convex progression in his original work and much of the literature builds on this notion. In particular, generalizations of convex progressions were used implicitly in results of Bourgain [31] and Green and Sanders [84] (see Sander's extensive survey [133] for more information). We note that many authors require convex progressions to be symmetric, but here we impose no such assumptions.

Motivated by Freiman's theorem, much of the study of inverse sumset problems has focused on the following quantitative question.

Question 1.1.3. In Freiman's theorem, what are the *optimal* parameters $b_1(\lambda), b_2(\lambda), b_3(\lambda)$?

Of course, the optimal parameters should vary, depending on how we prioritize them. Moreover, they might vary between the two formulations of Freiman's theorem, i.e., using generalized arithmetic progressions or using convex progressions.

Firstly, say we want to minimize the maximum of $b_1(\lambda), b_2(\lambda), \exp(b_3(\lambda))$. In Freiman's formulation using generalized arithmetic progressions, Lovett and Regev [118] showed that the parameters $b_1(\lambda)$, $b_2(\lambda)$, and $\exp(b_3(\lambda))$ cannot all be brought down to polynomial. However, in Freiman's formulation using (symmetric) convex progressions, Freiman and Ruzsa [131] conjectured that such a polynomial bound exists. In spite of several serious attempts, the *polynomial Freiman–Ruzsa conjecture* remains open. Green and Ruzsa [83], Chang [42], Bourgain [31] and Green and Tao [86] showed the parameters could be reduced to $b_1(\lambda) = b_2(\lambda) = \exp(b_3(\lambda)) = \exp(O(\lambda^C))$ for some constant C. This was improved by Schoen [137] to $\exp(\exp(O(\sqrt{\log(\lambda)})))$, and finally improved by Sanders [133] to $\exp(O(\log^{3+o(1)} \lambda))$.

Secondly, say we want to reduce the dimension $b_3(\lambda)$ and the size $b_2(\lambda)$ at the cost of the number of translates $b_1(\lambda)$. In Freiman's formulation using generalized arithmetic progressions, Green and Tao [85] showed that for $\lambda \in [2^r, 2^{r+1} - \delta)$ we may take $b_2(\lambda) = 1$ and $b_3(\lambda) = \lfloor \log_2(\lambda) \rfloor$, which are optimal, and $b_1(\lambda) = \exp(\lambda^{O_\delta(1)})$. This builds on the earlier work of Freiman and Bilu [23, 80].

The Green–Tao improvement of Freiman–Bilu 2^k-inequality essentially states that for any $\delta > 0$, there is a constant $m_k(\delta)$ such that given $A \subset \mathbb{Z}^k$, either $|A + A| \geqslant (2^k - \delta)|A|$ or A is covered by $m_k(\delta)$ parallel hyperplanes.

This suggests that the correct notion of degeneracy for a set A in $\mathbb{Z}^k$ is being covered by a bounded number of parallel hyperplanes, and non-degenerate sets A have doubling constant bounded below by roughly 2^k. This reflects the continuous analogue for $A \subset \mathbb{R}^k$. There is a large literature of classifications of subsets $A \subset \mathbb{Z}^k$ with doubling at most $2^k - \delta$ (see, e.g., Fishburn [77], Freiman [80], Grynkiewicz and Serra [87], and Stanchescu [144–147]). However, little is known beyond the 2^k threshold.

Finally, say we want to reduce the dimension $b_3(\lambda)$ and the number of translates $b_1(\lambda)$ at the cost of the size $b_2(\lambda)$. Observe that we always have $b_1(\lambda) \geqslant 1$ and $b_3(\lambda) \geqslant \lfloor \log_2(\lambda) \rfloor$. Indeed, fix $r \in \mathbb{N}$ and choose $n \in \mathbb{N}$ large. Note that the set $[n]^r$ has doubling constant less than 2^r. Also, $[n]^r$ is not covered by a bounded number of parallel hyperplanes and thus by a bounded number of translates of a generalized arithmetic progressions of dimension $r-1$. Restricting the scope to non-degenerate sets $A \subset \mathbb{Z}^k$ with doubling $\lambda \approx 2^k$, gives rise to the following qualitative and quantitative questions.

Question 1.1.4. In Freiman's theorem, is it possible to take $b_1(\lambda) = 1$ and $b_3(\lambda) = \lfloor \log_2(\lambda) \rfloor$? If so what is the optimal value of $b_2(\lambda)$?

Marking the beginning of the study of inverse problems in additive combinatorics, the classical Freiman $3|A|-4$ theorem [80] answers Question 1.1.4 in $\mathbb{Z}$, opening the path to $\mathbb{Z}^k$.

Freiman's $3|A|-4$ theorem states that if A is a subset of the integers with $|A+A| \leqslant 3|A|-4$, then A is contained in an arithmetic progression P with $|P \setminus A| \leqslant |A+A| - 2|A| + 1$.

Results

Our main result in the discrete setting [96] gives a complete answer to Question 1.1.4 in $\mathbb{Z}^k$, i.e., we establish a sharp stability result for Freiman–Bilu 2^k-inequality (improved by Green and Tao [85]) and we obtain a natural generalization of Freiman's $3|A|-4$ theorem in arbitrary dimension.

Theorem 1.1.5.

a) For $k \in \mathbb{N}$, there exist positive constants $\Delta_k, m_k, \varepsilon_k$ such that the following holds. Given a subset A of $\mathbb{Z}^k$, if $|A+A| \leqslant (2^k + \Delta_k)|A|$, then either A is covered by m_k parallel hyperplanes or A is contained in a non-degenerate k-dimensional generalized arithmetic progression $B = B(n_1, \ldots, n_k; v_1, \ldots, v_k; b)$ with $v_1, \ldots, v_k$ linearly independent and $|A| \geqslant \varepsilon_k |B|$.

b) For $k \in \mathbb{N}$ and $\varepsilon_0 > 0$, there exist positive constants $\Delta_k(\varepsilon_0)$, $g_k(\varepsilon_0)$ and $c_k < (4k)^{5k}$ such that the following holds. Given a subset A of a non-degenerate k-dimensional generalized arithmetic progression $B = B(n_1, \ldots, n_k; v_1, \ldots, v_k; b) \subset \mathbb{Z}^k$ with $v_1, \ldots, v_k$ linearly independent, if $|A| \geqslant \varepsilon_0 |B|$ and also $|A+A| \leqslant (2^k + \Delta_k(\varepsilon_0))|A|$, then A is contained inside a non-degenerate k-dimensional convex progression P with

$$|P \setminus A| \leqslant c_k(|A+A| - 2^k|A|) + g_k(\varepsilon_0) \min\{n_i\}^{-\frac{1}{1+\frac{1}{2}(k-1)\lfloor k/2 \rfloor}} |A|.$$

Corollary 1.1.6. For $k \in \mathbb{N}$, there exist positive constants $c_k < (4k)^{5k}$, Δ_k and positive constants $m_k(\delta)$ for all $\delta \in (0, \Delta_k]$ such that the following holds. Given a subset A of $\mathbb{Z}^k$, if $|A+A| \leqslant (2^k+\delta)|A|$, then either A is covered by $m_k(\delta)$ parallel hyperplanes or A is contained inside a non-degenerate k-dimensional convex progression P with

$$|P \setminus A| \leqslant c_k \delta |A|.$$

Corollary 1.1.7. For $k \in \mathbb{N}$, there exist positive constants $c_k < (4k)^{5k}$ and Δ_k such that for all $\delta \in (0, \Delta_k]$ the following holds. Given a bounded open subset $\widetilde{A}$ of $\mathbb{R}^k$, if $|\widetilde{A}+\widetilde{A}| \leqslant (2^k+\delta)|\widetilde{A}|$, then $\widetilde{A}$ is contained inside a convex set $\widetilde{C}$ with

$$|\widetilde{C} \setminus \widetilde{A}| \leqslant c_k \delta |\widetilde{A}|.$$

Theorem 1.1.5, together with its corollaries, is the subject of Chapter 2, which is adapted from [96].

1.1.2 Continuous Setting

In the continuous setting, the Brunn–Minkowski inequality says that if A, B are measurable[3] subsets of $\mathbb{R}^k$ with positive measure, then

$$|A+B|^{\frac{1}{k}} \geqslant |A|^{\frac{1}{k}} + |B|^{\frac{1}{k}},$$

and equality holds if and only if A and B are *homothetic*[4] *convex*[5] sets (less a measure 0 set). Stability results for the Brunn–Minkowski inequality quantify how close A, B are to homothetic convex sets when $|A+B|^{\frac{1}{k}} \approx |A|^{\frac{1}{k}} + |B|^{\frac{1}{k}}$. The most natural formulations of these results rely on the following notions:

- The *Brunn–Minkowski deficit* is defined to be

$$\delta = \delta(A,B) = \frac{|A+B|^{\frac{1}{k}}}{|A|^{\frac{1}{k}} + |B|^{\frac{1}{k}}} - 1.$$

- The *normalised volume ratio* is defined to be

$$t = t(A,B) = \frac{\min(|A|^{\frac{1}{k}}, |B|^{\frac{1}{k}})}{|A|^{\frac{1}{k}} + |B|^{\frac{1}{k}}}.$$

- The *Fraenkel asymmetry index* is defined to be

$$\alpha(A,B) = \inf_{x \in \mathbb{R}^n} \frac{|A \triangle (s \cdot \mathrm{co}(B) + x)|}{|A|},$$

  where s satisfies $|A| = |s \cdot \mathrm{co}(B)|$ and $\mathrm{co}(B)$ is the convex hull of B, i.e., the smallest convex set containing B.

- The ω*-index* introduced by Figalli and Jerison [69] is defined to be

$$\omega = \omega(A,B) = \inf_{\substack{K_A \supset A,\ K_B \supset B \\ K_A, K_B \text{ homothetic convex sets}}} \frac{|K_A \setminus A|}{|A|} + \frac{|K_B \setminus B|}{|B|}.$$

Question 1.1.8 below, regarding the stability of the Brunn–Minkowski inequality for sets A, B restricted to various sub-classes of geometric objects, is one of the central open problems in the study of geometric inequalities. It has been explored intensively in recent years by Barchiesi and Julin [11], Carlen and Maggi [39], Christ [44], Figalli and Jerison [67, 69, 71], Figalli, Maggi and Mooney [72] and Figalli, Maggi and Pratelli [73, 74].

[3] It generally suffices to consider finite unions of axis aligned cubes

[4] Sets $X, Y \subset \mathbb{R}^k$ are *homothetic* if $X = v + \lambda Y$ for some $v \in \mathbb{R}^k, \lambda > 0$

[5] Set $X \subset \mathbb{R}^k$ is *convex* if $pX + (1-p)X = X$ for all $p \in [0,1]$

Question 1.1.8. Do there exist functions $f, g : (\mathbb{R}^+)^2 \to \mathbb{R}^+$ such that

$$\lim_{x\to 0} f(x, y_0) = 0 \text{ and } \lim_{x\to 0} g(x, y_0) = 0,$$

$$\alpha \leqslant f(\delta, t) \text{ and } \omega \leqslant g(\delta, t),$$

and if so, what are the optimal functions f, g?

Christ [44] pioneered a qualitative stability result for the Brunn–Minkowski inequality for arbitrary sets A, B. He proved via a compactness argument that for fixed $t > 0$ we have $\alpha \to 0$ as $\delta \to 0$. When A and B are convex, the optimal result $\alpha \leqslant C_k \tau^{-\frac{1}{2}} \delta^{\frac{1}{2}}$ was obtained by Figalli, Maggi, and Pratelli in [73, 74]. When B is a ball and A is arbitrary, the optimal result $\alpha \leqslant C_k \tau^{-\frac{1}{2}} \delta^{\frac{1}{2}}$ was obtained by Figalli, Maggi, and Mooney in [72]. We note that this particular case is intimately connected with stability for the isoperimetric inequality. When just B is convex the (sub-optimal) result $\alpha \leqslant C_k \tau^{-(k+\frac{3}{4})} \delta^{\frac{1}{4}}$ was obtained by Carlen and Maggi in [39]. Finally, Barchiesi and Julin [11] showed that when just B is convex, we have the optimal result $\alpha \leqslant C_k \tau^{-\frac{1}{2}} \delta^{\frac{1}{2}}$, subsuming these previous results. As it turns out, providing an upper bound for ω is stronger than providing an upper bound for α as we always have $\alpha \leqslant 2\omega$.

In a landmark paper [69], Figalli and Jerison established the first (sub-optimal) quantitative stability result for the Brunn–Minkowski inequality for arbitrary sets A, B.

They showed that there exist computable constants $a_k(\tau), b_k, c_k, d_k(\tau) > 0$ such that the following is true. Given measurable sets A, B in $\mathbb{R}^k$ with $t \in [\tau, 1-\tau]$ and $\delta \leqslant d_k(\tau)$ we have

$$\omega \leqslant c_k \tau^{-b_k} \delta^{a_k(\tau)}.$$

The exponents a_k, b_k obtained in [69] are double exponential in the dimension k. This naturally gives rise to the following question proposed by Figalli and Jerison, which attracted considerable attention lately.

Question 1.1.9. What are the optimal exponents a_k, b_k, prioritized in this order?

The exponents a_k, b_k are prioritized in this order because if the inequality holds for (a_k, b_k), then the inequality also holds for (a'_k, b'_k) whenever $a_k > a'_k$ by taking $d'_k(\tau)$ sufficiently small.

Before their general result [69] for arbitrary sets A, B, Figalli and Jerison [67] had considered the case of equal sets $A = B$ and gave a polynomial upper bound $\omega \leqslant C_k \delta^{a_k}$. Later, in [71], they conjectured the sharp bound $\omega \leqslant C_k \delta$ when $A = B$, and proved it in dimensions 2 and 3 using an intricate analysis which unfortunately does not extend to higher dimensions. Afterwards, Figalli and Jerison suggested a stronger conjecture that $\omega \leqslant C_k \tau^{-1} \delta$ for A, B homothetic regions [70].

Results

Our first result in the continuous setting [98] proves this last conjecture by Figalli and Jerison [71], i.e., we establish a sharp stability result for the Brunn–Minkowski inequality for homothetic sets.

Theorem 1.1.10. For all $k \geqslant 2$ there exist computable constants $c_k, d_k(\tau) > 0$ such that the following is true. Given measurable homothetic sets A, B in $\mathbb{R}^k$ with $t \in [\tau, 1-\tau]$ and $\delta \leqslant d_k(\tau)$, we have

$$\omega \leqslant c_k \tau^{-1} \delta.$$

For these optimal exponents we show that $\exp(\Omega(k)) \leqslant c_k \leqslant \exp(O(k))$.

The direct proof of Theorem 1.1.10 is not included in this thesis as it is included in Peter van Hintum's thesis. However, our main result in the discrete setting [96] can be seen as a serious extension of the conjecture of Figalli and Jerison [71]. In the case of equal sets, Theorem 1.1.10 is equivalent to Corollary 1.1.7 of Theorem 1.1.5, which is the subeject of Chapter 2, as we already mentioned.

The paper [98] has been accepted for publication in the Journal of the European Mathematical Society.

Our second result in the continuous setting [97] proves a strengthen version of this last conjecture by Figalli and Jerison [71] for a specific class of geometric objects, i.e., we establish a sharp stability result for a linear Brunn–Minkowski inequality for *hypographs* of functions in low dimensions.

Given a compact convex set C in $\mathbb{R}^k$ and a function $f : C \to \mathbb{R}^+$, the *hypograph* of f is defined to be the set

$$A_f = \{(x, y) \in C \times \mathbb{R} : 0 \leqslant y \leqslant f(x)\}.$$

If $f, g : C \to \mathbb{R}^+$ are bounded measurable functions, then the hypographs satisfy the following linear Brunn–Minkowski inequality

$$\left| \frac{A_f + A_g}{2} \right| \geqslant \frac{|A_f| + |A_g|}{2}.$$

Restricting to hypographs of functions defined on a convex domain in low dimensions, we extend the conjecture of Figalli and Jerison [71] in two ways. First, we establish an analogue for two distinct sets. Second, we establish an analogue for arbitrary many copies of the same set. This also relates to the study of how close a Minkowski sum is to its convex hull (see [149, 54, 129, 78]).

Theorem 1.1.11. If $f, g : C \to \mathbb{R}^+$ are bounded measurable functions with $C \subset \mathbb{R}^k$ a compact convex domain and $k \leqslant 3$ then

$$\left(\left| \frac{A_f + A_g}{2} \right| - \frac{|A_f| + |A_g|}{2} \right) \geqslant \frac{k+1}{2^{k+1}} |\operatorname{co}(A_f) \setminus A_f|,$$

and the constant $\frac{k+1}{2^{k+1}}$ is optimal.

Theorem 1.1.12. If $f : C \to \mathbb{R}^+$ is a bounded measurable function with $C \subset \mathbb{R}^k$ a compact convex domain and $k \leqslant 3$ then

$$\left(\left| \frac{A_f + \ldots + A_f}{n} \right| - |A_f| \right) \geqslant c_{k,n} |\operatorname{co}(A_f) \setminus A_f|,$$

with

$$c_{k,n} = \begin{cases} \frac{n-1}{n} & k = 1 \\ \frac{(2n-1)(n-1)}{2n^2} & k = 2 \\ \frac{(n-1)^2}{n^2} & k = 3, \end{cases}$$

and the constants $c_{k,n}$ are optimal.

Theorem 1.1.11 and Theorem 1.1.12 are the subject of Chapter 3, which is adapted from [97].

Our third result in the continuous setting [95] answers Question 1.1.9 in the plane, i.e., we establish a sharp stability result for the Brunn–Minkowski inequality for planar sets. The proof of this result is not included in this thesis as it is included in Peter van Hintum's thesis.

Theorem 1.1.13. There exist computable constants $c, d(\tau) > 0$ such that the following is true. Given measurable sets A, B in $\mathbb{R}^2$ with $t \in [\tau, 1-\tau]$ and $\delta \leqslant d(\tau)$, we have

$$\omega \leqslant c\tau^{-\frac{1}{2}} \delta^{\frac{1}{2}}.$$

The paper [95] has been accepted for publication in the Journal of the European Mathematical Society.

1.2 Covering systems

The results in Chapters 4, 5 and 6 are all joint with Paul Balister, Béla Bollobás, Robert Morris and Julian Sahasrabudhe [6, 3, 5].

The fundamental notion explored here is that of *covering systems*. A *covering system* is a finite collection $A_1, \dots, A_k$ of arithmetic progressions that cover the integers, i.e., that satisfy $\bigcup_{i=1}^{k} A_i = \mathbb{Z}$.

Throughout the thesis we exclude the trivial arithmetic progression $\mathbb{Z}$.

1.2.1 Density

The study of covering systems with distinct differences (or *moduli*) $1 < d_1 < \cdots < d_k$ was initiated in 1950 by Erdős [55], who used them to answer a question of Romanoff. Since their introduction, covering systems have been extensively studied, and numerous questions and conjectures have been posed regarding the existence of covering systems with various properties.

For example, Erdős [55] asked whether there exist such systems with minimum modulus arbitrarily large, Erdős and Selfridge (see, e.g., [75]) asked if there exists a covering system with all moduli distinct and odd, and Schinzel [136] conjectured that in any covering system there exist a pair of moduli, one of which divides the other. In 1980, Erdős and Graham [60] initiated the study of the density of the uncovered set; in particular, they conjectured that if the (distinct) moduli $d_1, \dots, d_k$ all lie in the interval $[n, Cn]$, where $n \geqslant n_0(C)$ is sufficiently large, then the uncovered set has density at least ε for some $\varepsilon = \varepsilon(C) > 0$.

The first significant progress on these problems was made by Filaseta, Ford, Konyagin, Pomerance and Yu [76] in 2007, who proved (in a strong form) the conjecture of Erdős and Graham, and took an important step towards solving Erdős' minimum modulus problem. Moreover, Filaseta, Ford, Konyagin, Pomerance and Yu proposed the following natural and beautiful question.

Question 1.2.1. Is it true that for each $C > 0$, there exist constants $M > 0$ and $\varepsilon > 0$ such that the following holds: for every covering system whose distinct moduli satisfy

$$d_1, \dots, d_k \geqslant M \qquad \text{and} \qquad \sum_{i=1}^{k} \frac{1}{d_i} < C, \tag{1.1}$$

the uncovered set has density at least ε?

Building on their work, and in a remarkable breakthrough, Hough [100] resolved the minimum modulus problem in 2015, showing that in every covering system with distinct moduli, the minimum modulus is at most 10^{16}. The method of [100] was further refined by Hough and Nielsen [101], who used it to prove that every covering system with distinct differences contains a difference that is divisible by either 2 or 3. However, Hough's method does not appear to be strong enough to resolve the problem of Erdős and Selfridge, and it moreover gives little information about the density of the uncovered set, as asked by Filaseta, Ford, Konyagin, Pomerance and Yu.

Results

The purpose of Chapter 4 and 5 is to develop a general method, for bounding the density of the set left uncovered by a collection of arithmetic progressions. The work in these chapters is joint with Paul Balister, Béla Bollobás, Robert Morris and Julian Sahasrabudhe [6, 5]. [6] has been published in Inventiones mathematicae and [5] has been published in Algebra and Number Theory. Our central result gives a complete answer to Question 1.2.1 of Filaseta, Ford, Konyagin, Pomerance and Yu [76]. Our method, which is based on that of Hough [100], but is actually somewhat simpler, turns out to be sufficiently powerful and flexible to allow us to also make further progress on the problem of Erdős and Selfridge [75], and to prove Schinzel's conjecture [136].

We answer Question 1.2.1 negatively for every $C \geqslant 1$, by constructing (a sequence of) families of arithmetic progressions with arbitrarily large moduli, for which the density of the set missed by these arithmetic progressions is arbitrarily small, and sum of the reciprocals of the moduli is less than 1. However, we answer a variation of Question 1.2.1 positively by showing that for a collection of arithmetic progressions with distinct and sufficiently large moduli, the density of the set missed by these arithmetic progressions is controlled by a weighted sum of the reciprocals of the moduli.

Theorem 1.2.2. Let $\varepsilon > 0$ and let μ be the multiplicative function defined by

$$\mu(p^i) = 1 + \frac{(\log p)^{3+\varepsilon}}{p} \tag{1.2}$$

for all primes p and integers $i \geqslant 1$. There exists $M > 0$ so that if $A_1, \dots, A_k$ are arithmetic progressions with distinct moduli $d_1, \dots, d_k \geqslant M$, and

$$C = \sum_{i=1}^{k} \frac{\mu(d_i)}{d_i},$$

then the density of the uncovered set $R := \mathbb{Z} \setminus \bigcup_{i=1}^{k} A_i$ is at least $e^{-4C}/2$.

We remark that the question of Filaseta, Ford, Konyagin, Pomerance and Yu [76] corresponds to replacing μ by the constant function 1. As noted above, we show that the conclusion of the theorem does not hold under this weaker condition; in fact, we prove that the theorem is close to best possible in the following much stronger sense.

Theorem 1.2.3. For every $M > 0$ and $\varepsilon > 0$, there exists a finite collection of arithmetic progressions $A_1, \dots, A_k$ with distinct moduli $d_1, \dots, d_k \geqslant M$, such that

$$\sum_{i=1}^{k} \frac{1}{d_i} < 1 \tag{1.3}$$

and the density of the uncovered set $R = \mathbb{Z} \setminus \bigcup_{i=1}^{k} A_i$ is less than ε.

Theorem 1.2.4. For every $\lambda > 0$, there exists $C = C(\lambda) > 0$ such that the following holds. Let μ be the multiplicative function defined by

$$\mu(p^i) = 1 + \frac{\lambda}{p}$$

for all primes p and integers $i \geqslant 1$. For every $M > 0$ and $\varepsilon > 0$, there exists a finite collection of arithmetic progressions $A_1, \dots, A_k$ with distinct square-free moduli $d_1, \dots, d_k \geqslant M$, such that

$$\sum_{i=1}^{k} \frac{\mu(d_i)}{d_i} \leqslant C,$$

and the density of the uncovered set $R = \mathbb{Z} \setminus \bigcup_{i=1}^{k} A_i$ is at most ε.

It would be extremely interesting to characterize the functions μ such that, under the conditions of Theorem 1.2.2, the density of the uncovered set is bounded from below by a constant $\delta(C) > 0$ depending only on C.

Note that Hough's theorem [100] is an immediate consequence of Theorem 1.2.2. Our method of sieving is inspired by that of Hough [100], but is simpler in various important ways (for example, we do not need to appeal to the Lovász Local Lemma, and do not need his notion of quasi-randomness), and as a result we obtain a somewhat simpler proof of his theorem, with a better bound on the minimum difference.

Theorem 1.2.5. Let $\mathscr{A}$ be a finite collection of arithmetic progressions with distinct moduli $d_1, \dots, d_k \geqslant 616000$. Then $\mathscr{A}$ does not cover the integers.

Although our sieve is developed to control the density of the uncovered set, it turns out that it can be used to prove a number of additional interesting results about covering systems. We focus here on the two classical examples mentioned above: the question of Erdős and Selfridge, and the conjecture of Schinzel.

Over 50 years ago, Erdős and Selfridge (see [75] or [136]) asked whether or not there exist covering systems with distinct odd moduli.[6] This question appears to have first been asked by Erdős [57] in 1965, and a few years later he conjectured (see [58]) that there does exist such a system. In 1977 he went further, conjecturing [59] that there exist covering systems with square-free moduli, all of whose prime factors are arbitrarily large. On the other hand (as recounted, for example, in [75]), Selfridge believed that there do *not* exist such systems, and (perhaps as a result) the question has become known as the Erdős–Selfridge problem. Apart from its intrinsic appeal, the problem is motivated by a theorem of Schinzel [136], who discovered a connection between the non-existence of such covering systems and the irreducibility of certain polynomials. More precisely, he showed that if no covering system with distinct, odd moduli exists, then for every polynomial $f(x) \in \mathbb{Z}[X]$ with

[6] Moreover, as recounted in [75], Erdős (who thought that such coverings are likely to exist) offered \$25 for a proof that there is no covering with these properties, and Selfridge (who expected the opposite) offered \$300 (later increased to \$2000) for a construction of such a covering.

$f \not\equiv 1$, $f(0) \neq 0$ and $f(1) \neq -1$, there exists an infinite arithmetic progression of values of $n \in \mathbb{Z}$ such that $x^n + f(x)$ is irreducible over the rationals. He also showed that this would imply the following statement: in any covering system, one of the moduli divides another. We prove this latter statement, known as Schinzel's conjecture.

Theorem 1.2.6. If $\mathscr{A}$ is a finite collection of arithmetic progressions that covers the integers, then at least one of the moduli divides another.

Unfortunately, our method does not seem to be strong enough to resolve the Erdős–Selfridge problem (see the discussion in Section 4.6). However, it does allow us to make some further progress towards a solution; in particular, we can prove that no such covering system exists under the additional constraint that the moduli are square-free. This application of our sieve requires several additional (somewhat technical) ideas.

Theorem 1.2.7. If $\mathscr{A}$ is a finite collection of arithmetic progressions with distinct square-free moduli that covers the integers, then at least one of the moduli is even.

A different strengthening of the condition in the Erdős–Selfridge problem was considered recently by Hough and Nielsen [101], who showed that in any covering system with distinct moduli, one of the moduli is divisible by either 2 or 3. Their proof required careful optimization of their techniques, and it seems difficult to use it to strengthen their result. Using our methods, we give a short proof of the following strenthening of their theorem.

Theorem 1.2.8. Let $\mathscr{A} = \{A_d : d \in D\}$ be a finite collection of arithmetic progressions with distinct moduli that covers the integers, and let $Q = \operatorname{lcm}(D)$ be the least common multiple of the moduli. Then either $2 \mid Q$, or $9 \mid Q$, or $15 \mid Q$.

In other words, either there is an even d, a d divisible by $3^2 = 9$, or there are $d_1, d_2 \in D$ (possibly equal) with $3 \mid d_1$ and $5 \mid d_2$. We remark that we are unable to prove that a *single* $d \in D$ has $15 \mid d$ in this last case.

1.2.2 Number

We turn our attention to another problem on covering systems, whose study was initiated by Erdős [56] in 1952:

Question 1.2.9. How many minimal covering systems of size n are there?

Erdős [56] gave a simple proof that there are only finitely many minimal[7] covering systems of size n, but the bound he obtained on their number was doubly exponential. A more reasonable upper bound follows from a result of Simpson [140], who proved in 1985 (see Section 6.2) that the largest

[7] A covering system $\mathscr{A}$ is *minimal* if no proper subset of it covers $\mathbb{Z}$. Without this restriction there are infinitely many covering systems of size 2, since we can take $\mathscr{A} = \{\mathbb{Z}, A\}$ for any arithmetic progression A.

modulus in a minimal covering system of size n is at most 2^{n-1}. Note that this bound is best possible, since $\mathscr{A} = \{2^{i-1} \pmod{2^i} : i \in [n-1]\} \cup \{0 \pmod{2^{n-1}}\}$ is a minimal covering system, and that it easily implies that there are at most $2^{O(n^2)}$ minimal covering systems of size n. As it turns out, there are in fact fewer such systems, which naturally gives rise to the problem of determining asymptotically the logarithm of their number.

Results

In Chapter 6, we resolve Erdős' problem [56], i.e., we determine the number of minimal covering systems of a given size up to a $1+o(1)$ factor in the exponent. In doing so we obtain a structural description of all covering systems that are close to optimal in an appropriate sense. The work in this chapter is joint with Paul Balister, Béla Bollobás, Robert Morris and Julian Sahasrabudhe [3].

Theorem 1.2.10. The number of minimal covering systems of $\mathbb{Z}$ of size n is

$$\exp\left(\left(\frac{4\sqrt{\tau}}{3}+o(1)\right)\frac{n^{3/2}}{(\log n)^{1/2}}\right) \tag{1.4}$$

as $n \to \infty$, where

$$\tau = \sum_{t=1}^{\infty}\left(\log\frac{t+1}{t}\right)^2.$$

1.3 Littlewood polynomials

In Chapter 7, we answer a question of Erdős from 1957 [63] and confirm a conjecture of Littlewood from 1966 [112] on the existence of *flat Littlewood polynomials*. The work in this chapter is joint with Paul Balister, Béla Bollobás, Robert Morris and Julian Sahasrabudhe and is published in Annals of Mathematics [4].

A *Littlewood polynomial* is a polynomial all of whose coefficients are 1 or -1. This notion has been extensively studied over the past century (see [123]), originating in the work of Hardy and Littlewood [91], Bloch and Pólya [24] and Littlewood and Offord [109, 115, 116]. Inspired by a list of problems posed by Erdős [63], Littlewood [110–113] wrote a series of papers on the extremal properties of polynomials with restricted coefficients, where he made the following conjecture.

Conjecture 1.3.1. There exist constants $\Delta > \delta > 0$ such that, for all $n \geqslant 2$, there exists a Littlewood polynomial $P(z)$ of degree n with $\delta\sqrt{n} \leqslant |P(z)| \leqslant \Delta\sqrt{n}$ for all $z \in \mathbb{C}$ with $|z| = 1$.

The bounds are almost as good as one can hope for, given that for a Littlewood polynomial $P(z)$ of degree n the average value of $|P(z)|^2$ over the complex unit circle is $n+1$. Soon after its publication, the conjecture acquired great notoriety and resisted several attempts of proof e.g., [114]. This chapter settles Conjecture 1.3.1 in full generality.

1.4 Judicious partitions

In Chapter 8, we resolve the next open case of a conjecture of Bollobás and Scott [28] from 2000 on judicious partitions [28]. The work in this chapter is joint with Hunter Spink and is published in Random Structures and Algorithms [142].

Numerous classical partitioning problems in combinatorics ask for a single quantity to be maximized or minimized over the set of partitions of a combinatorial object such as a hypergraph. Perhaps the best known example is the Max-Cut problem that asks for the largest bipartite subgraph of a given graph. In contrast, judicious partitioning problems ask for various quantities to be maximized or minimized simultaneously. Their study was initiated by Bollobás and Scott in 1993 [26], who posed a number of beautiful judicious partitioning problems that attracted considerable attention over time. A well known conjecture of Bollobás and Scott [28] from 2000 states the following.

Conjecture 1.4.1. Every r-uniform hypergraph with m hyperedges has a vertex-partition in k parts such that each part meets at least $(1-(1-\frac{1}{k})^r)m+o(m)$ hyperedges, and this number is optimal.

This is intuitive because in a random partition in k parts of the vertices of a r-uniform hypergraph with m hyperedges, the expected number of hyperedges each part meets is $(1-(1-\frac{1}{k})^r)m$. Nevertheless, even the case $r=k=3$ of this conjecture resisted several attempts of proof [25, 28, 93, 119]. Actually, the original motivation for the case $r=k=3$ came from a related conjecture of Bollobás and Thomason with the quantity $(1-(1-\frac{1}{3})^3)m+o(m)$ replaced by the quantity $\frac{3m}{5}$. This chapter settles the next open case of Conjecture 1.4.1, i.e., the case $r=k=3$.

1.5 Coloured structures in the Boolean lattice

In Chapter 9, we determine the number of coloured subsets of the Boolean lattice avoiding certain coloured chains patterns up to a $1+o(1)$ factor in the exponent. The work in this chapter is joint with Hunter Spink and is published in the European Journal of Combinatorics [143].

Numerous well studied problems in the literature concern families of combinatorial objects with forbidden structures. The method of hypergraph containers, developed independently by Balogh, Morris and Samotij [7] and Saxton and Thomason [134], inspired by the previous work of Conlon and Gowers [46], has had a tremendous success in producing enumerative, structural and extremal results in a wide variety of settings (see [8]). For example, a classical result of Kleitman [105] estimates the number of antichains in the Boolean lattice $\mathscr{P}([n])$. Using the hypergraph container algorithm, independently Collares and Morris [45] and Balogh, Mycroft, and Treglown [9] were able to count the number of antichains in a random subset of $\mathscr{P}([n])$, from which they were able to deduce the approximate size of the largest antichain therein. This gives rise to the following analogous questions in a coloured setting.

Question 1.5.1. Fix colours $c_1,\dots,c_m$ and fix a collection $\mathscr{G}$ of coloured chain *posets* [8]. What is the number of coloured subsets X of $\mathscr{P}([n])$ with the property that X does not induce coloured chains of $\mathscr{G}$?

Question 1.5.2. Fix colours $c_1,\dots,c_m$, fix parameters $p_0,\dots,p_m>0$ with $p_0+\ldots+p_m=1$ and fix a collection $\mathscr{G}$ of coloured chain posets. Assume we colour each element of $\mathscr{P}([n])$ independently with colour c_i with probability p_i and erase it with probability $1-\sum p_i$, in order to produce a random coloured subset Y of $\mathscr{P}([n])$. What is the expected number of induced coloured subsets X of Y with the property that X does not induce coloured chains of $\mathscr{G}$?

In this chapter we answer these two questions in the *sparse setting*, i.e., when $\mathscr{G}$ contains a monochromatic chain of each colour, complementing the recent result of Falgas-Ravry, O'Connell, and Uzzell [64] in the *dense setting*.

[8] *Poset* is the abbreviation for partial ordered set

1.6 Positional games

In Chapter 10, we make further progress on a question of Beck [19] and a question of Hefetz, Kusch, Narins, Pokrovskiy, Requilé and Sarid [94] on the existence of Ramsey Games in which the first player has a winning strategy but not in bounded time. The work in this chapter is joint with Stefan David and Ivailo Hartarsky and is published in the European Journal of Combinatorics [50].

The theory of positional games on graphs and hypergraphs goes back to the seminal work of Hales and Jewett [88] and Erdős and Selfridge [62], but it was turned into an important area of combinatorics by a series of papers of Beck [13–15, 17–20]. In recent years, the theory has seen exciting developments thanks to its connections with areas such as Ramsey theory, extremal graph theory and probabilistic combinatorics. There are several interesting types of positional games but perhaps the most natural ones are the so-called *strong games* of which the *Ramsey games* are a prominent example. In the Ramsey game $\mathscr{R}(B,H)$ on a graph B, called the *board*, with *target graph* H, two players alternate to take previously unclaimed edges of B. The earliest player to claim an isomorphic copy of H *wins* and if none does so in finite time, the game is declared a *draw*.

A simple strategy stealing argument due to Nash combined with Ramsey's theorem shows that in $\mathscr{R}(K_n,H)$, where H is a fixed graph and K_n is a complete graph with n sufficiently large, the first player has winning strategy. It might be tempting to go one step further and ask if the first player has a winning strategy in bounded time as $n \to \infty$ (see [94]). There are a few target graphs for which the answer is known to be positive and there are no target graphs for which the answer is known to be negative. In particular, even in the simple case when $H = K_5$, the answer is not known. Beck popularized this as one of his "7 most humiliating open problems" [19]. Surprisingly, Hefetz, Kusch, Narins, Pokrovskiy, Requilé and Sarid [94] considered the analogous question for 5-uniform hypergraphs and constructed a target 5-uniform hypergraph for which they showed the answer to be negative. However, going from hypergraphs to graphs poses several difficulties.

In this chapter we consider the analogous question for 4-uniform hypergraphs, and construct a target 4-uniform hypergraph for which we show the answer to be negative. Also, we consider the analogous question for graphs where we replace the board by two disjoint copies of the board, and construct a target graph for which we show the answer to be negative.

Chapter 2

Sets in $\mathbb{Z}^k$ with doubling $2^k + \delta$ are near convex progressions

The work in this section is joint with Peter van Hintum and Hunter Spink [96].

2.1 Introduction

In this chapter we prove Theorem 1.1.5, the sharp stability result for the Freiman–Bilu 2^k-inequality (improved by Green and Tao [85]) and the generalization of Freiman's $3k-4$ theorem [80] to arbitrary dimension. A corollary of this theorem is the sharp stability for the Brunn–Minkowski inequality for equal sets conjectured by Figalli and Jerison [71]. For convenience, we introduce some notations and conventions and reformulate Theorem 1.1.5 (and its corollaries) as Theorem 2.1.1 (and its corollaries). We include a brief introduction here (for a comprehensive introduction see Section 1.1).

Our starting point is Freiman's theorem which states that for any $\lambda > 0$ there exist $b_1(\lambda), b_2(\lambda), b_3(\lambda) > 0$ such that the following holds. Any finite subset $A \subset \mathbb{Z}^k$ with doubling constant less than λ can be covered by $b_1(\lambda)$ translates of a generalized arithmetic progression (or alternatively convex progression) (see Definition 1.1.1) of size at most $b_2(\lambda)|A|$, and of dimension at most $b_3(\lambda)$. There is a rich literature on optimizations of the parameters $b_1(\lambda), b_2(\lambda), b_3(\lambda)$.

A notorious open problem is the polynomial Freiman-Ruzsa conjecture [131] which says that we can take $b_1(\lambda), b_2(\lambda)$ and $\exp(b_3(\lambda))$ to be polynomial in λ, which if true would be best possible. In Freiman's formulation using generalized arithmetic progressions, Lovett and Regev [118] showed that these parameters cannot be simultaneously brought down to polynomial in λ. However, in Freiman's formulation using (symmetric) convex progressions, a succession of improvements by Green and Ruzsa [83], Chang [42], Bourgain [31], Green and Tao [86], Schoen [137] and Sanders [133] showed that these parameters can be simultaneously brought down to $\exp(O(\log^{3+o(1)} \lambda))$.

An important result of Green and Tao [85], building on a previous result of Freiman and Bilu [23, 80], says that if $2^{k-1} \leqslant \lambda \leqslant 2^k - \delta$ then we can take $b_2(\lambda) = 1$ and $b_3(\lambda) = \lfloor \log_2(\lambda) \rfloor$, which are essentially best possible, provided we allow $b_1(\lambda) = \exp(\lambda^{O_\delta(1)})$.

In particular, this last result says that for any $\delta > 0$ there is a constant $m_k(\delta)$ such that given $A \subset \mathbb{Z}^k$ either $|A+A| \leqslant 2^k - \delta$ or A is covered by $m_k(\delta)$ parallel hyperplanes. This suggests that the correct notion of degeneracy for sets $A \subset \mathbb{Z}^k$ is being covered by a bounded number of parallel hyperplanes. Thus, non-degenerate sets $A \subset \mathbb{Z}^k$ have doubling constant bounded below by roughly 2^k, which mirrors the fact that sets $A \subset \mathbb{R}^k$ have doubling constant (with respect to volume) bounded below by 2^k.

There is a large literature of classifications of subsets $A \subset \mathbb{Z}^k$ with doubling λ at most $2^k - \delta$ (see, e.g., Fishburn [77], Freiman [80], Grynkiewicz and Serra [87], and Stanchescu [144–147]). However, little is known beyond the 2^k threshold.

Our main result says that, restricting the scope to sets $A \subset \mathbb{Z}^k$ not covered by a bounded number of parallel hyperplanes, if $2^k \leqslant \lambda \leqslant 2^k + \Delta_k$ then we can take $b_1(\lambda) = 1$ and $b_3(\lambda) = \lfloor \log_2(\lambda) \rfloor$, which are essentially best possible, provided we allow $b_2(\lambda)$ to be sufficiently large. Moreover, our result says we can take $b_2(\lambda) = O_k(1)$ in Freiman's formulation using generalized arithmetic progressions, and $b_2(\lambda) = 1 + O_k(1)(\lambda - 2^k)$ in Freiman's formulation using generalized arithmetic progressions, which are optimal up to the constant factor $O_k(1)$.

In particular, our result says that for any $\delta > 0$ there is a constant $m_k(\delta)$ such that given $A \subset \mathbb{Z}^k$ either A is covered by $m_k(\delta)$ parallel hyperplanes or A is contained inside a convex progression C of dimension k and size at most $(1 + O_k(1)\delta)|A|$. This result can be regarded as an extension of Freiman's $3k-4$ theorem in arbitrary dimension.

Actually, we show our result using only non-degenerate generalized arithmetic progression and non-degenerate convex progressions.

We recall that a non-degenerate r-dimensional generalized arithmetic progression in $\mathbb{Z}^k$ is a subset of the form $B(n_1, \dots, n_r; v_1, \dots, v_r; b) := \{b + \sum_{i=1}^r \ell_i v_i : 0 \leqslant \ell_i < n_i\}$, where $b, v_1, \dots, v_r \in \mathbb{Z}^k$ and $v_1, v_2, \dots, v_r$ are independent.

We also recall that a non-degenerate r-dimensional convex progression in $\mathbb{Z}^k$ is a subset of the form $P = \Lambda \cap K$ where Λ is an r-dimensional affine subllatice in $\mathbb{Z}^k$ and K is a convex set in $\mathbb{R}^k$.

Any non-degenerate convex progression in $\mathbb{Z}^k$ is (at most) k-dimensional (and we can always view it as being exactly k-dimensional).

We adopt the convention that whenever we refer to generalized arithmetic progressions or to convex progressions we mean non-degenerate ones unless otherwise explicitly stated.

For $A \subset \mathbb{Z}^k$, define the *convex progression hull* $\widehat{\mathrm{co}}(A)$ to be the intersection of the real convex hull $\widetilde{\mathrm{co}}(A)$ with the affine sub-lattice Λ_A which is the translate of $\mathrm{span}(A-A)$ containing A. The convex progression hull $\widehat{\mathrm{co}}(A)$ is the smallest convex progression containing A as any affine sub-lattice containing A must contain Λ_A and any convex set containing A must contain $\widetilde{\mathrm{co}}(A)$. In particular, A is a convex progression if and only if $A = \widehat{\mathrm{co}}(A)$.

Finally, define $d_k(A) = |A+A| - 2^k|A|$. Our main result describes the structure of $A \subset \mathbb{Z}^k$ with $d_k(A) \leqslant \Delta_k|A|$, i.e., $|A+A| \leqslant (2^k + \Delta_k)|A|$.

Theorem 2.1.1.

a) For $k \in \mathbb{N}$, there are positive constants $\Delta_k, m_k, \varepsilon_k$ such that for $A \subset \mathbb{Z}^k$ with $d_k(A) \leqslant \Delta_k |A|$, either A is covered by m_k parallel hyperplanes, or A lies in some $B = B(n_1, \dots, n_k; v_1, \dots, v_k; b)$ with $v_1, \dots, v_k$ linearly independent and $|A| \geqslant \varepsilon_k |B|$.

b) For $k \in \mathbb{N}$ and $\varepsilon_0 > 0$, there are positive constants $\Delta_k(\varepsilon_0), g_k(\varepsilon_0)$ and $c_k < (4k)^{5k}$ such that, given a subset A of some $B = B(n_1, \dots, n_k; v_1, \dots, v_k; b)$ with $v_1, \dots, v_k$ linearly independent, if $|A| \geqslant \varepsilon_0 |B|$ and also $d_k(A) \leqslant \Delta_k(\varepsilon_0)|A|$, then

$$|\widehat{\mathrm{co}}(A) \setminus A| \leqslant c_k d_k(A) + g_k(\varepsilon_0) \min\{n_i\}^{-\frac{1}{1+\frac{1}{2}(k-1)\lfloor k/2 \rfloor}} |A|.$$

As we show in Section 2.6, Theorem 2.1.1 has the following corollaries.

Corollary 2.1.2. For $k \in \mathbb{N}$, there are positive constants $c_k < (4k)^{5k}$, Δ_k, $m_k(\delta)$ such that for $\delta \in (0, \Delta_k]$ and $A \subset \mathbb{Z}^k$ with $d_k(A) \leqslant \delta |A|$, either A is covered by $m_k(\delta)$ parallel hyperplanes, or

$$|\widehat{\mathrm{co}}(A) \setminus A| \leqslant c_k \delta |A|.$$

Corollary 2.1.3. For $k \in \mathbb{N}$, there are positive constants $c_k < (4k)^{5k}$, Δ_k such that for $\widetilde{A} \subset \mathbb{R}^k$ open and bounded with $|\widetilde{A} + \widetilde{A}| \leqslant (2^k + \Delta_k)|\widetilde{A}|$, we have

$$|\widetilde{\mathrm{co}}(\widetilde{A}) \setminus \widetilde{A}| \leqslant c_k (|\widetilde{A} + \widetilde{A}| - 2^k |\widetilde{A}|).$$

As it turns out, much of the work in proving the quantitative statement of Theorem 2.1.1 is devoted to proving the following qualitative statement which, surprisingly, was not previously known.

Theorem 2.1.4. Given $k \in \mathbb{N}$ and $\varepsilon_0 > 0$, there exist $n_k(\varepsilon_0, \delta), \omega(\varepsilon_0, \delta) > 0$ with $\omega(\varepsilon_0, \delta) \to 0$ as $\delta \to 0$ such that the following is true. If A lies in some $B = B(n_1, \dots, n_k; v_1, \dots, v_k; b)$ with $v_1, \dots, v_k$ linearly independent and $n_1, \dots, n_k \geqslant n_k(\varepsilon_0, \delta)$, such that $|A| \geqslant \varepsilon_0 |B|$ and $d_k(A) \leqslant \delta |A|$, then $|\widehat{\mathrm{co}}(A) \setminus A| \leqslant \omega(\varepsilon_0, \delta)|A|$.

We mention a particularly nice intermediate result we show during the proof of Theorem 2.1.4. For a real-valued function f on a convex progression A, i.e., a set satisfying $A = \widehat{\mathrm{co}}(A)$, we define the *infimum-convolution* (see, e.g., Strömberg's extensive survey [150]) $f^{\square} : A + A \to \mathbb{R}$ by

$$f^{\square}(z) = \min_{x+y=z} \{f(x) + f(y)\}.$$

Theorem 2.1.5. Given $k \in \mathbb{N}$ and $\varepsilon_0 > 0$, there exist positive constants $c'_{k+1} < (4(k+1))^{5(k+1)}$ and $g'_{k+1}(\varepsilon_0)$ such that the following is true. Let $A = \widehat{\mathrm{co}}(A) \subset B(n_1, \dots, n_k; v_1, \dots, v_k; b)$ with $v_1, \dots, v_k$ linearly independent, $|A| \geqslant \varepsilon_0 |B|$, and $f : A \to [0,1]$ a function. Then with $\widehat{f} : A \to [0,1]$ the lower

convex hull function,

$$\sum_{x\in A}(f-\widehat{f})(x) \leqslant c'_{k+1}\left(2^{k+1}\sum_{x\in A}f(x)-\sum_{x'\in A+A}f^{\square}(x')\right)+g'_{k+1}(\varepsilon_0)\min\{n_i\}^{-\frac{1}{1+\frac{1}{2}k\lfloor (k+1)/2\rfloor}}|A|. \ \square$$

Theorem 2.1.5 follows from our main results applied to the epigraph $A'_{f,N,M}=\{(a,x): a\in A,\, x\in[Nf(a),M]\}\subset A\times[0,M]$ for $1\ll N\ll M$ large constants for fixed A (note that the choice of M allows us to avoid having the condition given by Δ_{k+1}). Proposition 2.3.44 (used in the proof of Theorem 2.1.4) is essentially this statement for A a simplex with a smaller error term $g'_{k+1}(\varepsilon_0)\min\{n_i\}^{-1}|A|$ (and with $c'_k\mapsto c'^{-1}_k$).

Corollary 2.1.2 is a direct generalization of Freiman's $3|A|-4$ theorem to arbitrary dimension. It is the first characterization for sets A in $\mathbb{Z}^k$ with doubling constant beyond the 2^k threshold besides the coarse characterization given by Freiman [79] and its subsequent optimizations. Corollary 2.1.3, conjectured by Figalli and Jerison [71] and recently resolved by the present authors [98] without any digression to the discrete setting, follows by standard approximation techniques from Corollary 2.1.2.

Similar to Freiman's $3|A|-4$ theorem, Corollary 2.1.2 and Corollary 2.1.3 require a bound Δ_k on δ in all dimensions k. Consider for example the set $A=[n_0]^k\cup\{(n,0,0,\dots,0)\}$ where n_0 is a constant and $n\gg n_0$, which has $d_k(A)\leqslant|A|$ and $|\widehat{\mathrm{co}}(A)\setminus A|\geqslant n-n_0$. The additional condition of not being covered by $m_k(\delta)$ hyperplanes is also necessary, and has no analogue when $k=1$ (as subsets $A\subset\mathbb{Z}$ cannot exhibit lower dimensional degeneracies). But when $k\geqslant 2$, consider for example the set $A=(\{1,\dots,n_0\}\times\{1,\dots,2n\}^{k-1})\cup\{(-1,1,1,\dots,1)\}$, where n_0 is constant and $n\gg n_0\gg 1.$, which has $d_k(A)<0$ and $|\widehat{\mathrm{co}}(A)\setminus A|=\frac{|A|-1}{2^{k-1}n_0}$.

We remark that the constant c_k in Theorem 2.1.1 (as well as the constants from Corollary 2.1.2 and Corollary 2.1.3) can be taken to be less than $(4k)^{5k}$ (which was the constant found in [98]), and the constant c'_k can be taken to be at most c_k. We have the lower bound $c_k\geqslant\frac{1}{k}$. This lower bound is given by the following family of sets. Fix a simplex $\widetilde{T}$ in $\mathbb{R}^{k-1}$ having the set of vertices $V(\widetilde{T})$ in $\mathbb{Z}^{k-1}$. Fix a large constant s in $\mathbb{N}$. Construct the set $\widetilde{R}$ in $\mathbb{R}^k$ given by $R=((\widetilde{T}\times[-s,0])\cup(V(\widetilde{T})\times\{1\}))$. Finally, construct the family of sets A_n in $\mathbb{Z}^k$ given by $A_n=(nR)\cap\mathbb{Z}^k$. It follows that $|A_n|=(1+o(1))n^k|R|$, $|\widehat{\mathrm{co}}(A_n)|=(1+o(1))n^k|\widetilde{\mathrm{co}}(R)|$ and $|A_n+A_n|=(1+o(1))n^k|R+R|$ (since R is the union of a polyhedron and a finite number of integral points, k in this case). Finally note that $|R+R|-2^k|R|=k|\widetilde{\mathrm{co}}(R)\setminus R|\ll s^{-1}|R|$. The point is that each vertex in $V(\widetilde{T}\times\{1\})$ pulls out a copy of $\widetilde{\mathrm{co}}(R)\setminus R$. By taking s and then n sufficiently large, the hypothesis on d_k is satisfied and we get $c_k\geqslant\frac{1}{k}$. We also have $c'_k\geqslant\frac{1}{k}$ by taking the functional version of this example, namely with $f:(n\widetilde{T}\cap\mathbb{Z}^k)\to[0,1]$ whose value is 0 at the vertices and 1 elsewhere. We believe that the optimal values of c_k,c'_k lie closer to the lower bound $\frac{1}{k}$.

Finally, we remark on the exponent $-\frac{1}{1+\frac{1}{2}(k-1)\lfloor k/2\rfloor}$ in Theorem 2.1.1. Our proof of Theorem 2.1.1 reduces the case that $\widetilde{\mathrm{co}}(A)$ is a simplex. For a general A we first approximate $\widetilde{\mathrm{co}}(A)$ from within by a polytope $\widetilde{P}$, and then triangulate $\widetilde{P}$ into simplices via a triangulation of $\partial\widetilde{P}$. To approximate the volume of $\widetilde{\mathrm{co}}(A)$ by a polytope $\widetilde{P}$ with $|\widetilde{P}|\geqslant(1-\alpha)|\widetilde{\mathrm{co}}(A)|$ requires $\ell=O(\alpha^{-\frac{2}{k-1}})$ vertices by a

result of Gordon, Meyer, and Reisner [82], and the proof of the upper bound conjecture by Stanley [148] implies that a triangulation of $\partial\widetilde{P}$ has at most $O(\ell^{\lfloor k/2\rfloor})$ simplicies. These two bounds end up giving the exponent in Theorem 2.1.1.

2.1.1 Outline of the chapter

As mentioned before, most of the work is devoted to proving Theorem 2.1.4. The strategy is to construct in stages a highly structured set $A_\star$ from A with the properties that $|A \triangle A_\star| = o_{\varepsilon_0}(1)|A|$ and $d_k(A_\star) = o_{\varepsilon_0}(1)|A_\star| = o_{\varepsilon_0}(1)|A|$ (where for fixed ε_0 we have $o_{\varepsilon_0}(1) \to 0$ as $\delta \to 0$). The additional structure of $A_\star$ enables us to conclude that $|\widehat{\mathrm{co}}(A_\star) \setminus A_\star| = o_{\varepsilon_0}(1)|A_\star|$, which finally implies that $|\widehat{\mathrm{co}}(A) \setminus A| = o_{\varepsilon_0}(1)|A|$.

At each stage we produce a new set A_{new} from an existing set A_{old} which satisfies $|A \triangle A_{old}| = o_{\varepsilon_0}(1)|A|$ and $d_k(A_{old}) = o_{\varepsilon_0}(1)|A|$. With a single exception, this is done by throwing away rows R_x in the first coordinate direction which are are in some sense unstructured. If we can show $|A_{old} \setminus A_{new}| = o_{\varepsilon_0}(1)|A|$, then $|A \triangle A_{new}| = o_{\varepsilon_0}(1)|A|$ and hence we have $d_k(A_{new}) = o_{\varepsilon_0}(1)|A|$.

To bound $|A_{\mathrm{old}} \setminus A_{\mathrm{new}}|$ from above, we introduce an operation $(+)$ in order to create a "reference set" $A_{\mathrm{old}}(+)A_{\mathrm{old}} \subset A_{\mathrm{old}} + A_{\mathrm{old}}$, whose size we can guarantee to be approximately $2^k|A_{\mathrm{old}}|$. For sets $X, Y \subset \mathbb{Z}$ we define $X(+)Y := (X + \min Y) \cup (Y + \max X) \subset X + Y$. For sets $X \subset \mathbb{Z} \times \{x_2\} \times \ldots \times \{x_k\}$ and $Y \subset \mathbb{Z} \times \{y_2\} \times \ldots \times \{y_k\}$ we define $X(+)Y := (X + \min Y) \cup (Y + \max X) \subset X + Y$, by using the order on X and the order on Y induced by the first coordinate. In general, we define $A_{\mathrm{old}}(+)A_{\mathrm{old}} := \bigsqcup_{\vec{v} \in \{0\} \times \{0,1\}^{k-1}} \bigsqcup_{x \in \{0\} \times \mathbb{Z}^{k-1}} R_x(+)R_{x+\vec{v}} \subset A_{\mathrm{old}} + A_{\mathrm{old}}$. This union is disjoint[1] as each $y \in \{0\} \times \mathbb{Z}^{k-1}$ can be written uniquely as $y = x + (x + \vec{v})$. In particular, $|A_{\mathrm{old}}(+)A_{\mathrm{old}}| \geqslant \sum_{\vec{v} \in \{0\} \times \{0,1\}^{k-1}} \sum_{x \in \{0\} \times \mathbb{Z}^{k-1}} |R_x(+)R_{x+v}|$.

In order to control the size of the unstructured rows $U := \bigsqcup_{x \text{ - unstructured}} R_x$, we construct a set $D \subset U + A_{\mathrm{old}} \subset A_{\mathrm{old}} + A_{\mathrm{old}}$ of comparable size to U, disjoint from $A_{\mathrm{old}}(+)A_{\mathrm{old}}$. Then, we obtain $|A_{\mathrm{old}} \setminus A_{\mathrm{new}}| = |U| \approx |D| \leqslant |A_{\mathrm{old}} + A_{\mathrm{old}}| - |A_{\mathrm{old}}(+)A_{\mathrm{old}}| \approx d_k(A_{\mathrm{old}}) = o_{\varepsilon_0}(1)|A|$.

For this approach to work it is crucial that the reference set satisfies $|A_{\mathrm{old}}(+)A_{\mathrm{old}}| \approx 2^k|A_{\mathrm{old}}|$, or equivalently, $\sum_{\vec{v} \in \{0,1\}^{k-1}} \sum_x |R_x(+)R_{x+\vec{v}}| \approx 2^k \sum_x |R_x|$. It may seem surprising that this very restricted subset could account for such a big proportion. And indeed this is not true in general. However consider the following situation. Say that $A_{\mathrm{old}} \subset [n]^k$ has density at least ε_0 and that the projection of A_{old} in the first coordinate $\pi(A_{\mathrm{old}})$ is approximately a convex progression, say it satisfies $|\widehat{\mathrm{co}}(\pi(A_{\mathrm{old}})) \setminus \pi(A_{\mathrm{old}})| \leqslant o_{\varepsilon_0}(1)|\pi(A_{\mathrm{old}})|$. In this situation the above assertion is true. Indeed, note that for any $\vec{v} \in \{0,1\}^{k-1}$ and any x the quantity $|R_x(+)R_{x+\vec{v}}| - |R_x| - |R_{x+\vec{v}}|$ is 0 if both rows are empty, it is at least $-n$ if one row is empty and it is -1 if neither rows are empty. We deduce that $-2^k \sum_x |R_x| + \sum_{\vec{v} \in \{0,1\}^{k-1}} \sum_x |R_x(+)R_{x+\vec{v}}| \geqslant -np - q$ where p and q are the number of pairs $(x, \vec{v})$ such that $|\{x, x + (\vec{v})\} \cap \pi(A)| = 1$ and $|\{x, x + (\vec{v})\} \cap \pi(A)| = 2$, respectively. By the assumptions, it follows that $q = O_k(n^{k-1})$ and $p = o_{\varepsilon_0}(n^{k-1})$. Provided $n >> \varepsilon_0^{-1}$, we conclude that $|A_{\mathrm{old}}(+)A_{\mathrm{old}}| - 2^k|A_{\mathrm{old}}| = -2^k \sum_x |R_x| + \sum_{\vec{v} \in \{0,1\}^{k-1}} \sum_x |R_x(+)R_{x+\vec{v}}| \geqslant o_{\varepsilon_0}(1)|A_{\mathrm{old}}|$.

[1] We use the symbol $\bigsqcup$ to denote a disjoint union.

For the last step in the proof of Theorem 2.1.4 and for the proof of Theorem 2.1.1 b), we use two versions of an argument inspired by the one used in [98]. For the last part of Theorem 2.1.4, we prove that functions on convex domains with small infimum-convolution are close to their convex hulls (which is essentially Theorem 2.1.5). For Theorem 2.1.1 b), we proceed as follows. By choosing an appropriately small $\Delta_k(\varepsilon_0)$, Theorem 2.1.4 ensures $|\widehat{\mathrm{co}}(A)\setminus A|\cdot|A|^{-1}$ is as small as we like. This guarantees a large interior region of $\widehat{\mathrm{co}}(A+A)$ is contained in $A+A$. We control the size of $\widehat{\mathrm{co}}(A)\setminus A$ by inductively controlling the size of $\widehat{\mathrm{co}}(A)\setminus A$ restricted to certain homothetic copies of $\widetilde{\mathrm{co}}(A)$ used to cover a thickened boundary of $\widetilde{\mathrm{co}}(A)$. This allows us to show that $|\widehat{\mathrm{co}}(A+A)\setminus(A+A)|\leqslant(2^k-c_k')|\widehat{\mathrm{co}}(A)\setminus A|+o_{\varepsilon_0}(1)|B|$ for some constant c_k', which allows us to conclude Theorem 2.1.1 b).

Finally, for Theorem 2.1.1 a), we start with a result of Green and Tao [85] that A is covered by a bounded number of generalized arithmetic progressions of dimension k and size at most $|A|$, and show that we can reduce to a single generalized arithmetic progression of size $O(|A|)$.

In Section 2, we make some initial definitions, conventions and observations, which are used throughout the chapter. In Sections 3 and 4, we prove Theorem 2.1.4 and Theorem 2.1.1 b) with a simultaneous induction on dimension. In Section 5, we prove Theorem 2.1.1 a), whose proof is independent of Sections 3 and 4. Finally, for completeness, we include in Section 2.6 a proof of Corollary 2.1.2 and Corollary 2.1.3 from Theorem 2.1.1.

2.2 Definitions, Conventions, and Observations

In this section, we introduce our definitions and conventions, as well as observations we will be using throughout the remaining sections.

2.2.1 Definitions and Conventions

Convention 2.2.1. We use the symbol $\bigsqcup$ to denote a disjoint union. Generalized arithmetic progressions and convex progressions are assumed to be non-degenerate ones unless otherwise stated.

Definition 2.2.2. For $A'\subset\mathbb{Z}^k$, we denote by

- $\widetilde{\mathrm{co}}(A')\subset\mathbb{R}^k$ for the convex hull,
- $\mathrm{co}(A')=\widetilde{\mathrm{co}}(A')\cap\mathbb{Z}^k$,
- $\Lambda_{A'}=\langle A'-a\rangle+a\subset\mathbb{Z}^k$ for any $a\in A'$, the affine sublattice of $\mathbb{Z}^k$ spanned by A', and
- $\widehat{\mathrm{co}}(A')=\mathrm{co}(A')\cap\Lambda_{A'}$, the smallest convex progression containing A'.

Definition 2.2.3. We say that A' is *reduced* if $\Lambda_{A'}=\mathbb{Z}^k$, or equivalently $\mathrm{co}(A')=\widehat{\mathrm{co}}(A')$ and A' is not contained in a hyperplane.

We will typically denote regions of $\mathbb{R}^k$ with a tilde such as $\widetilde{A'}\subset\mathbb{R}^k$. By abuse of notation, we will use $|\cdot|$ to refer both to cardinality of sets, and for volumes of sets. It will be clear with the tilde

notation whether we intend to use discrete or continuous volume, and from context what dimension we are considering.

Convention 2.2.4. When we define a polytope or affine subspace $\widetilde{P} \subset \mathbb{R}^k$, we let $P = \widetilde{P} \cap \mathbb{Z}^k$.

Definition 2.2.5. Given numbers $n_1, \dots, n_k$, we define the *discrete box*

$$B(n_1, \dots, n_k) = \prod_{i=1}^{k} \{1, \dots, n_i\}.$$

We write B instead of $B(n_1, \dots, n_k)$ when $n_1, \dots, n_k$ are clear from context.

Convention 2.2.6. In Theorem 2.1.4 and Theorem 2.1.1 b), we assume without loss of generality that the vectors v_i coincide with basis vectors e_i of $\mathbb{R}^k$ and $b = \vec{1}$, writing the generalized arithmetic progression $B = B(n_1, \dots, n_k; v_1, \dots, v_k; b) = B(n_1, \dots, n_k)$.

Convention 2.2.7. In the proofs of Theorem 2.1.4 and Theorem 2.1.1 b), we shall assume A is reduced, as allowed by Observation 2.2.19 below.

Definition 2.2.8. We define the *projection*

$$\pi : \mathbb{Z}^k = \mathbb{Z} \times \mathbb{Z}^{k-1} \to \{0\} \times \mathbb{Z}^{k-1}$$

to be the projection away from the first coordinate.

Definition 2.2.9. For $\widetilde{A'} \subset \mathbb{R}^k$ a subset such that $\widetilde{\mathrm{co}}(\widetilde{A'})$ is a polytope with integral vertices, we define $V(\widetilde{A'}) \subset \mathbb{Z}^k$ to be the vertices of $\widetilde{\mathrm{co}}(\widetilde{A'})$, and $V_\pi(\widetilde{A'}) := \pi(V(\widetilde{A'})) \subset \{0\} \times \mathbb{Z}^{k-1}$.

Definition 2.2.10. A *row* of $A' \subset \mathbb{Z}^k$ is $R_x = \pi^{-1}(x) \cap A'$ for some $x \in \{0\} \times \mathbb{Z}^{k-1}$.

Convention 2.2.11. When talking about the rows of a set A', we will use the notation R_x without further clarification. It will always be clear from context which set A' is being referred to.

Definition 2.2.12. For $X, Y \subset \mathbb{Z}$, define the *restricted sumset* $X(+)Y := (X + \min Y) \cup (Y + \max X) \subset X + Y$ if X, Y are both nonempty, and $\emptyset$ otherwise. We extend this definition to sets $X \subset \mathbb{Z} \times \{x_2\} \times \dots \times \{x_k\}$ and $Y \subset \mathbb{Z} \times \{y_2\} \times \dots \times \{y_k\}$ by using the order on X and Y induced by the first coordinate. In general, for $A' \subset \mathbb{Z}^k$, we define *the restricted sumset*

$$A'(+)A' := \sum_{\vec{v} \in \{0\} \times \{0,1\}^{k-1}} \sum_{x \in \{0\} \times \mathbb{Z}^{k-1}} R_x(+)R_{x+\vec{v}} \subset A' + A'.$$

Definition 2.2.13. Given an increasing function $\zeta : \mathbb{R}_+^\star \to \mathbb{R}_+^\star$ with $\zeta(x) \to 0$ as $x \to 0$, and given two positive numbers a, b we write $a \ll_\zeta b$ if $a \leqslant \zeta(b)$.

In Theorem 2.1.4 we can assume without loss of generality that $\min\{n_i\}^{-1} \ll_\zeta \delta \ll_\zeta \varepsilon_0 \ll_\zeta 1$, for any given increasing function $\zeta : \mathbb{R}_+^\star \to \mathbb{R}_+^\star$ with $\zeta(x) \to 0$ as $x \to 0$. We can do so by passing to a smaller value of ε_0 if needed, by setting $\omega(\varepsilon_0, \delta) = 1/\varepsilon_0$ for large values of δ and by choosing $n_k(\varepsilon_0, \delta)$ sufficiently large.

Convention 2.2.14. In Theorem 2.1.4, whenever we have a statement that holds conditioned on the assumption that $\min\{n_i\}^{-1} \ll_\zeta \delta \ll_\zeta \varepsilon_0 \ll_\zeta 1$, for a given increasing function $\zeta : \mathbb{R}_+^\star \to \mathbb{R}_+^\star$ with $\zeta(x) \to 0$ as $x \to 0$, we automatically make this assumption in Theorem 2.1.4.

Convention 2.2.15. We omit the sentence "provided δ is sufficiently small in terms of ε_0 and $\min(n_i)$ sufficiently large in terms of δ and ε_0" inside the proofs of intermediate results, unless stated otherwise.

E.g. if we write $\delta + \delta^{1/2} < \delta^{1/3} < \varepsilon_0/2$, we understand that this statement holds provided δ is sufficiently small in terms of ε_0, which we can assume wlog.

Convention 2.2.16. The big and little o notations $O(1)$ and $o(1)$ are with respect to fixed ε_0 as $\delta \to 0$ throughout.

Finally, we introduce a small constant which we will use to absorb errors into exponents through the chapter.

Definition 2.2.17. We let $c = 10^{-10}$.

2.2.2 Observations

The first observation guarantees that hyperplanes $\widetilde{H}$ have small intersections with discrete boxes. In particular, large subsets of B cannot be covered by few hyperplanes.

Observation 2.2.18. Given a hyperplane $\widetilde{H}$ and a box $B = B(n_1, \dots, n_k)$, we have

$$|\widetilde{H} \cap B| \leqslant \min\{n_i\}^{-1}|B|.$$

In particular, a subset $A' \subset B$ with $|A'| > m\min\{n_i\}^{-1}|B|$ cannot be covered by m hyperplanes.

Proof. If coordinate basis vector e_i is not parallel to $\widetilde{H}$, then the projection of $\widetilde{H} \cap B$ away from the ith coordinate injects into the same projection for B, and therefore $|\widetilde{H} \cap B| \leqslant n_i^{-1}|B|$. □

The next observation is used to assume A is reduced so Convention 2.2.7 holds.

Observation 2.2.19. For all $\varepsilon_0 > 0$, the following holds. Given n_i sufficiently large in terms of ε_0, for a subset $A' \subset B = B(n_1, \dots, n_k)$ with $|A'| \geqslant \varepsilon_0|B|$, we can find a subset $A'' \subset B(2^k n_1, \dots, 2^k n_k)$ such that A'' is reduced, $|A'| = |A''|$, $d_k(A') = d_k(A'')$, and $|\widehat{\mathrm{co}}(A') \setminus A'| = |\mathrm{co}(A'') \setminus A''|$.

Proof. We first note that A' is not contained inside a hyperplane by Observation 2.2.18. Take some $a \in A'$, then $A' - a \in C := \prod_{i=1}^k \{-n_i + 1, \dots, n_i - 1\}$. The affine sub-lattice $\Lambda_{A'-a}$ is actually a subgroup $\langle v_1, \dots, v_k \rangle \subset \mathbb{Z}^k$ generated by linearly independent vectors $v_i = (v_{i,1}, \dots, v_{i,k})$. Note that this property is robust to operations of the form: v_a gets swapped with v_b or v_a gets replaced by $v_a \pm v_b$ for some $a \neq b$.

Without loss of generality, suppose $n_1 \leqslant \dots \leqslant n_k$. By performing a sequence of operations on the set of vectors $\{v_1, \dots, v_k\}$, we may assume that $|v_{j,i}| = 0$ for all $j > i$, and $|v_{j,i}| \leqslant |v_{i,i}|$ for

all $j \leqslant i$. Indeed, let us explain how to deal with the first coordinate. As long as there exist a, b such that $0 < |v_{a,1}| \leqslant |v_{b,1}|$, we replace v_b by $v_b - \text{sign}(v_{a,1}v_{b,1})v_a$. This operation strictly decreases $(|v_{1,1}|, |v_{2,1}|, \ldots, |v_{k,1}|)$ in lexicographic order, say. Thus, eventually this process must end. When this happens all but one of the vectors v_i must be null in the first coordinate. After we swap two of the vectors we can ensure that $v_{j,1} = 0$ for all $j \geqslant 2$. In a similar way, we can deal with all coordinates to ensure $|v_{j,i}| = 0$ for all $j > i$. Note that this property is robust to operations of the form: v_a gets replaced with $v_a \pm v_b$ for some $a < b$. Because $v_1, \ldots, v_i$ are linearly independent, we have $v_{i,i} \neq 0$. Finally, as long as there exist $a < b$ such that $|v_{a,b}| \geqslant |v_{b,b}| > 0$, we replace v_a by $v_a - \text{sign}(v_{a,b}v_{b,b})v_b$. Just as before, this process must end. When this happens we have $|v_{j,i}| \leqslant |v_{i,i}|$ for all $j \leqslant i$.

Now, consider a point $p = p_1v_1 + \ldots + p_kv_k \in C$. We show by induction on i that $|p_i| \leqslant 2^{i-1}(n_i - 1)$. Indeed, by considering the ith coordinate, we have that

$$|p_1v_{1,i} + p_2v_{2,i} + \ldots + p_iv_{i,i}| \leqslant n_i - 1$$

and hence

$$|p_i||v_{i,i}| \leqslant |p_1||v_{1,i}| + \ldots + |p_{i-1}||v_{i-1,i}| + n_i - 1 \leqslant (|p_1| + \ldots + |p_{i-1}| + n_i - 1)|v_{i,i}|.$$

This shows that $A' - a \subset \{p_1v_1 + \ldots + p_kv_k : |p_i| \leqslant 2^{i-1}(n_i - 1)\}$. If we let

$$A''' := \{(p_1, \ldots p_k) : p_1v_1 + \ldots + p_kv_k \in A' - a\} \subset \prod_{i=1}^{k} \{-2^{i-1}(n_i - 1), \ldots, 2^{i-1}(n_i - 1)\},$$

then A''' is reduced in $\mathbb{Z}^k$, and is obtained from A' by applying an element of $GL_n(\mathbb{Q})$ followed by a translation, so $|A'| = |A'''|$, $d_k(A''') = d_k(A')$, and $|\text{co}(A''') \setminus A'''| = |\widehat{\text{co}}(A') \setminus A'|$. We conclude by taking A'' a suitable translation of A'''. □

We now prove an easy corollary of a lemma of Green and Tao [85]. For convenience, we include the lemma below. But before that we need a couple of definitions which we use only here.

A subset $X \subset \mathbb{N}^d$ is said to be an *i-down-set* if whenever we have that $(x_1, \ldots, x_d) \in X$, then $(x_1, \ldots, x_{i-1}, y_i, x_{i+1}, \ldots, x_d) \in X$ for all integers $1 \leqslant y_i \leqslant x_i$. We say that X is a *down-set* if it is an i-down-set for each $1 \leqslant i \leqslant d$. In particular, the set $B = B(n_1, \ldots, n_d)$ is a down-set.

For each $I \subset \{1, 2, \ldots, d\}$ let $\pi_I : \mathbb{R}^d \to \mathbb{R}^d$ be the orthogonal projection $\pi_I(x_1e_1 + \ldots + x_de_d) = \sum_{i \in I} x_ie_i$ where $e_1, \ldots, e_d$ is the standard basis of $\mathbb{R}^d$.

Lemma 2.2.20. [Lemma 2.8 in [85]] Let X, Y be down-sets in $\mathbb{N}^d$ and let $A \subset X$ and $B \subset Y$. Then

$$|A + B| \geqslant 2^d \min(|A|, |B|) - \sum_{I \subsetneq \{1, \ldots, d\}} |\pi_I(X + Y)|.$$

Observation 2.2.21. For any subsets $X \subset B = B(n_1,\dots,n_k)$ and $Y \subset \pi(B)$ we have

$$d_k(X) \geqslant -2^{2k}\min\{n_i\}^{-1}|B|, \quad \text{and} \quad d_{k-1}(Y) \geqslant -2^{2(k-1)}\min\{n_i\}^{-1}n_1^{-1}|B|.$$

More generally, for $X_1, X_2 \subset B$ and $Y_1, Y_2 \subset \pi(B)$ we have

$$\begin{aligned}|X_1+X_2| &\geqslant 2^k\min(|X_1|,|X_2|) - 2^{2k}\min\{n_i\}^{-1}|B|, \text{ and}\\ |Y_1+Y_2| &\geqslant 2^{k-1}\min(|Y_1|,|Y_2|) - 2^{2(k-1)}\min\{n_i\}^{-1}n_1^{-1}|B|.\end{aligned}$$

Proof. Because B and $\pi(B)$ are downsets, the result follows from Lemma 2.2.20, and the trivial estimates that the size of each coordinate projection of $B+B$ and $\pi(B)+\pi(B)$ have sizes at most $2^k\min\{n_i\}^{-1}|B|$ and $2^{k-1}\min\{n_i\}^{-1}n_1^{-1}|B|$, respectively. □

We frequently need the following observation when considering $A'(+)A'$ to show it has size roughly $2^k|A'|$ as described in Section 2.1.1.

Observation 2.2.22. Let $Y \subset \pi(B)$ with $B = B(n_1,\dots,n_k)$, and let $0 \neq \vec{v} \in \{0\}\times\{0,1\}^{k-1}$. Then

$$\left|\{x \in \{0\}\times\mathbb{Z}^{k-1} : |\{x, x+\vec{v}\}\cap \mathrm{co}(Y)| = 1\}\right| \leqslant 2(k-1)\min\{n_i\}^{-1}n_1^{-1}|B|.$$

Proof. Consider all lines in the direction $\vec{v}$ intersecting $\pi(B)$. On each such line there are at most 2 values of x such that $|\{x,x+\vec{v}\}\cap\mathrm{co}(Y)| = 1$. Moreover, there are at most $(k-1)\min\{n_i\}^{-1}|\pi(B)|$ such lines which intersect $\pi(B)$ since each such line intersects (at least) two faces of the boundary of $\pi(B)$. Moreover, the intersection between the line and a face is a lattice point. Indeed, the i-th coordinate between two consecutive lattice points on such a line differs by one or zero. □

The next observation relates d_k between sets and subsets. In particular, it allows us to guarantee that all auxiliary sets we construct in the proof of Theorem 2.1.4 are reduced and have similar d_k solely because they are close in symmetric difference to the original set A.

Observation 2.2.23. If $X \subset Y$, then

$$d_k(X) \leqslant d_k(Y) + 2^k|Y \setminus X|. \tag{2.1}$$

In particular, for reduced $A \subset B = B(n_1,\dots,n_k)$ with $|A| \geqslant \varepsilon_0|B|$, $d_k(A) \leqslant \delta|B|$, δ sufficiently small in terms of ε_0, and $n_1,\dots,n_k$ sufficiently large in terms of ε_0,δ, if $A' \subset B$ has

$$|A \triangle A'| \leqslant 2^{-(k+1)}\varepsilon_0|B| \tag{2.2}$$

then A' is reduced.

Proof. For (2.1), we have

$$d_k(X) = |X+X| - 2^k|X| \leqslant |Y+Y| - 2^k|Y| + 2^k|Y \setminus X| = d_k(Y) + 2^k|Y \setminus X|.$$

For (2.2), it suffices to show $A \cap A'$ is reduced, so we may assume $A' \subset A$. Assume for the sake of contradiction that A' is not reduced. Then there is an $a \in A$, such that $a + A'$ is disjoint from $A' + A'$. Hence, we have $|A+A| \geqslant |A'+A'| + |A'|$, and in particular

$$\delta|B| \geqslant d_k(A) \geqslant d_k(A') - (2^k+1)|A \setminus A'| + |A| \geqslant \left(-2^{2k}\min\{n_i\}^{-1} + \left(\frac{1}{2} - \frac{1}{2^{k+1}}\right)\varepsilon_0\right)|B|,$$

a contradiction. □

We next have an observation which allows us to transition between convex sets and reduced convex progressions with a loss proportional to the surface area of a containing box.

Observation 2.2.24. Let $B = B(n_1, \dots, n_k)$, and suppose we have a convex polytope $\widetilde{P} \subset \widetilde{\mathrm{co}}(B)$. Then with $P = \widetilde{P} \cap \mathbb{Z}^k$, we have $\left||\widetilde{P}| - |P|\right| \leqslant 2k(k+1)\min\{n_i\}^{-1}|B|$. This is more generally true for any subset $\widetilde{P} \subset \widetilde{\mathrm{co}}(B)$ given as the intersection of finitely many open and closed half-spaces.

Proof. By perturbing the defining half-spaces slightly, we may replace $\widetilde{P}$ with a polytope without changing P, so we assume $\widetilde{P}$ is a polytope from now on.

Consider the set $X := \left\{z \in \mathbb{Z}^k : (z + [0,1]^k) \cap \partial\widetilde{P} \neq \emptyset\right\}$. We first show $|X|$ is small.

Claim 2.2.25. $|X| \leqslant 2k(k+1)\min\{n_i\}^{-1}|B|$

Proof of claim. For $1 \leqslant i \leqslant k$, let $\pi_i : \mathbb{Z}^k \to \mathbb{Z}^{i-1} \times \{0\} \times \mathbb{Z}^{k-i}$ be the projection away from the ith coordinate. For every $x = (x_1, \dots, x_{i-1}, 0, x_{i+1}, \dots, x_k)$, we order the set $\pi_i^{-1}(x)$ according to the i-th coordinate. Let $f_i^+, f_i^- : \pi_i(X) \to \mathbb{Z}$ be defined by

$$f_i^+ : x \mapsto \max(\pi_i^{-1}(x) \cap X) \qquad f_i^- : x \mapsto \min(\pi_i^{-1}(x) \cap X),$$

and for every $x = (x_1, \dots, x_{i-1}, 0, x_{i+1}, \dots, x_k) \in \pi_i(X)$, let

$$X_{i,x}^+ = \{(x_1, \dots, x_{i-1}, j, x_{i+1}, \dots, x_k) : f_i^+(x) - k \leqslant j \leqslant f_i^+(x)\}$$
$$X_{i,x}^- = \{(x_1, \dots, x_{i-1}, j, x_{i+1}, \dots, x_k) : f_i^-(x) \leqslant j \leqslant f_i^-(x) + k\}$$

be the $k+1$ elements of $\mathbb{Z}^k$ in the x-row in direction i of X below the maximum element and above the minimum element respectively. From these definitions it is immediate that

$$\left|\bigcup_{i=1}^{k} X_{i,x}^+ \cup X_{i,x}^-\right| \leqslant 2k(k+1)\min\{n_i\}^{-1}|B|,$$

so it suffices to show that $X \subset \bigcup X_{i,x}^+ \cup X_{i,x}^-$.

Suppose for the sake of contradiction that there is some $z \in X \setminus \left(\bigcup X_{i,x}^+ \cup X_{i,x}^-\right)$. Then

$$f_i^+(\pi_i(z)) \geqslant z_i + k + 1, \text{ and } f_i^-(\pi_i(z)) \leqslant z_i - k - 1$$

for all i, so there are $r_i^+, r_i^- \geqslant k+1$ such that $z + r_i^+ e_i + [0,1]^k$ and $z - r_i^- e_i + [0,1]^k$ intersect $\widetilde{P}$. As $z+[0,1]^k$ intersects $\widetilde{P}$, by convexity of $\widetilde{P}$ for all $i \in [k]$ there are points

$$y_i^+ \in (z + (k+1)e_i + [0,1]^k) \cap \widetilde{P}, \qquad y_i^- \in (z - (k+1)e_i + [0,1]^k) \cap \widetilde{P}.$$

We claim that

$$z + [0,1]^k \subset int(\widetilde{co}(\{y_1^+, \dots, y_k^+, y_1^-, \dots, y_k^-\})) \subset int(\widetilde{P}).$$

Write $y_i^+ = z + (\frac{1}{2}, \dots, \frac{1}{2}) + p_i^+$ and $y_i^- = z + (\frac{1}{2}, \dots, \frac{1}{2}) + p_i^-$ where $p_i^+ = (k+1)e_i + \varepsilon_i^+$ and $p_i^- = -(k+1)e_i + \varepsilon_i^-$ with $\varepsilon_i^\pm \in [-\frac{1}{2}, \frac{1}{2}]^k$. Then this is equivalent to showing

$$\left[-\frac{1}{2}, \frac{1}{2}\right]^k \subset int(\widetilde{\text{co}}(\{p_1^+, \dots, p_k^+, p_1^-, \dots, p_k^-\})).$$

We show that $\widetilde{\text{co}}(\{p_1^+, \dots, p_k^+, p_1^-, \dots, p_k^-\})$ has facets $\widetilde{\text{co}}(p_1^\pm, \dots, p_k^\pm)$ for the 2^k choices of $\pm$, and $\left[-\frac{1}{2}, \frac{1}{2}\right]^k$ lies on the same side of these facets as $\widetilde{\text{co}}(\{p_1^+, \dots, p_k^+, p_1^-, \dots, p_k^-\})$. To show this, let $\varepsilon \in [-\frac{1}{2}, \frac{1}{2}]^k$. We claim that it suffices to show that ε and p_1^- lie on the same side of the hyperplane $\widetilde{H}$ through $p_1^+, \dots, p_k^+$. Indeed, if this is the case, then by symmetry all vertices lie on the same side of $\widetilde{H}$, which implies that $\widetilde{\text{co}}(\{p_1^+, \dots, p_k^+\})$ is a facet, and $\widetilde{\text{co}}(\{p_1^+, \dots, p_k^+, p_1^-, \dots, p_k^-\})$ lies on the same side of this facet as ε. This is equivalent in turn to showing that for $w \in \{p_1^-, \varepsilon\}$, the determinants of the matrices whose columns are $p_i^+ - w$ for $1 \leqslant i \leqslant k$ have the same signs. We will in fact show that this sign is positive for both.

For $w = \varepsilon$, the matrix we are considering is $M + (k+1)I$ where M has as its ith column $\varepsilon_i^+ - \varepsilon$. Note that M has entries of magnitude at most 1, so the spectral radius of M is at most k. But if $\det(M + (k+1)I) \leqslant 0$, then as $\det(M + \lambda I) \to \infty$ as $\lambda \to \infty$ there must exist $\lambda \geqslant k+1$ with $\det(M + \lambda I) = 0$, a contradiction as $|-\lambda| > k$. Hence $\det(M + (k+1)I) > 0$ as desired.

For $w = p_1^-$, note that we have already shown that ε_1^- lies on the positive side of $\widetilde{H}$, and $p_1^- \in \varepsilon_1^- + \mathbb{R}_{\leqslant 0} e_1$. Hence it suffices to show that the point $\varepsilon_1^+ - Ne_1$ lies on the positive side of $\widetilde{H}$ for all $N > 0$ sufficiently large. This is equivalent to saying that the matrix M_N whose ith column is $p_i^+ + Ne_1 - \varepsilon_1^-$ has positive determinant for all $N > 0$ sufficiently large. Subtracting the first column from all subsequent columns and then considering the coefficient of N in $\det(M_N)$, this follows from an identical argument.

Hence we have $z + [0,1]^k \subset int(\widetilde{P})$, contradicting $(z + [0,1]^k) \cap \partial \widetilde{P} \neq \emptyset$.

□

Returning to the proof of Observation 2.2.24, consider the translates of $[0,1]^k$ by P, i.e., $P+[0,1]^k$. Each of these translates is either contained in $\widetilde{P}$ or intersects $\partial\widetilde{P}$. Hence, $|P| \leqslant |\widetilde{P}| + |X|$. On the other hand, consider the set of all integer translates of $[0,1]^k$ intersecting $\widetilde{P}$. All of these translates intersect $\partial\widetilde{P}$ or are of the form $a+[0,1]^k$ with $a \in P$. As these clearly cover $\widetilde{P}$, we find $|\widetilde{P}| \leqslant |P| + |X|$. □

Finally, the following observation implies A' being close to its convex hull implies $d_k(A')$ is small, which as mentioned in the introduction yields a weak converse to Theorem 2.1.1 b).

Observation 2.2.26. Given a set $A' \subset B = B(n_1,\dots,n_k)$, we have

$$d_k(A') \leqslant 2^k |\operatorname{co}(A') \setminus A'| + 2^{k+2}k(k+1)\min\{n_i\}^{-1}|B|$$

Proof. By Observation 2.2.23, we only need to show $d_k(\operatorname{co}(A')) \leqslant 2^{k+2}k(k+1)\min\{n_i\}^{-1}|B|$. This follows as by Observation 2.2.24 we have

$$\begin{aligned} d_k(\operatorname{co}(A')) &= |\operatorname{co}(A') + \operatorname{co}(A')| - 2^k|\operatorname{co}(A')| \\ &\leqslant |(\widetilde{\operatorname{co}}(A') + \widetilde{\operatorname{co}}(A')) \cap \mathbb{Z}^k| - 2^k|\operatorname{co}(A')| \\ &\leqslant |\widetilde{\operatorname{co}}(A') + \widetilde{\operatorname{co}}(A')| - 2^k|\operatorname{co}(A')| + 2^{k+1}k(k+1)\min\{n_i\}^{-1}|B| \\ &= 2^k|\widetilde{\operatorname{co}}(A')| - 2^k|\operatorname{co}(A')| + 2^{k+1}k(k+1)\min\{n_i\}^{-1}|B| \\ &\leqslant 2^{k+2}k(k+1)\min\{n_i\}^{-1}|B|. \end{aligned}$$

□

2.3 Proof of Theorem 2.1.4 for k given Theorem 2.1.1 b) for $k-1$

For $k=1$, Theorem 2.1.4 and Theorem 2.1.1 are implied by Freiman's $3|A|-4$ theorem [80], so we suppose from now on that $k \geqslant 2$. In this section, we prove Theorem 2.1.4 for dimension k given Theorem 2.1.1 b) for dimension $k-1$. We recall by Convention 2.2.7 that we assume that A is reduced and we may assume $\min\{n_i\}^{-1} \ll_\zeta \delta \ll_\zeta \varepsilon_0 \ll_\zeta 1$ for some increasing function $\zeta : \mathbb{R}_+^\star \to \mathbb{R}_+^\star$. Define $\varepsilon \geqslant \varepsilon_0$ to be the density of A in B, so we have

$$|A| = \varepsilon|B|, \text{ and } d_k(A) \leqslant \delta|B|. \tag{2.3}$$

2.3.1 Outline of the proof

We will create sets

$$A \supset A_1 \supset A_2 \supset A_3 \supset A_4 \supset A_5 \subset A_+ \supset A_\star$$

(note that $A_5 \subset A_+$) such that $|A \triangle A_\star|$ is small, and $A_\star$ has a large number of properties which allow us to show that $A_\star$ is close to $\operatorname{co}(A_\star)$. From this we will be able to conclude that A is close to $\operatorname{co}(A)$.

In Section 2.3.2, we derive a general reduction to sets for which the projection under π satisfies the induction hypothesis.

In Section 2.3.3, we construct $A \supset A_1 \supset A_2 \supset A_3$ such that A_3 is reduced, has large rows R_x close to $\widehat{\mathrm{co}}(R_x)$, and has $\pi(A_3)$ close to $\mathrm{co}(\pi(A_3))$.

In Section 2.3.4, we construct $A_3 \supset A_4 \supset A_5$ such that A_5 has the same properties as A_3 and the arithmetic progressions $\widehat{\mathrm{co}}(R_x)$ have the same step size d.

In Section 2.3.5, we show that $d = 1$, i.e., $\widehat{\mathrm{co}}(R_x) = \mathrm{co}(R_x)$ is an interval for all rows R_x of A_5.

In Section 2.3.6, we show that filling in the rows of A_5 to make a set $A_+ \supset A_5$ preserves the properties that A_5 had (this is the only step we deviate from throwing away a subset of rows).

In Section 2.3.7, we show that we can approximate A_+ with a subset $A_\star \subset A_+$ which has simultaneously

1. Few vertices on $\widetilde{\mathrm{co}}(A_\star)$
2. $\pi(A_\star)$ close to $\mathrm{co}(\pi(A_\star))$
3. The technical condition Observation 2.3.35.

Up to this point, we were able to show that $|A \triangle A_+| \leqslant \delta^{O(1)}|A|$. However obtaining $A_\star$ involves a double recursion, and we are only able to show $|A \triangle A_\star| = o(1)|A|$ where $o(1) \to 0$ as $\delta \to 0$.

In Section 2.3.8, we show that $A_\star$ is close to its convex hull. The key step is to convert the problem to one of bounding the size of the epigraph of a certain infimum-convolution of a function by the size of the epigraph of the original function.

Finally, in Section 2.3.9 we finish the proof of Theorem 2.1.4 by showing that $A_\star$ being close to its convex hull implies A is close to its convex hull.

2.3.2 Exploiting the inductive hypothesis

In this section we prove a result, relying on the inductive hypothesis, which we will frequently apply that allows us to remove a small number of rows from a set A' to ensure that the projection $\pi(A')$ is close to $\widehat{\mathrm{co}}(\pi(A'))$. Recall we introduced in Definition 2.2.17 a small constant c.

Proposition 2.3.1. Given $0 < \alpha < \lambda$, there exists an increasing function (depending on these parameters) $\zeta : \mathbb{R}_+^\star \to \mathbb{R}_+^\star$ with $\zeta(x) \to 0$ as $x \to 0$, such that if $\min\{n_i\}^{-1} \ll_\zeta \sigma \ll_\zeta \varepsilon_0 \ll_\zeta 1$, then the following holds. Let $A' \subset B$ with $|A'| = \varepsilon'|B| \geqslant \frac{\varepsilon_0}{2}|B|$ and $d_k(A') \leqslant \sigma^\lambda|B|$. Then there exists $A'' \subset A'$ formed as a union of rows R_x of A' with

$$|\widehat{\mathrm{co}}(\pi(A'')) \setminus \pi(A'')| \leqslant \sigma^\alpha|\pi(B)|, \quad |A' \setminus A''| \leqslant \sigma^{\lambda-\alpha-c}|B|.$$

Furthermore, if A' is reduced then A'' is reduced and in particular $\widehat{\mathrm{co}}(\pi(A'')) = \mathrm{co}(\pi(A''))$.

Proof of Proposition 2.3.1. Choose ζ tending to 0 sufficiently fast to guarantee the implicit inequalities below. We can take $A'' = \emptyset$ if $\lambda \leqslant \alpha + c$, so suppose $\lambda > \alpha + c$. Let

$$E_i = \{x \in \pi(A') : |\pi^{-1}(x) \cap A'| \geqslant i\}, \qquad F_i = \{x \in \pi(A'+A') : |\pi^{-1}(x) \cap (A'+A')| \geqslant i\}.$$

Note that $E_1 \supset E_2 \supset \ldots$ and $F_1 \supset F_2 \supset \ldots$, and we have

$$|A'| = \sum_{i=1}^{n_1} |E_i|, \text{ and } |A'+A'| = \sum_{i=1}^{2n_1-1} |F_i|. \tag{2.4}$$

We note that $E_i + E_i \subset F_{2i-1}, F_{2i-2}$, so we have

$$\begin{aligned} |A'+A'| &\geqslant -2^{k-1} n_1^{-1} |B| + 2\sum_{i=1}^{n_1} |E_i + E_i| \\ &\geqslant -\sigma^{\lambda} |B| + 2\sum_{i=1}^{n_1} |E_i + E_i|. \end{aligned}$$

Subtracting $2^k |A'| = 2\sum_{i=1}^{n_1} 2^{k-1} |E_i|$, we obtain $d_k(A') \geqslant -\sigma^{\lambda}|B| + 2\sum_{i=1}^{n_1} d_{k-1}(E_i)$, so by the hypothesis $\sigma^{\lambda}|B| \geqslant d_k(A')$ we see that

$$\sigma^{\lambda} |B| \geqslant \sum_{i=1}^{n_1} d_{k-1}(E_i). \tag{2.5}$$

Let i_0 be the first index with $d_{k-1}(E_{i_0}) \leqslant \sigma^{\alpha+c/2}|E_{i_0}|$, which exists as otherwise by (2.4),(2.5),

$$\sigma^{\lambda}|B| \geqslant \sigma^{\alpha+c/2}|A'| \geqslant \sigma^{\alpha+c/2}\frac{\varepsilon_0}{2}|B| > \sigma^{\lambda}|B|.$$

Let $A'' := \pi^{-1}(E_{i_0}) \cap A' \subset A'$ be the union of all rows of size at least i_0. By construction,

$$d_{k-1}(\pi(A'')) = d_{k-1}(E_{i_0}) \leqslant \sigma^{\alpha+c/2}|E_{i_0}| = \sigma^{\alpha+c/2}|\pi(A'')|. \tag{2.6}$$

Also as $|E_i|$ is decreasing in i, $\sum_{i=1}^{i_0-1} |E_i| \geqslant \frac{i_0-1}{n_1}|A'|$ by (2.4). Thus by (2.5) and Observation 2.2.21,

$$\begin{aligned} \sigma^{\lambda}|B| \geqslant \sum_{i=1}^{n_1} d_{k-1}(E_i) &\geqslant \sigma^{\alpha+c/2}\frac{i_0-1}{n_1}|A'| - n_1 2^{2(k-1)} \min\{n_i\}^{-1} n_1^{-1}|B| \\ &\geqslant \sigma^{\alpha+c/2}\frac{i_0-1}{n_1} \cdot \frac{\varepsilon_0}{2}|B| - \sigma^{\lambda}|B|. \end{aligned}$$

Thus we obtain

$$i_0 - 1 \leqslant 4\sigma^{\lambda-\alpha-c/2}\varepsilon_0^{-1} n_1 \leqslant \sigma^{\lambda-\alpha-3c/4} n_1.$$

As the $|\pi(A'\setminus A'')| \leqslant |\pi(B')| = n_1^{-1}|B|$ nonempty rows in $A'\setminus A''$ have size at most $\sigma^{\lambda-\alpha-3c/4}n_1$, we have

$$|A'\setminus A''| \leqslant \sigma^{\lambda-\alpha-3c/4}|B| < \sigma^{\lambda-\alpha-c}|B|.$$

We have $|A'\setminus A''| \leqslant 2^{-(k+2)}\varepsilon_0|B|$ so $|A''| \geqslant \frac{\varepsilon_0}{4}|B|$. In particular, $|\pi(A'')| \geqslant \frac{\varepsilon_0}{4}|\pi(B)|$. The set $\pi(A'')$ has $d_{k-1}(\pi(A'')) \leqslant \sigma^{\alpha+c/2}|\pi(A'')|$ by (2.6), and has density at least $\frac{\varepsilon_0}{4}$ in $\pi(B)$, which has side lengths at least $\min(n_i)$. By Observation 2.2.18, the number of parallel hyperplanes needed to cover $\pi(A'')$ is at least $\frac{\varepsilon_0}{4}\min(n_i) > m_{k-1}(\sigma^{\alpha+c/2})$, so by Corollary 2.1.2 for dimension $k-1$ we deduce that

$$|\widehat{\mathrm{co}}(\pi(A''))\setminus\pi(A'')| \leqslant c_{k-1}\sigma^{\alpha+c/2}|\pi(B)| \leqslant \sigma^{\alpha}|\pi(B)|.$$

By Observation 2.2.23 applied to A' (instead of A), and the fact that $|A'| \geqslant 2^{-1}\varepsilon_0|B|$, $d_k(A') < \sigma^{\lambda}|B|$ and $|A'\setminus A''| \leqslant 2^{-(k+2)}\varepsilon_0|B|$ we deduce that if A' is reduced then A'' is reduced. In particular, $\pi(A'')$ is reduced.

□

2.3.3 Reductions Part 1: All rows are dense in large APs

We start by constructing in a sequence of steps a set $A_3 \subset A$ such that $|A\setminus A_3|$ is small, $\pi(A_3)$ is close to $\mathrm{co}(\pi(A_3))$ and the rows R_x of A_3 are large and close to $\widehat{\mathrm{co}}(R_x)$. In the continuous setting, a similar preliminary reduction was carried out by Figalli and Jerison at the beginning of [67].

$A_1 \subset A$ has $\pi(A_1)$ close to its convex progression: Construction

There exists an increasing function $\zeta : \mathbb{R}_+^\star \to \mathbb{R}_+^\star$ with $\zeta(x) \to 0$ as $x \to 0$, such that if $\min\{n_i\}^{-1} \ll_\zeta \delta \ll_\zeta \varepsilon_0 \ll_\zeta 1$ (which we assume wlog by Convention 2.2.14), then the following holds. We can apply Proposition 2.3.1 to A with $\sigma = \delta$, $\alpha = \frac{1}{2}$, $\lambda = 1$ and $\varepsilon' = \varepsilon \geqslant \frac{\varepsilon_0}{2}$ (by (2.3)) to obtain a reduced set $A_1 \subset A$ with

$$|\mathrm{co}(\pi(A_1))\setminus\pi(A_1)| \leqslant \delta^{\frac{1}{2}}|\pi(B)|, \quad |A\setminus A_1| \leqslant \delta^{\frac{1}{2}-c}|B|. \tag{2.7}$$

By Observation 2.2.23, we have

$$d_k(A_1) \leqslant \delta|B| + 2^k\delta^{\frac{1}{2}-c}|B| \leqslant \delta^{\frac{1}{2}-2c}|B|. \tag{2.8}$$

A_2 has large rows close to their convex progressions: Setup

We show that assuming $\mathrm{co}(\pi(A'))\setminus\pi(A')$ is small, we can create a subset $A'' \subset A'$ by deleting rows with big doubling or small size without changing the size of A' too much.

Proposition 2.3.2. Given $0 < \gamma$ and $0 < \beta < \alpha < \lambda$, there exists an increasing function (depending on these parameters) $\zeta : \mathbb{R}_+^\star \to \mathbb{R}_+^\star$ with $\zeta(x) \to 0$ as $x \to 0$, such that if $\min\{n_i\}^{-1} \ll_\zeta \delta \ll_\zeta 1$, then

the following holds. Let $A' \subset B$ with

$$d_k(A') \leqslant \delta^{\lambda}|B|, \qquad |\operatorname{co}(\pi(A')) \setminus \pi(A')| \leqslant \delta^{\alpha}|\pi(B)|.$$

If $A'' \subset A'$ is the union all rows R_x which satisfy $d_1(R_x) \leqslant \delta^{\beta} n_1$ and $|R_x| \geqslant \delta^{\gamma} n_1$, then

$$|A' \setminus A''| \leqslant (\delta^{\alpha-\beta-c} + \delta^{\gamma})|B|.$$

Proof. Choose ζ tending to 0 sufficiently fast to guarantee the implicit inequalities below. Let A''' be the union all rows R_x of A' which satisfy $d_1(R_x) \leqslant \delta^{\beta} n_1$. For $0 \neq \vec{v} \in \{0\} \times \{0,1\}^{k-1}$ and $x \in \{0\} \times \mathbb{Z}^{k-1}$, we have

$$|R_x + R_{x+\vec{v}}| - |R_x| - |R_{x+\vec{v}}| \geqslant \begin{cases} 0 & |\{x, x+\vec{v}\} \cap \pi(A')| = 0 \\ -n_1 & |\{x, x+\vec{v}\} \cap \pi(A')| = 1 \\ -1 & |\{x, x+\vec{v}\} \cap \pi(A')| = 2 \end{cases}$$

From $|\operatorname{co}(\pi(A')) \setminus \pi(A')| \leqslant \delta^{\alpha} n_1^{-1}|B|$ and Observation 2.2.22, we have

$$\begin{aligned} \left| \left\{ x \in \{0\} \times \mathbb{Z}^{k-1} : |\{x, x+\vec{v}\} \cap \pi(A')| = 1 \right\} \right| &\leqslant 2\left|\operatorname{co}(\pi(A')) \setminus \pi(A')\right| + 2(k-1) n_1^{-1} \min(n_i)^{-1}|B| \\ &\leqslant \delta^{\alpha - c/4} n_1^{-1}|B| \end{aligned}$$

and

$$\left| \{ x \in \{0\} \times \mathbb{Z}^{k-1} : |\{x, x+\vec{v}\} \cap \pi(A')| = 2 \} \right| \leqslant |\pi(B)| \leqslant \min(n_i)^{-1}|B| \leqslant \delta^{\alpha - c/4}|B|.$$

Hence, as $\sum_{x \in \{0\} \times \mathbb{Z}^{k-1}} |R_x| = |A'|$, we have (taking $\vec{v} \in \{0\} \times \{0,1\}^{k-1}$ and $x \in \{0\} \times \mathbb{Z}^{k-1}$)

$$\begin{aligned} |A' + A'| &\geqslant \sum_{\vec{v}} \sum_{x} |R_x + R_{x+\vec{v}}| \\ &= \left(\sum_{j=0}^{2} \sum_{0 \neq \vec{v}} \sum_{|\{x, x+\vec{v}\} \cap \pi(A')| = j} |R_x + R_{x+\vec{v}}| \right) + \sum_{x \in \pi(A')} |R_x + R_x| \\ &\geqslant \left(\sum_{0 \neq \vec{v}} \sum_{x} |R_x| + |R_{x+\vec{v}}| \right) - 2(2^{k-1} - 1)\delta^{\alpha - c/4}|B| + \sum_{x \in \pi(A')} |R_x + R_x| \\ &\geqslant (2^k - 2)|A'| - \delta^{\alpha - c/2}|B| + \sum_{x \in \pi(A')} |R_x + R_x|. \end{aligned}$$

In particular, as $\sum_{x\in\pi(A')}|R_x| = |A'|$ and $d_1(R_x)\geqslant -1$ for all x, we have

$$\begin{aligned}\delta^\lambda|B| \geqslant d_k(A') &\geqslant -\delta^{\alpha-c/2}|B| + \sum_{x\in\pi(A')} d_1(R_x)\\ &\geqslant -\delta^{\alpha-c/2}|B| - \min(n_i)^{-1}|B| + \sum_{x\in\pi(A'\setminus A''')} d_1(R_x)\\ &\geqslant -\delta^{\alpha-3c/4}|B| + |\pi(A'\setminus A''')|\delta^\beta n_1,\end{aligned}$$

so

$$|A'\setminus A'''| \leqslant n_1|\pi(A'\setminus A''')| \leqslant (\delta^{\alpha-\beta-3c/4}+\delta^{\lambda-\beta})|B| \leqslant \delta^{\alpha-\beta-c}|B|.$$

Finally note that $A''\subset A'''$ satisfies $|A'''\setminus A''| \leqslant \delta^\gamma n_1|\pi(B)| \leqslant \delta^\gamma|B|$, from which the conclusion follows. □

A_2 has large rows close to their convex progression: Construction

There exists an increasing function $\zeta:\mathbb{R}_+^\star\to\mathbb{R}_+^\star$ with $\zeta(x)\to 0$ as $x\to 0$, such that if $\min\{n_i\}^{-1}\ll_\zeta \delta\ll_\zeta \varepsilon_0\ll_\zeta 1$ (which we assume wlog by Convention 2.2.14), then the following holds. We can apply Proposition 2.3.2 to A_1 with $\lambda=\frac{1}{2}-2c$, $\alpha=\frac{1}{2}-3c$, $\beta=\frac{3}{10}$, and $\gamma=\frac{1}{5}$ (by (2.7),(2.8)) to obtain a subset $A_2\subset A_1$ that is a union of rows. For all rows $R_x\subset A_2$, we have

$$d_1(R_x)\leqslant \delta^{\frac{3}{10}}n_1, \quad |R_x|\geqslant \delta^{\frac{1}{5}}n_1, \tag{2.9}$$

and by (2.7) we additionally have

$$|A\setminus A_2| \leqslant |A\setminus A_1| + |A_1\setminus A_2| \leqslant \left(\delta^{\frac{1}{2}-c}+\delta^{\frac{1}{5}-4c}+\delta^{\frac{1}{5}}\right)|B| \leqslant \delta^{\frac{1}{5}-5c}|B|. \tag{2.10}$$

By Observation 2.2.23, we have A_2 is reduced and

$$d_k(A_2)\leqslant \left(\delta+2^k\delta^{\frac{1}{5}-5c}\right)|B| \leqslant \delta^{\frac{1}{5}-6c}|B|. \tag{2.11}$$

Freiman's $3k-4$ theorem [80] says that for any $R\subset\mathbb{Z}$, we have

$$d_1(R)\geqslant \min(|R|-3, |\widehat{\mathrm{co}}(R)\setminus R|-1).$$

Therefore, because $\delta^{\frac{3}{10}}n_1 < \delta^{\frac{1}{5}}n_1-3$, we have by (2.9) that every row R_x of A_2 satisfies

$$|\widehat{\mathrm{co}}(R_x)\setminus R_x| \leqslant \delta^{\frac{3}{10}}n_1+1 \leqslant \delta^{\frac{1}{10}}|R_x|+1 \leqslant 2\delta^{\frac{1}{10}}|R_x|. \tag{2.12}$$

Remark 2.3.3. In particular, this means that for each non-empty row R_x we have $|R_x| > \left\lceil\frac{|\widehat{\mathrm{co}}(R_x)|}{2}\right\rceil$, so there exist two elements $z_1,z_2\in R_x$ with $z_1-z_2=(d_x,0,0\ldots,0)$, where d_x is the common difference in the arithmetic progression $\widehat{\mathrm{co}}(R_x)$.

$A_3 \subset A_2$ has $\pi(A_3)$ close to its convex progression: Construction

There exists an increasing function $\zeta : \mathbb{R}_+^\star \to \mathbb{R}_+^\star$ with $\zeta(x) \to 0$ as $x \to 0$, such that if $\min\{n_i\}^{-1} \ll_\zeta \delta \ll_\zeta \varepsilon_0 \ll_\zeta 1$ (which we assume wlog by Convention 2.2.14), then the following holds. We can apply Proposition 2.3.1 to A_2 with $\sigma = \delta$, $\alpha = \frac{1}{10}$, $\varepsilon' \geqslant \varepsilon - \delta^{\frac{1}{5}-5c} \geqslant \frac{\varepsilon_0}{2}$ and $\lambda = \frac{1}{5} - 6c$ (by (2.10),(2.11)) to obtain a reduced set $A_3 \subset A_2$ with

$$|\operatorname{co}(\pi(A_3)) \setminus A_3| \leqslant \delta^{\frac{1}{10}} |\pi(B)|, \qquad |A_2 \setminus A_3| \leqslant \delta^{\frac{1}{10}-7c}|B|. \tag{2.13}$$

By (2.10) and (2.13), we have

$$|A \setminus A_3| \leqslant |A \setminus A_2| + |A_2 \setminus A_3| \leqslant \left(\delta^{\frac{1}{5}-5c} + \delta^{\frac{1}{10}-7c}\right)|B| \leqslant \delta^{\frac{1}{10}-8c}|B|, \tag{2.14}$$

and by Observation 2.2.23, we have

$$d_k(A_3) \leqslant \left(\delta + 2^k \delta^{\frac{1}{10}-8c}\right)|B| \leqslant \delta^{\frac{1}{10}-9c}|B|. \tag{2.15}$$

Finally, as the rows of A_3 are a subset of the rows of A_2, by (2.9) and (2.12), we have

$$|R_x| \geqslant \delta^{\frac{1}{5}} n_1, \qquad |\widehat{\operatorname{co}}(R_x) \setminus R_x| \leqslant 2\delta^{\frac{1}{10}}|R_x|. \tag{2.16}$$

2.3.4 Reductions Part 2: All rows are in APs of the same step size

We now find a set $A_5 \subset A_3$ which has the same properties as A_3, and furthermore has the property that for each row R_x, the arithmetic progressions $\widehat{\operatorname{co}}(R_x)$ have the same step sizes. To do this, we carefully analyze a discrete analogue of Voronoi cells.

Let d_x be the smallest consecutive difference between two consecutive elements in R_x, which as noted in Remark 2.3.3 is also the common difference of $\widehat{\operatorname{co}}(R_x)$, and let $d = \min d_x$.

$A_4 \subset A_3$ has all rows in same step size APs: Setup

We now show that the rows with $d_x > d$ carry small weight.

Proposition 2.3.4. Given $0 < \alpha < \lambda$ there exists an increasing function (depending on these parameters) $\zeta : \mathbb{R}_+^\star \to \mathbb{R}_+^\star$ with $\zeta(x) \to 0$ as $x \to 0$, such that if $\min\{n_i\}^{-1} \ll_\zeta \delta \ll_\zeta \varepsilon_0 \ll_\zeta 1$, then the following holds. Let $A' \subset B$ with $d_k(A') \leqslant \delta^\lambda |B|$ and $|\operatorname{co}(\pi(A')) \setminus \pi(A')| \leqslant \delta^\alpha |\pi(B)|$. Let d_x be the smallest consecutive difference between two elements of row $R_x \subset A'$, and let $d = \min d_x$. If $A'' \subset A'$ is the subset of rows R_x with $d_x = d$, then

$$|A' \setminus A''| \leqslant \delta^{\alpha - c}|B|.$$

Choose ζ tending to 0 sufficiently fast to guarantee the implicit inequalities below. Before starting the proof of Proposition 2.3.4, we need to prove some claims. Claim 2.3.5 shows that $|R_x + R_y|$ is

large if $d_x \neq d_y$, and Claim 2.3.6 creates a large set of disjoint row sums of this form. Claim 2.3.7 is used to prove Claim 2.3.8, which shows that this set of disjoint row sums has small intersection with $A'(+)A'$. Finally, Claim 2.3.9 shows $A'(+)A'$ is large, and we can carry out the proof outline described in Section 2.1.1.

Claim 2.3.5. Let $X,Y \subset \mathbb{Z}$ with $|X| \geqslant 2$ such that the smallest differences d_X, d_Y between consecutive elements of X and Y respectively satisfy $d_X < d_Y$. Then

$$|X+Y| \geqslant |X|+2|Y|-2.$$

Proof. Consider elements $x,x' \in X$ such that $x'-x = d_X$. Let $X_{<x}$ be the set of elements less than x in X and analogously $X_{>x'}$ those elements greater than x'. Then the following four sets

$$X_{<x}+\min(Y), (Y+x), (Y+x'), X_{>x'}+\max(Y),$$

are disjoint subsets of $X+Y$. □

Now, we define

$$f : \pi(A') \setminus \pi(A'') \to \pi(A'')$$

by letting $f(x) \in \pi(A'')$ be a closest point to x in Euclidean distance (breaking ties arbitrarily). Fibers of f should be thought of as a discrete analogue of Voronoi cells associated to $\pi(A'')$.

Claim 2.3.6. We have $x+f(x) \neq y+f(y)$ for distinct $x,y \in \pi(A') \setminus \pi(A'')$. In particular,

$$Z_1 := \bigsqcup_{x_1 \in \pi(A') \setminus \pi(A'')} R_{x_1} + R_{f(x_1)} \subset A'+A'$$

is a disjoint union.

Proof. Indeed, otherwise $x,y,f(x),f(y)$ form a parallelogram with distinct vertices with diagonals $xf(x)$ and $yf(y)$. However, in any parallelogram (even degenerate as long as the vertices are distinct), the longest diagonal is longer than all sides. Hence, if say $xf(x)$ is the longest diagonal, then $|x-f(x)| > |x-f(y)|$, a contradiction. □

Let

$$Z := A'(+)A' = \bigsqcup_{\vec{v} \in \{0\}\times\{0,1\}^{k-1}} \bigsqcup_{x_2 \in \{0\}\times\mathbb{Z}^{k-1}} R_{x_2}(+)R_{x_2+\vec{v}} \subset A'+A'.$$

We now analyze when $R_{x_1}+R_{f(x_1)}$ and $R_{x_2}(+)R_{x_2+\vec{v}}$ can intersect.

Claim 2.3.7. If $x_1 \in \pi(A') \setminus \pi(A'')$, $x_2 \in \{0\}\times\mathbb{Z}^{k-1}$, $\vec{v} \in \{0\}\times\{0,1\}^{k-1}$ are such that $R_{x_1}+R_{f(x_1)} \cap R_{x_2}(+)R_{x_2+\vec{v}} \neq \emptyset$, then either $\{x_1,f(x_1)\} = \{x_2,x_2+\vec{v}\}$ or $x_2,x_2+\vec{v} \in \pi(A') \setminus \pi(A'')$.

Proof. The hypothesis implies $x_1+f(x_1) = 2x_2+\vec{v}$. Therefore, the points $x_1,x_2,f(x_1),x_2+\vec{v}$ form a (possibly degenerate) parallelogram with diagonals $x_1f(x_1)$ and $x_2(x_2+\vec{v})$. As $x_1 \in \pi(A') \setminus \pi(A'')$ and

$f(x_1) \in \pi(A'')$, we deduce $x_1 \neq f(x_1)$. Assuming that $\{x_1, f(x_1)\} \neq \{x_2, x_2+\vec{v}\}$, the parallelogram $x_1, x_2, f(x_1), x_2+\vec{v}$ has distinct diagonals, not both degenerate to points. (We could have a degenerate parallelogram where $x_2 = x_2+\vec{v}$ is the midpoint between x_1 and $f(x_1)$). In such a parallelogram, the longest diagonal is strictly larger than all the sides. Indeed, assume wlog the vertices of the parallelogram are $\pm a, \pm b$ where $b \neq 0$ and $\{\pm a\} \neq \{\pm b\}$. Then $\max(2|a|, 2|b|) \geqslant |a|+|b| \geqslant \max(|a \pm b|)$. Moreover, the only way equality can be attained in the triangle inequality is if $a = \lambda b$ for some $\lambda \in \mathbb{R}$, and equality in the first inequality gives $|a| = |b|$, but this forces $\{\pm a\} = \{\pm b\}$, giving a contradiction.

The number of odd coordinates of $x_1 - f(x_1)$ is the same as the number of odd coordinates of $x_1 + f(x_1) = 2x_2 + \vec{v}$, which is the same as the number of non-zero coordinates of v. Hence, $|x_1 - f(x_1)| \geqslant |\vec{v}|$, or equivalently $|x_1 - f(x_1)| \geqslant |x_2 - (x_2+\vec{v})|$. Therefore, $x_1 f(x_1)$ is the longest diagonal of the above parallelogram. As in this parallelogram, the largest diagonal is strictly longer than all sides, we deduce that the diagonal $x_1 f(x_1)$ is strictly longer than $x_1 x_2$ and $x_1(x_2+\vec{v})$. By definition of $f(x_1)$ this implies $x_2, x_2+\vec{v} \notin \pi(A'')$. As $R_{x_2}, R_{x_2+\vec{v}}$ are nonempty we also have $x_2, x_2+\vec{v} \in \pi(A')$ and the result follows. □

Claim 2.3.8. For any $x_1 \in \pi(A') \setminus \pi(A'')$ we have

$$\left|(R_{x_1} + R_{f(x_1)}) \setminus Z\right| \geqslant |R_{x_1}| - 1.$$

Proof. We have $|(R_{x_1}+R_{f(x_1)}) \setminus Z| = |(R_{x_1}+R_{f(x_1)}) \setminus (R_{x_2}(+)R_{x_2+\vec{v}})|$ for the unique $x_2 \in \{0\} \times \mathbb{Z}^{k-1}$ and $\vec{v} \in \{0\} \times \{0,1\}^{k-1}$ such that $x_1 + f(x_1) = x_2 + (x_2+\vec{v})$. Clearly $|(R_{x_1}+R_{f(x_1)})| \geqslant |R_{x_1}| - 1$, so assume $(R_{x_1}+R_{f(x_1)}) \cap (R_{x_2}(+)R_{x_2+\vec{v}}) \neq \emptyset$. By Claim 2.3.7 we have that either $\{x_1, f(x_1)\} = \{x_2, x_2+\vec{v}\}$ or $x_2, x_2+\vec{v} \in \pi(A') \setminus \pi(A'')$. In the former case, by Claim 2.3.5 we have that

$$\left|(R_{x_1}+R_{f(x_1)}) \setminus (R_{x_2}(+)R_{x_2+\vec{v}})\right| = |R_{x_1}+R_{f(x_1)}| - |R_{x_1}| - |R_{f(x_1)}| + 1 \geqslant |R_{x_1}| - 1.$$

Assume now we are in the latter case. Let $z_1, z_2 \in R_{f(x_1)}$ such that $z_1 - z_2 = (d, 0, \ldots, 0)$. As the smallest difference in $R_{x_2}(+)R_{x_2+\vec{v}}$ is strictly larger than d, for every element $z \in R_{x_1}$ either $z+z_1$ or $z+z_2$ is not in $R_{x_2}(+)R_{x_2+\vec{v}}$, and if there were $z, z' \in R_{x_1}$ with $z+z_1 = z'+z_2$ then $z'-z = (d,0,\ldots,0)$ contradicting $x_1 \notin \pi(A'')$. Hence

$$\left|(R_{x_1}+R_{f(x_1)}) \setminus (R_{x_2}(+)R_{x_2+\vec{v}})\right| \geqslant |R_{x_1}| > |R_{x_1}| - 1.$$

□

Here we begin to make use of the fact that ζ tends to 0 sufficiently fast.

Claim 2.3.9. We have $|Z| \geqslant 2^k |A'| - 2^k \delta^{\alpha - c/2} |B|$.

Proof. Note that for $x_2 \in \{0\}\times\mathbb{Z}^{k-1}$, $0 \neq \vec{v} \in \{0\}\times\{0,1\}^{k-1}$, we have

$$|R_{x_2}(+)R_{x_2+\vec{v}}| - |R_{x_2}| - |R_{x_2+\vec{v}}| \geqslant \begin{cases} 0 & |\{x_2, x_2+\vec{v}\}\cap\pi(A')| = 0 \\ -n_1 & |\{x_2, x_2+\vec{v}\}\cap\pi(A')| = 1 \\ -1 & |\{x_2, x_2+\vec{v}\}\cap\pi(A')| = 2. \end{cases}$$

From $|\operatorname{co}(\pi(A'))\setminus\pi(A')| \leqslant \delta^{\alpha}n_1^{-1}|B|$ and Observation 2.2.22, we have

$$\begin{aligned}\left|\{x\in\{0\}\times\mathbb{Z}^{k-1} : |\{x,x+\vec{v}\}\cap\pi(A')| = 1\}\right| &\leqslant 2|\operatorname{co}(\pi(A'))\setminus\pi(A')| + 2(k-1)n_1^{-1}\min(n_i)^{-1}|B| \\ &\leqslant \delta^{\alpha-c/4}n_1^{-1}|B|\end{aligned}$$

and

$$\left|\{x\in\{0\}\times\mathbb{Z}^{k-1} : |\{x,x+\vec{v}\}\cap\pi(A')| = 2\}\right| \leqslant |\pi(B)| \leqslant \min(n_i)^{-1}|B| \leqslant \delta^{\alpha-c/4}|B|.$$

As $\sum_{x\in\{0\}\times\mathbb{Z}^{k-1}}|R_x| = |A'|$, we have (taking $\vec{v}\in\{0\}\times\{0,1\}^{k-1}$ and $x\in\{0\}\times\mathbb{Z}^{k-1}$)

$$\begin{aligned}|Z| &= \sum_{\vec{v}}\sum_{x}|R_x(+)R_{x+\vec{v}}| \\ &= \left(\sum_{j=0}^{2}\sum_{0\neq\vec{v}}\sum_{|\{x,x+\vec{v}\}\cap\pi(A')|=j}|R_x(+)R_{x+\vec{v}}|\right) + \sum_{x\in\pi(A')}|R_x(+)R_x| \\ &\geqslant \left(\sum_{0\neq\vec{v}}\sum_{x}|R_x|+|R_{x+\vec{v}}|\right) - 2(2^{k-1}-1)\delta^{\alpha-c/4}|B| + \sum_{x\in\pi(A')}|R_x(+)R_x| \\ &\geqslant \left(\sum_{0\neq\vec{v}}\sum_{x}|R_x|+|R_{x+\vec{v}}|\right) - 2(2^{k-1}-1)\delta^{\alpha-c/4}|B| + \left(\sum_{x\in\pi(A')}|R_x|+|R_x|\right) - n_1^{-1}|B| \\ &\geqslant -2(2^{k-1}-1)\delta^{\alpha-c/4}|B| - \min(n_i)^{-1}|B| + \sum_{\vec{v}}\sum_{x}|R_x|+|R_{x+\vec{v}}| \\ &\geqslant 2^k|A'| - 2^k\delta^{\alpha-c/2}|B|.\end{aligned}$$

□

Proof of Proposition 2.3.4. Note that

$$|\pi(A')\setminus\pi(A'')| \leqslant |\pi(B)| \leqslant \min(n_i)^{-1}|B| \leqslant \delta^{\alpha-c/2}|B|.$$

Thus, by Claim 2.3.6, Claim 2.3.8, and Claim 2.3.9, we have

$$\begin{aligned}|A'+A'| &\geqslant |Z\cup Z_1|\\ &= |Z| + \sum_{x_1\in\pi(A')\setminus\pi(A'')} |(R_{x_1}+R_{f(x_1)})\setminus Z|\\ &\geqslant 2^k|A'| - 2^k\delta^{\alpha-c/2}|B| + \sum_{x_1\in\pi(A')\setminus\pi(A'')} (|R_{x_1}|-1)\\ &\geqslant 2^k|A'| - \delta^{\alpha-3c/4}|B| + |A'\setminus A''|.\end{aligned}$$

We conclude

$$|A'\setminus A''| \leqslant d_k(A') + \delta^{\alpha-3c/4}|B| \leqslant \delta^{\lambda}|B| + \delta^{\alpha-3c/4}|B| \leqslant \delta^{\alpha-c}|B|.$$

□

$A_4 \subset A_3$ has all rows in same step size APs: Construction

There exists an increasing function $\zeta : \mathbb{R}_+^\star \to \mathbb{R}_+^\star$ with $\zeta(x) \to 0$ as $x \to 0$, such that if $\min\{n_i\}^{-1} \ll_\zeta \delta \ll_\zeta \varepsilon_0 \ll_\zeta 1$ (which we assume wlog by Convention 2.2.14), then the following holds. We can apply Proposition 2.3.4 to A_3 with $\lambda = \frac{1}{10} - 9c$ $\alpha = \frac{1}{10} - 10c$ (by (2.13),(2.15)), to obtain a subset $A_4 \subset A_3$ that is a union of rows. This subset satisfies

$$|A_3 \setminus A_4| \leqslant \delta^{\frac{1}{10}-11c}|B|, \tag{2.17}$$

and thus by (2.14) and (2.17),

$$|A\setminus A_4| \leqslant |A\setminus A_3| + |A_3\setminus A_4| \leqslant (\delta^{\frac{1}{10}-8c} + \delta^{\frac{1}{10}-11c})|B| \leqslant \delta^{\frac{1}{10}-12c}|B|. \tag{2.18}$$

By Observation 2.2.23, A_4 is reduced and we have

$$d_k(A_4) \leqslant (\delta + 2^k\delta^{\frac{1}{10}-12c})|B| \leqslant \delta^{\frac{1}{10}-13c}|B|. \tag{2.19}$$

By construction and Remark 2.3.3, $\widehat{\mathrm{co}}(R_x)$ has the same step size d for all rows R_x of A_4. Finally, by (2.16), we have

$$|R_x| \geqslant \delta^{\frac{1}{5}}n_1, \qquad |\widehat{\mathrm{co}}(R_x)\setminus R_x| \leqslant 2\delta^{\frac{1}{10}}|R_x|. \tag{2.20}$$

$A_5 \subset A_4$ with $\pi(A_5)$ close to its convex progression: Construction

There exists an increasing function $\zeta : \mathbb{R}_+^\star \to \mathbb{R}_+^\star$ with $\zeta(x) \to 0$ as $x \to 0$, such that if $\min\{n_i\}^{-1} \ll_\zeta \delta \ll_\zeta \varepsilon_0 \ll_\zeta 1$ (which we assume wlog by Convention 2.2.14), then the following holds. We can apply Proposition 2.3.1 to A_4 with $\sigma = \delta$, $\lambda = \frac{1}{10} - 13c$, $\alpha = \frac{1}{20}$ and $\varepsilon' \geqslant \varepsilon - \delta^{\frac{1}{10}-12c} \geqslant \frac{\varepsilon_0}{2}$ (by (2.19),(2.18))

to obtain a reduced set $A_5 \subset A_4$ with

$$|\mathrm{co}(\pi(A_5)) \setminus \pi(A_5)| \leqslant \delta^{\frac{1}{20}}|\pi(B)| \qquad |A_4 \setminus A_5| \leqslant \delta^{\frac{1}{20}-14c}|B|. \tag{2.21}$$

By (2.18) and (2.21), we have

$$|A \setminus A_5| \leqslant |A \setminus A_4| + |A_4 \setminus A_5| \leqslant (\delta^{\frac{1}{10}-12c} + \delta^{\frac{1}{20}-14c})|B| \leqslant \delta^{\frac{1}{20}-15c}|B|, \tag{2.22}$$

and by Observation 2.2.23, we have

$$d_k(A_5) \leqslant (\delta + 2^k\delta^{\frac{1}{20}-15c})|B| \leqslant \delta^{\frac{1}{20}-16c}|B|. \tag{2.23}$$

Furthermore, all rows of A_5 are also rows of A_4, so have the same step size d and satisfy (2.20), so

$$|R_x| \geqslant \delta^{\frac{1}{5}}n_1 \geqslant 2, \qquad |\widehat{\mathrm{co}}(R_x) \setminus R_x| \leqslant 2\delta^{\frac{1}{10}}|R_x|. \tag{2.24}$$

2.3.5 Reductions Part 3: Showing the rows of A_5 are almost intervals

We now show that the arithmetic progressions $\widehat{\mathrm{co}}(R_x)$ containing the rows R_x of A_5 are in fact intervals, i.e., $\widehat{\mathrm{co}}(R_x) = \mathrm{co}(R_x)$. We suppose by way of contradiction that the rows R_x have convex progression $\widehat{\mathrm{co}}(R_x)$ with the same step size $d \neq 1$.

Definition 2.3.10. Let $\pi' : \mathbb{Z}^k \to \mathbb{Z}$ be the projection onto the second coordinate. For a set $A' \subset \mathbb{Z}^k$, we let a "hyperplane" H_y be $\pi'^{-1}(y) \cap A'$.

We shall make a series of temporary reductions in order to arrive at a contradiction, and we shall notate sets used in this proof by contradiction with the dagger symbol †.

Remark 2.3.11. The hyperplanes H_y of a set A' are unions of rows R_x.

$A_6^\dagger \subset A_5$ has big hyperplanes with small doubling: Setup

First, we show that assuming $\mathrm{co}(\pi(A')) \setminus \pi(A')$ is small, we can create a subset $A'' \subset A'$ by deleting hyperplanes with big doubling or small size without changing the size of A' too much. This is analogous to Proposition 2.3.2 for rows.

Proposition 2.3.12. Given $0 < \lambda$ and $0 < \beta < \alpha < \lambda$ there exists an increasing function (depending on these parameters) $\zeta : \mathbb{R}_+^\star \to \mathbb{R}_+^\star$ with $\zeta(x) \to 0$ as $x \to 0$, such that if $\min\{n_i\}^{-1} \ll_\zeta \delta \ll_\zeta \varepsilon_0 \ll_\zeta 1$, then the following holds. Let $A' \subset B$ with

$$d_k(A') \leqslant \delta^\lambda|B|, \qquad |\mathrm{co}(\pi(A')) \setminus \pi(A')| \leqslant \delta^\alpha|\pi(B)|.$$

If $A'' \subset A'$ is the union of all hyperplanes H_y with $d_{k-1}(H_y) \leqslant \delta^\beta n_2^{-1}|B|$ and $|H_y| \geqslant \delta^\gamma n_2^{-1}|B|$, then

$$|A' \setminus A''| \leqslant (\delta^{\alpha-\beta-c} + \delta^\gamma)|B|.$$

Proof. Choose ζ tending to 0 sufficiently fast to guarantee the implicit inequalities below. Let $A''' \subset A'$ be the union of the hyperplanes H_y with $d_{k-1}(H_y) \leqslant \delta^{\beta} n_2^{-1}|B|$. For $0 \neq \vec{v} \in \{0\} \times \{1\} \times \{0,1\}^{k-2}$ we have

$$|R_x + R_{x+\vec{v}}| - |R_x| - |R_{x+\vec{v}}| \geqslant \begin{cases} 0 & |\{x, x+\vec{v}\} \cap \pi(A')| = 0 \\ -n_1 & |\{x, x+\vec{v}\} \cap \pi(A')| = 1 \\ -1 & |\{x, x+\vec{v}\} \cap \pi(A')| = 2 \end{cases}$$

From $|\operatorname{co}(\pi(A')) \setminus \pi(A')| \leqslant \delta^{\alpha} n_1^{-1}|B|$ and Observation 2.2.22, we have

$$\begin{aligned} \left|\{x \in \{0\} \times \mathbb{Z}^{k-1} : |\{x, x+\vec{v}\} \cap \pi(A')| = 1\}\right| &\leqslant 2|\operatorname{co}(\pi(A')) \setminus \pi(A')| + 2(k-1)n_1^{-1} \min(n_i)^{-1}|B| \\ &\leqslant \delta^{\alpha - c/4} n_1^{-1}|B| \end{aligned}$$

and

$$\left|\{x \in \{0\} \times \mathbb{Z}^{k-1} : |\{x, x+\vec{v}\} \cap \pi(A')| = 2\}\right| \leqslant |\pi(B)| \leqslant \min(n_i)^{-1}|B| \leqslant \delta^{\alpha - c/4}|B|.$$

Hence as $\sum_{x \in \{0\} \times \mathbb{Z}^{k-1}} |R_x| = |A'|$, we have (taking $\vec{v} \in \{0\} \times \{1\} \times \{0,1\}^{k-2}$ and $x \in \{0\} \times \mathbb{Z}^{k-1}$)

$$\begin{aligned} |A' + A'| &\geqslant \left(\sum_{y \in \mathbb{Z}} |H_y + H_y|\right) + \sum_{j=0}^{2} \sum_{\vec{v}} \sum_{|\{x, x+\vec{v}\} \cap \pi(A')| = j} |R_x + R_{x+\vec{v}}| \\ &\geqslant \left(\sum_{y \in \mathbb{Z}} |H_y + H_y|\right) + \left(\sum_{\vec{v}} \sum_{x} |R_x| + |R_{x+\vec{v}}|\right) - 2 \cdot 2^{k-2} \delta^{\alpha - c/4}|B| \\ &\geqslant \left(\sum_{y \in \mathbb{Z}} |H_y + H_y|\right) + 2^{k-1}|A'| - \delta^{\alpha - c/2}|B|. \end{aligned}$$

Now, as $\sum_{y \in \mathbb{Z}} |H_y| = |A'|$, by Observation 2.2.21 with H_y and the box $\pi'^{-1}(y) \cap B$, we have

$$\begin{aligned} \delta^{\lambda}|B| \geqslant d_k(A') &\geqslant \left(\sum_{y \in \pi'(A')} d_{k-1}(H_y)\right) - \delta^{\alpha - c/2}|B| \\ &\geqslant \left(\sum_{y \in \pi'(A') \setminus \pi'(A''')} d_{k-1}(H_y)\right) - n_2 \cdot 2^{2(k-1)} \min(n_i)^{-1} n_2^{-1}|B| - \delta^{\alpha - c/2}|B| \\ &\geqslant |\pi'(A' \setminus A''')| \cdot \delta^{\beta} n_2^{-1}|B| - \delta^{\alpha - 3c/4}|B| \\ &\geqslant \delta^{\beta}|A' \setminus A'''| - \delta^{\alpha - 3c/4}|B|. \end{aligned}$$

Therefore,

$$|A' \setminus A'''| \leqslant (\delta^{\lambda - \beta} + \delta^{\alpha - \beta - 3c/4})|B| \leqslant \delta^{\alpha - \beta - c}|B|.$$

Note that $A'' \subset A'''$ satisfies $|A''' \setminus A''| \leqslant n_2 \delta^{\gamma} n_2^{-1}|B| = \delta^{\gamma}|B|$, from which the conclusion follows. □

$A_6^\dagger \subset A_5$ has big hyperplanes with small doubling: Construction

There exists an increasing function $\zeta : \mathbb{R}_+^\star \to \mathbb{R}_+^\star$ with $\zeta(x) \to 0$ as $x \to 0$, such that if $\min\{n_i\}^{-1} \ll_\zeta \delta \ll_\zeta \varepsilon_0 \ll_\zeta 1$ (which we assume wlog by Convention 2.2.14), then the following holds. We can apply Proposition 2.3.12 to A_5 with $\lambda = \frac{1}{20} - 16c$, $\alpha = \frac{1}{40}$, $\beta = \frac{1}{80}$, and $\gamma = \frac{1}{160} - c$, obtaining a set $A_6^\dagger \subset A_5$ that is a union of hyperplanes. This set satisfies

$$|A_5 \setminus A_6^\dagger| \leqslant (\delta^{\frac{1}{80}-c} + \delta^{\frac{1}{160}-c})|B| \leqslant \delta^{\frac{1}{160}-2c}|B| \tag{2.25}$$

and for every hyperplane $H_y \subset A_6^\dagger$ we have

$$d_{k-1}(H_y) \leqslant \delta^{\frac{1}{80}} n_2^{-1}|B|, \qquad |H_y| \geqslant \delta^{\frac{1}{160}-c} n_2^{-1}|B|. \tag{2.26}$$

We also have by (2.22) and (2.25) that

$$|A \setminus A_6^\dagger| \leqslant |A \setminus A_5| + |A_5 \setminus A_6^\dagger| \leqslant (\delta^{\frac{1}{20}-15c} + \delta^{\frac{1}{160}-2c})|B| \leqslant \delta^{\frac{1}{160}-3c}|B|, \tag{2.27}$$

and by Observation 2.2.23 we have $A_6^\dagger$ is reduced and

$$d_k(A_6^\dagger) \leqslant (\delta + 2^k\delta^{\frac{1}{160}-3c})|B| \leqslant \delta^{\frac{1}{160}-4c}|B|. \tag{2.28}$$

Consider the set H_y contained inside a box $B_y := \pi'^{-1}(y) \cap B$ with sides at least $\min(n_i)$. We have $|H_y| \geqslant \delta^{\frac{1}{160}-c}|B_y|$ and $d_{k-1}(H_y) \leqslant \delta^{\frac{1}{160}+c}|H_y|$. By Observation 2.2.18, the number of parallel hyperplanes needed to cover H_y is at least $\min(n_i)\delta^{\frac{1}{160}-c} > m_{k-1}(\delta^{\frac{1}{160}+c})$. Hence by Corollary 2.1.2 for dimension $k-1$, we deduce that

$$|\widehat{\mathrm{co}}(H_y) \setminus H_y| \leqslant c_{k-1}\delta^{\frac{1}{160}+c}|B_y| \leqslant \delta^{\frac{1}{160}}|B_y|. \tag{2.29}$$

Observation 2.3.13. For a hyperplane $H_y \subset A_6^\dagger$, the smallest affine sublattice $\Lambda_{H_y} \subset \pi'^{-1}(y) = \mathbb{Z} \times \{y\} \times \mathbb{Z}^{k-2}$ containing H_y has the property that the nonempty rows of Λ_{H_y} have step size d.

Proof. For each row R_x contained in a hyperplane H_y, the arithmetic progression $\widehat{\mathrm{co}}(R_x)$ has step size d. Let d' be the uniform step size of the nonempty rows of Λ_{H_y} (which exists by Lagrange's theorem), and hence of the nonempty rows of $\widehat{\mathrm{co}}(H_y)$. Assume for the sake of contradiction $d' \neq d$. As d' divides d, for every row R_x of H_y and corresponding row R'_x of $\widehat{\mathrm{co}}(H_y)$, we have $|R'_x \setminus R_x| \geqslant \frac{1}{2}|R_x|$ (as each row R_x has at least 2 elements by (2.24)). Adding this over all rows R_x of H_y, we obtain from (2.26)

$$|\widehat{\mathrm{co}}(H_y) \setminus H_y| \geqslant \frac{1}{2}|H_y| \geqslant \frac{1}{2}\delta^{\frac{1}{160}-c}|B_y|$$

contradicting (2.29) that $\delta^{\frac{1}{160}}|B_y| \geqslant |\widehat{\mathrm{co}}(H_y) \setminus H_y|$. □

$A_7^\dagger \subset A_6^\dagger$ with $\pi(A_7^\dagger)$ close to its convex progression: Construction

There exists an increasing function $\zeta : \mathbb{R}_+^\star \to \mathbb{R}_+^\star$ with $\zeta(x) \to 0$ as $x \to 0$, such that if $\min\{n_i\}^{-1} \ll_\zeta \delta \ll_\zeta \varepsilon_0 \ll_\zeta 1$ (which we assume wlog by Convention 2.2.14), then the following holds. We can apply Proposition 2.3.1 to $A_6^\dagger$ with $\sigma = \delta$, $\lambda = \frac{1}{160} - 4c$, $\alpha = \frac{1}{320}$, $\varepsilon' \geqslant \varepsilon - \delta^{\frac{1}{160}-3c} \geqslant \frac{\varepsilon_0}{2}$ (by (2.27),(2.28)) to obtain a reduced set $A_7^\dagger \subset A_6^\dagger$ which is a union of rows. This set satisfies

$$|A_6^\dagger \setminus A_7^\dagger| \leqslant \delta^{\frac{1}{320}-5c}|B|, \qquad |\operatorname{co}(\pi(A_7^\dagger)) \setminus \pi(A_7^\dagger)| \leqslant \delta^{\frac{1}{320}}|\pi(B)|. \tag{2.30}$$

By (2.27) and (2.30), we have

$$|A \setminus A_7^\dagger| \leqslant |A \setminus A_6^\dagger| + |A_6^\dagger \setminus A_7^\dagger| \leqslant (\delta^{\frac{1}{160}-3c} + \delta^{\frac{1}{320}-5c})|B| \leqslant \delta^{\frac{1}{320}-6c}|B|, \tag{2.31}$$

and by Observation 2.2.23 we have

$$d_k(A_7^\dagger) \leqslant (\delta + 2^k\delta^{\frac{1}{320}-6c})|B| \leqslant \delta^{\frac{1}{320}-7c}|B|. \tag{2.32}$$

As we pass from $A_6^\dagger$ to $A_7^\dagger$, the affine sub-lattice Λ_{H_y} shrinks, so now by Observation 2.3.13 the nonempty rows of Λ_{H_y} have step size at least d. As the nonempty rows R_x of $A_7^\dagger$ coincide with the rows of A_5 and thus, have step size exactly d, i.e., $\operatorname{co}(R_x)$ is an arithmetic progression with at least two elements and step size exactly d, this forces the nonempty rows of Λ_{H_y} to have step size exactly d.

We note that we do not know that the hyperplanes of $A_7^\dagger$ have big size or small doubling.

$A_8^\dagger \subset A_7^\dagger$ has $\pi(H)$ reduced for all hyperplanes H: Construction

There exists an increasing function $\zeta : \mathbb{R}_+^\star \to \mathbb{R}_+^\star$ with $\zeta(x) \to 0$ as $x \to 0$, such that if $\min\{n_i\}^{-1} \ll_\zeta \delta \ll_\zeta \varepsilon_0 \ll_\zeta 1$ (which we assume wlog by Convention 2.2.14), then the following estimates hold.

Let $A_8^\dagger \subset A_7^\dagger$ be the union of all hyperplanes H_y such that $\pi(H_y)$ is reduced in $\{0\} \times \{y\} \times \mathbb{Z}^{k-2}$. Recall we let $B_y = \pi'^{-1}(y) \cap B$, so $H_y \subset B_y$.

If $\pi(H_y)$ is not reduced inside $\{0\} \times \{y\} \times \mathbb{Z}^{k-2}$, then there is a direction e_j with $j \in \{3, \dots, k\}$ such that $\pi(H_y) \cap (\pi(H_y) + e_j) = \emptyset$. Hence, letting $\pi_j : \mathbb{Z}^k \to \mathbb{Z}^{k-1}$ be the projection away from the jth coordinate and $S_x = \pi_j^{-1}(x) \cap \pi(H_y)$, we have $|\operatorname{co}(S_x) \setminus S_x| \geqslant \frac{1}{2}(|S_x| - 1)$. Summing the above inequality over all $x \in \pi_j(\pi(H_y)) \subset \pi_j(\pi(B_y))$, we deduce

$$|\operatorname{co}(\pi(H_y)) \setminus \pi(H_y)| \geqslant \sum_{x \in \pi_j(\pi(H_y))} |\operatorname{co}(S_x) \setminus S_x| \geqslant \frac{1}{2}(|\pi(H_y)| - \min(n_i)^{-1}|\pi(B_y)|).$$

Adding this over all y with $\pi(H_y)$ not reduced, we obtain by (2.30) that

$$\begin{aligned} |\pi(A_7^\dagger \setminus A_8^\dagger)| &\leqslant 2\min(n_i)^{-1}|\pi(B)| + 2 \sum_{\pi(H_y) \text{ not reduced}} |\operatorname{co}(\pi(H_y)) \setminus \pi(H_y)| \\ &\leqslant 2\min(n_i)^{-1}|\pi(B)| + 2|\operatorname{co}(\pi(A_7^\dagger)) \setminus \pi(A_7^\dagger)| \leqslant 3\delta^{\frac{1}{320}}|\pi(B)| \leqslant \delta^{\frac{1}{320}-c}|\pi(B)|, \end{aligned} \tag{2.33}$$

so in particular we have

$$|A_7^\dagger \setminus A_8^\dagger| \leqslant \delta^{\frac{1}{320}-c}|B|.$$

Hence by (2.31) we have

$$|A \setminus A_8^\dagger| \leqslant |A \setminus A_7^\dagger| + |A_7^\dagger \setminus A_8^\dagger| \leqslant \left(\delta^{\frac{1}{320}-6c} + \delta^{\frac{1}{320}-c}\right)|B| \leqslant \delta^{\frac{1}{320}-7c}|B|, \tag{2.34}$$

and by Observation 2.2.23 we have $A_8^\dagger$ is reduced and

$$d_k(A_8^\dagger) \leqslant \left(\delta + 2^k\delta^{\frac{1}{320}-7c}\right)|B| \leqslant \delta^{\frac{1}{320}-8c}|B|. \tag{2.35}$$

Note that $|\mathrm{co}(\pi(A_8^\dagger)) \setminus \pi(A_8^\dagger)| \leqslant |\mathrm{co}(\pi(A_7^\dagger)) \setminus \pi(A_7^\dagger)| + |\pi(A_7^\dagger \setminus A_8^\dagger)|$, so we have by (2.30),(2.33) that

$$|\mathrm{co}(\pi(A_8^\dagger)) \setminus \pi(A_8^\dagger)| \leqslant \left(\delta^{\frac{1}{320}} + \delta^{\frac{1}{320}-c}\right)|\pi(B)| \leqslant \delta^{\frac{1}{320}-2c}|\pi(B)|. \tag{2.36}$$

As we pass from $A_7^\dagger$ to $A_8^\dagger$, the affine sub-lattice Λ_{H_y} remains the same (or becomes empty), so the nonempty rows of Λ_{H_y} have step size exactly d. Furthermore, the reducedness of $\pi(H_y)$ implies $\pi(\Lambda_{H_y}) = \{0\} \times \{y\} \times \mathbb{Z}^{k-2}$.

Contradiction

We now derive a contradiction to the assumption $d > 1$. Recall d is the common step size of the convex progressions $\mathrm{co}(R_x)$ for all non-empty rows R_x of A_5 and d is also the common step size of the nonempty rows of Λ_{H_y} for all non-empty hyperplanes H_y of $A_8^\dagger$.

Let $H_{y_1}, \ldots, H_{y_\ell}$ be the nonempty hyperplanes of $A_8^\dagger$ with $y_1 < \ldots < y_\ell$, and for notational convenience set $H_i := H_{y_i}$ and $\Lambda_i := \Lambda_{H_{y_i}}$. Let

$$\Phi_i : \pi(\Lambda_i) = \{0\} \times \{y_i\} \times \mathbb{Z}^{k-2} \to \mathbb{Z}/d\mathbb{Z}$$

be the affine-linear function defined by taking $\Phi_i(0, y_i, z) = z' \mod d$ where $(z' + d\mathbb{Z}, y_i, z)$ is a row in Λ_i.

Before we begin let us give an outline of the strategy. For simplicity assume $y_i = i$. As usual, the reference frame $A_8^\dagger(+)A_8^\dagger = \sqcup_{\vec{v}} \sqcup_x R_x(+)R_{x+\vec{v}}$ turns out to have the correct size $|A_8^\dagger(+)A_8^\dagger| \approx 2^k|A_8^\dagger|$. The aim is to construct a set $D \subset A_8^\dagger + A_8^\dagger$ disjoint from $A_8^\dagger(+)A_8^\dagger$ of size comparable to the size of $A_8^\dagger$.

Consider any cover $[\ell] = [a_1, b_1] \cup [a_2, b_2] \cup \ldots \cup [a_r, b_r]$ with $a_1 \leqslant b_1 \leqslant a_2 \leqslant b_2 \leqslant \ldots \leqslant a_r \leqslant b_r$ such that for $1 \leqslant t \leqslant r-1$ we have $\sqcup_{a_t \leqslant j \leqslant b_t} H_j$ is not reduced but $\sqcup_{a_t \leqslant j \leqslant b_t+1} H_j$ is reduced. Also assume $|\sqcup_{a_r \leqslant j \leqslant b_r} H_j| \leqslant 2^{-1}|A_8^\dagger|$. By construction, there exists a point $p_{t+1} \in H_{b_t+1}$ such that $p_{t+1} + \sqcup_{a_t \leqslant j \leqslant b_t} H_j$ and $2(\sqcup_{a_t \leqslant j \leqslant b_t} H_j)$ are disjoint. Define $K_t = \sqcup_{a_t \leqslant j \leqslant b_t - 1} H_j$. Note that on the one hand $p_{t+1} + K_t$ are disjoint for different t's and on other hand note that $p_{t+1} + K_t$ and $A_8^\dagger(+)A_8^\dagger$ are disjoint. Indeed, the first part follows from the fact that $p_{t+1} + K_t \subset \pi'^{-1}[a_t + b_t + 1, 2b_t]$ and the second part follows from the fact that $(A_8^\dagger(+)A_8^\dagger) \cap \pi'^{-1}[a_t + b_t + 1, 2b_t]$ is contained inside

$2(\sqcup_{a_t \leqslant j \leqslant b_t} H_j)$. Therefore, we can construct the set $Z_1 = \sqcup_t (p_t + K_t)$ inside $A_8^\dagger + A_8^\dagger$ disjoint from $A_8^\dagger(+)A_8^\dagger$ of size at least $2^{-1}|A_8^\dagger| - \sum_{t\,:\,b_t \neq a_{t+1}} |H_t|$. If the cover satisfies $a_{t+1} = b_t$ for all $1 \leqslant t \leqslant r-1$, then we conclude $|Z_1| \geqslant 2^{-1}|A_8^\dagger|$.

Unfortunately, it is not always possible to find a cover that satisfies $a_{t+1} = b_t$ for all $1 \leqslant t \leqslant r-1$. The obstruction appears exactly when two consecutive hyperplanes say H_i and H_{i+1} have $H_i \sqcup H_{i+1}$ reduced. In this case we are forced to end the previous block with H_i and begin the next block with H_{i+1} (rather than H_i). However, in this case one can show that $H_i + H_{i+1}$ is much bigger than $H_i(+)H_{i+1}$. Indeed, assume for simplicity dimension $k = 3$. Consider the rows R_x of the set $H_i \sqcup H_{i+1}$. We have

$$H_i(+)H_{i+1} = \bigsqcup_z \left(R_{(0,i,z)}(+)R_{(0,i+1,z)} \right) \bigsqcup_z \left(R_{(0,i,z)}(+)R_{(0,i+1,z+1)} \right).$$

Construct the set

$$Z_2^i = \bigsqcup_z \left(R_{(0,i,z+1)}(+)R_{(0,i+1,z)} \right).$$

One expects the set Z_2^i to have size comparable to $|H_i| + |H_{i+1}|$. The idea is to show $Z_2^i \subset H_i + H_{i+1}$ is disjoint from $H_i(+)H_{i+1}$ and hence from $A_8^\dagger(+)A_8^\dagger$. Assume this is not the case. Then, there exists z such that $R_{(0,i,z+1)}(+)R_{(0,i+1,z)}$ intersects $R_{(0,i,z)}(+)R_{(0,i+1,z+1)}$. Therefore, the affine linear functions Φ_i and Φ_{i+1} satisfy $\Phi_i(0,i,z+1) + \Phi_{i+1}(0,i+1,z) = \Phi_i(0,i,z) + \Phi_{i+1}(0,i+1,z+1)$. Writing $\Phi_i = c_i + \theta_i$ and $\Phi_{i+1} = c_{i+1} + \theta_{i+1}$ where θ_i and θ_{i+1} are linear functions, we deduce $\theta_i(0,i,1) = \theta_{i+1}(0,i+1,1)$. We get $\theta_i = \theta_{i+1}$ and $\Phi_i = \Phi_{i+1} + c_i - c_{i+1}$, i.e., Λ_i is a translate of Λ_{i+1}. We conclude $\Lambda_i \sqcup \Lambda_{i+1}$ is not reduced which in particular means $H_i \sqcup H_{i+1}$ is not reduced. This gives the desired contradiction.

Consider any cover that satisfies $a_{t+1} = b_t$ for all $1 \leqslant t \leqslant r-1$ except those t for which $H_t \sqcup H_{t+1}$ is reduced. Then we can construct the set $Z_2 = \sqcup_{t\,:\,a_{t+1} \neq b_t} Z_2^t$ inside $A_8^\dagger + A_8^\dagger$ disjoint from $A_8^\dagger(+)A_8^\dagger$. Moreover, Z_2 is disjoint from Z_1 as $Z_2 \subset \pi'^{-1}(\cup_t \{2b_t + 1\})$ and $Z_1 \subset \pi'^{-1}(\cup_t [a_t + b_t + 1, 2b_t])$. From a standard argument, it turns out that Z_2 has size comparable to $\sum_{t\,:\,a_{t+1} \neq b_t} H_t$. We conclude that the set $D = Z_1 \sqcup Z_2$ is contained inside $A_8^\dagger + A_8^\dagger$ and disjoint from $A_8^\dagger(+)A_8^\dagger$ and has size comparable to the size of $A_8^\dagger$.

We create r subintervals $I_i \subset \{1, \ldots, \ell\}$ satisfying the following properties.

- $1 \in I_1$
- $\bigsqcup_{j \in I_i} H_j$ is not reduced in $\mathbb{Z}^k$, and $H_{1+\max I_i} \sqcup \bigsqcup_{j \in I_i} H_j$ is reduced, for $1 \leqslant i \leqslant r-1$.
- $\min I_{i+1} = \max I_i + \begin{cases} 0 & H_{\max I_i} \sqcup H_{1+\max I_i} \text{ is not reduced.} \\ 1 & H_{\max I_i} \sqcup H_{1+\max I_i} \text{ is reduced.} \end{cases}$

These conditions uniquely determine intervals $I_1, \ldots, I_r$ which cover $\{1, \ldots, \ell\}$, and as $A_8^\dagger$ is reduced we have $r \geqslant 2$.

Remark 2.3.14. If $\max I_i = \min I_{i+1}$ then $|I_i| \geqslant 2$. If instead $\max I_i + 1 = \min I_{i+1}$, then with $j = \max I_i$ we have $y_j + 1 = y_{j+1}$, and $\Phi_j(w) - \Phi_{j+1}(w + e_2) : \{0\} \times \{y_j\} \times \mathbb{Z}^{k-2} \to \mathbb{Z}$ is non-constant.

For $1 \leqslant i \leqslant r-1$, let $z_i \in H_{1+\max I_i}$ be a point not in the affine sub-lattice containing $\bigsqcup_{j\in I_i} H_j$. Let

$$f : \bigsqcup_{j\in[1,\min I_r-1]} H_j \to A_8^\dagger$$

be defined by setting

$$f\left(\bigsqcup_{j\in[\min I_i,\min I_{i+1}-1]} H_j\right) = z_i.$$

Claim 2.3.15. If $z', z'' \in \bigsqcup_{j\in[1,\min I_r-1]} H_j$ are distinct, then we have

$$z' + f(z') \neq z'' + f(z'').$$

Proof. Indeed, if they were equal then

$$\pi'(z') + \pi'(f(z')) = \pi'(z'') + \pi'(f(z'')),$$

and if without loss of generality $\pi'(z') \leqslant \pi'(z'')$, then $\pi'(f(z')) \leqslant \pi'(f(z''))$, so we must have $\pi'(z') = \pi'(z'')$. Therefore $f(z) = f(z')$, so $z' = z''$, a contradiction. □

Hence the set

$$Z_1 = \bigsqcup_{i=1}^{r-1} \bigsqcup_{j\in[\min I_i,\max I_i-1]} (H_j + z_i) \subset A_8^\dagger + A_8^\dagger$$

is a disjoint union as $[\min I_i, \max I_i - 1] \subset [\min I_i, \min I_{i+1} - 1]$.

Claim 2.3.16. For $\vec{v} \in \{0\} \times \{0,1\}^{k-1}$ and $x \in \{0\} \times \mathbb{Z}^{k-1}$, and $z \in H_t$ with $t \in [\min I_i, \max I_i - 1]$ for some $1 \leqslant i \leqslant r-1$, we have

$$z + f(z) \notin R_x + R_{x+\vec{v}}.$$

Proof. Assume to the contrary that $z + f(z) \in R_x + R_{x+\vec{v}}$. First note that $\pi'(x) \geqslant y_{\min I_i}$ since

$$2y_{\min I_i} \leqslant \pi'(z) + \pi'(f(z)) = \pi'(x) + \pi'(x+\vec{v}) \leqslant 2\pi'(x) + 1.$$

Next, we note that $\pi'(x+\vec{v}) \leqslant y_{\max I_i}$. Indeed, if $\pi'(x+\vec{v}) > y_{\max I_i}$, then as R_x and $R_{x+\vec{v}}$ are non-empty, we have $\pi'(x+\vec{v}) \geqslant y_{1+\max I_i}$ and $\pi'(x) \geqslant y_{\max I_i}$. However, then

$$\pi'(x+\vec{v}) + \pi'(x) \geqslant y_{1+\max I_i} + y_{\max I_i} > \pi'(f(z)) + \pi'(z),$$

a contradiction. Hence, $z, R_x, R_{x+\vec{v}}$ are all contained in the affine sub-lattice containing $\sqcup_{j\in I_i} H_j$, and $f(z)$ is not in this affine sub-lattice by construction, contradicting $z + f(z) \in R_x + R_{x+\vec{v}}$. □

Hence the sets

$$Z := A_8^\dagger(+)A_8^\dagger = \bigsqcup_{\vec{v}\in\{0\}\times\{0,1\}^{k-1}} \bigsqcup_{x\in\{0\}\times\mathbb{Z}^{k-1}} R_x(+)R_{x+\vec{v}} \subset A_8^\dagger + A_8^\dagger$$

and Z_1 are disjoint.

The set of indices $\mathscr{I} = [1, \min I_r - 1] \setminus \bigcup_{i=1}^{r-1} [\min I_i, \max I_i - 1]$ whose corresponding hyperplanes H_j were not accounted for by Z_1 are precisely those indices j such that there exists $1 \leqslant i \leqslant r-1$ with $j = \max I_i = \min I_{i+1} - 1$. We will now find a third set Z_2 disjoint from Z, and Z_1 which accounts for the hyperplanes with indices in $\mathscr{I}$.

Consider two consecutive hyperplanes H_j, and H_{j+1} with $j \in \mathscr{I}$, and let i be such that $j = \max I_i$ and $j+1 = \min I_{i+1}$. Note that by Remark 2.3.14, we have $y_j + 1 = y_{j+1}$ and the affine-linear function $\Phi_j(w) - \Phi_{j+1}(w+e_2) : \{0\} \times \{y_j\} \times \mathbb{Z}^{k-2} \to \mathbb{Z}/d\mathbb{Z}$ is non-constant. In particular, there is an index $s_j \in \{3, \dots, k\}$ so that the standard basis vector e_{s_j} satisfies $\Phi_j(w+e_{s_j}) - \Phi_{j+1}(w+e_{s_j}+e_2) \neq \Phi_j(w) - \Phi_{j+1}(w+e_2)$ for all w. Rearranging,

$$\Phi_j(w+e_{s_j}) + \Phi_{j+1}(w+e_2) \neq \Phi_j(w) + \Phi_{j+1}(w+e_{s_j}+e_2)$$

for all w. Hence we have

$$\left(R_{w+e_2} + R_{w+e_{s_j}}\right) \cap \left(R_w + R_{w+e_2+e_{s_j}}\right) = \emptyset$$

for all w since they lie in different translated $d\mathbb{Z}$-progressions.

Claim 2.3.17. For $w \in \{0\} \times \{y_j\} \times \mathbb{Z}^{k-2}$, we have

$$(R_{w+e_2} + R_{w+e_{s_j}}) \cap Z = \emptyset.$$

Proof. If $(w+e_2) + (w+e_{s_j}) = x + (x+\vec{v})$ for some $x \in \{0\} \times \mathbb{Z}^{k-1}$ and $\vec{v} \in \{0\} \times \{0,1\}^{k-1}$, then by looking at the odd coordinates we see that $\vec{v} = e_2 + e_{s_j}$ and hence $x = w$. But then

$$\left(R_{w+e_2} + R_{w+e_{s_j}}\right) \cap Z = \left(R_{w+e_2} + R_{w+e_{s_j}}\right) \cap \left(R_w(+)R_{w+e_2+e_{s_j}}\right) = \emptyset.$$

□

Hence the disjoint union

$$Z_2 := \bigsqcup_{j\in\mathscr{I}} \bigsqcup_{w\in\{0\}\times\{y_j\}\times\mathbb{Z}^{k-2}} R_{w+e_2}(+)R_{w+e_{s_j}}$$

is disjoint from Z. Finally, we prove a claim which implies Z_1 is disjoint from Z_2.

Claim 2.3.18. For any $1 \leqslant s \leqslant r-1$ and for any $z \in H_t$ with $t \in [\min I_s, \max I_s - 1]$ and for any $w \in \{0\} \times \{y_j\} \times \mathbb{Z}^{k-2}$ with $j \in \mathscr{I}$, we have

$$z + f(z) \notin R_{w+e_2} + R_{w+e_{s_j}}.$$

Proof. Assume for the sake of contradiction that $z + f(z) \in R_{w+e_2} + R_{w+e_{s_j}}$. Let i be such that $j = \max I_i = \min I_{i+1} - 1$. First, suppose that $\pi'(z) = y_t \leqslant y_j - 1$. Then $s \leqslant i$ as if $i < s$ then $j = \min I_{i+1} - 1 < \min I_s \leqslant t$, and therefore $y_j < y_t$, a contradiction. Therefore $\pi'(f(z)) = y_{\max I_s + 1} \leqslant y_{\max I_i + 1} = y_{j+1} = y_j + 1$ by Remark 2.3.14, so we obtain the contradiction

$$\pi'(z + f(z)) \leqslant y_j - 1 + (y_j + 1) < 2y_j + 1 = \pi'(w + e_2) + \pi'(w + e_{s_j}).$$

Next, suppose $\pi'(z) = y_t = y_j$. Then $j = t$, contradicting that $\mathscr{I}$ is disjoint from $[\min I_s, \max I_s - 1]$ by construction of $\mathscr{I}$. Finally, suppose that $\pi'(z) = y_t \geqslant y_j + 1$. Then as $\pi'(f(z)) > \pi'(z)$, we have the contradiction

$$\pi'(z + f(z)) > 2y_j + 2 > 2y_j + 1 = \pi'(w + e_2) + \pi'(w + e_{s_j}).$$

□

Hence Z, Z_1 and Z_2 are disjoint subsets of $A_8^\dagger + A_8^\dagger$.

There exists an increasing function $\zeta : \mathbb{R}_+^\star \to \mathbb{R}_+^\star$ with $\zeta(x) \to 0$ as $x \to 0$, such that if $\min\{n_i\}^{-1} \ll_\zeta \delta \ll_\zeta \varepsilon_0 \ll_\zeta 1$ (which we assume wlog by Convention 2.2.14), then the rest of the estimates hold. Note that for $x_1 \neq x_2$ we have

$$|R_{x_1}(+)R_{x_2}| - |R_{x_1}| - |R_{x_2}| \geqslant \begin{cases} 0 & |\{x_1, x_2\} \cap \pi(A')| = 0 \\ -n_1 & |\{x_1, x_2\} \cap \pi(A')| = 1 \\ -1 & |\{x_1, x_2\} \cap \pi(A')| = 2 \end{cases}$$

For $0 \neq \vec{v} \in \{-1, 0, 1\}^k$ we have from (2.36) and Observation 2.2.22 (the fact that we allow coordinates of $\vec{v}$ to be in $\{-1, 0, 1\}$ does not change anything as we can always take a reflection and make all coordinates of $\vec{v}$ in $\{0, 1\}$)

$$\left|\{x \in \mathbb{Z}^{k-1} : |\{x, x + \vec{v}\} \cap \pi(A_8^\dagger)| = 1\}\right| \leqslant 2|\operatorname{co}(\pi(A_8^\dagger)) \setminus \pi(A_8^\dagger)| + 2(k-1)n_1^{-1}\min(n_i)^{-1}|B| \leqslant \delta^{\frac{1}{320} - 3c} n_1^{-1}|B|$$

and

$$\left|\{x \in \mathbb{Z}^{k-1} : |\{x, x + \vec{v}\} \cap \pi(A_8^\dagger)| = 2\}\right| \leqslant |\pi(B)| \leqslant \min(n_i)^{-1}|B| \leqslant \delta^{\frac{1}{320} - 3c}|B|.$$

Therefore, we have (taking $\vec{v} \in \{0\} \times \{0,1\}^{k-1}$ and $x \in \{0\} \times \mathbb{Z}^{k-1}$)

$$\begin{aligned}\sum_{\vec{v}} \sum_{x} |R_x(+)R_{x+\vec{v}}| - |R_x| - |R_{x+\vec{v}}| \geqslant & \sum_{\vec{v}} -n_1 |\{x \in \mathbb{Z}^{k-1} : |\{x, x+\vec{v}\} \cap \pi(A_8^\dagger)| = 1\}| \\ & \sum_{\vec{v}} -|\{x \in \mathbb{Z}^{k-1} : |\{x, x+\vec{v}\} \cap \pi(A_8^\dagger)| = 2\}| \\ \geqslant & -2^{k-1}(n_1 \delta^{\frac{1}{320}-3c} n_1^{-1}|B| + \delta^{\frac{1}{320}-3c}|B|) \\ \geqslant & -2^k \delta^{\frac{1}{320}-3c}|B|\end{aligned}$$

and

$$\begin{aligned}\sum_{j \in \mathscr{I}} \sum_{w \in \{0\} \times \{y_j\} \times \mathbb{Z}^{k-2}} |R_{w+e_2}(+)R_{w+e_{s_j}}| - |R_{w+e_2}| + |R_{w+e_{s_j}}| \geqslant & -n_1 |\{x \in \mathbb{Z}^{k-1} : |\{x, x+\vec{v}\} \cap \pi(A_8^\dagger)| = 1\}| \\ & -|\{x \in \mathbb{Z}^{k-1} : |\{x, x+\vec{v}\} \cap \pi(A_8^\dagger)| = 2\}| \\ \geqslant & -(n_1 \delta^{\frac{1}{320}-3c} n_1^{-1}|B| - \delta^{\frac{1}{320}-3c}|B|) \\ \geqslant & -2\delta^{\frac{1}{320}-3c}|B|.\end{aligned}$$

Therefore, we deduce (taking $\vec{v} \in \{0\} \times \{0,1\}^{k-1}$ and $x \in \{0\} \times \mathbb{Z}^{k-1}$)

$$\begin{aligned}|A_8^\dagger + A_8^\dagger| \geqslant & |Z| + |Z_1| + |Z_2| \\ = & \left(\sum_{\vec{v}} \sum_{x} |R_x(+)R_{x+\vec{v}}|\right) + \sum_{i=1}^{r-1} \sum_{j \in [\min I_i, \max I_i - 1]} |H_j| \\ & + \sum_{j \in \mathscr{I}} \sum_{w \in \{0\} \times \{y_j\} \times \mathbb{Z}^{k-2}} |R_{w+e_2}(+)R_{w+e_{s_j}}| \\ = & \left(\sum_{\vec{v}} \sum_{x} |R_x| + |R_{x+\vec{v}}|\right) + \sum_{i=1}^{r-1} \sum_{j \in [\min I_i, \max I_i - 1]} |H_j| \\ & + \left(\sum_{j \in \mathscr{I}} \sum_{w \in \{0\} \times \{y_j\} \times \mathbb{Z}^{k-2}} |R_{w+e_{s_j}}|\right) + \left(\sum_{\vec{v}} \sum_{x} |R_x(+)R_{x+\vec{v}}| - |R_x| - |R_{x+\vec{v}}|\right) \\ & + \left(\sum_{j \in \mathscr{I}} \sum_{w \in \{0\} \times \{y_j\} \times \mathbb{Z}^{k-2}} |R_{w+e_2}(+)R_{w+e_{s_j}}| - |R_{w+e_2}| + |R_{w+e_{s_j}}|\right) \\ \geqslant & \left(\sum_{\vec{v}} \sum_{x} |R_x| + |R_{x+\vec{v}}|\right) + \sum_{i=1}^{r-1} \sum_{j \in [\min I_i, \max I_i - 1]} |H_j| \\ & + \left(\sum_{j \in \mathscr{I}} \sum_{w \in \{0\} \times \{y_j\} \times \mathbb{Z}^{k-2}} |R_{w+e_{s_j}}|\right) - \delta^{\frac{1}{320}-4c}|B| = 2^k |A_8^\dagger| + \left(\sum_{j \in [1, \min I_r - 1]} |H_j|\right) - \delta^{\frac{1}{320}-4c}|B|.\end{aligned}$$

If we consider the same process run in reverse, we produce another collection of intervals $I'_1,\dots I'_{r'} \subset \{1,\dots \ell\}$ with $\ell \in I'_1$ such that

$$|A_8^\dagger + A_8^\dagger| \geqslant 2^k |A_8^\dagger| + \left(\sum_{j\in[\max I'_{r'}+1,l]} |H_j|\right) - \delta^{\frac{1}{320}-4c}|B|.$$

As $A_8^\dagger$ is reduced, we have that $r' \geqslant 2$. As $\sqcup_{i\in I_r} H_i$ is not reduced, we have that $I_r \subset I'_1 \subset [\max I'_r + 1, \ell]$. Averaging the two inequalities we get by (2.34) that

$$\begin{aligned}
|A_8^\dagger + A_8^\dagger| &\geqslant 2^k|A_8^\dagger| - \delta^{\frac{1}{320}-4c}|B| + 2^{-1}\left(\sum_{j\in[\max I'_{r'}+1,l]} |H_j|\right) + 2^{-1}\left(\sum_{j\in[1,\min I_r-1]} |H_j|\right)\\
&\geqslant 2^k|A_8^\dagger| - \delta^{\frac{1}{320}-4c}|B| + 2^{-1}\left(\sum_{j\in[\min(I_r),l]} |H_j|\right) + 2^{-1}\left(\sum_{j\in[1,\min I_r-1]} |H_j|\right)\\
&= 2^k + 2^{-1}|A_8^\dagger| - \delta^{\frac{1}{320}-4c}|B|.
\end{aligned}$$

Rearranging the terms, we get by (2.34) that

$$d_k(A_8^\dagger) \geqslant \frac{1}{2}|A_8^\dagger| - \delta^{\frac{1}{320}-4c}|B| \geqslant \frac{\varepsilon_0}{4}|B|,$$

contradicting (2.35) that $d_k(A_8^\dagger) \leqslant \delta^{\frac{1}{320}-8c}|B|$. We conclude $d = 1$, i.e., for every non-empty row R_x of A_5 the convex progression $\mathrm{co}(R_x)$ is an interval.

2.3.6 Reductions Part 4: Filling in the rows to create $A_+ \supset A_5$

We recall that we have just shown that all rows R_x of A_5 satisfy $\mathrm{co}(R_x) = \widehat{\mathrm{co}}(R_x)$. We now show that filling in all of the rows of A_5 does not change the size of A_5 or $d_k(A_5)$ too much.

$A_+ \supset A_5$ with all rows filled in

Let $A_+ \supset A_5$ be obtained by replacing each row R_x of A_5 with $\mathrm{co}(R_x)$. We have A_+ is reduced as A_5 is reduced. Also by (2.22) and (2.24), provided $\delta > 0$ is sufficently small in terms of ε_0 (which we assume wlog by Convention 2.2.14), we have

$$|A \triangle A_+| \leqslant |A \setminus A_5| + |A_+ \setminus A_5| \leqslant \delta^{\frac{1}{20}-15c}|B| + 2\delta^{\frac{1}{10}}|A_5| \leqslant \delta^{\frac{1}{20}-16c}|B|. \tag{2.37}$$

Furthermore, $\pi(A_+) = \pi(A_5)$, so by (2.21) we have

$$|\mathrm{co}(\pi(A_+)) \setminus \pi(A_+)| = |\mathrm{co}(\pi(A_5)) \setminus \pi(A_5)| \leqslant \delta^{\frac{1}{20}}|\pi(B)|.$$

Observation 2.3.19. Provided $\delta > 0$ is sufficently small in terms of ε_0 (which we assume wlog by Convention 2.2.14), we have $d_k(A_+) \leqslant \delta^{\frac{1}{20}-17c}|B|$.

Proof. We begin the proof with the following general claim.

Claim 2.3.20. Given a family of finite subsets $Z_i \subset \mathbb{Z}$ indexed by $i \in \mathscr{H}$ and a parameter ρ such that $|\mathrm{co}(Z_i)| \leqslant (1+\rho)|Z_i|$, we have

$$\left|\bigcup \mathrm{co}(Z_i)\right| \leqslant (1+2\rho)\left|\bigcup Z_i\right|.$$

Proof. We can assume wlog that for any index $i_0 \in \mathscr{H}$, the set $\mathrm{co}(Z_{i_0})$ is not contained inside $\cup_{i\neq i_0}\mathrm{co}(Z_i)$. This means that for any $i,j \in \mathscr{H}$ the intervals $\mathrm{co}(Z_i)$ and $\mathrm{co}(Z_j)$ are distinct and satisfy either $\min(\mathrm{co}(Z_i)) \leqslant \min(\mathrm{co}(Z_j))$ and $\max(\mathrm{co}(Z_i)) \leqslant \max(\mathrm{co}(Z_j))$ or alternatively $\min(\mathrm{co}(Z_i)) \geqslant \min(\mathrm{co}(Z_j))$ and $\max(\mathrm{co}(Z_i)) \geqslant \max(\mathrm{co}(Z_j))$. This induces a natural linear order $\preceq$ on the family $\mathscr{H}$. This linear order has the additional property that if $i \prec j \prec k$, then $\mathrm{co}(Z_i)$ and $\mathrm{co}(Z_k)$ are disjoint. Indeed, otherwise $\mathrm{co}(Z_j) \subset \mathrm{co}(Z_i) \cup \mathrm{co}(Z_k)$. Partition $\mathscr{H} = \mathscr{F} \sqcup \mathscr{G}$ in the odd and even positions in the linear order $\prec$. Then $\mathscr{F},\mathscr{G}$ are such that $\bigcup \mathrm{co}(Z_i) = (\bigsqcup_{i\in\mathscr{F}} \mathrm{co}(Z_i)) \cup (\bigsqcup_{i\in\mathscr{G}} \mathrm{co}(Z_i))$, where both are disjoint unions. Then

$$\left|\bigcup \mathrm{co}(Z_i) \setminus \bigcup Z_i\right| \leqslant \sum_{i\in\mathscr{F}\cup\mathscr{G}} |\mathrm{co}(Z_i)\setminus Z_i| \leqslant \sum_{i\in\mathscr{F}\cup\mathscr{G}} \rho|Z_i| \leqslant 2\rho\left|\bigcup Z_i\right|.$$

□

Now, for nonempty rows R_{x_1}, R_{x_2} of A_5, we have by (2.24) that $|\mathrm{co}(R_{x_1})| \leqslant (1+2\delta^{\frac{1}{10}})|R_{x_1}|$ and $|\mathrm{co}(R_{x_2})| \leqslant (1+2\delta^{\frac{1}{10}})|R_{x_2}|$, so

$$|\mathrm{co}(R_{x_1}+R_{x_2})| = |\mathrm{co}(R_{x_1})| + |\mathrm{co}(R_{x_2})| - 1 \leqslant (1+2\delta^{\frac{1}{10}})(|R_{x_1}|+|R_{x_2}|) - 1 \leqslant (1+4\delta^{\frac{1}{10}})|R_{x_1}+R_{x_2}|.$$

Taking the sets Z_i to be the pairwise row sums $R_{x_1}+R_{x_2}$ with $x_1+x_2 = x$ fixed, and summing the inequality in Claim 2.3.20 over all $x \in \{0\}\times\mathbb{Z}^{k-1}$, we obtain

$$|A_+ + A_+| \leqslant (1+8\delta^{\frac{1}{10}})|A_5+A_5|.$$

Hence by (2.23), we thus have

$$d_k(A_+) \leqslant (1+8\delta^{\frac{1}{10}})d_k(A_5) + 2^k\cdot 8\delta^{\frac{1}{10}}|A_5| \leqslant \delta^{\frac{1}{20}-17c}|B|.$$

□

2.3.7 Reductions Part 5: Approximating A_+ by $A_\star \subset A_+$ with few vertices in $\mathrm{co}(A_\star)$ and an extra technical condition

We now construct a set $A_\star \subset A_+$ with $|A_+ \setminus A_\star| = o(1)|B|$, $|V(A_\star)$ with the following properties. $|V(A_\star)|$, which we recall is the number of vertices of $\widetilde{\mathrm{co}}(A_\star)$, is bounded by a function of δ and $|\mathrm{co}(\pi(A_\star)) \setminus \pi(A_\star)| = o(1)|\pi(B)|$. We would like $A_\star$ to posses an additional technical property. Intuitively, the desired property is that whenever the interval rows $R_x, R_y, R_z, R_{z+\vec{v}}$ of $A_\star$ satisfy $x+y=z+\vec{v}$, R_x+R_y does not sit completely above or completely below $R_z+R_{z+\vec{v}}$ of the reference frame. This property is desirable because it allows us to switch between evaluating the sumset of a set to evaluating the infimum-convolution of a function. As usual we are not interested the in the entire sumset of $A_\star$ with itself, but rather in a very specific restricted sumset where each row R_x is summed with a bounded family of rows R_y with $y \in Y(x)$ and thus we are not interested in this property to hold globally but rather locally. To make this precise we need a few definitions.

Definition 2.3.21. For every $A' \subset B$, let $\mathscr{T}^+(A')$ (resp. $\mathscr{T}^-(A')$) be a triangulation of the upper (resp. lower) convex hull of $\widetilde{\mathrm{co}}(A')$ with respect to the e_1 direction, projected under π to $\{0\}\times\mathbb{R}^{k-1}$, so in particular every $\widetilde{T} \in \mathscr{T}^+(A')$ has $\widetilde{T} \subset \{0\}\times\mathbb{R}^{k-1}$. We ensure that if $\widetilde{\mathrm{co}}(A'_1) = \widetilde{\mathrm{co}}(A'_2)$, then $\mathscr{T}^+(A'_1) = \mathscr{T}^+(A'_2)$ and $\mathscr{T}^-(A'_1) = \mathscr{T}^-(A'_2)$.

Notation 2.3.22. For a simplex $\widetilde{T} \subset \{0\}\times\mathbb{R}^{k-1}$, we will write

$$T := \widetilde{T} \cap \{0\}\times\mathbb{Z}^{k-1}, \text{ and } T^\circ := \widetilde{T}^\circ \cap (\{0\}\times\mathbb{Z}^{k-1})$$

where $\widetilde{T}^\circ$ is the interior of $\widetilde{T}$.

Definition 2.3.23. Given $\widetilde{T} \subset \{0\}\times\mathbb{R}^{k-1}$ with integral vertices, and a set $\mathscr{W} \subset \{0\}\times\mathbb{Z}^{k-1}$, for every $x \in T$ we define the set

$$Y_{\mathscr{W}}(x) := ((x+\mathscr{W}) \cap T) \cup V(T) \subset T.$$

Definition 2.3.24. Given a collection $\mathscr{M}$ of sets $\mathscr{W}_T \subset \{0\}\times\mathbb{Z}^{k-1}$ constructed for each simplex $\widetilde{T} \in \{0\}\times R^{k-1}$ with integral vertices, we say that $A'' \subset B$ has *Property* $P(\mathscr{M})$ if the following holds. For every $\widetilde{T} \in \mathscr{T}^+(A'') \cup \mathscr{T}^-(A'')$, if $x \in T^\circ \setminus \pi(V(A''))$, $y \in Y_{\mathscr{W}_T}(x)$, $z \in \pi(B)$ and $\vec{v} \in \{0\}\times\{0,1\}^{k-1}$ with $x+y=z+z+\vec{v}$ and $R_x+R_y, R_z+R_{z+\vec{v}}$ nonempty, then $(R_x+R_y)\cap(R_z+R_{z+\vec{v}}) \neq \emptyset$.

Thus, the plan is to construct a set $A_\star \subset A_+$ with $|A_+ \setminus A_\star| = o(1)|B|$, which has simultaneously

1. $|V(A_\star)|$, which we recall is the number of vertices of $\widetilde{\mathrm{co}}(A_\star)$, is bounded by a function of δ
2. $|\mathrm{co}(\pi(A_\star)) \setminus \pi(A_\star)| = o(1)|\pi(B)|$
3. $A_\star$ has Property $P(\mathscr{M})$ for a suitable family $\mathscr{M}$ of bounded sets $\mathscr{W}_T$ constructed independently for all simplices $\widetilde{T} \in \{0\}\times\mathbb{R}^{k-1}$ with integral vertices (see Definition 2.3.34).

We show this using a double recursion, and the bounds we obtain will no longer be powers of δ.

In Part 1 of Section 2.3.7, we prove Proposition 2.3.25, which shows that we can ensure that 1 holds. This is accomplished by showing an analogous approximation result for (continuous) polytopes, and then transitioning to the discrete setting using Observation 2.2.24.

In Part 2 of Section 2.3.7, we prove Proposition 2.3.28, which shows that we can ensure that both 1 and 2 hold. Proposition 2.3.1 by itself shows that 2 holds, so we alternate applications of Proposition 2.3.25 and Proposition 2.3.1, and show that at some point both 1 and 2 hold simultaneously.

In Part 3 of Section 2.3.7, we prove Proposition 2.3.31, which shows that we can ensure that all of 1,2,3 hold. To do this, we show Proposition 2.3.30, which shows that we can ensure 1 and 3 hold. Similarly to the proof of Proposition 2.3.28, we alternate applications of Proposition 2.3.28 and Proposition 2.3.30, and show that at some point 1,2,3 hold simultaneously.

Finally, in the last part of Section 2.3.7, we apply Proposition 2.3.31 to A_+ to construct $A_\star$.

$A_\star \subset A_+$ with $|V(A_\star)|$ small: Setup Part 1

In the next result we drop Convention 2.2.15.

Proposition 2.3.25. For any $\alpha \geqslant 16\varepsilon_0^{-1}k(k+1)\min\{n_i\}^{-1}$, there exists an $\ell = \ell(\alpha)$ such that the following is true. For any set of points $C \subset B$ with $|\mathrm{co}(C)| \geqslant \frac{\varepsilon_0}{2}|B|$, there exists $Q = \mathrm{co}(Q) \cap C \subset C$ with $|V(Q)| \leqslant \ell$ and $V(Q) \subset V(C)$, such that $|\mathrm{co}(Q)| \geqslant (1-\alpha)|\mathrm{co}(C)|$. Moreover, if C has all rows intervals then Q has all rows intervals. Furthermore, if C is reduced then Q is reduced. Finally, there exists a constant τ_k such that for n_i sufficiently large and $\alpha = \min\{n_i\}^{-\frac{2}{(k-1)\lfloor k/2\rfloor+2}}$, we can take $\ell = \tau_k \min\{n_i\}^{\frac{k-1}{(k-1)\lfloor k/2\rfloor+2}}$.

To do this, we first consider a continuous analogue, which was proved constructively by Gordon, Meyer, and Reisner [82].

Lemma 2.3.26. For any $\alpha > 0$, there exists $\ell' = \ell'(\alpha)$ such that the following is true. For any polytope $\widetilde{C}$, there is a polytope $\widetilde{Q}$ which is the convex hull of at most ℓ' vertices of $\widetilde{C}$ with $|\widetilde{Q}| \geqslant (1-\alpha)|\widetilde{C}|$. There is an absolute constant τ independent of k such that for α sufficiently small in terms of k we can take $\ell' = (\frac{\tau}{k}\alpha)^{-\frac{k-1}{2}}$.

Proof of Lemma 2.3.26. It is enough to find such a polytope $\widetilde{Q}$ with vertices contained inside $\widetilde{C}$. Indeed, a simple convexity argument shows that as we vary the vertices of $\widetilde{Q}$ the maximum volume is attained when all vertices of $\widetilde{Q}$ are among the vertices of $\widetilde{C}$. The result, together with the bound on ℓ', then follow directly from [82, Theorem 3]. □

Proof of Proposition 2.3.25. Let $\widetilde{C} = \widetilde{\mathrm{co}}(C)$ be the continuous convex hull of C. By Lemma 2.3.26, there exists a polytope $\widetilde{P}$ with $|V(\widetilde{P})| \leqslant \ell'(\frac{\alpha}{2})$, $V(\widetilde{P}) \subset V(\widetilde{C}) = V(C)$, and $|\widetilde{P}| \geqslant (1-\frac{\alpha}{2})|\widetilde{C}|$. Let $Q = \widetilde{P} \cap C$, and note that $\widetilde{\mathrm{co}}(Q) = \widetilde{P}$.

We thus have by Observation 2.2.24 that

$$\begin{aligned}|\operatorname{co}(Q)| &\geqslant |\widetilde{P}| - 2k(k+1)\min\{n_i\}^{-1}|B| \\ &\geqslant \left(1-\frac{\alpha}{2}\right)|\widetilde{C}| - 2k(k+1)\min\{n_i\}^{-1}|B| \\ &\geqslant \left(1-\frac{\alpha}{2}\right)|\operatorname{co}(C)| - 4k(k+1)\min\{n_i\}^{-1}|B| \\ &\geqslant (1-\alpha)|\operatorname{co}(C)|.\end{aligned}$$

For $\alpha = \min\{n_i\}^{-\frac{2}{(k-1)\lfloor k/2\rfloor+2}}$ (sufficiently small in terms of k as n_i is sufficiently large) we see that $\ell = \ell'(\frac{\alpha}{2})$, yielding $\ell = \tau_k \min\{n_i\}^{\frac{k-1}{(k-1)\lfloor k/2\rfloor+2}}$. By construction, it is easy to see that if the rows of C are intervals then so are the rows of Q. Moreover, by including in addition a bounded number of rows of C in Q (which changes $V(Q)$ by at most a constant number), we can guarantee that Q is reduced if C is reduced. □

$A_\star \subset A_+$ with $|V(A_\star)|$ and $|\operatorname{co}(\pi(A_\star))\setminus\pi(A_\star)|$ small: Setup part 2

At this point in the proof, we will lose polynomial control over the doubling constant, so for convenience we will work with purely qualitative statements from now on. The following proposition is the qualitative analogue of Proposition 2.3.1. Crucially, this qualitative version holds for all $\delta > 0$, rather than for δ sufficiently small.

Proposition 2.3.27. Given $\varepsilon_0 > 0$, there are functions $h_1(t), h_2(t)$ with $h_1, h_2 \to 0$ as $t \to 0$ such that for every $f > 0$ the following is true, provided $\min(n_i)$ is sufficiently large in terms of ε_0 and f. If $A' \subset B$ is reduced, has all rows intervals and $|A'| \geqslant \frac{\varepsilon_0}{2}$ with $d_k(A') \leqslant f|B|$, then there is a reduced subset $A'' \subset A'$ formed as a union of rows R_x of A' which are intervals such that

$$|A'\setminus A''| \leqslant h_1(f)|B|, \qquad |\operatorname{co}(\pi(A''))\setminus\pi(A'')| \leqslant h_2(f)|\pi(B)|.$$

Proof of Proposition 2.3.27. Let $\Delta'(\varepsilon_0)$ denote the implicit bound on f required to apply Proposition 2.3.1 with $\sigma = f$, $\lambda = 1$ and $\alpha = \frac{1}{2}$. For $f \geqslant \Delta'(\varepsilon_0)$, set $A'' = \emptyset$ and $h_1(f) = h_2(f) = 1$ and for $f < \Delta'(\varepsilon_0)$ Proposition 2.3.1 gives the result with $h_1(f) = f^{\frac{1}{2}-c}$ and $h_2(f) = f^{\frac{1}{2}}$.

□

Proposition 2.3.28. Given $\varepsilon_0 > 0$, there are functions $h_3(t), h_4(t)$ with $h_3, h_4 \to 0$ as $t \to 0$ such that for every $f > 0$ the following is true, provided $\min(n_i)$ is sufficiently large in terms of ε_0 and f. If $A' \subset B$ is reduced, has all rows intervals, $|A'| \geqslant \frac{2\varepsilon_0}{3}|B|$ and $d_k(A') \leqslant f|B|$, then there exists a reduced subset $A'' \subset A'$ with each row an interval, such that

1. $|\mathrm{co}(\pi(A'')) \setminus \pi(A'')| \leqslant h_3(f)|\pi(B)|$
2. $|A' \setminus A''| \leqslant h_4(f)|B|$
3. $V(A'') \leqslant \ell(f)$ with ℓ as in the first part of Proposition 2.3.25.

Proof. We start by giving a high level outline of how the proof will work. Starting with $A' = A'_0$, we will create a nested sequence of sets $A'_0 \supset A''_1 \supset A'_1 \supset A''_2 \supset A'_2 \supset \ldots \supset A'_{\gamma(f)}$, where γ is some function, $A''_{i+1} \subset A'_i$ is obtained through an application of Proposition 2.3.27, and $A'_i \subset A''_i$ is obtained through an application of Proposition 2.3.25. All A'_i, A''_i are reduced and have all rows intervals and automatically satisfy the second point, and all of the A'_i automatically satisfy the third point, so it suffices to show there is an A'_i which satisfies the first point. But $|\mathrm{co}(\pi(A''_i)) \setminus \pi(A''_i)|$ is always extremely small by construction, so if $|\mathrm{co}(\pi(A'_i)) \setminus \pi(A'_i))| > h_3(f)|\pi(B)|$, then $|\pi(A''_i) \setminus \pi(A'_i)|$ must be at least roughly $h_3(f)|\pi(B)|$. This inequality can only happen approximately $h_3(f)^{-1}$ times however before the A'_i become empty, which is smaller than $\gamma(f)$.

Let h_1, h_2 be as in Proposition 2.3.27. We recursively define functions with $g_0(t) = t$ and

$$g_i(t) = g_{i-1}(t) + 2^k h_1(g_{i-1}(t)) + 2^k t.$$

For fixed i we have $g_i(t) \to 0$ as $t \to 0$.

We define a decreasing sequence of real numbers $r_1, r_2, \ldots \to 0$ with the properties

$$\bullet \sup_{t \in (0, r_i]} \frac{2}{i} + \sum_{j=0}^{i-1} h_2(g_j(t)) \to 0, \text{ as } i \to \infty, \tag{2.38}$$

$$\bullet \sup_{t \in (0, r_i]} it + \sum_{j=0}^{i-1} h_1(g_j(t)) \to 0, \text{ as } i \to \infty, \text{ and} \tag{2.39}$$

$$\bullet \sup_{t \in (0, r_i]} it + \sum_{j=0}^{i-1} h_1(g_j(t)) \leqslant \frac{\varepsilon_0}{6}. \tag{2.40}$$

For $t \leqslant r_1$ define $\gamma(t) = \max\{i : t \leqslant r_i\}$, and note $t \in (0, r_{\gamma(t)}]$, and $\gamma(t) \to \infty$ as $t \to 0$. Let

$$h_3(t) := \begin{cases} \frac{2}{\gamma(t)} + \sum_{j=0}^{\gamma(t)-1} h_2(g_j(t)) & \text{if } t \leqslant r_1 \\ 1 & \text{if } t > r_1 \end{cases}, \quad h_4(t) := \begin{cases} \gamma(t)t + \sum_{j=0}^{\gamma(t)-1} h_1(g_j(t)) & \text{if } t \leqslant r_1 \\ 1 & \text{if } t > r_1 \end{cases},$$

and note that $h_3, h_4 \to 0$ as $t \to 0$ by (2.38),(2.39).

If $f > r_1$, then the conclusion trivially holds with $A'' = \emptyset$. Otherwise, if $f \leqslant r_1$, we will see that we can iterate the below construction $\gamma(f)$ times while always satisfying the conditions in Proposition 2.3.27 and Proposition 2.3.25 that the corresponding set under consideration has size at least $\frac{\varepsilon_0}{2}|B|$.

We will now recursively construct reduced sets $A' = A'_0 \supset A'_1 \supset \ldots \supset A'_{\gamma(f)}$ with all rows intervals such that

$$d_k(A'_i) \leqslant g_i(f)|B|, \qquad |A' \setminus A'_i| \leqslant \left[if + \sum_{j=0}^{i-1} h_1(g_j(f)) \right] |B|. \tag{2.41}$$

Suppose that for some $i \leqslant \gamma(f)$, we have constructed A'_{i-1}, we will now construct A'_i. By (2.41) for $i-1$, (2.40) for $i = \gamma(f)$, and the fact that $f \leqslant r_{\gamma(f)}$, we have

$$|A' \setminus A'_{i-1}| \leqslant \left[(i-1)f + \sum_{j=0}^{i-2} h_1(g_j(f)) \right] |B| \leqslant \left[\gamma(f)f + \sum_{j=0}^{\gamma(f)-1} h_1(g_j(f)) \right] |B| \leqslant \frac{\varepsilon_0}{6}|B|,$$

so $|A'_{i-1}| \geqslant (\frac{2}{3} - \frac{1}{6})\varepsilon_0|B| = \frac{\varepsilon_0}{2}|B|$. Applying Proposition 2.3.27 to A'_{i-1} we obtain a reduced subset $A''_i \subset A'_{i-1}$ with all rows intervals such that

$$|A'_{i-1} \setminus A''_i| \leqslant h_1(g_{i-1}(f))|B|, \qquad |\operatorname{co}(\pi(A''_i)) \setminus \pi(A''_i)| \leqslant h_2(g_{i-1}(f))|\pi(B)|. \tag{2.42}$$

An identical proof shows $|A''_i| \geqslant (\frac{2}{3} - \frac{1}{6})\varepsilon_0|B| \geqslant \frac{\varepsilon_0}{2}|B|$. We now apply Proposition 2.3.25 with $\alpha = f$, to find a reduced subset $A'_i \subset A''_i$ with all rows intervals such that $|V(A'_i)| \leqslant \ell(f)$ and $A'_i = \operatorname{co}(A'_i) \cap A''_i$, with the property that

$$|A''_i \setminus A'_i| = |A''_i \setminus \operatorname{co}(A'_i)| \leqslant \left|\operatorname{co}(A''_i) \setminus \operatorname{co}(A'_i)\right| \leqslant f|\operatorname{co}(A''_i)| \leqslant f|B|, \tag{2.43}$$

so together with (2.41) for $i-1$ and (2.42), we deduce

$$|A' \setminus A'_i| \leqslant \left[(i-1)f + \sum_{j=0}^{i-2} h_1(g_j(f)) + f + h_1(g_{i-1}(f)) \right] |B| = \left[if + \sum_{j=0}^{i-1} h_1(g_j(f)) \right] |B|.$$

By (2.41) for $i-1$, (2.42),(2.43), and Observation 2.2.23,

$$d_k(A'_i) \leqslant d_k(A'_{i-1}) + 2^k(h_1(g_{i-1}(f)) + f)|B| \leqslant (g_{i-1}(f) + 2^k h_1(g_{i-1}(f)) + 2^k f)|B| = g_i(f)|B|,$$

verifying (2.41) for A'_i.

Writing γ for $\gamma(f)$, we have $|A' \setminus A'_\gamma| \leqslant (\gamma f + \sum_{j=0}^{\gamma-1} h_1(g_j(f)))|B| = h_4(f)|B|$. If all of $A'_1, \ldots, A'_\gamma$ have the property that $|\operatorname{co}(\pi(A'_i)) \setminus \pi(A'_i)| > h_3(f)|\pi(B)|$, then noting that $\pi(A'_i) \subset \pi(A''_i)$ implies $|\operatorname{co}(\pi(A'_i)) \setminus \pi(A'_i)| - |\operatorname{co}(\pi(A''_i)) \setminus \pi(A''_i)| = |\pi(A''_i) \setminus \pi(A'_i)| - |\operatorname{co}(\pi(A''_i)) \setminus \operatorname{co}(\pi(A'_i))|$, we deduce

$$\begin{aligned}
|\operatorname{co}(\pi(A_1')) \setminus \pi(A_\gamma')| &= |\operatorname{co}(\pi(A_1')) \setminus \pi(A_1')| + \sum_{i=2}^{\gamma} |\pi(A_{i-1}') \setminus \pi(A_i')| \\
&\geqslant |\operatorname{co}(\pi(A_1')) \setminus \pi(A_1')| + \sum_{i=2}^{\gamma} |\pi(A_i'') \setminus \pi(A_i')| \\
&\geqslant |\operatorname{co}(\pi(A_1')) \setminus \pi(A_1')| + \sum_{i=2}^{\gamma} |\operatorname{co}(\pi(A_i')) \setminus \pi(A_i')| - |\operatorname{co}(\pi(A_i'')) \setminus \pi(A_i'')| \\
&\geqslant \sum_{i=1}^{\gamma} |\operatorname{co}(\pi(A_i')) \setminus \pi(A_i')| - |\operatorname{co}(\pi(A_i'')) \setminus \pi(A_i'')| \\
&\geqslant \sum_{i=1}^{\gamma} (h_3(f) - h_2(g_{i-1}(f)))|\pi(B)| \\
&= 2|\pi(B)|,
\end{aligned}$$

a contradiction. Here the first inequality follows from the fact that $\pi(A_i'') \subset \pi(A_{i-1}')$ and the second inequality follows from the fact that $|\pi(A_i'') \setminus \pi(A_i')| \leqslant |\operatorname{co}(\pi(A_i')) \setminus \pi(A_i')| - |\operatorname{co}(\pi(A_i'')) \setminus \pi(A_i'')|$.

Let $1 \leqslant i_0 \leqslant \gamma$ be an index such that $|\operatorname{co}(\pi(A_{i_0}')) \setminus \pi(A_{i_0}')| \leqslant h_3(f)|\pi(B)|$. Then, as $i_0 \leqslant \gamma$, we have by (2.41) and the definition of h_4 that $|A' \setminus A_{i_0}'| \leqslant |A' \setminus A_\gamma'| \leqslant h_4(f)|B|$. Moreover, $|V(A_{i_0}')| \leqslant \ell(f)$ vertices. We thus conclude by setting $A'' = A_{i_0}'$. □

$A_\star \subset A_+$ with $|V(A_\star)|$ and $|\operatorname{co}(\pi(A_\star)) \setminus \pi(A_\star)|$ small and one further technical condition: Setup part 3

In this subsection we discuss the technical property that we would additionally like $A_\star$ to posses. Before we state our next result we need one more notation which we shall make use of in Proposition 2.3.30 and Proposition 2.3.31.

Notation 2.3.29. Let $\theta_0(\varepsilon_0)$ be the bound on δ from Observation 2.2.23. Let $\theta(\varepsilon_0) = \min(\theta_0(\varepsilon_0), 2^{-(k+2)}\varepsilon_0)$.

Proposition 2.3.30. Given $\varepsilon_0, \nu > 0$, for all $f_1, f_2 > 0$ the following holds, provided $\min(n_i)$ is sufficiently large in terms of f_1, f_2 and ε_0, ν. Let $A' \subset B$ be reduced with all rows intervals such that $|A'| \geqslant \frac{2\varepsilon_0}{3}|B|$, $d_k(A') \leqslant f_1|B|$, $|\operatorname{co}(\pi(A')) \setminus \pi(A')| \leqslant f_2|\pi(B)|$. Suppose we have a collection $\mathscr{M}$ of sets $\mathscr{W}_T \subset \{0\} \times \mathbb{Z}^{k-1}$ with $|\mathscr{W}_T| \leqslant \nu$ for every $\widetilde{T} \in \{0\} \times \mathbb{R}^{k-1}$ with integral vertices. Then there exists a reduced subset $A'' \subset A'$ formed as a union of rows R_x of A' which are intervals such that

$$|A' \setminus A''| \leqslant \theta^{-1}(\varepsilon_0/2)(2\nu+2)(f_1 + 2^{k+1}f_2)|B|$$

which satisfies the following additional properties.

1. $\operatorname{co}(A'') = \operatorname{co}(A')$.
2. A'' has Property $P(\mathscr{M})$

Proof. Here is the strategy of the proof. Each instance when Property $P(\mathscr{M})$ is violated is given by some bad quadruple of nonempty rows $R_x, R_y, R_z, R_{z+\vec{v}}$ of A' (where $\vec{v} \in \{0\} \times \{0,1\}^{k-1}$) such that $R_x + R_y$ is disjoint from $R_z + R_{z+\vec{v}}$. Note that for each bad quadruple of rows $R_x, R_y, R_z, R_{z+\vec{v}}$, if we delete the row R_x we do not produce any new bad quadruples and we do not change the convex hull of the set. The hope is that the total mass of the rows that we delete is small. To show this, we note that, as usual, the reference frame $A'(+)A'$ has size $|A'(+)A'| \approx 2^k|A'|$. The key point is that any bad quadruples of rows satisfies $R_x + R_y$ is disjoint $A'(+)A'$. The final point is that, though two bad quadruple of rows could satisfy $R_{x_1} + R_{y_1}$ intersects $R_{x_2} + R_{y_2}$, this does not happen too often because $\mathscr{W}_T$ is bounded for each T.

For every $\widetilde{T}$, write $\mathscr{W}_T := \{\vec{w}_{T,i} : 1 \leqslant i \leqslant \nu\}$ (we can assume wlog that $\mathscr{W}_T$ is maximal). Let $X_{T,i}$ consist of those $x \in T^\circ \setminus \pi(V(A'))$ such that $x + \vec{w}_{T,i} \in Y_{\mathscr{W}_T}(x)$, and writing $x + x + \vec{w}_{T,i} = z + z + \vec{v}$ with $\vec{v} \in \{0\} \times \{0,1\}^{k-1}$, we have

$$R_x + R_{x+\vec{w}_{T,i}}, R_z + R_{z+\vec{v}} \text{ nonempty and } (R_x + R_{x+\vec{w}_{T,i}}) \cap (R_z + R_{z+\vec{v}}) = \emptyset.$$

For $\star \in \{+,-\}$, let $X_i^\star := \bigsqcup_{\widetilde{T} \in \mathscr{T}^\star(A')} X_{T,i}$, let $y_i^\star$ be defined on $X_i^\star$ by setting $y_i^\star(x) = x + \vec{w}_{T,i}$ for $x \in X_{T,i}$, and set the disjoint union

$$Z_i^\star := \bigsqcup_{\widetilde{T} \in \mathscr{T}^\star(A')} \bigsqcup_{x \in X_{T,i}} R_x + R_{y_i^\star(x)} \subset A' + A'.$$

Here we note the union is disjoint as $\frac{1}{2}(x + y_i^\star(x)) \in \widetilde{T}^\circ$, which are disjoint for distinct $\widetilde{T}$, and for a given $\widetilde{T}$ we have $\{2x + \vec{w}_{T,i}\}_{x \in T}$ are distinct.

For $\star \in \{+,-\}$ let $X_0^\star$ consist of those $x \in \pi(B) \setminus \pi(V(A'))$ such that there exists $\widetilde{T} \in \mathscr{T}^\star(A')$ with $x \in T^\circ$ and $y_0^\star(x) \in V(T)$ (note $R_{y_0^\star(x)} \neq \emptyset$ as this represents a row containing a vertex of $\mathrm{co}(A')$), such that writing $x + y_0^\star(x) = z + z + \vec{v}$ with $\vec{v} \in \{0\} \times \{0,1\}^{k-1}$,

$$R_x, R_z + R_{z+\vec{v}} \text{ nonempty, and } (R_x + R_{y_0^\star(x)}) \cap (R_z + R_{z+\vec{v}}) = \emptyset.$$

Set the disjoint union

$$Z_0^\star := \bigsqcup_{x \in X_0^\star} R_x + R_{y_0^\star(x)} \subset A' + A'.$$

Here, the union is disjoint because $\frac{1}{2}(x + y_0^\star(x)) \in \widetilde{T}^\circ$, which are disjoint for distinct $\widetilde{T}$, and for a given $\widetilde{T}$ we have $\{\frac{1}{2}(\widetilde{T}^\circ + y)\}_{y \in V(T)}$ are disjoint.

Finally, set $X := \bigcup_{(\star,i) \in \{+,-\} \times \{0,\ldots,\nu\}} X_i^\star$, and let $A'' = A' \setminus \bigsqcup_{x \in X} R_x$, so that $|A' \setminus A''| = \sum_{x \in X} |R_x|$. By construction A'' satisfies the properties 1 and 2. Indeed, A'' satisfies property 1 as X is disjoint from $\pi(V(A'))$ which means that for $x \in X$ we have R_x is disjoint from $V(A')$ which gives $V(A'') = V(A')$. A'' satisfies property 2 as for every $\widetilde{T} \in \mathscr{T}^+(A') \cup \mathscr{T}^-(A') = \mathscr{T}^+(A'') \cup \mathscr{T}^-(A'')$, X contains all those $x \in T^\circ \setminus \pi(V(A'')) = T^\circ \setminus \pi(V(A'))$ for which there exists $y \in Y_{\mathscr{W}_T}(x)$, $z \in \pi(B)$ and $\vec{v} \in \{0\} \times \{0,1\}^{k-1}$ with $x + y = z + z + \vec{v}$ and $R_x + R_y, R_z + R_{z+\vec{v}}$ nonempty and $(R_x + R_y) \cap (R_z + R_{z+\vec{v}})$ empty. Note that

by removing the rows R_x with $x \in X$ we eradicate all these instances and moreover in doing so we don't create any new such instances.

So all that remains is to check A'' is reduced, has all rows intervals and satisfies

$$|A' \setminus A''| \leqslant \theta^{-1}(\varepsilon_0/2)(2\nu+2)(f_1+2^{k+1}f_2)|B|.$$

Set the disjoint union

$$Z := A'(+)A' = \bigsqcup_{\vec{v}\in\{0\}\times\{0,1\}^{k-1}} \bigsqcup_{x\in\{0\}\times\mathbb{Z}^{k-1}} R_x(+)R_{x+\vec{v}} \subset A'+A',$$

and note that by construction $Z \cap Z_i^\star = \emptyset$ for all $\star, i$. Choose $(\star, i) \in \{+,-\} \cup \{0,\dots,\nu\}$ so that

$$|A' \setminus A''| = \sum_{x\in X} |R_x| \leqslant (2\nu+2) \sum_{x\in X_i^\star} |R_x|.$$

Note that for $0 \neq \vec{v} \in \{0\}\times\{0,1\}^{k-1}$ and $x \in \{0\}\times\mathbb{Z}^{k-1}$ we have

$$|R_x(+)R_{x+\vec{v}}| - |R_x| - |R_{x+\vec{v}}| \geqslant \begin{cases} 0 & |\{x_1,x_2\} \cap \pi(A')| = 0 \\ -n_1 & |\{x_1,x_2\} \cap \pi(A')| = 1 \\ -1 & |\{x_1,x_2\} \cap \pi(A')| = 2 \end{cases}$$

By Observation 2.2.22, we have for $0 \neq \vec{v} \in \{0\}\times\{0,1\}^{k-1}$ that

$$\begin{aligned}\left|\{x \in \{0\}\times\mathbb{Z}^{k-1} : |\{x, x+\vec{v}\} \cap \pi(A')| = 1\}\right| &\leqslant 2|\operatorname{co}(\pi(A')) \setminus \pi(A')| + 2(k-1)n_1^{-1}\min(n_i)^{-1}|B| \\ &\leqslant 3f_2 n_1^{-1}|B|\end{aligned}$$

and

$$\left|\{x \in \{0\}\times\mathbb{Z}^{k-1} : |\{x, x+\vec{v}\} \cap \pi(A')| = 2\}\right| \leqslant |\pi(B)| \leqslant \min(n_i)^{-1}|B| \leqslant f_2|B|.$$

We therefore have (taking $\vec{v} \in \{0\}\times\{0,1\}^{k-1}$ and $x \in \{0\}\times\mathbb{Z}^{k-1}$)

$$\begin{aligned}|A'+A'| &\geqslant |Z| + |Z_i^\star| \\ &= \left(\sum_{\vec{v}}\sum_{x} |R_x(+)R_{x+\vec{v}}|\right) + \sum_{x\in X_i^\star} |R_x + R_{y_i^\star(x)}| \\ &\geqslant 2^k|A'| - (2^{k-1}-1)(3f_2+f_2)|B| - |\pi(B)| + \sum_{x\in X_i^\star} |R_x| \\ &\geqslant 2^k|A'| - 2^{k+1}f_2|B| + \frac{1}{2\nu+2}|A' \setminus A''|.\end{aligned}$$

Note that we used the fact that for $x \in X_i^\star$ we have $R_{y_i^\star(x)}$ is non-empty. This is because by construction we have $X_i^\star := \bigsqcup_{\widetilde{T}\in\mathscr{T}^\star(A')} X_{T,i}$ and for $x \in X_{T,i}$ we have $y_i^\star(x) = x+\vec{w}_{T,i}$ and $R_x + R_{x+\vec{w}_{T,i}}$ is non-empty. Hence,

$$|A' \setminus A''| \leqslant (2\nu+2)(d_k(A') + 2^{k+1}f_2|B|) \leqslant (2\nu+2)(f_1+2^{k+1}f_2)|B|.$$

By construction, A'' is a subset of rows of A' and hence all rows of A'' are intervals. Using Notation 2.3.29, by the second part of Observation 2.2.23, if A' is reduced, $|A'| \geqslant 2^{-1}\varepsilon_0|B|$, $d_k(A') \leqslant \theta(2^{-1}\varepsilon_0)|B|$ and $|A'\setminus A''| \leqslant \theta(2^{-1}\varepsilon_0)|B|$ then A'' is reduced. By hypothesis, we have A' is reduced, $|A'| \geqslant 2^{-1}\varepsilon_0|B|$, $d_k(A') \leqslant f_1|B|$ and $|A'\setminus A''| \leqslant (2\nu+2)(f_1+2^{k+1}f_2)|B|$. Therefore, in the case $(2\nu+2)(f_1+2^{k+1}f_2) \leqslant \theta(2^{-1}\varepsilon_0)$ we deduce A'' is reduced. In the case $\theta^{-1}(2^{-1}\varepsilon_0)(2\nu+2)(f_1+2^{k+1}f_2) \geqslant 1$ we can just take $A''=\emptyset$. □

Proposition 2.3.31. Given $\varepsilon_0, \nu > 0$, there exist functions $h_5(t), h_6(t), H_7(t)$ such that $h_5, h_6, \to 0$ as $t \to 0$ and $H_7(t) \to \infty$ as $t \to 0$ such that for every $e > 0$ the following is true, provided $\min(n_i)$ is sufficiently large in terms of ε_0, e, ν. Say $A' \subset B$ is reduced with all rows intervals and $|A'| \geqslant \frac{3\varepsilon_0}{4}|B|$, $d_k(A') \leqslant e|B|$. Say we have a collection $\mathscr{M}$ of sets $\mathscr{W}_T \subset \{0\}\times\mathbb{Z}^{k-1}$ with $|\mathscr{W}_T| \leqslant \nu$ for every simplex $\widetilde{T} \subset \{0\}\times\mathbb{R}^{k-1}$ with integral vertices. Then there exists a reduced subset $A'' \subset A'$ with all rows intervals and

$$|A'\setminus A''| \leqslant h_5(e)|B|, \qquad |\operatorname{co}(\pi(A''))\setminus\pi(A'')| \leqslant h_6(e)|\pi(B)|$$

which satisfies the following additional properties.

1. $|V(A'')| \leqslant H_7(e)$
2. A'' has Property $P(\mathscr{M})$.

Proof. The high level overview of this proof is identical to that given at the start of the proof of Proposition 2.3.28, except with h_3 replaced with h_6, Proposition 2.3.25 replaced with Proposition 2.3.30 and Proposition 2.3.27 replaced with Proposition 2.3.28.

We recursively define functions $g_0(t) = t$ and

$$\begin{aligned} g_i(t) =& \theta^{-1}(\varepsilon_0/2)(2\nu+2)\Bigg[\left(2^k + \frac{1}{\theta^{-1}(\varepsilon_0/2)(2\nu+2)}\right)g_{i-1}(e) + \\ &+ \left(2^{2k} + \frac{2^k}{\theta^{-1}(\varepsilon_0/2)(2\nu+2)}\right)h_4(g_{i-1}(e)) + 2^{2k+1}h_3(g_{i-1}(e))\Bigg]. \end{aligned}$$

For fixed i we have $g_i(t) \to 0$ as $t \to 0$.

We define a decreasing sequence of real numbers $r_1, r_2, \ldots \to 0$ with the following properties:

$$\bullet \sup_{t\in(0,r_i]} \frac{2}{i} + \sum_{j=0}^{i-1} h_3(g_j(t)) \to 0 \text{ as } i \to \infty \tag{2.44}$$

$$\bullet \sup_{t\in(0,r_i]} 2\theta^{-1}(\varepsilon_0/2)(\nu+1)\left[\sum_{j=0}^{i-1} g_j(t) + \left(2^k + \frac{\theta(\varepsilon_0/2)}{2(\nu+1)}\right) h_4(g_j(t)) + 2^{k+1}h_3(g_j(t))\right] \to 0 \text{ as } i \to \infty \tag{2.45}$$

$$\bullet \sup_{t\in(0,r_i]} 2\theta^{-1}(\varepsilon_0/2)(\nu+1)\left[\sum_{j=0}^{i-1} g_j(t) + \left(2^k + \frac{\theta(\varepsilon_0/2)}{2(\nu+1)}\right) h_4(g_j(t)) + 2^{k+1}h_3(g_j(t))\right] \leqslant \frac{1}{12}\varepsilon_0. \tag{2.46}$$

For $t \leqslant r_1$ set $\gamma(t) = \max\{i : t \leqslant r_i\}$. Note that $t \in (0, r_{\gamma(t)}]$ and that $\gamma(t) \to \infty$ as $t \to 0$. Set

$$h_5(t) := \begin{cases} \theta^{-1}(\varepsilon_0/2)(2\nu+2)\left[\sum_{j=0}^{\gamma(t)-1} g_j(t) + \left(2^k + \frac{\theta(\varepsilon_0/2)}{2\nu+2}\right) h_4(g_j(t)) + 2^{k+1}h_3(g_j(t))\right] & \text{if } t \leqslant r_1 \\ 1 & \text{if } t > r_1 \end{cases},$$

$$h_6(t) := \begin{cases} \frac{2}{\gamma(t)} + \sum_{j=0}^{\gamma(t)-1} h_3(g_j(t)) & \text{if } t \leqslant r_1 \\ 1 & \text{if } t > r_1 \end{cases},$$

and with ℓ as in Proposition 2.3.25

$$H_7(t) := \max_{0\leqslant j\leqslant \gamma(t)-1} (\ell(g_j(t))$$

and note that $h_5, h_6 \to 0$ as $t \to 0$ by (2.44),(2.45).

If $e > r_1$, then the conclusion trivially holds with $A'' = \emptyset$. Otherwise, if $e \leqslant r_1$, we will see that we can iterate the below construction $\gamma(e)$ times while always satisfying the conditions in Proposition 2.3.28 and Proposition 2.3.30 that the corresponding set under consideration has size at least $\frac{2\varepsilon_0}{3}|B|$.

We will now recursively construct reduced sets $A' = A'_0 \supset A'_1 \supset \ldots \supset A'_{\gamma(e)}$ with all rows intervals, such that

$$d_k(A'_i) \leqslant g_i(e)|B|, \tag{2.47}$$

$$|A' \setminus A'_i| \leqslant \theta^{-1}(\varepsilon_0/2)(2\nu+2)\left[\sum_{j=0}^{i-1} g_j(e) + \left(2^k + \frac{\theta(\varepsilon_0/2)}{2\nu+2}\right) h_4(g_j(e)) + 2^{k+1}h_3(g_j(e))\right]|B|. \tag{2.48}$$

Suppose that for some $i \leqslant \gamma(e)$, we have constructed A'_{i-1}, we will now construct A'_i. By (2.48) for $i-1$, (2.46) for $i = \gamma(e)$, and the fact that $e \leqslant r_{\gamma(e)}$, we have $|A'_{i-1}| \geqslant (\frac{3}{4} - \frac{1}{12})\varepsilon_0|B| = \frac{2}{3}\varepsilon_0|B|$. Applying Proposition 2.3.28 to A'_{i-1}, we find reduced A''_i with all rows intervals, $|V(A''_i)| \leqslant \ell(g_{i-1}(e))$,

and

$$|A'_{i-1}\setminus A''_i| \leqslant h_4(g_{i-1}(e))|B|, \qquad |\operatorname{co}(\pi(A''_i))\setminus \pi(A''_i)| \leqslant h_3(g_{i-1}(e))|\pi(B)|. \tag{2.49}$$

By (2.47) for $i-1$,(2.49), and Observation 2.2.23, we have

$$d_k(A''_i) \leqslant \left[g_{i-1}(e)+2^k h_4(g_{i-1}(e))\right]|B|. \tag{2.50}$$

An identical proof shows $|A''_i| \geqslant (\frac{3}{4}-\frac{1}{12})\varepsilon_0|B| = \frac{2}{3}\varepsilon_0|B|$. Now by (2.49) and (2.50), applying Proposition 2.3.30 to A''_i gives a reduced set $A'_i \subset A''_i$ with all rows intervals, $\operatorname{co}(A'_i)=\operatorname{co}(A''_i)$, property 2 with $A'=A'_i$, and

$$|A''_i\setminus A'_i| \leqslant \theta^{-1}(\varepsilon_0/2)(2\nu+2)\left[g_{i-1}(e)+2^k h_4(g_{i-1}(e))+2^{k+1}h_3(g_{i-1}(e))\right]|B|, \tag{2.51}$$

so together with (2.48) for $i-1$ and (2.49), we deduce

$$|A'\setminus A'_i| \leqslant \theta^{-1}(\varepsilon_0/2)(2\nu+2)\left[\sum_{j=0}^{i-1} g_j(e)+\left(2^k+\frac{\theta(\varepsilon_0/2)}{2\nu+2}\right)h_4(g_j(e))+2^{k+1}h_3(g_j(e))\right]|B|. \tag{2.52}$$

By (2.50),(2.51) and Observation 2.2.23, we have

$$\begin{aligned} d_k(A'_i) &\leqslant \theta^{-1}(\varepsilon_0/2)(2\nu+2)\left[\left(2^k+\frac{1}{\theta^{-1}(\varepsilon_0/2)(2\nu+2)}\right)g_{i-1}(e)+\right.\\ &\quad\left.+\left(2^{2k}+\frac{2^k}{\theta^{-1}(\varepsilon_0/2)(2\nu+2)}\right)h_4(g_{i-1}(e))+2^{2k+1}h_3(g_{i-1}(e))\right]|B|\\ &= g_i(e)|B|. \end{aligned}$$

Write γ for $\gamma(e)$. If for $1\leqslant i\leqslant\gamma$ we have $|\operatorname{co}(\pi(A'_i))\setminus\pi(A'_i)|>h_6(e)|\pi(B)|$, then we have removed at least

$$\begin{aligned} |\operatorname{co}(\pi(A'_1))\setminus\pi(A'_\gamma)| &\geqslant |\operatorname{co}(\pi(A'_1))\setminus\pi(A'_1)|+\sum_{j=2}^{\gamma}|\pi(A''_j)\setminus\pi(A'_j)|\\ &\geqslant \sum_{j=1}^{\gamma}|\operatorname{co}(\pi(A'_j))\setminus\pi(A'_j)|-|\operatorname{co}(\pi(A''_j))\setminus\pi(A''_j)|\\ &\geqslant \sum_{j=1}^{\gamma}(h_6(e)-h_3(g_{j-1}(e)))|\pi(B)|\\ &= 2|\pi(B)| \end{aligned}$$

rows in $\pi(B)$, a contradiction. Let $1 \leqslant i_0 \leqslant \gamma$ be an index such that $|\operatorname{co}(\pi(A'_{i_0})) \setminus \pi(A'_{i_0})| \leqslant h_6(e)|\pi(B)|$. As $i_0 \leqslant \gamma$, we have by (2.48) and the definition of h_5 that

$$|A' \setminus A'_{i_0}| \leqslant |A' \setminus A'_{\gamma}| \leqslant h_5(e)|B|.$$

Moreover, $V(A'_{i_0}) = V(A''_{i_0}) \leqslant \ell(g_{i_0-1}(e)) \leqslant H_7(e)$, and as remarked earlier, A'_{i_0} satisfies property 2. We thus conclude by setting $A'' = A'_{i_0}$. □

$A_\star \subset A_+$ with $|V(A_\star)|$ and $|\operatorname{co}(\pi(A_\star)) \setminus \pi(A_\star)|$ small and one further technical condition: Construction

Before we proceed we need to introduce the following definition.

Definition 2.3.32. Given a simplex $\widetilde{T} \subset \{0\} \times \mathbb{R}^{k-1}$ with vertices $x_0, \dots, x_{k-1}$ construct inductively a dense family of translates of $\frac{1}{2^i}\widetilde{T}$ inside $\widetilde{T}$ as follows. Set

$$\mathscr{S}_{i,0}(\widetilde{T}) := \left\{ \left(1 - \frac{1}{2^i}\right) x_r + \frac{1}{2^i}\widetilde{T} : 0 \leqslant r \leqslant k-1 \right\}$$
$$\mathscr{S}_{i,j+1}(\widetilde{T}) := \left\{ \frac{\widetilde{S}_1 + \widetilde{S}_2}{2} : \widetilde{S}_1, \widetilde{S}_2 \in \mathscr{S}_{i,j} \right\}$$

Definition 2.3.33. Given a simplex $\widetilde{T} \subset \{0\} \times \mathbb{R}^{k-1}$ we define

$$\mathscr{U}_{i,j}(\widetilde{T}) := \{\vec{u} : \exists \widetilde{S}_1, \widetilde{S}_2 \in \mathscr{S}_{i,j} \text{ with } \widetilde{S}_1 + \vec{u} = \widetilde{S}_2\}.$$

Before preceeding, we remark that we will now need a future result, Proposition 2.3.52, to define certain constants μ_1 and μ_2 depending only on k. The proof is entirely self-contained, and while we could include the result and its proof at this point, we feel it is better to defer them.

Definition 2.3.34. We define constants $\mu_1 = \mu_1(k), \mu_2 = \mu_2(k)$ as those produced by Proposition 2.3.52. We construct a collection $\mathscr{M}$ of sets $\mathscr{W}_T \subset \{0\} \times \mathbb{Z}^{k-1}$ for every simplex $\widetilde{T} \subset \{0\} \times \mathbb{R}^{k-1}$ with integral vertices. Given a simplex $\widetilde{T}\{0\} \times \mathbb{Z}^{k-1}$ with integral vertices, we set

$$\mathscr{W}_T := \bigcup_{\vec{u} \in \mathscr{U}_{\mu_1,\mu_2}(\widetilde{T})} \mathscr{R}(\vec{u}) \cup (-\mathscr{R}(\vec{u})),$$

where $\mathscr{R}(\vec{u}) = \lfloor \vec{u} \rceil + \{0\} \times \{0,1\}^{k-1}$ is obtained by rounding the coordinates of $\vec{u}$ to integers. This satisfies $|\mathscr{W}_T| \leqslant \nu$ for $\nu = \nu(k)$ the constant $2^k|\mathscr{U}_{\mu_1,\mu_2}|$, which is independent of the simplex $\widetilde{T}$. Note that $0 \in \mathscr{U}_{\mu_1,\mu_2}(\widetilde{T})$ so $\{0\} \times \{0,1\}^{k-1} \subset \mathscr{W}_T$, and $\mathscr{W}_T = -\mathscr{W}_T$.

Provided $\varepsilon_0 > 0$ is sufficiently small, $\delta > 0$ is sufficiently small in terms of ε_0 and $\min(n_i)$ is sufficiently large in terms of ε_0 and δ (which we assume wlog by Convention 2.2.14), we can apply

Proposition 2.3.31, obtaining a reduced subset $A_\star \subset A_+$ with all rows intervals satisfying the following properties. By (2.37), we have

$$|A \triangle A_\star| \leqslant |A \triangle A_+| + |A_+ \setminus A_\star| \leqslant (\delta^{\frac{1}{20}-16c} + h_5(\delta^{\frac{1}{20}-17c}))|B| =: h_8(\delta)|B|, \tag{2.53}$$

and by Observation 2.2.23, we have $A_\star$ is reduced and

$$d_k(A_\star) \leqslant (\delta^{\frac{1}{20}-17c} + 2^k h_8(\delta))|B| =: h_9(\delta)|B|. \tag{2.54}$$

Also,

$$|\operatorname{co}(\pi(A_\star)) \setminus \pi(A_\star)| \leqslant h_6(\delta^{\frac{1}{20}-17c})|\pi(B)|, \text{ and} \tag{2.55}$$

$$V(A_\star) \leqslant H_7(\delta^{\frac{1}{20}-17c}). \tag{2.56}$$

Here, $h_6, h_8, h_9 \to 0$ as $\delta \to 0$.

Observation 2.3.35. By Proposition 2.3.31, $A_\star$ has Property $P(\mathscr{M})$.

2.3.8 $A_\star$ is close to $\operatorname{co}(A_\star)$

In this section, we show that $|\operatorname{co}(A_\star) \setminus A_\star| = o(1)|B|$. It is easy to show that

$$2^k|\operatorname{co}(A_\star) \setminus A_\star| \leqslant \left|\operatorname{co}(A_\star + A_\star) \setminus (A_\star + A_\star)\right| + o(1)|B|.$$

Indeed, by (2.54) we have $|A_\star + A_\star| \leqslant 2^k|A_\star| + o(1)|B|$ and by Observation 2.2.21 we have $2^k|\operatorname{co}(A_\star)| - o(1)|B| \leqslant |\operatorname{co}(A_\star) + \operatorname{co}(A_\star)|$. Adding the two inequalities and rearranging the terms gives the desired inequality.

Therefore, it suffices to show that for some constant $c_k' > 0$ we have

$$\left|\operatorname{co}(A_\star + A_\star) \setminus (A_\star + A_\star)\right| \leqslant (2^k - c_k')|\operatorname{co}(A_\star) \setminus A_\star| + o(1)|B| \tag{2.57}$$

Indeed, by combining the two inequalities above, it follows that $2^k|\operatorname{co}(A_\star) \setminus A_\star| \leqslant (2^k - c_k')|\operatorname{co}(A_\star) \setminus A_\star| + o(1)|B|$ which forces $|\operatorname{co}(A_\star) \setminus A_\star| = c_k'^{-1} o(1)|B|$. We now give a motivating outline.

For $\widetilde{T} \in \mathscr{T}^+(A_\star) \cup \mathscr{T}^-(A_\star)$, we will define functions $g_T^+, g_T^- : T \to [0, n_1]$ (actually we will need $[0, 2n_1]$ for technical reasons), which encode the distances from the nonempty rows of $A_\star$ in T to the upper and lower convex hulls of $A_\star$ respectively. Then we can estimate

$$|\operatorname{co}(A_\star) \setminus A_\star| = o(1)|B| + \sum_{*\in\{+,-\}} \sum_{\widetilde{T}\in\mathscr{T}^*} \sum_{x\in T} g_T^*(x). \tag{2.58}$$

Here is why one might expect the error term $o(1)|B|$. For a typical $x \in \operatorname{co}(\pi(A))$ we expect the interval row R_x of $A_\star$ to be non-empty and we also expect for each $* \in \{+,-\}$ to exist a unique

$\widetilde{T^*} \in \mathscr{T}^*$ such that $x \in T^*$. For such a typical x, the contribution from x in $|\operatorname{co}(A_\star) \setminus A_\star|$ is the total number of integer points above and below R_x contained inside $\widetilde{\operatorname{co}}(A_\star)$; the contribution from x in $\sum_{*\in\{+,-\}} \sum_{\widetilde{T}\in\mathscr{T}^*} \sum_{x\in T} g_T^*(x)$ is the total length of the intervals extending above and below R_x inside $\widetilde{\operatorname{co}}(A_\star)$. Note that for an interval, the length and the number of integer points inside it differ by at most 1. Therefore, a typical x contributes $O(1)$ to the discrepancy. Hence, all the typical x contribute $O(1)|\pi(B)| = o(1)|B|$ in total to the discrepancy.

An atypical x could potentially contribute up to $O(n_1)$ to the discrepancy. However, it turns out there are $o(1)|\pi(B)|$ such atypical x. To bound the number of points x contained inside at least two simplices $\widetilde{T}_1, \widetilde{T}_2 \in \mathscr{T}^+$ note that x must be actually contained on the boundary of the two simplices. Moreover, the number of simplices in a triangulation of $\partial\widetilde{\operatorname{co}}(A_\star)$ is bounded in terms of δ because the number of vertices of $\widetilde{\operatorname{co}}(A_\star)$ is bounded in terms of δ. Furthermore, the number of points x in $\pi(B)$ contained inside a hyperplane is at most $\min(n_i)^{-1} n_1^{-1}|B|$. Therefore, all the atypical x contribute $n_1 o(1)|\pi(B)| = o(1)|B|$ in total to the discrepancy.

We will define functions $g_T^{+\square}, g_T^{-\square}$ as certain restricted infimum convolutions of g_T^+ and g_T^- with themselves. These will encode the distance between the rows of a certain subset of $A_\star + A_\star$ (which we will guarantee to be intervals by Observation 2.3.35) to the upper and lower convex hulls of $A_\star + A_\star$ respectively. This subset accounts for almost all rows. Then we can similarly estimate

$$\left|\operatorname{co}(A_\star + A_\star) \setminus (A_\star + A_\star)\right| \leqslant o(1)|B| + \sum_{*\in\{+,-\}} \sum_{\widetilde{T}\in\mathscr{T}^*} \sum_{x'\in 2\widetilde{T}\cap\{0\}\times\mathbb{Z}^{k-1}} g_T^{*\square}(x'). \tag{2.59}$$

Therefore, in order to show (2.57) it suffices to prove

$$\sum_{*\in\{+,-\}} \sum_{\widetilde{T}\in\mathscr{T}^*} \sum_{x'\in 2\widetilde{T}\cap\{0\}\times\mathbb{Z}^{k-1}} g_T^{*\square}(x') \leqslant o(1)|B| + (2^k - c_k') \sum_{*\in\{+,-\}} \sum_{\widetilde{T}\in\mathscr{T}^*} \sum_{x\in T} g_T^*(x)$$

To prove this inequality, it will suffice to show for every $* \in \{+,-\}$ and $\widetilde{T} \in \mathscr{T}^*$, given a function $g : T \to [0, 2n_1]$ which is 0 at the vertices of T, that

$$\sum_{x'\in 2\widetilde{T}\cap\{0\}\times\mathbb{Z}^{k-1}} g^{\square}(x') \leqslant o(1)|B| + (2^k - c_k') \sum_{x\in T} g(x). \tag{2.60}$$

In Section 2.3.8, we properly define the functions $g_T^*(x)$ and $g_T^{*\square}(x)$ and show (2.58) in Observation 2.3.39 and (2.59) in Proposition 2.3.42, thus reducing the problem to showing (2.60).

In Section 2.3.8, we prove (2.60) in Proposition 2.3.44.

Finally, in Section 2.3.8, we combine these results and conclude (2.57) in Proposition 2.3.53.

Transitioning from $A_\star$ to functions and their infimum convolutions

We focus on the gaps in the e_1-direction between $A_\star$ and the convex hull of $A_\star$ via functions on $\pi(A_\star)$.

Notation 2.3.36. We denote by $V_\pi = \pi(V(A_\star))$ the projection of the vertices of $\widetilde{\mathrm{co}}(A_\star)$ under π to $\{0\}\times\mathbb{Z}^{k-1}$. We denote the empty rows by $E := \mathrm{co}(\pi(A_\star))\setminus\pi(A_\star)$. Finally, we write $\mathscr{T}^+ := \mathscr{T}^+(A_\star)$ and $\mathscr{T}^- := \mathscr{T}^-(A_\star)$.

Recall that for a simplex $\widetilde{T}\subset\widetilde{\mathrm{co}}(\pi(B))$, we denote $T = \widetilde{T}\cap\mathbb{Z}^k = \widetilde{T}\cap(\{0\}\times\mathbb{Z}^{k-1})$.

Definition 2.3.37. Denoting $\Psi^+_{A_\star},\Psi^-_{A_\star} : \widetilde{\mathrm{co}}(\pi(A_\star))\to\mathbb{R}$ the upper and lower convex hull function of $A_\star$ in the e_1-direction respectively, we define for $\widetilde{T}^+\in\mathscr{T}^+$ and $\widetilde{T}^-\in\mathscr{T}^-$ the functions

$$g^+_{T^+} : T^+\to[0,2n_1], \qquad g^-_{T^-} : T^-\to[0,2n_1]$$

according to the formulas

$$g^+_{T^+}(x) = \begin{cases} \Psi^+_{A^\star}(x) - \max R_x & \text{if } x\in V(T^+) \text{ or } x\notin V_\pi\cup E \\ 2n_1 & \text{otherwise.}\end{cases},$$

and

$$g^-_{T^-}(x) = \begin{cases} \min R_x - \Psi^-_{A^\star}(x) & \text{if } x\in V(T^-) \text{ or } x\notin V_\pi\cup E \\ 2n_1 & \text{otherwise.}\end{cases}.$$

Remark 2.3.38. For $x\in V(T^*)$ or $x\notin V_\pi\cup E$, $g^*_{T^*}(x)\in[0,n_1-1]$ is the distance in the e_1-direction from the row R_x to the upper convex hull of $A_\star$ for $*=+$ and lower convex hull of $A_\star$ for $*=-$, and we always have $g^*_{T^*}(x)\leqslant 2n_1$. In particular, for $x\in V(T^*)$ we note that $g^*_{T^*}(x)=0$.

Observation 2.3.39. Provided $\min(n_i)$ is sufficiently large in terms of δ (which we assume wlog by Convention 2.2.14), we have the following estimate for some function $h_{10}(\delta)\to 0$ as $\delta\to 0$ that

$$\left(\sum_{*\in\{+,-\}}\sum_{\widetilde{T}^*\in\mathscr{T}^*}\sum_{x\in T^*} g^*_{T^*}(x)\right) - |\mathrm{co}(A_\star)\setminus A_\star| \leqslant h_{10}(\delta)|B|.$$

Actually, a similar lower bound holds but we only need the upper bound.

Proof. Writing $|\mathrm{co}(A_\star)\setminus A_\star| = \sum_{x\in\mathrm{co}(\pi(A_\star))}|(\mathrm{co}(A_\star)\setminus A_\star)\cap\pi^{-1}(x)|$, we upper bound the contribution separately to the left hand side for each $x\in\mathrm{co}(\pi(A_\star))$. The idea is that the typical $x\in\mathrm{co}(\pi(A_\star))$ lies outside $V_\pi\cup E$ and inside a unique simplex $\widetilde{T}^+\in\mathscr{T}^+$ and $\widetilde{T}^-\in\mathscr{T}^-$. For such a typical x, the contribution to the left hand side is $O(1)$. The contribution from an atypical x is always $O(n_1)$. However, there are $o(1)n_1^{-1}|B|$ atypical x. This is because x is atypical if either x lies inside $V_\pi\cup E$ or x lies on the boundary of a simplex $\widetilde{T^*}\in\mathscr{T}^*$. Note that V_π and E are small by the hypothesis on $A_\star$. Also note that the number of simplices is bounded in terms of δ and the number of integer points on the face of a simplex (or any co-dimension 2 hyperplane) is bounded by $\min(n_i)^{-1}|\pi(B)|$.

- We estimate the contribution of $x \in \partial \widetilde{T}^+$ for some $\widetilde{T}^+ \in \mathscr{T}^+$ or with $x \in \partial \widetilde{T}^-$ for some $\widetilde{T}^- \in \mathscr{T}^-$. There are at most $\binom{|V_\pi|}{k}$ simplices, each with k facets, with each facet having at most $\min(n_i)^{-1}|\pi(B)|$ integral points by Observation 2.2.18, each of which can in turn contribute $2n_1 + 2n_1$ to the left hand side.
- We estimate the contribution of $x \in V_\pi \cup E$ that lie in the interior of at most one simplex in $\mathscr{T}^+$ and at most one simplex in $\mathscr{T}^-$. There are at most $|E| + |V_\pi|$ of these points, each of which can in turn contribute $2n_1 + 2n_1$ to the left hand side.
- Finally, each remaining x lies in a unique simplex of $\widetilde{T}^+ \in \mathscr{T}^+$ and $\widetilde{T}^- \in \mathscr{T}^-$, and $x \notin V_\pi \cup E$. For such x, we have $|(\mathrm{co}(A_\star) \setminus A_\star) \cap R_x| = \lfloor g^+_{T^+}(x) \rfloor + \lfloor g^-_{T^-}(x) \rfloor$. This discrepancy with $g^+_{T^+}(x) + g^-_{T^-}(x)$ is crudely bounded by $2|\pi(B)|$.

Combining these errors, and noting that $|E|$ and $|V_\pi|$ are bounded by (2.55) and (2.56), we conclude by choosing $h_{10}(\delta)$ so that

$$4k\binom{|V_\pi|}{k} \min(n_i)^{-1}|B| + (|E| + |V_\pi|)4n_1 + 2|\pi(B)| \leqslant h_{10}(\delta)|B|.$$

□

Recall in Definition 2.3.23, we introduced for a simplex $\widetilde{T} \subset \{0\} \times \mathbb{R}^{k-1}$ with integral vertices and a subset $\mathscr{W} \subset \{0\} \times \mathbb{Z}^{k-1}$ the notation $Y_{\mathscr{W}}(x) = ((x + \mathscr{W}) \cap T) \cup V(T)$. We now define a restricted infimum convolution with respect to $\mathscr{W}$.

Definition 2.3.40. Given a simplex $\widetilde{T} \subset \{0\} \times \mathbb{R}^{k-1}$ with integral vertices, a subset $\mathscr{W} \subset \{0\} \times \mathbb{Z}^{k-1}$, and a function $g : T \to \mathbb{R}_{\geqslant 0}$, we define the restricted infimum convolution $g^{\square}_{\mathscr{W}} : T + T \to \mathbb{R}_{\geqslant 0}$ by

$$g^{\square}_{\mathscr{W}}(x) = \begin{cases} \min\{g(x_1) + g(x_2)\} & \text{over all } x_1 + x_2 = x \text{ with } x_1 \in T^\circ \text{ and } x_2 \in Y_{\mathscr{W}}(x_1) \\ 0 & \text{no such } x_1, x_2 \text{ exist.} \end{cases}$$

Notation 2.3.41. For $g = g^*_{T^*}$ for some $* \in \{+,-\}$, we will always take $\mathscr{W} = \mathscr{W}_T$ as defined in Definition 2.3.34 in the infimum convolution $g^{*\square}_{T^*,\mathscr{W}}$. We will always omit the subscript $\mathscr{W}$, writing $g^{*\square}_{T^*}$ instead.

Proposition 2.3.42. Provided $\min(n_i)$ is sufficiently large in terms of δ (which we assume wlog by Convention 2.2.14), we have the following estimate for some function $h_{11}(\delta) \to 0$ as $\delta \to 0$ that

$$\Big|\mathrm{co}(A_\star + A_\star) \setminus (A_\star + A_\star)\Big| - \sum_{* \in \{+,-\}} \sum_{\widetilde{T}^* \in \mathscr{T}^*} \sum_{x' \in (T^* + T^*)} g^{*\square}_{T^*}(x') \leqslant h_{11}(\delta)|B|.$$

Proof. Writing $|\mathrm{co}(A_\star + A_\star) \setminus (A_\star + A_\star)| = \sum_{x' \in \mathrm{co}(\pi(A_\star + A_\star))} |(\mathrm{co}(A_\star + A_\star) \setminus (A_\star + A_\star)) \cap \pi^{-1}(x')|$, we upper bound the contribution for each $x' \in \mathrm{co}(\pi(A_\star + A_\star))$, which we express uniquely as $x' = x + x + \vec{v}$ with $\vec{v} \in \{0\} \times \{0,1\}^{k-1}, x \in \{0\} \times \mathbb{Z}^{k-1}$.

Claim 2.3.43. Fix $x' = x+x+\vec{v}$ as above. Suppose for each $* \in \{+,-\}$ there exists $\widetilde{T}^* \in \mathscr{T}^*$ with $x, x+\vec{v} \in T^{*\circ} \setminus (V_\pi \cup E)$. Then

$$\left|(\mathrm{co}(A_\star+A_\star)\setminus(A_\star+A_\star))\cap\pi^{-1}(x')\right| \leqslant g_{T^+}^{+\square}(x') + g_{T^-}^{-\square}(x').$$

Proof. Note that we can write $x' = x+(x+\vec{v})$, with $x \in T^{*\circ}$ and $x+\vec{v} \in Y_{\mathscr{W}_{T^*}}(x)$ since $\vec{v} \in \mathscr{W}_{T^*}$. Hence by Definition 2.3.40 there exists $x_1^* \in T^{*\circ}$ and $x_2^* \in Y_{\mathscr{W}_{T^*}}(x_1^*)$ so that $x_1^*+x_2^* = x'$ and $g_{T^*}^{*\square}(x') = g_{T^*}^*(x_1^*) + g_{T^*}^*(x_2^*)$. If either x_1^* or x_2^* are in $(V_\pi \cup E)\setminus V(T^*)$, the corresponding term on the right hand side is at least $2n_1$ and the inequality is trivially true. Hence, we may assume $x_1^*, x_2^* \notin (V_\pi \cup E)\setminus V(T^*)$. In particular, we have $x_1^* \in T^{*\circ}\setminus V_\pi$ and both $x_1^*, x_2^* \notin E$.

Let $\Psi^+_{A_\star+A_\star}, \Psi^-_{A_\star+A_\star} : \widetilde{\mathrm{co}}(\pi(A_\star+A_\star)) \to \mathbb{R}$ be the upper and lower convex hull function on $A_\star+A_\star$ in the e_1-direction respectively. Because $\widetilde{\mathrm{co}}(A_*)+\widetilde{\mathrm{co}}(A_*) = 2\widetilde{\mathrm{co}}(A_*)$, we can write $\Psi^+_{A_\star+A_\star}(2x) = 2\Psi^+_{A_\star}(x)$ for all $x \in 0\times\mathbb{R}^{k-1}$. Moreover, because $\widetilde{T}^+ \in \mathscr{T}^+$ is a simplex in the triangulation of upper convex hull of $\widetilde{\mathrm{co}}(A_\star)$ projected under π to $0\times\mathbb{R}^{k-1}$, the restricted function $\Psi^+_{A_\star} : \widetilde{T} \to \mathbb{R}$ is an affine linear function. Hence, for $x_1^+, x_2^+ \in \widetilde{T}$ we have $\Psi^+_{A_\star+A_\star}(x_1^+ + x_2^+) = 2\Psi^+_{A_\star}(2^{-1}(x_1^+ + x_2^+)) = \Psi^+_{A_\star}(x_1^+) + \Psi^+_{A_\star}(x_2^+)$. Therefore, we have

$$\begin{aligned}\Psi^+_{A_\star+A_\star}(x') - \max(R_{x_1^+}+R_{x_2^+}) &= \Psi_{A_\star}(x_1^+) + \Psi_{A_\star}(x_2^+) - \max R_{x_1^+} - \max R_{x_2^+}\\ &= g^+_{T^+}(x_1^+) + g^+_{T^+}(x_2^+) = g^{+\square}_{T^+}(x')\end{aligned}$$

and

$$\begin{aligned}\min(R_{x_1^-}+R_{x_2^-}) - \Psi^-_{A_\star+A_\star}(x') &= \min R_{x_1^-} + \min R_{x_2^-} - \Psi_{A_\star}(x_1^-) - \Psi_{A_\star}(x_2^-)\\ &= g^-_{T^-}(x_1^-) + g^-_{T^-}(x_2^-) = g^{-\square}_{T^-}(x').\end{aligned}$$

By Observation 2.3.35, as $x_1^* \in T^\circ \setminus V_\pi$, $x_2^* \in Y_{\mathscr{W}_{T^*}}(x_1^*)$, and $x_1^*, x_2^*, x, x+\vec{v} \notin E$, the intervals $R_{x_1^+}+R_{x_2^+}$ and $R_{x_1^-}+R_{x_2^-}$ both overlap $R_x + R_{x+\vec{v}}$, so

$$I_{x'} := (R_{x_1^-}+R_{x_2^-}) \cup (R_x + R_{x+\vec{v}}) \cup (R_{x_1^+}+R_{x_2^+})$$

is an interval. Therefore

$$\begin{aligned}&\left|(\mathrm{co}(A_\star+A_\star)\setminus(A_\star+A_\star))\cap\pi^{-1}(x')\right|\\ \leqslant&\left|(\mathrm{co}(A_\star+A_\star)\cap\pi^{-1}(x'))\setminus I_{x'}\right|\\ \leqslant&\lfloor\Psi^+_{A_\star+A_\star}(x')\rfloor - \max(R_{x_1^+}+R_{x_2^+}) + \min(R_{x_1^-}+R_{x_2^-}) - \lceil\Psi^-_{A_\star+A_\star}(x')\rceil\\ \leqslant&g^{+\square}_{T^+}(x') + g^{-\square}_{T^-}(x').\end{aligned}$$

□

Returning to the proof of Proposition 2.3.42, we have the following estimates. Recall that we uniquely write $x' = x + x + \vec{v}$ with $x' \in \mathrm{co}(\pi(A_\star + A_\star))$, $x \in \{0\} \times \mathbb{Z}^{k-1}$ and $\vec{v} \in \{0\} \times \{0,1\}^{k-1}$.

- We estimate the contribution of $x, \vec{v}$ such that for some $* \in \{+,-\}$, there is no $\widetilde{T}^* \in \mathscr{T}^*$ with $x, x+\vec{v} \in T^{*\circ}$. For the simplex $\widetilde{T}^*$ containing $\frac{1}{2}(x+(x+\vec{v})) = \frac{1}{2}x' \in \pi(\widetilde{\mathrm{co}}(A_\star))$, there is a hyperplane $\widetilde{H}$ containing a facet of $\widetilde{T}^*$ that separates (or contains one of) x and $x+\vec{v}$. For each $\vec{v}$, there are at most $\binom{|V_\pi|}{k}$ simplices, each with k facets, and each facet separating (or containing) at most 2 such $x, x+\vec{v}$ pairs on each of the at most $(k-1)\min(n_i)^{-1}|\pi(B)|$ $\vec{v}$-fibers of $\pi(B)$, and each such $x, \vec{v}$ contributes at most $2n_1$ to the left hand side.

- We now estimate the contribution for those $x, \vec{v}$ such that one of $x, x+\vec{v}$ lies in $V_\pi \cup E$. There are 2^{k-1} choices of v, and for each of these choices there are at most $2(|V_\pi| + |E|)$ such values of x, each of which contributes at most $2n_1$ to the right hand side.

- The remaining $x, \vec{v}$ have $x, x+\vec{v} \in T^{+o} \setminus (V_\pi \cup E), T^{-o} \setminus (V_\pi \cup E)$ for unique simplices $\widetilde{T}^+ \in \mathscr{T}^+$ and $\widetilde{T}^- \in \mathscr{T}^-$. But the above claim shows that the contribution of such x' is non-positive.

Combining these errors, and noting that $|E|$ and $|V_\pi|$ are bounded by (2.55) and (2.56), we conclude by taking $h_{11}(\delta)$ so that

$$2^{k+1}(k-1)\binom{|V_\pi|}{k}\min(n_i)^{-1}|B| + 2^{k+1}(|V_\pi| + |E|)n_1 \leqslant h_{11}(\delta)|B|.$$

□

Infimum convolution of functions

In this section we prove a general result about the infimum convolution (see Definition 2.3.40) of functions, related to the fact that small doubling implies being close to the convex hull.

Proposition 2.3.44. There exist constants $c_k', c_k'' > 0$ such that the following is true. Let $T \subset \pi(B)$ be a discrete simplex with integral vertices $x_0, \dots, x_{k-1} \in \{0\} \times \mathbb{Z}^{k-1}$, and let $g : T \to [0, 2n_1]$ with $g(x_i) = 0$ for all i. Then

$$\sum_{x' \in T+T} g^{\square}_{\mathscr{W}_T}(x') \leqslant (2^k - c_k') \sum_{x \in T} g(x) + c_k'' \min\{n_i\}^{-1}|B|$$

We omit the subscript $\mathscr{W}_T$ from now on. Throughout the entire proof we shall consider the sets $\mathscr{S}_{i,j} = \mathscr{S}_{i,j}(\widetilde{T})$ in Definition 2.3.32 with parameters i, j bounded above by μ_1, μ_2, respectively.

Definition 2.3.45. For a subset $\widetilde{P} \subset \widetilde{T}$, we define $g(\widetilde{P}) = \sum_{x \in \widetilde{P} \cap T} g(x)$, and for a subset $\widetilde{Q} \subset \widetilde{T} + \widetilde{T}$, define $g^{\square}(\widetilde{Q}) = \sum_{x \in \widetilde{Q} \cap (T+T)} g^{\square}(\widetilde{Q})$.

Definition 2.3.46. For a subset $\widetilde{P} \subset \widetilde{T}$, we define $d_k'(\widetilde{P}) = -g^{\square}(\widetilde{P} + \widetilde{P}) + 2^k g(\widetilde{P})$.

Let us give the high level overview of the proof Proposition 2.3.44. We want to show that

$$g(\widetilde{T}) \leqslant c_k'^{-1} d_k'(\widetilde{T}) + c_k'^{-1} c_k'' \min\{n_i\}^{-1}|B|.$$

The first idea is to control the weight of g in (a dense family of) translates $\widetilde{P}$ of $2^{-i}\widetilde{T}$. We show

$$g(\widetilde{P}) \leqslant 2^{-ik} g(\widetilde{T}) + O_i(1) d_k'(\widetilde{T}) + O_i(1) \min\{n_i\}^{-1}|B|.$$

Note that the factor in front of the dominant term one might initially guess is $2^{-i(k-1)}$ rather than 2^{-ik}, as the size of the domain over which we are summing is scaled by a factor of $2^{-i(k-1)}$ but the function on this domain is unchanged. The bigger i, the bigger this anomaly is.

The second idea is to produce an efficient covering of $\widetilde{T}$ with translates $\widetilde{P}_1,\dots,\widetilde{P}_s$ of $2^{-i}\widetilde{T}$. Of course, we always have $\sum_j |\widetilde{P}_j| \geqslant |\widetilde{T}|$. We show there exist a covering such that

$$\sum_j |\widetilde{P}_j| = O(1)|\widetilde{T}|$$

Note that the inefficiency $O(1)$ does not depend on the scale i and we get that the number of translates is $s = O(1)2^{i(k-1)}$.

Combining these two ideas we get

$$g(\widetilde{T}) \leqslant \sum_j g(\widetilde{P}_j) \leqslant O(1)2^{i(k-1)}(2^{-ik} g(\widetilde{T}) + O_i(1) d_k'(\widetilde{T}) + O_i(1)\min\{n_i\}^{-1}|B|).$$

Provided the scale 2^{-i} is chosen sufficiently small, we get

$$g(\widetilde{T}) \leqslant 2^{-1} g(\widetilde{T}) + O(1) d_k'(\widetilde{T}) + O(1)\min\{n_i\}^{-1}|B|,$$

from which the conclusion follows immediately.

Here we sketch the proof of the second idea. We may assume wlog $\widetilde{T} \subset \{0\}\times\mathbb{R}^{k-1}$ is a regular simplex. Let $\widetilde{C} \subset \{0\}\times\mathbb{R}^{k-1}$ be a cube with $|\widetilde{C}| = |\widetilde{T}|$. Construct a cover of $\widetilde{C}$ with $O(1)$ translates of $\widetilde{T}$. This gives a cover of $2^{-i}\widetilde{C}$ with $O(1)$ translates of $2^{-i}\widetilde{T}$. In addition, construct a cover of $\widetilde{T}$ with $(1+o(1))2^{i(k-1)}$ translates of $2^{-i}\widetilde{C}$ (which is possible since $\widetilde{C}$ is a cube with $|\widetilde{T}| = |\widetilde{C}|$). By combining the two covers, we obtain a cover of $\widetilde{T}$ with $O(1)2^{i(k-1)}$ translates of $2^{-i}\widetilde{T}$.

Here we sketch the proof of the first idea in case $k=2$. In this case $T = \{0\}\times[a,b]$ is an interval in $\{0\}\times\mathbb{Z}$. Assume for simplicity a and b are even such that $P = \{0\}\times[a,\frac{a+b}{2}]$ is a half size interval. In addition, assume for simplicity $\mathscr{W} = \{0\}\times\mathbb{Z}^{k-1}$.

Before we write the exact calculation let us give the intuition. Note that $g^{\square}$ is defined on the interval $T+T$ which can be regarded as two copies of the interval T. Because g is 0 at the endpoints of T, it turns out that on the second copy of T, $g^{\square}$ is bounded by g; on the first copy of T, $g^{\square}$ is bounded by a scaled version (by a factor of 2) of the restriction of g on P (here the scaling is in all

directions). Thus, we expect $g^{\square}(T+T) \leqslant g(T)+4g(P)$ which is what we want. More precisely, by definition, we have

$$\begin{aligned}
\sum_{x\in\{0\}\times[2a,2b]} g^{\square}(x) &= \sum_{x\in\{0\}\times[a,b]} g^{\square}(a+x) + \sum_{x\in\{0\}\times[\frac{a+b}{2}+1,b]} g^{\square}(2x) + \sum_{x\in\{0\}\times[\frac{a+b}{2},b-1]} g^{\square}(2x+1) \\
&\leqslant \sum_{x\in\{0\}\times[a,b]} g(a)+g(x) + \sum_{x\in\{0\}\times[\frac{a+b}{2}+1,b]} 2g(x) + \sum_{x\in\{0\}\times[\frac{a+b}{2},b-1]} g(x)+g(x+1) \\
&= \sum_{x\in\{0\}\times[a,b]} g(x) + \sum_{x\in\{0\}\times[\frac{a+b}{2}+1,b]} 2g(x) + \sum_{x\in\{0\}\times[\frac{a+b}{2},b-1]} g(x)+g(x+1) \\
&= \sum_{x\in\{0\}\times[a,b]} g(x) + 4\sum_{x\in\{0\}\times[\frac{a+b}{2}+1,b]} g(x) + g(\frac{a+b}{2}) \\
&= \sum_{x\in\{0\}\times[a,b]} g(x) - 4\sum_{x\in\{0\}\times[a,\frac{a+b}{2}]} g(x) + 4\sum_{x\in\{0\}\times[a,b]} g(x) + g(\frac{a+b}{2}).
\end{aligned}$$

After rearranging we get

$$4\sum_{x\in\{0\}\times[a,\frac{a+b}{2}]} g(x) \leqslant \sum_{x\in\{0\}\times[a,b]} g(x) + 4\sum_{x\in\{0\}\times[a,b]} g(x) - \sum_{x\in\{0\}\times[2a,2b]} g^{\square}(x) + g(\frac{a+b}{2})$$

which is equivalent to

$$4g(\widetilde{P}) \leqslant g(\widetilde{T}) + d'_k(\widetilde{T}) + 2n_1.$$

Observation 2.3.47. For a polytope $\widetilde{P} \subset \widetilde{T}$, we have $d'_k(\widetilde{P}) \geqslant -2^{2k+5}\min\{n_i\}^{-1}|B|$. In particular, with $P = \widetilde{P} \cap (\{0\}\times\mathbb{Z}^{k-1})$ we have $d'_k(P) \geqslant -2^{2k+5}\min\{n_i\}^{-1}|B|$. More generally, the same conclusion holds for any region $\widetilde{P} \subset \widetilde{T}$ defined as the intersection of open and closed half-spaces.

Proof. Note that if $\widetilde{P} \subset \widetilde{T}$ is defined as the intersection of open and closed half-spaces, then we can perturb the open half-spaces to closed ones without changing the lattice points in $\widetilde{P}$ or $\widetilde{P}+\widetilde{P}$, so we may assume that $\widetilde{P}$ is a polytope. Recall that $\{0\}\times\{0,1\}^{k-1} \subset \mathscr{W}_T$. Note that if we write $z \in (\widetilde{P}+\widetilde{P}) \cap T = 2\widetilde{P} \cap T$ as $z = x + (x+\vec{v})$ with $\vec{v} \in \{0\}\times\{0,1\}^{k-1}$, then as $\widetilde{P}$ is convex, either $x, x+\vec{v} \in P^{\circ}$ or the segment $[x, x+\vec{v}]$ intersects $\partial\widetilde{P}$. Therefore with $\vec{v}$ ranging over $\{0\}\times\{0,1\}^{k-1}$

we have

$$\begin{aligned}
d'_k(\widetilde{P}) &= 2^k g(\widetilde{P}) - g^{\square}(\widetilde{P}+\widetilde{P}) \\
&\geqslant 2^k \sum_{x\in P} g(x) - \sum_{\vec{v}} \sum_{x,x+\vec{v}\in P^\circ} g^{\square}(x+x+\vec{v}) - \sum_{\vec{v}} \sum_{[x,x+\vec{v}]\cap\partial\widetilde{P}\neq\emptyset} g^{\square}(x+x+\vec{v}) \\
&\geqslant 2^k \sum_{x\in P} g(x) - \sum_{\vec{v}} \sum_{x,x+\vec{v}\in P^\circ} g(x)+g(x+\vec{v}) - \sum_{\vec{v}} \sum_{[x,x+\vec{v}]\cap\partial\widetilde{P}\neq\emptyset} 4n_1 \qquad (2.61)\\
&\geqslant -\sum_{\vec{v}} \sum_{[x,x+\vec{v}]\cap\partial\widetilde{P}\neq\emptyset} 4n_1 \\
&\geqslant -8n_1 \sum_{\vec{v}} |(\widetilde{P}\triangle(\widetilde{P}-\vec{v}))\cap\{0\}\times\mathbb{Z}^{k-1}| \\
&\geqslant -8n_1 \sum_{\vec{v}} |(\widetilde{P}\triangle(\widetilde{P}-\vec{v}))| - 8n_1 2^{k-1}\cdot 3\cdot 2(k-1)k\min\{n_i\}^{-1}|\pi(B)| \qquad (2.62)\\
&\geqslant -8n_1 \sum_{\vec{v}} |\vec{v}||\partial\widetilde{P}| - 3\cdot 2^{k+3}(k-1)k\min\{n_i\}^{-1}|B| \\
&\geqslant -8n_1 2^{k-1}\sqrt{k-1}\cdot 2(k-1)\min\{n_i\}^{-1}|\pi(B)| - 3\cdot 2^{k+3}(k-1)k\min\{n_i\}^{-1}|B| \qquad (2.63)\\
&\geqslant -2^{2k+5}\min\{n_i\}^{-1}|B|.
\end{aligned}$$

In (2.61) we used the fact that $\vec{v}\in\mathscr{W}_T$ so $x+\vec{v}\in Y_{\mathscr{W}_T}(x)$, and $g^{\square}\leqslant 2\max g\leqslant 4n_1$. In (2.62) we have used Observation 2.2.24 to upper bound

$$\left|(\widetilde{P}\triangle(\widetilde{P}-\vec{v}))\cap\{0\}\times\mathbb{Z}^{k-1}\right| = \left|\widetilde{P}\cap\{0\}\times\mathbb{Z}^{k-1}\right| + \left|(\widetilde{P}-\vec{v})\cap\{0\}\times\mathbb{Z}^{k-1}\right| - \left|(\widetilde{P}\cap(\widetilde{P}-\vec{v}))\cap\{0\}\times\mathbb{Z}^{k-1}\right|,$$

and in (2.63) the fact that $|\partial\widetilde{P}|\leqslant|\widetilde{\partial\pi(B)}|\leqslant 2(k-1)\min\{n_i\}^{-1}|\pi(B)|$. □

Lemma 2.3.48. For a vertex $x\in V(\widetilde{T})$ and simplices $\widetilde{S}=(1-2^{-i})x+2^{-i}\widetilde{T}\in\mathscr{S}_{i,0}$ and $\widetilde{S}'=\frac{1}{2}(x+\widetilde{S})\in\mathscr{S}_{i+1,0}$, we have

$$g(S')\leqslant 2^{-k}g(S)+2^{-k}d'_k(T)+2^{k+6}\min\{n_i\}^{-1}|B|$$

Proof. By Observation 2.3.47 we have (letting $\widetilde{S}'^c$ be the complement of $\widetilde{S}'$ inside $\widetilde{T}$) that

$$\begin{aligned}
d'_k(T)=d'_k(\widetilde{T}) &= 2^k g(\widetilde{T}) - g^{\square}(\widetilde{T}+\widetilde{T}) \\
&\geqslant 2^k g(\widetilde{S}') - g^{\square}(\widetilde{S}'+\widetilde{S}') + 2^k g(\widetilde{S}'^c) - g^{\square}(\widetilde{S}'^c+\widetilde{S}'^c) \\
&= 2^k g(\widetilde{S}') - g^{\square}(\widetilde{S}'+\widetilde{S}') + d'_k(\widetilde{S}'^c) \\
&\geqslant 2^k g(\widetilde{S}') - g^{\square}(\widetilde{S}'+\widetilde{S}') - 2^{2k+5}\min\{n_i\}^{-1}|B| \\
&\geqslant 2^k g(\widetilde{S}') - g^{\square}(\widetilde{S}^\circ+x) - g^{\square}(\partial\widetilde{S}+x) - 2^{2k+5}\min\{n_i\}^{-1}|B| \\
&\geqslant 2^k g(\widetilde{S}') - g(\widetilde{S}^\circ) - k(4n_1)\min\{n_i\}^{-1}|\pi(B)| - 2^{2k+5}\min\{n_i\}^{-1}|B| \qquad (2.64)\\
&\geqslant 2^k g(S') - g(S) - 2^{2k+6}\min\{n_i\}^{-1}|B|.
\end{aligned}$$

where in (2.64) we used that $x \in Y_{\mathscr{W}_T}(y)$ for all $y \in \widetilde{T}^\circ$, Observation 2.2.18 to estimate the number of lattice points on each facet of $\widetilde{S}$, and the fact that $\max g^\square \leqslant 4n_1$.

□

Corollary 2.3.49. For $\widetilde{S}' \in \mathscr{S}_{i,0}$, we have

$$\begin{aligned} g(S') &\leqslant 2^{-ik} g(T) + \max\left(0, \frac{1}{2^k-1} d'_k(T)\right) + \frac{2^k}{2^k-1} 2^{k+6} \min\{n_i\}^{-1} |B| \\ &\leqslant 2^{-ik} g(T) + \frac{1}{2^k-1} d'_k(T) + 2^{2k+6} \min\{n_i\}^{-1} |B| \end{aligned}$$

Lemma 2.3.50. Let $i \leqslant \mu_1$ and $j \leqslant \mu_2 - 1$. For $\widetilde{S}'_1, \widetilde{S}'_2 \in \mathscr{S}_{i,j}$ and $\widetilde{S}'' = \frac{1}{2}(\widetilde{S}'_1 + \widetilde{S}'_2) \in \mathscr{S}_{i,j+1}$, we have

$$g(S'') \leqslant \frac{1}{2}(g(S'_1) + g(S'_2)) + 2^{-k} d'_k(T) + 2^{5k} \min\{n_i\}^{-1} |B|$$

Proof. Let $\vec{u}$ be the vector such that $\widetilde{S}'_1 + \vec{u} = \widetilde{S}'_2$. Then by Definition 2.3.34, we have $\mathscr{R}(\vec{u}) = \lfloor \vec{u} \rfloor + \{0\} \times \{0,1\}^{k-1} \subset \mathscr{W}_T$. Let

$$\widetilde{S}''^c = \widetilde{P}_1 \sqcup \cdots \sqcup \widetilde{P}_k$$

be a partition into convex regions, the intersection of open and closed half-spaces. Indeed this can be obtained by taking the defining equations $x \cdot \vec{c}_i \leqslant \vec{b}_i$ for $1 \leqslant i \leqslant k$ of $\widetilde{S}''$, and defining $\widetilde{P}_j$ by setting $x \cdot \vec{c}_i \leqslant \vec{b}_i$ for $1 \leqslant i < j$ and $x \cdot \vec{w}_j > \vec{b}_j$ inside $\widetilde{T}$. Hence, by Observation 2.3.47 we find

$$\begin{aligned} d'_k(T) \geqslant d'_k(\widetilde{T}) = 2^k g(\widetilde{T}) - g^\square(\widetilde{T} + \widetilde{T}) &= 2^k g(\widetilde{S}'') - g^\square(\widetilde{S}'' + \widetilde{S}'') + \sum_j 2^k g(\widetilde{P}_j) - g^\square(\widetilde{P}_j + \widetilde{P}_j) \\ &= 2^k g(\widetilde{S}'') - g^\square(\widetilde{S}'_1 + \widetilde{S}'_2) + \sum_j d'_k(\widetilde{P}_j) \\ &\geqslant 2^k g(\widetilde{S}'') - g^\square(\widetilde{S}'_1 + \widetilde{S}'_2) - k 2^{2k+5} \min\{n_i\}^{-1} |B|. \end{aligned}$$

Note that every point $x' \in (\widetilde{S}'_1 + \widetilde{S}'_2) \cap \{0\} \times \mathbb{Z}^{k-1}$ we can write uniquely as $x' = x + x + \vec{w}$ for some $\vec{w} \in \mathscr{R}(\vec{u})$ (this is true in fact for every $x' \in \{0\} \times \mathbb{Z}^{k-1}$), and for this $\vec{w}$ (in fact for any $\vec{w} \in \mathscr{R}(u)$) we have $\vec{w} - \vec{u} \in \{0\} \times (-1,1]^{k-1}$. We have $\vec{u} \in \{0\} \times \prod_{i=2}^k [-n_i + 1, n_i - 1]$ and $x' \in (\pi(\widetilde{B}) + \pi(\widetilde{B})) \cap \{0\} \times \mathbb{Z}^{k-1} = \{0\} \times \prod_{i=2}^k \{2, \ldots, 2n_i\}$, so

$$x = \left\lfloor \frac{x' - \lfloor \vec{u} \rfloor}{2} \right\rfloor \in B' := \{0\} \times \prod_{i=2}^k \left\{ \left\lfloor \frac{3 - n_i}{2} \right\rfloor, \ldots, \left\lfloor \frac{3n_i - 1}{2} \right\rfloor \right\},$$

where B' is a translate of $B(2n_2 - 1, \ldots, 2n_k - 1)$. Also, as the midpoint $\frac{1}{2}(x + (x + (\vec{w} - \vec{u})))$ lies in $\widetilde{S}'_1$, either $x \in \widetilde{S}'_1$ and $x + (\vec{w} - \vec{u}) \in \widetilde{S}'_1$ (equivalently $x + \vec{w} \in \widetilde{S}'_2$), or x and $x + (\vec{w} - \vec{u})$ are separated by some hyperplane $\widetilde{H}$ containing one of the k facets of $\widetilde{S}'_1$.

Given $\vec{w} \in \mathscr{R}(\vec{u})$ and a hyperplane $\widetilde{H}$, there are at most $2^{2k} \min\{n_i\}^{-1} |\pi(B)|$ many choices of $x \in B'$ with $x, x + (\vec{w} - \vec{u})$ separated by $\widetilde{H}$. Indeed, set $\widetilde{G}_{\vec{w},\widetilde{H}}$ to be the convex region of the box $\widetilde{B}'$

between the hyperplanes $\widetilde{H}$ and $\widetilde{H}-(\vec{w}-\vec{u})$. Note that

$$|\widetilde{G}_{\vec{w},\widetilde{H}}| \leqslant |\vec{w}-\vec{u}|\cdot|\partial\widetilde{B}'| \leqslant (k-1)^{\frac{1}{2}}2(k-1)2^{k-2}\min\{n_i\}^{-1}|\pi(B)| \leqslant 2^{2k-1}\min\{n_i\}^{-1}|\pi(B)|.$$

By Observation 2.2.24 applied to B',

$$|\widetilde{G}_{\vec{w},\widetilde{H}}\cap(\{0\}\times\mathbb{Z}^{k-1})| \leqslant (2^{2k-1}+2(k-1)k2^{k-2})\min\{n_i\}^{-1}|\pi(B)| \leqslant 2^{2k}\min\{n_i\}^{-1}|\pi(B)|. \quad (2.65)$$

From the above discussion, if $x+x+\vec{w}\in\widetilde{S}_1'+\widetilde{S}_2'$, and either $x\notin\widetilde{S}_1'$ or $x+\vec{w}\notin\widetilde{S}_2'$, then $x\in\widetilde{G}_{\vec{w},\widetilde{H}}$ for some $\widetilde{H}$ containing a facet of $\widetilde{S}_1'{}'$. Hence from (2.65) (taking $\vec{w}\in\mathscr{R}(\vec{u})$ and $x\in\{0\}\times\mathbb{Z}^{k-1}$) we deduce

$$\sum_{\vec{w}}\sum_{\substack{x+x+\vec{w}\in\widetilde{S}_1'+\widetilde{S}_2'\\ x\notin\widetilde{S}_1' \text{ or } x+\vec{w}\notin\widetilde{S}_2'}} g^{\square}(x+x+\vec{w}) \leqslant 2^{k-1}k2^{2k}\min\{n_i\}^{-1}|\pi(B)|\max g^{\square} \leqslant k2^{3k+1}\min\{n_i\}^{-1}|B|.$$

Also, as $\mathscr{R}(u)\subset\mathscr{W}_T$ and $\max g^{\square}\leqslant 4n_1$, we have

$$\begin{aligned}\sum_{\vec{w}}\sum_{\substack{x+x+\vec{w}\in\widetilde{S}_1'+\widetilde{S}_2'\\ x\in\widetilde{S}_1' \text{ and } x+\vec{w}\in\widetilde{S}_2'}} g^{\square}(x+x+\vec{w}) &\leqslant \sum_{\vec{w}}\sum_{\substack{x+x+\vec{w}\in\widetilde{S}_1'+\widetilde{S}_2'\\ x\in(\widetilde{S}_1')^{\circ} \text{ and } x+\vec{w}\in\widetilde{S}_2'}}(g(x)+g(x+\vec{w}))+\sum_{x\in\partial\widetilde{S}_1'}4n_1\\ &\leqslant 2^{k-1}(g(\widetilde{S}_1')+g(\widetilde{S}_2'))+4k\min\{n_i\}^{-1}|B|, \quad (2.66)\end{aligned}$$

where in (2.66) we used Observation 2.2.18 on each of the facets of $\widetilde{S}_1'$. Putting it all together,

$$\begin{aligned}d_k'(T)\geqslant d_k'(\widetilde{T}) \geqslant& 2^k g(\widetilde{S}'')-k2^{2k+5}\min\{n_i\}^{-1}|B|\\ &-\sum_{\substack{x+x+\vec{w}\in\widetilde{S}_1'+\widetilde{S}_2'\\ x\in\widetilde{S}_1' \text{ and } x+\vec{w}\in\widetilde{S}_2'}} g^{\square}(x+x+\vec{w})-\sum_{\substack{x+x+\vec{w}\in\widetilde{S}_1'+\widetilde{S}_2'\\ x\notin\widetilde{S}_1' \text{ or } x+\vec{w}\notin\widetilde{S}_2'}} g^{\square}(x+x+\vec{w})\\ \geqslant& 2^k g(\widetilde{S}'')-2^{k-1}(g(\widetilde{S}_1')+g(\widetilde{S}_2'))-(4k+k2^{2k+5}+k2^{3k+1})\min\{n_i\}^{-1}|B|\\ \geqslant& 2^k g(S'')-2^{k-1}(g(S_1')+g(S_2'))-2^{6k}\min\{n_i\}^{-1}|B|.\end{aligned}$$

□

Corollary 2.3.51. For $\widetilde{S}'\in\mathscr{S}_{\mu_1,\mu_2}$ we have

$$g(S')\leqslant 2^{-\mu_1 k}g(T)+\left(\frac{1}{2^k-1}+\mu_2 2^{-k}\right)d_k'(T)+(2^{2k+6}+\mu_2 2^{5k})\min\{n_i\}^{-1}|B|$$

Finally, before we prove Proposition 2.3.44, we prove the following result Proposition 2.3.52 which as mentioned before constructs the constants μ_1,μ_2.

Proposition 2.3.52. Let $\widetilde{T}$ be a simplex in $\mathbb{R}^{k-1}$. There exist $\mu_1=\mu_1(k)$ and $\mu_2=\mu_2(k)$ and a family $\mathscr{F}\subset\mathscr{S}_{\mu_1,\mu_2}(\widetilde{T})$ such that $\widetilde{T}\subset\bigcup_{\widetilde{S}\in\mathscr{F}}\widetilde{S}$ and $\sum_{\widetilde{S}\in\mathscr{F}}|\widetilde{S}|\leqslant 2^{\mu_1-1}|\widetilde{T}|$, i.e., $|\mathscr{F}|\leqslant 2^{\mu_1 k-1}$.

Proof. Without loss of generality assume $\widetilde{T}$ is regular of volume 1 centered at the origin. Extend a finite covering of $[0,1]^{k-1}$ with q_k translates of $\widetilde{T}$ to a periodic covering $\mathscr{C}$ of $\mathbb{R}^{k-1}$ with average density q_k. We mention without proof that the following stronger statement holds. For any $q_k' > q_k$ there exist μ_1, μ_2 depending on q_k' and there exists a family $\mathscr{F} \subset \mathscr{S}_{\mu_1,\mu_2}(\widetilde{T})$ such that $\widetilde{T} \subset \bigcup_{\widetilde{S}\in\mathscr{F}} \widetilde{S}$ and $\sum_{\widetilde{S}\in\mathscr{F}} |\widetilde{S}| \leqslant q_k'|\widetilde{T}|$. Returning to the original statement, we fix $\mu_1(k) := \lceil \log_2(q_k) \rceil + 2k - 1$ (which, in particular, satisfies $2^{\mu_1-1} > q_k$).

We have that $2^{-\mu_1-1}\mathscr{C}$ is a periodic covering of $\mathbb{R}^{k-1}$ by translates of $2^{-\mu_1-1}\widetilde{T}$ with average density q_k, so for any polytope $\widetilde{P}$ there exists a $\vec{u}$ with $\sum_{\widetilde{S}\in\vec{u}+2^{-\mu_1+1}\mathscr{C}} |\widetilde{S}\cap\widetilde{P}| \leqslant q_k|\widetilde{P}|$. Take $\widetilde{P} = 2\widetilde{T}$, and let $\mathscr{C}' \subset \vec{u} + 2^{-\mu_1-1}\mathscr{C}$ be the set of simplices which intersect $\widetilde{T}$, so that $\widetilde{T} \subset \bigcup_{\widetilde{S}\in\mathscr{C}'} \widetilde{S}$. Each $\widetilde{S} \subset \mathscr{C}'$ is contained in $\widetilde{T} + 2^{-\mu_1-1}\widetilde{T} - 2^{-\mu_1-1}\widetilde{T} \subset \widetilde{T} + 2^{-\mu_1-1}\widetilde{T} + 2^{-\mu_1-1}(k-1)\widetilde{T} \subset 2\widetilde{T}$, so

$$\sum_{\widetilde{S}\in\mathscr{C}'} |\widetilde{S}| = \sum_{\widetilde{S}\in\mathscr{C}'} |\widetilde{S}\cap 2\widetilde{T}| \leqslant q_k|2\widetilde{T}| = 2^{k-1}q_k.$$

For each $\widetilde{S} \in \mathscr{C}'$, there exists a translate $f(\widetilde{S})$ of $2^{-\mu_1-1}\widetilde{T}$ such that $\widetilde{S}\cap\widetilde{T} \subset f(\widetilde{S}) \subset \widetilde{T}$, and we construct $\mathscr{C}'' := \{f(S) : \widetilde{S} \in \mathscr{C}'\}$. Then $\sum_{\widetilde{S}'\in\mathscr{C}''} |\widetilde{S}'| \leqslant 2^{k-1}q_k$, all simplices in $\mathscr{C}''$ are contained in $\widetilde{T}$, and $\widetilde{T} \subset \bigcup_{\widetilde{S}'\in\mathscr{C}''} \widetilde{S}'$.

The collection $\cup_{j\geqslant 0}\mathscr{S}_{\mu_1,j}(\widetilde{T})$ is a dense collection of translates of $2^{-\mu_1}\widetilde{T}$ contained inside $\widetilde{T}$, and in fact for every (possibly lower dimensional) face $\widetilde{F}$ of $\widetilde{T}$, the sub-collection of simplices in $\cup_{j\geqslant 0}\mathscr{S}_{\mu_1,j}(\widetilde{T})$ intersecting $\widetilde{F}$ is dense among all translates of $2^{-\mu_1-1}\widetilde{T}$ contained in $\widetilde{T}$ which intersect $\widetilde{F}$. Therefore for each element $\widetilde{S} \in \mathscr{C}''$, there exist a translate $h(\widetilde{S}) \in \cup_{j\geqslant 0}\mathscr{S}_{\mu_1,j}(\widetilde{T}')$ which contains $\widetilde{S}$. Finally, we can construct the family $\mathscr{F} := \{h(\widetilde{S}) : \widetilde{S} \in \mathscr{C}''\}$. As $\mathscr{C}''$ is a fixed finite set, there exist $\mu_2 = \mu_2(k)$ such that $\mathscr{F} \subset \mathscr{S}_{\mu_1,\mu_2}(\widetilde{T}')$. Hence, $\sum_{\widetilde{S}\in\mathscr{F}} |\widetilde{S}| \leqslant 2^{2k-2}q_k \leqslant 2^{\mu_1-1}$ as desired. □

Proof of Proposition 2.3.44. Recall by Proposition 2.3.52 we find a family $\mathscr{F} \subset \mathscr{S}_{\mu_1,\mu_2}$ such that $\widetilde{T} \subset \bigcup_{\widetilde{S}\in\mathscr{F}} \widetilde{S}$ and $\sum_{\widetilde{S}\in\mathscr{F}} |\widetilde{S}| \leqslant 2^{\mu_1-1}|\widetilde{T}|$, i.e., $|\mathscr{F}| \leqslant 2^{\mu_1 k-1}$. By Corollary 2.3.51, we conclude that

$$\begin{aligned} g(T) &\leqslant \sum_{\widetilde{S}\in\mathscr{F}} g(S) \\ &\leqslant \sum_{\widetilde{S}\in\mathscr{F}} \left[2^{-\mu_1 k} g(T) + \left(\frac{1}{2^k-1} + \mu_2 2^{-k}\right) d_k'(T) + (2^{2k+6} + \mu_2 2^{5k}) \min\{n_i\}^{-1}|B|\right] \\ &\leqslant 2^{\mu_1 k-1}\left[2^{-\mu_1 k} g(T) + \left(\frac{1}{2^k-1} + \mu_2 2^{-k}\right) d_k'(T) + (2^{2k+6} + \mu_2 2^{5k}) \min\{n_i\}^{-1}|B|\right] \end{aligned}$$

Hence as $d_k'(T) = -g^{\square}(T+T) + 2^k g(T)$, we have

$$g^{\square}(T+T) \leqslant \left(2^k - \frac{2^{-\mu_1 k}}{\frac{1}{2^k-1} + \mu_2 2^{-k}}\right) g(T) + \frac{2^{2k+6} + \mu_2 2^{5k}}{\frac{1}{2^k-1} + \mu_2 2^{-k}} \min\{n_i\}^{-1}|B|.$$

□

$A_\star$ is close to $\mathrm{co}(A_\star)$: Construction

In this section we prove that $|\mathrm{co}(A_\star)\setminus A_\star||B|^{-1}\to 0$ as $\delta\to 0$.

Proposition 2.3.53. Provided $\min(n_i)$ is sufficiently large in terms of δ (which we assume wlog by Convention 2.2.14), there exists a function $h_\star$ with $h_\star(\delta)\to 0$ as $\delta\to 0$ such that $|\mathrm{co}(A_\star)\setminus A_\star|\leqslant h_\star(\delta)|B|$.

Proof. Note that we have the identity

$$2^k|\mathrm{co}(A_\star)\setminus A_\star| = d_k(A_\star)-d_k(\mathrm{co}(A_\star))+\Big|\mathrm{co}(A_\star+A_\star)\setminus(A_\star+A_\star)\Big|.$$

By (2.54), we have

$$d_k(A_\star)\leqslant h_9(\delta)|B|.$$

By Observation 2.2.21, we have

$$d_k(\mathrm{co}(A_\star))\geqslant -2^{2k}\min(n_i)^{-1}|B|.$$

By Proposition 2.3.42, we have

$$\Big|\mathrm{co}(A_\star+A_\star)\setminus(A_\star+A_\star)\Big| - \sum_{*\in\{\pm\}}\sum_{T^*\in\mathscr{T}^*}\sum_{x\in(T^*+T^*)} g_{T^*}^{*\square}(x)\leqslant h_{11}(\delta)|B|.$$

Combining the above inequalities, we get

$$2^k|\mathrm{co}(A_\star)\setminus A_\star|\leqslant h_9(\delta)|B|+2^{2k}\min(n_i)^{-1}|B|+h_{11}(\delta)|B|+\sum_{*\in\{\pm\}}\sum_{T^*\in\mathscr{T}^*}\sum_{x\in(T^*+T^*)} g_{T^*}^{*\square}(x).$$

Because the number of vertices of $A_\star$ satisfies $|V(A_\star)|\leqslant H_7(\delta^{\frac{1}{20}-17c})$ (by (2.56)), we deduce that the number of simplices in any triangulation of $\partial\widetilde{\mathrm{co}(A_\star)}$ is at most $\binom{H_7(\delta^{\frac{1}{20}-17c})}{k}$ and hence $|\mathscr{T}^+|+|\mathscr{T}^-|\leqslant\binom{H_7(\delta^{\frac{1}{20}-17c})}{k}$. Hence, by applying Proposition 2.3.44 to each simplex in $\mathscr{T}^+\cup\mathscr{T}^-$, we deduce

$$\sum_{*\in\{\pm\}}\sum_{T^*\in\mathscr{T}^*}\sum_{x\in(T^*+T^*)} g_{T^*}^{*\square}(x)\leqslant\binom{H_7(\delta^{\frac{1}{20}-17c})}{k}c_k''\min(n_i)^{-1}|B|+(2^k-c_k')\sum_{*\in\{\pm\}}\sum_{T^*\in\mathscr{T}^*}\sum_{x\in(T^*)} g_{T^*}^{*}(x).$$

By Observation 2.3.39, we have

$$\Bigg(\sum_{*\in\{+,-\}}\sum_{\widetilde{T}^*\in\mathscr{T}^*}\sum_{x\in T^*} g_{T^*}^{*}(x)\Bigg)-|\mathrm{co}(A_\star)\setminus A_\star|\leqslant h_{10}(\delta)|B|.$$

Combining the last three inequalities we get

$$2^k|\operatorname{co}(A_\star)\setminus A_\star| \leqslant h_9(\delta)|B|+2^{2k}\min(n_i)^{-1}|B|+h_{11}(\delta)|B|+\binom{H_7(\delta^{\frac{1}{20}-17c})}{k}c_k''\min(n_i)^{-1}|B|+(2^k-c_k')h_{10}(\delta).$$

We conclude that $|\operatorname{co}(A_\star)\setminus A_\star| \leqslant h_\star(\delta)|B|$, for a function $h_\star \to 0$ as $\delta \to 0$. □

2.3.9 A is close to $\operatorname{co}(A)$

Recall that we have $d_k(A) \leqslant \delta|B|, |A| \geqslant \varepsilon_0|B|$, and for some functions $h_8, h_9, h_\star \to 0$ as $\delta \to 0$ that

$$|\operatorname{co}(A_\star)\setminus A_\star| \leqslant h_\star(\delta)|B|, \quad |A\triangle A_\star| \leqslant h_8(\delta)|B|, \text{ and } \quad d_k(A_\star) \leqslant h_9(\delta)|B|.$$

Provided $\min(n_i)$ is sufficiently large in terms of δ (which we assume wlog by Convention 2.2.14), we will now show that $|\operatorname{co}(A)\setminus A| \leqslant h(\delta)|B|$ for some function $h \to 0$ as $\delta \to 0$.

Lemma 2.3.54. Given a polytope $\widetilde{Q}$ and $\lambda > 0$, let o be the barycenter of the largest volume simplex $\widetilde{T} \subset \widetilde{Q}$. Consider the dilation with respect to o. If $p \notin (1+\lambda)\widetilde{Q}$, then there is a subpolytope $\widetilde{P} \subset \widetilde{Q}$ with $|\widetilde{P}| = \left(\frac{\lambda}{2k^2}\right)^k |\widetilde{Q}|$ such that

$$\frac{\widetilde{P}+p}{2} \cap \widetilde{Q} = \emptyset.$$

Proof. We may assume $p \in \partial(1+\lambda)\widetilde{Q}$, and that $\widetilde{T}$ is regular with inradius 1, with o at the origin. Then $\widetilde{T} \subset \widetilde{Q} \subset -k\widetilde{T}$, and we estimate the diameter of $\widetilde{Q}$ is strictly less than $2k^2$, as this is an upper bound for the side length of $-k\widetilde{T}$ (as the distance from a vertex to o is k^2). Also, note that the distance s from o to $\partial\widetilde{Q}$ is at least 1, the inradius of $\widetilde{T}$. Let $q = op \cap \partial\widetilde{Q}$, let H be the homothety with center q and ratio $\frac{\lambda}{2k^2}$, and let $\widetilde{P} = H(\widetilde{Q})$. Clearly $\widetilde{P}$ has the desired volume, and has diameter strictly less than λ. Let H' be the homothety with center q and ratio $-\frac{\lambda}{2}$. Then $H'(\widetilde{Q})$ and $\widetilde{Q}$ are separated by the supporting hyperplane at the point q. It is enough to show that $\frac{\widetilde{P}+p}{2}$ is contained in the interior of $H'(\widetilde{Q})$.

Note that the distance from $H'(o)$ to $\partial H'(\widetilde{Q})$ is at least $\frac{\lambda}{2}$. As $\frac{\widetilde{P}+p}{2}$ is a set of diameter strictly less than $\frac{\lambda}{2}$ containing $H'(o) = \frac{p+q}{2}$, it is contained in the interior of $H'(\widetilde{Q})$ as desired. □

Proof of Theorem 2.1.4. Note that $|\widetilde{\operatorname{co}}(A_\star)| \geqslant \frac{\varepsilon_0}{2}|B|$ by Observation 2.2.24. Let $\widetilde{T} \subset \widetilde{\operatorname{co}}(A_\star)$ be the largest volume simplex, and let o be its barycenter. Consider the homothety H with center o and ratio $1+\lambda(\delta)$ where $\lambda(\delta)$ is a function which goes to 0 as $\delta \to 0$ which will be chosen later. Let $\widetilde{R} = H(\widetilde{\operatorname{co}}(A_\star))$. We show now that $A \subset \widetilde{R}$. Indeed, suppose not, let $x \in A \setminus \widetilde{R}$. Then by Lemma 2.3.54, there is a subset $\widetilde{P} \subset \widetilde{\operatorname{co}}(A_\star)$ with volume $(\frac{\lambda}{2k^2})^k|\widetilde{\operatorname{co}}(A_\star)|$ such that $\widetilde{P}+x$ is disjoint from $2\widetilde{\operatorname{co}}(A_\star)$.

Then, by Observation 2.2.23 and Observation 2.2.24,

$$\begin{aligned}
|A+A| &\geqslant \left|x+(\widetilde{P}\cap A_\star)\right| + \left|A_\star + A_\star\right| \\
&\geqslant \left|x+(\widetilde{P}\cap \operatorname{co}(A_\star))\right| - h_\star(\delta)|B| + 2^k|A_\star| - d_k(A_\star) \\
&\geqslant |\widetilde{P}\cap\mathbb{Z}^k| - h_\star(\delta)|B| + 2^k(|A| - h_8(\delta)|B|) - h_9(\delta)|B| \\
&\geqslant |\widetilde{P}| + 2^k|A| - h_{12}(\delta)|B| \\
&\geqslant \left(\frac{\lambda}{2k^2}\right)^k |\widetilde{\operatorname{co}}(A_\star)| + 2^k|A| - h_{12}(\delta)|B| \\
&\geqslant 2^k|A| + \left(\frac{\varepsilon_0}{2}\left(\frac{\lambda}{2k^2}\right)^k - h_{12}(\delta)\right)|B|.
\end{aligned}$$

where $h_{12}(\delta)$ is the function $h_{12}(\delta) = 2^k h_8(\delta) + h_\star(\delta) + 2k(k+1)\min(n_i)^{-1}$ which satisfies $h_{12}\to 0$ as $\delta\to 0$. Hence,

$$\delta|B| \geqslant d_k(A) \geqslant \left(\frac{\varepsilon_0}{2}\left(\frac{\lambda}{2k^2}\right)^k - h_{12}(\delta)\right)|B|.$$

Choosing $\lambda(\delta)$ such that $\frac{\varepsilon_0}{2}(\frac{\lambda}{2k^2})^k - h_{12}(\delta) > \delta$ for all sufficiently small δ yields the desired contradiction. Therefore $A\subset(1+\lambda)\widetilde{\operatorname{co}}(A_\star)$, so $\widetilde{\operatorname{co}}(A)\subset(1+\lambda)\widetilde{\operatorname{co}}(A_\star)$. Hence by Observation 2.2.24,

$$\begin{aligned}
|\operatorname{co}(A)| &\leqslant (1+\lambda)^k|\operatorname{co}(A_\star)| + (1+(1+\lambda)^k)2k(k+1)\min(n_i)^{-1}|B| \\
&\leqslant (1+2\lambda)^k|\operatorname{co}(A_\star)| \\
&\leqslant |\operatorname{co}(A_\star)| + ((1+2\lambda)^k-1)|B| \\
&\leqslant |A_\star| + ((1+2\lambda)^k - 1 + h_\star(\delta))|B| \\
&\leqslant |A| + (h_8(\delta) + (1+2\lambda)^k - 1 + h_\star(\delta))|B|.
\end{aligned}$$

Defining $\omega(\varepsilon_0,\delta) = h_8(\delta) + (1+2\lambda)^k - 1 + h_\star(\delta)$, we have $|\operatorname{co}(A)\setminus A| \leqslant \omega(\varepsilon_0,\delta)|B|$ with $\omega(\varepsilon_0,\delta)\to 0$ as $\delta\to 0$ for any fixed $\varepsilon_0>0$. □

2.4 Proof of Theorem 2.1.1 b) for k given Theorem 2.1.4 for k

To prove Theorem 2.1.1 b), we first prove the following closely related proposition.

Proposition 2.4.1. For $\varepsilon_0>0$ there are positive constants c_k (we can take $c_k=(4k)^{5k}$), f_k not depending on ε_0 and positive constants $\rho_k(\varepsilon_0), n_k(\varepsilon_0)$ such that the following is true. For every box $B=B(n_1,\dots,n_k)$ with $n_1,\dots,n_k\geqslant n_k(\varepsilon_0)$, and for $A'\subset B$ a reduced set with $|A'|\geqslant\varepsilon_0|B|$, $|\operatorname{co}(A')\setminus A'|\leqslant\rho_k(\varepsilon_0)|A'|$, and a triangulation $\mathscr{T}$ of $\partial\widetilde{\operatorname{co}}(A')$, we have that

$$|\operatorname{co}(A')\setminus A'| \leqslant c_k d_k(A') + f_k|\mathscr{T}|\min\{n_i\}^{-1}|B|.$$

We will see that this result follows from the following result.

Proposition 2.4.2. There are constants $c_k^1, c_k^2, \eta_k > 0$ (we can take $c_k^1 \leqslant 2^{2k}(2k)^{5k}$) such that the following is true. For every box $B = B(n_1,\dots,n_k)$ and for $\widetilde{T} \subset \widetilde{B}$ a simplex with vertices $o = x_0, x_1, \dots, x_k$, and $A \subset T = \widetilde{T} \cap \mathbb{Z}^k$ with $\{o, x_1, \dots, x_k\} \subset A$, scaling with respect to o, we have

$$\left|(T \setminus (1-\eta_k)\widetilde{T}) \setminus A\right| \leqslant \frac{1}{2}|T \setminus A| + c_k^1 d_k(A) + c_k^2 \min\{n_i\}^{-1}|B|.$$

Notation 2.4.3. We shall write $A_S := A \cap S$.

For the next couple of lemmas fix T and A as in Proposition 2.4.2. Recall Definition 2.3.32 in Section 2.3.7, which recursively constructs a family of simplices $\mathscr{S}_{i,j}(\widetilde{T})$ such that $\mathscr{S}_{0,0} = \{\widetilde{T}\}$, $\mathscr{S}_{i,0}$ are the averages of a vertex in $V(T)$ with a simplex in $\mathscr{S}_{i-1,0}$, and $\mathscr{S}_{i,j}$ are the averages of two simplices in $\mathscr{S}_{i,j-1}$.

Lemma 2.4.4. For x a vertex of $\widetilde{T}$ and $\widetilde{S} = (1-2^{-i})x + 2^{-i}\widetilde{T} \in \mathscr{S}_{i,0}$ and $\widetilde{S}' = \frac{1}{2}(x + \widetilde{S}) \in \mathscr{S}_{i+1,0}$, we have

$$|A_{S'}| \geqslant 2^{-k}|A_S| - 2^{-k} d_k(A) - 2^k \min\{n_i\}^{-1}|B|.$$

Proof. By Observation 2.2.21, with $\widetilde{S}'^c$ the complement of $\widetilde{S}'$ in $\widetilde{T}$,

$$\begin{aligned} d_k(A) = |A+A| - 2^k|A| &= \left|(A+A) \cap 2\widetilde{S}'\right| - 2^k|A_{S'}| + \left|(A+A) \cap 2\widetilde{S}'^c\right| - 2^k|A_{S'^c}| \\ &\geqslant \left|(A+A) \cap 2\widetilde{S}'\right| - 2^k|A_{S'}| + \left|A_{S'^c} + A_{S'^c}\right| - 2^k|A_{S'^c}| \\ &\geqslant |x + A_S| - 2^k|A_{S'}| - 2^{2k}\min\{n_i\}^{-1}|B| \\ &= |A_S| - 2^k|A_{S'}| - 2^{2k}\min\{n_i\}^{-1}|B|. \end{aligned}$$

□

Corollary 2.4.5. For $\widetilde{S} \in \mathscr{S}_{i,0}$ we have

$$|A_S| \geqslant 2^{-ik}|A| - \frac{1-2^{-ik}}{2^k-1} d_k(A) - 2^{k+1}\min\{n_i\}^{-1}|B|.$$

Lemma 2.4.6. For $\widetilde{S}_1, \widetilde{S}_2 \in \mathscr{S}_{i,j}$, and $\widetilde{S}' = \frac{1}{2}(\widetilde{S}_1 + \widetilde{S}_2) \in \mathscr{S}_{i,j+1}$, we have

$$|A_{S'}| \geqslant \min(|A_{S_1}|, |A_{S_2}|) - 2^{-k} d_k(A) - (k+2)2^k \min\{n_i\}^{-1}|B|.$$

Proof. Let $\widetilde{P}_1,\dots,\widetilde{P}_{k+1}$ be a partition of $\widetilde{S}'^c$ into convex sets as in the proof of Lemma 2.3.50. Then by Observation 2.2.21 applied twice (once to sets A_{S_1} and A_{S_2} and once to sets $A_{\widetilde{P}_1}$ and $A_{\widetilde{P}_2}$), we have

$$\begin{aligned}
d_k(A) &= |A+A| - 2^k|A| \\
&= \left|(A+A)\cap 2\widetilde{S}'\right| - 2^k|A_{S'}| + \left|(A+A)\cap 2\widetilde{S}'^c\right| - 2^k|A_{S'^c}| \\
&\geqslant \left|A_{S_1}+A_{S_2}\right| - 2^k|A_{S'}| + \sum_{i=1}^{k+1}\left|A_{\widetilde{P}_i}+A_{\widetilde{P}_i}\right| - 2^k|A_{\widetilde{P}_i}| \\
&\geqslant 2^k\left(\min(|A_{S_1}|,|A_{S_2}|\right) - 2^k|A_{S'}| - (k+2)2^{2k}\min\{n_i\}^{-1}|B|.
\end{aligned}$$

□

Corollary 2.4.7. For $\widetilde{S}\in\mathscr{S}_{i,j}$ we have

$$|A_S| \geqslant 2^{-ik}|A| - \left(\frac{1-2^{-ik}}{2^k-1} + j2^{-k}\right)d_k(A) - \left(2^{k+1}+j(k+2)2^k\right)\min\{n_i\}^{-1}|B|.$$

In particular, by Observation 2.2.24 applied to S and T, we have

$$|S\setminus A_S| \leqslant 2^{-ik}|T\setminus A| + c^1_{i,j}d_k(A) + c^2_{i,j}\min\{n_i\}^{-1}|B|,$$

with $c^1_{i,j} = \frac{1-2^{-ik}}{2^k-1} + j2^{-k}$ and $c^2_{i,j} = (1+2^{-ik})2k(k+1) + 2^{k+1} + j(k+2)2^k$.

Proof of Proposition 2.4.2. Let $i = \left\lceil \log_{\frac{1}{2}}\left(\frac{k^{1/k}}{(2k)^5}\right)\right\rceil$ and $j = 16k\log(2k)$. Let $c^1_k = (2k)^{5k}c^1_{i,j} \leqslant 2^{2k}(2k)^{5k}$ and $c^2_k = (2k)^{5k}c^2_{i,j}$ where $c^1_{i,j}$ and $c^2_{i,j}$ are as in Corollary 2.4.7. By [98, Claim 4.2], there exists a constant η_k and a family of simplices $\mathscr{F}\subset\mathscr{S}_{i,j}$ with $|\mathscr{F}|\leqslant(2k)^{5k}$ with

$$\sum_{\widetilde{S}\in\mathscr{F}}|\widetilde{S}| \leqslant \frac{1}{2}|\widetilde{T}|,$$

and

$$\widetilde{T}\setminus(1-\eta_k)\widetilde{T} \subset \bigcup\mathscr{F}.$$

We prove Proposition 2.4.2 with parameters c^1_k, c^2_k, η_k as above. Noting that $2^{-ik} = \frac{|\widetilde{S}|}{|\widetilde{T}|}$, we have

$$\begin{aligned}
|(T\setminus A)\setminus(1-\eta_k)\widetilde{T}| &\leqslant \sum_{\widetilde{S}\in\mathscr{F}}|S\setminus A_S| \\
&\leqslant \sum_{\widetilde{S}\in\mathscr{F}}\left(2^{-ik}|T\setminus A| + c^1_{i,j}d_k(A) + c^2_{i,j}\min\{n_i\}^{-1}|B|\right) \\
&\leqslant \frac{1}{2}|T\setminus A| + c^1_k d_k(A) + c^2_k\min\{n_i\}^{-1}|B|.
\end{aligned}$$

□

We fix η_k as in Proposition 2.4.2. We need one final lemma to prove Proposition 2.4.1.

Lemma 2.4.8. For every $\varepsilon_0 > 0$, there exists a constant $\rho_k(\varepsilon_0) > 0$ such that if $A' \subset B$ with $|A'| \geqslant \varepsilon_0|B|$, $|\operatorname{co}(A') \setminus A'| \leqslant \rho_k(\varepsilon_0)|B|$, then there exists $o \in \mathbb{Z}^k$, such that (scaling with respect to o and recalling we take η_k as in Proposition 2.4.2) we have

$$(1-\eta_k)\widetilde{\operatorname{co}}(A'+A') \cap \mathbb{Z}^k \subset A'+A'.$$

Proof. By Observation 2.2.24, $|\widetilde{\operatorname{co}}(A')| \geqslant \left(\varepsilon_0 - 2k(k+1)\min\{n_i\}^{-1}\right)|B| \geqslant \frac{\varepsilon_0}{2}|B|$, so by John's Lemma [103], there exists an ellipsoid $\widetilde{F}' \subset \widetilde{\operatorname{co}}(A')$ with $|\widetilde{F}'| \geqslant k^{-k}\frac{\varepsilon_0}{2}|B|$. Let o' be the centre of this ellipsoid $\widetilde{F}'$. Let $o \in \mathbb{Z}^k$ be a point closest to o' and let $p \in \partial\widetilde{F}'$ be the intersection of the ray $o'o$ with $\partial\widetilde{F}'$. Let H' be the homothety centred at p with ratio $\frac{|op|}{|o'p|} \geqslant 1 - \frac{\sqrt{k}}{|o'p|}$, so that $H'(o') = o$. If $|o'p| \leqslant 2\sqrt{k}$, then as the cross-sectional area of $\widetilde{F}'$ perpendicular to $o'p$ is at most $|\partial\widetilde{\operatorname{co}}(B)|$, we see that $|\widetilde{F}'| \leqslant 2\sqrt{k}|\partial\widetilde{\operatorname{co}}(B)| < k^{-k}\frac{\varepsilon_0}{2}|B|$, a contradiction. Hence $\widetilde{F} = H'(\widetilde{F}') \subset \widetilde{\operatorname{co}}(A')$ is an ellipse with center o and $|\widetilde{F}| \geqslant (2k)^{-k}\frac{\varepsilon_0}{2}|B|$. Taking a point $x' \in (1-\eta_k)\widetilde{\operatorname{co}}(A'+A') \cap \mathbb{Z}^k$, where the scaling is with respect to o, our goal is to show that $x' \in A'+A'$.

Let $x = \frac{1}{2}(x'+o) \in \mathbb{R}^k$, and let y be the intersection of the ray ox with $\partial\widetilde{\operatorname{co}}(A')$. Note that the ratio $r = |xy|/|oy| \geqslant \eta_k$. Let H be the homothety with center y and ratio r. This homothety sends o to x and $\widetilde{\operatorname{co}}(A')$ to $H(\widetilde{\operatorname{co}}(A'))$. Note that $H(\widetilde{\operatorname{co}}(A')) \subset \widetilde{\operatorname{co}}(A')$ because $\widetilde{\operatorname{co}}(A')$ is convex. Note that $H(\widetilde{F})$ is symmetric around x and satisfies $|H(\widetilde{F})| = r^k|\widetilde{F}|$. By Observation 2.2.24,

$$|H(\widetilde{F}) \cap \mathbb{Z}^k| \geqslant r^k|\widetilde{F}| - 2k(k+1)\min\{n_i\}^{-1}|B| \geqslant \eta_k^k(2k)^{-k}\frac{\varepsilon_0}{4}|B| > 2\rho_k(\varepsilon_0).$$

for $\rho_k(\varepsilon_0)$ sufficiently small. In particular, as $H(\widetilde{F}) \subset \widetilde{\operatorname{co}}(A')$,

$$|H(\widetilde{F}) \cap A'| \geqslant |H(\widetilde{F}) \cap \mathbb{Z}^k| - |\operatorname{co}(A') \setminus A'| > \frac{1}{2}|H(\widetilde{F}) \cap \mathbb{Z}^k|.$$

By the symmetry of $H(\widetilde{F})$ around x, we have that $z \in H(\widetilde{F}) \cap \mathbb{Z}^k$ implies that also $x'-z \in H(\widetilde{F}) \cap \mathbb{Z}^k$. Hence, as $H(\widetilde{F}) \cap A'$ contains more than half the elements in $H(\widetilde{F}) \cap \mathbb{Z}^k$, we can find $z, z' \in H(\widetilde{F}) \cap A'$, such that $x' = z + z'$ and thus $x' \in A'+A'$. □

Proof of Proposition 2.4.1. Let $c_k = 2c_k^1 + 2^{1-k} \leqslant (4k)^{5k}$ and $f_k = 2(k+1)(\frac{1}{2} + 2^k c_k^1) + 8k(k+1) + 2^{k+2} + 2c_k^2$. Let o be the point supplied by Lemma 2.4.8. Note that as $o + A' \subset A'+A'$, we find $d_k(A' \cup \{o\}) \leqslant d_k(A')$ and $|\operatorname{co}(A' \cup \{o\}) \setminus (A' \cup \{o\})| \geqslant |\operatorname{co}(A') \setminus A'| - 1$, so we may assume $o \in A'$. Let $\mathscr{T}'$ be a triangulation of $\widetilde{\operatorname{co}}(A)$ obtained by coning off the simplices in $\mathscr{T}$ at o, so in particular $|\mathscr{T}| = |\mathscr{T}'|$. For each $\widetilde{T} \in \mathscr{T}'$ all vertices are in A. By Lemma 2.4.8, we have

$$\begin{aligned}
|\operatorname{co}(A+A)\setminus(A+A)| &\leqslant \sum_{\widetilde{T}\in\mathscr{T}'}\left|\operatorname{co}(T+T)\setminus(A+A)\right| \\
&= \sum_{\widetilde{T}\in\mathscr{T}'}\left|(\operatorname{co}(T+T)\setminus(A+A))\setminus(1-\eta_k)2\widetilde{T}\right| \\
&\leqslant \sum_{\widetilde{T}\in\mathscr{T}'}\left|\left((2\widetilde{T}\setminus(1-\eta_k)2\widetilde{T})\cap\mathbb{Z}^k\right)\setminus\left(A_{\widetilde{T}\setminus(1-\eta_k)\widetilde{T}}+A_{\widetilde{T}\setminus(1-\eta_k)\widetilde{T}}\right)\right| \\
&\leqslant \sum_{\widetilde{T}\in\mathscr{T}'}\left|\left(2\widetilde{T}\setminus(1-\eta_k)2\widetilde{T}\right)\cap\mathbb{Z}^k\right| - \left|A_{\widetilde{T}\setminus(1-\eta_k)\widetilde{T}}+A_{\widetilde{T}\setminus(1-\eta_k)\widetilde{T}}\right|.
\end{aligned}$$

By Observation 2.2.21, Observation 2.2.24, and Proposition 2.4.2, this is

$$\begin{aligned}
&\leqslant \sum_{\widetilde{T}\in\mathscr{T}'}\left(\left|2\widetilde{T}\setminus(1-\eta_k)2\widetilde{T}\right| - 2^k\left|A_{\widetilde{T}\setminus(1-\eta_k)\widetilde{T}}\right| + \left(2^k2k(k+1)+2^{2k}\right)\min\{n_i\}^{-1}|B|\right) \\
&=2^k\sum_{\widetilde{T}\in\mathscr{T}'}\left(\left|\widetilde{T}\setminus(1-\eta_k)\widetilde{T}\right| - \left|A_{\widetilde{T}\setminus(1-\eta_k)\widetilde{T}}\right|\right) + |\mathscr{T}|\left(2^{k+1}k(k+1)+2^{2k}\right)\min\{n_i\}^{-1}|B| \\
&\leqslant 2^k\sum_{\widetilde{T}\in\mathscr{T}'}\left(\left|(T\setminus A)\setminus(1-\eta_k)\widetilde{T}\right|\right) + |\mathscr{T}|\left(2^{k+2}k(k+1)+2^{2k}\right)\min\{n_i\}^{-1}|B|. \\
&\leqslant 2^k\sum_{\widetilde{T}\in\mathscr{T}'}\left(\frac{1}{2}|T\setminus A| + c_k^1 d_k(A\cap T)\right) + |\mathscr{T}|\left(2^{k+2}k(k+1)+2^{2k}+2^kc_k^2\right)\min\{n_i\}^{-1}|B| \\
&\leqslant 2^{k-1}|\operatorname{co}(A)\setminus A| + 2^kc_k^1d_k(A) \\
&\quad + |\mathscr{T}|\left(2^k(k+1)\left(\frac{1}{2}+2^kc_k^1\right)+2^{k+2}k(k+1)+2^{2k}+2^kc_k^2\right)\min\{n_i\}^{-1}|B|.
\end{aligned}$$

Note that we have identity

$$|\operatorname{co}(A)\setminus A| = 2^{-k}d_k(A) - 2^{-k}d_k(\operatorname{co}(A)) + 2^{-k}\left|\operatorname{co}(A+A)\setminus(A+A)\right|.$$

By Observation 2.2.21 we have the inequality

$$d_k(\operatorname{co}(A)) \geqslant -2^{2k}\min\{n_i\}^{-1}|B|.$$

Combining the last two equations we get

$$|\operatorname{co}(A)\setminus A| \leqslant 2^{-k}d_k(A) + 2^k\min\{n_i\}^{-1}|B| + 2^{-k}|\operatorname{co}(A+A)\setminus(A+A)|.$$

Using the bound on $|\mathrm{co}(A+A)\setminus(A+A)|$, we now deduce

$$\begin{aligned}|\mathrm{co}(A)\setminus A| \leqslant & 2^{-k}d_k(A)+2^k\min\{n_i\}^{-1}|B|+2^{-1}|\mathrm{co}(A)\setminus A|+c_k^1 d_k(A)\\ &+|\mathscr{T}|\left((k+1)\left(\frac{1}{2}+c_k^1\right)+2^2k(k+1)+2^k+c_k^2\right)\min\{n_i\}^{-1}|B|.\end{aligned}$$

Scaling both sides by a factor 2 and reducing both sides by $|\mathrm{co}(A)\setminus A|$, we conclude

$$\begin{aligned}|\mathrm{co}(A)\setminus A| \leqslant & (2c_k^1+2^{1-k})d_k(A)\\ &+|\mathscr{T}|\left(2(k+1)\left(\frac{1}{2}+2^k c_k^1\right)+8k(k+1)+2^{k+2}+2c_k^2\right)\min\{n_i\}^{-1}|B|\\ =& c_k d_k(A)+f_k|\mathscr{T}|\min\{n_i\}^{-1}|B|\end{aligned}$$

□

Proof Theorem 2.1.1 b). Let $\alpha=\min\{n_i\}^{-\gamma}$ with $\gamma=\frac{1}{1+\frac{1}{2}(k-1)\lfloor k/2\rfloor}$ and $\ell=\tau_k\alpha^{-\frac{k-1}{2}}$ with τ_k as in Proposition 2.3.25. Note that this γ satisfies $-\gamma=\frac{k-1}{2}\lfloor k/2\rfloor\gamma-1$.

Take $\Delta_k(\varepsilon_0)$ sufficiently small so that by Theorem 2.1.4, we have $|\mathrm{co}(A)\setminus A|\leqslant\rho_k(\frac{\varepsilon_0}{2})|B|$ where ρ_k is the constant from Proposition 2.4.1. By Proposition 2.3.25, we find a subset $A'\subset A$ such that $|\mathrm{co}(A)\setminus\mathrm{co}(A')|\leqslant\alpha|B|$ and $A'=A\cap\mathrm{co}(A')$ such that $\mathrm{co}(A')$ has at most ℓ vertices. In particular, we have $|A\setminus A'|\leqslant\alpha|B|$. By Observation 2.2.23, we have $d_k(A')\leqslant 2^k\alpha|B|+d_k(A)$.

Now, by Stanley's resolution of the upper bound conjecture [148], as $\partial\widetilde{\mathrm{co}}(A')$ is a combinatorial sphere with ℓ vertices, if we take a triangulation $\mathscr{T}$ of $\partial\widetilde{\mathrm{co}}(A')$ we have $|\mathscr{T}|\leqslant f_k'\ell^{\lfloor k/2\rfloor}=f_k'\tau_{k-1}\alpha^{-\frac{k-1}{2}\lfloor k/2\rfloor}$ for some constant f_k'. By Proposition 2.4.1 applied to $A'=\mathrm{co}(A')\cap A\subset A$, we thus have

$$|\mathrm{co}(A')\setminus A'|\leqslant c_k(2^k\alpha|B|+d_k(A))+\alpha^{-\frac{k-1}{2}\lfloor k/2\rfloor}f_k'\tau_k f_k\min\{n_i\}^{-1}|B|.$$

Thus, dropping Convention 2.2.15, provided that $\min\{n_i\}$ is bounded from below by a function of k,ε_0 given by Theorem 2.1.4 and Proposition 2.3.25, we get as $-\gamma=\frac{k-1}{2}\lfloor k/2\rfloor\gamma-1$ that

$$\begin{aligned}|\mathrm{co}(A)\setminus A| &\leqslant c_k(2^k+1)\alpha|B|+c_k d_k(A)+\alpha^{-\frac{k-1}{2}\lfloor k/2\rfloor}f_k'\min\{n_i\}^{-1}|B|\\ &\leqslant c_k d_k(A)+g_k\min\{n_i\}^{-\frac{1}{1+\frac{1}{2}(k-1)\lfloor k/2\rfloor}}|A|.\end{aligned}$$

By taking $g_k=g_k(\varepsilon_0)$ sufficiently large in terms of ε_0 we can guarantee that the above inequality actually holds for any choice of $\{n_i\}$, which concludes the proof. □

2.5 Proving Theorem 2.1.1 a)

We will now prove Theorem 2.1.1 a). To do this, we will need the following special case of the main result of Green and Tao [85].

Theorem 2.5.1 (Special case of [85])**.** There exist constants w_k such that for any $A \subset \mathbb{Z}^k$ with $d_k(A) \leqslant |A|$, there exists a (possibly degenerate) generalized arithmetic progression P of dimension at most k and size at most $|A|$, along with vectors $x_1,\dots,x_{w_k}$ such that

$$A \subset \bigcup_{i=1}^{w_k} P + x_i.$$

Proof of Theorem 2.1.1 a). We apply Theorem 2.5.1, obtaining a (possibly degenerate) generalized arithmetic progression P and $x_1,\dots,x_{w_k} \in \mathbb{Z}^k$ such that $A \subset \bigcup_{i=1}^{w_k} P + x_i$. Take n_k^0 to be a large threshold, chosen later. If P is contained inside a hyperplane, then A is covered by w_k parallel hyperplanes and we are done. Similarly, if one of the side lengths of P is at most n_k^0, then we can cover P by n_k^0 parallel hyperplanes, so A can be covered by $w_k n_k^0$ parallel hyperplanes. Therefore, we may assume that

$$P = B(n_1,\dots,n_k; v_1,\dots,v_k; 0)$$

is non-degenerate and $n_i \geqslant n_k^0$ for all i. By applying a linear transformation from $GL_k(\mathbb{Q})$ taking v_i to the standard basis vectors e_i and then scaling up to clear denominators, we may assume that $v_i = be_i$, where $b \in \mathbb{N}$.

Claim 2.5.2. There exist a factor b' of b such that $b' \geqslant w_k!^{-w_k} b$ and the following holds. If we consider the decomposition

$$A = A_1 \sqcup \ldots \sqcup A_r$$

associated to the cosets $y_1,\dots,y_r \in (\mathbb{Z}/b'\mathbb{Z})^k$, then after possibly relabeling we have $|A_1| \geqslant |A_j|$ for all j, and for every $p \neq 1$ there exists $j_p \neq 1$ such that

$$y_p + y_{j_p} \neq y_1 + y_k$$

for $k \in \{1,\dots,r\}$.

Proof. Set $b_0' = b$, and $y_{1,0},\dots,y_{r_0,0} \in (\mathbb{Z}/b_0'\mathbb{Z})^k = (\mathbb{Z}/b\mathbb{Z})^k$ the distinct representatives of A, or equivalently the distinct representatives of $x_1,\dots,x_{w_k}$. We note that in particular, this implies that $r_0 \leqslant w_k$.

We recursively construct factors b'_{j+1} of b'_j with $b'_{j+1} \geqslant w_k!^{-1} b'_j$ such that if $y_{1,j},\dots,y_{r_j,j} \in (\mathbb{Z}/(b'_j\mathbb{Z})^k$ are the distinct representatives of A, then the following is true. If we consider the associated coset decomposition

$$A = A_{1,j} \sqcup \ldots \sqcup A_{r_j,j},$$

possibly relabeling so that $|A_{1,j}| \geqslant |A_{p,j}|$ for $1 \leqslant p \leqslant r_j$, then either for every $p \neq 1$ there exists a $\lambda(p,j) \neq 1$ such that

$$y_{p,j} + y_{\lambda(p,j),j} \neq y_{1,j} + y_{\ell,j}$$

for $1 \leqslant \ell \leqslant r_j$, or else we have $r_{j+1} < r_j$.

Suppose that $y_{p,j}$ does not have the property that there exists $\lambda(p,j)$ such that $y_{p,j}+y_{\lambda(p,j),j} \neq y_{1,j}+y_{\ell,j}$ for all ℓ. This is equivalent to saying that $y_{p,j}$ has the property that for all λ, there is an ℓ such that $(y_{p,j}-y_{1,j})+(y_{\lambda,j}-y_{1,j}) = y_{\ell,j}-y_{1,j}$. Then the cyclic group generated by $y_{p,j}-y_{1,j}$ lies entirely inside $\{0, y_{2,j}-y_{1,j}, \dots, y_{r_j,j}-y_{1,j}\}$, so has order at most $r_j \leqslant w_k$. Setting $b_{j+1} = b_j/\gcd(b_j, w_k!)$, we obtain that $y_{p,j}-y_{1,j} = 0$ in $(\mathbb{Z}/b_{j+1}\mathbb{Z})^k$, so $r_{j+1} < r_j$.

As r_j can decrease at most w_k times from $r_0 \leqslant w_k$, there exists a $j \leqslant w_k$ for which $r_j = r_{j+1}$. Taking $j_p = \lambda(p,j)$, $b' = b_j$ and $y_p = y_{p,j}$ the distinct representatives of A in $(\mathbb{Z}/b_j\mathbb{Z})^k = (\mathbb{Z}/b'\mathbb{Z})^k$, we obtain the desired result. □

Returning to the proof of Theorem 2.1.1 a), let $P' = B(n_1, \dots, n_k; b'e_1, \dots, b'e_k; 0)$ where b' is furnished by Claim 2.5.2. Let $w'_k = w_k \cdot w_k!^{k \cdot w_k}$, and $x'_1, \dots, x'_{w'_k}$ be translation vectors such that

$$A \subset \bigcup_{i=1}^{w'_k} P' + x'_i.$$

Also note that $|P'| \leqslant |A|$.

Claim 2.5.3. There exists an x such that $A \subset x + (b'\mathbb{Z})^k$.

Before we begin the proof of the claim we need the following lemma.

Lemma 2.5.4. For $X, Y \subset A$, then

$$|X+Y| \geqslant 2^k \min(|X|,|Y|) - 2^{2k}(w'_k n_k^0)^{-1} w_k'^k |A|.$$

Proof. Let $C(X), C(Y)$ be obtained by compressing X, Y in each of the coordinate directions. Then $C(X), C(Y)$ are contained in the down-set $C(A) \subset C(\bigcup P' + x'_i)$, which in turn is contained inside a box of side lengths $w'_k n_1, \dots, w'_k n_k \geqslant w'_k n_k^0$, which has volume at most $w_k'^k \prod n_i \leqslant w_k'^k |A|$. Therefore, by applying Observation 2.2.21 to a box B of side lenghts $w'_k n_1, \dots, w'_k n_k \geqslant w'_k n_k^0$, we obtain

$$|X+Y| \geqslant |C(X)+C(Y)| \geqslant 2^k \min(|X|,|Y|) - 2^{2k}(w'_k n_k^0)^{-1} w_k'^k |A|.$$

□

Proof of Claim 2.5.3. Let $A = A_1 \sqcup \dots \sqcup A_r$ be the coset decomposition as in the claim with $|A_1|$ maximal, and $r \leqslant w'_k$. We want to show that $r = 1$. We have for any $p \neq 1$ that

$$\begin{aligned}
|A+A| &\geqslant |A_p + A_{j_p}| + \sum_i |A_1 + A_i| \\
&\geqslant |A_p| + \sum_{i=1}^{r} (2^k |A_i| - 2^{2k}(w'_k n_k^0)^{-1} w_k'^k |A|) \\
&\geqslant |A_p| + 2^k |A| - 2^{2k}(n_k^0)^{-1} w_k'^k |A|
\end{aligned}$$

so averaging over all $p \neq 1$ we obtain

$$d_k(A) \geqslant \frac{1}{w_k'-1}|A \setminus A_1| - 2^{2k}(n_k^0)^{-1}w_k'^k|A|. \tag{2.67}$$

On the other hand, assuming $r \geqslant 2$ we have

$$|A+A| \geqslant |A_1+A_2|+|A_1+A_1| \geqslant |A_1|+2^k|A_1| - 2^{2k}(w_k'n_k^0)^{-1}w_k'^k|A|$$

so

$$d_k(A) \geqslant |A| - (2^k+1)|A \setminus A_1| - 2^{2k}(w_k'n_k^0)^{-1}w_k'^k|A|. \tag{2.68}$$

Adding $(w_k'-1)(2^k+1)$ of (2.67) to (2.68), and using $\Delta_k|A| \geqslant d_k(A)$, we obtain

$$((w_k'-1)(2^k+1)+1)\Delta_k|A| \geqslant |A| - ((w_k'-1)(2^k+1)+w_k'^{-1})2^{2k}(n_k^0)^{-1}w_k'^k|A|,$$

which gives the desired contradiction provided Δ_k is sufficiently small and n_k^0 is sufficiently large. □

Returning to the proof of Theorem 2.1.1 a), after translating A we can assume that $A \subset (b'\mathbb{Z})^k$, so we may scale down and assume that $b' = 1$.

We now show that the boxes are in some sense "near" each other.

Claim 2.5.5. There exists a universal constant f_k so that for $P'' = B(f_kn_1,\dots,f_kn_k;e_1,\dots,e_k;0)$ we have that $A \subset P''+x$ for some x.

Proof. Recall that $x_1',\dots,x_{w_k'}'$ are the translation vectors for $P' = B(n_1,\dots,n_k;e_1,\dots,e_k;0)$ which cover A. Suppose that $|A\cap(P'+x_1)|$ is maximal, so $|A\cap(P'+x_1)| \geqslant \frac{1}{w_k'}|A|$. Let $A_1 = A\cap(P'+x_1)$.

We first show that the width in the j-direction is bounded by a fixed multiple of n_j for each j. So fix a j and let π_j be the projection in the jth coordinate. For a subset $\mathscr{B} \subset \{1,\dots,w_k'\}$, let $h_j(\mathscr{B})$ be the difference between the largest and smallest values in $\{\pi_j(x_i')\}_{i\in\mathscr{B}}$. Suppose we have a set $\mathscr{B} \subset \{1,\dots,w_k'\}$ containing 1. If $\mathscr{B}$ is not the whole set and there is no $i \notin \mathscr{B}$ such that $h_j(\mathscr{B}\cup\{i\}) \leqslant 100h_j(\mathscr{B})+100n_j$, then taking $\mathscr{B}'$ to be either $\{x_\ell' : \pi_j(x_\ell') \geqslant \min\{\pi_j(x_i')\}_{i\in\mathscr{B}}\}$ or $\{x_\ell' : \pi_j(x_\ell') \leqslant \max\{\pi_j(x_i')\}_{i\in\mathscr{B}}\}$ (whichever has $\mathscr{B}'^c$ non-empty), then the sets

$$Z_1 = \bigcup_{i\in\mathscr{B}'} P'+x_i', \quad Z_2 = \bigcup_{i\in(\mathscr{B}')^c} P+x_i'$$

have the following property. Let z be the closest point of $Z_2\cap A$ in the e_j-direction to Z_1. Then by considering the projections under π_j, the sets

$$(Z_1\cap A)+(Z_1\cap A),(Z_2\cap A)+(Z_2\cap A),z+A_1$$

are disjoint. By Lemma 2.5.4 applied to these sets we obtain

$$\begin{aligned}|A+A| &\geqslant \Big|(Z_1\cap A)+(Z_1\cap A)\Big| + \Big|(Z_2\cap A)+(Z_2\cap A)\Big| + |z+A_1|\\ &\geqslant 2^k|Z_1\cap A| + 2^k|Z_2\cap A| + |A_1| - 2^{2k+1}(w_k'n_k^0)^{-1}w_k'^k|A|\\ &\geqslant 2^k|A| + \frac{1}{w_k'}|A| - 2^{2k+1}(w_k'n_k^0)^{-1}w_k'^k|A|,\end{aligned}$$

a contradiction provided Δ_k is sufficiently small and n_k^0 sufficiently large. Hence, if $\mathscr{B}$ is not the whole set $\{1,\dots,w_k'\}$, then there is an $i\notin\mathscr{B}$ such that $h_j(\mathscr{B}\cup\{i\})\leqslant 100h_j(\mathscr{B})+100n_j$.

Start with $\mathscr{B}=\{1\}$ and $h_j(\{1\})=0$. Repeatedly applying this, we see that $h_j(\{1,\dots,w_k'\})\leqslant f_kn_j-1$ for some universal integer constant $f_k\leqslant 101^{w_k'}$, independent of j.

We deduce there is a translate x such that $P'+x_i'\subset f_kP'+x$ for all i. □

Returning to the proof of Theorem 2.1.1 a), taking $\varepsilon_k\leqslant f_k^{-k}$, then as $|P'|\leqslant|A|$ we find A is a set of density at least ε_k inside the generalised arithmetic progression $f_kP'+x$. □

2.6 Proof of Corollary 2.1.2 and Corollary 2.1.3

Proof of Corollary 2.1.2. Decorate constants from Theorem 2.1.1 with a dash to distinguish them from constants with the same name in Corollary 2.1.2.

Let $\Delta_k=\min\{\Delta_k',\Delta_k'(\varepsilon_k')\}$ as given by Theorem 2.1.1. Let $c_k=c_k'+\frac{1}{2}((4k)^{5k}-c_k')$, and $m_k(\delta)=\max\left\{m_k',\left(\frac{2g_k'(\varepsilon_k')\delta^{-1}}{(4k)^{5k}-c_k'}\right)^{1+\frac{1}{2}(k-1)\lfloor k/2\rfloor}=\left(\frac{g_k'(\varepsilon_k')\delta^{-1}}{c_k-c_k'}\right)^{1+\frac{1}{2}(k-1)\lfloor k/2\rfloor}\right\}$.

As $d_k(A)\leqslant\delta|A|\leqslant\Delta_k'|A|$, by Theorem 2.1.1 a), either A is covered by $m_k'\leqslant m_k(\delta)$ parallel hyperplanes, or there is some generalized arithmetic progression $B=B(n_1,\dots,n_k;v_1,\dots,v_k;b)$ with the v_i linearly independent, $A\subset B$ and $|A|\geqslant\varepsilon_k'|B|$.

If $n_i\leqslant m_k(\delta)$ for some i, then B (and thus also A) is covered by $n_i\leqslant m_k(\delta)$ parallel hyperplanes. Hence, we may assume $\min\{n_i\}\geqslant m_k(\delta)$.

As $d_k(A)\leqslant\Delta_k'(\varepsilon_k')$, A and B satisfy the conditions of Theorem 2.1.1 b), so that

$$\begin{aligned}|\widehat{\mathrm{co}}(A)\setminus A| &\leqslant c_k'd_k(A)+g_k'(\varepsilon_k')\min\{n_i\}^{-\frac{1}{1+\frac{1}{2}(k-1)\lfloor k/2\rfloor}}|A|\\ &\leqslant c_k'\delta|A|+g_k'(\varepsilon_k')m_k(\delta)^{-\frac{1}{1+\frac{1}{2}(k-1)\lfloor k/2\rfloor}}|A|\\ &= c_k\delta|A|.\end{aligned}$$

That concludes the proof. □

Proof of Corollary 2.1.3. Let $m_k(\delta),\Delta_k',c_k'$ be the constants from Corollary 2.1.2. Let $(4k)^{5k}>c_k>c_k'$ be any constant and $\Delta_k\leqslant\frac{c_k'}{c_k}\Delta_k'$. By standard approximations (see, e.g., [71, p.3 footnote 2]) we may

assume that $\widetilde{A}$ is a finite union of positive measure convex polytopes, and $\widetilde{A}$ satisfies the condition $|\widetilde{A}+\widetilde{A}| \leqslant (2^k+\Delta_k)|\widetilde{A}|$ from Corollary 2.1.3. Define $\delta = \frac{|\widetilde{A}+\widetilde{A}|-2^k|\widetilde{A}|}{|\widetilde{A}|}$.

Let $N \in \mathbb{N}$, let $A_N = (\frac{1}{N}\mathbb{Z})^k \cap \widetilde{A}$, and let $(A+A)_N = (\frac{1}{N}\mathbb{Z})^k \cap (\widetilde{A}+\widetilde{A})$. Note that for each $r \in \mathbb{N}$, A_N contains a translate of the combinatorial box $B(r,\dots,r)$ for N sufficiently large. In particular, this implies A_N is reduced.

As $N \to \infty$ we have $\frac{1}{N^k}|A_N| \to |\widetilde{A}|$, $\frac{1}{N^k}|\operatorname{co}(A_N)| \to |\widetilde{\operatorname{co}}(\widetilde{A})|$, and $\frac{1}{N^k}|(A+A)_N| \to |\widetilde{A}+\widetilde{A}|$. Thus for N sufficiently large $\frac{1}{N^k}d_k(A_N) \leqslant \frac{1}{N^k}(|(A+A)_N| - 2^k|A_N|) \leqslant \frac{1}{N^k}\frac{c_k}{c_k'}\delta|A_N|$, so

$$d_k(A_N) \leqslant \frac{c_k}{c_k'}\delta|A_N| \leqslant \frac{c_k}{c_k'}\Delta_k|A_N| \leqslant \Delta_k'|A_N|.$$

By Observation 2.2.18, the number of hyperplanes needed to cover A_N is larger than $m_k(\frac{c_k}{c_k'}\delta)$ for N sufficiently large. Therefore by Corollary 2.1.2, we have $|\operatorname{co}(A_N)\setminus A_N| \leqslant c_k'(\frac{c_k}{c_k'}\delta)|A_N| = c_k\delta|A_N|$ for N sufficiently large. Dividing by N^k and taking the limit as $N \to \infty$ of both sides yields

$$|\operatorname{co}(\widetilde{A})\setminus\widetilde{A}| \leqslant c_k|\widetilde{A}|.$$

□

Chapter 3

Sharp L^1 inequalities for Sup-Convolution

The work in this section is joint with Peter van Hintum and Hunter Spink [97].

3.1 Introduction

In this chapter we prove Theorem 1.1.11 and Theorem 1.1.12, the sharp stability result for a linear Brunn–Minkowski inequality for hypographs of functions in low dimensions. For this class of geometric objects, Theorem 1.1.11 and Theorem 1.1.12 are strengthen versions of a conjecture of Figalli and Jerison [71]. For convenience, we introduce some notation and reformulate Theorem 1.1.11 and Theorem 1.1.12 as corollaries of Corollary 3.1.3 and Corollary 3.1.4, respectively.

Let $C \subset \mathbb{R}^k$ be a compact convex domain. For a bounded function $f : C \to \mathbb{R}$, the *upper convex hull of* f, denoted by $\mathrm{co}(f)$, is defined to be the infimum of all concave functions larger than f, and for bounded measurable functions $f_1, \dots, f_n : C \to \mathbb{R}$, the *sup-convolution* is defined to be

$$(f_1 * \dots * f_n)(z) := \sup\left\{ \frac{f_1(x_1) + \dots + f_n(x_n)}{n} : \frac{x_1 + \dots + x_n}{n} = z \right\}.$$

The operation of sup-convolution, or in its equivalent form *inf-convolution* $-((-f_1) * \dots * (-f_n))$, naturally appears in problems of optimization, with $f_1, \dots, f_n$ utility functions and C representing a cost domain [53]. For a general survey, see [150]. Clearly $f_1 * \dots * f_n \geqslant \frac{f_1 + \dots + f_n}{n}$, and equality is attained when for example $f_1, \dots, f_n$ coincide with the same concave function $f = \mathrm{co}(f)$.

We can view the sup-convolution operation geometrically in terms of the Minkowski sum of regions in $\mathbb{R}^{k+1}$. Indeed, consider the *hypograph*

$$A_{f,\lambda} = \{(x,y) \in C \times \mathbb{R} : \lambda \leqslant y \leqslant f(x)\}.$$

Then we have the closed convex hull $\overline{\mathrm{co}(A_{f,\lambda})} = A_{\mathrm{co}(f),\lambda}$, and for λ sufficiently negative we have

$$A_{f_1 * \dots * f_n, \lambda} = \frac{1}{n}(A_{f_1,\lambda} + \dots + A_{f_n,\lambda}).$$

The study of how close a Minkowski sum is to its convex hull was started by Starr-Shapley-Folkman [149] and Emerson-Greenleaf [54], who showed that if $A_1,\ldots,A_n$ are subsets of the unit ball in $\mathbb{R}^k$, then the Hausdorff distance between the Minkowski averages $\frac{1}{n}(A_1+\ldots+A_n)$ and $\frac{1}{n}(\mathrm{co}(A_1)+\ldots+\mathrm{co}(A_n))$ is bounded above by $\sqrt{k}n^{-1}$. Of particular interest for us will be when $A_1=\ldots=A_n=A$, where we are concerned with how close $\frac{1}{n}(A+\ldots+A)$ is to $\mathrm{co}(A)$; for this equal sets case we refer the reader to the extensive survey [78]. Ruzsa [129, Theorem 5] showed that there is a constant D_k such that for $A\subset\mathbb{R}^k$ and $n>D_k\frac{|\mathrm{co}(A)|}{|A|}$, we have $|\frac{1}{n}(A+\ldots+A)|\geqslant\left(1-\frac{D_k}{n}\cdot\frac{|\mathrm{co}(A)|}{|A|}\right)^k|\mathrm{co}(A)|$ (taking the outer Lebesgue measure everywhere).

The study of how close a Minkowski sum is to its convex hull is also closely related to the stability of the Brunn–Minkowski inequality. The Brunn–Minkowski inequality says that if A and B are measurable subsets of $\mathbb{R}^k$ with positive measure, then $|A+B|^{\frac{1}{k}}\geqslant|A|^{\frac{1}{k}}+|B|^{\frac{1}{k}}$. Equality holds if and only if A and B are homothetic convex sets (less a measure 0 set). Stability results for the Brunn–Minkowski inequality quantify how close A,B are to homothetic convex sets when $|A+B|^{\frac{1}{k}}\approx|A|^{\frac{1}{k}}+|B|^{\frac{1}{k}}$.

The precise stability of the Brunn–Minkowski inequality for sets A,B restricted to various subclasses of geometric objects, is one of the central open problems in the study of geometric inequalities. It has been explored intensively in recent years by Barchiesi and Julin [11], Carlen and Maggi [39], Christ [44], Figalli and Jerison [67, 69, 71], Figalli, Maggi and Mooney [72] and Figalli, Maggi and Pratelli [73, 74].

Resolving a conjecture of Figalli and Jerison [71, 68] on the stability of the Brunn–Minkowski inequality for homothetic sets, we [98] showed that for $t\in(0,1)$ there are $c_k(t),d_k(t)>0$ such that for subsets $A\subset\mathbb{R}^k$ of positive measure, $|(tA+(1-t)A)\setminus A|\geqslant c_k(t)|\mathrm{co}(A)\setminus A|$ provided $|(tA+(1-t)A)\setminus A|\leqslant d_k(t)|A|$. In particular, this implies that $|\mathrm{co}(A)\setminus\frac{1}{n}(A+\ldots+A)|\to 0$ as $n\to\infty$ polynomially fast, provided $|\mathrm{co}(A)\setminus A|<<|A|$.

By applying the stability result, to the hypograph $A=A_{f,\lambda}$ with $t=1/n$ we get

$$\left|(\frac{1}{n}A+\frac{n-1}{n}A)\setminus A\right|\geqslant c_{k,n}|\mathrm{co}(A)\setminus A|$$

provided

$$\left|(\frac{1}{n}A+\frac{n-1}{n}A)\setminus A\right|\leqslant d_{k,n}|A|.$$

By taking λ sufficiently negative the hypothesis is automatically satisfied and the conclusion can be rewritten as

$$\int_C f^{*n}(x)-f(x)dx\geqslant c_{k,n}\int_C\mathrm{co}(f)(x)-f(x)dx.$$

(see Section 3.7, where we also give an alternate self-contained proof of this particular inequality).

The constants $c_{k,n}$ obtained in this way are not optimal, which naturally gives rise to the following question. What are the optimal constants? Theorem 3.1.1, answers this question for one function in dimension $k\leqslant 3$. Theorem 3.1.2, answers the analogous question for two distinct functions in dimension $k\leqslant 3$.

Theorem 3.1.1. If $f : C \to \mathbb{R}$ is a bounded measurable function with $C \subset \mathbb{R}^k$ a compact convex domain and $k \leqslant 3$ then

$$\int_C f^{*n}(x) - f(x)dx \geqslant c_{k,n} \int_C \mathrm{co}(f)(x) - f(x)dx$$

with

$$c_{k,n} = \begin{cases} \frac{n-1}{n} & k = 1 \\ \frac{(2n-1)(n-1)}{2n^2} & k = 2 \\ \frac{(n-1)^2}{n^2} & k = 3. \end{cases}$$

Theorem 3.1.2. If $f, g : C \to \mathbb{R}$ are bounded measurable functions with $C \subset \mathbb{R}^k$ a compact convex domain and $k \leqslant 3$ then

$$\int_C f * g(x) - \frac{f(x)+g(x)}{2} dx \geqslant \frac{k+1}{2^{k+1}} \int_C \mathrm{co}(f)(x) - f(x)dx.$$

If $f, g : C \to \mathbb{R}^+$ are bounded measurable functions, then the hypographs corresponding to $\lambda = 0$ satisfy the following linear Brunn–Minkowski inequality

$$\left| \frac{A_f + A_g}{2} \right| \geqslant \frac{|A_f| + |A_g|}{2}.$$

Restricting to hypographs of functions defined on a convex domain in low dimensions, Corollary 3.1.3 and Corollary 3.1.4 extend the conjecture of Figalli and Jerison [71] on the stability of the Brunn–Minkowski inequality, by providing an analogue for two distinct sets and for arbitrary many copies of the same set, with optimal constants.

Corollary 3.1.3. If $f : C \to \mathbb{R}^+$ is a bounded measurable function with $C \subset \mathbb{R}^k$ a compact convex domain and $k \leqslant 3$ then

$$\left(\left| \frac{A_f + \ldots + A_f}{n} \right| - |A_f| \right) \geqslant c_{k,n} |\mathrm{co}(A_f) \setminus A_f|,$$

with

$$c_{k,n} = \begin{cases} \frac{n-1}{n} & k = 1 \\ \frac{(2n-1)(n-1)}{2n^2} & k = 2 \\ \frac{(n-1)^2}{n^2} & k = 3 \end{cases}$$

□

Corollary 3.1.4. If $f, g : C \to \mathbb{R}^+$ are bounded measurable functions with $C \subset \mathbb{R}^k$ a compact convex domain and $k \leqslant 3$ then

$$\left(\left| \frac{A_f + A_g}{2} \right| - \frac{|A_f| + |A_g|}{2} \right) \geqslant \frac{k+1}{2^{k+1}} |\mathrm{co}(A_f) \setminus A_f|.$$

□

In both theorems the constants are optimal, as can be seen by taking f and g to be the indicator function on the vertices of $C = T$ a simplex. We define the *m'th k-dimensional hypersimplex* to be the set $P_{k,m} := [0,1]^{k+1} \cap \{\sum x_i = m\}$. In any dimension k, letting $e_1, \ldots, e_{k+1}$ be the standard basis vectors in $\mathbb{R}^{k+1}$ and identifying $C = T$ with the convex hull of $ne_1, \ldots, ne_{k+1}$ we show that the level sets of this particular f^{*n} induce a subdivision of C into hypersimplices, where translates of the m'th k-dimensional hypersimplex $P_{k,m}$ appear $\binom{n+k-m}{k}$ times (see Section 3.3).

For example for $k = 2$ (depicting the case $n = 4$ below), f^{*n} takes value $\frac{n-1}{n}$ in the shaded region, the union of $\binom{n+1}{2}$ translates of the triangle $P_{2,1}$ ▲, and $\frac{n-2}{n}$ in the unshaded region, the union of $\binom{n}{2}$ translates of the triangle $P_{2,2}$ ▽:



The shaded regions are precisely those parts of T whose points can be expressed as $\frac{x_1+\ldots+x_n}{n}$ with all but one of the x_i a vertex of T, and the remaining unshaded regions can be expressed with all but two of the x_i a vertex of T.

For $k = 3$, we can subdivide T into $\binom{n+2}{3}$ translates of $\frac{1}{n}T = P_{3,1}$, $\binom{n+1}{3}$ translates of the octahedron $P_{3,2}$, and $\binom{n}{3}$ translates of $-P_{3,1} = P_{3,3}$, on which f^{*n} takes the values $\frac{n-1}{n}$, $\frac{n-2}{n}$, $\frac{n-3}{n}$ respectively. The partition is according to whether the maximum number of vertices of T which can be used to express the point as an n-average is $n-1, n-2$, or $n-3$ respectively.

To prove Theorem 3.1.1, we pass to a piecewise-linear approximation and then triangulate according to the domains of linearity of $\mathrm{co}(f)$. On each simplex T we prove a sharp inequality relating $\int_R \mathrm{co}(f)(x) - f^{*n}(x)dx$ and $\int_T \mathrm{co}(f)(x) - f(x)dx$ for R ranging over the hypersimplices in the subdivision alluded to above. This in turn is encompassed in our notion of an "m-averageable" subset of T (Section 4), and showing certain hypersimplices are "m-averageable" allows us to conclude.

We make the following conjecture for arbitrary $k, n \geqslant 1$. In what follows, write $A(k,\ell)$ for the Eulerian number counting permutations of S_k with ℓ descents; the number of descents of a permutation $\sigma : [k] \to [k]$ is equal to the number of positions $1 \leqslant i \leqslant k-1$ such that $\sigma(i) > \sigma(i+1)$.

Conjecture 3.1.5. If $f : C \to \mathbb{R}$ is a bounded measurable function with $C \subset \mathbb{R}^k$ a compact convex domain then

$$\int_C f^{*n}(x) - f(x)dx \geqslant c_{k,n} \int_C \mathrm{co}(f)(x) - f(x)dx,$$

where

$$c_{k,n} = \frac{1}{n^k} \sum_{m=1}^{k} \frac{n-m}{n} \binom{n+k-m}{k} A(k, m-1) = \frac{k+1}{n^{k+1}}(1^k + \ldots + (n-1)^k).$$

If true, this would be sharp by taking f the indicator function on the vertices of $C = T$ a simplex (see Conjecture 3.1.5). Here $\frac{n-m}{m}$ is the value of f^{*n} on each hypersimplex $P_{k,m} + x$, $\binom{n+k-m}{k}$ is the number of such hypersimplices, and $\frac{A(k,m-1)}{n^k}$ is the volume ratio of $P_{k,m}$ to T (see Fact 5.3 of [139]).

Remark 3.1.6. Omitting the $\frac{n-m}{m}$ factor, we obtain a geometric proof of the Worpitzky identity $n^k = \sum \binom{n+k-m}{k} A(k,m-1)$. A similar observation was recently exploited by Early [52, Section 3] to categorify the Worpitzky identity via the representation theory of the symmetric group.

We also show the following asymptotic result for fixed k as $n \to \infty$.

Theorem 3.1.7. For any $k \geqslant 1$ and $n \geqslant k+1$, we have $c_{k,n} \geqslant 1 - \binom{n}{k} \frac{k^{k+1}}{n^{k+1}} = 1 - O(\frac{1}{n})$.

This is optimal up to the constant on $\frac{1}{n}$, which this theorem shows can be taken to be $\frac{k^{k+1}}{k!} = e^{O(k)}$, though our conjectured extremal example gives a constant of $\frac{k+1}{2}$.

In Section 2 we show that Theorem 3.1.1, Conjecture 3.1.5, Theorem 3.1.7, and Theorem 3.1.2 reduce to the case that $C = T$ is a simplex, $f = 0$ on the vertices and $f \leqslant 0$ otherwise. In Section 3 we construct our hypersimplex subdivision of T. In Section 4 we introduce a new geometric notion of "m-averageable subsets of T", and reduce to showing certain hypersimplices in T are "m-averageable". In Section 5 we show that the relevant hypersimplices up to dimension 3 are "m-averageable" and conclude Theorem 3.1.1 and Theorem 3.1.2. In Section 6 we prove Theorem 3.1.7. Finally, in Section 7 we show how the existence of a non-sharp constant in Conjecture 3.1.5 can be derived from [98], and we also give a quick self-contained proof. We end the chapter with a brief discussion about Conjecture 3.1.5 in Section 8.

3.2 Reduction to Simplices

Here we reduce Theorem 3.1.1, Conjecture 3.1.5, Theorem 3.1.7, and Theorem 3.1.2 to the case $C = T$ is a simplex, $f \leqslant 0$, and $f = 0$ at the vertices.

Proposition 3.2.1. The statements of Theorem 3.1.1, Conjecture 3.1.5, Theorem 3.1.7, and Theorem 3.1.2, respectively, are equivalent to the corresponding statements with the additional assumption that $C = T$ is a simplex, $f \leqslant 0$, and $f = 0$ at the vertices.

Proof. The reduction is divided in the following three steps. We always focus on the reduction of Conjecture 3.1.5, as the others follow in a similar way.

Claim 3.2.2. Suppose that Theorem 3.1.1 (resp. Conjecture 3.1.5, Theorem 3.1.7, Theorem 3.1.2) is true when the domain C is a polytope P, $f \geqslant 0$, and $f = 0$ on a neighborhood of ∂C. Then Theorem 3.1.1 (resp. Conjecture 3.1.5, Theorem 3.1.7, Theorem 3.1.2) is true.

Proof. We do this for Conjecture 3.1.5, the other cases are similar. The inequality doesn't change if we scale f or add a constant so assume that $f(x) \in [n, n+1]$ for all $x \in C$. Let $P_1, P_2, \ldots$ be a sequence of polytopes with $C \subset P_i^\circ$ and $|P_i| \to |C|$. We extend f to a function f_i on P_i by setting $f_i = 0$ on $P_i \setminus C$. Then we note that for any $x \in C$, $f_i^{*n}(x) \geqslant f_i(x) \geqslant n$, but

$$\frac{f_i(x_1) + \ldots + f_i(x_n)}{n} \leqslant \frac{(n+1)(n-1)}{n} < n$$

provided any $x_j \in P_i \setminus C$, so we conclude that $f_i^{*n}|_C = f^{*n}$. Thus as $\mathrm{co}(f_i) \geqslant \mathrm{co}(f)$,

$$\int_C f^{*n}(x) - f(x)dx \geqslant \int_{P_i} f_i^{*n}(x) - f_i(x)dx - |P_i \setminus C| \cdot ||f_i^{*n}||_\infty \geqslant c_{k,n} \int_{P_i} \mathrm{co}(f_i)(x) - f_i(x)dx - |P_i \setminus C| \cdot ||f_i^{*n}||_\infty$$
$$\geqslant c_{k,n} \int_C \mathrm{co}(f)(x) - f(x)dx - |P_i \setminus C| \cdot ||f_i^{*n}||_\infty \to c_{k,n} \int_C \mathrm{co}(f)(x) - f(x)dx,$$

where in the last step we used that $||f_i^{*n}||_\infty = ||f||_\infty \leqslant n+1$. □

Claim 3.2.3. Suppose that Theorem 3.1.1 (resp. Conjecture 3.1.5, Theorem 3.1.2) is true when the domain C is a polytope P and $\mathrm{co}(f)$ is the upper convex hull of finitely many points $(r_i, f(r_i))$ (so is in particular piecewise linear). Then it is true when the domain C is a polytope P, $f \geqslant 0$, and $f = 0$ on a neighborhood of ∂C.

Proof. We do this for Conjecture 3.1.5, the other cases are similar. Suppose $C = P$, $f \geqslant 0$, and $f = 0$ on a neighborhood of ∂C (but we do not necessarily know that $\mathrm{co}(f)$ is piecewise linear).

We'll show that $\mathrm{co}(f)$ is continuous at all points $x \in C$. First, suppose that $x \in C^\circ$. Then for $y \in C^\circ$, let z_1, z_2 be the points on ∂C such that z_1, x, y, z_2 are collinear in that order. We have

$$\mathrm{co}(f)(x) \geqslant \frac{||x-z_1||}{||y-z_1||}\mathrm{co}(f)(y) + \frac{||x-y||}{||y-z_1||}\mathrm{co}(f)(z_1) \geqslant \frac{||x-z_1||}{||y-z_1||}\mathrm{co}(f)(y)$$
$$\mathrm{co}(f)(y) \geqslant \frac{||y-z_2||}{||x-z_2||}\mathrm{co}(f)(x) + \frac{||x-y||}{||x-z_2||}\mathrm{co}(f)(z_2) \geqslant \frac{||y-z_2||}{||x-z_2||}\mathrm{co}(f)(x),$$

so $\mathrm{co}(f)(y) \to \mathrm{co}(f)(x)$ as $y \to x$. Next, instead suppose that $x \in \partial C$. Then take any linear function L which is 0 at x and positive with $\inf_{\mathrm{supp}(f)} L > 0$, which exists as $f(x)$ is supported on a compact subset of the interior of C. We may further assume that $L|_{\mathrm{supp}(f)} > ||f||_\infty$ by replacing L with $(1 + ||f||_\infty)(\inf_{\mathrm{supp}(f)} L)^{-1} L$. Then $\mathrm{co}(f)$ is sandwiched between the constant function 0 and the continuous function L which agree at x, which implies that $\mathrm{co}(f)(x) = 0$ and $\mathrm{co}(f)$ is continuous at x.

In particular, because $\mathrm{co}(f)$ is continuous and concave, it is approximated in the supremum norm by concave piecewise-linear functions from above. Let c be a concave piecewise-linear approximation to $\mathrm{co}(f)$ with $c \geqslant \mathrm{co}(f)$, and $||c - \mathrm{co}(f)||_\infty \leqslant \varepsilon$ for some fixed ε. Let $x_1, \ldots, x_N \in C$ be a finite collection of points for which the graph of c is the upper convex hull of the points $(x_i, c(x_i))$ (note that here we use the fact that the domain is a polytope).

We note that

$$\mathrm{co}(f)(x) = \sup\{\lambda_1 f(x_1) + \ldots + \lambda_\ell f(x_\ell) : \ell \in \mathbb{N}, \lambda_1, \ldots, \lambda_\ell \in [0,1], \sum \lambda_i = 1, \sum \lambda_i x_i = x\}.$$

Hence, there exists M, points $\{x_{i,j}\}_{1 \leqslant i \leqslant N, 1 \leqslant j \leqslant M}$ and parameters $\lambda_{i,j} \in [0,1]$ with $\sum_{j=1}^M \lambda_{i,j} = 1$, $\sum_{j=1}^M \lambda_{i,j} x_{i,j} = x_i$, and

$$\mathrm{co}(f)(x_i) \leqslant \sum_{j=1}^M \lambda_{i,j} f(x_{i,j}) + \varepsilon.$$

Let

$$f'(x) = \begin{cases} f(x) + 2\varepsilon & \text{if } x = x_{i,j} \text{ for some } i, j, \text{ and} \\ f(x) & \text{otherwise.} \end{cases}$$

We remark that

$$\sum_{j=1}^{M} \lambda_{i,j} f'(x_{i,j}) = 2\varepsilon + \sum_{j=1}^{M} \lambda_{i,j} f(x_{i,j}) \geqslant \mathrm{co}(f)(x_i) + \varepsilon \geqslant c(x_i).$$

Hence letting g be the upper convex hull of the points $(x_{i,j}, f'(x_{i,j}))$, we have $g \geqslant c \geqslant f$.

We claim that $\mathrm{co}(f')(x) = g$. Indeed, we trivially have $g \leqslant \mathrm{co}(f')$, so it suffices to show $g \geqslant \mathrm{co}(f')$. For $x = x_{i,j}$ we clearly have $g(x) \geqslant f'(x)$ and for $x \neq x_{i,j}$, we have $g(x) \geqslant f(x) = f'(x)$. Hence $g \geqslant f'$, so as g is concave, $g \geqslant \mathrm{co}(f')$.

Hence, $\mathrm{co}(f')$ is the upper convex hull of finitely many points $(r_i, f'(r_i))$. As $||f' - f||_\infty \leqslant 2\varepsilon$ and $f' \geqslant f$, we have by our hypothesis,

$$\begin{aligned} \int_C f^{*n}(x) - f(x)dx + 2\varepsilon|C| \geqslant \int_C (f')^{*n}(x) - f'(x) &\geqslant c_{k,n} \int_C \mathrm{co}(f')(x) - f'(x)dx \\ &\geqslant c_{k,n} \int_C \mathrm{co}(f)(x) - f(x)dx - 2\varepsilon c_{k,n}|C|, \end{aligned}$$

where the first inequality follows from the fact that $f^{*n} + 2\varepsilon = (f + 2\varepsilon)^{*n} \geqslant (f')^{*n}$. Letting $\varepsilon \to 0$ we conclude.

□

Claim 3.2.4. Suppose that Theorem 3.1.1 (resp. Conjecture 3.1.5, Theorem 3.1.7, Theorem 3.1.2) is true when the domain C is a simplex T, $f = 0$ at the vertices of T and $f \leqslant 0$. Then it is true when the domain C is a polytope P and $\mathrm{co}(f)$ is the upper convex hull of finitely many points $(r_i, f(r_i))$.

Proof. We do this for Conjecture 3.1.5, the other cases are similar. Let f be defined on a polytopal domain $C = P$ with $\mathrm{co}(f)$ the upper convex hull of finitely many points $(r_i, f(r_i))$. The domains of linearity of $\mathrm{co}(f)$ decompose C into convex polytopes with vertices a subset of the r_i. Further subdivide this decomposition into triangulation $\mathscr{T}$. Then $\mathrm{co}(f|_T T) = \mathrm{co}(f)|_T$ for all $T \in \mathscr{T}$, so $\int_C \mathrm{co}(f)(x) - f(x)dx = \sum_{T \in \mathscr{T}} \int_T \mathrm{co}(f|_T)(x) - f|_T(x)dx$ and

$$\int_C f^{*n}(x) - f(x)dx \geqslant \sum_T \int_T (f|_T)^{*n}(x) - f|_T(x)dx.$$

Hence it suffices to prove for every $T \in \mathscr{T}$ that

$$\int_T (f|_T)^{*n}(x) - f|_T(x)dx \geqslant c_{k,n} \int \mathrm{co}(f|_T)(x) - f|_T(x)dx.$$

As $\mathrm{co}(f|_T)$ is linear, and the inequality is preserved by subtracting linear functions from f, we may subtract $\mathrm{co}(f|_T)$ from f, after which $f = 0$ at the vertices of T and $f \leqslant 0$ on T. Thus by hypothesis we are done. □

The above sequence of reductions gives the desired conclusion.

□

3.3 Hypersimplex Covering

We take T to be the convex hull of the standard basis vectors $e_1, \ldots, e_{k+1} \in \mathbb{R}^{k+1}$. Recall from Section 1 that the m'th k-dimensional hypersimplex for $1 \leqslant m \leqslant k$ is defined to be the region in $\mathbb{R}^{k+1}$ given by

$$P_{k,m} := \left\{(x_1, \ldots, x_{k+1}) \in [0,1]^{k+1} : \sum x_i = m\right\}.$$

Definition 3.3.1. Let

$$\mathscr{B}_{k,\ell} = \left\{(x_1, \ldots, x_{k+1}) \in \mathbb{Z}^{k+1}_{\geqslant 0} : \sum x_i = \ell\right\}.$$

Proposition 3.3.2. We have a polytopal subdivision

$$T = \bigcup_{m=1}^{\min(k,n)} \bigcup_{v \in \mathscr{B}_{k,n-m}} \frac{1}{n} P_{k,m} + \frac{1}{n} v.$$

Proof. Note that because $\bigcup_{v \in \mathbb{Z}^{k+1}} [0,1]^{k+1} + v$ subdivide $\mathbb{R}^{k+1}$, the intersections $\bigcup_{v \in \mathbb{Z}^{k+1}} ([0,1]^{k+1} + v) \cap nT$ form a polytopal subvidision of nT. Let $\mathscr{B}$ be the set of such v for which $([0,1]^{k+1} + v) \cap nT$ is k-dimensional, such that $\bigcup_{v \in \mathscr{B}} ([0,1]^{k+1} + v) \cap nT$ also forms a polytopal subvidision of nT.

Let $v \in \mathscr{B}$, and set $m = n - \sum v_i$. We first claim that $1 \leqslant m \leqslant k$ and

$$([0,1]^{k+1} + v) \cap \left\{\sum x_i = n\right\} = P_{k,m} + v.$$

Indeed, as nT lies in the $\sum x_i = n$ hyperplane and $([0,1]^{k+1} + v) \cap nT$ is k-dimensional, we must have $n - k \leqslant \sum v_i \leqslant n - 1$, i.e., $1 \leqslant m \leqslant k$. Then it is easy to see that $([0,1]^{k+1} + v) \cap \{\sum x_i = n\} = P_{k,m} + v$ by definition.

Next, we claim that $v \in \mathscr{B}_{k,n-m}$, i.e., that v has no negative coordinates. Indeed, suppose that $v_1 \leqslant -1$. Then $([0,1]^{k+1} + v) \cap nT \subset \{x_1 = 0\} \cap nT$ which is at most $k-1$-dimensional.

Finally, we claim that

$$([0,1]^{k+1} + v) \cap \left\{\sum x_i = n\right\} \subset nT.$$

Indeed, as $v \in \mathscr{B}_{k,n-m}$, all coordinates are non-negative.

Conversely, if $v \in \mathscr{B}_{k,n-m}$, then $\frac{1}{n} P_{k,m} + \frac{1}{n} v = ([0,1]^{k+1} + v) \cap \{\sum x_i = n\} = ([0,1]^{k+1} + v) \cap nT$, where the last equality is because all coordinates are non-negative.

Therefore we have the polytopal subdivision $nT = \bigcup_{m=1}^{k} \bigcup_{v \in \mathscr{B}_{k,n-m}} P_{k,m} + v$, and as $\mathscr{B}_{k,n-m} = \emptyset$ for $m > n$, we conclude. □

Proposition 3.3.3. For $1 \leqslant m \leqslant \min(k,n)$ and $v \in \mathscr{B}_{k,n-m}$, the points in $\frac{1}{n}P^{\circ}_{k,m} + \frac{1}{n}v$ can be written as $\frac{x_1+\ldots+x_n}{n}$ with $n-m$ of the x_i being vertices of T and the rest contained inside T, but not with at least $n-m+1$ of the x_i being vertices of T and the rest contained inside T.

Proof. For $y = \frac{1}{n}w + \frac{1}{n}v \in \frac{1}{n}P^{\circ}_{k,m} + \frac{1}{n}v$, we can write $y = \frac{m\cdot(\frac{1}{m}w)+\sum v_i e_i}{n}$, and $\frac{1}{m}w \in \frac{1}{m}P^{\circ}_{k,m} \subset T$.

Conversely, suppose that we can write $y = \frac{x_1+\ldots+x_n}{n}$ with $x_1, \ldots, x_{n-m+1}$ vertices of T and $x_{n-m+2}, \ldots, x_n \in T$. Then $\lfloor ny \rfloor = v$, so we obtain the contradiction

$$n-m = \sum_{i=1}^{k+1} \lfloor ny_i \rfloor \geqslant \sum_{i=1}^{k+1} \sum_{j=1}^{n} \lfloor (x_j)_i \rfloor \geqslant \sum_{i=1}^{k+1} \sum_{j=1}^{n-m+1} \lfloor (x_j)_i \rfloor = n-m+1.$$

□

3.4 m-averageable sets

We now define a new notion of "m-averageable" subset of a simplex T.

Definition 3.4.1. Given a simplex T, say that a subset $S \subset T$ is "m-averageable" if there are piecewise linear maps $H_1, \ldots, H_m : T \to T$ which are generically bijective of Jacobian 1 such that $\frac{H_1+\ldots+H_m}{m}$ is a piecewise linear map $T \to S$ which is generically bijective with constant Jacobian $\frac{|S|}{|T|}$. We call a piecewise linear map $H : T \to S$ generically bijective if there exist finite union of hyperplanes T_0 and S_0 such that the restricted map $H : (T \setminus T_0) \to (S \setminus S_0)$ is bijective. The Jacobian of a linear map $H : T \to S$ is the ratio $|H(R)|/|R|$ which is constant for all measurable region $R \subset T$.

The key property of an m-averageable set $S \subset T$ is the observation that

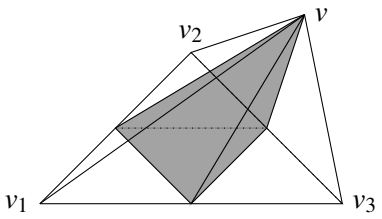
$$\int_S f_1 * \ldots * f_m(x)dx = \frac{|S|}{|T|} \int_T f_1 * \ldots * f_m\left(\frac{H_1(y) + \ldots + H_m(y)}{m}\right)dy \geqslant$$

$$\frac{|S|}{|T|} \int_T \frac{f_1(H_1(y)) + \ldots + f_m(H_m(y))}{m} dy = \frac{|S|}{|T|} \frac{\sum_{i=1}^{m} \int_T f_i(x)dx}{m}.$$

This observation is used directly, and in a slightly modified form, in the propositions below.

Example 3.4.2. In three dimensions, the subsimplex $S \subset T$ defined by a vertex of T and the opposite medial triangle is 2-averageable.

Indeed, we can take H_1 to be the identity map and H_2 to be the linear map fixing the common vertex v of S and T and cycling the remaining vertices $v_1 \mapsto v_2 \mapsto v_3 \mapsto v_1$. Then $\frac{H_1+H_2}{2}$ is a linear map sending the vertices of T to the vertices of S, and is thus a constant Jacobian $\frac{1}{4}$ map $T \to S$.



Recall that for $m \in \{1,\dots,k\}$, we denote by $P_{k,m} = [0,1]^{k+1} \cap \{\sum x_i = m\}$ for the m'th k-dimensional hypersimplex . The following two propositions reduce the theorems from the introduction to showing that certain hypersimplices embedded in T are m-averageable.

Proposition 3.4.3. Let T be the convex hull of the standard basis vectors in $\mathbb{R}^{k+1}$, and suppose that $\frac{1}{2}P_{k,2}$ is 2-averageable. Then for bounded functions $f,g : T \to \mathbb{R}$ with $f \leqslant 0$ and $f(x_i) = 0$ for the vertices x_i of T, we have

$$\int_T f * g(x) - \frac{f(x)+g(x)}{2} dx \geqslant \frac{k+1}{2^{k+1}} \int_T \mathrm{co}(f)(x) - f(x) dx.$$

Proposition 3.4.4. Let T be the convex hull of the standard basis vectors in $\mathbb{R}^{k+1}$, and suppose that $\frac{1}{m}P_{k,m}$ is m-averageable for $m \leqslant \min(k,n)$. Then for a bounded function $f : T \to \mathbb{R}_{\leqslant 0}$ with $f(x_i) = 0$ for the vertices x_i of T, we have

$$\int_T f^{*n}(x) - f(x) dx \geqslant c_{k,n} \int_T \mathrm{co}(f)(x) - f(x) dx$$

where $c_{k,n}$ is as in Conjecture 3.1.5.

Proof of Proposition 3.4.3. Consider the $n = 2$ polytope decomposition from Proposition 3.3.2

$$T = \frac{1}{2}P_{k,2} \cup \bigcup_{i=1}^{k+1} \frac{e_i + T}{2}.$$

Because by hypothesis $\frac{1}{2}P_{k,2}$ is 2-averageable, there are functions H_1, H_2 such that $H_1, H_2 : T \to T$ are generically bijective with Jacobian 1, and $\frac{H_1+H_2}{2} : T \to S$ is generically bijective with Jacobian $\frac{|\frac{1}{2}P_{k,2}|}{|T|} = 1 - \frac{k+1}{2^k}$. Then the result follows by adding the inequality

$$\begin{aligned}
\int_{\frac{1}{2}P_{k,2}} (f*g)(x) dx &= \left(1 - \frac{k+1}{2^k}\right) \int_T f * g \left(\frac{H_1(x)+H_2(x)}{2}\right) dx \\
&\geqslant \left(1 - \frac{k+1}{2^k}\right) \frac{\int_T f(H_1(x))dx + \int_T g(H_2(x))dx}{2} \\
&= \left(1 - \frac{k+1}{2^k}\right) \frac{\int_T f(x)dx + \int_T g(x)dx}{2}
\end{aligned}$$

to the inequalities

$$\int_{\frac{e_i+T}{2}} (f*g)(x)dx = \frac{1}{2^k}\int_T f*g\left(\frac{e_i+x}{2}\right)dx \geqslant \frac{1}{2^{k+1}}\int_T f(e_i)+g(x)dx = \frac{1}{2^{k+1}}\int_T g(x)dx$$

for $i = 1,\ldots,k+1$. □

Proof of Proposition 3.4.4. By Proposition 3.3.2, there is a polytope subdivision

$$T = \bigcup_{m=1}^{\min(k,n)} \bigcup_{x\in\mathscr{B}_{k,n-m}} \frac{1}{n}P_{k,m} + \frac{1}{n}x$$

where

$$\mathscr{B}_{k,\ell} = \left\{x = (x_1,\ldots,x_{k+1}) \in \mathbb{Z}_{\geqslant 0}^{k+1} : \sum x_i = \ell\right\}.$$

Let $H_{k,m}^1,\ldots,H_{k,m}^m$ be the functions associated to the m-averageable set $\frac{1}{m}P_{k,m}$. Then

$$\begin{aligned}
\int_{\frac{1}{n}P_{k,m}+\frac{1}{n}x} f^{*n}(x)dx &= \frac{1}{n^k}\cdot\frac{|P_{k,m}|}{|T|}\int_T f^{*n}\left(\frac{H_{k,m}^1(x)+\ldots+H_{k,m}^m(x)+x_1e_1+\ldots+x_{k+1}e_{k+1}}{n}\right)dx \\
&\geqslant \frac{1}{n^k}\cdot\frac{|P_{k,m}|}{|T|}\int_T \frac{f(H_{k,m}^1(x))+\ldots+f(H_{k,m}^m(x))+x_1f(e_1)+\ldots+x_{k+1}f(e_{k+1})}{n}dx \\
&= A(k,m-1)\cdot\frac{m}{n^{k+1}}\int_T f(x)dx.
\end{aligned}$$

The last equality uses the fact that $|P_{k,m}|/|T| = |P_{k,m}|/|P_{k,1}| = A(k,m-1)$ (see Fact 5.3 of [139]), where $A_{k,m-1}$ represents the number of permutation of $\{1,2,\ldots,k\}$ with $m-1$ descents.

Recalling that $\mathrm{co}(f) = 0$, summing these inequalities and using the Worpitzky identity that $\sum_m |\mathscr{B}_{k,n-m}|A(k,m-1) = \sum_m \binom{n+k-m}{k}A(k,m-1) = n^k$ yields the desired result. □

3.5 Proofs of Theorem 3.1.1 and Theorem 3.1.2

By the propositions in the previous section, it suffices to show that

$$\begin{aligned}
P_{1,1} &\subset \mathbb{R}^2 \\
P_{2,1}, \frac{1}{2}P_{2,2} &\subset \mathbb{R}^3 \\
P_{3,1}, \frac{1}{2}P_{3,2}, \frac{1}{3}P_{3,3} &\subset \mathbb{R}^4
\end{aligned}$$

are all m-averageable in the corresponding convex hull of standard basis vectors T for $m = 1$, $m = 1,2$, and $m = 1,2,3$ respectively.

The following lemma handles all cases except for $\frac{1}{2}P_{3,2}$.

Lemma 3.5.1. $P_{k,1}$ is 1-averageable and $\frac{1}{k}P_{k,k}$ is k-averageable for all k.

Proof. For $n = 1$, $P_{k,1} = T$ so we may take H to be the identity map and there is nothing to prove.

For $n = k$, let σ be the linear map taking $e_1 \mapsto e_2 \mapsto \ldots \mapsto e_{k+1} \mapsto e_1$. Then σ is an isometry, and so $H_i = \sigma^i$ is also an isometry. The average

$$\frac{H_1 + \ldots + H_k}{k}$$

is the linear map taking $e_i \mapsto \frac{1}{k}\sum_{j \neq i} e_j$, which is a linear bijection from the simplex T to the simplex $\frac{1}{k}P_{k,k}$. □

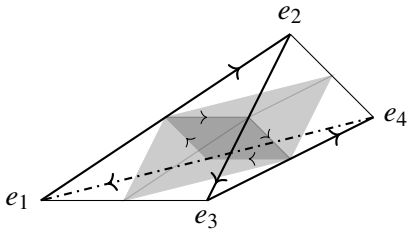
The following lemma therefore completes the proofs of Theorem 3.1.1 and Theorem 3.1.2.

Lemma 3.5.2. $\frac{1}{2}P_{3,2}$ is 2-averageable.

Proof. Decompose $T = R_{12} \cup R_{23} \cup R_{34} \cup R_{41}$ where $R_{i(i+1)}$ is the simplex

$$R_{i(i+1)} = \mathrm{co}\left(e_i, e_{i+1}, \frac{e_1 + e_3}{2}, \frac{e_2 + e_4}{2}\right).$$

Indeed, viewing T as the 1-dimensional cycle connecting $e_1 \to e_2 \to e_3 \to e_4 \to e_1$ coned off at the points $\frac{e_1+e_3}{2}$ and $\frac{e_2+e_4}{2}$, $R_{i(i+1)}$ corresponds to the line segment connecting $e_i \to e_{i+1}$ coned off at the points $\frac{e_1+e_3}{2}$ and $\frac{e_2+e_4}{2}$.



Let H_1 be the identity map and $H_2 : T \to T$ be the piecewise linear local isometry defined by taking $R_{i(i+1)} \mapsto R_{(i+1)(i+2)}$, sending the vertices $e_i, e_{i+1}, \frac{e_1+e_3}{2}, \frac{e_2+e_4}{2}$ to $e_{i+1}, e_{i+2}, \frac{e_1+e_3}{2}, \frac{e_2+e_4}{2}$, respectively.

Then $\frac{H_1+H_2}{2}$ takes $R_{i(i+1)}$ to

$$S_{i(i+1)(i+2)} = \mathrm{co}\left(\frac{e_i + e_{i+1}}{2}, \frac{e_{i+1} + e_{i+2}}{2}, \frac{e_1 + e_3}{2}, \frac{e_2 + e_4}{2}\right),$$

and the simplices $S_{123}, S_{234}, S_{341}, S_{412}$ subdivide $\frac{1}{2}P_{3,2}$. Indeed, the octahedron $\frac{1}{2}P_{3,2}$ can be described as the one-dimensional cycle around the boundary of the square $\frac{e_1+e_2}{2} \to \frac{e_2+e_3}{2} \to \frac{e_3+e_4}{2} \to \frac{e_4+e_1}{2} \to \frac{e_1+e_2}{2}$ coned off at the points $\frac{e_1+e_3}{2}$ and $\frac{e_2+e_4}{2}$, and $S_{i(i+1)(i+2)}$ is the segment connecting $\frac{e_i+e_{i+1}}{2}$ and $\frac{e_{i+1}+e_{i+2}}{2}$ coned off at the points $\frac{e_1+e_3}{2}$ and $\frac{e_2+e_4}{2}$.

Hence $\frac{H_1+H_2}{2}$ is a bijection, and by symmetry has almost everywhere constant Jacobian. This shows $\frac{1}{2}P_{3,2}$ is 2-averageable as desired. □

3.6 Asymptotics for $c_{n,k}$ for k fixed and n large

In this section we prove Theorem 3.1.7 that for $n \geqslant k+1$ we have

$$c_{k,n} \geqslant 1 - \binom{n}{k} \frac{k^{k+1}}{n^{k+1}}.$$

Proof of Theorem 3.1.7. Indeed, it suffices to show this $c_{k,n}$ works for functions f on a simplex $C = T$ with $f = 0$ at the vertices and $f \leqslant 0$ everywhere by Section 3.2. Set T to be the convex hull of the standard basis vectors $e_1, \ldots, e_{k+1}$ in $\mathbb{R}^{k+1}$.

First, using the notation from Definition 3.3.1, we claim that we have a covering

$$T = \bigcup_{v \in \mathscr{B}_{k,n-k}} \frac{kT + v}{n}.$$

Indeed, take $y \in T$, and consider ny. We can write $ny = w_1 + \lfloor ny \rfloor$, and $\sum(\lfloor ny \rfloor)_i \geqslant n-k$. Write $\lfloor ny \rfloor = v + w_2$ with v, w_2 non-negative integral vectors such that $\sum v_i = n-k$. Then

$$y = \frac{(w_1 + w_2) + v}{n},$$

with $w_1 + w_2 \in kT$ and $v \in \mathscr{B}_{k,n-k}$.

We can then write

$$\begin{aligned}
\int_T f^{*n} &\geqslant \sum_{v \in \mathscr{B}_{k,n-k}} \int_{\frac{kT+v}{n}} f^{*n}(x)dx \\
&= \sum_{v \in \mathscr{B}_{k,n-k}} \left(\frac{k}{n}\right)^k \int_T f^{*n}\left(\frac{kx + v_1 e_1 + \ldots + v_n e_n}{n}\right) dx \\
&\geqslant \sum_{v \in \mathscr{B}_{k,n-k}} \left(\frac{k}{n}\right)^k \int_T \frac{kf(x) + v_1 f(e_1) + \ldots + v_{k+1} f(e_{k+1})}{n} dx \\
&= \binom{n}{k} \frac{k^{k+1}}{n^{k+1}} \int_T f(x)dx.
\end{aligned}$$

As $\mathrm{co}(f) = 0$ we can rearrange this to

$$\int_T f^{*n}(x) - f(x)dx \geqslant \left(1 - \binom{n}{k} \frac{k^{k+1}}{n^{k+1}}\right) \int_T \mathrm{co}(f)(x) - f(x)dx.$$

□

3.7 Non-sharp $c_{k,n}$ for f^{*n} for all k,n

We now discuss the existence of a non-sharp constant $c_{k,n} > 0$ in all dimensions, i.e., that for all compact convex $C \subset \mathbb{R}^k$ and bounded measurable $f : C \to \mathbb{R}$, we have

$$\int f^{*n}(x) - f(x) \geqslant c_{k,n} \int_C \mathrm{co}(f)(x) - f(x) dx.$$

We note that we have $f^{*n}(x) \geqslant f(x)$ for all $x \in C$. We can immediately deduce the existence of such constants from following result on the stability of Brunn–Minkowski for homothetic regions.

Theorem 3.7.1 ([98])**.** For any $k \in \mathbb{N}$ and $t \in (0,1)$, there are constants $c(k,t), d(k,t) > 0$ such that for any $A \subset \mathbb{R}^{k+1}$ of positive measure if $|tA + (1-t)A| - |A| \leqslant d(k,t)|A|$, then

$$|tA + (1-t)A| - |A| \geqslant c(k,t)|\mathrm{co}(A) \setminus A|,$$

where we write $\mathrm{co}(A)$ for the convex hull of A.

Indeed, the existence of the constant $c_{k,n}$ for sup-convolution then follows by applying this theorem to the set $A = A_{f,-N}$ where

$$A_{f,\lambda} = \{(x,y) \in C \times \mathbb{R} : \lambda \leqslant y \leqslant f(x)\},$$

$t = \frac{1}{n}$, and $-N \leqslant \min(f)$ is sufficiently small so that the $d(k, \frac{1}{n})$ bound is satisfied (in fact this shows that we have a lower bound even if we restricted in the n-fold sup-convolution that $x_1 = \ldots = x_{n-1}$).

For the benefit of the reader, we present here a simpler, more direct argument. Recall from Section 3.2 that it suffices to prove the theorem when we have the domain $C = T$ is a simplex, $f = 0$ at the vertices of T, and $f \leqslant 0$.

Call a translate of $T' \subset T$ of $\frac{1}{n^\ell} T$ "good" if there exists an absolute constant $C_{T'}$ (independent of f) such that

$$\int_{T'} f(x) dx \geqslant \frac{1}{n^{\ell(k+1)}} \int_T f(x) dx - C_{T'} \int_T f^{*n}(x) - f(x) dx.$$

We make the following observations

1. T is good.
2. If T' is good and v is a vertex of T, then $\frac{(n-1)v+T'}{n}$ is good
3. If T', T'' are good and of the same size, then $\frac{T'+(n-1)T''}{n}$ is good.

The first observation is trivial. For the second, we note that

$$\begin{aligned}\int_{\frac{(n-1)v+T'}{n}} f(x)dx &\geqslant \int_{\frac{(n-1)v+T'}{n}} f^{*n}(x)dx - \int_T f^{*n}(x) - f(x)dx \\ &\geqslant \frac{1}{n^k}\int_{T'} \frac{(n-1)f(v)+f(x)}{n}dx - \int_T f^{*n}(x) - f(x)dx \\ &= \frac{1}{n^{k+1}}\int_{T'} f(x)dx - \int_T f^{*n}(x) - f(x)dx \\ &\geqslant \frac{1}{n^{(\ell+1)(k+1)}}\int_T f(x)dx - \left(1+\frac{C_{T'}}{n^{k+1}}\right)\int_T f^{*n}(x) - f(x)dx\end{aligned}$$

For the third observation, we note that

$$\begin{aligned}\int_{\frac{(n-1)T'+T''}{n}} f(x)dx &\geqslant \int_{\frac{(n-1)T'+T''}{n}} f^{*n}(x)dx - \int_T f^{*n}(x) - f(x)dx \\ &\geqslant \frac{(n-1)\int_{T'} f(x)dx + \int_{T''} f(x)dx}{n} - \int_T f^{*n}(x) - f(x)dx \\ &\geqslant \frac{1}{n^{\ell(k+1)}}\int_T f(x)dx - \left(1+\frac{(n-1)C_{T'}+C_{T''}}{n}\right)\int_T f^{*n}(x) - f(x)dx.\end{aligned}$$

Assume that for some $\ell \in \mathbb{N}$ we have a family $\mathscr{A}$ of good translates of $\frac{1}{n^\ell}T$ which cover T, such that $|\mathscr{A}| \leqslant 2^{-1}n^{\ell(k+1)}$. By definition, for each $T' \in \mathscr{A}$ we have

$$\int_{T'} f(x)dx \geqslant \frac{1}{n^{\ell(k+1)}}\int_T f(x)dx - C_{T'}\int_T f^{*n}(x) - f(x)dx.$$

Therefore, we deduce

$$\sum_{T'\in\mathscr{A}}\int_{T'} f(x)dx \geqslant \frac{|\mathscr{A}|}{n^{\ell(k+1)}}\int_T f(x)dx - \Big(\sum_{T'\in\mathscr{A}} C_{T'}\Big)\int_T f^{*n}(x) - f(x)dx$$

Recalling $f \leqslant 0$ and $\mathscr{A}$ covers T, we deduce

$$\int_T f(x)dx \geqslant \frac{|\mathscr{A}|}{n^{\ell(k+1)}}\int_T f(x)dx - \Big(\sum_{T'\in\mathscr{A}} C_{T'}\Big)\int_T f^{*n}(x) - f(x)dx.$$

Recalling $|\mathscr{A}| \leqslant 2^{-1}n^{\ell(k+1)}$ after rearranging we get

$$\Big(\sum_{T'\in\mathscr{A}} C_{T'}\Big)\int_T f^{*n}(x) - f(x)dx \geqslant -\frac{1}{2}\int_T f(x)dx.$$

Finally, recalling that $\mathrm{co}(f) = 0$ as f is 0 at the vertices of T and $f \leqslant 0$, we conclude

$$\Big(\sum_{T'\in\mathscr{A}} C_{T'}\Big)\int_T f^{*n}(x) - f(x)dx \geqslant \frac{1}{2}\int_T \mathrm{co}(f)(x) - f(x)dx.$$

Ir remains to explain why such a family $\mathscr{A}$ exists. From the second and third observations, for every face F of T (including T), the set of good translates of $\frac{1}{n^\ell}T$ is dense among the set of all translates of $\frac{1}{n^\ell}T$ incident to F. Together with the fact that simplices have a bounded inefficiency of covering space, we are able to accomplish this task for a sufficiently large ℓ. Indeed, as each simplex $\frac{1}{n^\ell}T$ covers a $\frac{1}{n^{\ell k}}$ volume of T, standard results from covering theory imply that we can find a family $\mathscr{A}$ with $|\mathscr{A}| = O(n^{\ell k}) < 2^{-1}n^\ell \cdot n^{\ell k} = 2^{-1}n^{\ell(k+1)}$ for ℓ sufficiently large.

3.8 Conjecture 3.1.5

We begin by noting that, if true, Conjecture 3.1.5 is sharp. Indeed, let $C = T$ be a simplex and take $f : T \to \mathbb{R}$ be the indicator function on the vertices of T such that $\mathrm{co}(f) = 1$. Proposition 3.3.2 gives a polytopal subdivision $T = \bigcup_{m=1}^{\min(k,n)} \bigcup_{v\in\mathscr{B}_{k,n-m}} \frac{1}{n}P_{k,m} + \frac{1}{n}v$ where $P_{k,m}$ appears $\binom{n+k-m}{k}$ times. Fact 5.3 of [139] states that $|P_{k,m}|/|T| = A(k,m-1)$. Proposition 3.3.3 shows that $f^{*n}(x) = (n-m)/n$ for $x \in \frac{1}{n}P_{k,m} + \frac{1}{n}v$ where $v \in \mathscr{B}_{k,n-m}$. We deduce

$$\begin{aligned}
\int_T f^{*n}(x) - f(x)dx &= \int_T f^{*n}(x)dx \\
&= \sum_{m=1}^{\min(k,n)} \sum_{v\in\mathscr{B}_{k,n-m}} \int_{\frac{1}{n}P_{k,m}+\frac{1}{n}v} f^{*n}(x)dx \\
&= \sum_{m=1}^{k} \binom{n+k-m}{k} \left(\frac{1}{n^k}A(k,m-1)|T|\right)\frac{n-m}{n} \\
&= c_{k,n}|T| = c_{k,n}\int_T \mathrm{co}(f)(x) - f(x)dx.
\end{aligned}$$

We proved Theorem 3.1.1 for $k \leqslant 3$ by proving that $\frac{1}{m}P_{k,m}$ are m-averageble for $k \leqslant 3$ and $m \leqslant k$. In general, one could potentially prove Conjecture 3.1.5 for all $k \in \mathbb{N}$ by proving that $\frac{1}{m}P_{k,m}$ are m-averageble for all $k \in \mathbb{N}$ and $m \leqslant k$.

However, we do not believe that $\frac{1}{m}P_{k,m}$ are m-averageble for all $k \in \mathbb{N}$ and $m \leqslant k$. To give some motivation for this, let us rule out the following natural candidates for the piecewise linear functions $H_1,\ldots,H_m$. Consider a polytopal subdivision $T = T_1 \cup \ldots \cup T_s$ obtained by conning of at points $x_1,\ldots,x_t$ which are the centers of some disjoint (lower dimensional) faces $F_1,\ldots,F_t$, respectively. Assume that the piecewise linear function H_i is constructed such that H_i fixes the points $x_1,\ldots,x_t$ and the restrictions $H_i : T_j \to T_{\sigma_i(j)}$ are isometries, for some permutation $\sigma_i : [s] \to [s]$. Construct the function $H = \frac{H_1+\ldots+H_m}{m}$. Assuming that $H : T \to \frac{1}{m}P_{k,m}$ is a generically bijective linear function, we get that the vertices of $\frac{1}{m}P_{k,m}$ are among $H(v_1),\ldots,H(v_{k+1}),H(x_1),\ldots,H(x_t)$, where $v_1,\ldots,v_{k+1}$ are the verices of T and $x_1,\ldots,x_t$ are the points where we cone off T. However, the number of vertices of $\frac{1}{m}P_{k,m}$ is $\binom{(k+1)}{m}$ and the maximal set of points from which we can cone off is $(k+1)/2$. When k is odd and large and $m = (k+1)/2$ we obtain a contradiction from the fact that $\binom{(k+1)}{(k+1)/2}$ is much larger than $\frac{3}{2}(k+1)$.

Chapter 4

On the Erdős Covering Problem: The density of the uncovered set

The work in this section is joint with Paul Balister, Béla Bollobás, Robert Morris and Julian Sahasrabudhe and has been published in Inventiones mathematicae [6].

4.1 Introduction

In this chapter we develop a general method, which is based on that of Hough [100], for bounding the density of the set left uncovered by a collection of arithmetic progressions, which we use to prove Theorems 1.2.2 to 1.2.8 (we defer Theorem 1.2.7 to the next chapter). These results treat several well known open problems on covering systems of Erdős [55], Erdős and Selfridge [75], Schinzel [136] and Filaseta, Ford, Konyagin, Pomerance and Yu [76]. For convenience, we restate Theorems 1.2.2 to 1.2.8 here as Theorems 4.1.1 to 4.1.7.

Throughout the thesis we exclude the trivial arithmetic progression $\mathbb{Z}$.

Theorem 4.1.1. Let $\varepsilon > 0$ and let μ be the multiplicative function defined by

$$\mu(p^i) = 1 + \frac{(\log p)^{3+\varepsilon}}{p} \tag{4.1}$$

for all primes p and integers $i \geqslant 1$. There exists $M > 0$ so that if $A_1, \ldots, A_k$ are arithmetic progressions with distinct moduli $d_1, \ldots, d_k \geqslant M$, and

$$C = \sum_{i=1}^{k} \frac{\mu(d_i)}{d_i},$$

then the density of the uncovered set $R := \mathbb{Z} \setminus \bigcup_{i=1}^{k} A_i$ is at least $e^{-4C}/2$.

Theorem 4.1.2. For every $M > 0$ and $\varepsilon > 0$, there exists a finite collection of arithmetic progressions $A_1, \ldots, A_k$ with distinct moduli $d_1, \ldots, d_k \geqslant M$, such that

$$\sum_{i=1}^{k} \frac{1}{d_i} < 1 \tag{4.2}$$

and the density of the uncovered set $R = \mathbb{Z} \setminus \bigcup_{i=1}^{k} A_i$ is less than ε.

Theorem 4.1.3. For every $\lambda > 0$, there exists $C = C(\lambda) > 0$ such that the following holds. Let μ be the multiplicative function defined by

$$\mu(p^i) = 1 + \frac{\lambda}{p}$$

for all primes p and integers $i \geqslant 1$. For every $M > 0$ and $\varepsilon > 0$, there exists a finite collection of arithmetic progressions $A_1, \ldots, A_k$ with distinct square-free moduli $d_1, \ldots, d_k \geqslant M$, such that

$$\sum_{i=1}^{k} \frac{\mu(d_i)}{d_i} \leqslant C,$$

and the density of the uncovered set $R = \mathbb{Z} \setminus \bigcup_{i=1}^{k} A_i$ is at most ε.

Theorem 4.1.4. Let $\mathscr{A}$ be a finite collection of arithmetic progressions with distinct moduli $d_1, \ldots, d_k \geqslant 616000$. Then $\mathscr{A}$ does not cover the integers.

In the following results we do not allow a modulus to be 1.

Theorem 4.1.5. If $\mathscr{A}$ is a finite collection of arithmetic progressions that covers the integers, then at least one of the moduli divides another.

Theorem 4.1.6. If $\mathscr{A}$ is a finite collection of arithmetic progressions with distinct square-free moduli that covers the integers, then at least one of the moduli is even.

Theorem 4.1.7. Let $\mathscr{A} = \{A_d : d \in D\}$ be a finite collection of arithmetic progressions with distinct moduli that covers the integers, and let $Q = \operatorname{lcm}(D)$ be the least common multiple of the moduli. Then either $2 \mid Q$, or $9 \mid Q$, or $15 \mid Q$.

The rest of this chapter is organized as follows: in Section 4.2 we outline the sieve that we use in the proofs, and in Section 4.3 we state and prove our main technical results, Theorems 4.3.1 and 4.3.2. In Section 4.4 we complete the proof of Theorem 4.1.1, and in Section 4.5 we prove a variant of the main result of [76]. In Section 4.6 we explain how our sieve can be used to prove the non-existence of coverings sets with certain properties, and in Sections 4.7–4.9 we use this method to prove Theorems 4.1.4, 4.1.5 and 4.1.7 (we leave the proof of Theorem 4.1.6 to the next chapter). Finally, in Section 4.10, we provide the constructions for Theorem 4.1.2 and Theorem 4.1.3.

4.2 The Sieve

In this section we outline the proof of Theorem 4.1.1. We consider a finite set of moduli denoted by $D \subseteq \mathbb{N} \setminus \{1\}$ and a finite collection $\mathscr{A} = \{A_d : d \in D\}$ of arithmetic progressions, where $A_d = a_d + d\mathbb{Z}$ is an arithmetic progression with modulus d. The goal is to estimate the density of the uncovered set

$$R := \mathbb{Z} \setminus \bigcup_{d \in D} A_d.$$

Rather than considering the entire collection of progressions $\mathscr{A}$ all at once, we expose the progressions 'prime by prime' and track how the density of the uncovered set evolves. To be more precise, let $p_1, p_2, \ldots, p_n$ be the distinct prime factors of $Q := \operatorname{lcm}(D)$ (usually, but not necessarily, listed in increasing order) so that

$$Q = \prod_{j=1}^{n} p_j^{\gamma_j}$$

for some integers $\gamma_j \geqslant 1$. Define, for each $0 \leqslant i \leqslant n$,

$$Q_i := \prod_{j=1}^{i} p_j^{\gamma_j}$$

and write

$$D_i := \big\{d \in D : d \mid Q_i\big\} \qquad \text{and} \qquad \mathscr{A}_i := \big\{A_d : d \in D_i\big\}$$

for the set of differences and the corresponding arithmetic progressions whose prime factorization only includes the first i of these primes. (In particular, $Q_0 = 1$ and $D_0 = \mathscr{A}_0 = \emptyset$.) Note that, although $\operatorname{lcm}(D_i) \mid Q_i$, we do not necessarily have $\operatorname{lcm}(D_i) = Q_i$, since earlier primes can occur to higher powers in later moduli. Let

$$R_i := \mathbb{Z} \setminus \bigcup_{d \in D_i} A_d = \mathbb{Z} \setminus \bigcup_{A_d \in \mathscr{A}_i} A_d,$$

be the set of elements not contained in any of the progressions of $\mathscr{A}_i$, so that $R_0 = \mathbb{Z}$ and $R_n = R$. We also write $N_i := D_i \setminus D_{i-1}$ for the set of "new" differences at the ith stage, and define

$$B_i := \bigcup_{d \in N_i} A_d \tag{4.3}$$

to be the union of the arithmetic progressions exposed at step i, so that $R_i = R_{i-1} \setminus B_i$.

It is convenient to consider R_i as a subset of the cyclic group $\mathbb{Z}_{Q_i}$ (or of $\mathbb{Z}_Q$), which is possible because for each $d \in D_i$ the set A_d is periodic with period $d \mid Q_i$. In particular, note that the density of R_i in $\mathbb{Z}$ is equal to the measure of the set R_i in the uniform probability measure on the finite set $\mathbb{Z}_{Q_i}$. During the proof we in fact need to consider *non-uniform* probability measures $\mathbb{P}_i$ on $\mathbb{Z}_{Q_i}$; note that each such measure can be extended (uniformly on each congruence class mod Q_i) to a probability

measure on $\mathbb{Z}_Q$.[1] We borrow (and abuse) terminology from measure theory by calling a subset $S \subseteq \mathbb{Z}_Q$ (or a Q-periodic set $S \subseteq \mathbb{Z}$) *Q_i-measurable* if S is a union of congruence classes mod Q_i.

4.2.1 A sketch of the method

The basic idea is quite simple. We construct measures $\mathbb{P}_i$ in such a way that $\mathbb{P}_i(B_i)$ is small, but without changing the measure of B_j for any $j < i$. It follows that the measure of $\mathbb{Z} \setminus R$ in the final measure $\mathbb{P}_n$ is at most $\sum_i \mathbb{P}_i(B_i)$, and thus if this quantity is less than 1, it follows that the arithmetic progressions do not cover $\mathbb{Z}$.

To bound $\mathbb{P}_i(B_i)$, we use the 1st and 2nd moment methods (see Lemma 4.3.3, below). More precisely, we bound the expectation (in the measure $\mathbb{P}_{i-1}$) of the proportion of the 'fibre' $F(x) = \left\{(x,y) : y \in \mathbb{Z}_{p_i^{\gamma_i}}\right\}$ of $x \in \mathbb{Z}_{Q_{i-1}}$ removed in step i, and the expectation of the square of this quantity. Bounding these moments is not too difficult, see Lemmas 4.3.6 and 4.3.7, below.

Finally, to deduce a lower bound on the uncovered set in the *uniform* measure, $\mathbb{P}_0$, we need to bound the average distortion $\mathbb{P}_n(x)/\mathbb{P}_0(x)$. We design the measures $\mathbb{P}_i$ so that the 'average' here (which we take in the $\mathbb{P}_n$-measure) is enough (by a convexity argument) to give such a lower bound, see Lemma 4.3.5.

4.2.2 The probability measures $\mathbb{P}_i$

We next define the non-uniform probability measures $\mathbb{P}_i$, which are inspired by (but different in important ways to) a sequence of measures used in [100], and which play a key role in the proof of Theorem 4.1.1. The rough idea is that we would like to distort the space so as to 'blow up' the uncovered set, but without increasing the measure of any single point too much. More precisely, we define the measures inductively, choosing $\mathbb{P}_i$ so that it agrees with $\mathbb{P}_{i-1}$ on Q_{i-1}-measurable sets, is not too much larger that $\mathbb{P}_{i-1}$ anywhere, and (subject to these conditons) is as small as possible on the set B_i, the union of the arithmetic progressions removed at step i.

First, let $\mathbb{P}_0$ be the trivial probability measure on $\mathbb{Z}_{Q_0} = \mathbb{Z}_1$, or (equivalently, by the comments above) the uniform measure on $\mathbb{Z}_Q$. Now fix $\delta_1, \dots, \delta_n \in [0, 1/2]$, let $i \in [n]$, and suppose we have already defined a probability measure $\mathbb{P}_{i-1}$ on $\mathbb{Z}_{Q_{i-1}}$. Our task is to define a probability measure $\mathbb{P}_i$ on $\mathbb{Z}_{Q_i}$.

In order to view R_{i-1} and R_i as subsets of the same set, let us (by the Chinese Remainder Theorem) represent $\mathbb{Z}_{Q_i}$ as $\mathbb{Z}_{Q_{i-1}} \times \mathbb{Z}_{p_i^{\gamma_i}}$, and denote elements of $\mathbb{Z}_{Q_i}$ by pairs (x,y), where $x \in \mathbb{Z}_{Q_{i-1}}$ and $y \in \mathbb{Z}_{p_i^{\gamma_i}}$. We may view R_{i-1} as a collection of 'fibres' of the form $F(x) = \{(x,y) : y \in \mathbb{Z}_{p_i^{\gamma_i}}\}$, noting that $\mathbb{P}_{i-1}$ is uniform on each fibre, and view R_i as being obtained from R_{i-1} by removing the points that are contained in the new progressions of $\mathscr{A}_i \setminus \mathscr{A}_{i-1}$.

[1] To be precise, we can set $Q \cdot \mathbb{P}_i(x + Q\mathbb{Z}) := Q_i \cdot \mathbb{P}_i(x + Q_i\mathbb{Z})$. Note that, since $\gcd(Q_i, Q/Q_i) = 1$, we can (via the Chinese Remainder Theorem) consider $\mathbb{P}_i$ on $\mathbb{Z}_Q \cong \mathbb{Z}_{Q_i} \times \mathbb{Z}_{Q/Q_i}$ as a product measure of $\mathbb{P}_i$ on $\mathbb{Z}_{Q_i}$ with the uniform measure on $\mathbb{Z}_{Q/Q_i}$.

Now, for each $x \in \mathbb{Z}_{Q_{i-1}}$, define

$$\alpha_i(x) = \frac{\mathbb{P}_{i-1}\big(F(x) \cap B_i\big)}{\mathbb{P}_{i-1}(x)} = \frac{\big|\big\{y \in \mathbb{Z}_{p_i^{\gamma_i}} : (x,y) \in B_i\big\}\big|}{p_i^{\gamma_i}}, \tag{4.4}$$

that is, the proportion of the fibre $F(x)$ that is removed at stage i. The probability measure $\mathbb{P}_i$ on $\mathbb{Z}_{Q_i}$ is defined as follows:

$$\mathbb{P}_i(x,y) := \begin{cases} \max\left\{0, \dfrac{\alpha_i(x)-\delta_i}{\alpha_i(x)(1-\delta_i)}\right\} \cdot \mathbb{P}_{i-1}(x,y), & \text{if } (x,y) \in B_i; \\ \min\left\{\dfrac{1}{1-\alpha_i(x)}, \dfrac{1}{1-\delta_i}\right\} \cdot \mathbb{P}_{i-1}(x,y), & \text{if } (x,y) \notin B_i. \end{cases} \tag{4.5}$$

This is an important (and slightly technical) definition, and therefore deserves some additional explanation. First, observe that if $\alpha_i(x) \leqslant \delta_i$, then $\mathbb{P}_i(x,y) = 0$ for every element of Q_i that is covered in step i, and that the measure is increased proportionally elsewhere to compensate. On the other hand, for those $x \in \mathbb{Z}_{Q_{i-1}}$ for which $\alpha_i(x) > \delta_i$, we 'cap' the distortion by increasing the measure at each point not covered in step i by a factor of $1/(1-\delta_i)$, and decreasing the measure on removed points by a corresponding factor.

The following simple properties of the measure $\mathbb{P}_i$ will be useful in the proof.

Lemma 4.2.1. $\mathbb{P}_i(S) = \mathbb{P}_{i-1}(S)$ for any Q_{i-1}-measurable set S.

Proof. Let $x \in \mathbb{Z}_{Q_{i-1}}$, and observe first that if $\alpha_i(x) \leqslant \delta_i$, then

$$\mathbb{P}_i(x) = \Big(\alpha_i(x) \cdot 0 + \big(1-\alpha_i(x)\big) \cdot \frac{1}{1-\alpha_i(x)}\Big) \cdot \mathbb{P}_{i-1}(x) = \mathbb{P}_{i-1}(x),$$

where $\mathbb{P}_i(x) = \sum_{(x,y)\in F(x)} \mathbb{P}_i(x,y)$. On the other hand, if $\alpha_i(x) > \delta_i$ then

$$\mathbb{P}_i(x) = \Big(\alpha_i(x) \cdot \frac{\alpha_i(x)-\delta_i}{\alpha_i(x)(1-\delta_i)} + \big(1-\alpha_i(x)\big) \cdot \frac{1}{1-\delta_i}\Big) \cdot \mathbb{P}_{i-1}(x) = \mathbb{P}_{i-1}(x).$$

Summing over $x \in S$, we obtain $\mathbb{P}_i(S) = \mathbb{P}_{i-1}(S)$, as claimed. □

Lemma 4.2.2. For any set $S \subseteq \mathbb{Z}_Q$, we have

$$\mathbb{P}_i(S) \leqslant \frac{1}{1-\delta_i} \cdot \mathbb{P}_{i-1}(S). \tag{4.6}$$

Moreover, if $S \subseteq B_i$ then

$$\mathbb{P}_i(S) \leqslant \mathbb{P}_{i-1}(S). \tag{4.7}$$

Proof. Both inequalities follow immediately (for each 'atom' $S = \{(x,y)\}$, and hence also in general, by additivity) from (4.5). □

Given a function $f\colon \mathbb{Z}_Q \to \mathbb{R}_{\geqslant 0}$, let us define the expectation of f with respect to $\mathbb{P}_i$ to be

$$\mathbb{E}_i\big[f(x)\big] := \sum_{x\in\mathbb{Z}_Q} f(x)\,\mathbb{P}_i(x).$$

By the observations above, we have

$$\mathbb{E}_i\big[f(x)\big] \leqslant \frac{1}{1-\delta_i}\cdot \mathbb{E}_{i-1}\big[f(x)\big],$$

and moreover $\mathbb{E}_i[f(x)] \leqslant \mathbb{E}_{i-1}[f(x)]$ if f is supported on B_i, and $\mathbb{E}_i[f(x)] = \mathbb{E}_{i-1}[f(x)]$ if f is Q_{i-1}-measurable.

4.2.3 A more precise sketch of the method

Having defined the non-uniform probability measures $\mathbb{P}_i$, we can give a high level overview of why the method works, for example, showing that in any covering system $\mathscr{A} = \{a_d \bmod d : d \in D\}$ with distinct square free moduli, the minimum modulus is bounded. In this subsection we write p_i for the i-th prime for all i. Fix any small constant $\delta \geqslant 0$ and set parameters $\delta_i = \delta$ for all i. The strategy is divided into three parts.

First we note that we always have $\mathbb{P}_i(B_i) \leqslant \mathbb{E}_{i-1}[\alpha_i(x)]$ (this follows immediately from the definition). Moreover, if we take $\delta > 0$ we can show that $\mathbb{P}_i(B_i) \leqslant O_\delta(1)\mathbb{E}_{i-1}[\alpha_i(x)^2]$.

Second we show that $\mathbb{E}_{i-1}[\alpha_i(x)^2] \leqslant O(1)p_i^{-2}\prod_{j<i}(1+O(p_j^{-1}))$. For simplicity, consider the hypothetical case when we take $\delta = 0$. In this case $\alpha_i(x) \leqslant \sum_{d=mp_i\in N_i} p_i^{-1}\cdot \mathbb{1}[x\equiv a_d \bmod m]$ and also $\alpha_i(x)^2 \leqslant \sum_{\substack{d_1=m_1p_i\in N_i\\ d_2=m_2p_i\in N_i}} p_i^{-2}\cdot \mathbb{1}[x\equiv a_{d_1} \bmod m_1 \text{ and } x\equiv a_{d_2} \bmod m_2]$. Hence, $\mathbb{E}_{i-1}[\alpha_i(x)^2] \leqslant p_i^{-2}\sum_{\substack{d_1=m_1p_i\in N_i\\ d_2=m_2p_i\in N_i}} \mathbb{P}_{i-1}\mathbb{1}[x\equiv a_{d_1} \bmod m_1 \text{ and } x\equiv a_{d_2} \bmod m_2] \leqslant p_i^{-2}\sum_{m_1,m_2\mid Q_{i-1}} \operatorname{lcm}(m_1,m_2)^{-1}$. By noting the sum is a multiplicative function, we easily get $\mathbb{E}_{i-1}[\alpha_i(x)^2] \leqslant p_i^{-2}\prod_{j<i}(1+3p_j^{-1})$.

Third we note that $\sum_i O(1)p_i^{-2}\prod_{j<i}(1+O(p_j^{-1}))$ is convergent. This is because the sum is essentially $\sum_i O(1)p_i^{-2}\log(p_i)^{O(1)}$ which is convergent.

Putting everything together, we deduce that there exists $i_0 \in \mathbb{N}$ such that $\sum_{i\geqslant i_0}\mathbb{P}_i(B_i) < 1$. Moreover, if all moduli are square free and are greater than $p_1p_2\dots p_{i_0}$ then $B_1 = \dots = B_{i_0} = \emptyset$. In this case we conclude $\sum_i \mathbb{P}_i(B_i) < 1$, i.e., $\mathscr{A}$ does not cover $\mathbb{Z}$.

4.2.4 Novelty of the method

As mentioned in the introduction, our method is based on that of Hough [100], but is actually simpler in various important ways. The technical difference is that the sieve used here processes the primes one at a time whereas the sieve used in [100] processes the primes in large groups; the measures constructed here allow covered points to have some small weight whereas the measures constructed in [100] are supported on the uncovered set. As a consequence, our analysis is simpler: we do not need to appeal to the Lovász Local Lemma to estimate the density of the surviving set in each fiber and

we do not need to track any notion of quasi-randomness of the surviving set at each step. Moreover, because we distort the measure in incremental steps, we are able to track the distortion between the uniform measure $\mathbb{P}_0$ and the final measure $\mathbb{P}_n$. More precisely, we are able to track the 'average' distortion $\mathbb{P}_n(x)/\mathbb{P}_0(x)$ (which we take in the $\mathbb{P}_n$-measure) and thus we are able to estimate the density of the uncovered set not only in the measure $\mathbb{P}_n$ but also in the uniform measure which was not done in [100].

4.3 A general theorem

In this section we will prove our main technical results, Theorems 4.3.1 and 4.3.2, below, which together imply Theorems 4.1.1, 4.1.5 and 4.1.7, and also Hough's theorem. In each case the deduction involves little more than choosing a suitable sequence $(\delta_1,\dots,\delta_n)$.

Given a finite collection $\mathscr{A} = \{A_d : d \in D\}$ of arithmetic progressions, let n be the number of distinct prime factors of $Q = \mathrm{lcm}(D)$, and for each sequence $\delta_1,\dots,\delta_n \in [0,1/2]$, let the probability distributions $\mathbb{P}_i$ and functions $\alpha_i\colon \mathbb{Z}_{Q_{i-1}} \to [0,1]$ be as defined in (4.4) and (4.5). Set

$$M_i^{(1)} := \mathbb{E}_{i-1}\big[\alpha_i(x)\big] \qquad \text{and} \qquad M_i^{(2)} := \mathbb{E}_{i-1}\big[\alpha_i(x)^2\big],$$

and define a multiplicative function ν, defined on factors of Q, by setting

$$\nu(d) = \prod_{p_j \mid d} \frac{1}{1-\delta_j}. \tag{4.8}$$

for each $d \mid Q$.

Theorem 4.3.1. Let $\mathscr{A} = \{A_d : d \in D\}$ be a finite collection of arithmetic progressions, and let $\delta_1,\dots,\delta_n \in [0,1/2]$. If

$$\eta := \sum_{i=1}^{n} \min\left\{ M_i^{(1)}, \frac{M_i^{(2)}}{4\delta_i(1-\delta_i)} \right\} < 1, \tag{4.9}$$

then $\mathscr{A}$ does not cover the integers. Moreover, the uncovered set R has density at least

$$\mathbb{P}_0(R) \geqslant \big(1-\eta\big) \exp\bigg(-\frac{2}{1-\eta} \sum_{d\in D} \frac{\nu(d)}{d} \bigg). \tag{4.10}$$

In order to show that (4.9) holds in our applications, we need to bound the moments of $\alpha_i(x)$. The following technical theorem provides general bounds that are sufficient in most cases.

Theorem 4.3.2. Let $\mathscr{A} = \{A_d : d \in D\}$ be a finite collection of arithmetic progressions, and let $\delta_1,\dots,\delta_n \in [0,1/2]$. Then

$$M_i^{(1)} \leqslant \sum_{m p_i^j \in N_i, m \mid Q_{i-1}} p_i^{-j} \cdot \frac{\nu(m)}{m} \leqslant \frac{1}{p_i - 1} \prod_{j<i} \bigg(1 + \frac{1}{(1-\delta_j)(p_j-1)} \bigg),$$

and

$$M_i^{(2)} \leqslant \sum_{\substack{m_1 p_i^{j_1}, m_2 p_i^{j_2} \in N_i \\ m_1, m_2 \mid Q_{i-1}}} p_i^{-(j_1+j_2)} \cdot \frac{\nu\big(\operatorname{lcm}(m_1,m_2)\big)}{\operatorname{lcm}(m_1,m_2)} \leqslant \frac{1}{(p_i-1)^2} \prod_{j<i} \left(1 + \frac{3p_j - 1}{(1-\delta_j)(p_j-1)^2}\right).$$

The proofs of Theorems 4.3.1 and 4.3.2 are both surprisingly simple. Let us begin with the following easy lemma, which is the first step in the proof of Theorem 4.3.1. We assume throughout this section that $\mathscr{A} = \{A_d : d \in D\}$ is a given finite collection of arithmetic progressions such that $Q = \operatorname{lcm}(D)$ has exactly n distinct prime factors, and fix a sequence $\delta_1, \dots, \delta_n \in [0, 1/2]$, and hence a function α_i and measure $\mathbb{P}_i$ for each $i \in [n]$.

Lemma 4.3.3.

$$\mathbb{P}_i(B_i) \leqslant \min\left\{ \mathbb{E}_{i-1}\left[\alpha_i(x)\right], \ \frac{\mathbb{E}_{i-1}\left[\alpha_i(x)^2\right]}{4\delta_i(1-\delta_i)} \right\}.$$

Proof. Observe first that

$$\mathbb{P}_i(B_i) \leqslant \mathbb{P}_{i-1}(B_i) = \mathbb{E}_{i-1}\left[\alpha_i(x)\right],$$

where the inequality holds by (4.7), and the equality by (4.4).

For the other upper bound, we will use the elementary inequality $\max\{a-d, 0\} \leqslant a^2/4d$, which is easily seen to hold for all $a, d > 0$ by rearranging the inequality $(a-2d)^2 \geqslant 0$. By (4.4) and (4.5) (the definitions of α_i and $\mathbb{P}_i$), we have

$$\begin{aligned}
\mathbb{P}_i(B_i) &= \sum_{x \in \mathbb{Z}_{Q_{i-1}}} \max\left\{0, \frac{\alpha_i(x) - \delta_i}{\alpha_i(x)(1-\delta_i)}\right\} \cdot \mathbb{P}_{i-1}\big(F(x) \cap B_i\big) \\
&= \frac{1}{1-\delta_i} \sum_{x \in \mathbb{Z}_{Q_{i-1}}} \max\big\{0, \alpha_i(x) - \delta_i\big\} \cdot \mathbb{P}_{i-1}(x) \\
&\leqslant \frac{1}{1-\delta_i} \sum_{x \in \mathbb{Z}_{Q_{i-1}}} \frac{\alpha_i(x)^2}{4\delta_i} \cdot \mathbb{P}_{i-1}(x) = \frac{\mathbb{E}_{i-1}\left[\alpha_i(x)^2\right]}{4\delta_i(1-\delta_i)},
\end{aligned}$$

as required. □

It is already straightforward to deduce from Lemma 4.3.3 and inequality (4.9) that $\mathscr{A}$ does not cover the integers. In order to deduce the bound (4.10) on the density of the uncovered set, we will need to work slightly harder. First, we need the following easy bound on the $\mathbb{P}_i$-measure of an arithmetic progression.

Lemma 4.3.4. For each $0 \leqslant i \leqslant n$, and all $b, d \in \mathbb{Z}$ such that $d \mid Q$, we have

$$\mathbb{P}_i\big(b + d\mathbb{Z}\big) \leqslant \frac{1}{d} \prod_{p_j \mid d,\, j \leqslant i} \frac{1}{1-\delta_j} = \frac{\nu\big(\gcd(d, Q_i)\big)}{d}. \tag{4.11}$$

Proof. The proof is by induction on i. Note first that $\mathbb{P}_0$ is just the uniform measure, so $\mathbb{P}_0(b + d\mathbb{Z}) = 1/d$. So let $i \in [n]$, and assume that the claimed bound holds for $\mathbb{P}_{i-1}$. Suppose first that $p_i \mid d$. Then,

by (4.6) and the induction hypothesis, we have

$$\mathbb{P}_i\left(b+d\mathbb{Z}\right) \leqslant \frac{1}{1-\delta_i}\cdot\mathbb{P}_{i-1}\left(b+d\mathbb{Z}\right) \leqslant \frac{1}{d}\cdot\frac{1}{1-\delta_i}\prod_{p_j\mid d,\, j<i}\frac{1}{1-\delta_j} = \frac{1}{d}\prod_{p_j\mid d,\, j\leqslant i}\frac{1}{1-\delta_j},$$

as required. On the other hand, if $p_i \nmid d$ then we may write $d = m\ell$, where $m = \gcd(d,Q_i) = \gcd(d,Q_{i-1})$. It follows that

$$\mathbb{P}_i\left(b+d\mathbb{Z}\right) = \frac{\mathbb{P}_i\left(b+m\mathbb{Z}\right)}{\ell} = \frac{\mathbb{P}_{i-1}\left(b+m\mathbb{Z}\right)}{\ell} \leqslant \frac{1}{\ell m}\prod_{p_j\mid d,\, j<i}\frac{1}{1-\delta_j} = \frac{1}{d}\prod_{p_j\mid d,\, j\leqslant i}\frac{1}{1-\delta_j},$$

as required. □

Let us define the *distortion* $\Delta_i(x)$ of a point $x\in\mathbb{Z}_{Q_i}$ by

$$\Delta_i(x) := \max\left\{0, \log\frac{\mathbb{P}_i(x)}{\mathbb{P}_0(x)}\right\}. \tag{4.12}$$

The following bound on the average distortion will allow us to prove (4.10).

Lemma 4.3.5. For each $0\leqslant i\leqslant n$, we have

$$\mathbb{E}_i\left[\Delta_i(x)\right] \leqslant 2\cdot\sum_{d\in D_i}\frac{\nu(d)}{d}.$$

Proof. We claim first that

$$\log\frac{\mathbb{P}_j(x)}{\mathbb{P}_{j-1}(x)} \leqslant 2\cdot\alpha_j(x)$$

for every $j\in[n]$ and $x\in\mathbb{Z}_{Q_j}$. Indeed, observe that $\mathbb{P}_{j-1}(x)/\mathbb{P}_j(x) \geqslant \max\{1-\alpha_j(x), 1-\delta_j\}$, by (4.5), and use the inequality $-\log(1-z)\leqslant 2z$, which holds for $z\leqslant 1/2$, and the fact that $\delta_j\leqslant 1/2$. It follows that

$$\mathbb{E}_i\left[\Delta_i(x)\right] \leqslant \sum_{j=1}^{i}\mathbb{E}_i\left[\max\left\{0,\log\frac{\mathbb{P}_j(x)}{\mathbb{P}_{j-1}(x)}\right\}\right] \leqslant 2\cdot\sum_{j=1}^{i}\mathbb{E}_i\left[\alpha_j(x)\right].$$

Now, by (4.4) and Lemma 4.2.1, we have

$$\mathbb{E}_i\left[\alpha_j(x)\right] = \mathbb{E}_{j-1}\left[\alpha_j(x)\right] = \mathbb{P}_{j-1}(B_j)$$

for each $j\in[i]$, since the function α_j is Q_{j-1}-measurable. Moreover, by (4.3) (the definition of B_i), the union bound, and Lemma 4.3.4, we have

$$\mathbb{P}_{j-1}(B_j) \leqslant \sum_{d\in N_j}\mathbb{P}_{j-1}(A_d) \leqslant \sum_{d\in N_j}\frac{\nu(\gcd(d,Q_{j-1}))}{d} \leqslant \sum_{d\in N_j}\frac{\nu(d)}{d}.$$

Hence we obtain

$$\mathbb{E}_i\left[\Delta_i(x)\right] \leqslant 2\cdot\sum_{j=1}^{i}\sum_{d\in N_j}\frac{\nu(d)}{d} = 2\cdot\sum_{d\in D_i}\frac{\nu(d)}{d},$$

as claimed. □

Theorem 4.3.1 now follows easily from Lemmas 4.3.3 and 4.3.5.

Proof of Theorem 4.3.1. We claim first that

$$1-\mathbb{P}_n(R) \leqslant \sum_{i=1}^{n}\mathbb{P}_n(B_i) = \sum_{i=1}^{n}\mathbb{P}_i(B_i) \leqslant \eta. \tag{4.13}$$

Indeed, the first inequality is just the union bound; the equality holds by Lemma 4.2.1, since B_i is Q_i-measurable, so $\mathbb{P}_i(B_i) = \mathbb{P}_{i+1}(B_i) = \cdots = \mathbb{P}_n(B_i)$; and the final inequality follows from Lemma 4.3.3 and (4.9), the definition of η. It follows that $\mathbb{P}_n(R) \geqslant 1-\eta > 0$ if $\eta < 1$, and hence R is non-empty, i.e., $\mathscr{A}$ does not cover the integers.

To prove the claimed lower bound on the density of the uncovered set, we will use Lemma 4.3.5. Indeed, by the definition (4.12) of $\Delta_n(x)$, we have

$$\mathbb{P}_0(R) = \mathbb{E}_0\left[\mathbb{1}_{x\in R}\right] \geqslant \mathbb{E}_n\left[\mathbb{1}_{x\in R}\exp\left(-\Delta_n(x)\right)\right].$$

Now, by the convexity of e^{-z}, and noting that $\mathbb{P}_n(R)\cdot\mathbb{E}_n\left[\Delta_n(x)\mid x\in R\right] \leqslant \mathbb{E}_n\left[\Delta_n(x)\right]$,

$$\begin{aligned}\mathbb{E}_n\left[\mathbb{1}_{x\in R}\exp\left(-\Delta_n(x)\right)\right] &= \mathbb{P}_n(R)\cdot\mathbb{E}_n\left[\exp\left(-\Delta_n(x)\right)\,\middle|\, x\in R\right]\\ &\geqslant \mathbb{P}_n(R)\cdot\exp\left(-\mathbb{E}_n\left[\Delta_n(x)\mid x\in R\right]\right)\\ &\geqslant \mathbb{P}_n(R)\cdot\exp\left(-\frac{\mathbb{E}_n\left[\Delta_n(x)\right]}{\mathbb{P}_n(R)}\right).\end{aligned}$$

Hence, by Lemma 4.3.5, and since $\mathbb{P}_n(R) \geqslant 1-\eta$, by (4.13), we obtain

$$\mathbb{P}_0(R) \geqslant \left(1-\eta\right)\exp\left(-\frac{2}{1-\eta}\sum_{d\in D}\frac{\nu(d)}{d}\right),$$

as required. □

4.3.1 Bounding the moments of $\alpha_i(x)$

The proof of Theorem 4.3.2 is also quite straightforward. First, recall that $N_i := D_i \setminus D_{i-1}$ is the set of new differences at step i, and note that any $d \in N_i$ can be represented in the form $d = mp_i^j$, where $m \mid Q_{i-1}$ and $1 \leqslant j \leqslant \gamma_i$. The first step is the following general bound on the moments of $\alpha_i(x)$.

Lemma 4.3.6. For each $k \in \mathbb{N}$,

$$\mathbb{E}_{i-1}\left[\alpha_i(x)^k\right] \leqslant \sum_{\substack{m_1 p_i^{j_1},\dots,m_k p_i^{j_k} \in N_i \\ m_1,\dots,m_k \mid Q_{i-1}}} \frac{1}{p_i^{j_1+\cdots+j_k}} \cdot \frac{\nu\big(\operatorname{lcm}(m_1,\dots,m_k)\big)}{\operatorname{lcm}(m_1,\dots,m_k)} \tag{4.14}$$

$$\leqslant \frac{1}{(p_i-1)^k} \sum_{m_1,\dots,m_k \mid Q_{i-1}} \frac{\nu\big(\operatorname{lcm}(m_1,\dots,m_k)\big)}{\operatorname{lcm}(m_1,\dots,m_k)}. \tag{4.15}$$

Proof. Recall the definitions (4.3) and (4.4) of the set B_i and the function α_i, respectively. Applying the union bound, we obtain, for each $x \in \mathbb{Z}_{Q_{i-1}}$,

$$\alpha_i(x) = \sum_{(x,y)\in\mathbb{Z}_{Q_i}} p_i^{-\gamma_i} \cdot \mathbb{1}\left[(x,y) \in B_i\right] \leqslant \sum_{(x,y)\in\mathbb{Z}_{Q_i}} \sum_{d\in N_i} p_i^{-\gamma_i} \cdot \mathbb{1}\left[(x,y) \in A_d\right].$$

Now, observe that if $d = mp_i^j \in N_i$, where $p_i \nmid m$, then there are either zero or $p^{\gamma_i - j}$ values of $y \in \mathbb{Z}_{p_i^{\gamma_i}}$ with $(x,y) \in A_d$. Indeed, $(x,y) \in A_d = a_d + d\mathbb{Z}$ iff $x \equiv a_d \bmod m$ and $y \equiv a_d \bmod p_i^j$. It follows that

$$\alpha_i(x) \leqslant \sum_{\substack{d = mp_i^j \in N_i \\ m \mid Q_{i-1}}} p_i^{-j} \cdot \mathbb{1}\left[x \equiv a_d \bmod m\right],$$

and hence

$$\mathbb{E}_{i-1}\left[\alpha_i(x)^k\right] \leqslant \sum_{\substack{d_1 = m_1 p_i^{j_1} \in N_i \\ m_1 \mid Q_{i-1}}} \cdots \sum_{\substack{d_k = m_k p_i^{j_k} \in N_i \\ m_k \mid Q_{i-1}}} \frac{1}{p_i^{j_1+\cdots+j_k}} \mathbb{P}_{i-1}\left(x \equiv a_{d_j} \bmod m_j \text{ for } j \in [k]\right).$$

Now, note that the intersection of the events $\{x \equiv a_{d_j} \bmod m_j\}$ over $j \in [k]$ is either empty (if the congruences are incompatible), or is equivalent to $x \equiv b \bmod \operatorname{lcm}(m_1,\dots,m_k)$ for some b. Therefore, by Lemma 4.3.4, we have

$$\mathbb{P}_{i-1}\left(x \equiv a_{d_j} \bmod m_j \text{ for } j \in [k]\right) \leqslant \frac{\nu\big(\operatorname{lcm}(m_1,\dots,m_k)\big)}{\operatorname{lcm}(m_1,\dots,m_k)},$$

and hence

$$\mathbb{E}_{i-1}\left[\alpha_i(x)^k\right] \leqslant \sum_{\substack{m_1 p_i^{j_1},\dots,m_k p_i^{j_k} \in N_i \\ m_1,\dots,m_k \mid Q_{i-1}}} \frac{1}{p_i^{j_1+\cdots+j_k}} \cdot \frac{\nu\big(\operatorname{lcm}(m_1,\dots,m_k)\big)}{\operatorname{lcm}(m_1,\dots,m_k)}$$

$$\leqslant \sum_{j_1,\dots,j_k \geqslant 1} \frac{1}{p_i^{j_1+\cdots+j_k}} \sum_{m_1,\dots,m_k \mid Q_{i-1}} \frac{\nu\big(\operatorname{lcm}(m_1,\dots,m_k)\big)}{\operatorname{lcm}(m_1,\dots,m_k)}.$$

This proves (4.14); to obtain (4.15), simply note that

$$\sum_{j_1,\dots,j_k\geqslant 1}\frac{1}{p_i^{j_1+\dots+j_k}}=\left(\sum_{j\geqslant 1}\frac{1}{p_i^j}\right)^k=\frac{1}{(p_i-1)^k}.$$ □

To complete the proof of Theorem 4.3.2, it only remains to prove the following bounds.

Lemma 4.3.7. For each $i\in[n]$,

$$\sum_{m|Q_{i-1}}\frac{\nu(m)}{m}\leqslant\prod_{j<i}\left(1+\frac{1}{(1-\delta_j)(p_j-1)}\right)$$

and

$$\sum_{m_1,m_2|Q_{i-1}}\frac{\nu\big(\operatorname{lcm}(m_1,m_2)\big)}{\operatorname{lcm}(m_1,m_2)}\leqslant\prod_{j<i}\left(1+\frac{3p_j-1}{(1-\delta_j)(p_j-1)^2}\right).$$

Proof. Recall from (4.8) that ν is a multiplicative function. It follows that

$$\sum_{m|Q_{i-1}}\frac{\nu(m)}{m}=\prod_{j<i}\sum_{t=0}^{\gamma_j}\frac{\nu(p_j^t)}{p_j^t},$$

and by (4.8), we have

$$\prod_{j<i}\sum_{t=0}^{\gamma_j}\frac{\nu(p_j^t)}{p_j^t}=\prod_{j<i}\left(1+\frac{1}{1-\delta_j}\sum_{t=1}^{\gamma_j}\frac{1}{p_j^t}\right)\leqslant\prod_{j<i}\left(1+\frac{1}{(1-\delta_j)(p_j-1)}\right).$$

To prove the second inequality, let us write $\chi(m)$ for the number of ways of representing a number $m>1$ as the least common multiple of two numbers, i.e.,

$$\chi(m):=\big|\big\{(m_1,m_2):\operatorname{lcm}(m_1,m_2)=m\big\}\big|,$$

so that

$$\sum_{m_1,m_2|Q_{i-1}}\frac{\nu\big(\operatorname{lcm}(m_1,m_2)\big)}{\operatorname{lcm}(m_1,m_2)}=\sum_{m|Q_{i-1}}\frac{\chi(m)\nu(m)}{m}.$$

Observe that the function χ is multiplicative and satisfies $\chi(p^t)=2t+1$ for all primes p and integers $t\geqslant 0$. It follows that

$$\sum_{m|Q_{i-1}}\frac{\chi(m)\nu(m)}{m}=\prod_{j<i}\sum_{t=0}^{\gamma_i}\frac{\chi(p_j^t)\nu(p_j^t)}{p_j^t}\leqslant\prod_{j<i}\left(1+\sum_{t=1}^{\infty}\frac{2t+1}{(1-\delta_j)p_j^t}\right).\tag{4.16}$$

Finally, note that for any $p>1$, we have

$$\sum_{t\geqslant 1}\frac{2t+1}{p^t}=\frac{1}{(1-p^{-1})}\left(\frac{3}{p}+\frac{2}{p^2}+\frac{2}{p^3}+\dots\right)=\frac{1}{(1-p^{-1})^2}\left(\frac{3}{p}-\frac{1}{p^2}\right)=\frac{3p-1}{(p-1)^2}.\tag{4.17}$$

This completes the proof of the lemma. □

Theorem 4.3.2 is an almost immediate consequence of Lemmas 4.3.6 and 4.3.7.

Proof of Theorem 4.3.2. To bound $M_i^{(1)}$ we apply Lemma 4.3.6 with $k = 1$ and the first inequality in Lemma 4.3.7. This gives

$$\mathbb{E}_{i-1}\big[\alpha_i(x)\big] \leqslant \sum_{\substack{mp_i^j \in N_i \\ m \mid Q_{i-1}}} \frac{1}{p_i^j} \cdot \frac{\nu(m)}{m} \leqslant \frac{1}{p_i - 1} \sum_{m \mid Q_{i-1}} \frac{\nu(m)}{m} \leqslant \frac{1}{p_i - 1} \prod_{j<i} \left(1 + \frac{1}{(1-\delta_j)(p_j - 1)}\right),$$

as claimed. To bound $M_i^{(2)}$ we apply Lemma 4.3.6 with $k = 2$ and Lemma 4.3.7.

$$\begin{aligned}\mathbb{E}_{i-1}\big[\alpha_i(x)^2\big] &\leqslant \sum_{\substack{m_1 p_i^{j_1}, m_2 p_i^{j_2} \in N_i \\ m_1, m_2 \mid Q_{i-1}}} p_i^{-(j_1+j_2)} \cdot \frac{\nu\big(\operatorname{lcm}(m_1, m_2)\big)}{\operatorname{lcm}(m_1, m_2)} \\ &\leqslant \frac{1}{(p_i-1)^2} \sum_{m_1, m_2 \mid Q_{i-1}} \frac{\nu\big(\operatorname{lcm}(m_1, m_2)\big)}{\operatorname{lcm}(m_1, m_2)} \leqslant \frac{1}{(p_i-1)^2} \prod_{j<i} \left(1 + \frac{3p_j - 1}{(1-\delta_j)(p_j-1)^2}\right),\end{aligned}$$

as required. □

4.4 Proof of the Main Theorem

In order to deduce Theorem 4.1.1 from Theorems 4.3.1 and 4.3.2, it will suffice to show that there is an appropriate choice of M and $\delta_1, \delta_2, \ldots, \delta_n$.

Proof of Theorem 4.1.1. Let $p_1, \ldots, p_n$ be the primes that divide Q, listed in increasing order, and fix an integer k^*, to be determined later. We set $\delta_i = 0$ for $i \leqslant k^*$ and

$$\delta_i = \frac{\mu(p_i) - 1}{\mu(p_i)} = \frac{(\log p_i)^{3+\varepsilon}}{p_i} \left(1 + \frac{(\log p_i)^{3+\varepsilon}}{p_i}\right)^{-1}$$

for $i > k^*$. Note that we have $\delta_1, \ldots, \delta_n \in [0, 1/2]$ if k^* is chosen sufficiently large.

In order to apply Theorem 4.3.1, we will bound $M_i^{(1)}$ for each $i \leqslant k^*$, and $M_i^{(2)}$ for $i > k^*$. We will do so using Theorem 4.3.2.

Claim 1. For any choice of k^*, if M is sufficiently large then

$$\sum_{i \leqslant k^*} M_i^{(1)} \leqslant \frac{1}{4}.$$

Proof of Claim 1. Note first that $\nu(d) = 1$ for every $d \in D_{k^*}$, by (4.8) and our choice of $\delta_1, \dots, \delta_n$. Thus, by Theorem 4.3.2, we have

$$\sum_{i \leqslant k^*} M_i^{(1)} \leqslant \sum_{i \leqslant k^*} \sum_{m p_i^j \in N_i} p_i^{-j} \cdot \frac{\nu(m)}{m} = \sum_{i \leqslant k^*} \sum_{d \in N_i} \frac{1}{d} = \sum_{d \in D_{k^*}} \frac{1}{d}.$$

Now, let $\mathscr{S}(q)$ denote the set of *q-smooth* numbers, i.e., numbers all of whose prime factors are at most q, and note that $D_{k^*} \subseteq \mathscr{S}(p_{k^*})$, and moreover

$$\sum_{d \in \mathscr{S}(q)} \frac{1}{d} = \prod_{p \leqslant q} \left(1 + \frac{1}{p-1}\right) < \infty$$

for every q, where the product is over primes $p \leqslant q$. Hence, by choosing M to be sufficiently large, it follows that

$$\sum_{i \leqslant k^*} M_i^{(1)} \leqslant \sum_{d \in \mathscr{S}(p_{k^*}), d \geqslant M} \frac{1}{d} \leqslant \frac{1}{4},$$

as claimed. □

Bounding $M_i^{(2)}$ for $i > k^*$ is only slightly less trivial.

Claim 2. If k^* is sufficiently large, then

$$\sum_{i > k^*} M_i^{(2)} \leqslant \frac{1}{4}.$$

Proof of Claim 2. Recall from Theorem 4.3.2 that

$$M_i^{(2)} \leqslant \frac{1}{(p_i-1)^2} \prod_{j<i} \left(1 + \frac{3p_j - 1}{(1-\delta_j)(p_j-1)^2}\right) \leqslant \frac{1}{(p_i-1)^2} \exp\left(\sum_{j<i} \frac{3p_j-1}{(1-\delta_j)(p_j-1)^2}\right).$$

Now, by our choice of $\delta_1, \dots, \delta_n$, we have

$$\frac{3p_j - 1}{(1-\delta_j)(p_j-1)^2} \leqslant \frac{3}{p_j} + \frac{O\big((\log p_j)^{3+\varepsilon}\big)}{p_j^2}$$

for every $j < i$, and hence, using a weak form of Merten's theorem to deduce that $\sum_{j<i} \frac{1}{p_j} \leqslant \log\log p_i + O(1)$, we obtain

$$M_i^{(2)} \leqslant \frac{1}{(p_i-1)^2}\exp\big(3\log\log p_i + O(1)\big) \leqslant \frac{C_0(\log p_i)^3}{p_i^2}$$

for some absolute constant $C_0 > 0$. It follows that, if $i > k^*$, then

$$\frac{M_i^{(2)}}{4\delta_i(1-\delta_i)} = \frac{\mu(p_i)^2}{4\big(\mu(p_i)-1\big)} \cdot M_i^{(2)} \leqslant \frac{C_1}{p_i(\log p_i)^{\varepsilon}},$$

for some absolute constant $C_1 > 0$. Now, using the prime number theorem $\pi(x) \sim x/\log x$ to crudely bound the sum of the right-hand side over all primes p, we obtain

$$\sum_{p\geqslant 3} \frac{C_1}{p(\log p)^{\varepsilon}} = \sum_{t\geqslant 1} \sum_{e^t < p \leqslant e^{t+1}} \frac{C_1}{p(\log p)^{\varepsilon}} \leqslant \sum_{t\geqslant 1} \frac{C_1 \cdot \pi\big(e^{t+1}\big)}{e^t \cdot t^{\varepsilon}} \leqslant \sum_{t\geqslant 1} \frac{C_2}{t^{1+\varepsilon}} < \infty$$

for any $\varepsilon > 0$. It follows that if k^* is sufficiently large, then

$$\sum_{i>k^*} \frac{M_i^{(2)}}{4\delta_i(1-\delta_i)} \leqslant \frac{1}{4}.$$

as claimed. □

By Claims 1 and 2, it follows that $\eta \leqslant 1/2$, and hence, by Theorem 4.3.1, the uncovered set has density at least

$$\frac{1}{2}\exp\bigg(-4\sum_{d\in D}\frac{\nu(d)}{d}\bigg).$$

Finally, observe that $\mu(d) \geqslant \nu(d)$ for every $d \in \mathbb{N}$, since both functions are multiplicative,

$$\mu(p) = 1 + \frac{(\log p)^{3+\varepsilon}}{p} = \frac{1}{1-\delta_i} = \nu(p)$$

for $p > p_{k^*}$, and $\mu(p) \geqslant 1 = \nu(p)$ for $p \leqslant p_{k^*}$. It follows that the uncovered set has density at least

$$\frac{1}{2}\exp\bigg(-4\sum_{d\in D}\frac{\mu(d)}{d}\bigg) = \frac{e^{-4C}}{2},$$

as required. □

4.5 The Erdős–Graham Conjecture

As a simple consequence of Theorem 4.1.1, we will next give a new strengthening of the conjecture of Erdős and Graham [60] mentioned in the Introduction. Recall that in the original conjecture, which was confirmed by Filaseta, Ford, Konyagin, Pomerance and Yu [76] in 2007, the lower bound M on n was allowed to depend on K. The result proved in [76] required such a bound ($n > K^{\omega}$ for some

$\omega = \omega(K) \to \infty$ as $K \to \infty$), but gave an asymptotically optimal bound on the density of the uncovered set. Our result gives a non-optimal bound on this density, but does not require n to grow with K.

Theorem 4.5.1. There exists M such that for any $K \geqslant 1$, there exists $\delta = \delta(K) > 0$ such that the following holds. If $A_1, \dots, A_k$ are arithmetic progressions with distinct moduli $d_1, \dots, d_k \in [n, Kn]$, $n \geqslant M$, then the uncovered set $R := \mathbb{Z} \setminus \bigcup_{i=1}^{k} A_i$ has density at least δ.

Proof. We apply Theorem 4.1.1 with $\varepsilon = 1$. In order to prove the corollary, it will suffice to show that there exists a constant $C > 0$, depending only on K, such that

$$\sum_{d \in [n, Kn]} \frac{\mu(d)}{d} \leqslant C \tag{4.18}$$

for all $n \in \mathbb{N}$. Indeed, if we set $\delta := e^{-4C}/2$ then it follows from Theorem 4.1.1 that, if (4.18) holds and $n \geqslant M$, then the density of R is at least δ.

We will prove, by induction on $t \geqslant 0$, that (4.18) holds for all $n \leqslant 2^t$. This is clearly the case when $t = 0$ (assuming that $C = C(K)$ is sufficiently large), so let $t \geqslant 1$, and let us assume that (4.18) holds for all $n \leqslant 2^{t-1}$. We will use the following telescopic series, which holds for any i_0 and $d \mid Q$ by (4.1):

$$\begin{aligned}\mu(d) &= \mu\big(\gcd(d, Q_{i_0})\big) + \sum_{i > i_0} \Big(\mu\big(\gcd(d, Q_i)\big) - \mu\big(\gcd(d, Q_{i-1})\big)\Big) \\ &= \mu\big(\gcd(d, Q_{i_0})\big) + \sum_{p_i \mid d, i > i_0} \big(\mu(p_i) - 1\big)\mu\big(\gcd(d, Q_{i-1})\big).\end{aligned}$$

Summing over $d \in [n, Kn]$, we obtain

$$\sum_{d \in [n, Kn]} \frac{\mu(d)}{d} \leqslant \sum_{d \in [n, Kn]} \frac{\mu(Q_{i_0})}{d} + \sum_{i > i_0} \frac{\mu(p_i) - 1}{p_i} \sum_{d \in [n/p_i, Kn/p_i]} \frac{\mu(d)}{d}.$$

Now, since $n/p \leqslant 2^{t-1}$ for every prime p, it follows by the induction hypothesis that

$$\sum_{d \in [n, Kn]} \frac{\mu(d)}{d} \leqslant \mu(Q_{i_0})\big(\log K + 1\big) + C \sum_{i > i_0} \frac{\mu(p_i) - 1}{p_i}.$$

Finally, note that the sum over all primes

$$\sum_{p \text{ prime}} \frac{\mu(p) - 1}{p} = \sum_{p \text{ prime}} \frac{(\log p)^4}{p^2}$$

converges, and therefore

$$\sum_{d \in [n, Kn]} \frac{\mu(d)}{d} \leqslant \mu(Q_{i_0})\big(\log K + 1\big) + \frac{C}{2} \leqslant C$$

if i_0 and then $C = C(K)$ are chosen sufficiently large, as required. □

4.6 A general method of applying the sieve

In this section we describe a practical method of applying our method to problems involving covering systems, such as Schinzel's conjecture and the Erdős–Selfridge problem. More precisely, we will show how one can choose the constants δ_i sequentially and optimally via a simple recursion which may be run on a computer. We will also give a simple criterion (see Theorem 4.6.1, below) which we prove is sufficient to deduce that the collection $\mathscr{A}$ does not cover the integers. Combining these (that is, running the recursion until the criterion is satisfied), we reduce the problems to finite calculations, which in some cases are tractable. To demonstrate the power of this approach, we will use it in Sections 4.7–4.9 to prove Theorems 4.1.5 and 4.1.7, and to significantly improve the bound on M in Hough's theorem.

We begin by defining a sequence of numbers $f_k = f_k(\mathscr{A})$, which will (roughly speaking) encode how "well" we are doing after k steps of our sieve. We remark that, from now on, we will perform a step of the sieve for *every* prime, whether or not it divides Q. We will therefore write p_k for the kth prime, i.e., $p_1 = 2$, $p_2 = 3$, etc. Accordingly, we write Q_i for the maximal divisor of Q with prime factors $p_1, \dots, p_i$. Let p_n be the largest prime factor of Q.

Fix $i_0 \in \mathbb{N}$ and fix a probability measure $\mathbb{P}_{i_0}$ on $\mathbb{Z}_{Q_{i_0}}$. Given parameters $\delta_{i_0+1}, \dots, \delta_n \in [0, 1/2]$ and given probability measures $\mathbb{P}_{i_0+1}, \dots, \mathbb{P}_n$ on $\mathbb{Z}_{Q_{i_0+1}}, \dots, \mathbb{Z}_{Q_n}$ constructed recursively using (4.4) and (4.5) from the probability measure $\mathbb{P}_{i_0}$, define

$$f_k = f_k(\mathscr{A}) := \frac{\kappa}{\mu_k} \prod_{i_0 < i \leqslant k} \left(1 + \frac{3p_i - 1}{(1-\delta_i)(p_i-1)^2}\right) \tag{4.19}$$

for each $k \geqslant i_0$, where

$$\mu_k := 1 - \sum_{i_0+1 \leqslant i \leqslant k} \mathbb{P}_i(B_i) - \sum_{i \leqslant i_0} \mathbb{P}_{i_0}(B_i),$$

and $\kappa > 0$ and $i_0 \in \mathbb{N}$ are chosen so that

$$M_k^{(2)} \leqslant \frac{\kappa}{(p_k-1)^2} \prod_{i_0 < i < k} \left(1 + \frac{3p_i - 1}{(1-\delta_i)(p_i-1)^2}\right) = \frac{\mu_{k-1} f_{k-1}}{(p_k-1)^2} \tag{4.20}$$

for every $k > i_0$. For example, say in addition we are also given parameters $\delta_1, \dots, \delta_{i_0}$ and we are given the probability measures $\mathbb{P}_1, \dots, \mathbb{P}_{i_0}$ constructed recursively using (4.4) and (4.5) from the trivial probability measure $\mathbb{P}_0$ on $\mathbb{Z}_1$. Then $\mu_k = 1 - \sum_{i \leqslant k} \mathbb{P}_i(B_i)$ and, by Theorem 4.3.2,

$$\kappa = \prod_{i \leqslant i_0,\, p_i \mid Q} \left(1 + \frac{3p_i - 1}{(1-\delta_i)(p_i-1)^2}\right) \tag{4.21}$$

is a valid choice, although in some cases we can prove a stronger bound. We remark that, in practice, we will choose the constants κ and $\delta_1, \dots, \delta_{i_0}$, and show that (4.20) holds for any sequence $(\delta_{i_0+1}, \dots, \delta_n)$. We will then choose each subsequent δ_i so as to minimize f_i.

The following theorem gives a sufficient condition (at step k) for our sieve to be successful. The bound we prove gives an almost optimal termination criterion when k is large.

Theorem 4.6.1. Let $i_0 \in \mathbb{N}$ and let $\mathbb{P}_{i_0}$ be a probability measure on $\mathbb{Z}_{Q_{i_0}}$. Let $\kappa > 0$ be such that for any choice of parameters $\delta_{i_0+1}, \dots, \delta_n \in [0, 1/2]$, if we construct the probability measures $\mathbb{P}_{i_0+1}, \dots, \mathbb{P}_n$, recursively using (4.4) and (4.5) from the probability measure $\mathbb{P}_{i_0}$, then (4.20) holds for any $i_0 < k \leqslant n$. Fix $k \geqslant \max(i_0, 10)$ and fix $\delta_{i_0+1}, \dots, \delta_k \in [0, 1/2]$ and let probability measures $\mathbb{P}_{i_0+1}, \dots, \mathbb{P}_k$ be constructed recursively using (4.4) and (4.5) from the probability measure $\mathbb{P}_{i_0}$. If $\mu_k > 0$ and $f_k(\mathscr{A}) \leqslant (\log k + \log\log k - 3)^2 k$, then the system of arithmetic progressions $\mathscr{A}$ does not cover $\mathbb{Z}$.

In this section, it will be convenient to define, for each $i \in \mathbb{N}$,

$$a_i = \frac{3p_i - 1}{(p_i - 1)^2} \qquad \text{and} \qquad b_i = \frac{1}{4(p_i - 1)^2}. \tag{4.22}$$

The first step in the proof of Theorem 4.6.1 is the following simple (but key) lemma.

Lemma 4.6.2. Let $i > i_0$, and assume that $\mu_{i-1} > 0$. If $b_i f_{i-1} < \delta_i(1 - \delta_i)$, then $\mu_i > 0$, and

$$f_i \leqslant \left(1 + \frac{a_i}{1 - \delta_i}\right)\left(1 - \frac{b_i f_{i-1}}{\delta_i(1 - \delta_i)}\right)^{-1} f_{i-1}. \tag{4.23}$$

Proof. Recall from Lemma 4.3.3 and (4.20), that

$$\mu_{i-1} - \mu_i = \mathbb{P}_i(B_i) \leqslant \frac{M_i^{(2)}}{4\delta_i(1 - \delta_i)} \leqslant \frac{\mu_{i-1} f_{i-1}}{4\delta_i(1 - \delta_i)(p_i - 1)^2}. \tag{4.24}$$

It follows that $\mu_i \geqslant \mu_{i-1}\big(1 - b_i f_{i-1}/\delta_i(1 - \delta_i)\big) > 0$, and moreover

$$\begin{aligned} \frac{f_i}{f_{i-1}} &= \frac{\mu_{i-1}}{\mu_i}\left(1 + \frac{3p_i - 1}{(1 - \delta_i)(p_i - 1)^2}\right) \\ &\leqslant \left(1 + \frac{3p_i - 1}{(1 - \delta_i)(p_i - 1)^2}\right)\left(1 - \frac{f_{i-1}}{4\delta_i(1 - \delta_i)(p_i - 1)^2}\right)^{-1}, \end{aligned}$$

as claimed. □

To deduce Theorem 4.6.1, we will use the main result (which is also the title) of [51], which states that for each $k \geqslant 2$, the kth prime is greater than $k(\log k + \log\log k - 1)$.

Proof of Theorem 4.6.1. The plan is to choose $\delta_{k+1}, \dots, \delta_n \in [0, 1/2]$ conveniently and construct the probability measures $\mathbb{P}_{k+1}, \dots, \mathbb{P}_n$ recursively using (4.4) and (4.5) from the probability measure $\mathbb{P}_k$, in order to show that $\mu_n > 0$. To do so, we will use Lemma 4.6.2 to show, by induction on i, that $\mu_i > 0$ for every $k \leqslant i \leqslant n$. As part of the induction, we will also prove that $f_i \leqslant \lambda_i^2 i$ for each $k \leqslant i \leqslant n$, where $\lambda_i = \log i + \log\log i - 3$.

Note that the base case, $i = k$, follows from our assumptions, and set $\delta_i = 1/2$ for each $k < i \leqslant n$. By Lemma 4.6.2, for the induction step it will suffice to show that $4b_i f_{i-1} < 1$ and

$$\lambda_{i-1}^2(i-1)\left(\frac{1+2a_i}{1-4b_i f_{i-1}}\right) \leqslant \lambda_i^2 i,$$

where a_i and b_i were defined in (4.22). We claim that, writing $\lambda = \lambda_{i-1}$, we have

$$2a_i \leqslant \frac{6(\lambda+2)i+4}{(\lambda+2)^2 i^2}, \qquad 4b_i f_{i-1} \leqslant \frac{\lambda^2(i-1)}{(\lambda+2)^2 i^2} < 1, \qquad \text{and} \qquad \lambda_i^2 i \geqslant \lambda(\lambda i+2).$$

To see these inequalities, note that $p_i \geqslant (\lambda_i+2)i$ for all $i \geqslant k$, by the result of [51] stated above, and $\lambda_i - \lambda_{i-1} \geqslant \log(i/(i-1)) > 1/i$, so $p_i - 1 \geqslant (\lambda_{i-1}+1/i+2)i - 1 = (\lambda+2)i$.

It is therefore sufficient to show that

$$\lambda^2(i-1)\left(1+\frac{6(\lambda+2)i+4}{(\lambda+2)^2 i^2}\right) \leqslant \lambda(\lambda i+2)\left(1-\frac{\lambda^2(i-1)}{(\lambda+2)^2 i^2}\right),$$

which (after multiplying by $(\lambda+2)^2 i^2/\lambda$, expanding and rearranging) becomes

$$0 \leqslant 8i^2 + \lambda^3 i + 4\lambda^2 i + 8\lambda i + 2\lambda^2 + 4\lambda.$$

This clearly holds for all $\lambda > 0$ and $i \in \mathbb{N}$, and so the theorem follows. □

Now, combining Theorem 4.6.1 and Lemma 4.6.2, we can deduce from (sufficiently strong) bounds on f_k, for any $k \in \mathbb{N}$, that the uncovered set is non-empty. To do so, observe first that, given f_{i-1}, the bound on f_i given by Lemma 4.6.2 is just a function of a *single* δ_i. Elementary calculus shows that the optimal choice of δ_i occurs when

$$\delta_i = \frac{1+a_i}{1+\sqrt{1+a_i(1+a_i)/(b_i f_{i-1})}}. \tag{4.25}$$

This expression for δ_i allows for very fast numerical computation of the bounds on the f_i.

Let us therefore, for each $i \in \mathbb{N}$, define g_i to be the largest value of $f_i(\mathscr{A})$ such that, by repeatedly applying the recursion (4.23) with δ_j given by (4.25), we eventually satisfy the conditions of Theorem 4.6.1 for some $k \geqslant 10$. In Table 4.1 we list the bounds on g_k given by performing this calculation, which was implemented as follows: starting with a potential value of f_3, we ran the iteration given in (4.23) using the value of δ_i given in (4.25) until either the conditions of Theorem 4.6.1 were satisfied, or the condition $b_i f_{i-1} < \delta_i(1-\delta_i)$ failed. The optimal value g_3 of f_3 was determined by binary chop, and the other bounds g_k were read off by taking the largest successful f_3 and listing the corresponding bounds on f_k.

Let us state, for future reference, the conclusion of this section.

k	p_k	$g_k \geqslant$
2	3	1.260997
3	5	3.007888
4	7	5.860938
5	11	9.032082
6	13	13.30344
7	17	17.99687
8	19	23.90973
9	23	30.38722
10	29	36.72372
21	73	138.877
100	541	1691.365
1000	7919	42420.78
10000	104729	802133.7
51000	625187	5821999

Table 4.1 Upper bounds on f_k that ensure that the system does not cover the integers. All the bounds are rounded down in the last decimal digit. At each stage in the calculation, f_k was increased by a factor of $1+10^{-15}$ to account for rounding errors in the floating point arithmetic. The choice of 51000 is needed to make $p_k > 616000$, as used in Section 4.8, below.

Corollary 4.6.3. Let $i_0 \in \mathbb{N}$ and let $\mathbb{P}_{i_0}$ be a probability measure on $\mathbb{Z}_{Q_{i_0}}$. Let $\kappa > 0$ be such that for any choice of parameters $\delta_{i_0+1}, \dots, \delta_n \in [0, 1/2]$, if we construct the probability measures $\mathbb{P}_{i_0+1}, \dots, \mathbb{P}_n$, recursively using (4.4) and (4.5) from the probability measure $\mathbb{P}_{i_0}$, then (4.20) holds for any $i_0 < k \leqslant n$. Fix $k \geqslant i_0$ and fix $\delta_{i_0+1}, \dots, \delta_k \in [0, 1/2]$ and let probability measures $\mathbb{P}_{i_0+1}, \dots, \mathbb{P}_k$ be constructed recursively using (4.4) and (4.5) from the probability measure $\mathbb{P}_{i_0}$. If $f_k(\mathscr{A}) \leqslant g_k$, then the system of arithmetic progressions $\mathscr{A}$ does not cover $\mathbb{Z}$.

We remark that the bound on g_1 given by our sieve is less than 1, and for this reason we are unable to resolve the Erdős–Selfridge problem.

4.7 The Erdős–Selfridge Problem

In this section we will prove Theorem 4.1.7, which is a simple consequence of the method described in the previous section. First, however, let us show how to prove the following (only slightly weaker) theorem, which was first proved by Hough and Nielsen [101].

Theorem 4.7.1. Let $\mathscr{A}$ be a finite collection of arithmetic progressions with distinct moduli, none of which is divisible by 2 or 3. Then $\mathscr{A}$ does not cover the integers.

Proof. Let $\mathbb{P}_0$ be the trivial measure on $\mathbb{Z}_1$. Set $\delta_1 = \delta_2 = 0$ and construct probability measures $\mathbb{P}_1$ and $\mathbb{P}_2$ recursively using (4.4) and (4.5) from the probability measure $\mathbb{P}_0$. In order to apply Corollary 4.6.3, set $k = i_0 = 2$ and $\kappa = 1$. Note that $\mu_2 = 1$, since there are no moduli divisible by 2 or 3, and recall

from (4.21) that this is a valid choice of κ, by Theorem 4.3.2. Recalling from (4.19) that $f_2 = \kappa/\mu_2$, and using Table 4.1, we see that $f_2 = 1 < 1.26 < g_2$, and hence, by Corollary 4.6.3, the system $\mathscr{A}$ does not cover the integers. □

We remark that we did not actually need the full strength of Corollary 4.6.3 to prove this theorem; in fact, we could have just run our sieve with $\delta_1 = \cdots = \delta_n = 1/4$, say, and applied Theorems 4.3.1 and 4.3.2, together with Theorem 4.6.1. In order to prove Theorem 4.1.7, we will need a slightly more complicated version of the proof above.

Proof of Theorem 4.1.7. By Theorem 4.7.1, we may assume that $Q = \mathrm{lcm}(D)$ satisfies $Q = 3Q'$, where Q' is not divisible by 2, 3 or 5. Recall that Q_i is the largest divisor of Q with prime factors $p_1, \dots, p_i$. Let $\mathbb{P}_0$ be the trivial measure on $\mathbb{Z}_1$. Set $\delta_1 = \delta_2 = \delta_3 = 0$ and construct probability measures $\mathbb{P}_1, \mathbb{P}_2$ and $\mathbb{P}_3$ recursively using (4.4) and (4.5) from the probability measure $\mathbb{P}_0$. Thus, $\mathbb{P}_1, \mathbb{P}_2$ and $\mathbb{P}_3$ are uniform probability measures on $\mathbb{Z}_{Q_1}$, $\mathbb{Z}_{Q_2}$ and $\mathbb{Z}_{Q_3}$, respectively, i.e., $\mathbb{Z}_1, \mathbb{Z}_3$ and $\mathbb{Z}_3$, respectively. Observe that $\mu_3 \geqslant 2/3$, by the (trivial) first moment bound $M_2^{(1)} \leqslant 1/3$, and since there are no moduli divisible by 2 or 5. In order to apply Corollary 4.6.3, set $i_0 = k = 3$ and $\kappa = 2$. To see that this is a valid choice of κ, we need to improve (4.21) slightly, using the fact that $3^2 \nmid Q$. To be precise, in the proof of Lemma 4.3.7, in the last expression in (4.16), when $j = 2$ we only need to include the term $t = 1$ in the sum. Keeping the rest of the proof of Theorem 4.3.2 the same, this implies that

$$M_i^{(2)} \leqslant \frac{1}{(p_i-1)^2}\left(1+\frac{3}{(1-\delta_2)p_2}\right)\prod_{3<j<i}\left(1+\frac{3p_j-1}{(1-\delta_j)(p_j-1)^2}\right),$$

and hence (4.20) holds with $\kappa = 1 + 3/p_2 = 2$, as claimed. Using Table 4.1, it follows that

$$f_3 = \frac{\kappa}{\mu_3} \leqslant 3 < 3.007 \leqslant g_3,$$

and therefore, by Corollary 4.6.3, the system $\mathscr{A}$ does not cover the integers. □

4.8 The Minimum Modulus Problem

In this section we improve the bound on the minimum modulus given in [100, Theorem 1].

Theorem 4.8.1. Let $\mathscr{A}$ be a finite collection of arithmetic progressions with distinct moduli $d_1, \dots, d_k \geqslant 616000$. Then $\mathscr{A}$ does not cover the integers.

Proof. We seek to apply Corollary 4.6.3 with $i_0 = 1, k = 51000$ and $\kappa = 1$. Recall (4.21) that this is a valid choice of κ by Theorem 4.3.2. It remains to choose the values of δ_i for $i \leqslant 51000$. We apply Theorem 4.3.1, using the first moment $M_i^{(1)}$ and setting $\delta_i = 0$ for $i \leqslant 51$ (note that $p_{51} = 233$). After

the first 51 primes we have

$$\mu_{51} \geqslant 1 - \sum_{\substack{d \geqslant 616000 \\ p_{51}\text{-smooth}}} \frac{1}{d} \geqslant 0.654258 \tag{4.26}$$

and

$$f_{51} \leqslant \frac{1}{\mu_{51}} \prod_{j \leqslant 51} \left(1 + \frac{3p_j - 1}{(p_j - 1)^2}\right) \leqslant 886.56.$$

For $51 < i \leqslant 51000$ we apply the second moment bound using

$$M_i^{(2)} \leqslant \hat{M}_i^{(2)} := \sum_{\substack{m_1 p_i^j, m_2 p_i^k \geqslant 616000 \\ m_1, m_2 \ p_{i-1}\text{-smooth}}} p^{-j-k} \cdot \frac{\nu\big(\operatorname{lcm}(m_1, m_2)\big)}{\operatorname{lcm}(m_1, m_2)}. \tag{4.27}$$

The values of δ_i were not optimized, but instead defined by the following equation

$$\delta_i = \left(1 - \frac{1}{\sqrt{p_i}}\right) \cdot \frac{1 + a_i}{1 + \sqrt{1 + 4\hat{\mu}_i a_i (1 + a_i)/\hat{M}_i^{(2)}}},$$

which is based on (4.25), but with $b_i f_{i-1}$ replaced by the bound on $M_i^{(2)}/(4\mu_i)$ implied by (4.27). The (rather arbitrary) factor of $(1 - 1/\sqrt{p_i})$ was included to improve the bounds obtained, and $\hat{\mu}_i$ is the lower bound on μ_i defined inductively by

$$\hat{\mu}_i = \hat{\mu}_{i-1} - \frac{\hat{M}_i^{(2)}}{4\delta_i(1 - \delta_i)}.$$

Finally, after processing $p_{51000} = 625187 > 616000$ we calculated the bound $f_{51000} \leqslant 5589593$ from (4.19) using $i_0 = 0$ and $\kappa = 1$. This is less than the bound g_{51000} given in Table 1, and hence, by Corollary 4.6.3, the system $\mathscr{A}$ does not cover the integers.

It only remains to describe an efficient way of calculating the expressions (4.26) and (4.27). For (4.26) we note that the sum of $1/d$ over all p_i-smooth d is given by the product

$$\sum_{d \ p_i\text{-smooth}} \frac{1}{d} = \prod_{j \leqslant i} \left(1 + \frac{1}{p_j - 1}\right)$$

and the sum in (4.26) can then be calculated by subtracting the finite sum of $1/d$ over all p_i-smooth $d < 616000$. For (4.27) the procedure is somewhat more complicated. First we define

$$\Theta_i(s,t) := \sum_{\substack{m_1 \geqslant s, m_2 \geqslant t \\ m_1, m_2 \ p_i\text{-smooth}}} \frac{\nu\big(\operatorname{lcm}(m_1, m_2)\big)}{\operatorname{lcm}(m_1, m_2)},$$

which can be calculated inductively using the identity

$$\Theta_i(s,t) = \Theta_{i-1}(s,t) + \frac{1}{1-\delta_i} \sum_{j,k\geqslant 0,\, j+k>0} p_i^{-\max\{j,k\}} \cdot \Theta_{i-1}\big(\lceil s/p_i^j \rceil, \lceil t/p_i^k \rceil\big),$$

which, despite its appearance, can be calculated as a finite sum. Indeed, $\lceil s/p_i^j \rceil = 1$ for sufficiently large j, and so there are only finitely many terms $\Theta_{i-1}(s',t')$ that occur, and these are multiplied (when s' or $t' = 1$) by geometric series that can be summed exactly. Finally, the calculation of

$$M_i^{(2)} \leqslant \sum_{j,k\geqslant 1} p_i^{-j-k} \cdot \Theta_{i-1}\big(\lceil K/p_i^j \rceil, \lceil K/p_i^k \rceil\big),$$

where $K = 616000$, can similarly be reduced to a finite sum. □

4.9 Schinzel's Conjecture

In this section we use the method of Section 4.6 to prove Schinzel's Conjecture [136], restated below.

Theorem 4.9.1. If $1 < d_1 < d_2 < \cdots < d_k$ are the moduli of a finite collection of arithmetic progressions that covers the integers, then $d_i \mid d_j$ for some $i < j$.

We will argue by contradiction, assuming that we have a set $\{d_1, \ldots, d_k\}$ of moduli of a covering system of the integers that forms an antichain under divisibility. We call an antichain of natural numbers *p-smooth* if all its elements are p-smooth, i.e., have no prime factor greater than p. In order to apply our sieve, we will need the following three simple lemmas about 5-smooth antichains.

Lemma 4.9.2. If D is a 5-smooth antichain containing no prime power, then

$$\sum_{d\in D} \frac{1}{d} \leqslant \frac{1}{3},$$

with equality if and only if $D = \{6, 10, 15\}$.

Proof. Suppose first that D' is a 3-smooth antichain (possibly containing a prime power), and observe that

$$D' = \big\{2^{a_1}3^{b_1}, 2^{a_2}3^{b_2}, \ldots, 2^{a_k}3^{b_k}\big\}$$

for some $a_1 > a_2 > \cdots > a_k$ and $b_1 < b_2 < \cdots < b_k$. We claim that $\sum_{d\in D'} \frac{1}{d}$ is maximized when D' is 'compressed', that is, $a_i = a_{i+1} + 1$ and $b_{i+1} = b_i + 1$ for all $i \in [k-1]$. Indeed, if $a_i > a_{i+1} + 1$ then we can reduce a_i, and if $b_{i+1} > b_i + 1$ then we can reduce b_{i+1}, in each case increasing $\sum_{d\in D'} \frac{1}{d}$ while maintaining the property that D' is an antichain.

Now, let us write the 5-smooth antichain D as a union of sets of the form $\{5^i d : d \in D_i\}$, where each D_i is a (possibly empty) 3-smooth antichain. Since D contains no prime power, neither can D_0, so $\sum_{d\in D_0} \frac{1}{d}$ is maximized when $D_0 = \{2^a \cdot 3, 2^{a-1} \cdot 3^2, \ldots, 2 \cdot 3^a\}$. In this case a simple calculation

shows that $\sum_{d\in D_0}\frac{1}{d} = 2^{-a}-3^{-a}$, which attains a unique maximum when $a=1$. Thus $\sum_{d\in D_0}\frac{1}{d}\leqslant\frac{1}{6}$, with equality if and only if $D_0=\{6\}$.

Next, observe that $1\notin D_i$ for every $i\geqslant 1$, and therefore $\sum_{d\in D_i}\frac{1}{d}$ is maximized when $D_i = \{2^a, 2^{a-1}3^1,\ldots,3^a\}$, with $a\geqslant 1$. In this case we have $\sum_{d\in D_i}\frac{1}{d} = 6(2^{-1-a}-3^{-1-a})$ and the unique extremal case is $a=1$, so $\sum_{d\in D_i}\frac{1}{d}\leqslant\frac{5}{6}$, with equality if and only if $D_i=\{2,3\}$. If we additionally assume that $D_i\neq\{2,3\}$, then $\sum_{d\in D_i}\frac{1}{d}\leqslant\frac{11}{18}$, with the unique maximum occurring for the (uncompressed) antichain $D_i=\{2,9\}$.

Finally, if $D_1=\{2,3\}$ then $D_i=\emptyset$ for every $i>1$, and therefore

$$\sum_{d\in D}\frac{1}{d} = \sum_{d\in D_0}\frac{1}{d}+\frac{1}{5}\sum_{d\in D_1}\frac{1}{d}\leqslant\frac{1}{6}+\frac{1}{5}\cdot\frac{5}{6}=\frac{1}{3},$$

with equality only when $D=\{6,10,15\}$. On the other hand, if $D_1\neq\{2,3\}$ then

$$\sum_{d\in D}\frac{1}{d} = \sum_{i=0}^{\infty}\frac{1}{5^i}\sum_{d\in D_i}\frac{1}{d}\leqslant\frac{1}{6}+\frac{11}{5\cdot 18}+\sum_{i=2}^{\infty}\frac{1}{5^{i-1}\cdot 6}=\frac{119}{360}<\frac{1}{3},$$

as required. □

Lemma 4.9.3. If A and B are two 3-smooth antichains, then

$$\sum_{a\in A}\sum_{b\in B}\frac{1}{\operatorname{lcm}(a,b)}\leqslant\frac{31}{36}, \tag{4.28}$$

except in the cases $A=B=\{1\}$ and $A=B=\{2,3\}$.

Note that the sum in (4.28) is equal to 1 if $A=B=\{1\}$, and to $7/6$ if $A=B=\{2,3\}$.

Proof. Suppose first that $|A|\leqslant|B|$, and that $B=\{2^{k-1},2^{k-2}3^1,\ldots,3^{k-1}\}$ for some $k\geqslant 5$. Then

$$\sum_{a\in A}\sum_{b\in B}\frac{1}{\operatorname{lcm}(a,b)}\leqslant\sum_{a\in A}\sum_{b\in B}\frac{1}{b}=|A|\sum_{b\in B}\frac{1}{b}=6k\big(2^{-k}-3^{-k}\big)<\frac{31}{36}.$$

However, if $|B|=k\geqslant 5$ but $B\neq\{2^{k-1},2^{k-2}3^1,\ldots,3^{k-1}\}$, then by compressing (as in the proof of Lemma 4.9.2) we may increase the left-hand side of (4.28), so we are also done in this case. The lemma therefore reduces to a finite check of families with $\max\{|A|,|B|\}\leqslant 4$, and in fact (using compression once again) it is sufficient to consider the antichains $\{1\}$, $\{2,3\}$, $\{2,9\}$, $\{3,4\}$, $\{4,6,9\}$, and $\{8,12,18,27\}$. The lemma now follows from a trivial case analysis, which can be done by hand. □

Lemma 4.9.4. If A and B are two 5-smooth antichains, then

$$\sum_{a\in A}\sum_{b\in B}\frac{1}{\operatorname{lcm}(a,b)}\leqslant 1.7,$$

with equality if and only if $A = B = \{2,3,5\}$.

Proof. We decompose A and B as a union of sets $5^i \cdot A_i$ and $5^j \cdot B_j$, where A_i and B_j are 3-smooth antichains, as in the proof of Lemma 4.9.2. Suppose first that there is no pair (i,j) with $A_i = B_j = \{1\}$ or $A_i = B_j = \{2,3\}$. Then, by Lemma 4.9.3 and (4.17), we have

$$\sum_{a\in A}\sum_{b\in B}\frac{1}{\operatorname{lcm}(a,b)} = \sum_{i=0}^{\infty}\sum_{j=0}^{\infty}\frac{1}{5^{\max\{i,j\}}}\sum_{a\in A_i}\sum_{b\in B_j}\frac{1}{\operatorname{lcm}(a,b)}$$
$$\leqslant \frac{31}{36}\left(1+\frac{3}{5}+\frac{5}{5^2}+\frac{7}{5^3}+\cdots\right) = \frac{31}{36}\cdot\frac{15}{8} < 1.7 - \frac{1}{12}.$$

Next, suppose that $A_i = B_j = \{1\}$ for some pair (i,j), and observe that $A_{i'} = B_{j'} = \emptyset$ for every $i' > i$ and $j' > j$, so the pair (i,j) is unique. If there is no pair (s,t) with $A_s = B_t = \{2,3\}$, then the bound above increases by at most $(1-\frac{31}{36})5^{-\max\{i,j\}}$, and this is less that $\frac{1}{12}$ if $\max\{i,j\} \geqslant 1$. On the other hand, if $A_0 = B_0 = \{1\}$, then (since $A_i = B_i = \emptyset$ for all $i > 0$) we have $A = B = \{1\}$, and so $\sum_{a\in A}\sum_{b\in B}\frac{1}{\operatorname{lcm}(a,b)} = 1 < 1.7$.

We may therefore assume that $A_i = B_j = \{2,3\}$ for some pair (i,j), which implies that $A_{i'}, B_{j'} \subseteq \{1\}$ for every $i' > i$ and $j' > j$, and (as above) at most one of the sets in each sequence is non-empty. The bound above increases by at most

$$\left(\frac{7}{6}-\frac{31}{36}\right)\frac{1}{5^{\max\{i,j\}}} + \left(1-\frac{31}{36}\right)\frac{1}{5^{\max\{i,j\}+1}} = \frac{1}{3}\cdot\frac{1}{5^{\max\{i,j\}}} < \frac{1}{12}$$

if $\max\{i,j\} \geqslant 1$. However, if $A_0 = B_0 = \{2,3\}$, then it is easy to see that $\sum_{a\in A}\sum_{b\in B}\frac{1}{\operatorname{lcm}(a,b)}$ is maximized by taking $A = B = \{2,3,5\}$, and in that case it is equal to 1.7. □

Having completed the easy preliminaries, we are ready to prove Schinzel's conjecture.

Proof of Theorem 4.1.5. We first observe that we may assume that none of the moduli d_i are prime powers. Indeed, we may assume that the covering is minimal, so the removal of any A_{d_i} results in a set of progressions that do not cover $\mathbb{Z}$. If $d_i = p^j$ for some prime p and $j > 0$, then the prime can appear at most to the $(j-1)$st power in any other moduli. Thus the other progressions fail to cover some congruence class mod Q/p, where $Q = \operatorname{lcm}\{d_1,\ldots,d_k\}$. But this congruence class cannot be covered by A_{d_i} as $d_i \nmid Q/p$, a contradiction.

We now apply our sieve with $\delta_1 = \delta_2 = \delta_3 = 0$, so that $\mathbb{P}_1, \mathbb{P}_2, \mathbb{P}_3$ are equal to the uniform measure when processing the primes $p_1 = 2$, $p_2 = 3$ and $p_3 = 5$, respectively. Set $k = i_0 = 3$ and $\kappa = 1.7$. We claim this is a valid choice for κ and that $f_3 \leqslant 2.55 < g_3$. Observe that, by Lemma 4.9.2, the total measure of $B_1 \cup B_2 \cup B_3$ is at most $1/3$. Now we improve the bound on $\mathbb{E}[\alpha_i(x)^2]$ for $i \geqslant 4$ as follows. By Theorem 4.3.2, we have

$$M_i^{(2)} \leqslant \sum_{j_1,j_2\geqslant 1}\frac{1}{p_i^{j_1+j_2}}\sum_{m_1,m_2\in S_i}\frac{\nu\big(\operatorname{lcm}(m_1,m_2)\big)}{\operatorname{lcm}(m_1,m_2)}\sum_{a\in D(m_1,j_1)}\sum_{b\in D(m_2,j_2)}\frac{1}{\operatorname{lcm}(a,b)},$$

where S_i is the set of integers whose prime factors all lie between 7 and p_{i-1}, and for each $m \in S_i$ and $j \geqslant 1$, we define

$$D(m,j) = \big\{a : mp_i^j a \in D \text{ and } a \text{ is 5-smooth}\big\}.$$

Since $D(m,j)$ is a 5-smooth antichain, it follows from Lemmas 4.3.7 and 4.9.4 that

$$M_i^{(2)} \leqslant 1.7 \sum_{j_1,j_2\geqslant 1} \frac{1}{p_i^{j_1+j_2}} \sum_{m_1,m_2\in S_i} \frac{\nu\big(\mathrm{lcm}(m_1,m_2)\big)}{\mathrm{lcm}(m_1,m_2)} = \frac{1.7}{(p_i-1)^2} \prod_{j=4}^{i-1} \left(1 + \frac{3p_j-1}{(1-\delta_j)(p_j-1)^2}\right).$$

Therefore, setting $i_0 = 3$ and $\kappa = 1.7$, it follows that (4.20) holds. Hence, recalling from above that $\mu_3 \geqslant 2/3$, we obtain $f_3 = 1.7 \cdot 3/2 = 2.55 < g_3$ (see Table 4.1), so, by Corollary 4.6.3, the system $\mathscr{A} = \{A_d : d \in D\}$ does not cover the integers. □

4.10 Constructions

In this section we will provide constructions of families of arithmetic progressions that answer (negatively) the question of Filaseta, Ford, Konyagin, Pomerance and Yu [76] mentioned in the Introduction, and show that Theorem 4.1.1 is not far from best possible. To be precise, we will prove Theorem 4.1.2 and Theorem 4.1.3.

Note that the bound (4.2) in Theorem 4.1.2 is (obviously) best possible; we remark that the moduli in our construction will moreover be square-free. Theorem 4.1.3 shows that the function $(\log p)^{3+\varepsilon}$ in the statement of Theorem 4.1.1 cannot be replaced by a constant.

The proof of Theorem 4.1.2 is relatively simple, while the proof of Theorem 4.1.3 will require somewhat more work.

Proof of Theorem 4.1.2. We will choose a collection $P_1, \dots, P_N$ of disjoint sets of primes, and define

$$Q_i := \prod_{j\leqslant i} \prod_{p\in P_j} p \qquad \text{and} \qquad D_i := \big\{p \cdot Q_{i-1} : p \in P_i\big\}$$

for each $i \in [N]$, where $Q_0 := 1$. We will show that, for a suitable choice of $P_1, \dots, P_N$, the set $D = D_1 \cup \cdots \cup D_N$ has the following properties:

$$\sum_{d\in D} \frac{1}{d} \leqslant 1 + \frac{\varepsilon}{3}, \tag{4.29}$$

and there exists a collection of arithmetic progressions, with distinct moduli in D, such that the uncovered set has density at most $\varepsilon/3$. By removing a few of the progressions from this family, we will obtain the claimed collection $A_1, \dots, A_k$.

We construct the sets $P_1, \dots, P_N$ of primes as follows. First, let us fix some positive constants c_0, c and δ such that

$$c_0 := 1 + \frac{\varepsilon}{3}, \qquad \text{and} \qquad c = \frac{\delta}{1-e^{-\delta}} \in (1, c_0).$$

Indeed, $\delta/(1-e^{-\delta})$ is a continuous increasing function of δ which tends to 1 as $\delta \to 0$, so for sufficiently small δ we have $1 < \delta/(1-e^{-\delta}) < c_0$. Assume (without loss) that $M > 3/\varepsilon$, and choose N sufficiently large so that $e^{-\delta N} < \varepsilon/3$. Now let P_1 be any set of primes such that $p \geqslant M$ for every $p \in P_1$, and

$$\delta \leqslant \sum_{p \in P_1} \frac{1}{p} \leqslant \delta + \frac{c_0 - c}{N}.$$

In general, if we have already constructed $P_1, \dots, P_j$, then let P_{j+1} be any set of primes (disjoint from $P_1 \cup \cdots \cup P_j$) such that $p \geqslant M$ for every $p \in P_{j+1}$, and

$$\delta e^{-\delta j} \leqslant \sum_{p \in P_{j+1}} \frac{1}{p \cdot Q_j} \leqslant \delta e^{-\delta j} + \frac{c_0 - c}{N}. \tag{4.30}$$

As the sum $\sum 1/p$ over prime p diverges, it is clear that sets $P_1, \dots, P_N$ exist with these properties. It follows that

$$\sum_{d \in D} \frac{1}{d} = \sum_{j=0}^{N-1} \sum_{p \in P_{j+1}} \frac{1}{p \cdot Q_j} \leqslant \sum_{j=0}^{N-1} \left(\delta e^{-\delta j} + \frac{c_0 - c}{N} \right) \leqslant c_0 = 1 + \frac{\varepsilon}{3},$$

where in the final inequality we used the identity $\sum_{j=0}^{\infty} \delta e^{-\delta j} = \delta \big(1 - e^{-\delta}\big)^{-1} = c$.

Now, to construct the arithmetic progressions, simply choose (for each $d \in D$ in turn) any arithmetic progression with modulus d that has at least the expected intersection with the (as yet) uncovered set. To be more precise, for each $j \in [N]$ let $\mathscr{A}_j$ denote the collection of arithmetic progressions whose modulus lies in D_j, and write $\varepsilon_j := \mathbb{P}_0(R_j)$ for the density of the uncovered set $R_j := \mathbb{Z}_{Q_j} \setminus \bigcup_{A \in \mathscr{B}_j} A$, where $\mathscr{B}_j := \mathscr{A}_1 \cup \cdots \cup \mathscr{A}_j$. Now observe that if $d = p \cdot Q_j \in D_{j+1}$, then there are $p \cdot |R_j|$ congruence classes mod d that completely cover R_j. Hence there is a congruence class that covers at least a fraction $1/(p \cdot |R_j|)$ of the as yet uncovered set. It follows that

$$\varepsilon_{j+1} \leqslant \varepsilon_j \prod_{p \in P_{j+1}} \left(1 - \frac{1}{p \cdot |R_j|} \right) \leqslant \varepsilon_j \cdot \exp\left(-\frac{1}{|R_j|} \sum_{p \in P_{j+1}} \frac{1}{p} \right)$$

for each $0 \leqslant j \leqslant N-1$ (where $\varepsilon_0 := 1$), and hence, by (4.30),

$$\varepsilon_{j+1} \leqslant \varepsilon_j \cdot \exp\left(-\frac{Q_j}{|R_j|} \cdot \delta e^{-\delta j} \right) = \varepsilon_j \cdot \exp\left(-\frac{\delta e^{-\delta j}}{\varepsilon_j} \right).$$

It now follows immediately by induction that $\varepsilon_j \leqslant e^{-\delta j}$ for every $j \in [N]$, and in particular $\varepsilon_N \leqslant e^{-\delta N} \leqslant \varepsilon/3$, by our choice of N.

We have therefore constructed a collection of arithmetic progressions whose set D of (distinct) moduli satisfies (4.29), and whose uncovered set has density at most $\varepsilon/3$. To complete the construction, simply choose a maximal subset $D' \subset D$ such that $\sum_{d \in D'} \frac{1}{d} < 1$, and observe that the density of the set

uncovered by $\{A_d : d \in D'\}$ is at most

$$\frac{\varepsilon}{3} + \sum_{d \in D \setminus D'} \frac{1}{d} \leqslant \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{1}{M} \leqslant \varepsilon,$$

as required. □

The proof of Theorem 4.1.3 is similar to that of Theorem 4.1.2, but the details are somewhat more complicated.

Proof of Theorem 4.1.3. We will again choose a large collection of disjoint sets of primes, but this time we will arrange them in a tree-like structure, and our common differences will be formed by taking products of certain subsets of the primes along paths in the tree. To begin, let us choose $t > 2$ sufficiently large so that $t^{\lambda} < e^{t-3}$, and let P_1 be a set of primes such that $p \geqslant M$ for every $p \in P_1$, and

$$t - 1 \leqslant \prod_{p \in P_1} \left(1 + \frac{1}{p}\right) \leqslant t.$$

This is possible as the product $\prod_p (1 + 1/p)$ over all primes is infinite, and by taking only primes greater than t we can ensure that some finite product lands in $[t-1, t]$. Set

$$Q_1 := \prod_{p \in P_1} p \qquad \text{and} \qquad D_1 := \{d > 1 : d \mid Q_1\},$$

and choose a collection of arithmetic progressions $\mathscr{A}_1 = \{a_d + d\mathbb{Z} : d \in D_1\}$ so as to minimize the density of the uncovered set $R_1 := \mathbb{Z}_{Q_1} \setminus \bigcup_{A \in \mathscr{A}_1} A$. As in the previous proof, this can be done (greedily) so that

$$\mathbb{P}_0(R_1) \leqslant \prod_{d \in D_1} \left(1 - \frac{1}{d}\right) \leqslant \exp\left(-\sum_{d \in D_1} \frac{1}{d}\right) \leqslant \exp\left(1 - \prod_{p \in P_1} \left(1 + \frac{1}{p}\right)\right) \leqslant e^{2-t}.$$

Now, for each $x_1 \in R_1$, let $P_2^{x_1}$ be a set of new primes (i.e., disjoint for each choice of x_1, and disjoint from P_1) such that $p \geqslant M$ for every $p \in P_2^{x_1}$, and

$$t - 1 \leqslant \prod_{p \in P_2^{x_1}} \left(1 + \frac{1}{p}\right) \leqslant t.$$

Set

$$Q_2^{x_1} := \prod_{p \in P_2^{x_1}} p \qquad \text{and} \qquad D_2^{x_1} := \{d \cdot Q_1 : d > 1 \text{ and } d \mid Q_2^{x_1}\},$$

and choose a collection of arithmetic progressions $\mathscr{A}_2^{x_1} = \{a_d + d\mathbb{Z} : d \in D_2^{x_1}\}$ so as to minimize the density of the uncovered set

$$R_2^{x_1} := \left\{(x_1, y) : y \in \mathbb{Z}_{Q_2^{x_1}}\right\} \setminus \bigcup_{A \in \mathscr{A}_2^{x_1}} A,$$

where (as usual) each $A \in \mathscr{A}_2^{x_1}$ is viewed as a subset of $\mathbb{Z}_{Q_1} \times \mathbb{Z}_{Q_2^{x_1}}$. Note that

$$\mathbb{P}_0\left(R_2^{x_1}\right) \leqslant \frac{1}{Q_1} \prod_{dQ_1 \in D_2^{x_1}} \left(1 - \frac{1}{d}\right) \leqslant \frac{1}{Q_1} \cdot \exp\left(1 - \prod_{p \in P_2^{x_1}} \left(1 + \frac{1}{p}\right)\right) \leqslant \frac{e^{2-t}}{Q_1},$$

and hence, setting $R_2 := \bigcup_{x_1 \in R_1} R_2^{x_1}$ and summing over $x_1 \in R_1$, we have

$$\mathbb{P}_0(R_2) \leqslant |R_1| \cdot \frac{e^{2-t}}{Q_1} = \mathbb{P}_0(R_1) \cdot e^{2-t} \leqslant e^{2(2-t)}.$$

To describe a general step of this construction, suppose that we have already defined the tree of primes and progressions to depth $i-1$, and for each $x_1 \in R_1$, $x_2 \in R_2^{x_2}$, $\dots$, $x_{i-1} \in R_{i-1}^{x_1,\dots,x_{i-2}}$, choose a set $P_i^{x_1,\dots,x_{i-1}}$ of new primes (disjoint from all previously chosen sets) such that $p \geqslant M$ for every $p \in P_i^{x_1,\dots,x_{i-1}}$, and

$$t - 1 \leqslant \prod_{p \in P_i^{x_1,\dots,x_{i-1}}} \left(1 + \frac{1}{p}\right) \leqslant t.$$

Set

$$Q_i^{x_1,\dots,x_{i-1}} := \prod_{p \in P_i^{x_1,\dots,x_{i-1}}} p$$

and

$$D_i^{x_1,\dots,x_{i-1}} := \left\{d \cdot Q_1 \cdot Q_2^{x_1} \cdots Q_{i-1}^{x_1,\dots,x_{i-2}} : d > 1 \text{ and } d \mid Q_i^{x_1,\dots,x_{i-1}}\right\},$$

and choose a collection of arithmetic progressions $\mathscr{A}_i^{x_1,\dots,x_{i-1}} = \{a_d + d\mathbb{Z} : d \in D_i^{x_1,\dots,x_{i-1}}\}$ so as to minimize the density of the uncovered set

$$R_i^{x_1,\dots,x_{i-1}} := \left\{(x_{i-1}, y) : y \in \mathbb{Z}_{Q_i^{x_1,\dots,x_{i-1}}}\right\} \setminus \bigcup_{A \in \mathscr{A}_i^{x_1,\dots,x_{i-1}}} A,$$

where, as before, each $A \in \mathscr{A}_i^{x_1,\dots,x_{i-1}}$ is viewed as a subset of $\mathbb{Z}_{Q_1} \times \cdots \times \mathbb{Z}_{Q_i^{x_1,\dots,x_{i-1}}}$. Setting $R_i := \bigcup_{x_1 \in R_1} \cdots \bigcup_{x_{i-1} \in R_{i-1}^{x_1,\dots,x_{i-2}}} R_i^{x_1,\dots,x_{i-1}}$, and repeating the calculation above, we obtain

$$\mathbb{P}_0\left(R_i^{x_1,\dots,x_{i-1}}\right) \leqslant \frac{1}{Q_1 \cdots Q_{i-1}^{x_1,\dots,x_{i-2}}} \prod_{dQ_1 \cdots Q_{i-1}^{x_1,\dots,x_{i-2}} \in D_i^{x_1,\dots,x_{i-1}}} \left(1 - \frac{1}{d}\right) \leqslant \frac{e^{2-t}}{Q_1 \cdots Q_{i-1}^{x_1,\dots,x_{i-2}}},$$

and therefore

$$\mathbb{P}_0(R_i) \leqslant \mathbb{P}_0(R_{i-1}) \cdot e^{2-t} \leqslant e^{(2-t)i},$$

by induction. Hence, defining $\mathscr{A}_i$ to be the union of all $\mathscr{A}_i^{x_1,\dots,x_{i-1}}$ and $\mathscr{A} = \mathscr{A}_1 \cup \cdots \cup \mathscr{A}_n$, it follows that the uncovered set R has density

$$\mathbb{P}_0(R) = \mathbb{P}_0(R_n) \leqslant e^{(2-t)n} \to 0$$

as $n \to \infty$.

It remains to show that

$$\sum_{d\in D} \frac{\mu(d)}{d} \leqslant C$$

for every $n \in \mathbb{N}$, where D is the set of moduli of progressions in $\mathscr{A}$. To prove this, observe first that

$$\mu(Q_1) = \prod_{p\in P_1} \left(1+\frac{\lambda}{p}\right) \leqslant \prod_{p\in P_1} \left(1+\frac{1}{p}\right)^{\lambda} \leqslant t^{\lambda},$$

and, assuming (as we may) that $M \geqslant \lambda$,

$$\sum_{d\in D_1} \frac{\mu(d)}{d} = \prod_{p\in P_1} \left(1+\frac{\mu(p)}{p}\right) - 1 \leqslant \prod_{p\in P_1} \left(1+\frac{1}{p}+\frac{\lambda}{p^2}\right) \leqslant \prod_{p\in P_1} \left(1+\frac{1}{p}\right)^{2} \leqslant t^2.$$

Similarly, we have

$$\sum_{d\in D_2^{x_1}} \frac{\mu(d)}{d} = \frac{\mu(Q_1)}{Q_1} \sum_{dQ_1\in D_2^{x_1}} \frac{\mu(d)}{d} \leqslant \frac{t^{\lambda+2}}{Q_1}$$

for each $x_1 \in R_1$, and, more generally,

$$\sum_{d\in D_i^{x_1,\dots,x_{i-1}}} \frac{\mu(d)}{d} \leqslant \frac{\mu\big(Q_1\cdots Q_{i-1}^{x_1,\dots,x_{i-2}}\big)}{Q_1\cdots Q_{i-1}^{x_1,\dots,x_{i-2}}} \sum_{d|Q_i^{x_1,\dots,x_{i-1}}} \frac{\mu(d)}{d} \leqslant \frac{t^{\lambda(i-1)+2}}{Q_1\cdots Q_{i-1}^{x_1,\dots,x_{i-2}}}.$$

Hence, summing over $i \in [n]$ and sequences $x_1 \in R_1, \dots, x_{i-1} \in R_{i-1}^{x_1,\dots,x_{i-2}}$, we obtain

$$\sum_{d\in D} \frac{\mu(d)}{d} \leqslant \sum_{i=0}^{n-1} t^{\lambda i+2} \cdot \mathbb{P}_0(R_i) \leqslant \sum_{i=0}^{n-1} t^{\lambda i+2} \cdot e^{(2-t)i} < 2t^2,$$

since $t^{\lambda} < e^{t-3}$, as required. □

Chapter 5

The Erdős–Selfridge problem with square-free moduli

The work in this section is joint with Paul Balister, Béla Bollobás, Robert Morris and Julian Sahasrabudhe and is published in Algebra and Number Theory [5].

5.1 Introduction

In this chapter we refine the method from the previous chapter in order to prove Theorem 1.2.7, which states that in any covering system with distinct square free moduli there exists an even modulus. This resolves, in the square free case, a problem of Erdős and Selfridge [75] from 1965. For convenience, we restate Theorem 1.2.7 here as Theorem 5.1.1.

Theorem 5.1.1. In any finite collection of arithmetic progressions with distinct square-free moduli greater than 1 that covers the integers, at least one of the moduli is even.

We prove Theorem 5.1.1 in a (slightly more general) geometric setting; a second aim of this chapter is to investigate covering systems in this setting. Let $S_1, \dots, S_n$ be finite sets with at least two elements, and set

$$Q = S_1 \times \cdots \times S_n.$$

If $A = A_1 \times \cdots \times A_n \subseteq Q$ with each A_k either equal to S_k or a singleton element of S_k, then we say that A is a *hyperplane*. We write $A = [x_1, \dots, x_n]$, where $x_k \in S_k \cup \{*\}$ for each $k \in [n]$, and $*$ indicates that $A_k = S_k$. Let us write $F(A) = \{k : x_k \in S_k\}$ for the set of *fixed coordinates* of A, and say that two hyperplanes A and A' are *parallel* if $F(A) = F(A')$. Let us also say that A is *non-trivial* if $F(A) \neq \emptyset$.

Theorem 5.1.1 is equivalent to the following theorem in this geometric setting.

Theorem 5.1.2. For each $k \in [n]$, let p_k be the kth prime, and set $S_k = [p_{k+1}]$. Any collection of non-trivial hyperplanes that covers $Q := S_1 \times \cdots \times S_n$ contains two parallel hyperplanes.

To see the equivalence of Theorems 5.1.1 and 5.1.2, observe that, by the Chinese Remainder Theorem, there is a natural equivalence[1] between finite collections $\mathscr{A}$ of arithmetic progressions with square-free, odd, p_{n+1}-smooth moduli that cover the integers, and finite collections $\mathscr{H}$ of hyperplanes that cover the box $Q = [p_2] \times \cdots \times [p_{n+1}]$. Moreover, if the moduli of $\mathscr{A}$ are distinct then the hyperplanes in $\mathscr{H}$ are non-parallel.

In order to motivate our second main theorem, let us next state, in this geometric setting, a special case (for square-free moduli) of the breakthrough result of Hough [100] which answered the Erdős minimum modulus problem.

Theorem 5.1.3 (Hough, 2015)**.** Let $p_1, \dots, p_n$ be the first n primes. There exists a constant C such that if $\mathscr{A}$ is a collection of hyperplanes that cover $Q := [p_1] \times \cdots \times [p_n]$, then either two of the hyperplanes are parallel, or there exists a hyperplane $A \in \mathscr{A}$ with $F(A) \subseteq [C]$.

To deduce Theorem 5.1.3 from the main result of [100], simply note that if every covering system with distinct, square-free moduli contains an arithmetic progression with modulus at most $M \leqslant p_C$, then the progression with minimum modulus corresponds to a hyperplane A with $F(A) \subseteq [C]$. In the other direction, it follows from Theorem 5.1.3 that a covering system with distinct, square-free moduli contains a progression corresponding to a hyperplane A with $F(A) \subseteq [C]$, and the modulus d of this progression satisfies $d \leqslant \prod_{i=1}^{C} p_i$.

Using our method, we are able to prove the following strengthening of Theorem 5.1.3.

Theorem 5.1.4. For every sequence of integers $(q_k)_{k\geqslant 1}$ such that $q_k \geqslant 2$ for each $k \in \mathbb{N}$ and

$$\liminf_{k\to\infty} \frac{q_k}{k} > 3,$$

there exists a constant C such that the following holds. Let $\mathscr{A}$ be a collection of hyperplanes that cover $Q := [q_1] \times \cdots \times [q_n]$ for some $n \in \mathbb{N}$. Then either two of the hyperplanes are parallel, or there exists a hyperplane $A \in \mathscr{A}$ with $F(A) \subseteq [C]$.

Note that in Theorem 5.1.3 the sequence $(p_k)_{k\geqslant 1}$ grows asymptotically as $k \log k$, whereas in Theorem 5.1.4 we allow the sequence $(q_k)_{k\geqslant 1}$ to grow only linearly. We show (see Section 5.4) that Theorem 5.1.4 is close to best possible, since there exists an example with $\lim q_k/k = 1$ for which the conclusion of the theorem fails.

The rest of the chapter is organized as follows. In Section 5.2 we outline the sieve that we use in the proofs, and in Section 5.3 we state and prove our main technical results, Theorems 5.3.1 and 5.3.2. In Section 5.4 we deduce Theorem 5.1.4. Finally, we dedicate Section 5.5 to the proof of our main result, Theorem 5.1.1.

[1] To be precise, the progression $a + d\mathbb{Z}$ with $d = \prod_{i\in I} p_i$ corresponds to the hyperplane $A = [x_1, \dots, x_n]$ where $x_i = a \bmod p_i$ if $i \in I$, and $x_i = *$ otherwise.

5.2 Definition of the Sieve

In this section we outline the proofs of Theorems 5.1.2 and 5.1.4. In particular, we generalize the method developed in Chapter 4 to the geometric setting, and while doing so we introduce several new ideas that turn out to be crucial in the proofs. For the convenience of the reader and for completeness, we include full proofs of all intermediate results, even though several of them are direct adaptations of the corresponding results in Chapter 4.

As in the Introduction, let $S_1, \dots, S_n$ be finite sets with at least two elements, set

$$Q := S_1 \times \cdots \times S_n,$$

and let $\mathscr{A}$ be a collection of non-trivial hyperplanes, no two of which are parallel. Set

$$\mathscr{F} = \mathscr{F}(\mathscr{A}) := \big\{F(A) : A \in \mathscr{A}\big\} \subseteq \mathscr{P}([n]) \setminus \{\emptyset\},$$

and (recalling that $F(A) \neq F(A')$ for distinct $A, A' \in \mathscr{A}$) let us index the hyperplanes in $\mathscr{A}$ by the corresponding set of fixed coordinate indices, so $\mathscr{A} = \{A_F : F \in \mathscr{F}\}$. Our goal is to estimate the density (under some probability measure) of the uncovered set

$$R := Q \setminus \bigcup_{F \in \mathscr{F}} A_F.$$

Rather than considering the entire collection of hyperplanes $\mathscr{A}$ all at once, we expose the hyperplanes dimension by dimension and track how the density of the uncovered set evolves. To be more precise, define, for each $1 \leqslant k \leqslant n$,

$$\mathscr{F}_k := \big\{F \in \mathscr{F} : F \subseteq [k]\big\} \qquad \text{and} \qquad \mathscr{A}_k := \big\{A_F : F \in \mathscr{F}_k\big\}$$

for the family of sets of fixed coordinate indices and the corresponding hyperplanes that are contained in the initial segment $[k]$. Let

$$R_k := Q \setminus \bigcup_{F \in \mathscr{F}_k} A_F = Q \setminus \bigcup_{A_F \in \mathscr{A}_k} A_F$$

be the set of elements not contained in any of the hyperplanes of $\mathscr{A}_k$, so in particular $R_n = R$. We also write $\mathscr{N}_k := \mathscr{F}_k \setminus \mathscr{F}_{k-1}$ for the family of "new" sets of fixed coordinate indices at the kth stage, i.e., those sets that contain k and are contained in $[k]$, and define

$$B_k := \bigcup_{F \in \mathscr{N}_k} A_F \tag{5.1}$$

to be the union of the hyperplanes exposed at step k, so that $R_k = R_{k-1} \setminus B_k$.

It is often convenient to consider R_k, B_k and A_F with $F \in \mathscr{F}_k$ as subsets of

$$Q_k := S_1 \times \cdots \times S_k$$

by identifying $X \subseteq Q_k$ with $X \times S_{k+1} \times \cdots \times S_n \subseteq Q$. We call a set of this form *Q_k-measurable*.

5.2.1 The probability measures $\mathbb{P}_k$

The construction of the probability measures is similar to that in Chapter 4, and no significant new ideas are needed. The main difference from Chapter 4 is that instead of starting with the uniform measure as our $\mathbb{P}_0$, we allow for possible optimization of the measure on the first few coordinates. In general we start with some measure $\mathbb{P}_a$, to be determined, which is supported on $R_a \subseteq Q_a = S_1 \times \cdots \times S_a$.[2]

Our aim is to construct, for each $a < k \leqslant n$, a measure $\mathbb{P}_k$ on Q_k in such a way that $\mathbb{P}_k(B_k)$ is small, but without changing the measure of B_i for any $i < k$. Fix a sequence of constants $\delta_{a+1}, \ldots, \delta_n \in [0, 1/2]$, and assume that we have already defined a probability measure $\mathbb{P}_{k-1}$ on Q_{k-1}. Recall that $Q_k = Q_{k-1} \times S_k$, and hence the elements of Q_k can be written as pairs (x,y), where $x \in Q_{k-1}$ and $y \in S_k$. We may view R_{k-1} as a collection of fibres of the form $F_x = \{(x,y) : y \in S_k\} \subseteq Q_k$, where $\mathbb{P}_{k-1}$ is extended uniformly to a measure on Q_k (so is uniform on each fibre), and view R_k as being obtained from R_{k-1} by removing B_k, i.e., by removing the points that are contained in the new hyperplanes of $\mathscr{A}_k \setminus \mathscr{A}_{k-1}$.

Now, for each $x \in Q_{k-1}$, define

$$\alpha_k(x) = \frac{\mathbb{P}_{k-1}\left(F_x \cap B_k\right)}{\mathbb{P}_{k-1}(x)} = \frac{\left|\left\{y \in S_k : (x,y) \in B_k\right\}\right|}{|S_k|}, \tag{5.2}$$

that is, the proportion of the fibre F_x that is removed at stage k. The probability measure $\mathbb{P}_k$ on Q_k is defined as follows:

$$\mathbb{P}_k(x,y) := \begin{cases} \max\left\{0, \dfrac{\alpha_k(x) - \delta_k}{\alpha_k(x)(1-\delta_k)}\right\} \cdot \mathbb{P}_{k-1}(x,y), & \text{if } (x,y) \in B_k; \\ \min\left\{\dfrac{1}{1-\alpha_k(x)}, \dfrac{1}{1-\delta_k}\right\} \cdot \mathbb{P}_{k-1}(x,y), & \text{if } (x,y) \notin B_k. \end{cases} \tag{5.3}$$

To motivate the definition above, note that if $\alpha_k(x) \leqslant \delta_k$, then $\mathbb{P}_k(x,y) = 0$ for every element of Q_k that is covered in step k, and that the measure is increased proportionally elsewhere to compensate. On the other hand, for those $x \in Q_{k-1}$ for which $\alpha_k(x) > \delta_k$, we 'cap' the distortion by increasing the measure at each point not covered in step k by a factor of $1/(1-\delta_k)$, and decreasing the measure on removed points by a corresponding factor.

The measure $\mathbb{P}_k$ satisfies the following simple properties, see Lemma 4.2.1 and Lemma 4.2.2.

[2] We emphasize that here we allow $a = 0$, in which case $\mathbb{P}_0$ is the trivial probability measure on the empty product.

Lemma 5.2.1. For any $k > a$ and any Q_{k-1}-measurable set S we have

$$\mathbb{P}_k(S) = \mathbb{P}_{k-1}(S). \tag{5.4}$$

For any set $S \subseteq Q$, we have

$$\mathbb{P}_k(S) \leqslant \frac{1}{1-\delta_k} \cdot \mathbb{P}_{k-1}(S). \tag{5.5}$$

Moreover, if $S \subseteq B_k$ then

$$\mathbb{P}_k(S) \leqslant \mathbb{P}_{k-1}(S). \tag{5.6}$$

In particular, it follows from Lemma 5.2.1 that if

$$\sum_{k=a+1}^{n} \mathbb{P}_k(B_k) < 1 \tag{5.7}$$

then $\mathscr{A}$ does not cover Q, since B_k is a Q_k-measurable set, so by (5.4) we have $\mathbb{P}_n(B_k) = \mathbb{P}_k(B_k)$. For each $a \leqslant k \leqslant n$, define

$$\mu_k := 1 - \sum_{i=a+1}^{k} \mathbb{P}_i(B_i),$$

and observe that $\mu_k \leqslant \mathbb{P}_k(R_k)$.

5.3 Bounding the density of the covered set

In this section we prove two technical results, Theorems 5.3.1 and 5.3.2, which together imply Theorems 5.1.2 and 5.1.4. We remark that Theorem 5.3.1 essentially follows from Theorem 4.3.1, but Theorem 5.3.2 introduces a new bound that is motivated geometrically, which turns out to be crucial in the proof of Theorem 5.1.2.

Given a collection $\mathscr{A}$ of hyperplanes in $Q = S_1 \times \cdots \times S_n$, a probability distribution $\mathbb{P}_a$ supported on R_a, and constants $\delta_{a+1}, \ldots, \delta_n \in [0, 1/2]$, let the probability distributions $\mathbb{P}_k$ and functions $\alpha_k \colon Q_{k-1} \to [0,1]$ be defined as in (5.2) and (5.3), and set

$$M_k^{(1)} := \mathbb{E}_{k-1}\left[\alpha_k(x)\right] \qquad \text{and} \qquad M_k^{(2)} := \mathbb{E}_{k-1}\left[\alpha_k(x)^2\right], \tag{5.8}$$

where we write $\mathbb{E}_{k-1}$ to denote expectation with respect to the measure $\mathbb{P}_{k-1}$. In order to show that $\mathscr{A}$ does not cover Q, it is sufficient, by (5.7), to show that $\mu_n > 0$. To do so, we will bound $\mathbb{P}_k(B_k)$ in terms of the moments $M_k^{(1)}$ and $M_k^{(2)}$. As noted, the following theorem is (essentially) proved in Chapter 4.

Theorem 5.3.1. Let $\mathscr{A}$ be a collection of non-trivial hyperplanes in $Q = S_1 \times \cdots \times S_n$, no two of which are parallel. If

$$\sum_{k=a+1}^{n} \min\left\{ M_k^{(1)}, \frac{M_k^{(2)}}{4\delta_k(1-\delta_k)} \right\} < 1, \tag{5.9}$$

then $\mathscr{A}$ does not cover Q.

In order to show that (5.9) holds in our applications, we need to bound the moments of $\alpha_k(x)$. To state our bounds on $M_k^{(1)}$ and $M_k^{(2)}$, we will need some additional notation. Define a function $c\colon \mathscr{P}([a]) \to [0,1]$ by setting

$$c(I) = \max\big\{\, \mathbb{P}_a(H) : H \text{ is a hyperplane in } Q_a \text{ with } F(H) = I \big\} \tag{5.10}$$

for each $I \subseteq [a]$, and define a function $\nu\colon \mathscr{P}([a+1,n]) \to \mathbb{R}_{>0}$, by setting

$$\nu(J) = \prod_{j\in J} \frac{1}{(1-\delta_j)|S_j|} \tag{5.11}$$

for each $J \subseteq [a+1,n]$. Note that $c(\emptyset) = \nu(\emptyset) = 1$. For each $k \geqslant a$ and $x \in \mathbb{R}$, set

$$c_k(x) = \sum_{I\subseteq[a]} \sum_{J\subseteq[a+1,k]} c(I)\nu(J)x^{|I|+|J|} = \sum_{I\subseteq[a]} c(I)x^{|I|} \prod_{j=a+1}^{k} \left(1 + \frac{x}{(1-\delta_j)|S_j|}\right). \tag{5.12}$$

The following technical theorem provides general bounds on $M_k^{(1)}$ and $M_k^{(2)}$.

Theorem 5.3.2. Let $\mathscr{A}$ be a collection of non-trivial hyperplanes in $Q = S_1 \times \cdots \times S_n$, no two of which are parallel. Let $\mathbb{P}_a$ be a probability measure supported on R_a. Let $\delta_{a+1}, \dots, \delta_n \in [0, 1/2]$. Then, with $M_k^{(1)}$ and $M_k^{(2)}$ as in (5.8) and c_{k-1} as in (5.12), for each $a < k \leqslant n$,

$$M_k^{(1)} \leqslant \frac{c_{k-1}(1)}{|S_k|} \qquad \text{and} \qquad M_k^{(2)} \leqslant \frac{c_{k-1}(3)}{|S_k|^2}. \tag{5.13}$$

Moreover, if none of the hyperplanes in $\mathscr{N}_k$ has co-dimension 1, then

$$M_k^{(2)} \leqslant \frac{1}{|S_k|^2}\big(c_{k-1}(3) - 2c_{k-1}(1) + 1\big). \tag{5.14}$$

Before embarking on the (straightforward) proofs of Theorems 5.3.1 and 5.3.2, let us briefly discuss the bound (5.14), which will play an important role in the proof of Theorem 5.1.1. In order to apply it, we first need to remove from $\mathscr{A}$ each of the codimension 1 hyperplanes, each of which is of the form $S_1 \times \cdots \times S_{i-1} \times \{s\} \times S_{i+1} \times \cdots \times S_n$ for some $i \in [n]$ and $s \in S_i$. Note that in doing so we remove the point s from the possible values of the ith coordinate, effectively replacing S_i by $S_i' := S_i \setminus \{s\}$. After removing these hyperplanes, the remaining elements of $\mathscr{A}$ will all have at least two fixed coordinates, and can be assumed to be hyperplanes in $Q' = S_1' \times \cdots \times S_n'$, where $S_i' = S_i$ if $\{i\} \notin \mathscr{F}(\mathscr{A})$. Removing the codimension 1 hyperplanes in this way makes a significant difference to our estimate on $M_k^{(2)}$, at the expense of (possibly) reducing each $|S_i|$ by 1. In practice, this turns out to often give better bounds on the removed measure.

Proof of Theorem 5.3.1. Observe first that

$$\mathbb{P}_n(B_k) = \mathbb{P}_k(B_k) \leqslant \mathbb{P}_{k-1}(B_k) = \mathbb{E}_{k-1}\left[\alpha_k(x)\right],$$

where the first two steps follow by Lemma 5.2.1 (since B_k is Q_k-measurable), and the third follows by the definition (5.2) of $\alpha_k(x)$. Moreover, by (5.2) and (5.3) (the definitions of α_k and $\mathbb{P}_k$), we have

$$\begin{aligned}\mathbb{P}_k(B_k) &= \sum_{x\in Q_{k-1}} \max\left\{0, \frac{\alpha_k(x)-\delta_k}{\alpha_k(x)(1-\delta_k)}\right\} \cdot \mathbb{P}_{k-1}\left(F_x \cap B_k\right) \\ &= \frac{1}{1-\delta_k} \sum_{x\in Q_{k-1}} \max\left\{0, \alpha_k(x)-\delta_k\right\} \cdot \mathbb{P}_{k-1}(x) \\ &\leqslant \frac{1}{1-\delta_k} \sum_{x\in Q_{k-1}} \frac{\alpha_k(x)^2}{4\delta_k} \cdot \mathbb{P}_{k-1}(x) = \frac{\mathbb{E}_{k-1}\left[\alpha_k(x)^2\right]}{4\delta_k(1-\delta_k)}, \end{aligned} \tag{5.15}$$

where we used the elementary inequality $\max\{a-d,0\} \leqslant a^2/4d$, which is easily seen to hold for all $a,d>0$ by rearranging the inequality $(a-2d)^2 \geqslant 0$.

It follows that the uncovered set R satisfies

$$\mathbb{P}_n(R) \geqslant 1 - \sum_{k=a+1}^{n} \mathbb{P}_n(B_k) \geqslant 1 - \sum_{k=a+1}^{n} \min\left\{M_k^{(1)}, \frac{M_k^{(2)}}{4\delta_k(1-\delta_k)}\right\} > 0,$$

by (5.9), and hence $\mathscr{A}$ does not cover Q, as required. □

In the proof of Theorem 5.3.2 we will use the following notation. Given a hyperplane $A = [x_1,\dots,x_n]$ and $X \subseteq [n]$, we define $A^X = [y_1,\dots,y_n]$ to be the hyperplane with $y_i = x_i$ for all $i \in F(A)\cap X$, and $y_i = *$ otherwise. Note that $(A^X)^Y = A^{X\cap Y}$ for every $X,Y \subseteq [n]$.

The first step in the proof of Theorem 5.3.2 is the following easy bound on the $\mathbb{P}_k$-measure of a Q_k-measurable hyperplane.

Lemma 5.3.3. Let $a \leqslant k \leqslant n$, and let A be a Q_k-measurable hyperplane. If $F(A) = I \cup J$, where $I \subseteq [a]$ and $J \subseteq [a+1,k]$, then

$$\mathbb{P}_k(A) \leqslant c(I)\nu(J). \tag{5.16}$$

Proof. The proof is by induction on k. Note first that for $k = a$ the conclusion follows immediately from the definition (5.10) of the function c, since $\nu(\emptyset) = 1$. So let $k \in [a+1,n]$, and assume that the claimed bound holds for $\mathbb{P}_{k-1}$.

Note first that if $k \notin F(A)$ then A is Q_{k-1}-measurable, and so the claimed bound follows immediately by (5.4) and the induction hypothesis. So assume that $k \in F(A)$, and observe that, by (5.5), we have

$$\mathbb{P}_k(A) \leqslant \frac{1}{1-\delta_k}\,\mathbb{P}_{k-1}(A) = \frac{1}{(1-\delta_k)|S_k|}\,\mathbb{P}_{k-1}\left(A^{[k-1]}\right),$$

since the probability measure $\mathbb{P}_{k-1}$ is extended uniformly on each fibre. Since $A^{[k-1]}$ is Q_{k-1}-measurable, by the induction hypothesis we have

$$\mathbb{P}_{k-1}\left(A^{[k-1]}\right) \leqslant c(I)\nu(J\setminus\{k\}),$$

and so, recalling the definition (5.11) of the function ν, the claimed bound follows. □

We will next prove the following bound on the tth moments $\mathbb{E}_{k-1}\left[\alpha_k(x)^t\right]$.

Lemma 5.3.4. For each $a < k \leqslant n$ and $t \in \mathbb{N}$ we have

$$\mathbb{E}_{k-1}\left[\alpha_k(x)^t\right] \leqslant \frac{1}{|S_k|^t}\sum_{F_1,\dots,F_t\in\mathscr{N}_k} c\big((F_1\cup\cdots\cup F_t)\cap[a]\big)\cdot\nu\big((F_1\cup\cdots\cup F_t)\cap[a+1,k-1]\big).$$

Proof. Observe first that, for each $x \in Q_{k-1}$, we have[3]

$$\alpha_k(x) = \frac{1}{|S_k|}\sum_{y\in S_k}\mathbb{1}\left[(x,y)\in B_k\right] \leqslant \frac{1}{|S_k|}\sum_{y\in S_k}\sum_{F\in\mathscr{N}_k}\mathbb{1}\left[(x,y)\in A_F\right],$$

by the union bound, and the definitions (5.1) and (5.2) of B_k and α_k. Note that, given $x \in Q_{k-1}$ and $F \in \mathscr{N}_k$, there exists $y \in S_k$ with $(x,y) \in A_F$ if and only if $x \in A_F^{[k-1]}$, and moreover such a y (if it exists) is unique. It follows that

$$\alpha_k(x) \leqslant \frac{1}{|S_k|}\sum_{F\in\mathscr{N}_k}\mathbb{1}\left[x\in A_F^{[k-1]}\right].$$

Note also that if A_1 and A_2 are hyperplanes, then $A_1\cap A_2$ is either the empty set, or a hyperplane whose set of fixed coordinate indices is $F(A_1)\cup F(A_2)$. Therefore, by Lemma 5.3.3,

$$\begin{aligned}\mathbb{E}_{k-1}\left[\alpha_k(x)^t\right] &\leqslant \frac{1}{|S_k|^t}\sum_{F_1,\dots,F_t\in\mathscr{N}_k}\mathbb{P}_{k-1}\left(A_{F_1}^{[k-1]}\cap\cdots\cap A_{F_t}^{[k-1]}\right)\\ &\leqslant \frac{1}{|S_k|^t}\sum_{F_1,\dots,F_t\in\mathscr{N}_k} c\big((F_1\cup\cdots\cup F_t)\cap[a]\big)\cdot\nu\big((F_1\cup\cdots\cup F_t)\cap[a+1,k-1]\big),\end{aligned}$$

as required. □

The claimed bounds on $M_k^{(1)}$ and $M_k^{(2)}$ will now follow easily.

[3]Here we write $\mathbb{1}[E]$ for the indicator function of an event E, which takes the value 1 if the event holds, and 0 otherwise.

Proof of Theorem 5.3.2. By Lemma 5.3.4, we have

$$\begin{aligned}\mathbb{E}_{k-1}\left[\alpha_i(x)^t\right] &\leqslant \frac{1}{|S_k|^t}\sum_{F_1,\dots,F_t\in\mathscr{N}_k} c\big((F_1\cup\dots\cup F_t)\cap[a]\big)\cdot\nu\big((F_1\cup\dots\cup F_t)\cap[a+1,k-1]\big)\\ &\leqslant \frac{1}{|S_k|^t}\sum_{I\subseteq[a]}\sum_{J\subseteq[a+1,k-1]}\sum_{\substack{X_1,\dots,X_t\subseteq[k-1]\\ X_1\cup\dots\cup X_t=I\cup J}} c(I)\nu(J)\\ &= \frac{1}{|S_k|^t}\sum_{I\subseteq[a]}\sum_{J\subseteq[a+1,k-1]}(2^t-1)^{|I|+|J|}c(I)\nu(J) = \frac{c_{k-1}(2^t-1)}{|S_k|^t},\end{aligned}$$

which proves (5.13). To prove (5.14), suppose that $\mathscr{F}(\mathscr{N}_k)$ contains no singletons (i.e., $\{k\}\notin\mathscr{F}(\mathscr{N}_k)$), and observe that, by Lemma 5.3.4, we have

$$\begin{aligned}|S_k|^2\,\mathbb{E}_{k-1}\left[\alpha_k(x)^2\right] &\leqslant \sum_{F_1,F_2\in\mathscr{N}_k} c\big((F_1\cup F_2)\cap[a]\big)\cdot\nu\big((F_1\cup F_2)\cap[a+1,k-1]\big)\\ &\leqslant \sum_{I\subseteq[a]}\sum_{J\subseteq[a+1,k-1]}\sum_{\substack{\emptyset\neq X_1,X_2\subseteq[k-1]\\ X_1\cup X_2=I\cup J}} c(I)\nu(J)\\ &= 1+\sum_{I\subseteq[a]}\sum_{J\subseteq[a+1,k-1]}\big(3^{|I|+|J|}-2\big)c(I)\nu(J)\\ &= c_{k-1}(3)-2c_{k-1}(1)+1,\end{aligned}$$

as required. □

5.4 Proof of Theorem 5.1.4

In order to deduce Theorem 5.1.4 from Theorems 5.3.1 and 5.3.2, it will suffice to show that there is an appropriate choice of C and $\delta_1,\delta_2,\dots,\delta_n$ such that $\mu_n>0$.

Proof of Theorem 5.1.4. Let $(q_k)_{k\geqslant1}$ be a sequence of integers with $\liminf_{k\to\infty}q_k/k>3$, and let $N\in\mathbb{N}$ and $\varepsilon>0$ be such that $q_k>(3+\varepsilon)k$ for all $k\geqslant N$. Let $C=C(N,\varepsilon)$ be sufficiently large, let $n\in\mathbb{N}$, and for each $k\in[n]$, let S_k be a set of size q_k. We will show that if $\mathscr{A}=\{A_F:F\in\mathscr{F}\}$ is a finite collection of hyperplanes in $Q=S_1\times\dots\times S_n$, no two of which are parallel, and $F(A)\not\subseteq[C]$ for every $A\in\mathscr{A}$, then $\mathscr{A}$ does not cover Q.

Fix $\delta_1=\dots=\delta_n=\varepsilon/6$, and assume (without loss of generality) that ε is sufficiently small. We will start with the trivial probability measure $\mathbb{P}_0$ on the empty product Q_0 (which we also think of as the uniform probability measure on Q), and construct inductively the probability measures $\mathbb{P}_k$ as described in Section 5.2. By Theorem 5.3.1 it suffices to show that

$$\sum_{k=1}^{n}\frac{M_k^{(2)}}{4\delta_k(1-\delta_k)}<1.$$

To prove this, note first that $M_k^{(2)} = 0$ for all $1 \leqslant k \leqslant C$, since $F(A) \not\subseteq [C]$ for every $A \in \mathscr{A}$. So let $C < k \leqslant n$, and observe that, by Theorem 5.3.2, we have

$$M_k^{(2)} \leqslant \frac{c_{k-1}(3)}{|S_k|^2} = \frac{1}{|S_k|^2} \prod_{j=1}^{k-1} \left(1 + \frac{3}{(1-\delta_j)|S_j|}\right).$$

Now, since $|S_j| = q_j \geqslant (3+\varepsilon)j$ for all $j \geqslant N$, and by our choice of δ_j, it follows that[4]

$$\prod_{j=N}^{k-1} \left(1 + \frac{3}{(1-\delta_j)|S_j|}\right) \leqslant \exp\left(\sum_{j=N}^{k-1} \frac{3}{(1-\varepsilon/6)(3+\varepsilon)j}\right) \leqslant k^{1-\varepsilon/9}.$$

Moreover, $\prod_{j=1}^{N-1} \left(1 + \frac{3}{(1-\delta_j)|S_j|}\right) \leqslant 3^N$. Thus, assuming that $C \geqslant N$ (so $|S_k| \geqslant 3k$), we have

$$M_k^{(2)} \leqslant \frac{3^N}{|S_k|^2} \cdot k^{1-\varepsilon/9} \leqslant \frac{3^{N-2}}{k^{1+\varepsilon/9}}$$

for every $C < k \leqslant n$, and hence

$$\sum_{k=1}^{n} \frac{M_k^{(2)}}{4\delta_k(1-\delta_k)} \leqslant \frac{3^N}{\varepsilon} \sum_{k=C}^{n} \frac{1}{k^{1+\varepsilon/9}} < 1$$

if $C = C(N,\varepsilon)$ is sufficiently large, as required. □

We will next show that the condition on the sequence $(q_k)_{k\geqslant 1}$ in Theorem 5.1.4 is close to best possible. To be precise, we will prove the following proposition.

Proposition 5.4.1. There exists a sequence of integers $(q_k)_{k\geqslant 1}$ with $q_k \geqslant 2$ for all $k \in \mathbb{N}$ and

$$\lim_{k\to\infty} \frac{q_k}{k} = 1$$

such that the following holds. For each $C > 0$, there exists $n \in \mathbb{N}$ and a collection $\mathscr{A}$ of non-trivial hyperplanes that cover $Q := [q_1] \times \cdots \times [q_n]$, no two of which are parallel, and with $F(A) \cap [C] = \emptyset$ for every $A \in \mathscr{A}$.

The first step is the following simple lemma; all hyperplanes are assumed to be non-trivial.

Lemma 5.4.2. Let $n \geqslant 3$, and let $q_1, \ldots, q_n \geqslant 2$ be a sequence of integers such that

$$\prod_{k=1}^{n} \left(1 + \frac{1}{q_k}\right) \geqslant n \log n.$$

Then $Q = [q_1] \times \cdots \times [q_n]$ can be covered with hyperplanes, no two of which are parallel.

[4] To see the final inequality, note that $\sum_{j=N}^{k-1} 1/j \leqslant \log k$ and $(1-\varepsilon/6)(3+\varepsilon)(1-\varepsilon/9) \geqslant 3$, since we assumed that ε is sufficiently small.

Proof. The proof is by induction on n, so first let $n = 3$, and note that if $2 \leqslant q_1 \leqslant q_2 \leqslant q_3$ satisfy $\prod_{k=1}^{3}(1+q_k^{-1}) \geqslant 3\log 3$, then $q_1 = q_2 = 2$. Now observe that $[2] \times [2]$ (and hence $[2] \times [2] \times [q_3]$) can be covered by three hyperplanes, no two of which are parallel.

For the induction step, observe first that, by the induction hypothesis, if $\prod_{k=1}^{n-1}(1+q_k^{-1}) \geqslant (n-1)\log(n-1)$ then we can find hyperplanes (with fixed coordinates in $[n-1]$) which cover Q. We may therefore assume that

$$1+\frac{1}{q_n} > \frac{n\log n}{(n-1)\log(n-1)} > 1+\frac{1}{n},$$

and hence (without loss of generality) that $2 \leqslant q_1 \leqslant \ldots \leqslant q_n < n$.

We now cover Q greedily: for each set $\emptyset \neq F \subseteq [n]$ in turn we choose a hyperplane A_F with fixed coordinates F so as to cover as much of the remaining (uncovered) subset of Q as possible. Since Q can be partitioned into exactly $\prod_{k\in F} q_k$ such hyperplanes, there must exist some choice of A_F that covers at least a proportion $\prod_{k\in F} q_k^{-1}$ of the remaining set. Thus, after all the hyperplanes have been chosen, the remaining set has size at most

$$\begin{aligned}|Q| \prod_{\emptyset\neq F\subseteq[n]} \left(1-\prod_{k\in F}\frac{1}{q_k}\right) &\leqslant |Q| \exp\left(-\sum_{\emptyset\neq F\subseteq[n]}\prod_{k\in F}\frac{1}{q_k}\right)\\ &= \exp\left(1+\sum_{k=1}^{n}\log q_k - \prod_{k=1}^{n}\left(1+\frac{1}{q_k}\right)\right).\end{aligned}$$

Now simply observe that

$$1+\sum_{k=1}^{n}\log q_k - \prod_{k=1}^{n}\left(1+\frac{1}{q_k}\right) < 0,$$

since $1+\sum_{k=1}^{n}\log q_k \leqslant 1+n\log(n-1) < n\log n$, whereas $\prod_{k=1}^{n}(1+q_k^{-1}) \geqslant n\log n$, by assumption. It follows that the number of uncovered points is less than 1, as required. □

We can now easily deduce Proposition 5.4.1.

Proof of Proposition 5.4.1. Assume that C is sufficiently large, and set

$$q_k := \left\lfloor \left(1-\frac{2}{\log k}\right)k \right\rfloor$$

for each $k > C$. Observe that $\lim_{k\to\infty} q_k/k = 1$, and that

$$\begin{aligned}\prod_{k=C+1}^{n}\left(1+\frac{1}{q_k}\right) &= \exp\left(\sum_{k=C+1}^{n}\frac{1}{q_k}+\frac{O(1)}{q_k^2}\right) = \exp\left(\sum_{k=C+1}^{n}\left(\frac{1}{k}+\frac{2}{k\log k}\right)+O(1)\right)\\ &= \exp\Big(\log n + 2\log\log n + O_C(1)\Big) = \Omega\big(n(\log n)^2\big).\end{aligned}$$

Thus, for all sufficiently large n, we have

$$\prod_{k=C+1}^{n}\left(1+\frac{1}{q_k}\right) \geqslant (n-C)\log(n-C),$$

and hence, by Lemma 5.4.2, we can cover $[q_{C+1}] \times \cdots \times [q_n]$ with hyperplanes, no two of which are parallel. But this implies that we can cover $[q_1] \times \cdots \times [q_n]$ with hyperplanes whose fixed coordinates do not intersect $[C]$, as required. □

5.5 The Erdős–Selfridge problem

In this section we will prove Theorem 5.1.2 (and hence also Theorem 5.1.1). To do so, we will again apply the sieve introduced in Section 5.2, but this time we will need to choose the various parameters much more carefully. In particular, we will deal with the primes in three groups: first the set $\{3,5,7,11\}$, then the primes between 13 and 73, and finally the primes larger than 73. We will discuss these in reverse order, so as to motivate the bounds we prove.

Let $\mathscr{B}$ be a collection of non-trivial hyperplanes in $P := [3] \times [5] \times \cdots \times [p_n]$, no two of which are parallel. Our aim is to show that $\mathscr{B}$ does not cover P. To do so, we will in fact apply our sieve to a modified collection, obtained by removing the co-dimension 1 hyperplanes, as described after the statement of Theorem 5.3.2, for the primes $p \leqslant p_{21} = 73$.[5] After doing so, we obtain a collection $\mathscr{A}$ of hyperplanes in $Q = S_2 \times \cdots \times S_n$, where $|S_k| = p_k - 1$ for each $2 \leqslant k \leqslant 21$, and $|S_k| = p_k$ for each $22 \leqslant k \leqslant n$, such that no two hyperplanes in $\mathscr{A}$ are parallel, and if $F(A) = \{i\}$ for some $A \in \mathscr{A}$ then $i \geqslant 22$. We remark that we will use some results from Chapter 4 to deal with the large primes, and our indexing of the sets S_k is chosen to avoid a conflict with the notation used there. It will also be convenient (see Section 5.5.3, below) to assume (as we may) that $A \not\subseteq B$ for any $A, B \in \mathscr{A}$ with $A \neq B$.

5.5.1 The primes greater than 73

For large primes, it will suffice to apply the results of Section 4.6. To state the results we will use, let us first recall some notation. Assume that we have chosen $\delta_6, \ldots, \delta_{21}$ and some probability distribution $\mathbb{P}_a = \mathbb{P}_5$ supported on

$$R_5 \subseteq Q_5 := S_2 \times \cdots \times S_5.$$

Now, noting that $p_{21} = 73$, set $\kappa := c_{21}(3)$ and define

$$f_k = f_k(\mathscr{A}) := \frac{\kappa}{\mu_k} \prod_{21<i\leqslant k} \left(1 + \frac{3p_i - 1}{(1-\delta_i)(p_i-1)^2}\right) \tag{5.17}$$

for each $k \geqslant 21$, where the constants $\{\delta_i : i > 21\}$ will be chosen later, and recall that

$$\mu_k = 1 - \sum_{i=6}^{k} \mathbb{P}_i(B_i)$$

[5] For simplicity, we will assume (without loss of generality) that $\mathscr{B}$ contains a co-dimension 1 hyperplane with fixed set $\{i\}$ for each i.

for each $5 \leqslant k \leqslant n$ (as $\mathbb{P}_5$ is supported on R_5). In order to apply the results stated below, we need to check condition (20) of Chapter 4, which states that for any $k > 21$ and any choice of $\delta_{22}, \delta_{23}, \ldots, \delta_k \in [0,1/2]$ we have

$$M_k^{(2)} \leqslant \frac{\kappa}{(p_k-1)^2} \prod_{21<i<k} \left(1 + \frac{3p_i - 1}{(1-\delta_i)(p_i-1)^2}\right) = \frac{\mu_{k-1} f_{k-1}}{(p_k-1)^2}. \tag{5.18}$$

To see this, note that

$$c_k(3) = c_{k-1}(3)\left(1 + \frac{3}{(1-\delta_k)p_k}\right) \leqslant c_{k-1}(3)\left(1 + \frac{3p_k - 1}{(1-\delta_k)(p_k-1)^2}\right)$$

for every $k > 21$, and observe that therefore, by Theorem 5.3.2, we have

$$M_k^{(2)} \leqslant \frac{c_{k-1}(3)}{|S_k|^2} \leqslant \frac{c_{21}(3)}{|S_k|^2} \prod_{21<i<k} \left(1 + \frac{3p_i - 1}{(1-\delta_i)(p_i-1)^2}\right),$$

as required, since $|S_k| = p_k > p_k - 1$. Indeed, the bound (5.18) is the bound given by the arguments of Chapter 4 when the primes p_k are allowed to occur to higher powers in the moduli when $k > 21$. The following theorem, which gives an almost optimal termination criterion when k is large, was proved in Chapter 4.

Theorem 5.5.1. Let $\mathbb{P}_5$ be a probability measure on R_5. Let $\kappa > 0$ be such that for any choice of parameters $\delta_6, \ldots, \delta_n \in [0,1/2]$, if we construct the probability measures $\mathbb{P}_6, \ldots, \mathbb{P}_n$, recursively using (5.2) and (5.3) from the probability measure $\mathbb{P}_5$, then (5.18) holds for any $5 < k \leqslant n$. Fix $k \geqslant \max(i_0, 10)$ and fix $\delta_6, \ldots, \delta_k \in [0,1/2]$ and let probability measures $\mathbb{P}_6, \ldots, \mathbb{P}_k$ be constructed recursively using (5.2) and (5.3) from the probability measure $\mathbb{P}_5$. If $\mu_k > 0$ and $f_k(\mathscr{A}) \leqslant (\log k + \log\log k - 3)^2 k$, then the system of arithmetic progressions $\mathscr{A}$ does not cover $\mathbb{Z}$.

Using Theorem 5.5.1, one can now compute (see the discussion in Section 4.6) weaker sufficient conditions on f_k for the event that the uncovered set is non-empty. In particular, we will use the following result Corollary 4.6.3.[6]

Corollary 5.5.2. Let $\mathbb{P}_5$ be a probability measure on R_5. Let $\kappa > 0$ be such that for any choice of parameters $\delta_6, \ldots, \delta_n \in [0,1/2]$, if we construct the probability measures $\mathbb{P}_6, \ldots, \mathbb{P}_n$, recursively using (5.2) and (5.3) from the probability measure $\mathbb{P}_5$, then (5.18) holds for any $5 < k \leqslant n$. Fix $\delta_6, \ldots, \delta_{21} \in [0,1/2]$ and let probability measures $\mathbb{P}_6, \ldots, \mathbb{P}_{21}$ be constructed recursively using (5.2) and (5.3) from the probability measure $\mathbb{P}_5$. If $f_{21}(\mathscr{A}) \leqslant 138.877$, then the collection of hyperplanes $\mathscr{A}$ does not cover Q.

In order to prove Theorem 5.1.2, it will therefore suffice to show that we can choose the probability distribution $\mathbb{P}_5$ and constants $\delta_6, \ldots, \delta_{21}$ such that $f_{21}(\mathscr{A}) \leqslant 138.877$, where we take $\kappa = c_{21}(3)$.

[6] In order to prove Corollary 5.5.2, it suffices to set $\delta_i = (1+a_i)/(1+\sqrt{1+a_i(1+a_i)/b_i f_{i-1}})$ for each $i > 21$, where $a_i = (3p_i - 1)/(p_i - 1)^2$ and $b_i = 1/4(p_i - 1)^2$, and apply Lemma 4.6.2 and Theorem 5.5.1 (which are both relatively straightforward consequences of (5.15) and Theorem 5.3.2), see Chapter 4, and also [2].

5.5.2 The primes between 13 and 73

We will next deduce from Corollary 5.5.2 a sufficient condition[7] on $\mathbb{P}_5$ for the event that the uncovered set is non-empty.

Lemma 5.5.3. If $\mathbb{P}_5$ satisfies $c_5(3) - 3c_5(1)/4 \leqslant 9.019$, then there exists a choice of the constants $\delta_6, \dots, \delta_{21}$ such that $f_{21}(\mathscr{A}) < 138.874$.

Proof. For a given choice of $\delta_6, \delta_7, \dots, \delta_{21} \in [0, 1/2]$ we make the following construction. Set $\hat{\mu}_5 := \mu_5 = 1$ and define, for $k = 6, \dots, 21$,

$$\hat{\mu}_k := \hat{\mu}_{k-1} - \frac{c_{k-1}(3) - 2c_{k-1}(1) + 1}{4\delta_k(1-\delta_k)|S_k|^2} \tag{5.19}$$

for each $k \in \{6, \dots, 21\}$. Recall that, by (5.15) and Theorem 5.3.2, we have

$$\mathbb{P}_k(B_k) \leqslant \frac{M_k^{(2)}}{4\delta_k(1-\delta_k)} \leqslant \frac{c_{k-1}(3) - 2c_{k-1}(1) + 1}{4\delta_k(1-\delta_k)|S_k|^2},$$

so $\hat{\mu}_k \leqslant \mu_k$, for each $k \in \{6, \dots, 21\}$, and hence $f_{21}(\mathscr{A}) = c_{21}(3)/\mu_{21} \leqslant c_{21}(3)/\hat{\mu}_{21}$.

Now, observe that

$$c_k(x) = c_{k-1}(x)\left(1 + \frac{x}{(1-\delta_k)(p_k - 1)}\right) \tag{5.20}$$

for each $x \in \{1, 3\}$ and $k \in \{6, \dots, 21\}$, and therefore (for a given choice of δ_k) we may write $c_{21}(3) = g\big(c_5(3)\big)$ and $\hat{\mu}_{21} = h\big(c_5(1), c_5(3)\big)$ as affine linear functions of $c_5(1)$ and $c_5(3)$. Moreover, the function $g(x)$ is increasing, and it follows from (5.19) and (5.20) that the function $h(x, y)$ is increasing in x and decreasing in y. It follows that $c_{21}(3)/\hat{\mu}_{21}$ is increasing in $c_5(3)$ and decreasing in $c_5(1)$, provided $\hat{\mu}_{21} > 0$. Thus, to bound $f_{21}(\mathscr{A})$ from above for all pairs $(c_5(1), c_5(3))$ with $c_5(3) - 3c_5(1)/4 \leqslant 9.019$, it is enough to bound $c_{21}(3)/\hat{\mu}_{21}$ for a finite set of pairs $(c_5(1), c_5(3))$ that 'dominate' the region $c_5(3) - 3c_5(1)/4 \leqslant 9.019$, and satisfy $\hat{\mu}_{21} > 0$.

To do this, note that $c_5(1) \geqslant 1$ and $c_5(3) - 1 \geqslant 3(c_5(1) - 1)$, by (5.12). We may therefore assume that $c_5(1) \leqslant 5$, since otherwise $c_5(3) - 3c_5(1)/4 \geqslant 9c_5(1)/4 - 2 > 9.019$. We therefore only need to cover the part of the region $c_5(3) - 3c_5(1)/4 \leqslant 9.019$ with $1 \leqslant c_5(1) \leqslant 5$. We do so by looping through values of $c_5(1)$ from 1 to 5 in steps of 10^{-4}, i.e., we check the point

$$\big(u(i), v(i)\big) := \left(\frac{i}{10^4}, 9.019 + \frac{3(i+1)}{4 \cdot 10^4}\right)$$

in the $(c_5(1), c_5(3))$-plane for each $10^4 \leqslant i < 5 \cdot 10^4$. Note that $\big(u(i), v(i)\big)$ dominates the set

$$I(i) := \Big\{\big(c_5(1), c_5(3)\big) : c_5(3) - 3c_5(1)/4 \leqslant 9.019 \text{ and } 10^{-4}i \leqslant c_5(1) \leqslant 10^{-4}(i+1)\Big\},$$

[7] A rough diagram of the pairs $(c_5(1), c_5(3))$ that were sufficient to prove the required bound on $f_{21}(\mathscr{A})$ was determined. Based on this, the linear combination $c_5(3) - 3c_5(1)/4$ appears to give the best 'figure of merit' among simple linear combinations.

in the sense that if $(x,y) \in I(i)$ then $x \geqslant u(i)$ and $y \leqslant v(i)$, so (by the monotonicity properties proved above) any upper bound on $c_{21}(3)/\hat{\mu}_{21}$ that holds at the point $(u(i),v(i))$ applies to all points of $I(i)$. Hence, if an upper bound for $c_{21}(3)/\hat{\mu}_{21}$ holds for each pair $(u(i),v(i))$ in the range above, then it holds whenever $\mathbb{P}_5$ satisfies $c_5(3) - 3c_5(1)/4 \leqslant 9.019$.

Now for each $10^4 \leqslant i < 5 \cdot 10^4$ we choose the constants $\delta_6, \ldots, \delta_{21} \in (0,1/2]$ so as to minimize the ratio $c_{21}(3)/\hat{\mu}_{21}$ after processing the prime 73. The optimization of the δ_k was made by first taking an initial choice $\delta_k = 1/4$. Then the δ_k were changed by performing coordinate-wise optimization: we minimized $c_{21}(3)/\hat{\mu}_{21}$ with respect to each δ_k in turn from $k = 6$ to 21, and then repeated this process a second time, after which it was seen that $c_{21}(3)/\hat{\mu}_{21}$ had converged adequately. We also checked that $\hat{\mu}_{21} > 0$ in each case.

The maximum value of $c_{21}(3)/\hat{\mu}_{21}$ obtained for any of these points was about 138.873682, and hence $f_{21} \leqslant 138.874$ for any $\mathbb{P}_5$ such that $c_5(3) - 3c_5(1)/4 \leqslant 9.019$, as required. Source code for these calculations can be found at [2]. □

5.5.3 Constructing the measure $\mathbb{P}_5$

It remains to construct a measure on the uncovered set $R_5 \subseteq S_2 \times \cdots \times S_5$ obtained after removing all hyperplanes corresponding to arithmetic progressions whose moduli involve only the primes 3, 5, 7 and 11. By Corollary 5.5.2 and Lemma 5.5.3, it will suffice to prove the following lemma.

Lemma 5.5.4. For each $\mathscr{A}$ as above, there exists $\mathbb{P}_5$ such that $c_5(3) - 3c_5(1)/4 \leqslant 9.019$.

Proof. Let us write $\mathscr{F}$ for the collection of sets $F \subseteq \{2,3,4,5\}$ with $|F| \geqslant 2$. For each set $F \in \mathscr{F}$, we need to choose a hyperplane A_F with fixed set F, and for each such family we need to construct a measure $\mathbb{P}_5$ supported on the uncovered set. Not surprisingly, there are far too many configurations to deal with easily, so we need to make a few reductions.

Recall first that (by assumption) no hyperplane in $\mathscr{A}$ is contained in another. Also, we may assume there is a hyperplane for each F with $|F| \geqslant 2$ as including extra hyperplanes only makes covering easier. However, it is not clear that we can make both assumptions simultaneously. Let us write $F' \prec F$ if $\sum_{i \in F'} 2^i < \sum_{i \in F} 2^i$ (i.e., F' precedes F in colexicographic order). In studying all possible configurations, we shall go through the elements of $\mathscr{F}$ in colexicographic order and for each $F \in \mathscr{F}$ we choose the hyperplane A_F. In particular, if F' appears before F then $A_{F'}$ cannot be contained in A_F. Moreover, if the hyperplanes $A_{F'}$ preceding A_F do not form a cover everything, then there exists a choice for A_F such that A_F covers a yet uncovered point, i.e., A_F is not contained in any prior $A_{F'}$. Therefore, at each stage in order to ensure that there exists a valid choice for the hyperplane A_F, we just need to check that the hyperplanes chosen so far do not cover everything. Moreover, we only need to study configurations 'up to isomorphism', in the following sense. For each $F \subseteq \{2,3,4,5\}$ write $A_F = [a_{2,F}, a_{3,F}, a_{4,F}, a_{5,F}]$ with $a_{i,F} \in S_i \cup \{*\}$ so that $A_F \subseteq \{x_i = a_{i,F}\}$ when $i \in F$ and $a_{i,F} = *$ otherwise. Next, for a given F, order the choices of hyperplane A_F lexicographically, so that $A'_F \prec' A_F$ if $a'_{i,F} < a_{i,F}$ when $i = \min\{j \in F : a'_{j,F} \neq a_{j,F}\}$. Now, suppose there exists a pair (F,i) such that

$i \in F$ and $a_{i,F} \geqslant a_{i,F'} + 2$ for every $F' \prec F$ with $i \in F'$. Then we can transpose $a_{i,F}$ and $a_{i,F} - 1$ in S_i to obtain an isomorphic configuration $\mathscr{A}'$ which is lexicographically smaller (with respect to the orders $\prec$ and $\prec'$), since $A'_F \prec' A_F$, but $A'_{F'} = A_{F'}$ for all $F' \prec F$.

Applying these reductions reduces the number of configurations to 6,025,640,717 which, while it represents substantial progress, is still too large to conveniently construct optimized probability distributions for each configuration. However, the main contribution to the large number of configurations comes from the choice of the 'last' few hyperplanes, namely A_{45}, A_{245}, A_{345}, and A_{2345}.[8] For example, we might have as many as $2 \times 4 \times 6 \times 10 = 480$ choices for A_{2345}, as we are selecting a single point in Q_5 (although in practice the number of choices is reduced somewhat by the comments above). Ignoring the choices for A_{45}, A_{245}, A_{345}, and A_{2345} reduces the number of configurations to just 7637, which is far more manageable.

Our strategy is therefore as follows. We first consider a choice of the hyperplanes, A_{23}, A_{24}, A_{25}, A_{34}, A_{35}, A_{234} and A_{235}, without including the last four hyperplanes A_{45}, A_{245}, A_{345} and A_{2345}. In order to optimize the probability distribution on the uncovered region $R := Q_5 \setminus (A_{23} \cup \cdots \cup A_{235})$, we construct a linear programming problem with variables x_r for each $r \in R$ representing the probability of the atom $\{r\}$. For each non-empty set $I \subseteq \{2,3,4,5\}$, and each hyperplane H in Q_5 with $F(H) = I$, we include the constraint

$$\sum_{r \in R \cap H} x_r \leqslant c_I,$$

where the c_I are new variables giving upper bounds on the $c(I)$, cf. (5.10). (Note that we include the constraints corresponding to sets not in $\mathscr{F}$, since we need to bound $c(I)$ for all subsets $I \subseteq \{2,3,4,5\}$.) We also add the constraints

$$x_r \geqslant 0 \qquad \text{and} \qquad \sum_{r \in R} x_r = 1$$

to ensure that we have a probability measure supported on R, and then minimize

$$\sum_{I \subseteq \{2,3,4,5\}} \big(3^{|I|} - 3/4\big) c_I, \tag{5.21}$$

where we define $c_\emptyset = 1$. We define $\mathbb{P}_5$ to be the probability distribution corresponding to this minimum, i.e., we set $\mathbb{P}_5(r) = x_r$ for each $r \in R$. Recalling (5.12), note that

$$c_5(3) - \frac{3c_5(1)}{4} = \sum_{I \subseteq \{2,3,4,5\}} \big(3^{|I|} - 3/4\big) c(I),$$

and observe that the minimum occurs when $c_I = c(I)$.

We are therefore done, except for the (important) fact that we have not restricted the measure to be zero on the set

$$U := A_{45} \cup A_{245} \cup A_{345} \cup A_{2345}.$$

[8] We have for brevity denoted e.g., $A_{\{4,5\}}$ by A_{45}.

Configuration	$c_5(3)-3c_5(1)/4$	$p\leqslant$	Bound
11**, 2*1*, *22*, 121*, 1**1, *3*2, 13*3	8.772328	0.043227	9.157362

Table 5.1 One of the 90 partial configurations on Q_5 using just $A_{23},\dots,A_{235}$ with the bound given by (5.22) greater than 9.018. The full list is given in [2].

Configuration	$c_5(3)-3c_5(1)/4$
11**, 2*1*, *22*, 121*, 1**1, *3*2, 13*3, **34, 2*31, *232, 1233	9.018070
11**, 2*1*, *22*, 121*, 1**1, *3*2, 13*3, **34, 2*33, *232, 1233	9.018070

Table 5.2 Full configurations on Q_5 with $c_5(3)-3c_5(1)/4\geqslant 9.018$.

To do so, we simply remove the measure from the (unknown) set U, and uniformly rescale the measure to again give a probability measure. We claim that this can increase the value of $c_5(3)-3c_5(1)/4$ to at most

$$\frac{c_5(3)-3c_5(1)/4-p/4}{1-p}, \tag{5.22}$$

where p is the probability assigned to U. To see this, observe that removing the measure on U does not increase any $c(I)$, and decreases $c(\emptyset)$ by p. Since $c_5(3)-3c_5(1)/4$ is a positive linear combination of the $c(I)$, with $c(\emptyset)$ occurring with coefficient $1/4$, it follows that $c_5(3)-3c_5(1)/4$ decreases by at least $p/4$. Renormalizing the measure then increases each $c(I)$ (and hence this linear combination of the $c(I)$) by a factor of $1-p$.

To complete the proof, we bound the probability p of the unspecified set U by

$$p\leqslant c_{45}+c_{245}+c_{345}+c_{2345},$$

where the c_I are the bounds on $c(I)$ given by the linear programming problem. We then check if the bound in (5.22) is less than 9.018.[9] If so, then we proceed to the next configuration. There are 90 (out of 7637) configurations of $(A_{23},\dots,A_{235})$ where this fails. For these, we loop through all choices of A_{45} and perform the above calculation with just A_{245}, A_{345}, and A_{2345} unspecified. From these 90 configurations we obtain 1083 configurations with A_{45} included, but for only 12 of these does our bound still exceed 9.018. These 12 give rise to 312 configurations including A_{245}, of which 3 still exceed our bound. These 3 give rise to 216 configurations where we are forced to include A_{345}, but only 2 which still exceed our bound. Finally, these 2 give 142 configurations where we are forced to include all the A_F, but only two have $c_5(3)-3c_5(1)/4\geqslant 9.018$, and these are listed in Table 5.2. We deduce that for all choices of the hyperplanes $\{A_F : F\in\mathscr{F}\}$ in Q_5 we can find a probability measure $\mathbb{P}_5$ on R_5 such that $c_5(3)-3c_5(1)/4<9.018071$.

[9]This bound was chosen, after some experimentation, to be just below the worst case value of $c_5(3)-3c_5(1)/4$ given in Table 5.2 for configurations using all the hyperplanes.

All calculations were performed in `C` using the Gurobi linear optimization package to solve the LP minimizations. With the strategy as described above, the calculations to determine the worst case configurations took about 40 seconds on a laptop. Source code and results can be found at [2]. □

As noted above, Theorem 5.1.2 follows immediately from Corollary 5.5.2 and Lemmas 5.5.3 and 5.5.4. In fact, since the results from Chapter 4 (which we used to deal with the large primes) did not require the assumption that the moduli are square-free, we actually proved something slightly stronger: if the moduli are distinct and each prime $p \leqslant 73$ in their prime factorization occurs to a power at most 1, then at least one of the moduli must be even. In other words, the 'square-free' condition is only needed on the 73-smooth part of the moduli. Of course, if we could reduce "$p \leqslant 73$" to "$p < 3$" then the Erdős–Selfridge problem would be solved, so it would be interesting to see to what extent the bound 73 could be reduced.

Chapter 6

The structure and number of Erdős covering systems

The work in this section is joint with Paul Balister, Béla Bollobás, Robert Morris and Julian Sahasrabudhe [3].

6.1 Introduction

In this chapter we prove Theorem 1.2.10, the asymptotic formula for the logarithm of the number of minimal covering systems of a given size. This answers a question of Erdős [56] from 1952. For convenience, we restate Theorem 1.2.10 here as Theorem 6.1.1.

Theorem 6.1.1. The number of minimal covering systems of $\mathbb{Z}$ of size n is

$$\exp\left(\left(\frac{4\sqrt{\tau}}{3}+o(1)\right)\frac{n^{3/2}}{(\log n)^{1/2}}\right) \tag{6.1}$$

as $n\to\infty$, where

$$\tau=\sum_{t=1}^{\infty}\left(\log\frac{t+1}{t}\right)^2.$$

We remark that proving a weaker upper bound, with a different constant in the exponent, is significantly easier, and we give a short proof of such a bound in Section 6.6. Let us also note here that we prove the lower bound under the additional restriction that the moduli are distinct, and so the conclusion of Theorem 6.1.1 also holds for such systems.

In order to motivate the form of the formula (6.1), let us begin by describing a simple construction that gives a slightly weaker lower bound. Let $p_1<\ldots<p_k$ be the first k primes, and for each $i\in[k]$, choose p_i-1 arithmetic progressions $A_1^{(i)},\ldots,A_{p_i-1}^{(i)}$ with the following properties: for each $j\in[p_i-1]$, the modulus of $A_j^{(i)}$ is divisible by p_i and divides $Q_i:=p_1\cdots p_i$, and $A_j^{(i)}$ contains $j\cdot Q_{i-1}$. It is not hard to see that, for each such choice, by adding the progression $\{0 \pmod{Q_k}\}$ we obtain a

distinct minimal covering system of size $n = \sum_{i=1}^{k}(p_i - 1) + 1 \approx k^2 \log k$. Since we have 2^{i-1} choices for the progression $A_j^{(i)}$ for each $i \in [k]$ and $j \in [p_i - 1]$, this implies that there are at least

$$\prod_{i=1}^{k} 2^{(i-1)(p_i-1)} = \exp\Big(\Omega\big(k^3 \log k\big)\Big) = \exp\left(\frac{\Omega\big(n^{3/2}\big)}{(\log n)^{1/2}}\right)$$

minimal covering systems of $\mathbb{Z}$ of size n. In Section 6.5 we describe a somewhat more complicated construction that proves the lower bound in Theorem 6.1.1.

We refer to collections of progressions as in the construction above as “frames" (see Section 6.2 for a precise definition). The second main result of this chapter, and the key step in the proof of Theorem 6.1.1, is a structural description of all “efficient" covering systems; roughly speaking, we show that every such covering system contains a large “approximate frame". The purpose of the next section is to state this structural theorem.

6.2 The structure of efficient coverings

In this section we state our main structural theorem. In order to do so, it is convenient to shift our attention to the following (slightly more general) *geometric* setting. Let $S_1, \dots, S_k$ be finite sets with at least two elements and set $S_I := \prod_{i \in I} S_i$ for each $I \subseteq [k]$. If $H = H_1 \times \cdots \times H_k \subseteq S_{[k]}$ with each H_i either equal to S_i or a singleton element of S_i, then we say that H is a *hyperplane*. We write $F(H) := \{i \in [k] : |H_i| = 1\}$ for the *fixed coordinates* of H, and $F(\mathscr{A}) := \bigcup_{H \in \mathscr{A}} F(H)$ if $\mathscr{A}$ is a collection of hyperplanes. We also write $H = [x_1, \dots, x_k]$, where $x_i \in S_i \cup \{*\}$ for each $i \in [k]$, and $*$ indicates that $H_i = S_i$.

Definition 6.2.1. A *simple frame* centred at an element $(s_1, \dots, s_k) \in S_{[k]}$ (which we call the *axis*), is a sequence $(\mathscr{F}_1, \dots, \mathscr{F}_k)$, where $\mathscr{F}_i$ is a collection of $|S_i| - 1$ hyperplanes of the form

$$\big[x_1, \dots, x_{i-1}, a, *, \cdots, *\big], \tag{6.2}$$

one for each $a \in S_i \setminus \{s_i\}$, with $x_j \in \{s_j, *\}$ for each $j \in [i-1]$.

A *frame* is obtained from a simple frame by permuting the order of the sets $S_1, \dots, S_k$.

Observe that if $(\mathscr{F}_1, \dots, \mathscr{F}_k)$ is a frame centred at $(s_1, \dots, s_k)$, then the collection

$$\mathscr{A} := \mathscr{F}_1 \cup \cdots \cup \mathscr{F}_k \cup \big\{[s_1, \dots, s_k]\big\}$$

is a minimal cover of $S_{[k]}$. Indeed, if we remove the hyperplane $\big[x_1, \dots, x_{i-1}, a, *, \cdots, *\big]$ from $\mathscr{A}$, then the element $(s_1, \dots, s_{i-1}, a, s_{i+1}, \dots, s_k)$ is uncovered by the remaining hyperplanes. Note that if we set $S_i = \{0, \dots, p_i - 1\}$ for each $i \in [k]$, then the construction given in the introduction is equivalent to a frame centred at $(0, \dots, 0)$. When we (for now informally, but later on precisely) discuss frames in $\mathbb{Z}$, we always mean that each set $S_i = \{0, \dots, p-1\}$ for some prime p (these primes are not generally distinct), and we map $S_1 \times \cdots \times S_k$ into $\mathbb{Z}_N$, where $N = \prod_{i=1}^{k} |S_i|$, using the Chinese Remainder

Theorem to identify $\mathbb{Z}_N$ with the product of groups $\mathbb{Z}_{p^\gamma}$, and then expanding base p.[1] Note that every arithmetic progression in $\mathbb{Z}$ corresponds to a hyperplane, but not every hyperplane corresponds to an arithmetic progression if primes are repeated (see Sections 6.5 and 6.7).

The key idea behind the proof of Theorem 6.1.1 is the following (imprecise) conjecture:

"Almost every minimal covering system of $\mathbb{Z}$ of size n is close to a frame."

We do not prove a result of this form; instead, we use a slightly weaker notion, which we call a *δ-generalized frame*. These objects differ from frames in two key ways: the fixed elements "to the left" of i in a hyperplane $H \in \mathscr{F}_i$ are allowed to vary with i, and instead of insisting that "all coordinates to the right are free" (as in (6.2)), we allow a few "small" coordinates to be fixed (with the product of their sizes bounded by $1/\delta$).

The next definition is both important and somewhat technical, and requires some additional notation. Given a hyperplane H, we write H_i for its ith coordinate, and for any $I \subseteq [k]$ we write $H_I = \prod_{i\in I} H_i$ for the hyperplane in S_I obtained by restricting H to the coordinates of I, and define $\mu_I(H) := |H_I| \cdot |S_I|^{-1}$ when $I \neq \emptyset$, and $\mu_\emptyset(H) := 1$.

Definition 6.2.2 (δ-generalized frames). Let $\delta > 0$, and let $S_1, \dots, S_k$ be finite sets with at least two elements. A *simple δ-generalized frame* in $S_{[k]}$ is a sequence $(\mathscr{F}_1, \dots, \mathscr{F}_k)$, where $\mathscr{F}_i$ is a collection of at most $|S_i| - 1$ hyperplanes, satisfying the following conditions. For each $i \in [k]$, there exists a set $I(i) \supseteq \{i+1, \dots, k\}$, and for each $j \notin I(i) \cup \{i\}$, there exists an element $s_j(i) \in S_j$, such that, for each $H \in \mathscr{F}_i$,

$$i \in F(H), \qquad \mu_{I(i)}(H) > \delta \qquad \text{and} \qquad H_j \in \big\{s_j(i), S_j\big\}.$$

Moreover, if $\min\big\{|S_i|, |S_j|\big\} \geqslant \delta^{-1}$ and $i \neq j$, then $\mathscr{F}_i$ and $\mathscr{F}_j$ are disjoint. A *δ-generalized frame* is obtained from a simple δ-generalized frame by permuting the sets $S_1, \dots, S_k$.

We are now ready to state our main structural theorem for covering systems that contain roughly (up to a constant factor) the same number of elements as a frame.

Theorem 6.2.3. For every $C, \varepsilon > 0$ there exists $\delta = \delta(C, \varepsilon)$ so that for every collection of finite sets $S_1, \dots, S_k$ with at least two elements, the following holds. If $\mathscr{A}$ is a minimal cover of $S_{[k]}$ with hyperplanes such that $F(\mathscr{A}) = [k]$ and

$$|\mathscr{A}| \leqslant C \sum_{i=1}^{k} \big(|S_i| - 1\big), \tag{6.3}$$

then $\mathscr{A}$ contains a δ-generalized frame $(\mathscr{F}_1, \dots, \mathscr{F}_k)$, with

$$\sum_{i=1}^{k} |\mathscr{F}_i| \geqslant (1-\varepsilon) \sum_{i=1}^{k} \big(|S_i| - 1\big). \tag{6.4}$$

[1] For example, $(a_0, a_1, \dots, a_{\gamma-1}) \in \mathbb{Z}_p \times \cdots \times \mathbb{Z}_p$ corresponds to the element $\sum_{i=0}^{\gamma-1} a_i p^i \in \mathbb{Z}_{p^\gamma}$.

The theorem above can be thought of as an inverse theorem for the following extremal result of Simpson [140]. If $\mathscr{A}$ is a collection of arithmetic progressions, then we write $\mathrm{lcm}(\mathscr{A})$ for the least common multiple of the moduli of the progressions in $\mathscr{A}$.

Theorem 6.2.4 (Simpson's theorem)**.** If $\mathscr{A}$ is a minimal cover of $S_{[k]}$ with hyperplanes such that $F(\mathscr{A}) = [k]$, then

$$|\mathscr{A}| \geqslant \sum_{i=1}^{k} \big(|S_i| - 1\big) + 1.$$

In particular, if $\mathscr{A}$ is a minimal covering system of $\mathbb{Z}$ with $\mathrm{lcm}(\mathscr{A}) = p_1^{\gamma_1} \cdots p_m^{\gamma_m}$, then

$$|\mathscr{A}| \geqslant \sum_{i=1}^{m} \gamma_i \big(p_i - 1\big) + 1.$$

In the Section 6.8, we provide (for the reader's convenience) a proof of Simpson's theorem. Let us also remark here that, while the form of the function $\delta(C,\varepsilon)$ does not matter for our purposes, we prove that Theorem 6.2.3 holds with $\delta = (\varepsilon/C)^{O(\log(1/\varepsilon))}$.

In order to deduce Theorem 6.1.1 from Theorem 6.2.3, we need to count δ-generalized frames quite precisely, and show that there are relatively few choices for the remaining elements; we also need to show that there are few minimal covering systems that fail to satisfy (6.3). These calculations are carried out in Sections 6.6 and 6.7.

6.2.1 An outline of the proof of the structural theorem

The proof of Theorem 6.2.3 requires a few somewhat technical definitions, and to prepare the reader for these we begin by giving an outline of the argument. The idea is to construct a tree that encodes the structure of the covering system by 'exploring' it coordinate by coordinate. To be more precise, given a minimal cover $\mathscr{A}$ of $S_{[k]}$, let us choose a coordinate $i \in [k]$ to explore, and observe (see Section 6.3.2 for the details) that for each $s \in S_i$ we obtain a covering system of

$$S_1 \times \cdots \times S_{i-1} \times \{s\} \times S_{i+1} \times \cdots \times S_k,$$

which we identify with $S_{[k]\setminus\{i\}}$. (Note that hyperplanes $H \in \mathscr{A}$ with $H_i = S_i$ appear in each of the $|S_i|$ covering systems corresponding to coordinate i.) These covering systems may not be minimal, but for each $s \in S_i$ we can take a minimal sub-covering $\mathscr{A}_s$.

Now, some of the systems $\mathscr{A}_s$ may be trivial (i.e., may consist of a single hyperplane), and when this occurs we are happy, because such hyperplanes can be used in the frame that we are trying to construct. For the remaining elements $s \in S_i$, we consider the set of fixed coordinates $F(\mathscr{A}_s)$ of $\mathscr{A}_s$, and observe (see Lemma 6.3.7) that every coordinate (except i) is in $F(\mathscr{A}_s)$ for some $s \in S_i$. We may now choose, for each s such that $F(\mathscr{A}_s)$ is non-empty, a coordinate $j \in F(\mathscr{A}_s)$, and repeat the above construction, exploring the minimal covering system $\mathscr{A}_s$, starting with the coordinate j. Iterating this process produces a rooted tree (which we call an 'index tree', see Definition 6.3.1), each of whose vertices is labelled with a set $I \subseteq [k]$ and a coordinate $i \in I$, which are the fixed coordinates of the corresponding minimal covering system, and the coordinate 'explored' at that vertex, respectively.

So far, we have not said anything about how to construct the sets $\mathscr{F}_i$, or how to choose the coordinate i that we explore in a given step. For simplicity, let us explain this only for the first step (the choice for later steps is similar). First, if there exists $i \in [k]$ such that there are at least $(1-\varepsilon)\big(|S_i|-1\big)$ hyperplanes $H \in \mathscr{A}$ with $i \in F(H)$ and $\mu_{[k]\setminus\{i\}}(H) > \delta$, then we choose such a coordinate i to explore, and associate this collection of 'frame-like' hyperplanes with the current vertex (in this case, the root of the tree). One of the key ideas of the chapter is that, if such a collection of hyperplanes does not exist for any $i \in [k]$, then we may use the Lovász Local Lemma to deduce (see Lemma 6.3.5) that there exists a coordinate j (which we will choose to explore), and a 'large' collection $\mathscr{G}$ of hyperplanes in the current collection, such that j is a fixed coordinate of each. This collection of 'garbage' hyperplanes will later be used, together with (6.3), to show that this case does not occur too often.

To see why the Lovász Local Lemma is useful in this context, consider the following simplified problem. Say we want to find an efficient cover $\mathscr{A}$ of $[s]^k$ with hyperplanes each with at least two fixed coordinates. Then there exists a coordinate j such that $\mathscr{A}_j = \{H \in \mathscr{A} : j \in F(H)\}$ satisfies $|A_j| >> s^2$. View $[s]^k$ as a probability space with the uniform probability measure μ. Note that two hyperplanes H_1, H_2 are independent events if and only if $F(H_1) \cap F(H_2) = \emptyset$. If this is not the case write $H_1 \sim H_2$. For each $H \in \mathscr{A}$ let $x(H) = \exp(|F(H)|)\mu(H)$. Then the local lemma implies that $\mathscr{A}$ is a cover only if there exists $H \in \mathscr{A}$ such that $\mu(H) \geqslant x(H)\prod_{H' \sim H}(1-x(H'))$. We deduce $\mu(H) \geqslant x(H)\prod_{H' \sim H}\exp(-2x(H))$ (if $s >> e$ then $x(H) << 1$). We further deduce $\mu(H) \geqslant \exp(|F(H)|)\mu(H)\exp(-2\sum_{H' \sim H}\exp(|F(H')|)\mu(H'))$. This is equivalent to $\sum_{H' \sim H}\exp(|F(H')|)\mu(H') \geqslant 2^{-1}|F(H)|$. Finally, we deduce $\sum_{j \in F(H)}\sum_{F(H') \ni j}\exp(|F(H')|)\mu(H') \geqslant 2^{-1}|F(H)|$. By averaging, we get that there exist $j \in F(H)$ such that $|A_j|\max_{H'}\exp(|F(H')|)\mu(H') \geqslant 2^{-1}$. As every hyperplane has at least two fixed point we conclude that $\exp(|F(H')|)\mu(H') \leqslant (e/s)^2$ and $|A_j| \geqslant 2^{-1}e^{-2}s^2$.

The plan described above is carried out in Section 6.3, the main result being Lemma 6.3.3, which states that if δ is sufficiently small, then there exists a suitable 'exploration tree' $\mathbb{T}$ of $\mathscr{A}$ (see Definition 6.3.2). This exploration tree can be very large, however, and to extract our δ-generalized frame from it we need to choose a suitable sub-tree T. To do so, we choose k 'special' vertices of $\mathbb{T}$, one for each coordinate, and take the union of the paths from these vertices to the root. If almost all of these special vertices are 'good' (that is, we found a large collection of frame-like hyperplanes when exploring them), then we obtain a sufficiently large δ-generalized frame. On the other hand, if a positive proportion of them are 'bad', then we use the 'garbage' hyperplanes to show that inequality (6.3) cannot hold. In order to carry out this argument, we need to choose the special vertices carefully; it turns out that it is sufficient to choose them via a depth-first search, see Section 6.4 for the details.

6.3 Exploring the cover

In this section we take the first step towards Theorem 6.2.3 by describing a much larger object that is somewhat easier to construct, the *exploration tree*. To define these, we first need to introduce the following simpler objects, which we call *index trees*. Let us fix, for the rest of the proof of

Theorem 6.2.3, a collection of finite sets $S_1, \dots, S_k$ with at least two elements, and let us write $N(u)$ for the set of out-neighbours of u in a rooted tree, where we orient the edges away from the root.

Definition 6.3.1. An *index tree* $\mathbb{T}$ of $[k]$ is a rooted tree, equipped with a labelling of its vertices $u \mapsto (I_u, i_u)$, where $I_u \subseteq [k]$ and $i_u \in I_u$, that satisfies the following conditions:

(i) the root of $\mathbb{T}$ has label $([k], i)$ for some $i \in [k]$;

(ii) for each vertex $v \in V(\mathbb{T})$,

$$\bigcup_{u \in N(v)} I_u = I_v \setminus \{i_v\}. \tag{6.5}$$

We can now define the exploration tree of a collection of hyperplanes in $S_{[k]}$. Given a rooted tree T and vertices $u, v \in V(T)$, let us write $u \prec_T v$ to indicate that u lies on the path from v to the root (so, in particular, $v \prec_T v$).

Definition 6.3.2. Let $\lambda, \varepsilon, \delta > 0$, and let $\mathscr{A}$ be a collection of hyperplanes in $S_{[k]}$. An $(\lambda, \varepsilon, \delta)$-*exploration tree* of $\mathscr{A}$ is an index tree $\mathbb{T}$ of $[k]$ such that, for each vertex $u \in V(\mathbb{T})$:

(a) there exists a collection $\mathscr{A}_u \subseteq \mathscr{A}$ such that $\mathscr{A}'_u := \{H_{I_u} : H \in \mathscr{A}_u\}$ is a minimal cover of S_{I_u} with $F(\mathscr{A}'_u) = I_u$, and if $u \in N(v)$, then:

  (i) $\mathscr{A}_u \subseteq \mathscr{A}_v$;

  (ii) $F(\mathscr{A}_u) \subseteq \{i_w : w \prec_{\mathbb{T}} v\} \cup I_u$;

  (iii) there exists an element $s_u \in S_{i_v}$ such that $H_{i_v} \in \{s_u, S_{i_v}\}$ for each $H \in \mathscr{A}_u$.

Moreover, for each vertex $u \in V(\mathbb{T})$, one of the following holds:

(b) u is *good*, which means that there exists a collection of hyperplanes $\mathscr{F}_u \subseteq \mathscr{A}_u$, with

$$|\mathscr{F}_u| \geqslant (1-\varepsilon)\big(|S_{i_u}| - 1\big), \tag{6.6}$$

such that $i_u \in F(H)$ and $\mu_{I_u \setminus \{i_u\}}(H) > \delta$ for each $H \in \mathscr{F}_u$.

(c) u is *bad*, which means that there exists a collection of hyperplanes $\mathscr{G}_u \subseteq \mathscr{A}_u$, with

$$\sum_{H \in \mathscr{G}_u} 2^{-|F(H) \cap I_u|/4} \geqslant \frac{|S_{i_u}|}{\lambda}, \tag{6.7}$$

such that $i_u \in F(H)$ and $|F(H) \cap I_u| \geqslant 2$ for each $H \in \mathscr{G}_u$.

We think of the elements of $\mathscr{F}_u$ (when u is good) and $\mathscr{G}_u$ (when u is bad) as hyperplanes that (respectively) do and do not look like parts of a frame from the perspective of the vertex u. We will show (see Lemma 6.3.3, below) that exploration trees always exist, as long as we choose δ to be sufficiently small, depending on λ and ε. We will then, in Section 6.4, carefully choose a subtree T of our exploration tree $\mathbb{T}$, and one 'special' vertex for each coordinate $i \in [k]$, with the following three properties: the frames corresponding to good special vertices are disjoint (unless one

of the corresponding sets S_i is very small); if 'many' special vertices of T are bad, then $\mathscr{A}$ fails to satisfy (6.3); and if 'almost all' of the special vertices of T are good, then there exists a sufficiently large δ-generalized frame in $\mathscr{A}$.

The main aim of this section is to prove the following lemma.

Lemma 6.3.3. Let $\lambda, \varepsilon \in (0,1)$, and let $\mathscr{A}$ be a minimal cover of $S_{[k]}$ with hyperplanes such that $F(\mathscr{A}) = [k]$. If

$$0 < \delta < 2^{-9}\lambda^2 \varepsilon^{2\log_2(1/\lambda\varepsilon)+11}, \tag{6.8}$$

then there exists an $(\lambda, \varepsilon, \delta)$-exploration tree of $\mathscr{A}$.

Let us fix, for the rest of this section, constants $0 < \lambda, \varepsilon < 1$ and $\delta > 0$ satisfying (6.8). We will prove Lemma 6.3.3 by iteratively extending a 'partial' exploration tree by applying the following lemma to a leaf of the current tree.

Lemma 6.3.4. Let $\emptyset \neq I \subseteq [k]$, and let $\mathscr{A}$ be a minimal cover of S_I with $F(\mathscr{A}) = I$.

(*a*) For each $i \in I$, there exists a map $J\colon S_i \to \mathscr{P}(I \setminus \{i\})$, with

$$\bigcup_{s \in S_i} J(s) = I \setminus \{i\}, \tag{6.9}$$

and for each $s \in S_i$ there exists $\mathscr{A}_s \subseteq \mathscr{A}$ such that $\mathscr{A}'_s := \{H_{J(s)} : H \in \mathscr{A}_s\}$ is a minimal cover of $S_{J(s)}$, $F(\mathscr{A}_s) \setminus \{i\} = J(s)$, and $H_i \in \{s, S_i\}$ for each $H \in \mathscr{A}_s$.

Moreover, there exists $i \in I$ such that one of the following holds:

(*b*) there exists a collection $\mathscr{F} \subseteq \mathscr{A}$, with

$$|\mathscr{F}| \geqslant (1-\varepsilon)\big(|S_i| - 1\big),$$

such that $i \in F(H)$ and $\mu_{I\setminus\{i\}}(H) > \delta$ for each $H \in \mathscr{F}$.

(*c*) there exists a collection $\mathscr{G} \subseteq \mathscr{A}$, with

$$\sum_{H \in \mathscr{G}} 2^{-|F(H)|/4} \geqslant \frac{|S_i|}{\lambda}, \tag{6.10}$$

such that $i \in F(H)$ and $|F(H)| \geqslant 2$ for each $H \in \mathscr{G}$.

Let us fix a set $\emptyset \neq I \subseteq [k]$ until the end of the proof of Lemma 6.3.4. This section is organised as follows: in Section 6.3.1 we will prove two (straightforward) technical lemmas, in Section 6.3.2, we will introduce the operation that we will use to construct the map J and the families $\mathscr{A}_s$, and in Section 6.3.3, we will prove Lemma 6.3.4, and deduce Lemma 6.3.3.

6.3.1 Two technical lemmas

Our first technical lemma (Lemma 6.3.5, below) follows from a straightforward application of the Lovász Local Lemma. We will apply it, in the case that there does not exist a collection $\mathscr{F} \subseteq \mathscr{A}$ as in

Lemma 6.3.4(b) for any $i \in I$, to a certain subset $\prod_{i\in I} R_i \subseteq S_I$, in order to find an index such that (c) holds. Our second technical lemma (Lemma 6.3.6, below) will allow us to deduce the bound (6.10) from the condition given by the local lemma. Let us say that a hyperplane H in R_I is *non-trivial* if $H \neq R_I$.

Lemma 6.3.5. Let $0 < \eta < 1/5$ and let $\{R_i : i \in I\}$ be a collection of finite sets, each with at least two elements. Let $\mathscr{A}$ be a collection of non-trivial hyperplanes in R_I, and let $\tilde{\mu}$ denote the uniform measure on R_I. If

$$\sum_{H\in\mathscr{A}\,:\,i\in F(H)} e^{\eta|F(H)|}\tilde{\mu}(H) < \frac{\eta}{2} \tag{6.11}$$

for every $i \in [k]$, then R_I is not covered by the hyperplanes in $\mathscr{A}$.

Proof. We choose a point $y \in R_I$ uniformly at random and apply the local lemma. For each hyperplane $H \in \mathscr{A}$ we define E_H to be the ("bad") event that $y \in H$. Observe that $\mathbb{P}(E_H) = \tilde{\mu}(H)$, and define a dependency graph G on the events $\{E_H\}_{H\in\mathscr{A}}$ by setting $E_H \sim E_{H'}$ if $F(H)\cap F(H') \neq \emptyset$. Observe that if $F(H)\cap\big(F(H^{(1)})\cup\cdots\cup F(H^{(t)})\big) = \emptyset$, then E_H is independent of the collection $\big\{E_{H^{(1)}},\dots,E_{H^{(t)}}\big\}$, so G is a valid dependency graph.

Next, we define weights

$$x(H) = e^{\eta|F(H)|}\tilde{\mu}(H)$$

for each $H \in \mathscr{A}$. To apply the local lemma we need to show that

$$\mathbb{P}(E_H) \leqslant x(H)\prod_{E_H\sim E_{H'}}\big(1-x(H')\big).$$

To do so, we first claim that $1-x(H) \geqslant e^{-2x(H)}$ for every $H \in \mathscr{A}$. This holds because

$$x(H) = e^{\eta|F(H)|}\tilde{\mu}(H) \leqslant \big(e^{\eta}/2\big)^{|F(H)|} \leqslant \big(1-e^{-1}\big)^{|F(H)|} \leqslant 1-e^{-1},$$

where the first inequality is $\tilde{\mu}(H) \leqslant 2^{-|F(H)|}$, which holds because each set S_i has at least two elements, the second follows since $\eta < 1/5$, and the third since the hyperplanes in $\mathscr{A}$ are non-trivial, so $|F(H)| \geqslant 1$. Therefore, for each $H \in \mathscr{A}$, we have

$$\begin{aligned}\prod_{E_H\sim E_{H'}}\big(1-x(H')\big) &\geqslant \exp\bigg(-2\sum_{E_H\sim E_{H'}} x(H')\bigg) \geqslant \exp\bigg(-2\sum_{i\in F(H)}\sum_{H'\in\mathscr{A}\,:\,i\in F(H')} x(H')\bigg)\\ &= \exp\bigg(-2\sum_{i\in F(H)}\sum_{H'\in\mathscr{A}\,:\,i\in F(H')} e^{\eta|F(H')|}\tilde{\mu}(H')\bigg) \geqslant \exp\big(-\eta|F(H)|\big),\end{aligned}$$

where the last inequality follows from (6.11). This implies that

$$x(H)\prod_{E_H\sim E_{H'}}\big(1-x(H')\big) \geqslant x(H)e^{-\eta|F(H)|} = \tilde{\mu}(H) = \mathbb{P}(E_H),$$

as required. By the local lemma, it follows that the probability that none of the events E_H holds is non-zero, and hence there exists a point $y \in R_I$ that is not covered by $\mathscr{A}$. □

The second technical lemma is even more straightforward. Recall that $S_1,\dots,S_k$ are fixed finite sets with at least two elements, and that the (non-empty) set $I \subseteq [k]$ and positive constants λ and ε were fixed above.

Lemma 6.3.6. For each $j \in I$, let $R_j \subseteq S_j$ be such that $|R_j| \geqslant \varepsilon\big(|S_j|-1\big)+1$, and let $\tilde{\mu}$ denote the uniform measure on R_I. Let H be a hyperplane in S_I, and let $i \in F(H)$. If

$$\mu_{I\setminus\{i\}}(H) \leqslant 2^{-9}\lambda^2\varepsilon^{2\log_2(1/\lambda\varepsilon)+11},$$

then

$$\tilde{\mu}\big(H \cap R_I\big) \leqslant \frac{\lambda}{2^{|F(H)|/2+4}|S_i|}. \tag{6.12}$$

Proof. Set $\ell := |F(H)|$ and $\delta_0 := 2^{-9}\lambda^2\varepsilon^{2\log_2(1/\lambda\varepsilon)+11}$, and observe that

$$\tilde{\mu}\big(H \cap R_I\big)\cdot|R_i| \leqslant \mu_{I\setminus\{i\}}(H)\prod_{j\in F(H),\, i\neq j}\frac{|S_j|}{|R_j|} \leqslant \delta_0\varepsilon^{-(\ell-1)}.$$

Now, note that $|R_j| \geqslant 2$ for every $j \in I$, and suppose that (6.12) does not hold. Then

$$2^{-(\ell-1)} \geqslant \prod_{j\in F(H),\, j\neq i}\frac{1}{|R_j|} = \tilde{\mu}\big(H\cap R_I\big)\cdot|R_i| \geqslant \frac{\lambda}{2^{\ell/2+4}}\cdot\frac{|R_i|}{|S_i|} \geqslant \frac{\lambda\varepsilon}{2^{\ell/2+4}},$$

and hence $\ell \leqslant 2\log_2(1/\lambda\varepsilon)+10$. It follows that

$$2^{\ell/2+4}\cdot\tilde{\mu}\big(H\cap R_I\big) \leqslant \frac{2^{9/2}\delta_0}{|R_i|}\left(\frac{\sqrt{2}}{\varepsilon}\right)^{\ell-1} \leqslant \frac{2^{9/2}\delta_0}{|R_i|}\left(\frac{\sqrt{2}}{\varepsilon}\right)^{2\log_2(1/\lambda\varepsilon)+9} = \frac{\lambda\varepsilon}{|R_i|} \leqslant \frac{\lambda}{|S_i|},$$

as required. □

6.3.2 An operation on a covering system

We next introduce a simple operation that, given a minimal cover of S_I, produces a map J and a collection $\{\mathscr{A}_s : s \in S_i\}$ as required by Lemma 6.3.4(a). This operation is the basic tool we will use in the construction of our exploration trees. Recall that the (non-empty) set $I \subseteq [k]$ was fixed above, and let $\mathscr{A}$ be a minimal cover of S_I with $F(\mathscr{A}) = I$. For each $i \in I$ and $s \in S_i$, set

$$\mathscr{H}(i,s) := \big\{H \in \mathscr{A} : H_i \in \{s, S_i\}\big\},$$

and observe that the collection $\mathscr{H}'(i,s) := \big\{H_{I\setminus\{i\}} : H \in \mathscr{H}(i,s)\big\}$ is a cover of $S_{I\setminus\{i\}}$. Note that moreover, since $\mathscr{A}$ is minimal, there is a bijection between $\mathscr{H}(i,s)$ and $\mathscr{H}'(i,s)$. Let $\mathscr{A}_s^* \subseteq \mathscr{H}'(i,s)$ be an arbitrary minimal subcover of $S_{I\setminus\{i\}}$, and define

$$J(s) := F(\mathscr{A}_s^*) \qquad \text{and} \qquad \mathscr{A}_s := \big\{H \in \mathscr{H}(i,s) : H_{I\setminus\{i\}} \in \mathscr{A}_s^*\big\}.$$

Note that $\mathscr{A}'_s = \{H_{J(s)} : H \in \mathscr{A}_s\}$ is a minimal cover of $S_{J(s)}$, that $F(\mathscr{A}_s) \setminus \{i\} = J(s)$, and that $H_i \in \{s, S_i\}$ for each $H \in \mathscr{A}_s$. To verify that J and $\{\mathscr{A}_s : s \in S_i\}$ satisfy Lemma 6.3.4(a), it therefore only remains to check that (6.9) holds.

Lemma 6.3.7. Let $\mathscr{A}$ be a minimal cover of S_I with hyperplanes. If $F(\mathscr{A}) = I$, then

$$\bigcup_{s \in S_i} J(s) = I \setminus \{i\}$$

for each $i \in I$.

Proof. We will in fact show that for every $H \in \mathscr{A}$, there exists $s \in S_i$ with $H_{I\setminus\{i\}} \in \mathscr{A}^*_s$. Since $J(s) = F(\mathscr{A}^*_s) \subseteq I \setminus \{i\}$ and $F(\mathscr{A}) = I$, this will be enough to prove the lemma.

To prove the claim, let $x \in S_I$ be an element that is only covered by H (recall that $\mathscr{A}$ is minimal), and set $s := x_i$. We claim that $H_{I\setminus\{i\}} \in \mathscr{A}^*_s$. Indeed, since $\mathscr{A}^*_s$ is a cover of $S_{I\setminus\{i\}}$, it must cover the vector x' obtained from x by ignoring the ith coordinate, and $H_{I\setminus\{i\}}$ is the only potential element of $\mathscr{A}^*_s$ that can do so. □

6.3.3 Construction of the exploration tree

Having completed our preparations, we are now ready to prove Lemma 6.3.4, and deduce Lemma 6.3.3.

Proof of Lemma 6.3.4. Let $\emptyset \neq I \subseteq [k]$, and let $\mathscr{A}$ be a minimal cover of S_I such that $F(\mathscr{A}) = I$. To prove part (a), for each $i \in I$ we apply the construction defined in Section 6.3.2 to obtain a map $J\colon S_i \to \mathscr{P}(I \setminus \{i\})$ and a collection $\{\mathscr{A}_s : s \in S_i\}$, where $\mathscr{A}_s \subseteq \mathscr{A}$, as required. In particular, $\mathscr{A}'_s = \{H_{J(s)} : H \in \mathscr{A}_s\}$ is a minimal cover of $S_{J(s)}$, $F(\mathscr{A}_s) \setminus \{i\} = J(s)$, $H_i \in \{s, S_i\}$ for each $H \in \mathscr{A}_s$, and the map J satisfies (6.9) by Lemma 6.3.7.

To prove that there exists $i \in I$ such that either (b) or (c) holds, let us define an element $s \in S_i$ to be *special for* i if there exists a hyperplane $H \in \mathscr{A}$ such that

$$H_i = s \qquad \text{and} \qquad \mu_{I\setminus\{i\}}(H) > \delta.$$

If this holds, then we say that the hyperplane H is a *witness* for the pair (s,i). Now, for each $i \in I$ define S^*_i to be the set of elements $s \in S_i$ that are special for i, and set $R_i := S_i \setminus S^*_i$. We consider two cases, corresponding to conditions (b) and (c) of Definition 6.3.2, respectively.

Case 1: There exists $i \in I$ with $|R_i| < \varepsilon\big(|S_i| - 1\big) + 1$.

In this case we define

$$\mathscr{F} := \bigcup_{s \in S^*_i} \big\{H \in \mathscr{A} : H \textit{ is a witness for } (s,i)\big\}.$$

Since a hyperplane H cannot witness (s,i) for more than one element $s \in S^*_i$, we have

$$|\mathscr{F}| \geqslant |S^*_i| \geqslant |S_i| - \varepsilon\big(|S_i| - 1\big) - 1 = (1-\varepsilon)\big(|S_i| - 1\big),$$

and by definition $i \in F(H)$ and $\mu_{I\setminus\{i\}}(H) > \delta$ for each $H \in \mathscr{F}$.

Case 2: $|R_i| \geqslant \varepsilon\big(|S_i|-1\big)+1$ for every $i \in I$.

In this case we apply Lemma 6.3.5 to the set $R_I := \prod_{i\in I} R_i$ with $\eta = \log(2)/4$. Define $\mathscr{A}' \subseteq \mathscr{A}$ by removing all hyperplanes that are witness for (s,i) for some $i \in I$ and $s \in S_i$. Observe that none of the witness hyperplanes intersects R_I, so $\mathscr{A}'' := \{H \cap R_I : H \in \mathscr{A}'\}$ is a cover of R_I. We claim that there exists a coordinate $i \in I$ such that

$$\sum_{H\in\mathscr{A}' : i\in F(H)} e^{\eta|F(H)|}\tilde{\mu}\big(H\cap R_I\big) \geqslant \frac{\eta}{2}, \tag{6.13}$$

where $\tilde{\mu}$ denotes the uniform measure on R_I. Indeed, since $|S_i| \geqslant 2$ for every $i \in [k]$, it follows that $|R_i| \geqslant 2$ for every $i \in I$, and hence if $R_I \subseteq H$ for some $H \in \mathscr{A}'$, then also $S_I \subseteq H$. However, since $\mathscr{A}$ is a minimal cover of S_I, if $S_I \subseteq H$ for some $H \in \mathscr{A}$, then $\mathscr{A} = \{H\}$, and hence $F(\mathscr{A}) = \emptyset$. This contradicts our assumption that $F(\mathscr{A}) = I$, and so $R_I \not\subseteq H$ for each $H \in \mathscr{A}'$, i.e., $\mathscr{A}''$ is a collection of non-trivial hyperplanes in R_I. By Lemma 6.3.5, and recalling that $\mathscr{A}''$ is a cover of R_I, it follows that if (6.13) holds for some $i \in I$, as claimed.

Fix such an $i \in I$, and define $\mathscr{G} := \mathscr{G}^{(1)} \cup \cdots \cup \mathscr{G}^{(k)}$, where

$$\mathscr{G}^{(\ell)} := \big\{H \in \mathscr{A}' : i \in F(H) \text{ and } |F(H)| = \ell\big\}$$

for each $\ell \in [k]$. Observe that $\mathscr{G}^{(1)} = \emptyset$, since if $H \in \mathscr{A}$ and $F(H) = \{i\}$, then $\mu_{I\setminus\{i\}}(H) = 1$, and so H would have been removed when we formed $\mathscr{A}'$. Similarly, for each $H \in \mathscr{A}'$ with $i \in F(H)$ we have

$$\mu_{I\setminus\{i\}}(H) \leqslant \delta < 2^{-9}\lambda^2\varepsilon^{2\log_2(1/\lambda\varepsilon)+11},$$

since otherwise H would witness (s,i) for some $s \in S_i$, and so would been removed when we formed $\mathscr{A}'$. It follows, by Lemma 6.3.6, that

$$\tilde{\mu}\big(H\cap R_I\big) \leqslant \frac{\lambda}{2^{|F(H)|/2+4}|S_i|}$$

for every $H \in \mathscr{A}'$ with $i \in F(H)$. Finally, combining this with (6.13) gives

$$\sum_{\ell\geqslant 2}|\mathscr{G}^{(\ell)}|\frac{e^{\eta\ell}}{2^{\ell/2}} = \sum_{H\in\mathscr{A}' : i\in F(H)}\frac{e^{\eta|F(H)|}}{2^{|F(H)|/2}} \geqslant \sum_{H\in\mathscr{A}' : i\in F(H)} e^{\eta|F(H)|}\tilde{\mu}(H\cap R_I)\frac{16|S_i|}{\lambda} \geqslant \frac{8\eta|S_i|}{\lambda},$$

and hence

$$\sum_{H\in\mathscr{G}} 2^{-|F(H)|/4} = \sum_{\ell\geqslant 2}\frac{|\mathscr{G}^{(\ell)}|}{2^{\ell/4}} = \sum_{\ell\geqslant 2}|\mathscr{G}^{(\ell)}|\frac{e^{\eta\ell}}{2^{\ell/2}} \geqslant \frac{8\eta|S_i|}{\lambda} \geqslant \frac{|S_i|}{\lambda},$$

as required by (6.10). □

The deduction of Lemma 6.3.3 is now straightforward. Let $\partial(T)$ denote the set of vertices of a rooted tree T with no out-neighbours, and call $\partial(T)$ the *boundary* of T.

Proof of Lemma 6.3.3. We construct $\mathbb{T}$, our exploration tree, inductively, with Lemma 6.3.4 providing the induction step. We begin our induction by defining T_0 to be a single vertex v, and setting $I_v := [k]$, and $\mathscr{A}_v := \mathscr{A}$. For the induction step, suppose that we have constructed a rooted tree T_t (with root v), a

set $\emptyset \neq I_u \subseteq [k]$ and a collection $\mathscr{A}_u \subseteq \mathscr{A}$ for each vertex $u \in V(T_t)$, and an index $i_u \in I_u$ for each non-boundary vertex $u \in V(T_t) \setminus \partial(T_t)$, such that condition (a) of Definition 6.3.2 holds for every vertex $u \in V(T_t)$, and condition (ii) of Definition 6.3.1, and either condition (b) or (c) of Definition 6.3.2, hold for every non-boundary vertex $u \in V(T_t) \setminus \partial(T_t)$. Observe that, since $\mathscr{A}$ is a minimal cover of $S_{[k]}$ with hyperplanes such that $F(\mathscr{A}) = [k]$, these conditions are satisfied in the base case $t = 0$.

To construct T_{t+1}, choose a vertex $u \in \partial(T)$ such that $|I_u| \geqslant 2$, if one exists (we will deal with the other case below), and apply Lemma 6.3.4 to the set I_u and minimal cover $\mathscr{A}'_u = \{H_{I_u} : H \in \mathscr{A}_u\}$ of S_{I_u} (noting that $F(\mathscr{A}'_u) = I_u$, by the induction hypothesis). We obtain an index $i \in I_u$, a map $J\colon S_i \to \mathscr{P}(I \setminus \{i\})$ and a collection $\{\mathscr{A}_s : s \in S_i\}$ as in part (a) of the lemma, and either a collection $\mathscr{F} \subseteq \mathscr{A}'_u$ as in part (b), or a collection $\mathscr{G} \subseteq \mathscr{A}'_u$ as in part (c). In either case, we set $i_u := i$, add an out-neighbour of u for each element $s \in S_i$ such that $J(s) \neq \emptyset$, and to the new vertex $w(s)$ corresponding to s we assign the set $I_{w(s)} = J(s)$, the element $s_{w(s)} := s$, and the collection of hyperplanes

$$\mathscr{A}_{w(s)} := \big\{H \in \mathscr{A}_u : H_{I_u} \in \mathscr{A}_s\big\}.$$

Note that, by Lemma 6.3.4(a), condition (a) of Definition 6.3.2 holds for each vertex $w(s)$, since $\mathscr{A}_{w(s)} \subseteq \mathscr{A}_u \subseteq \mathscr{A}$, the family $\mathscr{A}'_{w(s)} = \{H_{J(s)} : H \in \mathscr{A}_{w(s)}\} = \{H_{J(s)} : H \in \mathscr{A}_s\}$ is a minimal cover of $S_{J(s)}$ with $F(\mathscr{A}'_{w(s)}) = F(\mathscr{A}_s) \setminus \{i\} = J(s)$, $H_{i_u} \in \{s, S_{i_u}\}$ for each $H \in \mathscr{A}_{w(s)}$, and

$$F(\mathscr{A}_{w(s)}) \subseteq \big(F(\mathscr{A}_u) \setminus I_u\big) \cup F(\mathscr{A}_s) \subseteq \big\{i_w : w \prec_{\mathbb{T}} u\big\} \cup J(s).$$

Note also that, by (6.9), u satisfies condition (ii) of Definition 6.3.1.

It remains to observe that u is either good or bad, i.e., satisfies either condition (b) or (c) of Definition 6.3.2. Indeed, if Lemma 6.3.4(b) holds then we set

$$\mathscr{F}_u := \big\{H \in \mathscr{A}_u : H_{I_u} \in \mathscr{F}\big\},$$

and if Lemma 6.3.4(c) holds then we set

$$\mathscr{G}_u := \big\{H \in \mathscr{A}_u : H_{I_u} \in \mathscr{G}\big\}.$$

In each case, the properties guaranteed by the lemma are exactly those that we require. Since the properties required of all vertices of T_{t+1} other than u and its out-neighbours continue to hold, it follows that T_{t+1} satisfies the same properties that we assumed for T_t.

Finally, observe that in passing from T_t to T_{t+1} we replace a vertex of the boundary by a finite number of boundary elements, each associated with with strictly smaller sets. This process must therefore eventually end, and when it does, it follows that $|I_u| = 1$ for every boundary vertex $u \in \partial(T_t)$. When this happens, we simply set i_u equal to the unique member of I_u for each $u \in \partial(T_t)$, and claim that u is good. Indeed, by the induction hypothesis, the collection $\mathscr{A}'_u$ forms a minimal cover of S_{i_u} with $i_u \in F(\mathscr{A}'_u)$. It follows that $\mathscr{A}'_u$ consists of exactly $|S_{i_u}|$ singleton hyperplanes, and so (6.6) holds with $\mathscr{F}_u := \mathscr{A}_u$. Since condition (ii) of Definition 6.3.1 holds automatically (with both sides equal to the empty set), it follows that the tree $\mathbb{T}$ that we have constructed is an $(\lambda, \varepsilon, \delta)$-exploration tree of $\mathscr{A}$, as required. □

6.4 Extraction of the frame, and the proof of Theorem 6.2.3

In order to prove Theorem 6.2.3, we will use the exploration tree $\mathbb{T}$ constructed in the previous section, together with the bound (6.3), to find a δ-generalized frame for $\mathscr{A}$. Roughly speaking, we would like to do this by choosing k vertices $\beta(1),\dots,\beta(k)$ such that the label of $\beta(i)$ in $\mathbb{T}$ is i, define a tree T to be the union of the paths (in $\mathbb{T}$) from $\beta(i)$ to the root, and define the elements $s_j(i)$ using the elements s_u. For each good vertex $\beta(i)$ we have a collection $\mathscr{F}_i$ of hyperplanes more or less as required, and for each bad vertex we obtain a large collection of 'garbage' hyperplanes. We might therefore hope to use (6.3) to show that there are few bad vertices, and thus to deduce the bound (6.4).

There are two main problems with the strategy described above: the frame elements obtained for good vertices might not be disjoint, and each hyperplane might be included in the garbage set $\mathscr{G}_u$ for a very large number of bad vertices. We overcome both obstacles in the same way: by choosing the vertices $\beta(i)$ via a depth-first search algorithm.

Here is the intuition. Assume that vertices $\beta(i)$ and $\beta(j)$ are good, $H \in \mathscr{F}_i \cap \mathscr{F}_j$ and $\min(|S_i|,|S_j|) \geqslant \delta^{-1}$. Say $\beta(i)$ comes before $\beta(j)$ in the depth-first search algorithm. Then coordinate j is not explored anywhere on the branch extending to $\beta(i)$. But coordinate j is in the set of fixed coordinates associated to $\beta(i)$ as H is in the family of hyperplanes associated to $\beta(i)$. This implies that j is explored on a branch extending further from $\beta(i)$. As $\beta(j)$ is the first point in the depth-first search algorithm that explores coordinate j and $\beta(i)$ comes before $\beta(j)$, this forces $\beta(i)$ to be on the branch extending to $\beta(j)$. Thus, with respect to vertex $\beta(i)$, hyperplane H has weight at most $|S_j|^{-1} < \delta$, which contradicts the assumption that $\beta(i)$ is a good vertex and H a frame element.

In Section 6.4.1 we will state precisely the object we will construct, and show that its existence implies the existence of a δ-generalized frame. In Section 6.4.2 we will describe how we choose the sub-tree $T \subseteq \mathbb{T}$, the frame elements $(\mathscr{F}_1,\dots,\mathscr{F}_k)$, and the 'garbage' sets $(\mathscr{G}_1,\dots,\mathscr{G}_k)$; in Section 6.4.3, we will prove two lemmas on the disjointness of the frame elements and garbage sets; and in Section 6.4.4 we will complete the proof of Theorem 6.2.3.

6.4.1 Tree-frames

The purpose of this section is to introduce the following somewhat complicated objects, which also provide significantly more information (though we will not need this) about the covering system. We will use these objects to construct our frames.

Definition 6.4.1. Let T be a rooted tree equipped with maps

$$\alpha\colon V(T) \to [k], \qquad \beta\colon [k] \to V(T) \qquad \text{and} \qquad \gamma\colon E(T) \to S_1 \cup \dots \cup S_k$$

such that

(*a*) $\alpha(u) \neq \alpha(v)$ if $u \prec_T v$;

(*b*) $\alpha\big(\beta(i)\big) = i$ for each $i \in [k]$;

(*c*) if $e \in E(T)$ and v is the endpoint of e that is closer to the root, then $\gamma(e) \in S_{\alpha(v)}$;

(d) there exists a permutation π of $[k]$ such that, if for each $i \in [k]$ we set

$$J(i) := \{\alpha(u) : u \prec_T \beta(i)\} \qquad \text{and} \qquad I(i) := [k] \setminus J(i),$$

then $J(\pi(i)) \subseteq \{\pi(1), \dots, \pi(i)\}$ for each $i \in [k]$.

Now, for each $\delta > 0$, a *δ-generalized tree-frame* centred at T is a sequence $(\mathscr{F}_1, \dots, \mathscr{F}_k)$, where $\mathscr{F}_i$ is a collection of at most $|S_i| - 1$ hyperplanes, satisfying

(i) $i \in F(H)$ for each $H \in \mathscr{F}_i$.

(ii) $\mu_{I(i)}(H) > \delta$ for each $H \in \mathscr{F}_i$.

(iii) $H_j \in \{\gamma(e), S_j\}$ for each $H \in \mathscr{F}_i$ and each $j \in J(i) \setminus \{i\}$, where $e \in E(T)$ is the edge leaving the unique vertex $v \prec_T \beta(i)$ with $\alpha(v) = j$ in the direction of $\beta(i)$;

(iv) If $\min\{|S_i|, |S_j|\} \geqslant \delta^{-1}$ and $i \neq j$, then $\mathscr{F}_i$ and $\mathscr{F}_j$ are disjoint.

In Sections 6.4.2–6.4.4 we will construct, for any $\mathscr{A}$ satisfying (6.3), a δ-generalized tree-frame satisfying (6.4). The next lemma shows that this will be sufficient to prove Theorem 6.2.3.

Lemma 6.4.2. If $(\mathscr{F}_1, \dots, \mathscr{F}_k)$ is a δ-generalized tree-frame centred at a rooted tree T, then $(\mathscr{F}_1, \dots, \mathscr{F}_k)$ is a δ-generalized frame.

Proof. Let T be a rooted tree equipped with maps α, β and γ satisfying conditions (a)–(d) of Definition 6.4.1. In particular, let π be the permutation given by condition (d), and (to simplify the notation) let us permute the sets $S_1, \dots, S_k$ so that π is the identity, and therefore $J(i) \subseteq \{1, \dots, i\}$ (and hence $I(i) \supseteq \{i+1, \dots, k\}$) for each $i \in [k]$. Now, for each $i \in [k]$ and $j \in J(i) \setminus \{i\}$, set $s_j(i) := \gamma(e) \in S_j$, where $e \in E(T)$ is the edge leaving the unique vertex $v \prec_T \beta(i)$ with $\alpha(v) = j$ in the direction of $\beta(i)$.

We claim that, for each $H \in \mathscr{F}_i$,

$$i \in F(H), \qquad \mu_{I(i)}(H) > \delta \qquad \text{and} \qquad H_j \in \{s_j(i), S_j\}.$$

Indeed, these follow directly from properties (i), (ii) and (iii) of Definition 6.4.1. Finally, observe that if $\min\{|S_i|, |S_j|\} \geqslant \delta^{-1}$ then $\mathscr{F}_i$ and $\mathscr{F}_j$ are disjoint, by property (iv). □

6.4.2 Constructing the frame

In this section we will construct the δ-generalized tree-frame $(\mathscr{F}_1, \dots, \mathscr{F}_k)$, along with the rooted tree T, and a collection $(\mathscr{G}_1, \dots, \mathscr{G}_k)$ of 'garbage' sets. Let $C > 0$ and $\varepsilon > 0$ be arbitrary, as in the statement of Theorem 6.2.3, and set

$$\lambda := \frac{\varepsilon}{2^4 C} \qquad \text{and} \qquad \delta := 2^{-23} \lambda^4 \varepsilon^{2\log_2(1/\lambda\varepsilon)+15}. \tag{6.14}$$

Recall that the sets $S_1, \dots, S_k$ were fixed earlier, and let us fix, for the rest of this section, a minimal cover $\mathscr{A}$ of $S_{[k]}$ with hyperplanes such that $F(\mathscr{A}) = [k]$. By Lemma 6.3.3, there exists an $(\lambda, \varepsilon/2, \delta)$-exploration tree of $\mathscr{A}$; let us also fix such a tree $\mathbb{T}$.

The first step is to observe that every $i \in [k]$ occurs as the label i_u of some vertex $u \in V(\mathbb{T})$. This is an immediate consequence of the following simple observation about index trees.

Observation 6.4.3. Let $\mathbb{T}$ be an index tree, let $u \in V(\mathbb{T})$, and let $j \in I_u$. Then there exists $v \in V(\mathbb{T})$, with $u \prec_{\mathbb{T}} v$, such that $i_v = j$.

Proof. This follows easily from (6.5): if $j \in I_v$ and $i_v \neq j$, then $j \in I_w$ for some $w \in N(u)$, and if $j \in I_v$ and $v \in \partial(\mathbb{T})$ then $I_v = \{j\}$, so $i_v = j$. □

To extract our δ-generalized tree-frame from the exploration tree $\mathbb{T}$, we will also need the notion of a *depth-first search ordering* $\prec$ on the vertices of a rooted tree T. This is defined by placing an arbitrary linear order on the out-neighbours of each vertex of T, and then setting $u \prec v$ if either $u \prec_T v$, or if the branch leading to u precedes the branch leading to v in the ordering of the neighbours of the last common ancestor of u and v.

Definition 6.4.4. Let $\prec$ be a depth-first search ordering on the vertices of $\mathbb{T}$. We define a rooted tree T and a δ-generalized tree-frame centred at T as follows:

1. For each $i \in [k]$, define $\beta(i)$ to be the $\prec$-minimal vertex u of $\mathbb{T}$ such that $i_u = i$.
2. Define T to be the union of the paths in $\mathbb{T}$ from $\beta(1), \dots, \beta(k)$ to the root.
3. For each $u \in V(T)$, define $\alpha(u) := i_u$.
4. For each edge $uv \in E(T)$, where $u \in N(v)$, define $\gamma(uv) := s_u \in S_{i_v}$.
5. (*a*) Set $\mathscr{F}_i := \mathscr{F}_{\beta(i)}$ and $\mathscr{G}_i := \emptyset$ for each $i \in [k]$ such that $\beta(i)$ is a good vertex of $\mathbb{T}$.
   (*b*) Set $\mathscr{G}_i := \mathscr{G}_{\beta(i)}$ and $\mathscr{F}_i := \emptyset$ for each $i \in [k]$ such that $\beta(i)$ is a bad vertex of $\mathbb{T}$.

Observe that several of the properties required by Definition 6.4.1 follow immediately from this construction. Indeed, $\alpha(u) \neq \alpha(v)$ if $u \prec_T v$ and $u \neq v$, because $\mathbb{T}$ is an index tree, and therefore satisfies (6.5) (so $\alpha(u) = i_u$ is not included in any of the sets associated with the descendants of u); $\alpha(\beta(i)) = i$ for each $i \in [k]$ by our choice of α and β; and if $uv \in E(T)$ and $u \in N(v)$, then $\gamma(uv) \in S_{i_v} = S_{\alpha(v)}$. We also have $i \in F(H)$ for each $i \in [k]$ and $H \in \mathscr{F}_i$, by Definition 6.3.2(*b*). Moreover, recalling that $J(i) = \big\{\alpha(u) : u \prec_T \beta(i)\big\}$ and $I(i) = [k] \setminus J(i)$ for each $i \in [k]$, the permutation π of $[k]$ given by the ordering $\prec$ restricted to $\{\beta(1), \dots, \beta(k)\}$ satisfies $J(\pi(i)) \subseteq \{\pi(1), \dots, \pi(i)\}$ for each $i \in [k]$, since if $u \prec_T \beta(\pi(i))$ then $u \prec \beta(\pi(i))$, so $\beta(\alpha(u)) \prec \beta(\pi(i))$, and hence $\alpha(u) \in \{\pi(1), \dots, \pi(i)\}$. Two further properties follow almost immediately, but for completeness we spell out the details.

Lemma 6.4.5. Let $i \in [k]$ and $H \in \mathscr{F}_i$, let $j \in J(i) \setminus \{i\}$, and let $e_j \in E(T)$ be the edge leaving the unique vertex $v \prec_T \beta(i)$ with $\alpha(v) = j$ in the direction of the vertex $\beta(i)$. Then

$$\mu_{I(i)}(H) > \delta \qquad \text{and} \qquad H_j \in \{\gamma(e_j), S_j\}.$$

Proof. Since $\mathscr{F}_i$ is non-empty, the vertex $\beta(i)$ is good, and therefore, by Definition 6.3.2(*b*), we have $\mu_{I_{\beta(i)} \setminus \{i\}}(H) > \delta$ for each $H \in \mathscr{F}_i$. Note also that, by (6.5), we have $I_{\beta(i)} \setminus \{i\} \subseteq I(i)$. Now, recall from Definition 6.3.2 that $\mathscr{F}_i \subseteq \mathscr{A}_{\beta(i)}$ and $F(\mathscr{A}_{\beta(i)}) \subseteq J(i) \cup I_{\beta(i)}$, and observe that therefore $H_j = S_j$ for every $j \in I(i) \setminus I_{\beta(i)}$. Since $i \notin I(i)$, it follows that $\mu_{I(i)}(H) > \delta$.

For the second part, let $e_j = uv$ (with $u \in N(v)$) and observe that, by Definition 6.3.2(a), we have $H'_j \in \{s_u, S_j\} = \{\gamma(e_j), S_j\}$ for every hyperplane $H' \in \mathscr{A}_u$. Since $\mathscr{A}_{\beta(i)} \subseteq \mathscr{A}_u$ (again by Definition 6.3.2(a)), it follows that $H_j \in \{\gamma(e_j), S_j\}$, as claimed. □

It therefore only remains to show that Definition 6.4.1(iv) and the inequality (6.4) hold. Both of these properties will follow from our choice of $\beta(1), \dots, \beta(k)$.

6.4.3 Lemmas on disjointness

In this section we will prove two straightforward but crucial lemmas; the first verifies condition (iv) of the definition of a δ-generalized frame.

Lemma 6.4.6. If $\min\{|S_i|, |S_j|\} \geqslant \delta^{-1}$ and $i \neq j$, then $\mathscr{F}_i$ and $\mathscr{F}_j$ are disjoint.

Proof. Suppose that $H \in \mathscr{F}_{\beta(i)} \cap \mathscr{F}_{\beta(j)}$, and suppose that $\beta(i) \prec \beta(j)$ in the depth-first search ordering. Recall that $i, j \in F(H)$, by Definition 6.3.2(b), and $\mu_{I(i)}(H) > \delta$, by Lemma 6.4.5. Since $|S_j| \geqslant \delta^{-1}$, it follows that $j \notin I(i)$, and hence $j \in J(i)$, i.e., there exists $u \prec_T \beta(i)$ with $\alpha(u) = j$. However, this is a contradiction, since $u \prec \beta(i) \prec \beta(j)$ in the depth-first search ordering, and $\beta(j)$ was chosen to be the $\prec$-minimal vertex u of $\mathbb{T}$ such that $\alpha(u) = j$. □

The final lemma we need shows that each garbage set only appears on a single path through T. Since the number of fixed coordinates of H_{I_u} decreases along the path (and decreases strictly whenever $\alpha(u) \in F(H)$), this will imply that each hyperplane contributes only $O(1)$ to the sum of the left-hand side of (6.7) over vertices $u \in \{\beta(1), \dots, \beta(k)\}$.

Lemma 6.4.7. Let $H \in \mathscr{G}_i \cap \mathscr{G}_j$. Then either $\beta(i) \prec_{\mathbb{T}} \beta(j)$, or $\beta(j) \prec_{\mathbb{T}} \beta(i)$.

Proof. Suppose (without loss of generality) that $\beta(i) \prec \beta(j)$ in the depth-first search ordering, and suppose that $\beta(i) \not\prec_{\mathbb{T}} \beta(j)$, which implies that $u \prec \beta(j)$ for every $u \in V(\mathbb{T})$ with $\beta(i) \prec_{\mathbb{T}} u$. Note that $j \in F(H) \subseteq I_{\beta(i)} \cup J(i)$, since $H \in \mathscr{G}_i \cap \mathscr{G}_j$ and by Definition 6.3.2.

If $j \in J(i)$, then $i_v = j$ for some $v \in V(\mathbb{T})$ with $v \prec_{\mathbb{T}} \beta(i)$, and hence $v \prec \beta(i) \prec \beta(j)$. On the other hand, if $j \in I_{\beta(i)}$, then by Observation 6.4.3 we have $i_v = j$ for some $v \in V(\mathbb{T})$ with $\beta(i) \prec_{\mathbb{T}} v$, and hence (by the observation above) $v \prec \beta(j)$. In either case, this contradicts our choice of $\beta(j)$ as the $\prec$-minimal vertex v of $\mathbb{T}$ such that $i_v = j$. □

Let us record here the following simple consequence of Lemma 6.4.7.

Lemma 6.4.8. If $H \in \mathscr{G}_i \cap \mathscr{G}_j$ and $i \neq j$, then $|F(H) \cap I_{\beta(i)}| \neq |F(H) \cap I_{\beta(j)}|$.

Proof. By Lemma 6.4.7, we have (without loss of generality) $\beta(i) \prec_{\mathbb{T}} \beta(j)$, which implies, by (6.5) and since $\beta(i) \neq \beta(j)$, that $I_{\beta(j)} \subseteq I_{\beta(i)} \setminus \{i\}$. Since $i \in F(H) \cap I_{\beta(i)}$, it follows that $F(H) \cap I_{\beta(j)} \subsetneq F(H) \cap I_{\beta(i)}$, as required. □

6.4.4 The proof of Theorem 6.2.3

We are ready to prove our main structural result, Theorem 6.2.3, by deriving inequality (6.4) from (6.3). We will use the following easy consequence of Lemma 6.4.8. Set $B := \{i \in [k] : \beta(i) \text{ is bad}\}$.

Lemma 6.4.9. $\left|\bigcup_{i\in B}\mathscr{G}_i\right| \geqslant \frac{1}{5\lambda}\sum_{i\in B}|S_i|$.

Proof. Summing (6.7) over $i \in B$, we obtain

$$\sum_{i\in B}\sum_{H\in\mathscr{G}_i} 2^{-|F(H)\cap I_{\beta(i)}|/4} \geqslant \frac{1}{\lambda}\sum_{i\in B}|S_i|.$$

Now, by Lemma 6.4.8, for each $\ell \geqslant 2$ there is at most one value of $i \in B$ such that $H \in \mathscr{G}_i$ and $|F(H)\cap I_{\beta(i)}| = \ell$, so for each $H \in \bigcup_{i\in B}\mathscr{G}_i$ we have

$$\sum_{i\in B:H\in\mathscr{G}_i} 2^{-|F(H)\cap I_{\beta(i)}|/4} \leqslant \sum_{\ell=2}^{\infty} 2^{-\ell/4} = \left(2^{1/2}-2^{1/4}\right)^{-1} < 5,$$

as required. □

Theorem 6.2.3 now follows easily from the lemmas above.

Proof of Theorem 6.2.3. We claim that the sequence $(\mathscr{F}_1,\dots,\mathscr{F}_k)$ constructed in Definition 6.4.4 is a δ-generalized tree-frame centred at T, and satisfies (6.4). By Lemma 6.4.2, it will follow that $(\mathscr{F}_1,\dots,\mathscr{F}_k)$ is also a δ-generalized frame, so this will be sufficient to prove the theorem. Note that properties (a)–(d) and (i)–(iv) of Definition 6.4.1 follow from the comments after Definition 6.4.4, and by Lemmas 6.4.5 and 6.4.6. Moreover, by discarding excess hyperplanes if necessary, we may assume that $|\mathscr{F}_i| \leqslant |S_i| - 1$ for each $i \in [k]$. It therefore only remains to show that (6.4) holds.

To do so, recall $\mathbb{T}$ is an $(\lambda,\varepsilon/2,\delta)$-exploration tree of $\mathscr{A}$, and hence $|\mathscr{F}_i| > \left(1-\varepsilon/2\right)\left(|S_i|-1\right)$ for each i such that $\beta(i)$ is a good vertex, i.e., for each $i \in [k]\setminus B$. Now, by Lemma 6.4.9 and the condition (6.3), we have

$$\frac{1}{5\lambda}\sum_{i\in B}|S_i| \leqslant |\mathscr{A}| \leqslant C\sum_{i=1}^{k}\left(|S_i|-1\right),$$

and hence, recalling from (6.14) that $\lambda = \varepsilon/(2^4C)$, we obtain

$$\begin{aligned}\sum_{i=1}^{k}|\mathscr{F}_i| &\geqslant \sum_{i\in[k]\setminus B}\left(1-\varepsilon/2\right)\left(|S_i|-1\right) \geqslant \left(1-\varepsilon/2\right)\sum_{i=1}^{k}\left(|S_i|-1\right) - \sum_{i\in B}|S_i| \\ &\geqslant \left(1-\varepsilon/2-5C\lambda\right)\sum_{i=1}^{k}\left(|S_i|-1\right) \geqslant (1-\varepsilon)\sum_{i=1}^{k}\left(|S_i|-1\right),\end{aligned}$$

as required. □

6.5 Arithmetic frames, and the proof of the lower bound

In order to deduce Theorem 6.1.1 from Theorem 6.2.3, we will need to bound the number of δ-generalized frames in the integers. In this section we will warm up for the calculation ahead by counting a simpler set of objects, which we call 'arithmetic frames', and thereby deducing a lower bound on the number of minimal covering systems of $\mathbb{Z}$. Recall that

$$\tau = \sum_{t=1}^{\infty} \left(\log \frac{t+1}{t} \right)^2 \approx 0.977.$$

The following proposition provides the lower bound in Theorem 6.1.1.

Proposition 6.5.1. The number of minimal covering systems of $\mathbb{Z}$ of size n is at least

$$\exp\left(\left(\frac{4\sqrt{\tau}}{3} + o(1) \right) \frac{n^{3/2}}{(\log n)^{1/2}} \right)$$

as $n \to \infty$

To begin, let us define precisely the collection of covering systems that we will count. Given an integer $N = p_1^{\gamma_1} \cdots p_m^{\gamma_m} > 1$ (which will be the least common multiple of the moduli in our family $\mathscr{A}$), let us write

$$\langle N \rangle := \bigcup_{i=1}^{m} \big\{ (p_i, j) : j \in [\gamma_i] \big\},$$

and say that a total ordering $\prec$ on the pairs $(p_i, j) \in \langle N \rangle$ is *arithmetic* if

$$(p_i, 1) \prec (p_i, 2) \prec \cdots \prec (p_i, \gamma_i) \tag{6.15}$$

for all $i \in [m]$. Note that (6.15) does not impose any constraint on $\prec$ for different primes, and in particular we may have $(p,i) \prec (q,j) \prec (p,i+1)$. We begin by observing the following lower bound on the number of minimal covering systems of $\mathbb{Z}$ of size n.

Lemma 6.5.2. Let $N = p_1^{\gamma_1} \cdots p_m^{\gamma_m} > 1$, and let $\prec$ be an arithmetic ordering of $\langle N \rangle$. There are at least

$$\exp\left(\sum_{(p,e) \in \langle N \rangle} (p-1) \sum_{\substack{(q,f) \prec (p,e) \\ q \neq p}} \log\left(\frac{f+1}{f} \right) \right)$$

minimal covering systems of $\mathbb{Z}$ of size $n := \sum_{i=1}^{m} \gamma_i (p_i - 1) + 1$.

Proof. To prove the lemma we count simple frames in $\prod_{(p,e) \in \langle N \rangle} S_{(p,e)}$ centred at $(0, \ldots, 0)$ that correspond to covering systems, where $S_{(p,e)} = \{0, \ldots, p-1\}$ and the sets $S_{(p,e)}$ are listed in the order $\prec$. Recall from Definition 6.2.1 that, for each $(p,e) \in \langle N \rangle$ and each $s \in \{1, \ldots, p-1\}$, we need to choose a hyperplane of the form $\big[x_1, \ldots, x_{i-1}, s, *, \cdots, *\big]$, with $x_j \in \{0, *\}$ for each $j \in [i-1]$,

where (p,e) is the ith element in the ordering $\prec$. However, when we map $\prod_{(p,e)\in\langle N\rangle} S_{(p,e)}$ into $\mathbb{Z}_N \cong \mathbb{Z}_{p_1^{\gamma_1}} \times \cdots \times \mathbb{Z}_{p_m^{\gamma_m}}$, only some of these hyperplanes correspond to arithmetic progressions: those which correspond to cosets of subgroups of $\mathbb{Z}_N$. In particular, we may choose the map so that, for each prime $q \neq p$, if we choose any initial segment (in the order $\prec$) of

$$J_q(p,e) := \big\{(q,f) \in \langle N\rangle : (q,f) \prec (p,e)\big\}$$

and set $x_j = 0$ for the corresponding coordinates (and set all later coordinates equal to $*$), then we obtain a subgroup of $\mathbb{Z}_{q^\gamma}$. Hence, if we set $x_j = 0$ for every coordinate (p,f) with $f < e$, we obtain a hyperplane corresponding to a coset, as required. By the comments after Definition 6.2.1, each such choice of hyperplanes, together with $[0,\dots,0]$, gives a minimal covering system of $\prod_{(p,e)\in\langle N\rangle} S_{(p,e)}$, and hence of $\mathbb{Z}$, of size n. To conclude, note the number of choices for each hyperplane is

$$\prod_{q\neq p} \big(|J_q(p,e)|+1\big) = \exp\Bigg(\sum_{\substack{(q,f)\prec(p,e)\\ q\neq p}} \log\left(\frac{f+1}{f}\right)\Bigg).$$

□

Now, for each arithmetic ordering $\prec$ of $\langle N\rangle$, let us define

$$Q(N,\prec) := \sum_{(p,e)\in\langle N\rangle} (p-1) \sum_{\substack{(q,f)\prec(p,e)\\ q\neq p}} \log\left(\frac{f+1}{f}\right). \tag{6.16}$$

We will use the following particular arithmetic ordering $<$ to prove Proposition 6.5.1.

Definition 6.5.3. For each prime p and integer $e \in \mathbb{N}$, set $y_{p,e} := (p-1)\big(\log\frac{e+1}{e}\big)^{-1}$. Now, given primes p and q, and integers $e, f \in \mathbb{N}$, define

$$(q,f) < (p,e) \qquad \Leftrightarrow \qquad y_{q,f} < y_{p,e}.$$

Moreover, if $x \in \mathbb{R}$ then we write $(p,e) < x$ if and only if $y_{p,e} < x$, and define

$$n(x) := 1 + \sum_{(p,e)<x} (p-1) \qquad \text{and} \qquad N(x) := \prod_{(p,e)<x} p.$$

Note that $n(x), N(x) < \infty$ for every $x \in \mathbb{R}$, and that for any $N \in \mathbb{N}$, the ordering $<$ on $\langle N\rangle$ is arithmetic. Our next lemma, combined with Lemma 6.5.2, implies Proposition 6.5.1.

Lemma 6.5.4. Let $x > 0$, and set $N = N(x)$ and $n = n(x)$. Then

$$Q(N,<) = \left(\frac{4\sqrt{\tau}}{3} + o(1)\right) \frac{n^{3/2}}{(\log n)^{1/2}}$$

as $x \to \infty$.

Proof. Recalling the definition of $Q(N,<)$, observe first that, for each $(p,e) \in \langle N \rangle$,

$$\sum_{\substack{(q,f)<(p,e)\\ q\neq p}} \log\left(\frac{f+1}{f}\right) = \sum_{f\geqslant 1} \left|\left\{q \neq p : q-1 < y_{p,e}\log\left(\frac{f+1}{f}\right)\right\}\right| \cdot \log\left(\frac{f+1}{f}\right). \tag{6.17}$$

Now, by the prime number theorem, for each fixed $f \in \mathbb{N}$ and as $y_{p,e} \to \infty$,

$$\left|\left\{q \neq p : q-1 < y_{p,e}\log\left(\frac{f+1}{f}\right)\right\}\right| = \big(1+o(1)\big)\frac{y_{p,e}\log\left(\frac{f+1}{f}\right)}{\log\left(y_{p,e}\log\left(\frac{f+1}{f}\right)\right)}.$$

Moreover, the sum in (6.17) of the terms with $f \geqslant f_0$ is $o\big(y_{p,e}/\log y_{p,e}\big)$ as $f_0 \to \infty$, so

$$\sum_{\substack{(q,f)<(p,e)\\ q\neq p}} \log\left(\frac{f+1}{f}\right) = \frac{\big(1+o(1)\big)y_{p,e}}{\log y_{p,e}} \sum_{f\geqslant 1} \left(\log\frac{f+1}{f}\right)^2 = \big(\tau+o(1)\big)\frac{y_{p,e}}{\log y_{p,e}}$$

as $y_{p,e} \to \infty$. We next fix $e \in \mathbb{N}$, and sum over primes p. We obtain

$$\begin{aligned}\sum_{p-1<x\log\frac{e+1}{e}} (p-1) \sum_{\substack{(q,f)<(p,e)\\ q\neq p}} \log\left(\frac{f+1}{f}\right) &= \sum_{p-1<x\log\frac{e+1}{e}} \big(\tau+o(1)\big)\frac{(p-1)^2}{\log\left(\frac{e+1}{e}\right)\cdot\log(p-1)} \\ &= \big(\tau+o(1)\big)\frac{x^3\left(\log\frac{e+1}{e}\right)^2}{3(\log x)^2} \end{aligned} \tag{6.18}$$

as $x \to \infty$, again using the prime number theorem. Thus, summing over e, and noting that the left-hand side of (6.18) is uniformly bounded from above by an absolute constant times the right-hand side (without the $o(1)$ term), we obtain

$$Q(N,<) = \big(\tau+o(1)\big)\frac{x^3}{3(\log x)^2}\sum_{e\geqslant 1}\left(\log\frac{e+1}{e}\right)^2 = \big(\tau^2+o(1)\big)\frac{x^3}{3(\log x)^2}$$

as $x \to \infty$. Finally, observe that

$$\begin{aligned} n(x) &= 1+\sum_{e\geqslant 1}\sum_{p-1<x\log\frac{e+1}{e}} (p-1) = \big(1+o(1)\big)\sum_{e\geqslant 1}\frac{\left(x\log\frac{e+1}{e}\right)^2}{2\log\left(x\log\frac{e+1}{e}\right)} \\ &= \big(1+o(1)\big)\frac{x^2}{2\log x}\sum_{e\geqslant 1}\left(\log\frac{e+1}{e}\right)^2 = \big(\tau+o(1)\big)\frac{x^2}{2\log x}, \end{aligned} \tag{6.19}$$

and hence $Q(N,<)\cdot n^{-3/2}\sqrt{\log n} \to \sqrt{2}(\tau^2/3)(\tau/2)^{-3/2} = 4\sqrt{\tau}/3$ as $x \to \infty$, as claimed. □

We can now easily deduce the lower bound in Theorem 6.1.1, the only remaining difficulty being to deal with those $n \in \mathbb{N}$ that are not of the form $n = n(x)$ for some $x \in \mathbb{R}$.

Proof of Proposition 6.5.1. It follows immediately from Lemmas 6.5.2 and 6.5.4 that the number of minimal covering systems of $\mathbb{Z}$ of size $n(x)$ at least

$$e^{Q(N(x),<)} = \exp\left(\left(\frac{4\sqrt{\tau}}{3}+o(1)\right)\frac{n(x)^{3/2}}{(\log n(x))^{1/2}}\right)$$

as $x \to \infty$. Let $x > 0$ be maximal such that $n(x) \leqslant n$, and set $t := n - n(x)$. Observe that $t < x = o(n)$, by (6.19), and that, by removing the hyperplane $[0,\dots,0]$ (i.e., the progression $\{0 \pmod N\}$) from the construction given in the proof of Lemma 6.5.2, we obtain a family of minimal covers of $\mathbb{Z}_N \setminus \{0\}$ of size $n(x)-1$. We complete each to a minimal cover of $\mathbb{Z}$ of size n by adding the progressions $\{2^{\ell-1}N \pmod{2^\ell N}\}$, for each $\ell \in [t]$, and $\{0 \pmod{2^t N}\}$. We obtain a family of

$$\exp\left(\left(\frac{4\sqrt{\tau}}{3}+o(1)\right)\frac{n^{3/2}}{(\log n)^{1/2}}\right)$$

minimal covering systems of $\mathbb{Z}$ of size $n(x)+t = n$, as required. □

6.6 Counting coverings that are far from frames

In this section we will begin the deduction of Theorem 6.1.1 from Theorem 6.2.3 by bounding the number of minimal covers that fail to satisfy (6.3). In the process, we will obtain a short proof of weaker version of Theorem 6.1.1, bounding the number of minimal covering systems of $\mathbb{Z}$ of size n up to a constant factor in the exponent.

Proposition 6.6.1. Let $C > 0$ be a constant, and let $n \in \mathbb{N}$ and $N = p_1^{\gamma_1} \cdots p_m^{\gamma_m}$ satisfy

$$n > C\sum_{i=1}^{m} \gamma_i(p_i - 1).$$

Then the number of minimal covering systems $\mathscr{A}$ of $\mathbb{Z}$ of size n with $\operatorname{lcm}(\mathscr{A}) = N$ is at most

$$\exp\left(\left(\frac{2\sqrt{\tau}}{\sqrt{C}}+o(1)\right)\frac{n^{3/2}}{(\log n)^{1/2}}\right)$$

as $n \to \infty$.

In order to bound the number of covering systems, we will need to bound the number of choices for the modulus d and shift a of each arithmetic progression in $\mathscr{A}$. The following simple but important lemma, which we will use again later, shows that, given the moduli, we have relatively few choices for the shifts.

Lemma 6.6.2. Let $d_1,\dots,d_n \in \mathbb{N}$. There are at most $(n!)^2$ minimal covering systems $\mathscr{A} = \{A_1,\dots,A_n\}$ of $\mathbb{Z}$ of size n such that, for each $i \in [n]$, the modulus of A_i is d_i.

Proof. Let $\mathscr{A} = \{A_1,\dots,A_n\}$ be a minimal covering system of $\mathbb{Z}$ with progressions, and observe that we may reorder the elements of $\mathscr{A}$ so that, for each $i \in [n]$, the progression A_i covers at least a $1/i$

proportion of the set

$$R_i := \mathbb{Z} \setminus \bigcup_{j>i} A_j.$$

Indeed, to see that this is possible we simply choose the sets one by one (in reverse order), letting A_i be the (remaining) progression in $\mathscr{A}$ whose intersection with R_i has largest density, observing that R_i is non-empty (since $\mathscr{A}$ is minimal) and recalling that $\mathscr{A}$ covers $\mathbb{Z}$. The total number of choices for $\mathscr{A}$ is therefore at most the sum over permutations of $(d_1,\dots,d_n)$ of the number of sequences $(A_1,\dots,A_n)$ with this additional property.

Now let $i \in [n]$, and suppose that we have already chosen progressions $(A_{i+1},\dots,A_n)$. We claim that we have at most i choices for the arithmetic progression A_i. Indeed, since the progressions $\{a \pmod{d_i}\}$ (for $a \in \{0,\dots,d_i-1\}$) are disjoint, there are at most i progressions with modulus d_i that cover at least a $1/i$ proportion of R_i. It follows that the number of choices for $\mathscr{A}$ is at most $(n!)^2$, as claimed. □

It therefore only remains to bound the number of choices of the moduli. Note that if $\operatorname{lcm}(\mathscr{A}) = p_1^{\gamma_1}\cdots p_m^{\gamma_m}$ then we have at most $\prod_i (\gamma_i+1)$ choices for each modulus. The following lemma, which we will use again later, provides a sharp bound on this product.

Lemma 6.6.3. Let $p_1, p_2, \dots$ be a sequence of distinct primes, let $\gamma_1, \gamma_2, \dots$ be a sequence of non-negative integers, and let $M \geqslant \sum_{i\geqslant 1} \gamma_i(p_i-1)$. Then

$$\sum_{i\geqslant 1} \log(\gamma_i+1) \leqslant \big(2\sqrt{\tau}+o(1)\big)\left(\frac{M}{\log M}\right)^{1/2} \tag{6.20}$$

as $M \to \infty$.

Proof. We may assume that $p_1 < p_2 < \cdots$, and reorder the γ_i so that $\gamma_1 \geqslant \gamma_2 \geqslant \cdots$, noting that this does not change the left-hand side of (6.20), and that the inequality $M \geqslant \sum_{i\geqslant 1}\gamma_i(p_i-1)$ still holds under the new ordering. Set $x_t := \max\{i : \gamma_i \geqslant t\}$, and observe that maximizing the left-hand side of (6.20) is equivalent to maximizing

$$X := \sum_{t\geqslant 1} x_t \cdot \log\left(\frac{t+1}{t}\right)$$

subject to the constraint

$$M \geqslant \sum_{i\geqslant 1}\gamma_i(p_i-1) = \sum_{t\geqslant 1}\sum_{i=1}^{x_t}(p_i-1) \geqslant \sum_{t\geqslant 1}\frac{x_t^2}{2}\max\{\log x_t - 3, 1\}, \tag{6.21}$$

where in the final step we used the following bound of Massias and Robin [120],

$$\sum_{i=1}^{x} p_i \geqslant \frac{x^2}{2}\left(\log x + \log\log x - \frac{3}{2} - \frac{3.568}{\log x}\right) \geqslant \frac{x^2}{2}\big(\log x - 2\big),$$

which holds for every $x \geqslant e^3$. Note that X is increasing in x_t for each $t \geqslant 1$, and so (by allowing $0 \leqslant x_t \in \mathbb{R}$) we may assume that M is equal to the right-hand side of (6.21).

Applying the method of Lagrange multipliers, it follows that there exists $\lambda \in \mathbb{R}$ such that, for each $t \geqslant 1$, either

$$\lambda \log \frac{t+1}{t} = x_t \left(\log x_t - \frac{5}{2} \right), \tag{6.22}$$

or $x_t \leqslant e^4$. We will first show that the contribution to X of those values of t such that $x_t = O(1)$ is small. To do so, note that $x_t = 0$ for all $t > M$, by (6.21), and observe that therefore

$$\sum_{t \geqslant 1} x_t \cdot \log \left(\frac{t+1}{t} \right) \mathbb{1}\big[x_t \leqslant \log M\big] \leqslant \big(\log(M+1)\big)^2. \tag{6.23}$$

We may therefore restrict our attention to those values of t for which $x_t > \log M > e^4$, so that, in particular, (6.22) holds. Let $T = \max\big\{t : x_t > \log M\big\}$ and observe that, by (6.21), (6.22) and (6.23), we have

$$X \leqslant \lambda \sum_{t=1}^{T} \frac{\big(\log \frac{t+1}{t}\big)^2}{\log x_t - 5/2} + O\big(\log M\big)^2 \qquad \text{and} \qquad M \geqslant \frac{\lambda^2}{2} \sum_{t \geqslant 1} \frac{\big(\log \frac{t+1}{t}\big)^2}{\log x_t}.$$

To bound these sums, observe that $\lambda \to \infty$ as $M \to \infty$ (by (6.22), and since $x_1 > \log M$), and that for each fixed $t \in [T]$,

$$\log x_t = \log \lambda + \log \log \frac{t+1}{t} - \log\big(\log x_t - 5/2\big) = \big(1 + o(1)\big) \log \lambda$$

as $\lambda \to \infty$, and that therefore, for each fixed $c \in \mathbb{R}$, we have

$$\sum_{t \geqslant 1} \frac{\big(\log \frac{t+1}{t}\big)^2}{\log x_t - c} = \frac{1+o(1)}{\log \lambda} \sum_{t \geqslant 1} \left(\log \frac{t+1}{t} \right)^2 = \frac{\tau + o(1)}{\log \lambda},$$

as $\lambda \to \infty$. It follows that $M \geqslant \big(\tau/2 + o(1)\big)\lambda^2/\log \lambda$, and hence

$$X \leqslant \big(\tau + o(1)\big) \frac{\lambda}{\log \lambda} \leqslant \big(2\sqrt{\tau} + o(1)\big) \left(\frac{M}{\log M} \right)^{1/2}$$

as $M \to \infty$, as required. □

We can now easily deduce Proposition 6.6.1.

Proof of Proposition 6.6.1. We first choose the moduli of the progressions in $\mathscr{A} = \{A_1, \dots, A_n\}$, and then the shifts. Since $\operatorname{lcm}(\mathscr{A}) = N = p_1^{\gamma_1} \cdots p_m^{\gamma_m}$, for each $j \in [n]$ we have at most

$$\prod_{i=1}^{m} (\gamma_i + 1) \leqslant \exp\left(\big(2\sqrt{\tau} + o(1)\big) \left(\frac{n}{C \log(n/C)} \right)^{1/2} \right)$$

choices for the modulus of the arithmetic progression A_j, where the inequality follows by applying Lemma 6.6.3 with $M = n/C$, and using our bound on n. By Lemma 6.6.2, it follows that the number

of choices for $\mathscr{A}$ is at most

$$(n!)^2 \cdot \exp\left(\left(\frac{2\sqrt{\tau}}{\sqrt{C}} + o(1)\right) \frac{n^{3/2}}{\left(\log(n/C)\right)^{1/2}}\right) = \exp\left(\left(\frac{2\sqrt{\tau}}{\sqrt{C}} + o(1)\right) \frac{n^{3/2}}{(\log n)^{1/2}}\right)$$

as $n \to \infty$, as required. □

Using Simpson's theorem (Theorem 6.2.4), we can now easily deduce an upper bound on the number of minimal covering systems that is sharp up to a constant factor in the exponent.

Corollary 6.6.4. The number of minimal covering systems of $\mathbb{Z}$ of size n is

$$\exp\left(\frac{\Theta\left(n^{3/2}\right)}{(\log n)^{1/2}}\right).$$

Proof. The lower bound follows by Proposition 6.5.1 (or by the simpler construction in the introduction). For the upper bound, recall that, by Simpson's theorem, we have

$$|\mathscr{A}| \geqslant \sum_{i=1}^{m} \gamma_i\left(p_i - 1\right) + 1 \geqslant \sum_{i=1}^{m} \gamma_i \log_2 p_i$$

for any minimal covering system $\mathscr{A}$ of $\mathbb{Z}$ with $\operatorname{lcm}(\mathscr{A}) = N = p_1^{\gamma_1} \cdots p_m^{\gamma_m}$, and hence $N \leqslant 2^n$. Thus, applying Proposition 6.6.1 with $C = 1$ (and summing over $N \leqslant 2^n$), there are at most

$$\exp\left(\left(2\sqrt{\tau} + o(1)\right) \frac{n^{3/2}}{(\log n)^{1/2}}\right)$$

minimal covering systems of $\mathbb{Z}$ of size n, as required. □

6.7 Proof of Theorem 6.1.1

In this section we will complete the proof of Theorem 6.1.1; we will begin by giving a sketch of the remaining part of the argument. Let $\mathscr{A}$ be a minimal covering system of $\mathbb{Z}$ of size n, let $N = \operatorname{lcm}(\mathscr{A})$, and set $S_{(p,e)} = \{0, \dots, p-1\}$ for each $(p,e) \in \langle N \rangle$. Let us map $\mathbb{Z}_N$ into $S_{\langle N \rangle} = \prod_{(p,e) \in \langle N \rangle} S_{(p,e)}$ in the natural way, i.e., using the Chinese Remainder Theorem, we map $x \in \mathbb{Z}_N$ to a vector $y \in S_{\langle N \rangle}$ where $y_{(p,e)}$ is the coefficient of p^{e-1} in the p-ary expansion of x modulo p^e. Observe that the image of each progression $H \in \mathscr{A}$ is a hyperplane in $S_{\langle N \rangle}$, and moreover, for each prime p, the set

$$\left\{(p,e) \in \langle N \rangle : (p,e) \in F(H)\right\}$$

forms a (possibly empty) initial segment of the sequence $(p,1), (p,2), \dots$. We will say that the hyperplane H is *arithmetic* if it satisfies this condition.

We will apply Theorem 6.2.3 to $\mathscr{A}$ (with $C = 4$ and $\varepsilon > 0$ an arbitrarily small constant), and deduce that either (6.3) fails to hold, *or* $\mathscr{A}$ contains an almost optimal δ-generalized frame $\left(\mathscr{F}_{(p,e)}\right.$:

$(p,e) \in \langle N \rangle\big)$. In the former case we are done by Proposition 6.6.1, so let us assume the latter. We will carefully count the number of choices for the fixed sets of the frame elements $\mathscr{F}_{(p,e)}$ such that $p > \delta^{-1}$. The bound we obtain will be sufficiently strong unless N is primarily composed of primes smaller than δ^{-1}; however, for such N it turns out that the simpler argument used in Section 6.6 suffices to give a sufficiently strong bound.

Next, we bound the number of choices for the fixed sets of the remaining hyperplanes: those in frame sets $\mathscr{F}_{(p,e)}$ for some prime $p \leqslant \delta^{-1}$, and those not used in the frame. Surprisingly, it turns out that we can again obtain a sufficiently strong bound using the method of Section 6.6. Roughly speaking, these 'extra' hyperplanes are being used inefficiently, and would be better off (in terms of increasing the number of choices) by contributing to the construction of a larger frame (and thus a different value of N).

Finally, noting that the fixed sets of the hyperplanes in $\mathscr{A}$ correspond to the moduli of the original arithmetic progressions, we will use Lemma 6.6.2 to bound the number of minimal covering systems of $\mathbb{Z}$ of size n with given moduli.

6.7.1 Choosing the fixed sets of δ-generalized frames

Let $N \in \mathbb{N}$ and $\delta > 0$, and suppose that $\big(\mathscr{F}_{(p,e)} : (p,e) \in \langle N \rangle\big)$ is a δ-generalized frame in $S_{\langle N \rangle}$ consisting of arithmetic hyperplanes. Recall that $\mathscr{F}_{(p,e)}$ is a collection of at most $p-1$ hyperplanes, and that there exists an ordering $\prec$ on $\langle N \rangle$, and for each $(p,e) \in \langle N \rangle$ a set

$$I(p,e) \supseteq \big\{(q,f) \in \langle N \rangle : (p,e) \prec (q,f) \text{ and } (q,f) \neq (p,e)\big\}, \tag{6.24}$$

such that $\mu_{I(p,e)}(H) > \delta$ for each $(p,e) \in \langle N \rangle$ and $H \in \mathscr{F}_{(p,e)}$. Recall also that $(p,e) \in F(H)$, and that the sets $\mathscr{F}_{(p,e)}$ with $p > \delta^{-1}$ are disjoint. We remark that the ordering $\prec$ might not be arithmetic, but the hyperplanes *are* arithmetic, and this will turn out to be sufficient.

In this subsection we will bound the number of choices for the fixed sets of the hyperplanes in $\big(\mathscr{F}_{(p,e)} : (p,e) \in \langle N \rangle\big)$ corresponding to primes larger than δ^{-1}. While doing so, it will be convenient to write $\langle N \rangle_\delta := \big\{(p,e) \in \langle N \rangle : p > \delta^{-1}\big\}$, and define

$$\Gamma(N) := \sum_{(p,e) \in \langle N \rangle} (p-1) \qquad \text{and} \qquad \Gamma_\delta(N) := \sum_{(p,e) \in \langle N \rangle_\delta} (p-1).$$

Given a collection $\mathscr{A}$ of hyperplanes, let us write $\mathscr{D}(\mathscr{A}) := \big(F(H) : H \in \mathscr{A}\big)$ for the corresponding collection of fixed sets. We begin by observing the following upper bound (cf. Lemma 6.5.2) on the number of choices for the sequence $\big(\mathscr{D}(\mathscr{F}_{(p,e)}) : (p,e) \in \langle N \rangle_\delta\big)$.

Lemma 6.7.1. Let $N \in \mathbb{N}$ and $\delta > 0$, and let $\prec$ be an ordering on $\langle N \rangle$. There are at most

$$\exp\left(\sum_{(p,e) \in \langle N \rangle_\delta} (p-1)\left(\sum_{(q,f) \prec (p,e)} \log\left(\frac{f+1}{f}\right) + \frac{1}{\delta^2}\right)\right)$$

sequences $\big(\mathscr{D}(\mathscr{F}_{(p,e)}) : (p,e) \in \langle N\rangle_\delta\big)$ such that $\big(\mathscr{F}_{(p,e)} : (p,e) \in \langle N\rangle\big)$ is a δ-generalized frame in $S_{\langle N\rangle}$ with ordering $\prec$ and consisting only of arithmetic hyperplanes.

Proof. Let $(p,e) \in \langle N\rangle_\delta$, let $H \in \mathscr{F}_{(p,e)}$, and let q be a prime with $q > \delta^{-1}$. Note that if $(p,e) \prec (q,f)$ and $(p,e) \neq (q,f)$ then $(q,f) \notin F(H)$, by (6.24) and since $\mu_{I(p,e)}(H) > \delta$. Since H is arithmetic, it follows that $F(H) \cap J_q(p,e)$ must be an initial segment (in increasing order of f) of the set $J_q(p,e) = \big\{(q,f) \in \langle N\rangle : (q,f) \prec (p,e)\big\}$, and there are at most

$$\big|\big\{(q,f) \in \langle N\rangle : (q,f) \prec (p,e)\big\}\big| + 1 = \exp\left(\sum_{f:(q,f)\prec(p,e)} \log\left(\frac{f+1}{f}\right)\right)$$

choices for this initial segment. (Note that for $q = p$ this is an overcount.)

Next, let q be a prime with $q \leqslant \delta^{-1}$, and observe that

$$\big|\big\{(q,f) \in F(H) : (p,e) \prec (q,f)\big\}\big| \leqslant |F(H) \cap I(p,e)| < \log_2(\delta^{-1}),$$

by (6.24) and the fact that $\mu_{I(p,e)}(H) > \delta$. Since H is arithmetic, $F(H)$ must induce an initial segment of the set $(q,1),(q,2),\ldots$ containing at most $\log_2(\delta^{-1})$ elements with $(p,e) \prec (q,f)$, and there are at most

$$\big|\big\{(q,f) \in \langle N\rangle : (q,f) \prec (p,e)\big\}\big| + \log_2(\delta^{-1}) + 1 \leqslant \exp\left(\sum_{f:(q,f)\prec(p,e)} \log\left(\frac{f+1}{f}\right) + \frac{1}{\delta}\right)$$

choices for this initial segment.

Finally, recall that there are at most $p-1$ hyperplanes in $\mathscr{F}_{(p,e)}$ for each $(p,e) \in \langle N\rangle_\delta$, and at most δ^{-1} primes $q \leqslant \delta^{-1}$. Hence, summing over $(p,e) \in \langle N\rangle_\delta$ and all primes q that divide N, it follows that we have at most

$$\exp\left(\sum_{(p,e)\in\langle N\rangle_\delta} (p-1)\left(\sum_{(q,f)\prec(p,e)} \log\left(\frac{f+1}{f}\right) + \frac{1}{\delta^2}\right)\right)$$

choices for the sequence $\big(\mathscr{D}(\mathscr{F}_{(p,e)}) : (p,e) \in \langle N\rangle_\delta\big)$, as claimed. □

For each $N \in \mathbb{N}$ and $\delta > 0$, and each ordering $\prec$ on $\langle N\rangle$, let us define

$$Q_\delta(N,\prec) := \sum_{(p,e)\in\langle N\rangle_\delta} (p-1) \sum_{(q,f)\prec(p,e)} \log\left(\frac{f+1}{f}\right).$$

The following lemma provides a sufficiently strong upper bound on $Q_\delta(N,\prec)$.

Lemma 6.7.2. Fix $\delta > 0$, let $N \in \mathbb{N}$, and let $\prec$ be an ordering on $\langle N \rangle$. If $\Gamma_\delta(N) > \delta \cdot \Gamma(N)$, then

$$Q_\delta(N, \prec) \leqslant \left(\frac{4\sqrt{\tau}}{3} + o(1) \right) \frac{\Gamma_\delta(N)^{3/2}}{\big(\log \Gamma_\delta(N)\big)^{1/2}} \tag{6.25}$$

as $N \to \infty$.

We first use Lemma 6.6.3 to obtain the following bound.

Lemma 6.7.3. Let $M \in \mathbb{N}$ and $\delta > 0$. Then

$$\sum_{(q,f) \in \langle M \rangle} \log\left(\frac{f+1}{f} \right) \leqslant \big(2\sqrt{\tau} + o(1)\big) \left(\frac{\Gamma_\delta(M)}{\log \Gamma_\delta(M)} \right)^{1/2} + \frac{1}{\delta} \log \Gamma(M)$$

as $\Gamma_\delta(M) \to \infty$.

Proof. Let $M = p_1^{\gamma_1} \cdots p_m^{\gamma_m}$ with $\Gamma_\delta(M) > 0$, and observe that

$$\begin{aligned} \sum_{(q,f) \in \langle M \rangle} \log\left(\frac{f+1}{f} \right) &= \sum_{i=1}^{m} \log\big(\gamma_i + 1\big) = \sum_{i \colon p_i > \delta^{-1}} \log\big(\gamma_i + 1\big) + \sum_{i \colon p_i \leqslant \delta^{-1}} \log\big(\gamma_i + 1\big) \\ &\leqslant \big(2\sqrt{\tau} + o(1)\big) \left(\frac{\Gamma_\delta(M)}{\log \Gamma_\delta(M)} \right)^{1/2} + \frac{1}{\delta} \log \Gamma(M) \end{aligned}$$

as $\Gamma_\delta(M) \to \infty$, as required, by Lemma 6.6.3 applied to the sequence $\gamma_i \cdot \mathbb{1}[p_i > \delta^{-1}]$, and by the bound $\gamma_i + 1 \leqslant \Gamma(M)$, which holds for every $i \in [m]$, since $\Gamma_\delta(M) > 0$. □

We will also need the following easy lemma.

Lemma 6.7.4. Let $e < m_0 < m_1 < \cdots < m_\ell \leqslant m$. Then

$$\sum_{i=0}^{\ell-1} \left(\frac{m_i}{\log m_i} \right)^{1/2} \big(m_{i+1} - m_i\big) \leqslant \left(\frac{2}{3} + o(1) \right) \frac{m^{3/2}}{(\log m)^{1/2}}$$

as $m \to \infty$.

We can now prove Lemma 6.7.2.

Proof of Lemma 6.7.2. For each $(p,e) \in \langle N \rangle_\delta$, define

$$M(p,e) := \frac{1}{p} \prod_{(q,f) \prec (p,e)} q,$$

and note that $\langle M(p,e) \rangle = \big\{ (q,f) \in \langle N \rangle : (q,f) \prec (p,e),\, (q,f) \neq (p,e) \big\}$. Thus, applying Lemma 6.7.3 with $M := M(p,e)$, we obtain

$$\sum_{(q,f) \prec (p,e)} \log\left(\frac{f+1}{f} \right) \leqslant \big(2\sqrt{\tau} + o(1)\big) \left(\frac{\Gamma_\delta\big(M(p,e)\big)}{\log \Gamma_\delta\big(M(p,e)\big)} \right)^{1/2} + \frac{1}{\delta} \log \Gamma\big(M(p,e)\big),$$

as $\Gamma_\delta(M(p,e)) \to \infty$, where we absorbed the term $\log \frac{e+1}{e}$ corresponding to $(q,f) = (p,e)$ into the error term. Summing over $(p,e) \in \langle N \rangle_\delta$, and noting that $\Gamma(M(p,e)) \leqslant \Gamma(N)$, it follows that

$$Q_\delta(N,\prec) \leqslant \sum_{(p,e)\in\langle N\rangle_\delta} (p-1)\left(\big(2\sqrt{\tau}+o(1)\big)\left(\frac{\Gamma_\delta\big(M(p,e)\big)}{\log\Gamma_\delta\big(M(p,e)\big)}\right)^{1/2} + \frac{1}{\delta}\log\Gamma(N)\right),$$

where the $o(1)$ terms depend on (p,e), each tending to zero as $\Gamma_\delta(M(p,e)) \to \infty$.

We will use Lemma 6.7.4 to bound the sum on the right. To do so, observe first that if (p',e') immediately follows (p,e) in the ordering $\prec$ restricted to $\langle N \rangle_\delta$, then

$$\Gamma_\delta\big(M(p',e')\big) - \Gamma_\delta\big(M(p,e)\big) = p-1,$$

since Γ_δ only counts the large primes. Recall also that $\Gamma_\delta(N) > \delta \cdot \Gamma(N)$, so $\Gamma_\delta(N) \to \infty$ as $N \to \infty$. Therefore, applying Lemma 6.7.4 with $(m_0,\dots,m_\ell) = \big(\Gamma_\delta\big(M(p,e)\big)\big)_{(p,e)\in\langle N\rangle_\delta}$ and $m = \Gamma_\delta(N)$, we obtain

$$Q_\delta(N,\prec) \leqslant \left(\frac{4\sqrt{\tau}}{3}+o(1)\right)\frac{\Gamma_\delta(N)^{3/2}}{\big(\log\Gamma_\delta(N)\big)^{1/2}} + \frac{\Gamma_\delta(N)\log\Gamma(N)}{\delta},$$

as $N \to \infty$. Recalling again that $\Gamma_\delta(N) > \delta \cdot \Gamma(N)$, we obtain (6.25), as required. □

Before continuing, let us observe that the condition $\Gamma_\delta(N) > \delta \cdot \Gamma(N)$ in the statement of Lemma 6.7.2 (which in any case could be weakened considerably) is not a serious restriction, since we can easily obtain, using the method of Section 6.6, a suitable bound on the number of minimal covering systems whose least common multiple has mostly small prime factors.

Lemma 6.7.5. Let $\beta,\delta > 0$ be constants and let $n \in \mathbb{N}$ and $N \in \mathbb{N}$ with $\Gamma_\delta(N) \leqslant \beta \cdot \Gamma(N)$. The number of minimal covering systems $\mathscr{A}$ of $\mathbb{Z}$ of size n with $\operatorname{lcm}(\mathscr{A}) = N$ is at most

$$\exp\left(\big(2\sqrt{\beta\tau}+o(1)\big)\frac{n^{3/2}}{(\log n)^{1/2}}\right)$$

as $n \to \infty$.

Proof. The proof is the same as that of Proposition 6.6.1, but we use Lemma 6.7.3 in place of Lemma 6.6.3 to count the choices of the moduli. To be precise, if $\mathscr{A} = \{A_1,\dots,A_n\}$ is to be a minimal covering system of $\mathbb{Z}$ of size n with $\operatorname{lcm}(\mathscr{A}) = N$, then for each $j \in [n]$ we have at most

$$\prod_{(p,e)\in\langle N\rangle}\left(\frac{e+1}{e}\right) \leqslant \exp\left(\big(2\sqrt{\tau}+o(1)\big)\left(\frac{\Gamma_\delta(N)}{\log\Gamma_\delta(N)}\right)^{1/2} + \frac{1}{\delta}\log\Gamma(N)\right)$$

choices for the modulus of the arithmetic progression A_j, by Lemma 6.7.3. Now, since (by assumption and by Simpson's theorem) we have $\Gamma_\delta(N) \leqslant \beta \cdot \Gamma(N) \leqslant \beta n$, it follows by Lemma 6.6.2 that the

number of choices for $\mathscr{A}$ is at most

$$(n!)^2 \cdot \exp\left(\left(2\sqrt{\beta\tau}+o(1)\right)\frac{n^{3/2}}{(\log \beta n)^{1/2}}+\frac{n\log n}{\delta}\right) = \exp\left(\left(2\sqrt{\beta\tau}+o(1)\right)\frac{n^{3/2}}{(\log n)^{1/2}}\right)$$

as $n \to \infty$, as claimed. □

6.7.2 Proof of Theorem 6.1.1

We are finally ready to put together the pieces and deduce our main counting result. We will need the following easy bound.

Lemma 6.7.6. For every $m \in \mathbb{N}$ and $x \geqslant 0$, we have

$$\frac{(m+x)^{3/2}}{(\log(m+x))^{1/2}} \geqslant \frac{m^{3/2}}{(\log m)^{1/2}} + \left(\frac{3}{2}+o(1)\right)\left(\frac{m}{\log m}\right)^{1/2} x,$$

as $m \to \infty$.

We can now deduce Theorem 6.1.1 from Theorem 6.2.3, Propositions 6.5.1 and 6.6.1, Lemma 6.6.2, Simpson's theorem, and the results of this section.

Proof of Theorem 6.1.1. The lower bound follows immediately from Proposition 6.5.1, so we will prove the upper bound. Observe first (cf. Section 6.6) that, by Simpson's theorem, if $\mathscr{A}$ is a minimal covering system of $\mathbb{Z}$ of size n, then $\operatorname{lcm}(\mathscr{A}) \leqslant 2^n$. We may therefore fix $N \leqslant 2^n$, and consider only covering systems $\mathscr{A}$ such that $\operatorname{lcm}(\mathscr{A}) = N$. We will consider each progression $H \in \mathscr{A}$ as an arithmetic hyperplane in $S_{\langle N\rangle}$, as described above.

Let $\varepsilon > 0$ be an arbitrarily small constant, set $C = 4$, and let $\delta = \delta(C,\varepsilon) > 0$ be the constant given by Theorem 6.2.3. Suppose first that either $n > 4\Gamma(N)$ or $\Gamma(N) > 4\Gamma_\delta(N)$. Then, by Proposition 6.6.1 and Lemma 6.7.5, there are at most

$$\exp\left(\left(\sqrt{\tau}+o(1)\right)\frac{n^{3/2}}{(\log n)^{1/2}}\right)$$

minimal covering systems $\mathscr{A}$ of $\mathbb{Z}$ of size n with $\operatorname{lcm}(\mathscr{A}) = N$, as required. By Simpson's theorem, let us therefore assume from now on that $\Gamma(N) \leqslant n \leqslant 4\Gamma(N) \leqslant 2^4\Gamma_\delta(N)$.

By Theorem 6.2.3 (and our choice of δ), every minimal covering system $\mathscr{A}$ of $\mathbb{Z}$ of size $n \leqslant 4\Gamma(N)$ with $\operatorname{lcm}(\mathscr{A}) = N$ contains a δ-generalized frame $\left(\mathscr{F}_{(p,e)} : (p,e) \in \langle N\rangle\right)$, with

$$\sum_{(p,e)\in\langle N\rangle} |\mathscr{F}_{(p,e)}| \geqslant (1-\varepsilon)\Gamma(N). \tag{6.26}$$

Since $\Gamma(N) \leqslant 4\Gamma_\delta(N)$, it follows by Lemmas 6.7.1 and 6.7.2 that the number of sequences $\big(\mathscr{D}(\mathscr{F}_{(p,e)}) : (p,e) \in \langle N\rangle_\delta\big)$ such that $\big(\mathscr{F}_{(p,e)} : (p,e) \in \langle N\rangle\big)$ is a δ-generalized frame in $S_{\langle N\rangle}$ is at most

$$\Gamma(N)! \cdot \exp\left(\left(\frac{4\sqrt{\tau}}{3} + o(1)\right) \frac{\Gamma_\delta(N)^{3/2}}{\big(\log \Gamma_\delta(N)\big)^{1/2}} + \frac{\Gamma_\delta(N)}{\delta^2}\right).$$

Recall (from Definition 6.2.2) that the sets $\mathscr{F}_{(p,e)}$ with $(p,e) \in \langle N\rangle_\delta$ are pairwise disjoint, so there are exactly

$$x := n - \sum_{(p,e)\in\langle N\rangle_\delta} |\mathscr{F}_{(p,e)}|,$$

hyperplanes in $\mathscr{A}$ that are not included in $\mathscr{F}_{(p,e)}$ for any $(p,e) \in \langle N\rangle_\delta$. We bound the number of choices for the fixed sets of these remaining hyperplanes in $\mathscr{A}$ using Lemma 6.7.3, which implies that we have at most

$$\exp\left(\big(2\sqrt{\tau} + o(1)\big)\left(\frac{\Gamma_\delta(N)}{\log \Gamma_\delta(N)}\right)^{1/2} + \frac{1}{\delta}\log\Gamma(N)\right)$$

choices for each. Recalling that $\Gamma(N) \leqslant n \leqslant 2^4\Gamma_\delta(N)$, it follows that we have at most

$$\exp\left(\left(\frac{4\sqrt{\tau}}{3} + o(1)\right) \frac{\Gamma_\delta(N)^{3/2}}{\big(\log \Gamma_\delta(N)\big)^{1/2}} + \big(2\sqrt{\tau} + o(1)\big)\left(\frac{\Gamma_\delta(N)}{\log \Gamma_\delta(N)}\right)^{1/2} x\right)$$

choices for the fixed sets in $\mathscr{A}$, given N and x. Applying Lemma 6.7.6 with $m = \Gamma_\delta(N)$, and noting that $0 \leqslant x \leqslant n$, and that $m \to \infty$ as $n \to \infty$, we obtain

$$\frac{\big(\Gamma_\delta(N)+x\big)^{3/2}}{\big(\log(\Gamma_\delta(N)+x)\big)^{1/2}} \geqslant \frac{\Gamma_\delta(N)^{3/2}}{\big(\log \Gamma_\delta(N)\big)^{1/2}} + \frac{3}{2}\left(\frac{\Gamma_\delta(N)}{\log \Gamma_\delta(N)}\right)^{1/2} x + \frac{o(n^{3/2})}{(\log n)^{1/2}}$$

as $n \to \infty$. Now, since $|\mathscr{F}_{(p,e)}| \leqslant p-1$ for each $(p,e) \in \langle N\rangle$, it follows from (6.26) that

$$n - x = \sum_{(p,e)\in\langle N\rangle_\delta} |\mathscr{F}_{(p,e)}| \geqslant \Gamma_\delta(N) - \varepsilon\Gamma(N) \geqslant \Gamma_\delta(N) - \varepsilon n,$$

so $\Gamma_\delta(N) + x \leqslant (1+\varepsilon)n$, and hence

$$\frac{\big(\Gamma_\delta(N)+x\big)^{3/2}}{\big(\log(\Gamma_\delta(N)+x)\big)^{1/2}} \leqslant (1+\varepsilon)^{3/2}\frac{n^{3/2}}{(\log n)^{1/2}}.$$

It follows that we have at most

$$\exp\left(\left(\frac{4\sqrt{\tau}}{3} + o(1)\right)(1+\varepsilon)^{3/2}\frac{n^{3/2}}{(\log n)^{1/2}}\right)$$

choices for the fixed sets of the hyperplanes in $\mathscr{A}$, and hence the same number of choices for the moduli of the progressions in $\mathscr{A}$. Hence, by Lemma 6.6.2, there are at most

$$\exp\left(\left(\frac{4\sqrt{\tau}}{3}+O(\varepsilon)\right)\frac{n^{3/2}}{(\log n)^{1/2}}\right)$$

minimal covering systems of $\mathbb{Z}$ of size n with $\operatorname{lcm}(\mathscr{A})=N$. Since $\varepsilon>0$ was arbitrarily small, this completes the proof of Theorem 6.1.1. □

6.8 Proof of the geometric Simpson's theorem

In this section we will provide, for the reader's convenience, a proof of the following slight generalization of Simpson's theorem [140].

Theorem 6.8.1 (Simpson's theorem)**.** Let $\mathscr{A}$ be a minimal cover of $S_{[k]}$ with hyperplanes, and let $I \subsetneq F(\mathscr{A})$. Then

$$\left|\left\{H\in\mathscr{A} : F(H)\not\subseteq I\right\}\right| \geqslant \sum_{i\in F(\mathscr{A})\setminus I}\left(|S_i|-1\right)+1.$$

Note that Theorem 6.2.4 follows from Theorem 6.8.1 by setting $I=\emptyset$.

Proof. The proof is by induction on $|F(\mathscr{A})|$. Set

$$L_s^{(i)} := S_1\times\cdots S_{i-1}\times\{s\}\times S_{i+1}\times\cdots S_k$$

for each $i\in[k]$ and $s\in S_i$, and note that if $F(\mathscr{A})=\{i\}$, then $\mathscr{A}$ (being minimal) must consist precisely of the $|S_i|$ hyperplanes $L_s^{(i)}$, one for each $s\in S_i$. Moreover, if $I\subsetneq F(\mathscr{A})$ then $I=\emptyset$, and hence

$$\left|\left\{H\in\mathscr{A} : F(H)\not\subseteq I\right\}\right| = |\mathscr{A}| = |S_i|,$$

as required. So suppose that $|F(\mathscr{A})|\geqslant 2$, and (recalling that $I\subsetneq F(\mathscr{A})$) choose an element $i\in F(\mathscr{A})\setminus I$. For each $s\in S_i$, let $\mathscr{A}_s\subseteq\mathscr{A}$ be a minimal cover of $L_s^{(i)}$, and observe that

$$\mathscr{A}=\bigcup_{s\in S_i}\mathscr{A}_s,$$

since $\mathscr{A}$ is minimal. For convenience, let us assume (without loss) that $S_i=\{1,\dots,p\}$.

Now, set $F_i(H):=F(H)\setminus\{i\}$ for each $H\in\mathscr{A}_s$ (and similarly for a family of hyperplanes), and define a sequence of sets $(R_0,\dots,R_p)$ by setting $R_0:=I$, and

$$R_s := R_{s-1}\cup F_i(\mathscr{A}_s)$$

for each $s \in S_i$, so in particular $R_p = F(\mathscr{A}) \setminus \{i\}$. Now set $I_s := R_{s-1} \cap F_i(\mathscr{A}_s)$, and define

$$\mathscr{Q}_s := \big\{H \in \mathscr{A}_s : F_i(H) \not\subseteq I_s\big\}.$$

We claim that, applying the induction hypothesis to the minimal cover $\mathscr{A}_s$ of $L_s^{(i)}$ (which we naturally identify with $S_1 \times \cdots S_{i-1} \times S_{i+1} \times \cdots S_k$), we have either $R_{s-1} = R_s$, or

$$|\mathscr{Q}_s| \geqslant \sum_{j \in R_s \setminus R_{s-1}} \big(|S_j| - 1\big) + 1 \tag{6.27}$$

for each $s \in S_i$. To see this, simply note that $R_s \setminus R_{s-1} = F_i(\mathscr{A}_s) \setminus I_s$, so if $R_{s-1} \neq R_s$ then $I_s \subsetneq F_i(\mathscr{A}_s)$, and that $F_i(\mathscr{A}_s) \subseteq F(\mathscr{A}) \setminus \{i\}$, so (since $i \in F(\mathscr{A})$) we have $|F_i(\mathscr{A}_s)| < |F(\mathscr{A})|$.

Set $J := \big\{s \in S_i : R_{s-1} \neq R_s\big\}$, and recall that (6.27) holds for each $s \in J$. We claim that

$$\left|\bigcup_{s \in J} \mathscr{Q}_s\right| = \sum_{s \in J} |\mathscr{Q}_s| \geqslant |J| - \big(|S_i| - 1\big) + \sum_{j \in F(\mathscr{A}) \setminus I} \big(|S_j| - 1\big).$$

The inequality follows from summing (6.27) over $s \in J$, and recalling that $i \in F(\mathscr{A}) \setminus I$, so it remains to show that the sets $\mathscr{Q}_s$ are disjoint. To see this, observe that, if $H \in \mathscr{A}_s$, then

$$F_i(H) \not\subseteq I_s \qquad \Leftrightarrow \qquad F_i(H) \subseteq R_s \quad \text{and} \quad F_i(H) \not\subseteq R_{s-1},$$

and so $H \in A_s$ for at most one element $s \in S_i$.

Finally, we claim that for each $s \in S_i \setminus J$, there exists a hyperplane $H \in \mathscr{A}$ such that $H \subseteq L_s^{(i)}$ and $F(H) \not\subseteq I$. To see this, observe first that $L_s^{(i)}$ is not covered by $\big\{H \in \mathscr{A} : i \notin F(H)\big\}$, as otherwise $S_{[k]}$ would be covered by $\big\{H \in \mathscr{A} : i \notin F(H)\big\}$, contradicting the minimality of $\mathscr{A}$ and the fact that $i \in F(\mathscr{A})$. It follows that there exists $H \in \mathscr{A}$ with $H \subseteq L_s^{(i)}$, and we have $F(H) \not\subseteq I$ because $i \in F(H) \setminus I$. Moreover, none of these $|S_i| - |J|$ distinct hyperplanes is included in $\mathscr{Q}_s$ for any $s \in J$, since they do not intersect the set $\bigcup_{s \in J} L_s^{(i)}$.

Hence, noting that $F(H) \not\subseteq I$ for each $H \in \mathscr{Q}_s$ (since $I \subseteq R_{s-1}$ and $F(H) \not\subseteq R_{s-1}$), we obtain

$$\big|\big\{H \in \mathscr{A} : F(H) \not\subseteq I\big\}\big| \geqslant \left|\bigcup_{s \in J} \mathscr{Q}_s\right| + |S_i| - |J| \geqslant \sum_{j \in F(\mathscr{A}) \setminus I} \big(|S_j| - 1\big) + 1,$$

as required. □

Chapter 7

Flat Littlewood Polynomials Exist

The work in this section is joint with Paul Balister, Béla Bollobás, Robert Morris and Julian Sahasrabudhe and is published in Annals of Mathematics [4].

7.1 Introduction

We say that a polynomial $P(z)$ of degree n is a *Littlewood polynomial* if

$$P(z) = \sum_{k=0}^{n} \varepsilon_k z^k,$$

where $\varepsilon_k \in \{-1, 1\}$ for all $0 \leqslant k \leqslant n$. The aim of this chapter is to prove the following theorem, which answers a question of Erdős [63, Problem 26] from 1957, and confirms a conjecture of Littlewood [112] from 1966.

Theorem 7.1.1. There exist constants $\Delta > \delta > 0$ such that, for all $n \geqslant 2$, there exists a Littlewood polynomial $P(z)$ of degree n with

$$\delta\sqrt{n} \leqslant |P(z)| \leqslant \Delta\sqrt{n} \tag{7.1}$$

for all $z \in \mathbb{C}$ with $|z| = 1$.

Polynomials satisfying (7.1) are known as *flat polynomials*, and Theorem 7.1.1 can therefore be more succinctly stated as follows: "flat Littlewood polynomials exist". It turns out that our main challenge will be to prove the lower bound on $|P(z)|$; indeed, explicit polynomials satisfying the upper bound in (7.1) have been known to exist since the work of Shapiro [138] and Rudin [128] over 60 years ago (see Section 7.3). In the 1980s a completely different (and non-constructive) proof of the upper bound was given by Spencer [141], who used a technique that had been developed a few years earlier by Beck [12] in his study of combinatorial discrepancy. We remark that the Rudin–Shapiro construction, and also ideas from discrepancy theory (see Section 7.4), will play key roles in our proof.

The study of Littlewood polynomials has a long and distinguished history (see, for example, [29] or [123]). The roots of their study go back to the work of Hardy and Littlewood [91] on Diophantine approximation over 100 years ago, the work of Bloch and Pólya [24] on the maximum number of real roots of polynomials with restricted coefficients, and the work of Littlewood and Offord [109, 115, 116] and others [61, 132] on random polynomials. Two important extremal problems that arose from these early investigations are Littlewood's L_1-problem [92], which was famously resolved (up to constant factors) in 1981 by McGehee, Pigno and Smith [122] and Konyagin [107], and Chowla's cosine problem [43], see [30, 130].

The question studied in this chapter was asked in 1957, by Erdős [63], and was then taken up (and extensively studied) by Littlewood [110, 111, 113, 112] in a series of papers on the extremal properties of polynomials with restricted coefficients. In particular, Littlewood conjectured in [112] that flat Littlewood polynomials exist. This problem was given particular consideration in his well-known 1968 monograph [114], where he lays out 30 of his favourite problems, which he had curated for himself and his students for more than 30 years. In the book, he details several failed approaches to the "fascinating" question of the existence of flat Littlewood polynomials, and notes that he tried in vain to construct such polynomials.

Another well-known open problem, also proposed by Erdős [63] in the same problem paper, asks whether there exists a constant $c > 0$ such that, for every polynomial $P_n(z) = \sum_{k=0}^{n} a_k z^k$ with $a_k \in \mathbb{C}$ and $|a_k| = 1$ for all $0 \leqslant k \leqslant n$, we have

$$|P_n(z)| \geqslant (1+c)\sqrt{n}$$

for some $z \in \mathbb{C}$ with $|z| = 1$. (Note that, by a simple application of Parseval's theorem, the conclusion holds with $c = 0$.) Let us write $\mathscr{F}_n$ for the family of Littlewood polynomials of degree n, and $\mathscr{G}_n$ for the (larger) family with coefficients satisfying $|a_k| = 1$. The class $\mathscr{G}_n$ is significantly richer than $\mathscr{F}_n$, and for polynomials in this richer class, significant progress was made in the years following Littlewood's work. It had been known since the work of Hardy and Littlewood [91] that the upper bound in (7.1) holds for the polynomial in $\mathscr{G}_n$ given by setting $a_k := k^{ik}$, and Littlewood [111] proved that the polynomial in $\mathscr{G}_n$ given by setting $a_k := \exp\left(\binom{k+1}{2}\pi i/(n+1)\right)$ satisfies the stronger upper and lower bounds

$$|P(z)| = \big(1+o(1)\big)\sqrt{n} \tag{7.2}$$

for all $z \in \mathbb{C}$ with $|z| = 1$ except in a small interval around $z = 1$. Following further progress in [21, 38], and building in particular on work of Körner [108], the second question of Erdős [63] mentioned above was answered by Kahane [104], who proved that there exist *ultra-flat* polynomials in $\mathscr{G}_n$, i.e., polynomials that satisfy (7.2) for all $z \in \mathbb{C}$ with $|z| = 1$. More recently, Bombieri and Bourgain [32] improved Kahane's bounds, and moreover gave an effective construction of an ultra-flat polynomial in $\mathscr{G}_n$.

For the more restrictive class of Littlewood polynomials, much less progress has been made over the past 50 years. The Rudin–Shapiro polynomials mentioned above satisfy the upper bound in (7.1)

with $\Delta = \sqrt{2}$ when $n = 2^t - 1$, and with $\Delta = \sqrt{6}$ in general (see [1]). However, the previously best-known lower bound, proved by Carroll, Eustice and Figiel [40] via a simple recursive construction, states that there exist Littlewood polynomials $P_n(z) \in \mathscr{F}_n$ with $|P_n(z)| \geqslant n^{0.431}$ for all sufficiently large $n \in \mathbb{N}$. Moreover, exhaustive search for small values of n (see [124]) suggests that ultra-flat Littlewood polynomials most likely do not exist. Let us mention one final interesting result in the direction of Littlewood's conjecture, due to Beck [16], who proved that there exist flat polynomials in $\mathscr{G}_n$ with $a_k^{400} = 1$ for every k.

In the next section, we outline the general strategy that we use to prove Theorem 7.1.1. Roughly speaking, our Littlewood polynomial will consist (after multiplication by a suitable negative power of z) of a real cosine polynomial that is based on the Rudin–Shapiro construction, and an imaginary sine polynomial that is designed to be large in the (few) places where the cosine polynomial is small. To be slightly more precise, we will attempt to "push" the sine polynomial far away from zero in these few dangerous places, using techniques from discrepancy theory to ensure that we can do so. In order to make this argument work, it will be important that the intervals on which the Rudin–Shapiro construction is small are "well-separated" (see Definition 7.2.2, below). The properties of the cosine polynomial that we need are stated in Theorem 7.2.3, and proved in Section 7.3; the properties of the sine polynomial are stated in Theorem 7.2.4, and proved in Section 7.5.

7.2 Outline of the proof

We may assume that n is sufficiently large, since the polynomial $1 - z - z^2 - \cdots - z^n$ has no roots with $|z| = 1$ if $n \geqslant 2$. It will also suffice to prove Theorem 7.1.1 for $n \equiv 0 \pmod 4$, since the addition of a constant number of terms of the form $\pm z^k$ can at worst only change $|P(z)|$ by an additive constant. We can also multiply the polynomial by $z^{-2n'}$ so that it becomes the centred 'Laurent polynomial'

$$\sum_{k=-2n'}^{2n'} \varepsilon_k z^k,$$

where $n = 4n'$. The following theorem therefore implies Theorem 7.1.1.

Theorem 7.2.1. For every sufficiently large $n \in \mathbb{N}$, there exists a Littlewood polynomial $P(z) = \sum_{k=-2n}^{2n} \varepsilon_k z^k$ such that

$$2^{-160}\sqrt{n} \leqslant |P(z)| \leqslant 2^{12}\sqrt{n}$$

for all $z \in \mathbb{C}$ with $|z| = 1$.

We remark that the constants in Theorem 7.2.1 could be improved somewhat, but we have instead chosen to (slightly) simplify the exposition wherever possible.

7.2.1 Strategy

Before embarking on the technical details of the proof, let us begin by giving a rough outline of the strategy that we will use to prove Theorem 7.2.1. The first idea is to choose a set $C \subseteq [2n] = \{1, \dots, 2n\}$, and set $\varepsilon_{-k} = \varepsilon_k$ for each $k \in C$, and $\varepsilon_{-k} = -\varepsilon_k$ for each $k \in S := [2n] \setminus C$. Setting $z = e^{i\theta}$, the polynomial $P(z)$ then decomposes as

$$\sum_{k=-2n}^{2n} \varepsilon_k z^k = \varepsilon_0 + 2\sum_{k\in C} \varepsilon_k \cos(k\theta) + 2i \sum_{k\in S} \varepsilon_k \sin(k\theta).$$

The real part of this expression is a cosine polynomial, while the imaginary part is a sine polynomial. Our aim is to choose the sine and cosine polynomials so that both are $O(\sqrt{n})$ for all θ, and so that the sine polynomial is large whenever the cosine polynomial is small.

Let us first describe our rough strategy for choosing the sine polynomial $s(\theta)$, given a suitable cosine polynomial $c(\theta)$. For each 'bad' interval $I \subseteq \mathbb{R}/2\pi\mathbb{Z}$ on which $|c(\theta)| < \delta\sqrt{n}$, we will choose a direction (positive or negative), and attempt to 'push' the sine polynomial in that direction on that interval. In other words, we pick a step function that is $\pm K\sqrt{n}$ on each of the bad intervals, and zero elsewhere, where K is a large constant. We then attempt to approximate this step function with a sine polynomial, the hope being that we can do so with an error of size $O(\sqrt{n})$ on each bad interval (independent of K).

In order to carry out this plan, we will use an old result[1] of Spencer [141] on combinatorial discrepancy (in the form of Corollary 7.4.2 below), first to choose the step function, and then to show that we can approximate it sufficiently closely. More precisely, the first application (see Lemma 7.5.3) provides us with a step function whose Fourier coefficients are all small, and the second application (see Lemmas 7.5.4 and 7.5.5) then produces a sine polynomial that does not deviate by more that $O(\sqrt{n})$ from this step function.

To make the sketch above rigorous, we will need the bad intervals to have a number of useful properties; roughly speaking, they should be 'few', 'small', and 'well-separated'. In particular, we will construct (see Definition 7.2.2 and Theorem 7.2.3) a set $\mathscr{I}$ of intervals, each of size $O(1/n)$, separated by gaps of size $\Omega(1/n)$, with $|c(\theta)| \geqslant \delta\sqrt{n}$ for all $\theta \notin \bigcup_{I\in\mathscr{I}} I$. Moreover, the number of intervals in $\mathscr{I}$ will be at most γn for some small constant $\gamma > 0$.

To see that these demands are not unreasonable, note first that if $C \subseteq [\gamma n]$ then the cosine polynomial has few roots, and the 'typical' value of the derivative of the cosine polynomial should be $\Theta((\gamma n)^{3/2})$. This means that, if we choose δ much smaller than γ, the polynomial should typically vary by more than $\delta\sqrt{n}$ over a distance of order $1/n$. In particular, we will show that if the set of bad intervals cannot be covered by a collection of small and well-separated intervals (in the sense described above), then several of the derivatives must be small simultaneously. For our cosine polynomial we shall use an explicit construction based on the Rudin–Shapiro polynomials (see Section 7.3), and we

[1] We will in fact find it convenient to use a variant of Spencer's theorem, due to Lovett and Meka [117].

will show (see Lemma 7.3.5) that the value and first three derivatives of this polynomial cannot all be simultaneously small.

7.2.2 The cosine polynomial

Let $n \in \mathbb{N}$ be sufficiently large, choose $2^{-43} < \gamma \leqslant 2^{-40}$ such that

$$\gamma n = 2^{t+11} + 2^t - 1 \tag{7.3}$$

for some *odd* integer t, and set

$$\delta := 2^{-8}\gamma^{7/2},$$

noting that $\delta > 2^{-160}$. Define $C \subseteq [2\gamma n]$ by setting $C = 2C'$, where

$$C' := \{2^{t+10}, \dots, 2^{t+10} + 2^t - 1\} \cup \{2^{t+11}, \dots, 2^{t+11} + 2^t - 1\},$$

so that C is a set of 2^{t+1} *even* integers. Our first aim is to construct a cosine polynomial

$$c(\theta) = \sum_{k \in C} \varepsilon_k \cos(k\theta),$$

that is only small on a few, well-separated intervals, and is never too large.

To state the two main steps in the proof of Theorem 7.2.1, we first need to define what we mean by a 'suitable' and 'well-separated' family of intervals.

Definition 7.2.2. Let $\mathscr{I}$ be a collection of disjoint intervals in $\mathbb{R}/2\pi\mathbb{Z}$. We will say that $\mathscr{I}$ is *suitable* if

(*a*) The endpoints of each interval in $\mathscr{I}$ lie in $\frac{\pi}{n}\mathbb{Z}$;

(*b*) $\mathscr{I}$ is invariant under the maps $\theta \mapsto \pi \pm \theta$;

(*c*) $|\mathscr{I}| = 4N$ for some $N \leqslant \gamma n$.

We say that a suitable collection $\mathscr{I}$ is *well-separated* if

(*d*) $|I| \leqslant 6\pi/n$ for each $I \in \mathscr{I}$;

(*e*) $d(I,J) \geqslant \pi/n$ for each $I, J \in \mathscr{I}$ with $I \neq J$;[2]

(*f*) $\bigcup_{I \in \mathscr{I}} I$ is disjoint from the set $(\pi/2)\mathbb{Z} + [-100\pi/n, 100\pi/n]$.

We will prove the following theorem about cosine polynomials.

[2]Given two sets $I, J \subseteq \mathbb{R}/2\pi\mathbb{Z}$, let us write $d(I,J) := \inf\{d(\theta,\theta') : \theta \in I,\ \theta' \in J\}$, where $d(\theta,\theta')$ is the distance between θ and θ' mod 2π.

Theorem 7.2.3. There exists a cosine polynomial

$$c(\theta) = \sum_{k \in C} \varepsilon_k \cos(k\theta),$$

with $\varepsilon_k \in \{-1, 1\}$ for every $k \in C$, and a suitable and well-separated collection $\mathscr{I}$ of disjoint intervals in $\mathbb{R}/2\pi\mathbb{Z}$, such that

$$|c(\theta)| \geqslant \delta\sqrt{n}$$

for all $\theta \notin \bigcup_{I \in \mathscr{I}} I$, and $|c(\theta)| \leqslant \sqrt{n}$ for all $\theta \in \mathbb{R}/2\pi\mathbb{Z}$.

The cosine polynomial we will use to prove Theorem 7.2.3 is a slight modification of the Rudin–Shapiro polynomial. We might remark here that one would expect almost any cosine polynomial whose absolute value is $O(\sqrt{n})$ to satisfy somewhat similar conditions, but this seems difficult to prove in general.

7.2.3 The sine polynomials

There will in fact be two sine polynomials; the first,

$$s_e(\theta) = \sum_{j \in S_e} \varepsilon_j \sin(j\theta), \tag{7.4}$$

will just be chosen to be small everywhere, more precisely at most $6\sqrt{n}$ for all $|z| = 1$ (see Lemma 7.3.3 below). It is defined on the set $S_e = 2S'_e$, where

$$S'_e := [n] \setminus C'$$

so that S_e is the set of remaining even integers in $[2n]$.

We write $S_o := \{1, 3, \ldots, 2n-1\}$ for the set of all the odd integers in $[2n]$, and our main task will be to construct an 'odd sine polynomial'

$$s_o(\theta) = \sum_{k \in S_o} \varepsilon_k \sin(k\theta)$$

that is large on each $I \in \mathscr{I}$, and not too large elsewhere. To be precise, we shall prove the following theorem.

Theorem 7.2.4. Let $\mathscr{I}$ be a suitable and well-separated collection of disjoint intervals in $\mathbb{R}/2\pi\mathbb{Z}$. There exists a sine polynomial

$$s_o(\theta) = \sum_{k \in S_o} \varepsilon_k \sin(k\theta),$$

with $\varepsilon_k \in \{-1, 1\}$ for every $k \in S_o$, such that

(*i*) $|s_o(\theta)| \geqslant 10\sqrt{n}$ for all $\theta \in \bigcup_{I \in \mathscr{I}} I$, and

(*ii*) $|s_o(\theta)| \leqslant 2^{10}\sqrt{n}$ for all $\theta \in \mathbb{R}$.

To deduce Theorem 7.2.1 from the results above, we simply set

$$P(e^{i\theta}) := \big(1 + 2c(\theta)\big) + 2i\big(s_e(\theta) + s_o(\theta)\big),$$

where $c(\theta)$ and $s_o(\theta)$ are the cosine and sine polynomials given by Theorems 7.2.3 and 7.2.4 respectively, and $s_e(\theta)$ is a sine polynomial as in (7.4) (see Section 7.5 for the details).

The rest of the chapter is organised as follows. First, in Section 7.3, we will define $c(\theta)$ and $s_e(\theta)$, and prove Theorem 7.2.3. In Section 7.4 we will recall the main lemma from [117] and deduce Corollary 7.4.2; this will be our main tool in the proof of Theorem 7.2.4, which is given in Section 7.5. Finally, we will conclude by completing the proof of Theorem 7.2.1.

7.3 Rudin–Shapiro Polynomials

In this section we will define the cosine polynomial that we will use to prove Theorem 7.2.3, and the sine polynomial that we will use on the remaining even integers. In both cases, we use the so-called Rudin–Shapiro polynomials, which were introduced independently by Shapiro [138] and Rudin [128] (and whose sequence of coefficients was also previously studied by Golay [81]). These polynomials have been extensively studied over the last few decades, see, e.g., [35–37, 127]. Let us begin by recalling their definition.

Definition 7.3.1 (Rudin–Shapiro polynomials). Set $P_0(z) = Q_0(z) = 1$ and inductively define

$$\begin{aligned} P_{t+1}(z) &= P_t(z) + z^{2^t} Q_t(z), \text{ and} \\ Q_{t+1}(z) &= P_t(z) - z^{2^t} Q_t(z), \end{aligned}$$

for each $t \geqslant 0$.

Observe that $P_t(z)$ and $Q_t(z)$ are both Littlewood polynomials of degree $2^t - 1$. A simple induction argument (see, e.g., [123]) shows that $P_t(z)P_t(1/z) + Q_t(z)Q_t(1/z) = 2^{t+1}$ for all $z \in \mathbb{C} \setminus \{0\}$. It follows that

$$|P_t(z)|^2 + |Q_t(z)|^2 = 2^{t+1}, \tag{7.5}$$

and hence $|P_t(z)|, |Q_t(z)| \leqslant 2^{(t+1)/2}$, for every $z \in \mathbb{C}$ with $|z| = 1$. Observing that the first 2^t terms of P_{t+1} are the same as for P_t, let us write $P_{<n}(z)$ for the polynomial of degree $n-1$ that agrees with $P_t(z)$ on the first n terms for all sufficiently large t, and note that $P_t(z) = P_{<2^t}(z)$. The following bound, which is a straightforward consequence of (7.5), was proved by Shapiro [138]. (Stronger bounds are known, see [1], but we shall not need them.)

Lemma 7.3.2. $|P_{<n}(z)| \leqslant 5\sqrt{n}$ for every $z \in \mathbb{C}$ with $|z| = 1$.

We now set

$$T := 2^{t+10},$$

and define our cosine polynomial to be

$$c(\theta) := \mathrm{Re}\left(z^T P_t(z) + z^{2T} Q_t(z)\right), \tag{7.6}$$

and our even sine polynomial to be

$$s_e(\theta) := \mathrm{Im}\left(P_{<(n+1)}(z) - z^T P_t(z) - z^{2T} P_t(z)\right), \tag{7.7}$$

where in both cases $z = e^{2i\theta}$ (note the factor of 2 in the exponent here). We claim first that[3] $\mathrm{supp}(c) = C$ and $\mathrm{supp}(s_e) = S_e$. This is clear for c, since $C = 2C'$ and $C' = \{T, \dots, T + 2^t - 1\} \cup \{2T, \dots, 2T + 2^t - 1\}$; for s_e it follows since the terms of $P_{<(n+1)}(z)$ corresponding to C' form the polynomial $z^T P_t(z) + z^{2T} P_t(z)$. (To see this, simply consider the first time that these terms appear in Definition 7.3.1, and note that the first 2^t terms of both P_{t+10} and Q_{t+10} are the same as for P_t.) We remark that the idea behind the definition of $c(\theta)$ is that the highly oscillatory factors z^T and z^{2T} allow us to show that c and its first three derivatives cannot all simultaneously be small (see Lemma 7.3.5, below).

The following lemma is an almost immediate consequence of Lemma 7.3.2 and (7.5).

Lemma 7.3.3. $|c(\theta)| \leqslant \sqrt{n}$ and $|s_e(\theta)| \leqslant 6\sqrt{n}$ for every $\theta \in \mathbb{R}$.

Proof. Observe first that, setting $z := e^{2i\theta}$, we have

$$|c(\theta)| \leqslant |P_t(z)| + |Q_t(z)| \leqslant 2^{(t+3)/2} \leqslant \sqrt{n},$$

where the first inequality follows from the definition of c, the second holds by (7.5), and the last holds by (7.3), since $\gamma \leqslant 1$. Similarly, we have

$$|s_e(\theta)| \leqslant |P_{<(n+1)}(z)| + 2|P_t(z)| \leqslant 5\sqrt{n+1} + 2^{(t+3)/2} \leqslant 6\sqrt{n},$$

where the first inequality follows from the definition of s_e, the second holds by Lemma 7.3.2 and (7.5), and the last holds by (7.3). □

In order to prove Theorem 7.2.3, it remains to show that $|c(\theta)| \geqslant \delta\sqrt{n}$ for all $\theta \notin \bigcup_{I \in \mathscr{I}} I$, for some suitable and well-separated collection $\mathscr{I}$ of disjoint intervals in $\mathbb{R}/2\pi\mathbb{Z}$. When doing so we will find it convenient to rescale the polynomial as follows: define a function $H\colon \mathbb{R} \to \mathbb{C}$ by setting

$$H(x) := e^{ix}\alpha(x) + e^{2ix}\beta(x),$$

[3] We define the *support*, $\mathrm{supp}(f)$, of a sine polynomial $f(\theta) = \sum_{k>0} \varepsilon_k \sin(k\theta)$ or cosine polynomial $f(\theta) = \sum_{k>0} \varepsilon_k \cos(k\theta)$ to be the set of k such that $\varepsilon_k \neq 0$.

where

$$\alpha(x) := 2^{-(t+1)/2} P_t(e^{ix/T}) \qquad \text{and} \qquad \beta(x) := 2^{-(t+1)/2} Q_t(e^{ix/T}),$$

and observe that

$$c(\theta) = 2^{(t+1)/2} \operatorname{Re}\big(H(2T\theta)\big).$$

Note that, by (7.5), we have

$$|\alpha(x)|^2 + |\beta(x)|^2 = 1.$$

We think of $\alpha(x)$ and $\beta(x)$ as being slowly varying functions, relative to the much more rapidly varying exponential factors in the definition of $H(x)$.

The key property of the polynomial $c(\theta)$ that we will need is given by the following lemma.

Lemma 7.3.4. Let $0 < \eta < 2^{-11}$. Every interval $I \subseteq \mathbb{R}$ of length 7η contains a sub-interval $J \subseteq I$ of length η such that

$$\big|\operatorname{Re}\big(H(x)\big)\big| \geqslant \frac{\eta^3}{2^7}$$

for every $x \in J$. Moreover, if $I = [a, a+7\eta]$ then we can take $J = [a+j\eta, a+(j+1)\eta]$ for some $j \in \{0,1,\dots,6\}$.

To prove Lemma 7.3.4, we will first need to prove the following lemma.

Lemma 7.3.5. For any $x \in \mathbb{R}$ there exists $k \in \{0,1,2,3\}$ such that

$$\big|\operatorname{Re}\big(H^{(k)}(x)\big)\big| \geqslant \frac{1}{4}.$$

The proof of Lemma 7.3.5 is not very difficult, but we will need to work a little. We will use Bernstein's classical inequality (see, e.g., [135]), which states that if $f(z)$ is a polynomial of degree n, then

$$\max_{|z|=1} |f'(z)| \leqslant n \cdot \max_{|z|=1} |f(z)|. \tag{7.8}$$

This easily implies the following bound on the derivatives of the Rudin–Shapiro polynomials.

Lemma 7.3.6. Let $0 \leqslant k, t \in \mathbb{Z}$. We have

$$\left|\frac{d^k}{d\theta^k} P_t(e^{i\theta})\right|, \left|\frac{d^k}{d\theta^k} Q_t(e^{i\theta})\right| \leqslant 2^{kt+(t+1)/2} \tag{7.9}$$

for every $\theta \in \mathbb{R}$. In particular,

$$|\alpha^{(k)}(x)|, |\beta^{(k)}(x)| \leqslant 2^{-10k} \tag{7.10}$$

for every $k \geqslant 1$ and $x \in \mathbb{R}$.

Note that (7.10) justifies our intuition that $\alpha(x)$ and $\beta(x)$ vary relatively slowly.

Proof. To prove (7.9) we simply apply (7.8) k times, and (7.5) once. It follows from (7.9) that

$$\max\left\{|\alpha^{(k)}(x)|, |\beta^{(k)}(x)|\right\} \leqslant 2^{-(t+1)/2} \cdot T^{-k} \cdot 2^{kt+(t+1)/2} = 2^{-10k}$$

for every $k \geqslant 1$ and $x \in \mathbb{R}$, as claimed. □

We will use the following easy consequences of Lemma 7.3.6.

Lemma 7.3.7. For each $0 \leqslant k \leqslant 4$, and every $x \in \mathbb{R}$, we have

$$\left|H^{(k)}(x) - \left(i^k e^{ix}\alpha(x) + (2i)^k e^{2ix}\beta(x)\right)\right| \leqslant \frac{1}{8}$$

and

$$|H^{(k)}(x)| \leqslant 2^k + 2.$$

Proof. Since $H(x) = e^{ix}\alpha(x) + e^{2ix}\beta(x)$, we have

$$H^{(k)}(x) = \sum_{j=0}^{k} \binom{k}{j} \left(i^{k-j} e^{ix}\alpha^{(j)}(x) + (2i)^{k-j} e^{2ix}\beta^{(j)}(x)\right),$$

and hence, using (7.10),

$$\left|H^{(k)}(x) - \left(i^k e^{ix}\alpha(x) + (2i)^k e^{2ix}\beta(x)\right)\right| \leqslant \sum_{j=1}^{k} \binom{k}{j} \left(1 + 2^{k-j}\right) 2^{-10j} \leqslant \frac{1}{8}$$

(with room to spare) since $k \leqslant 4$. Since $|i^k e^{ix}\alpha(x) + (2i)^k e^{2ix}\beta(x)| \leqslant 1 + 2^k$, it follows immediately that

$$|H^{(k)}(x)| \leqslant 2^k + 2,$$

as claimed. □

We can now easily deduce Lemma 7.3.5.

Proof of Lemma 7.3.5. Suppose that

$$\left|\operatorname{Re}\left(H^{(k)}(x)\right)\right| < \frac{1}{4}$$

for each $k \in \{0,1,2,3\}$. Setting

$$E_k := \operatorname{Re}\left(i^k e^{ix}\alpha(x) + (2i)^k e^{2ix}\beta(x)\right),$$

observe that

$$\operatorname{Re}\left(e^{ix}\alpha(x)\right) = \frac{4E_0+E_2}{3}, \qquad \operatorname{Re}\left(e^{2ix}\beta(x)\right) = -\frac{E_0+E_2}{3},$$
$$\operatorname{Im}\left(e^{ix}\alpha(x)\right) = -\frac{4E_1+E_3}{3}, \qquad \text{and} \quad \operatorname{Im}\left(e^{2ix}\beta(x)\right) = \frac{E_1+E_3}{6}.$$

Now, by Lemma 7.3.7, we have

$$|E_k| \leqslant \frac{1}{4}+\frac{1}{8} = \frac{3}{8}$$

for each $k \in \{0,1,2,3\}$, and therefore

$$\begin{aligned} 1 &= |\alpha(x)|^2 + |\beta(x)|^2 \\ &= \left|\operatorname{Re}\left(e^{ix}\alpha(x)\right)\right|^2 + \left|\operatorname{Im}\left(e^{ix}\alpha(x)\right)\right|^2 + \left|\operatorname{Re}\left(e^{2ix}\beta(x)\right)\right|^2 + \left|\operatorname{Im}\left(e^{2ix}\beta(x)\right)\right|^2 \\ &\leqslant \left(\frac{5^2}{3^2}+\frac{5^2}{3^2}+\frac{2^2}{3^2}+\frac{2^2}{6^2}\right)\cdot\frac{3^2}{8^2} = \frac{55}{9}\cdot\frac{9}{64} < 1, \end{aligned}$$

which is a contradiction. It follows that $|\operatorname{Re}(H^{(k)}(x_0))| \geqslant 1/4$ for some $0 \leqslant k \leqslant 3$. □

To deduce Lemma 7.3.4 from Lemmas 7.3.5 and 7.3.7, we shall use a generalization of Lagrange interpolation from [102, Theorem 2] that bounds the higher derivatives of a function in terms of its values at certain points.

Theorem 7.3.8. Let $f\colon I \to \mathbb{R}$ be a $k+1$ times continuously differentiable function and suppose $y_0,\dots,y_k \in I$ with $y_0 < y_1 < \cdots < y_k$. Then[4]

$$\left\| f^{(k)}(x) - \sum_{i=0}^{k} \frac{k!\,f(y_i)}{\prod_{j\neq i}(y_i-y_j)} \right\|_\infty \leqslant \left\| x - \frac{1}{k+1}\sum_{i=0}^{k} y_i \right\|_\infty \cdot \|f^{(k+1)}(x)\|_\infty.$$

Lemma 7.3.4 is a straightforward consequence of Lemmas 7.3.5 and 7.3.7 and Theorem 7.3.8.

Proof of Lemma 7.3.4. Let $I = [a, a+7\eta]$, and suppose (for a contradiction) that for each $0 \leqslant j \leqslant 6$, there exists a point

$$x_j \in I_j := \left[a+j\eta,\, a+(j+1)\eta\right]$$

such that $|\operatorname{Re}(H(x_j))| < 2^{-7}\eta^3$. We will show that $|\operatorname{Re}(H^{(k)}(x_0))| < 1/4$ for each $0 \leqslant k \leqslant 3$, which will contradict Lemma 7.3.5, and hence prove the lemma.

For $k = 0$, we have $|\operatorname{Re}(H^{(k)}(x_0))| < 2^{-7}\eta^3 < 1/4$ (by assumption), so let $k \in \{1,2,3\}$. By Lemma 7.3.7 and Theorem 7.3.8, applied with $f := \operatorname{Re}(H)$ and $y_j := x_{2j}$ for each $0 \leqslant j \leqslant k$ (so, in

[4] In the notation of [102], the sum in the first $\|\cdot\|_\infty$ expression is $L^{(k)}(x)$ where $L(x) = \sum_i f(y_i)\prod_{j\neq i}(x-y_j)/(y_i-y_j)$, and the second $\|\cdot\|_\infty$ expression is $\|\omega^{(k)}(x)/(k+1)!\|_\infty$, where $\omega(x) = \prod_i (x-y_i)$. Note that the inequality is tight when $f(x) = \omega(x)$.

particular, $|y_i - y_j| \geqslant \eta$ for all $i \neq j$), we have

$$\begin{aligned}\left|\operatorname{Re}\left(H^{(k)}(x_0)\right)\right| &\leqslant \sum_{i=0}^{k} \frac{k!}{\eta^k} \cdot \frac{\eta^3}{2^7} + 7\eta \cdot \left\|\operatorname{Re}\left(H^{(k+1)}(x)\right)\right\|_\infty \\ &\leqslant \frac{4 \cdot 3!}{2^7} + \frac{7(2^4+2)}{2^{11}} < \frac{1}{4},\end{aligned}$$

since $\eta < 2^{-11}$, as required. □

Finally, in order to show that $\bigcup_{I \in \mathscr{J}} I$ is disjoint from the set $(\pi/2)\mathbb{Z} + [-100\pi/n, 100\pi/n]$, we will need the following simple lemma.

Lemma 7.3.9. If $|x| \leqslant 1/8$ or $|x - T\pi| \leqslant 1/8$, then $\operatorname{Re}\left(H(x)\right) \geqslant 1/2$.

Proof. We will use the following facts (cf. [36, Theorem 5]), which can be easily verified by induction: for every $t \geqslant 0$,

$$P_{2t}(1) = P_{2t}(-1) = Q_{2t}(1) = -Q_{2t}(-1) = 2^t,$$

and

$$P_{2t+1}(1) = Q_{2t+1}(-1) = 2^{t+1}, \qquad P_{2t+1}(-1) = Q_{2t+1}(1) = 0.$$

Since t is odd, it follows that

$$\operatorname{Re}\left(H(0)\right) = 2^{-(t+1)/2}\left(P_t(1) + Q_t(1)\right) = 1$$

and

$$\operatorname{Re}\left(H(T\pi)\right) = 2^{-(t+1)/2}\left(P_t(-1) + Q_t(-1)\right) = 1.$$

Now, by Lemma 7.3.7 we have $|H'(x)| \leqslant 4$ for every $x \in \mathbb{R}$, and so

$$\operatorname{Re}\left(H(x)\right) \geqslant 1 - 4|x| \geqslant \frac{1}{2}$$

for all $x \in \mathbb{R}$ with $|x| \leqslant 1/8$. A similar argument works for those x near $T\pi$. □

Remark 7.3.10. Note that $x = T\pi$ corresponds to $\theta = \pi/2$ in the cosine polynomial $c(\theta)$. The reader may have noticed that we do not necessarily need the cosine polynomial to be large at this point, as the sine polynomial can be large there. However, for technical reasons, this will be useful later on, in the proof of Lemma 7.5.6.

We are finally ready to prove Theorem 7.2.3.

Proof of Theorem 7.2.3. Let $c(\theta)$ be the cosine polynomial defined in (7.6), and recall that $\operatorname{supp}(c) = C$, that $\varepsilon_k \in \{-1, 1\}$ for every $k \in C$, and that $|c(\theta)| \leqslant \sqrt{n}$ for every $\theta \in \mathbb{R}/2\pi\mathbb{Z}$, by Lemma 7.3.3. We will show that there exists a suitable and well-separated collection $\mathscr{I}$ of disjoint intervals in $\mathbb{R}/2\pi\mathbb{Z}$ such that $|c(\theta)| \geqslant \delta\sqrt{n}$ for all $\theta \notin \bigcup_{I \in \mathscr{I}} I$.

To prove this, set $\eta := 2T\pi/n$, and note that $\eta < \pi\gamma < 2^{-11}$. Partition $\mathbb{R}/4T\pi\mathbb{Z} = \mathbb{R}/2n\eta\mathbb{Z}$ into $2n$ intervals $I_j := [j\eta, (j+1)\eta]$, each of length η, and say that an interval I_j is *good* if

$$\left|\operatorname{Re}\left(H(x)\right)\right| \geqslant \frac{\eta^3}{2^7}$$

for all $x \in I_j$. Let $\mathscr{J}'$ be the collection of maximal unions of consecutive good intervals I_j, and let $\mathscr{I}'$ be the collection of remaining intervals (i.e., maximal unions of consecutive bad intervals). Thus $\mathscr{I}'$ and $\mathscr{J}'$ form interleaving collections of intervals decomposing $\mathbb{R}/4T\pi\mathbb{Z}$. Scaling from x to $\theta = x/2T$ gives corresponding collections of intervals $\mathscr{I}$ and $\mathscr{J}$; we claim that $\mathscr{I}$ is the required suitable and well-separated collection.

First, to see that $\mathscr{I}$ is suitable, note that each interval I_j (and hence each $I \in \mathscr{I}'$) starts and ends at a multiple of $\eta = 2T\pi/n$. Hence after scaling, each $I \in \mathscr{I}$ starts and ends at points of $\frac{\pi}{n}\mathbb{Z}$. The set $\mathscr{I}$ is invariant under the maps $\theta \mapsto \pi \pm \theta$ by the symmetries of the function $\cos(k\theta)$ when $k \in C \subseteq 2\mathbb{Z}$. To see that $|\mathscr{I}| \leqslant 4\gamma n$, note that since a cosine polynomial of degree d has at most $2d$ roots in its period, there are at most $4(2T + 2^t - 1) = 4\gamma n$ values of $x \in \mathbb{R}/4T\pi\mathbb{Z}$ where $\operatorname{Re}(H(x)) = 2^{-7}\eta^3$, and the same bound on the number where $\operatorname{Re}(H(x)) = -2^{-7}\eta^3$. Since each $I \in \mathscr{I}'$ must contain at least two such points (counted with multiplicity), we have $|\mathscr{I}| = |\mathscr{I}'| \leqslant 4\gamma n$, as required.

Next, let us show that $\mathscr{I}$ is well-separated. Recall first that, by Lemma 7.3.4, any set of 7 consecutive intervals I_j must contain a good interval. Thus $|I| \leqslant 6\eta$ for each $I \in \mathscr{I}'$, and so $|I| \leqslant 6\pi/n$ for each $I \in \mathscr{I}$. Now, $d(I,J) \geqslant \pi/n$ for distinct $I, J \in \mathscr{I}$ by construction, and the sets $[-100\eta, 100\eta]$ and $T\pi + [-100\eta, 100\eta]$ are each contained in an element of $\mathscr{J}'$ by Lemma 7.3.9, since $2^{-7}\eta^3 < 1/2$ and $100\eta < 1/8$. Scaling down, it follows that $\bigcup_{I\in\mathscr{J}} I$ is disjoint from the set $(\pi/2)\mathbb{Z} + [-100\pi/n, 100\pi/n]$, as required.

Finally, recalling that $\eta = 2T\pi/n$, $\gamma n = 2T + 2^t - 1$, $T = 2^{t+10}$, and that $|\operatorname{Re}(H(x))| \geqslant 2^{-7}\eta^3$ for each $x \in J \in \mathscr{J}'$, it follows that

$$|c(\theta)| \geqslant 2^{(t+1)/2} \cdot 2^{-7}\eta^3 = 2^{-12}\pi^3(2T)^{7/2}/n^3 \geqslant 2^{-8}\gamma^{7/2}\sqrt{n} = \delta\sqrt{n}$$

for every $\theta \notin \bigcup_{I\in\mathscr{J}} I$, as required. □

7.4 Minimising Discrepancy

In this section we recall the main 'partial colouring' lemma of Spencer [141] (whose proof, as noted in the introduction, was based on a technique of Beck [12]), which will play an important role in the proof of Theorem 7.2.4. In particular, we will use the results of this section both to choose in which direction we should 'push' the sine polynomial on each interval $I \in \mathscr{I}$, and to show that we can choose $\varepsilon_k \in \{-1, 1\}$ so that it is pushed (roughly) the correct distance. The following convenient

variant of Spencer's theorem was proved by Lovett and Meka [117, Theorem 4][5], who also gave a beautiful polynomial-time randomised algorithm for finding a colouring with small discrepancy.

Theorem 7.4.1 (Main Partial Colouring Lemma)**.** Let $v_1, \dots, v_m \in \mathbb{R}^n$ and $x_0 \in [-1,1]^n$. If $c_1, \dots, c_m \geqslant 0$ are such that

$$\sum_{j=1}^{m} \exp\left(-c_j^2/16\right) \leqslant \frac{n}{16},$$

then there exists an $x \in [-1,1]^n$ such that

$$|\langle x - x_0, v_j \rangle| \leqslant c_j \|v_j\|_2$$

for every $j \in [m]$, and moreover $x_i \in \{-1,1\}$ for at least $n/2$ indices $i \in [n]$.

We will in fact use the following corollary of Theorem 7.4.1.

Corollary 7.4.2. Let $v_1, \dots, v_m \in \mathbb{R}^n$ and $x_0 \in [-1,1]^n$. If $c_1, \dots, c_m \geqslant 0$ are such that

$$\sum_{j=1}^{m} \exp\left(-c_j^2/14^2\right) \leqslant \frac{n}{16}, \tag{7.11}$$

then there exists an $x \in \{-1,1\}^n$ such that

$$|\langle x - x_0, v_j \rangle| \leqslant (c_j + 30)\sqrt{n} \cdot \|v_j\|_\infty$$

for every $j \in [m]$.

Proof. We prove Corollary 7.4.2 by induction on n. Note first that the result is trivial for all $n \leqslant 900$, since we can choose $x \in \{-1,1\}^n$ with $\|x - x_0\|_\infty \leqslant 1$, and for such a vector we have $|\langle x - x_0, v_j \rangle| \leqslant n \cdot \|v_j\|_\infty \leqslant 30\sqrt{n} \cdot \|v_j\|_\infty$.

For $n > 900$, we apply Theorem 7.4.1 with constants $b_j := 2c_j/7$, noting that

$$\sum_{j=1}^{m} \exp\left(-b_j^2/16\right) = \sum_{j=1}^{m} \exp\left(-c_j^2/14^2\right) \leqslant \frac{n}{16}.$$

We obtain a vector $y \in [-1,1]^n$, with

$$|\langle y - x_0, v_j \rangle| \leqslant b_j \|v_j\|_2 \leqslant b_j\sqrt{n} \cdot \|v_j\|_\infty$$

for every $j \in [m]$, such that $y_i \in \{-1,1\}$ for at least $n/2$ indices $i \in [n]$.

[5] The theorem as stated in [117] only insists that $|x_i| \geqslant 1 - \delta$ for at least $n/2$ indices, due to the requirement that a fast algorithm exists. However, it is clear by continuity that we can take $\delta = 0$ if we are only interested in an 'existence proof'.

Now, let $U \subseteq [n]$ be a set of size $\lceil n/2 \rceil$ such that $y_i \in \{-1,1\}$ for every $i \in U$, and set $W := [n] \setminus U$. For each $j \in [m]$, define a constant $a_j \geqslant 0$ so that

$$a_j^2 := c_j^2 + 14^2 \log\left(\frac{n}{\lfloor n/2 \rfloor}\right),$$

and observe that

$$\sum_{j=1}^{m} \exp\left(-a_j^2/14^2\right) \leqslant \frac{\lfloor n/2 \rfloor}{16} = \frac{|W|}{16},$$

and that $a_j \leqslant c_j + 12$, since $14^2 \log(n/\lfloor n/2 \rfloor) < 196 \log 2.01 < 12^2$ for $n > 900$.

Let $\pi \colon \mathbb{R}^n \to \mathbb{R}^W$ be projection onto the coordinates of W. By the induction hypothesis, we obtain a vector $z \in \{-1,1\}^W$ with

$$|\langle z - \pi(y), \pi(v_j) \rangle| \leqslant (a_j + 30)\sqrt{|W|} \cdot \|\pi(v_j)\|_\infty \leqslant (a_j + 30)\sqrt{n/2} \cdot \|v_j\|_\infty.$$

Now, define $x \in \{-1,1\}^n$ by setting $x_i := y_i$ for $i \in U$ and $\pi(x) = z$, and observe that

$$\begin{aligned}
|\langle x - x_0, v_j \rangle| &\leqslant |\langle y - x_0, v_j \rangle| + |\langle z - \pi(y), \pi(v_j) \rangle| \\
&\leqslant \left(b_j + (a_j + 30)/\sqrt{2}\right)\sqrt{n} \cdot \|v_j\|_\infty \\
&\leqslant \left(\frac{2c_j}{7} + \frac{c_j + 42}{\sqrt{2}}\right)\sqrt{n} \cdot \|v_j\|_\infty \\
&\leqslant (c_j + 30)\sqrt{n} \cdot \|v_j\|_\infty,
\end{aligned}$$

as required, since $b_j = 2c_j/7$ and $a_j \leqslant c_j + 12$. This completes the induction step. □

Remark 7.4.3. The result is stated in terms of the ℓ^∞-norms $\|v_j\|_\infty$ because we cannot control the decrease in $\|v_j\|_2$ when we discard half of the coordinates.

Remark 7.4.4. It is important for our application that m can be much larger than n, and that the only restriction on m occurs via the condition (7.11). In particular, we will later apply Corollary 7.4.2 with m very large, but with the c_j increasing sufficiently rapidly so that (7.11) still holds.

7.5 The odd sine polynomial

The aim of this section is to prove Theorem 7.2.4. Let $\mathscr{I}$ be a collection of suitable well-separated intervals, and recall from Definition 7.2.2 that $|\mathscr{I}| = 4N$ for some $N \leqslant \gamma n$, and that $\mathscr{I}$ is invariant under the maps $\theta \mapsto \pi \pm \theta$. The collection $\mathscr{I}$ is therefore uniquely determined by the set $\mathscr{I}_0 \subseteq \mathscr{I}$ of N intervals that lie in $[0, \pi/2]$ (since no $I \in \mathscr{I}$ contains 0 or $\pi/2$).

As described in Section 7.2.1, our aim is to 'push' the sine polynomial away from zero (in either the positive or negative direction) on each interval in $\mathscr{I}$. Let us say that a colouring $\alpha \colon \mathscr{I} \to \{-1,1\}$ is *symmetric* if $\alpha(I') = \alpha(I)$ whenever $I' = \pi - I$, and $\alpha(I') = -\alpha(I)$ whenever $I' = \pi + I$. Note

that if α is symmetric, then it is uniquely determined by its values on the set $\mathscr{I}_0$. Finally, recall that $S_o = \{1,3,5,\dots,2n-1\}$, and set $K := 2^7$.

Definition 7.5.1. Given a colouring $\alpha\colon \mathscr{I} \to \{-1,1\}$, we define $g_\alpha\colon \mathbb{R}/2\pi\mathbb{Z} \to \{-1,0,1\}$ by

$$g_\alpha(\theta) := \sum_{I\in\mathscr{I}} \alpha(I)\mathbb{1}[\theta \in I].$$

We also define a vector $\hat{\varepsilon} = (\hat{\varepsilon}_1, \hat{\varepsilon}_3, \dots, \hat{\varepsilon}_{2n-1}) \in \mathbb{R}^{S_o}$ by setting

$$\hat{\varepsilon}_j := K\sqrt{n}\int_{-\pi}^{\pi} g_\alpha(\theta)\sin(j\theta)\,d\theta,$$

for each $j \in S_o$.

Remark 7.5.2. By Fourier inversion, one would expect the function $\hat{s}_\alpha(\theta) := \sum_{j\in S_o}\hat{\varepsilon}_j\sin(j\theta)$ to approximate $\pi K\sqrt{n}\,g_\alpha(\theta)$; in particular, it should be large on the intervals $I \in \mathscr{I}$. We will prove in Lemma 7.5.6, below, that this is indeed the case.

We will use $\hat{\varepsilon}$ as the starting point of an application of Corollary 7.4.2, so we need $|\hat{\varepsilon}_j| \leqslant 1$ for all $j \in S_o$. The following lemma, which we also prove using Corollary 7.4.2, shows that, since we chose γ sufficiently small, we can choose the colouring α so that this is the case.

Lemma 7.5.3. There exists a symmetric colouring $\alpha\colon \mathscr{I} \to \{-1,1\}$ such that $\hat{\varepsilon} \in [-1,1]^{S_o}$.

Proof. Write $\mathscr{I}_0 = \{I_1,\dots,I_N\}$ and recall that this collection determines $\mathscr{I}$. Now, for each $j \in [n]$, define a vector $v_j \in \mathbb{R}^N$ by setting

$$(v_j)_i := 4K\sqrt{n}\int_{I_i}\sin\big((2j-1)\theta\big)\,d\theta$$

for each $i \in [N]$, and observe that, for each $j \in [n]$, we have

$$\hat{\varepsilon}_{2j-1} = K\sqrt{n}\int_{-\pi}^{\pi} g_\alpha(\theta)\sin\big((2j-1)\theta\big)\,d\theta = \sum_{i=1}^{N}\alpha(I_i)(v_j)_i,$$

by the symmetry conditions on both α and $\mathscr{I}$. Our task is therefore to find a vector $x \in \{-1,1\}^N$ such that $|\langle x, v_j\rangle| \leqslant 1$ for all $j \in [n]$. Indeed, we will then be able to set $\alpha(I_i) = x_i$ for each $i \in [N]$, and deduce that $|\hat{\varepsilon}_k| \leqslant 1$ for all $k \in S_o$.

We do so by applying Corollary 7.4.2 with $x_0 := 0$ and $c_j := 14\sqrt{\log(16n/N)}$ for each $j \in [n]$. Noting that (7.11) is satisfied, it follows from Corollary 7.4.2 that there exists an $x \in \{-1,1\}^N$ such that

$$|\langle x, v_j\rangle| \leqslant \big(c_j + 30\big)\sqrt{N}\cdot\|v_j\|_\infty.$$

Now, since $\mathscr{I}$ is well-separated, by Definition 7.2.2(d) we have

$$|(v_j)_i| \leqslant 4K\sqrt{n}\cdot|I_i| \leqslant \frac{24\pi K}{\sqrt{n}}$$

for every $i \in [N]$ and $j \in [n]$. It follows that

$$|\langle x, v_j \rangle| \leqslant \left(14\sqrt{\log(16n/N)} + 30\right)\sqrt{N/n} \cdot 24\pi K.$$

Note that the right hand side is an increasing function of N for $N/n \leqslant \gamma < 1$ and so

$$|\langle x, v_j \rangle| \leqslant \left(14\sqrt{\log(16/\gamma)} + 30\right)\sqrt{\gamma} \cdot 24\pi K \leqslant 1,$$

where the last inequality follows from our choice of $K = 2^7$ and the inequality $\gamma \leqslant 2^{-40}$. □

For the rest of the proof fix this colouring α (and hence also the vector $\hat{\varepsilon}$). Recall that our aim is to choose a colouring $\varepsilon \colon S_o \to \{-1, 1\}$ so that the conclusion of Theorem 7.2.4 holds. Given such a colouring, define

$$s_o(\theta) := \sum_{j \in S_o} \varepsilon_j \sin(j\theta) \qquad \text{and} \qquad \hat{s}_\alpha(\theta) := \sum_{j \in S_o} \hat{\varepsilon}_j \sin(j\theta).$$

Our aim is to choose the ε_j so that $|s_o(\theta) - \hat{s}_\alpha(\theta)|$ is uniformly bounded for all $\theta \in \mathbb{R}/2\pi\mathbb{Z}$ (see Lemma 7.5.5, below). A naïve approach to controlling this difference on a sufficiently dense set of points would require imposing more constraints (with smaller values of c_j) than can be handled by Corollary 7.4.2. Instead we shall place constraints on the differences $|s_o^{(\ell)}(\theta) - \hat{s}_\alpha^{(\ell)}(\theta)|$ of the ℓth derivatives for each $\ell \geqslant 0$, but at many fewer values of θ, and then use Taylor's Theorem to bound $|s_o(\theta) - \hat{s}_\alpha(\theta)|$ at all other points. The advantage of this approach is that the constraints we need on the higher derivatives become rapidly weaker as ℓ increases, and in particular can be chosen so that (7.11) is satisfied.

Note that it is enough to bound $|s_o(\theta) - \hat{s}_\alpha(\theta)|$ on $[0, \frac{\pi}{2}]$ as both $s_o(\theta)$ and $\hat{s}_\alpha(\theta)$ have the same symmetries under $\theta \mapsto \pi \pm \theta$. Set $M := 16n$ and let $\theta_k := \frac{(2k-1)\pi}{4M}$ for $k = 1, \dots, M$. Then for any point $\theta \in [0, \frac{\pi}{2}]$ there exists $k \in [M]$ such that $|\theta - \theta_k| \leqslant \frac{\pi}{4M} = 2^{-6}\pi/n$. By Taylor's Theorem (and the fact that all sine polynomials are entire functions so their Taylor expansions converge), we have

$$s_o(\theta) - \hat{s}_\alpha(\theta) = \sum_{\ell=0}^{\infty} \left(s_o^{(\ell)}(\theta_k) - \hat{s}_\alpha^{(\ell)}(\theta_k)\right) \frac{(\theta - \theta_k)^\ell}{\ell!}. \tag{7.12}$$

We will bound the absolute value of the right-hand side using Corollary 7.4.2.

Lemma 7.5.4. There exists a colouring $\varepsilon \colon S_o \to \{-1, 1\}$ such that

$$\left|s_o^{(\ell)}(\theta_k) - \hat{s}_\alpha^{(\ell)}(\theta_k)\right| \leqslant (65 + 2\ell)\sqrt{n} \cdot (2n)^\ell$$

for every $k \in [M]$ and $\ell \geqslant 0$.

Proof. For each $k \in [M]$ and $\ell \geqslant 0$, define a vector $v_{(k,\ell)} \in \mathbb{R}^n$ by setting

$$(v_{(k,\ell)})_j = \frac{d^\ell}{d\theta^\ell} \sin\big((2j-1)\theta\big)\big|_{\theta=\theta_k}$$

for each $j \in [n]$, and observe that

$$s_o^{(\ell)}(\theta_k) - \hat{s}_\alpha^{(\ell)}(\theta_k) = \sum_{j=1}^{n} \big(\varepsilon_{2j-1} - \hat{\varepsilon}_{2j-1}\big)(v_{(k,\ell)})_j = \langle \varepsilon - \hat{\varepsilon}, v_{(k,\ell)} \rangle,$$

where we consider $\varepsilon - \hat{\varepsilon}$ and $v_{(k,\ell)}$ as vectors in $\mathbb{R}^{S_o}$.

We apply Corollary 7.4.2 with $x_0 := \hat{\varepsilon}$ and $c_{(k,\ell)} = 14\sqrt{(9+\ell)\log 2}$. Observe that

$$\sum_{k=1}^{M} \sum_{\ell=0}^{\infty} \exp\big(-c_{(k,\ell)}^2/14^2\big) = \sum_{k=1}^{M} \sum_{\ell=0}^{\infty} 2^{-(9+\ell)} = M \cdot 2^{-8} = \frac{n}{16},$$

and so (7.11) is satisfied. It follows[6] from Corollary 7.4.2 that there exists an $\varepsilon \in \{-1,1\}^n$ such that

$$|\langle \varepsilon - \hat{\varepsilon}, v_{(k,\ell)} \rangle| \leqslant \big(c_{(k,\ell)} + 30\big)\sqrt{n} \cdot \|v_{(k,\ell)}\|_\infty$$

for every $k \in [M]$ and $\ell \geqslant 0$. Now, observe that

$$\|v_{(k,\ell)}\|_\infty \leqslant (2n)^\ell,$$

and that $14^2(9+\ell)\log 2 \leqslant 35^2 + 140\ell \leqslant (35+2\ell)^2$, so

$$c_{(k,\ell)} + 30 \leqslant 65 + 2\ell.$$

Combining these bounds, we obtain

$$\big|s_o^{(\ell)}(\theta_k) - \hat{s}_\alpha^{(\ell)}(\theta_k)\big| = |\langle \varepsilon - \hat{\varepsilon}, v_{(k,\ell)} \rangle| \leqslant (65+2\ell)\sqrt{n} \cdot (2n)^\ell$$

for every $k \in [M]$ and $\ell \geqslant 0$, as required. □

The following bound on the magnitude of $s_o(\theta) - \hat{s}_\alpha(\theta)$ is a straightforward consequence.

Lemma 7.5.5. There exists a colouring $\varepsilon \colon S_o \to \{-1,1\}$ such that

$$|s_o(\theta) - \hat{s}_\alpha(\theta)| \leqslant 72\sqrt{n}$$

for every $\theta \in \mathbb{R}$.

[6] Note that we appear to be applying Corollary 7.4.2 with an infinite number of constraints, but in fact only finitely many of them are needed as the constraints vacuously hold when $\ell \geqslant n$.

Proof. Let us assume (without loss of generality) that $\theta \in [0, \frac{\pi}{2}]$, and let $k \in [M]$ be such that $|\theta - \theta_k| \leqslant 2^{-6}\pi/n$. By (7.12) and Lemma 7.5.4, we have

$$\big|s_o(\theta) - \hat{s}_\alpha(\theta)\big| \leqslant \sum_{\ell=0}^{\infty} \big|s_o^{(\ell)}(\theta_k) - \hat{s}_\alpha^{(\ell)}(\theta_k)\big| \frac{(2^{-6}\pi/n)^\ell}{\ell!} \leqslant \sum_{\ell=0}^{\infty} (65+2\ell)\sqrt{n} \cdot \frac{(2^{-5}\pi)^\ell}{\ell!}.$$

Now simply observe that

$$\sum_{\ell=0}^{\infty} (65+2\ell)\frac{(2^{-5}\pi)^\ell}{\ell!} = (65+2^{-4}\pi)e^{2^{-5}\pi} \leqslant 72,$$

and the lemma follows. □

We will prove that the conclusion of Theorem 7.2.4 holds for the colouring ε given by Lemma 7.5.5. To deduce this, it will suffice to show that $\hat{s}_\alpha(\theta)$ approximates the step function $\pi K\sqrt{n} \cdot g_\alpha(\theta)$ sufficiently well, and in particular that it is large on each interval $I \in \mathscr{I}$.

Lemma 7.5.6. For every $\theta \in \bigcup_{I\in\mathscr{I}} I$, we have

$$|\hat{s}_\alpha(\theta)| \geqslant \frac{2K\sqrt{n}}{3}.$$

Moreover, $|\hat{s}_\alpha(\theta)| \leqslant 5K\sqrt{n}$ for every $\theta \in \mathbb{R}$.

The proof of Lemma 7.5.6 follows from a standard (but somewhat technical) calculation, and to simplify things slightly we will find it convenient to renormalise, by defining

$$\tilde{s}_\alpha(\theta) := (K\sqrt{n})^{-1}\hat{s}_\alpha(\theta).$$

Fix $\theta_0 \in \mathbb{R}$, and observe that, by the symmetry conditions on both α and $\mathscr{I}$, we have

$$\begin{aligned}\tilde{s}_\alpha(\theta_0) &= \sum_{j=0}^{n-1} \sin\big((2j+1)\theta_0\big) \int_{-\pi}^{\pi} g_\alpha(\theta)\sin\big((2j+1)\theta\big)\,d\theta \\ &= 4\int_0^{\pi/2} g_\alpha(\theta) \sum_{j=0}^{n-1} \sin\big((2j+1)\theta_0\big)\sin\big((2j+1)\theta\big)\,d\theta. \end{aligned} \tag{7.13}$$

We can now use the following simple trigonometric fact.

Observation 7.5.7.

$$4\sum_{j=0}^{n-1} \sin\big((2j+1)\theta_0\big)\sin\big((2j+1)\theta\big) = \frac{\sin\big(2n(\theta-\theta_0)\big)}{\sin(\theta-\theta_0)} - \frac{\sin\big(2n(\theta+\theta_0)\big)}{\sin(\theta+\theta_0)}.$$

Proof. Simply note that both sides are equal to

$$2\sum_{j=0}^{n-1}\Big(\cos\big((2j+1)(\theta-\theta_0)\big)-\cos\big((2j+1)(\theta+\theta_0)\big)\Big),$$

using the addition formulae for $\sin(\alpha\pm\beta)$ and $\cos(\alpha\pm\beta)$ and the telescoping series

$$\begin{aligned}\sin(2n\varphi)&=\sum_{j=0}^{n-1}\Big(\sin\big((2j+1)\varphi+\varphi\big)-\sin\big((2j+1)\varphi-\varphi\big)\Big)\\&=\sum_{j=0}^{n-1}2\cos\big((2j+1)\varphi\big)\sin(\varphi)\end{aligned}$$

for $\varphi=\theta\pm\theta_0$. □

Combining (7.13) and Observation 7.5.7, and recalling the definition of $g_\alpha(\theta)$, it follows that

$$\tilde{s}_\alpha(\theta_0)=\sum_{I\in\mathscr{I}_0}\alpha(I)\int_I\left(\frac{\sin\big(2n(\theta-\theta_0)\big)}{\sin(\theta-\theta_0)}-\frac{\sin\big(2n(\theta+\theta_0)\big)}{\sin(\theta+\theta_0)}\right)d\theta. \tag{7.14}$$

Before bounding the right-hand side of (7.14), let us briefly discuss what is going on. Let $\theta_0\in[0,\pi/2]$, and recall from Definition 7.2.2(f) that no $I\in\mathscr{I}_0$ contains any point close to 0 or $\pi/2$. It follows that the integrand in (7.14) behaves roughly like a point mass placed at $\theta=\theta_0$, and hence $\tilde{s}_\alpha(\theta_0)$ should be approximately $\alpha(I)$ when $\theta_0\in I$, and small otherwise.

To make this rigorous, we will show that the integral of the first term over the interval $I\in\mathscr{I}_0$ containing θ_0 (if such an interval exists) is of order 1, and that the integral over the remaining intervals (and over the second term) is smaller. This will follow via a straightforward calculation from the fact that the endpoints of each interval in $\mathscr{I}$ lie in $\frac{\pi}{n}\mathbb{Z}$.

Instead of approximating the integral for an interval close to θ_0 directly, we will instead compare it to the following standard 'sine integral'.

Lemma 7.5.8. Let $I\in\mathscr{I}$ and let $\theta_0\in\mathbb{R}$.

(*a*) If $\theta_0\in I$, then

$$\frac{4}{3}\leqslant\int_I\frac{\sin\big(2n(\theta-\theta_0)\big)}{\theta-\theta_0}\,d\theta\leqslant 4.$$

(*b*) If $\theta_0\notin I$ then

$$-1\leqslant\int_I\frac{\sin\big(2n(\theta-\theta_0)\big)}{\theta-\theta_0}\,d\theta\leqslant 2.$$

Proof. Recall from Definition 7.2.2 that the endpoints of I are in $\frac{\pi}{n}\mathbb{Z}$, and let $I=[a\pi/n,b\pi/n]$, where $a,b\in\mathbb{Z}$ with $a<b$. Substituting $x=2n(\theta-\theta_0)$ gives us the integral

$$f(\theta_0):=\int_I\frac{\sin\big(2n(\theta-\theta_0)\big)}{\theta-\theta_0}\,d\theta=\int_{2a\pi-2n\theta_0}^{2b\pi-2n\theta_0}\frac{\sin x}{x}\,dx,$$

and we note that

$$f'(\theta_0) = (-2n)\left(\frac{\sin(2b\pi - 2n\theta_0)}{2b\pi - 2n\theta_0} - \frac{\sin(2a\pi - 2n\theta_0)}{2a\pi - 2n\theta_0}\right) = \frac{4\pi n(a-b)\sin(2n\theta_0)}{(2a\pi - 2n\theta_0)(2b\pi - 2n\theta_0)},$$

since $a, b \in \mathbb{Z}$, so $\sin(2a\pi - 2n\theta_0) = \sin(2b\pi - 2n\theta_0) = -\sin(2n\theta_0)$. Since $a \neq b$, it follows that the extremal values of $f(\theta_0)$ can occur only when $\sin(2n\theta_0) = 0$, i.e., when $2n\theta_0 \in \pi\mathbb{Z}$. These extremal values must therefore be of the form

$$u(\ell) + u(\ell+1) + \cdots + u(\ell + 2(b-a) - 1)$$

for some $\ell \in \mathbb{Z}$, where

$$u(j) := \int_{j\pi}^{(j+1)\pi} \frac{\sin x}{x}\, d\theta.$$

We claim first that if $\theta_0 \in I$, then

$$\int_0^{2\pi} \frac{\sin x}{x}\, d\theta \leqslant f(\theta_0) \leqslant \int_{-\pi}^{\pi} \frac{\sin x}{x}\, d\theta.$$

Indeed, if $\theta_0 \in I$ then $2a\pi \leqslant 2\theta_0 n \leqslant 2b\pi$, and so $\ell \leqslant 0 \leqslant \ell + 2(b-a)$. Note also that

$$u(2j) > 0, \qquad u(2j+1) < 0 \qquad \text{and} \qquad u(-j) = u(j-1)$$

for every non-negative $j \in \mathbb{Z}$, and moreover

$$u(2j-1) + u(2j) < 0 < u(2j) + u(2j+1)$$

for every $j \geqslant 1$. It follows that the maxima of $f(\theta_0)$ are at most $u(-1) + u(0)$, and the minima are at least $u(0) + u(1)$, as claimed. Similarly, if $\theta_0 \notin I$ then without loss of generality we have $\ell \geqslant 0$, and by the same argument as above we have

$$\int_{\pi}^{2\pi} \frac{\sin x}{x}\, d\theta \leqslant f(\theta_0) \leqslant \int_0^{\pi} \frac{\sin x}{x}\, d\theta.$$

It is now straightforward to obtain the claimed bounds by numerical integration. □

We will also use the following simple lemma to bound the integrals in (7.14).

Lemma 7.5.9. If $h\colon [a,b] \to \mathbb{R}$ is a monotonic function and $b - a \in \frac{\pi}{n}\mathbb{Z}$, then

$$\left| \int_a^b h(\theta)\sin(2n\theta)\, d\theta \right| \leqslant \frac{|h(b) - h(a)|}{n}.$$

Proof. Assume without loss of generality that h is increasing, and suppose first that $b = a + \frac{\pi}{n}$. Since $\sin(x+\pi) = -\sin(x)$ we have

$$\int_a^{a+\frac{\pi}{n}} h(\theta)\sin(2n\theta)\,d\theta = \int_a^{a+\frac{\pi}{2n}} \big(h(\theta) - h(\theta+\tfrac{\pi}{2n})\big)\sin(2n\theta)\,d\theta,$$

and therefore, since h is increasing,

$$\left|\int_a^{a+\frac{\pi}{n}} h(\theta)\sin(2n\theta)\,d\theta\right| \leqslant \big(h(b)-h(a)\big)\int_a^{a+\frac{\pi}{2n}} |\sin(2n\theta)|\,d\theta = \frac{h(b)-h(a)}{n},$$

as required. To deduce the general case, simply split the interval $[a,b]$ into sub-intervals of length $\frac{\pi}{n}$ and use the triangle inequality. □

We are now ready to prove Lemma 7.5.6.

Proof of Lemma 7.5.6. Recall that it is enough to prove the bounds when $\theta = \theta_0 \in [0,\pi/2]$, and that $\mathscr{I}_0 = \{I \in \mathscr{I} : I \subseteq [0,\pi/2]\}$. By (7.14), we have

$$\tilde{s}_\alpha(\theta_0) = \sum_{I\in\mathscr{I}_0} \alpha(I)\int_I \left(\frac{\sin\big(2n(\theta-\theta_0)\big)}{\sin(\theta-\theta_0)} - \frac{\sin\big(2n(\theta+\theta_0)\big)}{\sin(\theta+\theta_0)}\right)d\theta \tag{7.15}$$

for every $\theta_0 \in [0,\pi/2]$. We will deal with the second term first.

Claim 1: $\displaystyle\sum_{I\in\mathscr{I}_0}\left|\int_I \frac{\sin(2n(\theta+\theta_0))}{\sin(\theta+\theta_0)}\,d\theta\right| \leqslant \frac{1}{50\pi} + \frac{O(1)}{n}.$

Proof of Claim 1. Let $I \in \mathscr{I}_0$, and suppose first that $\sin\theta$ is monotonic on $I+\theta_0$. By Lemma 7.5.9, applied with $h(\theta) = 1/\sin\theta$, we have

$$\left|\int_I \frac{\sin(2n(\theta+\theta_0))}{\sin(\theta+\theta_0)}\,d\theta\right| \leqslant \frac{1}{n}\left(\max_{\theta\in I}\frac{1}{\sin(\theta+\theta_0)} - \min_{\theta\in I}\frac{1}{\sin(\theta+\theta_0)}\right),$$

since, by Definition 7.2.2, the endpoints of I are in $\frac{\pi}{n}\mathbb{Z}$. If $\sin\theta$ is not monotonic on $I+\theta_0$, then we instead use the trivial bound

$$\left|\int_I \frac{\sin(2n(\theta+\theta_0))}{\sin(\theta+\theta_0)}\,d\theta\right| \leqslant |I|\cdot\max_{\theta\in I}\frac{1}{\sin(\theta+\theta_0)} = \frac{O(1)}{n},$$

where the final inequality holds since $|I| = O(1/n)$, by Definition 7.2.2, and hence (since $\sin\theta$ is not monotonic on $I+\theta_0 \subseteq [0,\pi]$) we have $\sin(\theta+\theta_0) > 1/2$ for all $\theta \in I$.

Now, summing over intervals $I \in \mathscr{I}_0$, and partitioning into three classes according to whether $\sin\theta$ is increasing, decreasing, or neither on $I+\theta_0$, we obtain two alternating sums that are both bounded by their maximum terms, and possibly one additional term (for which we use the trivial bound). Recalling from Definition 7.2.2 that $\bigcup_{I\in\mathscr{I}} I$ is disjoint from the set $(\pi/2)\mathbb{Z} + [-100\pi/n, 100\pi/n]$, we

obtain

$$\sum_{I\in\mathscr{I}_0}\left|\int_I \frac{\sin(2n(\theta+\theta_0))}{\sin(\theta+\theta_0)}\,d\theta\right| \leqslant \frac{2}{n\sin(100\pi/n)}+\frac{O(1)}{n} = \frac{1}{50\pi}+\frac{O(1)}{n},$$

as claimed. □

The next claim will allow us to replace the first term in (7.15) by the integral in Lemma 7.5.8.

Claim 2: $\displaystyle\sum_{I\in\mathscr{I}_0}\left|\int_I \frac{\sin\big(2n(\theta-\theta_0)\big)}{\sin(\theta-\theta_0)}-\frac{\sin\big(2n(\theta-\theta_0)\big)}{\theta-\theta_0}\,d\theta\right| = \frac{O(1)}{n}.$

Proof of Claim 2. We again apply Lemma 7.5.9, this time with $h(\theta)=\frac{1}{\sin\theta}-\frac{1}{\theta}$, which is increasing on $[-\pi/2,\pi/2]$, to give

$$\left|\int_I \frac{\sin\big(2n(\theta-\theta_0)\big)}{\sin(\theta-\theta_0)}-\frac{\sin\big(2n(\theta-\theta_0)\big)}{\theta-\theta_0}\,d\theta\right| \leqslant \frac{1}{n}\Big(\max_{\theta\in I}h(\theta-\theta_0)-\min_{\theta\in I}h(\theta-\theta_0)\Big)$$

for every $I\in\mathscr{I}_0$ (note that $\theta-\theta_0\in[-\pi/2,\pi/2]$ for $\theta\in I\in\mathscr{I}_0$). Summing over intervals $I\in\mathscr{I}_0$, and noting that we again have an alternating sum, we obtain the bound

$$\sum_{I\in\mathscr{I}_0}\left|\int_I \frac{\sin\big(2n(\theta-\theta_0)\big)}{\sin(\theta-\theta_0)}-\frac{\sin\big(2n(\theta-\theta_0)\big)}{\theta-\theta_0}\,d\theta\right| \leqslant \frac{h(\pi/2)-h(-\pi/2)}{n} = \frac{O(1)}{n},$$

as claimed. □

It remains to bound $\int_I \frac{\sin(2n(\theta-\theta_0))}{\theta-\theta_0}\,d\theta$ for each $I\in\mathscr{I}$. When $d(\theta_0,I)<\pi/n$ we will apply Lemma 7.5.8 to bound this integral. However, in order to deal with the intervals that are far from θ_0 we will need the following stronger bound. Let $\mathscr{J}(\theta_0):=\big\{I\in\mathscr{I}_0 : d(\theta_0,I)\geqslant\pi/n\big\}$.

Claim 3: $\displaystyle\sum_{I\in\mathscr{J}(\theta_0)}\left|\int_I \frac{\sin\big(2n(\theta-\theta_0)\big)}{\theta-\theta_0}\,d\theta\right| \leqslant \frac{2}{\pi}.$

Proof of Claim 3. Once again we apply Lemma 7.5.9, this time with $h(\theta)=1/\theta$. We obtain

$$\left|\int_I \frac{\sin\big(2n(\theta-\theta_0)\big)}{\theta-\theta_0}\,d\theta\right| \leqslant \frac{1}{n}\left(\max_{\theta\in I}\frac{1}{\theta-\theta_0}-\min_{\theta\in I}\frac{1}{\theta-\theta_0}\right)$$

for every $I\in\mathscr{I}_0$ with $\theta_0\notin I$. Summing over intervals in $\mathscr{J}(\theta_0)$, and noting that we obtain two alternating sums (one on either side of θ_0), we obtain

$$\sum_{I\in\mathscr{J}(\theta_0)}\left|\int_I \frac{\sin\big(2n(\theta-\theta_0)\big)}{\theta-\theta_0}\,d\theta\right| \leqslant \frac{2}{n}\cdot\frac{1}{\pi/n} = \frac{2}{\pi}$$

as claimed. □

Note that $2/\pi+1/(50\pi)+O(1/n)\leqslant 2/3$ if n is sufficiently large, and suppose first that $\theta_0\in I$ for some $I\in\mathscr{I}_0$. Then $d(\theta_0,I')\geqslant\pi/n$ for all $I\neq I'\in\mathscr{I}_0$, by Definition 7.2.2. It follows, by (7.15),

Claims 1, 2 and 3, and Lemma 7.5.8, that

$$\frac{2}{3} = \frac{4}{3} - \frac{2}{3} \leqslant \left|\tilde{s}_\alpha(\theta_0)\right| \leqslant 4 + \frac{2}{3} < 5,$$

as required. On the other hand, if $\theta_0 \notin \bigcup_{I\in\mathscr{I}_0} I$ then there are at most two intervals $I \in \mathscr{I}_0$ such that $d(\theta_0, I) < \pi/n$. Therefore, by (7.15), Claims 1, 2 and 3, and Lemma 7.5.8, we have

$$|\tilde{s}_\alpha(\theta_0)| \leqslant 2\cdot 2 + \frac{2}{3} < 5.$$

Since $\hat{s}_\alpha(\theta) = K\sqrt{n}\,\tilde{s}_\alpha(\theta)$, this completes the proof of the lemma. □

Remark 7.5.10. We note that it is important that the lengths of the intervals $I \in \mathscr{I}$ are multiples of $\frac{\pi}{n}$. Without this assumption it is possible that the error term from the distant intervals $I \in \mathscr{I}_0$ in Claim 3 could be unbounded. Indeed, the reason it does not stems ultimately from the cancelation in the integrals provided by Lemma 7.5.9.

Theorem 7.2.4 is an almost immediate consequence of Lemmas 7.5.5 and 7.5.6.

Proof of Theorem 7.2.4. Let $\mathscr{I}$ be a suitable and well-separated collection of disjoint intervals in $\mathbb{R}/2\pi\mathbb{Z}$. By Lemma 7.5.5, there exists a colouring $\varepsilon\colon S_o \to \{-1,1\}$ such that, if α is the function given by Lemma 7.5.3, then

$$|s_o(\theta) - \hat{s}_\alpha(\theta)| \leqslant 72\sqrt{n}$$

for every $\theta \in \mathbb{R}$. Now observe that, by Lemma 7.5.6, we have

$$|s_o(\theta)| \geqslant |\hat{s}_\alpha(\theta)| - |s_o(\theta) - \hat{s}_\alpha(\theta)| \geqslant \left(\frac{2K}{3} - 72\right)\sqrt{n} > 10\sqrt{n}$$

for all $\theta \in \bigcup_{I\in\mathscr{I}} I$, and

$$|s_o(\theta)| \leqslant |\hat{s}_\alpha(\theta)| + |s_o(\theta) - \hat{s}_\alpha(\theta)| \leqslant \big(5K + 72\big)\sqrt{n} \leqslant 2^{10}\sqrt{n}$$

for all $\theta \in \mathbb{R}$, as required. □

Finally, let us put together the pieces and prove Theorem 7.2.1.

Proof of Theorem 7.2.1. Let $c(\theta)$ be the cosine polynomial, and $\mathscr{I}$ be the suitable and well-separated collection of disjoint intervals in $\mathbb{R}/2\pi\mathbb{Z}$, given by Theorem 7.2.3. Now, given $\mathscr{I}$, let $s_o(\theta)$ be the sine polynomial given by Theorem 7.2.4, and let $s_e(\theta)$ be the sine polynomial defined in (7.7). We claim that the polynomial

$$P(e^{i\theta}) := \big(1 + 2c(\theta)\big) + 2i\big(s_e(\theta) + s_o(\theta)\big)$$

has the properties required by the theorem.

To prove the claim, we should first observe that $P(z) = \sum_{k=-2n}^{2n} \varepsilon_k z^k$ with $\varepsilon_k \in \{-1,1\}$ for every $k \in [-2n,2n]$. Indeed the supports of $c(\theta)$, $s_e(\theta)$, and $s_o(\theta)$ are disjoint and cover the powers z^k with $k \in \{-2n,\dots,-2n\} \setminus \{0\}$, and the constant 1 provides the term corresponding to $k = 0$. Now, observe that

$$|P(e^{i\theta})|^2 \leqslant \big(2|c(\theta)|+1\big)^2 + 4|s_e(\theta)+s_o(\theta)|^2 \leqslant \big(2\sqrt{n}+1\big)^2 + 4\big(2^{10}+6\big)^2 n \leqslant \big(2^{12}\sqrt{n}\big)^2,$$

for every $\theta \in \mathbb{R}$, since $|c(\theta)| \leqslant \sqrt{n}$ and $|s_e(\theta)| \leqslant 6\sqrt{n}$, by Theorem 7.2.3 and Lemma 7.3.3, and $|s_o(\theta)| \leqslant 2^{10}\sqrt{n}$, by Theorem 7.2.4. Next, observe that if $\theta \notin \bigcup_{I \in \mathscr{J}} I$ then

$$|P(e^{i\theta})| \geqslant \big|\operatorname{Re}\big(P(e^{i\theta})\big)\big| \geqslant 2|c(\theta)| - 1 \geqslant \delta\sqrt{n}$$

for all sufficiently large n, by Theorem 7.2.3. Finally, if $\theta \in \bigcup_{I \in \mathscr{J}} I$, then

$$|P(e^{i\theta})| \geqslant \big|\operatorname{Im}\big(P(e^{i\theta})\big)\big| \geqslant 2\big(|s_o(\theta)| - |s_e(\theta)|\big) \geqslant 2\big(10\sqrt{n} - 6\sqrt{n}\big) = 8\sqrt{n},$$

by Theorem 7.2.4. Hence $|P(z)| \geqslant \delta\sqrt{n}$ for all $z \in \mathbb{C}$ with $|z| = 1$, as required. □

Chapter 8

Judiciously 3-partitioning 3-uniform hypergraphs

The work in this section is joint with Hunter Spink and is published in Random Structures and Algorithms [142].

8.1 Introduction

Judicious partitioning problems seek to partition the vertices of a hypergraph $\mathscr{H}$ with m edges such that various quantities are simultaneously maximized. The first such problem was treated by Bollobás and Scott [26], where they proved that one can partition the vertices of a graph with m edges into k parts such that each part contains at most $\frac{m}{k^2}+o(m)$ edges. They later proved [27] that the vertices of a 3-uniform hypergraph can be partitioned into k parts each of which contains at most $\frac{m}{k^3}+o(m)$ edges. Recently Hou, Wu, Zeng and Zhu [99] claimed to have shown the result for 4-uniform hypergraphs with $\frac{m}{k^4}+o(m)$ edges, but their key technical lemma is incorrect.

The judicious partitioning problem we consider involves partitioning the vertices of an r-uniform hypergraph with m edges into k parts so that the minimum number of edges touched by any part is maximized. Bollobás and Scott [28] conjectured that this maximum is $(1-(1-\frac{1}{k})^r)m+o(m)$,[1] which if true is optimal up to the error term by considering the complete r-uniform hypergraph. Special cases of this problem have garnered considerable interest, particularly the case $r=k=3$. Bollobás, Reed and Thomason [25] proved every 3-uniform hypergraph $\mathscr{H}$ has a vertex-partition $V(\mathscr{H})=V_1\sqcup V_2\sqcup V_3$ such that each part meets at least $\frac{1}{3}(1-\frac{1}{e})m$ edges, which was claimed to have been improved to $\frac{5m-1}{9}$ by Bollobás and Scott [28] (although their proof apparently contains a subtle error, see Haslegrave [93]). Halsegrave [93] proved the optimal bound $0.6m$, which is best possible for small values of m. For large values of m, Ma and Yu [119] improved asymptotically the bound to $0.65m+o(m)$.

[1] We note that this asymptotically strengthens a previously conjectured bound of $\frac{rm}{2r-1}$ for any m when $r=k$ by Bollobás and Thomason (see [25] and [28]).

Here we resolve the conjecture of Bollobás and Scott in this special case.

Theorem 8.1.1. Every 3-uniform hypergraph $\mathscr{H}$ has a vertex-partition $V(\mathscr{H}) = V_1 \sqcup V_2 \sqcup V_3$ such that each part meets at least $\frac{19}{27}m - O(m^{6/7})$ edges.

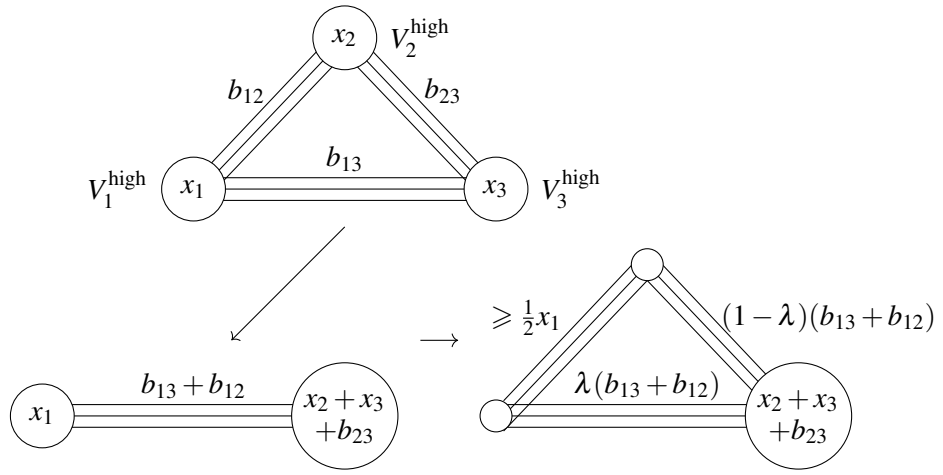
Our proof follows a strategy of Bollobás and Scott [27] and Ma and Yu [119]. The key technical lemma in [119] cannot be optimized sufficiently to attain $\frac{19}{27}$, in fact cannot be pushed past $0.7 < \frac{19}{27} = 0.\overline{703}$. We make an additional observation which imposes an additional inequality in this technical lemma.

We now outline the strategy. It would be nice if for a given 3-uniform hypergraph, the uniformly random 3-partition of the vertices worked. Each part would touch in expectation $\frac{19}{27}m + o(m)$ edges, but the presence of high degree vertices prevents these three quantities from simultaneously concentrating around the mean. For example, if there is a pair of vertices which belong to every edge, then a part which contains neither vertex touches with high probability only $\frac{1}{3}m + o(m)$ edges. To avoid this situation, we partition the vertices according to a two stage process.

We first partition "high" degree vertices $V^{\text{high}} \subseteq V(\mathscr{H})$ into three parts V_1, V_2, V_3, and then take the remaining "low" degree vertices $V^{\text{low}} = V(\mathscr{H}) \setminus V^{\text{high}}$ and assign each of these vertices independently to V_i with some probability p_i with $p_1 + p_2 + p_3 = 1$, so that the expected number of edges touched in each part is at least $\frac{19}{27}m + o(m)$. Because we exclude the high degree vertices from the random process, we have a tight concentration around the mean in each part by the Azuma–Hoeffding inequality (or more specifically the version of the inequality due to McDiarmid [121]), and we conclude.

The threshold for "high" degree vertices and the application of the Azuma–Hoeffding inequality are essentially automatic (if we only desire an $o(m)$ error, we only have to ensure that the threshold for "high" degree is $o(m)$). The challenge is to appropriately partition the "high" degree vertices so that the p_i can be chosen to make each part touch the correct number of edges in expectation.

As in [119], after choosing the high degree threshold, we partition V^{high} as follows. Denote by $E(\mathscr{H})^i$ for $i \in \{0,1,2,3\}$ to be the edges in $E(\mathscr{H})$ which touch i vertices in V^{high} and $3-i$ vertices in V^{low}. The edges in $E(\mathscr{H})^2$ induce a multi-graph G with vertex set V^{high} obtained by replacing each edge e with $e \cap V^{\text{high}}$. Denote by x_i the number of edges of G within V_i^{high}, and b_{ij} the number of edges of G with endpoints in V_i^{high} and V_j^{high}. We choose our partition $V^{\text{high}} = V_1^{\text{high}} \sqcup V_2^{\text{high}} \sqcup V_3^{\text{high}}$ so that $b_{23} + b_{13} + b_{12}$ is maximal (equivalently $x_1 + x_2 + x_3$ is minimal). The only facts used in [119] about this partition were that $b_{ij} \geqslant \max(2x_i, 2x_j)$. These inequalites follow by noting that the multigraph between V_i^{high} and V_j^{high} maximizes the number of edges between the two parts for fixed $V_i^{\text{high}} \sqcup V_j^{\text{high}}$.



However, these numerical constraints fail to rule out the possibility that G is (close to) a configuration which is complete in V_k^{high}, and complete bipartite between V_k^{high} and the other two parts, which prevents the method from proving a bound exceeding $0.7m$. We identify the additional inequality that if G maximizes $b_{23}+b_{13}+b_{12}$, then $b_{ij} \geqslant \frac{1}{2}x_k$, where k is the index $\neq i, j$. This arises from considering the effect of moving all vertices from V_i^{high} to V_j^{high}, decreasing $b_{23}+b_{13}+b_{12}$ by b_{ij}, and dividing V_k^{high} between V_k^{high} and V_i^{high} in such a way that b_{ik} is maximized, increasing $b_{23}+b_{13}+b_{12}$ by at least $\frac{1}{2}x_k$.

The core of this chapter is a rather technical lemma. This technical lemma is used to show that if we subdivide V^{high} so that $b_{23}+b_{13}+b_{12}$ is maximal, then we can find probabilities p_1, p_2, p_3 as described earlier. Given an instance of $\mathscr{H}$, finding appropriate values of p_i amount to simply solving cubic equations. Hence the laborious method used to prove the inequality is not reflected in the elegance of the final algorithm. We would be very interested in a less opaque derivation of the technical lemma.

The technical lemma is a system of inequalities with 13 variables and 11 degrees of freedom. We ultimately reduce this system to systems of inequalities with 6 variables and 2 degrees of freedom. For the systems which are not tight, we finely subdivide our parameter space and use a computer to check the system on each box. We prove all remaining systems of inequalities in the body of the chapter.

8.2 The technical lemma, its contrapositive, and our strategy

The following technical lemma is the crux of our proof.

Lemma 8.2.1. For any non-negative variables

$$x_1+x_2+x_3+a_1+a_2+a_3+b_{12}+b_{13}+b_{23}+c=1$$

with

$$b_{ij} \geqslant \max(2x_i, 2x_j, \frac{1}{2}x_k),$$

there exists $q_1, q_2, q_3 \in [0,1]$ with $q_1 + q_2 + q_3 = 2$ such that the following inequalities are satisfied:

$$q_1(b_{23} + x_2 + x_3) + q_1^2(a_2 + a_3) + q_1^3 c \leqslant \frac{8}{27}$$
$$q_2(b_{13} + x_1 + x_3) + q_2^2(a_1 + a_3) + q_2^3 c \leqslant \frac{8}{27}$$
$$q_3(b_{12} + x_1 + x_2) + q_3^2(a_1 + a_2) + q_3^3 c \leqslant \frac{8}{27}.$$

This lemma improves the lemma in [119], which does not impose the inequality $b_{ij} \geqslant \frac{1}{2}x_k$, and has the constant $\frac{8}{27}$ relaxed to 0.35. The following three sections are devoted to the proof of this lemma.

As the left hand sides of the inequalities in the system are non-decreasing in the q_i and are zero when $q_i = 0$, by symmetry the contrapositive of the lemma is equivalent to the following.

There does not exist $q_1, q_2, q_3 \in [0,1]$ with $q_1 + q_2 + q_3 = 2$ and non-negative variables

$$x_1 + x_2 + x_3 + b_{23} + b_{13} + b_{12} + a_1 + a_2 + a_3 + c = 1$$

with

$$b_{ij} \geqslant \max(2x_i, 2x_j, \frac{1}{2}x_k)$$

such that one of the following two systems holds.

System1

$$q_1(b_{23} + x_2 + x_3) + q_1^2(a_2 + a_3) + q_1^3 c > \frac{8}{27}$$
$$q_2(b_{13} + x_1 + x_3) + q_2^2(a_1 + a_3) + q_2^3 c > \frac{8}{27}$$
$$q_3(b_{12} + x_1 + x_2) + q_3^2(a_1 + a_2) + q_3^3 c > \frac{8}{27}.$$

System2 $q_1 = 1$ (so $q_2 + q_3 = 1$) and

$$q_2(b_{13} + x_1 + x_3) + q_2^2(a_1 + a_3) + q_2^3 c > \frac{8}{27}$$
$$q_3(b_{12} + x_1 + x_2) + q_3^2(a_1 + a_2) + q_3^3 c > \frac{8}{27}.$$

We will arrive at a contradiction assuming one of the two systems holds.

We may replace $(a_1, a_2, a_3, c) \mapsto (a_1 + (1-q_1)c, a_2 + (1-q_2)c, a_3 + (1-q_3)c, 0)$ and the corresponding system still holds, so we may assume from now on that $c = 0$. By working with the contrapositive we may assume q_1, q_2, q_3 are fixed and hence we are able to eliminate c in this fashion, which was not exploited previously, but drastically reduces the algebraic complexity of all future systems of inequalities.

Denoting $L = x_1+x_2+x_3+b_{23}+b_{13}+b_{12}+a_1+a_2+a_3+c$ and $L_i = q_i(b_{jk}+x_j+x_k)+q_i^2(x_j+x_k)+q_i^3c$, our strategy for **System1** is as follows. Keeping the q_i fixed, we repeatedly linearly perturb the remaining variables in such a way that L stays constant and L_1, L_2, L_3 do not decrease. We then perturb in this direction until one of the constraints attains equality. We then merge the associated variables (or set the corresponding variable to 0 if a variable hits zero), and repeat.

By linear algebra, it is clear we can do this (e.g., by choosing any perturbation fixing L, L_1, L_2 and choosing the sign so that L_3 does not decrease), until the number of variables besides q_1, q_2, q_3 is 3. However, we do some manual reductions first to reduce the final number of cases and simplify the exposition.

We handle **System2** in Section 3. **System1** will be dealt with in Sections 4, 5, and 6, completing the proof of Lemma 2.1.

8.3 System2

Recall we may assume $c = 0$. We may now replace

$$\begin{aligned}(x_1,x_2,x_3,a_1,a_2,a_3) &\mapsto (x_1,0,0,0,0,0)\\ (b_{23},b_{13},b_{12}) &\mapsto (\frac{1}{2}x_1, b_{13}+x_3+q_2(a_1+a_2+a_3),\\ &\qquad b_{12}+x_2+q_3(a_1+a_2+a_3)+b_{23}-\frac{1}{2}x_1)\end{aligned}$$

so we may assume that $x_2 = x_3 = a_1 = a_2 = a_3 = 0$ and $b_{23} = \frac{1}{2}x_1$. The system is now $b_{12}, b_{13} \geqslant 2x_1 \geqslant 0$,

$$b_{12}+b_{13}+\frac{3}{2}x_1 = 1$$

and

$$\begin{aligned}q_2(b_{13}+x_1) &> \frac{8}{27}\\ q_3(b_{12}+x_1) &> \frac{8}{27}.\end{aligned}$$

As $b_{12}, b_{13} \geqslant 2x_1$, we have $x_1 \leqslant \frac{2}{11}$. Hence,

$$\begin{aligned}1 &> \frac{8}{27}\left(\frac{1}{b_{13}+x_1}+\frac{1}{b_{12}+x_1}\right)\\ &\geqslant \frac{8}{27}\frac{4}{b_{13}+b_{12}+2x_1}\\ &= \frac{32}{27}\frac{1}{1+\frac{1}{2}x_1} \geqslant \frac{32}{27}\frac{1}{1+\frac{1}{11}} = \frac{88}{81},\end{aligned}$$

where the second inequality is by AM–HM, a contradiction.

8.4 System1 assuming $x_1 \geqslant 4x_2$ and $x_2 \geqslant x_3$

In this section, we prove the result assuming $x_1 \geqslant 4x_2$ and $x_2 \geqslant x_3$. The inequalities governing the b_{ij} are thus $b_{12}, b_{13} \geqslant 2x_1$ and $b_{23} \geqslant \frac{1}{2}x_1$. Our plan is to vary the variables, with the promise that we never break any of the constraints of **System1**. Recall we may assume $c = 0$. Then we have

$$x_1 + x_2 + x_3 + b_{23} + b_{13} + b_{12} + a_1 + a_2 + a_3 = 1$$

and

$$\begin{aligned} q_1(b_{23} + x_2 + x_3) + q_1^2(a_2 + a_3) &> \frac{8}{27} \\ q_2(b_{13} + x_1 + x_3) + q_2^2(a_1 + a_3) &> \frac{8}{27} \\ q_3(b_{12} + x_1 + x_2) + q_3^2(a_1 + a_2) &> \frac{8}{27}. \end{aligned}$$

Cases 1a and 1b First, suppose that $x_2 = x_3$. Call this common value x_{23}. Then we have

$$x_1 + 2x_{23} + b_{23} + b_{13} + b_{12} + a_1 + a_2 + a_3 = 1$$

and

$$\begin{aligned} q_1(b_{23} + 2x_{23}) + q_1^2(a_2 + a_3) &> \frac{8}{27} \\ q_2(b_{13} + x_1 + x_{23}) + q_2^2(a_1 + a_3) &> \frac{8}{27} \\ q_3(b_{12} + x_1 + x_{23}) + q_3^2(a_1 + a_2) &> \frac{8}{27}. \end{aligned}$$

Now, replace $(x_{23}, b_{23}) \mapsto (x_{23}, b_{23}) + \varepsilon(1, -2)$ starting with $\varepsilon = 0$, and increase ε until either $x_1 = 4x_{23}$ or $b_{23} = \frac{1}{2}x_1$.

Case1a Suppose first that $x_1 = 4x_{23}$, so that $b_{12}, b_{13} \geqslant 8x_{23}$, $b_{23} \geqslant 2x_{23}$, and $x_{23} \geqslant 0$. Then we have

$$6x_{23} + b_{23} + b_{13} + b_{12} + a_1 + a_2 + a_3 = 1$$

and

$$\begin{aligned} q_1(b_{23} + 2x_{23}) + q_1^2(a_2 + a_3) &> \frac{8}{27} \\ q_2(b_{13} + 5x_{23}) + q_2^2(a_1 + a_3) &> \frac{8}{27} \\ q_3(b_{12} + 5x_{23}) + q_3^2(a_1 + a_2) &> \frac{8}{27}. \end{aligned}$$

Now, replace $(x_{23}, b_{23}, b_{13}, b_{12}) \mapsto (x_{23}, b_{23}, b_{13}, b_{12}) + \varepsilon(1, 2, -4, -4)$ starting with $\varepsilon = 0$, and increase ε until either $b_{12} = 8x_{23}$ or $b_{13} = 8x_{23}$. Without loss of generality assume $b_{12} = 8x_{23}$. Then we have

System1a $b_{13} \geqslant 8x_{23}$ and $b_{23} \geqslant 2x_{23}$ and $x_{23} \geqslant 0$ and

$$14x_{23} + b_{23} + b_{13} + a_1 + a_2 + a_3 = 1$$

and

$$\begin{aligned} q_1(b_{23} + 2x_{23}) + q_1^2(a_2 + a_3) &> \frac{8}{27} \\ q_2(b_{13} + 5x_{23}) + q_2^2(a_1 + a_3) &> \frac{8}{27} \\ q_3(13x_{23}) + q_3^2(a_1 + a_2) &> \frac{8}{27}. \end{aligned}$$

Case1b Suppose now instead that $b_{23} = \frac{1}{2}x_1$. Then we have

$$\frac{3}{2}x_1 + 2x_{23} + b_{12} + b_{13} + a_1 + a_2 + a_3 = 1$$

and

$$\begin{aligned} q_1(\frac{1}{2}x_1 + 2x_{23}) + q_1^2(a_2 + a_3) &> \frac{8}{27} \\ q_2(b_{13} + x_1 + x_{23}) + q_2^2(a_1 + a_3) &> \frac{8}{27} \\ q_3(b_{12} + x_1 + x_{23}) + q_3^2(a_1 + a_2) &> \frac{8}{27}. \end{aligned}$$

Now, replace $(x_1, b_{12}, b_{13}) \mapsto (x_1, b_{12}, b_{13}) + \varepsilon(1, -\frac{3}{4}, -\frac{3}{4})$ starting with $\varepsilon = 0$, and increase ε until either $b_{12} = 2x_1$ or $b_{13} = 2x_1$. Without loss of generality assume $b_{12} = 2x_1$. Then we have

System1b $b_{13} \geqslant 2x_1 \geqslant 8x_{23} \geqslant 0$ and

$$\frac{7}{2}x_1 + 2x_{23} + b_{13} + a_1 + a_2 + a_3 = 1$$

and

$$\begin{aligned} q_1(\frac{1}{2}x_1 + 2x_{23}) + q_1^2(a_2 + a_3) &> \frac{8}{27} \\ q_2(b_{13} + x_1 + x_{23}) + q_2^2(a_1 + a_3) &> \frac{8}{27} \\ q_3(3x_1 + x_{23}) + q_3(a_1 + a_2) &> \frac{8}{27}. \end{aligned}$$

Cases 1c and 1d Now, we suppose that $x_2 \geqslant x_3$ and equality is not attained. We will do similar replacements of variables, but if we ever have $x_2 = x_3$ then we can reduce to either **System1a** or **System1b**, so we may suppose this never happens.

Replace $(x_3, b_{23}) \mapsto (x_3, b_{23}) + \varepsilon(1,-1)$ starting with $\varepsilon = 0$, and increase ε until $b_{23} = \frac{1}{2}x_1$. Now, replace $(x_3, b_{13}) \mapsto (x_3, b_{13}) + \varepsilon(1,-1)$ starting with $\varepsilon = 0$, and increase ε until $b_{13} = 2x_1$. Then we have

$$\frac{7}{2}x_1 + x_2 + x_3 + b_{12} + a_1 + a_2 + a_3 = 1$$

and

$$\begin{aligned} q_1(\frac{1}{2}x_1 + x_2 + x_3) + q_1^2(a_2 + a_3) &> \frac{8}{27} \\ q_2(3x_1 + x_3) + q_2^2(a_1 + a_3) &> \frac{8}{27} \\ q_3(b_{12} + x_1 + x_2) + q_3^2(a_1 + a_2) &> \frac{8}{27}. \end{aligned}$$

Replace $(x_2, b_{12}) \mapsto (x_2, b_{12}) + \varepsilon(1,-1)$ starting with $\varepsilon = 0$, and increase ε until either $b_{12} = 2x_1$ or $x_1 = 4x_2$.

Case1c Suppose first that $b_{12} = 2x_1$. Then we have

System1c $x_1 \geqslant 4x_2 \geqslant 4x_3 \geqslant 0$ and

$$\frac{11}{2}x_1 + x_2 + x_3 + a_1 + a_2 + a_3 = 1$$

and

$$\begin{aligned} q_1(\frac{1}{2}x_1 + x_2 + x_3) + q_1^2(a_2 + a_3) &> \frac{8}{27} \\ q_2(3x_1 + x_3) + q_2^2(a_1 + a_3) &> \frac{8}{27} \\ q_3(3x_1 + x_2) + q_3^2(a_1 + a_2) &> \frac{8}{27}. \end{aligned}$$

Case1d Finally, suppose that $x_1 = 4x_2$. Then we have

System1d $b_{12} \geqslant 8x_2 \geqslant 8x_3 \geqslant 0$ and

$$15x_2 + x_3 + b_{12} + a_1 + a_2 + a_3 = 1$$

and

$$\begin{aligned} q_1(3x_2 + x_3) + q_1^2(a_2 + a_3) &> \frac{8}{27} \\ q_2(12x_2 + x_3) + q_2^2(a_1 + a_3) &> \frac{8}{27} \\ q_3(b_{12} + 5x_2) + q_3^2(a_1 + a_3) &> \frac{8}{27}. \end{aligned}$$

8.4.1 Solving System1a,b,c,d

For each of these systems, we carry out the procedure described at the end of Section 2. This process terminates when we have 3 variables apart from q_1, q_2, q_3, and for each system there are $\binom{6}{3} = 20$ resulting cases. For each of **System1a,b,c,d** one of the 20 resulting cases will be when $a_1 = a_2 = a_3 = 0$, and one of the resulting cases will be when the remaining 3 variables are equal to zero.

In the former case, when $a_1 = a_2 = a_3 = 0$, we have

$$\begin{aligned} 2 &> \frac{8}{27}\left(\frac{1}{b_{23}+x_2+x_3} + \frac{1}{b_{13}+x_1+x_3} + \frac{1}{b_{12}+x_1+x_2}\right) \\ &\geqslant \frac{8}{27}\frac{9}{b_{23}+x_2+x_3+b_{13}+x_1+x_3+b_{12}+x_1+x_2} \\ &= \frac{8}{3(1+x_1+x_2+x_3)} \geqslant \frac{8}{3(1+\frac{1}{3})} = 2, \end{aligned}$$

where the second inequality is by AM–HM, a contradiction.

In the latter case, we have $a_1 + a_2 + a_3 = 1$ and

$$2 > \sqrt{\frac{8}{27}}\left(\frac{1}{\sqrt{a_2+a_3}} + \frac{1}{\sqrt{a_1+a_3}} + \frac{1}{\sqrt{a_1+a_2}}\right).$$

The right hand side is at least 2 by Jensen's inequality applied to $f(x) = \frac{1}{\sqrt{x}}$, a contradiction.

Aside from these two cases for each system, we handle all other cases by computer as follows. Let $\tilde{q}_i$ be the minimum of 1 and the solution to the quadratic equation $L_i = \frac{8}{27}$ (recall that $c = 0$). Then we may equivalently check that $\tilde{q}_1 + \tilde{q}_2 + \tilde{q}_3 \geqslant 2$ for any choice of the remaining variables. As it turns out, the infimum of $\tilde{q}_1 + \tilde{q}_2 + \tilde{q}_3$ is strictly larger than 2 in all remaining cases, which we can check by subdividing the space of the remaining variables very finely (boxes of dimension $0.001 \times 0.001 \times 0.001$ suffices in all cases), and crudely bounding in each box the smallest possible value of $\tilde{q}_1 + \tilde{q}_2 + \tilde{q}_3$. We report the results of these computations in Section 8.7.

8.5 System1 assuming $4x_2 \geqslant x_1 \geqslant x_2 \geqslant x_3$

We assume from now on that $4x_2 \geqslant x_1 \geqslant x_2 \geqslant x_3$, and so the inequalities with the b_{ij} become $b_{12}, b_{13} \geqslant 2x_1$, $b_{23} \geqslant 2x_2$. The symmetry of **System1**, along with the previous two sections, completes the verification of the technical lemma. As in the previous section, we will use the method described at the end of Section 2. However, if we ever attain equality in $4x_2 \geqslant x_1$, then we may stop and use Section 4, so we may assume from now on that we never attain equality in $4x_2 \geqslant x_1$. Recall we may assume $c = 0$. Then we have

$$x_1+x_2+x_3+b_{23}+b_{13}+b_{12}+a_1+a_2+a_3=1$$

and

$$\begin{aligned} q_1(b_{23}+x_2+x_3)+q_1^2(a_2+a_3) &> \frac{8}{27} \\ q_2(b_{13}+x_1+x_3)+q_2^2(a_1+a_3) &> \frac{8}{27} \\ q_3(b_{12}+x_1+x_2)+q_3^2(a_1+a_2) &> \frac{8}{27}. \end{aligned}$$

Replace $(x_1,x_2,x_3,b_{23},b_{13},b_{12}) \mapsto (x_1,x_2,x_3,b_{23},b_{13},b_{12})+\varepsilon(-1,-1,3,-2,-2,3)$ starting with $\varepsilon=0$, and increase ε until $x_2=x_3$. Call this common value x_{23}, so that $b_{12},b_{13} \geqslant 2x_1$, $b_{23} \geqslant 2x_{23}$ and $4x_{23} \geqslant x_1 \geqslant x_{23}$. Then we have

$$x_1+2x_{23}+b_{23}+b_{13}+b_{12}+a_1+a_2+a_3=1$$

and

$$\begin{aligned} q_1(b_{23}+2x_{23})+q_1^2(a_2+a_3) &> \frac{8}{27} \\ q_2(b_{13}+x_1+x_{23})+q_2^2(a_1+a_3) &> \frac{8}{27} \\ q_3(b_{12}+x_1+x_{23})+q_3^2(a_1+a_2) &> \frac{8}{27}. \end{aligned}$$

We now replace $(x_1,b_{13}) \mapsto (x_1,b_{13})+\varepsilon(1,-1)$ starting with $\varepsilon=0$, and increase ε until either $b_{13}=2x_1$ or $b_{12}=2x_1$. Without loss of generality, assume that $b_{13}=2x_1$. Finally, replace $(x_{23},b_{23}) \mapsto (x_{23},b_{23})+\varepsilon(1,-2)$ starting with $\varepsilon=0$, and increase ε until either $x_{23}=x_1$ or $b_{23}=2x_{23}$.

In the former case, let x be the common value of x_{23} and x_1. Set $A=4x$, $B=b_{23}+2x$, $C=b_{12}+2x$. Then (assuming $C \geqslant B$ without loss of generality) we have

System1e $C \geqslant B \geqslant A \geqslant 0$ and

$$\frac{1}{4}A+B+C+a_1+a_2+a_3=1$$

and

$$\begin{aligned} q_1B+q_1^2(a_2+a_3) &> \frac{8}{27} \\ q_2A+q_2^2(a_1+a_3) &> \frac{8}{27} \\ q_3C+q_3^2(a_1+a_2) &> \frac{8}{27}. \end{aligned}$$

In the latter case, set $A = 4x_{23}$, $B = 3x_1 + x_{23}$, and $C = b_{12} + x_1 + x_{23}$. Then we have

System1f $C \geqslant B \geqslant A \geqslant 0$ and $B \leqslant \frac{13}{4}A$ and

$$\frac{7}{12}A + \frac{2}{3}B + C + a_1 + a_2 + a_3 = 1$$

and

$$\begin{aligned} q_1A + q_1^2(a_2 + a_3) &> \frac{8}{27} \\ q_2B + q_2^2(a_1 + a_3) &> \frac{8}{27} \\ q_3C + q_3^2(a_1 + a_2) &> \frac{8}{27}. \end{aligned}$$

For each of these systems, we carry out the procedure described at the end of Section 2. This process terminates when we have 3 variables apart from q_1, q_2, q_3. Note in **System1f** that if we have $B = \frac{13}{4}A$ then $x_1 = 4x_{23}$ so this is handled by Section 4 and we can avoid analyzing all of these cases. Hence we are left with 20 cases for each system. The case $A = B = C = 0$ is handled as in Section 4.1.

Every case when $A = 0$ other than $A = B = C = 0$ we check by computer as before. Note that **System1e** is identical to **System1f** when $A = B$ as $1 + \frac{1}{4} = \frac{2}{3} + \frac{7}{12}$, so for these cases we only need to check one of the two systems. We report the results of these computations in Section 8.7.

8.5.1 Remaining cases of Systems1e,f

The only cases remaining are the 20 cases, 10 from **System1e** and 10 **System1f**, when three of $A = B, B = C, a_1 = 0, a_2 = 0, a_3 = 0$ occur. Note that $\frac{7}{12}A + \frac{2}{3}B + C + a_1 + a_2 + a_3 \leqslant \frac{1}{4}A + B + C + a_1 + a_2 + a_3$ when $A \leqslant B \leqslant C$, so it suffices to check the 10 cases as above in the following system.

System1f' $C \geqslant B \geqslant A \geqslant 0$ and

$$\frac{7}{12}A + \frac{2}{3}B + C + a_1 + a_2 + a_3 = 1$$

and

$$\begin{aligned} q_1A + q_1^2(a_2 + a_3) &> \frac{8}{27} \\ q_2B + q_2^2(a_1 + a_3) &> \frac{8}{27} \\ q_3C + q_3^2(a_1 + a_2) &> \frac{8}{27}. \end{aligned}$$

It is vitally important to note that there are solutions to **System1f'**, but we claim there are no solutions in the 10 cases mentioned above.

Suppose there is a solution in one of the cases above, we will derive a contradiction. We have

$$\begin{aligned}2 =& q_1+q_2+q_3\\ >& \frac{16}{27}\frac{1}{A+\sqrt{A^2+\frac{32}{27}(a_2+a_3)}}+\frac{1}{B+\sqrt{B^2+\frac{32}{27}(a_1+a_3)}}+\frac{1}{C+\sqrt{C^2+\frac{32}{27}(a_1+a_2)}}\\ \geqslant& \frac{16}{3}\frac{1}{A+B+C+\sqrt{A^2+\frac{32}{27}(a_2+a_3)}+\sqrt{B^2+\frac{32}{27}(a_1+a_3)}+\sqrt{C^2+\frac{32}{27}(a_1+a_2)}}.\end{aligned}$$

where the second inequality is by AM–HM, and therefore

$$A+B+C+\sqrt{A^2+\frac{32}{27}(a_2+a_3)}+\sqrt{B^2+\frac{32}{27}(a_1+a_3)}+\sqrt{C^2+\frac{32}{27}(a_1+a_2)}>\frac{8}{3}.$$

We will in fact show that

$$A+B+C+\sqrt{A^2+\frac{32}{27}(a_2+a_3)}+\sqrt{B^2+\frac{32}{27}(a_1+a_3)}+\sqrt{C^2+\frac{32}{27}(a_1+a_2)}\leqslant\frac{8}{3}$$

in each of the 10 cases, yielding a contradiction.

Suppose first we are in one of the three cases where $A=B=C$. Then $\frac{9}{4}A+a_1+a_2+a_3=1$, and by Jensen's inequality applied to $\sqrt{x}$, it suffices to show that

$$3A+\sqrt{3}\sqrt{3A^2+\frac{64}{27}(a_1+a_2+a_3)}\leqslant\frac{8}{3}.$$

As $A\leqslant\frac{4}{9}$, we have $\frac{8}{3}\geqslant 3A$. As $a_1+a_2+a_3=1-\frac{9}{4}A$, this inequality is

$$3A+|\frac{8}{3}-3A|\leqslant\frac{8}{3},$$

and as $\frac{8}{3}\geqslant 3A$, both sides are equal.

If $a_1=a_2=a_3=0$, then we have $2(A+B+C)\leqslant\frac{8}{3}(\frac{7}{12}A+\frac{2}{3}B+C)=\frac{8}{3}$ by the Chebyshev sum inequality.

In the remaining cases, we have one of $A=B$ and $B=C$. We have dealt with all cases where the maximum occurs on the boundary except when $A=0$, so we can assume the maximum is attained in the interior of the domain or $A=0$.

When $A=B$, we have by symmetry two distinct cases, when $a_1=a_2=0$ and when $a_1=a_3=0$.

When $a_1=a_2=0$ we have $0\leqslant A\leqslant C$, $\frac{5}{4}A+C+a_3=1$ and we want to show

$$2A+2C+2\sqrt{A^2+\frac{32}{27}a_3}\leqslant\frac{8}{3}.$$

By Lagrange multipliers, if the triple (A,C,a_3) attains a maximum on the interior of the domain then the gradient of the constraint $(\frac{5}{4},1,1)$ is parallel to the gradient $(2+\frac{2A}{\sqrt{A^2+\frac{32}{27}a_3}},2,\frac{32}{27}\frac{1}{\sqrt{A^2+\frac{32}{27}a_3}})$ of the objective function, so we have

$$\frac{4}{5}(2+\frac{2A}{\sqrt{A^2+\frac{32}{27}a_3}})=2$$
$$\frac{32}{27}\frac{1}{\sqrt{A^2+\frac{32}{27}a_3}}=2.$$

But then $\sqrt{A^2+\frac{32}{27}a_3}=\frac{16}{27}$, so $A=\frac{4}{27}$, $a_3=\frac{5}{18}$, and so $C=\frac{29}{54}$. These values make the left hand side $\frac{23}{9}<\frac{8}{3}$. If the maximum occurs when $A=0$, then the inequality is equivalent to $(\frac{4}{3}-C)^2\geqslant\frac{32}{27}(1-C)$, which is true.

When $a_1=a_3=0$ we have $0\leqslant A\leqslant C$, $\frac{5}{4}A+C+a_2=1$, and we want to show

$$3A+C+\sqrt{A^2+\frac{32}{27}a_2}+\sqrt{C^2+\frac{32}{27}a_2}\leqslant\frac{8}{3}.$$

If the maximum is attained in the interior of the domain, by Lagrange multipliers

$$3+\frac{A}{\sqrt{A^2+\frac{32}{27}a_2}}=\frac{5}{4}+\frac{5}{4}\frac{C}{\sqrt{C^2+\frac{32}{27}a_2}},$$

which cannot happen as $\frac{C}{\sqrt{C^2+\frac{32}{37}a_2}}\leqslant 1$. If the maximum occurs when $A=0$, then the left hand side is at most $C+\sqrt{2C^2+\frac{128}{27}a_2}$, and the inequality then becomes $(\frac{8}{3}-C)^2\geqslant 2C^2+\frac{128}{27}(1-C)$, which is true as $C\leqslant 1$.

Now, when $B=C$, we have by symmetry two distinct cases, when $a_2=a_3=0$ and when $a_1=a_3=0$.

When $a_2=a_3=0$, we have $0\leqslant A\leqslant C$, $\frac{7}{12}A+\frac{5}{3}C+a_1=1$, and we want to show

$$2A+2C+2\sqrt{C^2+\frac{32}{27}a_1}\leqslant\frac{8}{3}.$$

If the maximum is attained in the interior of the domain, by Lagrange multipliers

$$-\frac{120}{21}+2+\frac{2C}{\sqrt{C^2+\frac{32}{27}a_1}}=0$$

which cannot happen as $\frac{2C}{\sqrt{C^2+\frac{32}{27}a_1}} \leqslant 2$. If the maximum occurs when $A = 0$, then the inequality is equivalent to $(\frac{4}{3} - C)^2 \geqslant C^2 + \frac{32}{27}(1 - \frac{5}{3}C)$, which is true as $C \leqslant \frac{3}{5}$.

Finally, when $a_1 = a_3 = 0$, we have $0 \leqslant A \leqslant C$, $\frac{7}{12}A + \frac{5}{3}C + a_2 = 1$, and we want to show

$$A + 3C + \sqrt{A^2 + \frac{32}{27}a_2} + \sqrt{C^2 + \frac{32}{27}a_2} \leqslant \frac{8}{3}.$$

By Jensen's inequality applied to $\sqrt{x}$, it suffices to show that

$$A + 3C + \sqrt{2A^2 + 2C^2 + \frac{128}{27}a_2} \leqslant \frac{8}{3}.$$

As $A + 3C \leqslant \frac{8}{3}(\frac{7}{12}A + \frac{5}{3}C) \leqslant \frac{8}{3}$, we can move $A + 3C$ to the right hand side, square the equation, and then replace a_2 with $1 - \frac{7}{12}A - \frac{5}{3}C$ to get

$$(A + 3C - \frac{8}{3})^2 - (2A^2 + 2C^2 + \frac{128}{27}(1 - \frac{7}{12}A - \frac{5}{3}C)) \geqslant 0.$$

If the minimum is attained in the interior of the domain, then we have the partial derivatives with respect to A and C vanishing. This yields

$$2(A + 3C - \frac{8}{3}) - 4A + \frac{224}{81} = 0$$
$$6(A + 3C - \frac{8}{3}) - 4C + \frac{640}{81} = 0,$$

so $A = \frac{16}{81}, C = \frac{40}{81}$. Evaluating at this point yields $\frac{256}{2187} \geqslant 0$. The boundary case $A = C$ is identical to the case $A = B = C$ when $a_1 = a_3 = 0$. The boundary case $A = 0$ is equivalent to $(\frac{8}{3} - 3C)^2 \geqslant 2C^2 + \frac{128}{27}(1 - \frac{5}{3}C)$ for $0 \leqslant C \leqslant \frac{3}{5}$, and the boundary case $a_2 = 0$ is equivalent to $(\frac{8}{3} - 3C - (\frac{12}{7} - \frac{20}{7}C))^2 \geqslant 2C^2 + 2(\frac{12}{7} - \frac{20}{7}C)^2$ for $\frac{4}{9} \leqslant C \leqslant \frac{3}{5}$ (the lower bound on C when $a_2 = 0$ follows from $1 = \frac{7}{12}A + \frac{5}{3}C \leqslant (\frac{7}{12} + \frac{5}{3})C = \frac{9}{4}C$).

8.6 Proof of Theorem 1.1

In this section, we prove Theorem 1.1 using Lemma 2.1.

We can assume every vertex has positive degree. Order the vertices $v_1, \ldots, v_n$ such that $d(v_1) \geqslant d(v_2) \geqslant \ldots \geqslant d(v_n)$. Let V^{high} consist of the $t = m^\alpha$ vertices of highest degree with some fixed $0 < \alpha < \frac{1}{3}$. Then $|E(\mathscr{H})^3| < m$ and

$$m^\alpha d(v_{t+1}) < \sum_{v \in \mathscr{H}} d(v) = 3m,$$

so

$$d(v_{t+1}) \leqslant 3m^{1-\alpha}.$$

Hence in particular,

$$\sum_{i=t+1}^{n} d(v_i)^2 < 3m^{1-\alpha} \sum_{i=t+1}^{n} d(v_i) \leqslant 9m^{2-\alpha}.$$

Recall $x_1, x_2, x_3, b_{12}, b_{13}, b_{23}$ were defined in the introduction and satisfy the constraints $b_{ij} \geqslant \max(2x_i, 2x_j, \frac{1}{2}x_k)$ that appear in the hypothesis of Lemma 8.2.1. Further define a_i for $i \in \{1,2,3\}$ to be the number of edges in $E(\mathscr{H})^1$ such that the high degree vertex lies in V_i^{high}, and let c be the number of edges in $E(\mathscr{H})^0$. We have

$$x_1 + x_2 + x_3 + a_1 + a_2 + a_3 + b_{12} + b_{13} + b_{23} + c = m - |E(\mathscr{H})^3|$$

which, given $|E(\mathscr{H})^3| < m$, is equivalent to

$$\frac{x_1 + x_2 + x_3 + a_1 + a_2 + a_3 + b_{12} + b_{13} + b_{23} + c}{m - |E(\mathscr{H})^3|} = 1.$$

Apply Lemma 8.2.1 to all of our variables scaled by $\frac{1}{m-|E(\mathscr{H})^3|}$ to get $q_1, q_2, q_3 \in [0,1]$ with $q_1 + q_2 + q_3 = 2$ such that the following inequalities are satisfied:

$$\begin{aligned}
q_1(b_{23} + x_2 + x_3) + q_1^2(a_2 + a_3) + q_1^3 c &\leqslant \frac{8}{27}(m - |E(\mathscr{H})^3|) \\
q_2(b_{13} + x_1 + x_3) + q_2^2(a_1 + a_3) + q_2^3 c &\leqslant \frac{8}{27}(m - |E(\mathscr{H})^3|) \\
q_3(b_{12} + x_1 + x_2) + q_3^2(a_1 + a_2) + q_3^3 c &\leqslant \frac{8}{27}(m - |E(\mathscr{H})^3|).
\end{aligned}$$

Define $p_1, p_2, p_3 \in [0,1]$ by $p_i = 1 - q_i$. Place the low degree vertices independently into V_i with probability p_i. Note that there are $b_{12} + x_1 + x_2$ edges e in $E(\mathscr{H}^2)$ for which the two high degree vertices are outside V_3. For each such edge e, the probability that we place the low degree vertex outside V_3 is q_3, i.e., $\mathbb{P}(e \cap V_3 = \emptyset) = q_3$. Moreover, there are $a_1 + a_2$ edges in e in $E(\mathscr{H}^1)$ for which the only high degree vertices is outside V_3. For each such edge e, the probability that we place the two low degree vertices outside V_3 is q_3^2, i.e., $\mathbb{P}(e \cap V_3 = \emptyset) = q_3^2$. Furthermore, there are c edges e in $E(\mathscr{H}^0)$. For each such edge e, the probability that we place all three low degree vertices outside V_3 is q_3^3, i.e., $\mathbb{P}(e \cap V_3 = \emptyset) = q_3^3$. All the other $m - |E(\mathscr{H})^3|$ edges e, not counting those in $E(\mathscr{H})^3$, automatically have a vertex in V_3 regardless of how we place the low degree vertices. Then with $d(V_i)$ the number of edges with at least one vertex in V_i, we have

$$\mathbb{E}(d(V_3)) \geqslant m - |E(\mathscr{H})^3| - q_3(b_{12} + x_1 + x_2) - q_3^2(a_1 + a_2) - q_3^3 c \geqslant \frac{19}{27}(m - |E(\mathscr{H})^3|.$$

Similarly, we get $\mathbb{E}(d(V_i)) \geqslant \frac{19}{27}(m - |E(\mathscr{H})^3|)$ for all i.

We recall that the Azuma–Hoeffding inequality [121] asserts that if $X_1,\dots,X_n$ are independent random variables with values in $\{1,\dots,k\}$, and $f:\{1,\dots,k\}^n\to\mathbb{N}$ such that $|f(Y)-f(Y')|\leqslant d_i$ if Y and Y' differ only on the i'th coordinate, then for $t\geqslant 0$ we have $\mathbb{P}(f(X_1,\dots,X_n)\leqslant \mathbb{E}(f(X_1,\dots,X_n))-t)\leqslant e^{-\frac{2t^2}{\Sigma d_i^2}}$.

By encoding the distribution of the low degree vertices in the three parts we can construct the random variable $d(V_i):\{1,2,3\}^{|V^{\text{low}}|}\to\mathbb{N}$. Note that if $Y,Y'\in\{1,2,3\}^{|V^{\text{low}}|}$ differ only on the i'th coordinate, then $|d(V_i)(Y)-d(V_i)(Y')|\leqslant d(v_i)$.

By the Azuma–Hoeffding inequality with $k=3$ and $n=|V^{\text{low}}|$, we have

$$\mathbb{P}(d(V_i)<\mathbb{E}(d(V_i))-z)\leqslant e^{\frac{-2z^2}{\Sigma_{i=t+1}^{n} d(v_i)^2}}<e^{-\frac{2z^2}{9m^{2-\alpha}}}.$$

Taking $z=\sqrt{(9/2)\ln 3}m^{1-\frac{\alpha}{2}}$ yields $\mathbb{P}(d(V_i)<\mathbb{E}(d(V_i))-z)<\frac{1}{3}$. Hence there is a partition which has $d(V_i)>\frac{19}{27}(m-|E(\mathscr{H})^3|)-z$ simultaneously for $i=1,2,3$. Because $|E(\mathscr{H})^3|)=O(m^{3\alpha})$ and $z=O(m^{1-\frac{\alpha}{2}})$, taking $\alpha=\frac{2}{7}$ yields $d(V_i)>\frac{19}{27}m-O(m^{6/7})$ simultaneously for $i=1,2,3$ as desired.

8.7 Results of Computation

Tables 1–6 show the results of computation. We denote by ε the side length of the cubes used in the interval method (when applicable). We use $*$ to indicate that the minimum value was proved in the body of the chapter. We record a proven lower bound in the "Lower Bound" column. Finally, we note that when $B=A$ in **System1f**, the case is identical to the corresponding case in **System1e**.

Table 8.1 System1a

$x_{23}=0$	$b_{13}=8x_{23}$	$b_{23}=2x_{23}$	$a_1=0$	$a_2=0$	$a_3=0$	ε	Lower Bound
✓	✓	✓					2∗
✓	✓		✓			0.002	2.078
✓	✓			✓		0.002	2.077
✓	✓				✓	0.002	2.077
✓		✓	✓			0.002	2.077
✓		✓		✓		0.002	2.078
✓		✓			✓	0.002	2.077
✓			✓	✓		0.002	2.085
✓			✓		✓	0.002	2.086
✓				✓	✓	0.002	2.086
	✓	✓	✓			0.001	2.005
	✓	✓		✓		0.002	2.033
	✓	✓			✓	0.002	2.033
	✓		✓	✓		0.002	2.057
	✓		✓		✓	0.002	2.057
	✓			✓	✓	0.002	2.043
		✓	✓	✓		0.002	2.045
		✓	✓		✓	0.002	2.044
		✓		✓	✓	0.002	2.041
			✓	✓	✓	0.002	2∗ or 2.046

Table 8.2 System1b

$x_{23}=0$	$b_{13}=2x_1$	$x_1=4x_{23}$	$a_1=0$	$a_2=0$	$a_3=0$	ε	Lower Bound
✓	✓	✓					2∗
✓	✓		✓			0.002	2.042
✓	✓			✓		0.002	2.069
✓	✓				✓	0.002	2.069
✓		✓	✓			0.002	2.077
✓		✓		✓		0.002	2.078
✓		✓			✓	0.002	2.077
✓			✓	✓		0.002	2.072
✓			✓		✓	0.002	2.072
✓				✓	✓	0.002	2.070
	✓	✓	✓			0.001	2.005
	✓	✓		✓		0.002	2.033
	✓	✓			✓	0.002	2.033
	✓		✓	✓		0.002	2.026
	✓		✓		✓	0.002	2.026
	✓			✓	✓	0.002	2.024
		✓	✓	✓		0.002	2.045
		✓	✓		✓	0.002	2.044
		✓		✓	✓	0.002	2.041
			✓	✓	✓	0.002	2∗ or 2.025

Table 8.3 System1c

$x_3 = 0$	$x_2 = x_3$	$4x_2 = x_1$	$a_1 = 0$	$a_2 = 0$	$a_3 = 0$	ε	Lower Bound
✓	✓	✓					$2*$
✓	✓		✓			0.002	2.042
✓	✓			✓		0.002	2.069
✓	✓				✓	0.002	2.069
✓		✓	✓			0.001	2.019
✓		✓		✓		0.002	2.036
✓		✓			✓	0.002	2.037
✓			✓	✓		0.002	2.027
✓			✓		✓	0.002	2.028
✓				✓	✓	0.002	2.026
	✓	✓	✓			0.001	2.005
	✓	✓		✓		0.002	2.033
	✓	✓			✓	0.002	2.033
	✓		✓	✓		0.002	2.026
	✓		✓		✓	0.002	2.026
	✓			✓	✓	0.002	2.024
		✓	✓	✓		0.001	2.042
		✓	✓		✓	0.001	2.042
		✓		✓	✓	0.001	2.042
			✓	✓	✓	0.001	$2*$ or 2.033

Table 8.4 System1d

$b_{12} = 8x_2$	$x_2 = x_3$	$x_3 = 0$	$a_1 = 0$	$a_2 = 0$	$a_3 = 0$	ε	Lower Bound
✓	✓	✓					$2*$
✓	✓		✓			0.002	2.077
✓	✓			✓		0.002	2.077
✓	✓				✓	0.002	2.078
✓		✓	✓			0.001	2.019
✓		✓		✓		0.002	2.036
✓		✓			✓	0.002	2.037
✓			✓	✓		0.002	2.047
✓			✓		✓	0.002	2.047
✓				✓	✓	0.002	2.041
	✓	✓	✓			0.001	2.005
	✓	✓		✓		0.002	2.033
	✓	✓			✓	0.002	2.033
	✓		✓	✓		0.002	2.044
	✓		✓		✓	0.002	2.045
	✓			✓	✓	0.002	2.041
		✓	✓	✓		0.001	2.042
		✓	✓		✓	0.001	2.042
		✓		✓	✓	0.001	2.042
			✓	✓	✓	0.001	$2*$ or 2.042

Table 8.5 System1e

$A=0$	$B=A$	$C=B$	$a_1=0$	$a_2=0$	$a_3=0$	ε	Lower Bound
✓	✓	✓					2∗
✓	✓		✓			0.002	2.077
✓	✓			✓		0.002	2.077
✓	✓				✓	0.002	2.078
✓		✓	✓			0.002	2.075
✓		✓		✓		0.002	2.076
✓		✓			✓	0.002	2.076
✓			✓	✓		0.002	2.086
✓			✓		✓	0.002	2.085
✓				✓	✓	0.002	2.084
	✓	✓	✓				2∗
	✓	✓		✓			2∗
	✓	✓			✓		2∗
	✓		✓	✓			2∗
	✓		✓		✓		2∗
	✓			✓	✓		2∗
		✓	✓	✓			2∗
		✓	✓		✓		2∗
		✓		✓	✓		2∗
			✓	✓	✓		2∗

Table 8.6 System1f

$A=0$	$B=A$	$C=B$	$a_1=0$	$a_2=0$	$a_3=0$	ε	Lower Bound
✓	✓	✓					2∗
✓	✓		✓			0.002	2.077
✓	✓			✓		0.002	2.077
✓	✓				✓	0.002	2.078
✓		✓	✓			0.002	2.075
✓		✓		✓		0.002	2.076
✓		✓			✓	0.002	2.076
✓			✓	✓		0.002	2.086
✓			✓		✓	0.002	2.085
✓				✓	✓	0.002	2.084
	✓	✓	✓				2∗
	✓	✓		✓			2∗
	✓	✓			✓		2∗
	✓		✓	✓			2∗
	✓		✓		✓		2∗
	✓			✓	✓		2∗
		✓	✓	✓			2∗
		✓	✓		✓		2∗
		✓		✓	✓		2∗
			✓	✓	✓		2∗

Chapter 9

Multicolour chain avoidance in the Boolean lattice

The work in this section is joint with Hunter Spink and is published in the European Journal of Combinatorics [143].

9.1 Introduction

The method of hypergraph containers, recently developed by Balogh, Morris and Samotij [7] and independently by Saxton and Thomason [134], inspired by the previous work of Conlon and Gowers [46], is an essential tool for counting independent sets in hypergraphs. Many natural problems can be phrased in this way, with the most direct applications toward determining the number of graphs on n vertices which avoid a collection of subgraphs. We refer the reader to the survey [8].

Recently, Balogh, Treglown and Wagner [10] showed that graph containers could be used in the Boolean lattice $(\mathscr{P}([n]),\subseteq)$ to give an alternate proof of Kleitman's result [105] counting the number of antichains in $\mathscr{P}([n])$ (we note that the proof in [105] precedes the notion of graph containers, but has similar ideas). Using the hypergraph container algorithm, independently Collares and Morris [45] and Balogh, Mycroft, and Treglown [9] were able to further count the number of antichains in a random subset of $\mathscr{P}([n])$, from which they were able to deduce the approximate size of the largest antichain therein.

Because every k-chain free set can be partitioned into $k-1$ antichains, the graph container lemma (i.e., for 2-uniform hypergraphs) suffices not only to count k-chain free sets, but also to create a small collection of small sized containers for k-chain free sets via a product construction. However, [45] directly constructs a set of hypergraph containers without exploiting this observation as an application of the recent advances in the hypergraph container lemma through balanced supersaturation results.

In this chapter we answer analogous questions in a weighted coloured setting by building on [45]'s demonstration of the hypergraph container lemma through balanced supersaturation in $\mathscr{P}([n])$. Fix colours $1,2,\dots,m$. Given a poset $\mathscr{P}$, a *colouring* of $\mathscr{P}$ is a function $c:\mathscr{P}\to\{1,\dots,m\}$. A *coloured*

poset $(\mathscr{P},c)$ is a poset equipped with a colouring. A *coloured subset* of a poset $\mathscr{P}$ (resp. coloured poset $(\mathscr{P},c')$) is a pair (S,c) with $S\subset\mathscr{P}$ and c a colouring of S (resp. the colouring of S agreeing with c'). Let $\mathscr{G}$ be a collection of coloured chain posets of lengths at least 2. Denoting disjoint union by $\sqcup$, we write

$$\mathscr{G}\coloneqq\mathscr{G}_2\sqcup\mathscr{G}_3\sqcup\ldots\sqcup\mathscr{G}_k$$

where $\mathscr{G}_i$ contains exclusively chains of length i. We say that a coloured subset (S,c) of $\mathscr{P}([n])$ *avoids all configurations from* $\mathscr{G}$, or is *valid with respect to* $\mathscr{G}$, if no element of $\mathscr{G}$ appears as a coloured subchain of (S,c). The present work is motivated by the following questions.

Question 9.1.1. What is the cardinality of the collection $\Lambda(\mathscr{G},n)$ (where the dependency on the number of colours m is suppressed here and throughout the chapter) of validly coloured subsets (S,c) of $\mathscr{P}([n])$ with respect to $\mathscr{G}$?

Question 9.1.2. Let $p_1,\ldots,p_m\in(0,1]$ be such that $\sum_{i=1}^m p_i\leqslant 1$. If we colour each element of $\mathscr{P}([n])$ independently with colour c_i with probability p_i and leave it uncoloured with probability $1-\sum p_i$, then what is the expected number of validly coloured subsets (S,c) of (the coloured part of) $\mathscr{P}([n])$ with respect to $\mathscr{G}$?

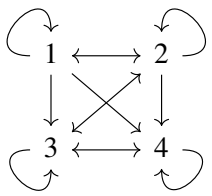
Multicoloured hypergraph container problems were only considered quite recently in the work of Falgas-Ravry, O'Connell, and Uzzell [64]. There it was shown that for a wide variety of coloured configuration avoidance problems, if there is a validly coloured subset using all but a $o(1)$ proportion of the vertices, then the number of validly coloured subsets can be estimated quite precisely (see Theorem 9.1.8).

The questions we consider are the first instances of coloured hypergraph container problems that we are aware of which work in the presence of a "sparse" extremal example. A separate interesting feature is that the hypergraph we work with is not uniform and no uniformity dominates, so when we iteratively apply the container algorithm we may have to use potentially different uniformities at each stage.

Example 9.1.3. We now describe some instructive examples. A coloured chain with colours $a_1,\ldots,a_\ell$ in increasing order is denoted $a_1\prec\ldots\prec a_\ell$.

1. Suppose we have 1 colour, i.e., $m=1$, and that we have only one forbidden chain $\mathscr{G}=\{(\underbrace{1\prec 1\prec\ldots\prec 1}_{k})\}$. Then $\Lambda(\mathscr{G},n)$ is the collection of k-chain free sets of $\mathscr{P}([n])$, and by [105, 10, 45] we have
$$|\Lambda(\mathscr{G},n)|=2^{(k-1)\binom{n}{n/2}(1+o(1))}.$$

2. Suppose that we have 4 colours, and let $\mathscr{G}$ be the 13 coloured 2-chains

$\mathscr{G} = \mathscr{G}_2 =$ 

, where $i \to j$ stands for $(i \prec j)$.

Equivalently for $(S,c) \in \Lambda(\mathscr{G},n)$, if $x,y \in S$ with $x \subsetneq y$, then we have $(c(x),c(y)) \in \{(3,1),(4,1),(4,2)\}$.

Notably in this example $\Lambda(\mathscr{G},n)$ contains two fundamentally different large families of valid configurations.

- Consider the family of coloured subsets (S,c) such that the elements of S have sizes in $\{\lfloor n/2 \rfloor, \lfloor n/2 \rfloor - 1\}$, and

$$c(x) = \begin{cases} 1 & |x| = \lfloor n/2 \rfloor \\ 3 \text{ or } 4 & |x| = \lfloor n/2 \rfloor - 1. \end{cases}$$

  In other words, we can colour all sets of size $\lfloor n/2 \rfloor$ with either 1 or nothing, and all sets of size $\lfloor n/2 \rfloor - 1$ with $3,4$ or nothing.
- Alternatively, we can colour all sets of size $\lfloor n/2 \rfloor$ with either $1,2$ or nothing, and all sets of size $\lfloor n/2 \rfloor - 1$ with either 4 or nothing.

In either case, the size of the family is

$$2^{\binom{n}{n/2}(1+o(1))} \cdot 3^{\binom{n}{n/2}(1+o(1))} = 6^{\binom{n}{n/2}(1+o(1))}.$$

As it turns out, by Theorem 9.1.10 it follows that this is also an upper bound to $|\Lambda(\mathscr{G},n)|$.

3. Suppose we have 2 colours, and $\mathscr{G} = \mathscr{G}_2 = \{(1 \prec 2)\}$. Then

$$|\Lambda(\mathscr{G},n)| \geqslant 2^{2^n(1+o(1))},$$

obtained by colouring each vertex either with 1 or nothing. As we will shortly see, this is also an upper bound by [64].

In general, the "dense" cases of Question 9.1.1 and Question 9.1.2 (see Remark 9.1.9) are solved by [64]. This occurs exactly when $\mathscr{G}$ does not contain a monochromatic chain of every colour. We are able to complete the analysis of these questions in the remaining "sparse" cases.

9.1.1 Preliminary definitions and main results

We recall from [64] some basic definitions of multicolour hypergraphs in our context.

Definition 9.1.4. For a poset $\mathscr{P}$, a *template* is a function

$$T\colon \mathscr{P} \to \mathscr{P}(\{1,\dots,m\})$$

from $\mathscr{P}$ to subsets of colours. Say that a template T is *supported on* $A \subset \mathscr{P}$ if $T(x) = \emptyset$ whenever $x \notin A$, and define $\mathrm{Supp}(T)$ to be the smallest set on which T is supported.

Definition 9.1.5. For a template T on a poset $\mathscr{P}$, we say a coloured subset (A,c) of $\mathscr{P}([n])$ is *contained in* T if $c(x) \in T(x)$ for all $x \in A$ (note that such an A is contained in $\mathrm{Supp}(T)$). We say that T is *valid with respect to* $\mathscr{G}$ if for every coloured subset (A,c) contained in T, no element of $\mathscr{G}$ appears as a coloured subchain of (A,c).

Definition 9.1.6. For a template T on a poset $\mathscr{P}$, denote by

$$\omega(T) = \sum_{x\in\mathscr{P}} \log(1+|T(x)|),$$

so that $e^{\omega(T)}$ is the number of coloured subsets contained in T.

We typically consider templates on the poset $\mathscr{P} = \mathscr{P}([n])$.

Remark 9.1.7. The reason we consider valid templates is that they provide a lower bound on the number of valid configurations

$$|\Lambda(\mathscr{G},n)| \geqslant \max_{T \text{ valid}} \prod_{x\in\mathscr{P}([n])} (1+|T(x)|) = e^{\max_{T \text{ valid}} \omega(T)}.$$

One would hope that this is the correct bound up to a $(1+o(1))$ factor in the exponent. The following theorem of [64], gives the sharp exponent in the "dense" case.

Theorem 9.1.8. [64, Corollary 3.34] Define the *maximal entropy* of $\mathscr{G}$ to be

$$\pi(\mathscr{G}) = \limsup_{n\to\infty} \frac{1}{2^n} \max_{T \text{ valid}} \omega(T).$$

We have

$$|\Lambda(\mathscr{G},n)| = e^{2^n(\pi(\mathscr{G})+o(1))}.$$

Remark 9.1.9. As mentioned earlier, this theorem correctly estimates $|\Lambda(\mathscr{G},n)|$ up to a $1+o(1)$ factor in the exponent when $\pi(\mathscr{G}) > 0$. We refer to this as the *dense* case, and it occurs precisely when $\mathscr{G}$ contains a monochromatic chain of every colour. In the *sparse* case when $\pi(\mathscr{G}) = 0$, the upper bound given by Theorem 9.1.8 is trivial and does not estimate the correct exponent up to a $1+o(1)$ factor.

We solve Question 9.1.1 in the sparse cases by estimating $|\Lambda(\mathscr{G},n)|$ up to a $1+o(1)$ factor in the exponent. This occurs when $\pi(\mathscr{G})=0$, i.e., $\mathscr{G}$ contains a monochromatic chain of every colour. Note that in this case, there exists $L=L(\mathscr{G})$ such that no validly coloured set (S,c) contains a chain of length L, and we may easily deduce a crude upper bound of $|\Lambda(\mathscr{G},n)|\leqslant(m+1)^{(L-1)\binom{n}{n/2}(1+o(1))}$ by using the L-chain containers constructed from either of [10, 45] (see Section 9.4).

The following is one of our main theorems, solving the sparse cases of Question 9.1.1.

Theorem 9.1.10. Suppose that $\mathscr{G}$ contains a monochromatic chain of every colour, and define the constant (independent of n for large enough n)

$$\omega_{crit}=\max_T\{\omega(T)\mid T \text{ a template on a chain poset, valid with respect to } \mathscr{G}\}.$$

Then we have

$$|\Lambda(\mathscr{G},n)|=e^{\omega_{crit}\binom{n}{n/2}(1+o(1))}.$$

Remark 9.1.11. Note that to compute the constant ω_{crit}, we only need to evaluate $\omega(T)$ for T ranging over the finite collection of valid templates on chain posets of length at most $L=L(\mathscr{G})$ as above, assigning each element a non-empty set.

As we will see later in Proposition 9.2.5, we can determine the exact maximum of $\omega(T)$ for valid templates T on $\mathscr{P}([n])$, and this value is $\omega_{crit}\binom{n}{n/2}(1+o(1))$. The lower bound in Theorem 9.1.10 will follow as in Remark 9.1.7.

Example 9.1.12.

- Suppose we are in the situation of Example 9.1.3 (1). The maximum ω_{crit} is attained by the template T on a length $k-1$ chain poset, sending every element to the set $\{1\}$. Then $\omega_{crit}=\omega(T)=\log 2^{k-1}$, and $|\Lambda(\mathscr{G},n)|=2^{(k-1)\binom{n}{n/2}(1+o(1))}$, recovering the results of [105, 10, 45].
- Suppose we are in the situation of Example 9.1.3 (2). Then the following two templates T on chain posets realize ω_{crit}. For $x\prec y$ a chain poset of length 2, the first such template has $T(x)=\{3,4\}$, $T(y)=\{1\}$, and the second has $T(x)=\{4\}$ and $T(y)=\{1,2\}$, attaining $\omega(T)=\omega_{crit}=\log 6$. We thus have $|\Lambda(\mathscr{G},n)|=6^{\binom{n}{n/2}(1+o(1))}$.

The answer to Question 9.1.2 requires a weighted version of $\omega(T)$.

Definition 9.1.13. Given $\mathscr{G}$ that contains a monochromatic chain of every colour, $\overline{\beta}=(\beta_1,\dots,\beta_m)\in(\mathbb{R}_{>0})^m$, and T a template on a poset $\mathscr{P}$, we denote by

$$|T(x)|_{\overline{\beta}}=\sum_{i\in T(x)}\beta_i$$
$$\omega(\overline{\beta},T)=\sum_{x\in\mathscr{P}}\log(1+|T(x)|_{\overline{\beta}})$$
$$\omega_{crit}(\overline{\beta})=\max_T\{\omega(\overline{\beta},T)\mid T \text{ a template on a chain poset, valid with respect to } \mathscr{G}\}.$$

The constant $\omega_{crit}(\overline{\beta})$ is also very easy to compute analogously to ω_{crit}, and yields the critical exponent in the following theorem.

Theorem 9.1.14. Suppose that $\mathscr{G}$ contains a monochromatic chain of every colour. With the probabilistic setup of Question 9.1.2, denoting $\overline{p} = (p_1, \dots, p_m)$ and V the number of validly coloured subsets (S, c) of (the coloured part of) $\mathscr{P}([n])$ with respect to $\mathscr{G}$, we have

$$\mathbb{E}(V) = e^{\omega_{crit}(\overline{p})\binom{n}{n/2}(1+o(1))}.$$

As before, the methods from [64] analogously answer the dense cases where there is some colour without a monochromatic forbidden chain. Theorem 9.1.14 thus completes the analysis of Question 9.1.2 in the sparse case.

For the remainder of the chapter, we write coloured sets as S rather than (S, c), suppressing the colouring c.

9.2 Strategy and auxiliary results

In this section we outline the strategy of the proof and present some auxiliary results. We start by formulating a weighted version of Theorem 9.1.10 and Theorem 9.1.14 which encompasses both of them. Throughout the rest of the chapter we always assume that $\mathscr{G}$ contains a monochromatic chain of each colour and refer to such a $\mathscr{G}$ as *sparse*. In this section, we let $\mathscr{G} := \mathscr{G}_2 \sqcup \ldots \sqcup \mathscr{G}_k$ by a fixed collection of forbidden coloured chains with colours $1, \dots, m$, and we let

$$\overline{\beta} = (\beta_1, \dots, \beta_m) \in (\mathbb{R}_{>0})^m$$

be a fixed sequence of positive real weights.

Definition 9.2.1. We define a measure μ on coloured subsets of $\mathscr{P}([n])$ by assigning to a subset $S \subset \mathscr{P}([n])$ with a colouring $c\colon S \to \{1, \dots, m\}$, the weight

$$\mu(\overline{\beta}, S) = \prod_{x \in S} \beta_{c(x)}$$

and extending it additively. In particular, for a collection Λ of coloured subsets of $\mathscr{P}([n])$, we have

$$\mu(\overline{\beta}, \Lambda) = \sum_{S \in \Lambda} \mu(\overline{\beta}, S).$$

Note that when $\beta_i = 1$ for all i we have that $\mu(\overline{\beta}, S) = 1$ for every coloured subset S, and $\mu(\overline{\beta}, \Lambda) = |\Lambda|$. We may now state the weighted reformulation of our main theorems.

Theorem 9.2.2. Suppose that $\mathscr{G}$ is sparse, then we have

$$\mu(\overline{\beta}, \Lambda(\mathscr{G}, n)) = e^{\omega_{crit}(\overline{\beta})\binom{n}{n/2}(1+o(1))}.$$

Remark 9.2.3. Theorem 9.2.2 specializes to Theorem 9.1.10 when all $\beta_i = 1$, and specializes to Theorem 9.1.14 when $\beta_i = p_i$.

9.2.1 Proof of the lower bound of Theorem 9.2.2

We now prove two propositions which we then use to prove the lower bound in Theorem 9.2.2. The first proposition relates the weight of a template to the measure of the collection of coloured subsets contained in the template. The second proposition relates the maximum weight of a template to the critical weight.

Proposition 9.2.4. Given a template T, let Λ_T be the collection of coloured subsets contained in T. Then

$$\mu(\overline{\beta}, \Lambda_T) = e^{\omega(\overline{\beta}, T)}.$$

Proof. Note that

$$e^{\omega(\overline{\beta}, T)} = \prod_{x \in \mathrm{Supp}(T)} (1 + |T(x)|_{\overline{\beta}}) = \prod_{x \in \mathrm{Supp}(T)} (1 + \sum_{i \in T(x)} \beta_i),$$

and expanding out the product yields $\mu(\overline{\beta}, \Lambda_T)$. □

Proposition 9.2.5. The maximum value of $\omega(\overline{\beta}, T)$ where T is a valid template is attained for some T with $\mathrm{Supp}(T)$ a consecutive block of layers of $\mathscr{P}([n])$ which intersects the middle layer(s), and with the property that $T(x)$ depends only on the size of x. In particular, we have that

$$\max_{\text{T valid}} \omega(\overline{\beta}, T) = (1 + O(\frac{1}{n}))\omega_{crit}(\overline{\beta})\binom{n}{n/2}.$$

Proof. Our strategy will be to consider a valid template T' which maximizes $\omega(\overline{\beta}, T')$, and construct from it another valid template T that satisfies the conclusions of Proposition 9.2.5. Choose a uniformly random maximal chain C and consider the random variable

$$Z(C) = \sum_{x \in C} \binom{n}{|x|} \log(1 + |T'(x)|_{\overline{\beta}}).$$

By linearity of expectation, it is easy to see that

$$\mathbb{E}Z(C) = \omega(\overline{\beta}, T').$$

Therefore, there exists a chain C such that $Z(C) \geqslant \omega(\overline{\beta}, T')$. But then the template T defined by $T(x) = T'(y)$ with $y \in C$ the unique element such that $|x| = |y|$ satisfies

$$\omega(\overline{\beta}, T) = Z(C) \geqslant \omega(\overline{\beta}, T').$$

By construction T is valid since $\mathscr{G}$ consists exclusively of chains, and by maximality of T', $\omega(\overline{\beta}, T) = \omega(\overline{\beta}, T')$ is maximal. Clearly the maximality of such a T further implies that $\mathrm{Supp}(T)$ is a consecutive block of layers containing the middle layer. □

We are now ready to prove the lower bound in Theorem 9.2.2.

Proof of the lower bound in Theorem 9.2.2. Let T be the extremal template from Proposition 9.2.5 and let Λ_T be the collection of coloured subsets contained in T. Then by Proposition 9.2.4 and Proposition 9.2.5 we have that

$$\mu(\overline{\beta}, \Lambda(\mathscr{G}, n)) \geqslant \mu(\overline{\beta}, \Lambda_T) = e^{\omega(\overline{\beta}, T)} = e^{(1+O(\frac{1}{n}))\omega_{crit}(\overline{\beta})\binom{n}{n/2}},$$

which gives the desired lower bound. □

9.2.2 Proof of the upper bound of Theorem 9.2.2 assuming a balanced supersaturation result

Now we describe the outline of the proof of the upper bound in Theorem 9.2.2. The main thrust of the proof is identical to that of [45]. The key new ideas are to create a balanced supersaturation result that works for templates and to implement the container lemma in a way which handles simultaneously the various uniformities of $\mathscr{G}$.

Our goal will be to find a collection $\mathscr{C}$ of $e^{o(1)\binom{n}{n/2}}$ templates with each template $T \in \mathscr{C}$ having $w(\overline{\beta}, T) \leqslant (\omega_{crit}(\overline{\beta}) + o(1))\binom{n}{n/2}$ such that every validly coloured subset of $\mathscr{P}([n])$ is contained in some template $T \in \mathscr{C}$. Then by a union bound we can conclude Theorem 9.2.2.

To accomplish this, we will use the following hypergraph container lemma.

Given a hypergraph $\mathscr{H}$, we denote $v(\mathscr{H})$ for the vertices of $\mathscr{H}$, $e(\mathscr{H})$ for the edges of $\mathscr{H}$, and we recall for $A \subset v(\mathscr{H})$ the standard notations $d_{\mathscr{H}}(A)$ for the number of hyperedges of $\mathscr{H}$ which contain A, and jth codegree $\Delta_j(\mathscr{H}) = \max_{|A|=j} d_{\mathscr{H}}(A)$.

Lemma 9.2.6. [7, 134] For every $K \in \mathbb{N}$ and $c > 0$ there exists $\varepsilon > 0$ such that the following holds. Let $\tau \in (0,1)$ and suppose that $\mathscr{H}$ is a K-uniform hypergraph on N vertices such that for $1 \leqslant j \leqslant K$ we have

$$\Delta_j(\mathscr{H}) \leqslant c \cdot \tau^{j-1} \frac{e(\mathscr{H})}{N}.$$

Then there exists a family $\mathscr{C}$ of subsets of $v(\mathscr{H})$, and a function $f\colon \mathscr{P}(v(\mathscr{H})) \to \mathscr{C}$ such that

1. For every $I \in \mathscr{I}(\mathscr{H})$, there exists $F \subset I$ with $|F| \leqslant K\tau N$ and $I \subset F \cup f(F)$
2. $|C| \leqslant (1-\varepsilon)N$ for every $C \in \mathscr{C}$.

In order to use the hypergraph container lemma we translate between validly coloured subsets of $\mathscr{P}([n])$ and independent subsets of a certain hypergraph. We consider the following ambient non-uniform hypergraph $\mathscr{A}$ defined by

$$v(\mathscr{A}) = \mathscr{P}([n]) \times \{1,\dots,m\},$$

$$e(\mathscr{A}) = \bigcup_{\ell=2}^{k} \{\{(x_1,i_1),\dots,(x_\ell,i_\ell)\} \mid x_1 \subsetneq \dots \subsetneq x_\ell \text{ in } \mathscr{P}([n]) \text{ and } (i_1 \prec \dots \prec i_\ell) \in \mathscr{G}_\ell\}.$$

By construction, a validly coloured subset of $\mathscr{P}([n])$ can be viewed as an independent set in $\mathscr{A}$, though we remark that this is not a 1-1 correspondence. Also, there is a natural 1-1 correspondence between templates T and subsets of $v(\mathscr{A})$, where we assign to a template T the subset of all (x,c) with $c \in T(x)$. By a slight abuse of notation we sometimes view T as a subset of vertices in $v(\mathscr{A})$ and sometimes view T as the induced sub-hypergraph

$$T = \mathscr{A}|_T \subset \mathscr{A}.$$

The notion of order of the sub-hypergraph associated to T is related to the notion of weight of T by

$$|T| = |v(\mathscr{A}|_T)| = \Theta(\omega(\overline{\beta},T)),$$

i.e., it is within a constant factor of $\omega(\overline{\beta},T)$.

Our desired set of hypergraph containers will correspond to a family of templates which efficiently contains validly coloured subsets of $\mathscr{P}([n])$.

Lemma 9.2.7. Suppose that $\mathscr{G}_k$ contains all coloured chains of length k. For every $\alpha > 0$ there exists a $\delta > 0$ such that the following holds. Let $n \in \mathbb{N}$ and suppose that T is a template of $\mathscr{P}([n])$ supported on sets of size between $\frac{n}{3}$ and $\frac{2n}{3}$ such that $\omega(\overline{\beta},T) \geqslant (\omega_{crit}(\overline{\beta}) + \alpha)\binom{n}{n/2}$. Then there exists $2 \leqslant \ell \leqslant k$ and there exists an l-uniform sub-hypergraph $\mathscr{H}_\ell$ of T such that

$$e(\mathscr{H}_\ell) \geqslant \delta^\ell n^{\ell-1}\binom{n}{n/2}, \text{ and}$$
$$\Delta_j(\mathscr{H}_\ell) \leqslant (\delta n)^{\ell-j}$$

for $1 \leqslant j \leqslant \ell$.

Corollary 9.2.8. For every $\alpha \in (0,1)$ there exists an $\varepsilon > 0$ such that the following holds. Let $n \in \mathbb{N}$ and suppose that T is a template of $\mathscr{P}([n])$ supported on sets of size between $\frac{n}{3}$ and $\frac{2n}{3}$ such that $\omega(\overline{\beta}, T) \geqslant (\omega_{crit}(\overline{\beta}) + \alpha)\binom{n}{n/2}$. Then there exists a family $\mathscr{C}$ of sub-templates of T such that

1. For every validly coloured subset I with respect to $\mathscr{G}$ contained in T, there is a $T' \in \mathscr{C}$ such that I is contained in T'.
2. We have $|\mathscr{C}| \leqslant e^{O(1)\frac{\log n}{n}|T|}$ for some constant $O(1)$ independent of α.
3. We have $|T'| \leqslant (1-\varepsilon)|T|$ for every $T' \in \mathscr{C}$.

Proof of Corollary 9.2.8 assuming Lemma 9.2.7. It suffices to prove this when $\mathscr{G}_k$ contains all coloured chains of length k (as we can always augment $\mathscr{G}$ with all coloured chains of length km without changing the valid configurations). Partition $T \subset v(\mathscr{A})$ into sets $T_0 \cup T_1 \cup \ldots \cup T_r$ for some $r \geqslant 1$ such that

$$\omega(\overline{\beta}, T_0) < (\omega_{crit}(\overline{\beta}) + \alpha)\binom{n}{n/2}, \text{ and}$$
$$\omega(\overline{\beta}, T_i) = (\omega_{crit}(\overline{\beta}) + \alpha)\binom{n}{n/2} + O(1).$$

By Lemma 9.2.7, there exists $\delta = \delta(\alpha)$ such that for each of $1 \leqslant i \leqslant r$ there exists an ℓ_i-uniform sub-hypergraph $\mathscr{H}^i = \mathscr{H}^i_{\ell_i}$ of the templates T_i for some $2 \leqslant \ell_i \leqslant k$ with the property that

$$e(\mathscr{H}^i) \geqslant \delta^{\ell_i} n^{\ell_i - 1}\binom{n}{n/2}, \text{ and}$$
$$\Delta_j(\mathscr{H}^i) \leqslant (\delta n)^{\ell_i - j}, \text{ for all } 1 \leqslant j \leqslant \ell_i.$$

Let $2 \leqslant \ell \leqslant k$ be the most frequent uniformity. Construct the ℓ-uniform sub-hypergraph $\mathscr{H}$ of T with $v(\mathscr{H}) = v(T)$ and

$$e(\mathscr{H}) \coloneqq \bigsqcup_{\ell_i = \ell} e(\mathscr{H}^i).$$

By construction we have

$$|e(\mathscr{H})| \geqslant \frac{r}{k}\delta^{\ell} n^{\ell-1}\binom{n}{n/2}$$
$$|v(\mathscr{H})| = |v(T)| = \sum_{i=0}^{r} |v(T_i)| = O(1)\sum_{i=0}^{r} \omega(\overline{\beta}, T_i) = O(1)r\binom{n}{n/2}$$
$$\Delta_j(\mathscr{H}) \leqslant (\delta n)^{\ell - j} \text{ for all } 1 \leqslant j \leqslant \ell.$$

Set $\tau = \frac{1}{n}$ and $c = O(1)k\delta^{-k}$, and apply Lemma 9.2.6 to the l-uniform hypergraph $\mathscr{H}$ to obtain the following. There exists ε depending only on c, k and there exists a collection $\mathscr{C}$ of subtemplates T' of T, and a function $f\colon \mathscr{P}(v(T)) \to \mathscr{C}$ such that

1. For every $I \in \mathscr{I}(T)$, there exists $F \subset I$ with $|F| \leqslant \frac{k}{n}|T|$ and $I \subset F \cup f(F)$
2. $|T'| \leqslant (1-\varepsilon)|T|$ for every $T' \in \mathscr{C}$.

Set $\mathscr{C}' \coloneqq \{F \cup f(F) : F \in \mathscr{C}\}$ and note that

1. For every $I \in \mathscr{I}(T)$, there exists $T' \in \mathscr{C}'$ such that $I \subset T'$
2. $|\mathscr{C}'| \leqslant \frac{k}{n}|T|\binom{|T|}{\frac{k}{n}|T|} = e^{O(1)\frac{\log(n)}{n}|T|}$.
3. $|T'| \leqslant (1-\varepsilon+o(1))|T|$ for every $T' \in \mathscr{C}'$.

□

Proof of upper bound of Theorem 9.2.2 assuming Corollary 9.2.8. Let $\mathscr{P}([n])'$ be all vertices in $x \in \mathscr{P}([n])$ with $|x| \in (\frac{n}{3}, \frac{2n}{3})$, and let $\Lambda'(\mathscr{G},n)$ be the validly coloured subsets of $\mathscr{P}([n])'$. We have

$$\mu(\overline{\beta},\Lambda(\mathscr{G},n)) \leqslant \left(\prod_{x \notin \mathscr{P}([n])'}(1+\sum_{i=1}^{m}\beta_i)\right)\mu(\overline{\beta},\Lambda'(\mathscr{G},n)) = e^{o(1)\binom{n}{n/2}}\mu(\overline{\beta},\Lambda'(\mathscr{G},n)).$$

Therefore it is enough to prove that for every $\alpha > 0$ we have

$$\mu(\overline{\beta},\Lambda'(\mathscr{G},n)) \leqslant e^{(\omega_{crit}(\overline{\beta})+\alpha+o(1))\binom{n}{n/2}}.$$

Fix a threshold value $1 > \alpha > 0$. Starting with $\mathscr{A}|_{\mathscr{P}([n])' \times \{1,\dots,m\}}$, we iteratively apply Corollary 9.2.8 until we obtain a family $\mathscr{C}$ of subtemplates T with $\omega(\overline{\beta},T) \leqslant (\omega_{crit}(\overline{\beta})+\alpha)\binom{n}{n/2}$. This is encoded by a branching process where a template T with $\omega(\overline{\beta},T) \geqslant (\omega_{crit}(\overline{\beta})+\alpha)\binom{n}{n/2}$ splits into subtemplates $T_1, T_2, \dots, T_s$ such that each validly coloured subset contained in T is contained in some T_i. By Corollary 9.2.8, there exists an $\varepsilon = \varepsilon(\alpha) > 0$ such that we have $s \leqslant e^{O(1)\frac{\log(n)}{n}|T|}$, and $|T_i| \leqslant (1-\varepsilon)|T|$.

Because our initial set has size at most $m2^n$, and the size of the templates decreases by a factor of $(1-\varepsilon)$ each iteration, each template at level i in this branching process splits into at most $e^{O(1)\frac{\log(n)}{n}(1-\varepsilon)^i m2^n}$ other templates. Therefore, the final collection $\mathscr{C}$ of templates has cardinality bounded above by

$$|\mathscr{C}| \leqslant \prod_{i=0}^{\infty} e^{O(1)\frac{\log(n)}{n}(1-\varepsilon)^i m2^n} = e^{O(1)\frac{log(n)}{n}\varepsilon^{-1}m2^n} = e^{o(1)\binom{n}{n/2}}.$$

Note that each set in $\Lambda'(\mathscr{G},n)$ is contained in some $T \in \mathscr{C}$. Therefore, letting Λ_T be the collection of coloured subsets contained in T we have by Proposition 9.2.4

$$\mu(\overline{\beta},\Lambda'(\mathscr{G},n)) \leqslant \sum_{T\in\mathscr{C}}\mu(\overline{\beta},\Lambda_T) = \sum_{T\in\mathscr{C}} e^{\omega(\overline{\beta},T)} \leqslant e^{o(1)\binom{n}{n/2}}e^{(\omega_{crit}(\overline{\beta})+\alpha)\binom{n}{n/2}}.$$

□

9.3 Balanced Supersaturation

In this section, we prove Lemma 9.2.7. To do this, we prove a series of technical results adapted from [45] for our purposes. We fix a template T for the remainder of this section, and recall that by hypothesis $\mathscr{G}_k$ contains all coloured chains of length k.

Definition 9.3.1. Define the random variables $\mathbf{X}$ and $\mathbf{Y}$ on a uniformly chosen random maximal chain C in $\mathscr{P}([n])$. Let

$$\mathbf{X}(C) = \sum_{x \in C} \log(1 + |T(x)|_{\overline{\beta}}),$$

and $\mathbf{Y}(C)$ be the total number of coloured subchains of C contained in T which appear as a coloured chain in $\mathscr{G}$.

Definition 9.3.2. For $x \in \mathscr{P}([n])$ define the constant $X^x = \log(1 + |T(x)|_{\overline{\beta}})$. Define the random variable $\mathbf{Y}^x$ on a uniformly chosen random maximal length chain C in $\mathscr{P}([n])$ whose top element is x by setting $\mathbf{Y}^x(C)$ to be the number of coloured subchains of C contained in T whose top element is x and which appear as a coloured chain in $\mathscr{G}$.

Lemma 9.3.3. There are constants $C_1, C_2 > 0$ independent of n such that the following is true. For any $\alpha \in (0, C_2)$, if T is a template with $\omega(\overline{\beta}, T) \geqslant (\omega_{crit}(\overline{\beta}) + \alpha)\binom{n}{n/2}$, then there exists a vertex $x \in \mathrm{Supp}(T)$ such that

$$\mathbb{E}\mathbf{Y}^x \geqslant C_1 \alpha.$$

Proof. Take $C_3 = \log(1 + \sum_{i=1}^m \beta_i)$, $C_4 = \log(1 + \min(\beta_i))$ and take C_1, C_2 to be

$$C_1 = C_3^{-1} C_4 \frac{1}{2\omega_{crit}(\overline{\beta})}$$
$$C_2 = \min\{C_3^{-1} C_1^{-1} C_4, \omega_{crit}(\overline{\beta})\} = \omega_{crit}(\overline{\beta}).$$

First note that

$$\omega_{crit}(\overline{\beta}) \geqslant \mathbf{X}(C) - C_3 \mathbf{Y}(C).$$

Indeed, while $\mathbf{X}(C) > \omega_{crit}(\overline{\beta})$ we can find a forbidden coloured subchain of C contained in T. Deleting one by one vertices of C from forbidden subchains, $\mathbf{Y}(C)$ decreases each time by at least 1 and $\mathbf{X}(C)$ decreases each time by at most C_3.

Now suppose for the sake of contradiction that $\mathbb{E}\mathbf{Y}^x < C_1 \alpha$ for all $x \in \mathrm{Supp}(T)$. If for any $x \in \mathrm{Supp}(T)$ we have $X^x \leqslant C_3 \mathbb{E}\mathbf{Y}^x$, we obtain the contradiction (recalling $\alpha < C_2$)

$$\log(1 + \min(\beta_i)) \leqslant X^x \leqslant C_3 \mathbb{E}\mathbf{Y}^x < C_3 C_1 \alpha < C_4 = \log(1 + \min(\beta_i)).$$

Hence we have $X^x > C_3 \mathbb{E}Y^x$ for all $x \in \mathrm{Supp}(T)$. Writing $\mathbf{X}$ and $\mathbf{Y}$ as a sum of indicator functions and using linearity of expectation we have $\mathbb{E}(\mathbf{X}) = \sum_{x \in \mathrm{Supp}(T)} \frac{1}{\binom{n}{|x|}} X^x$, and $\mathbb{E}\mathbf{Y} = \sum_{x \in \mathrm{Supp}(T)} \frac{1}{\binom{n}{|x|}} \mathbb{E}\mathbf{Y}^x$. Thus we obtain the contradiction

$$\begin{aligned}
\omega_{crit}(\overline{\beta}) \geqslant \mathbb{E}(\mathbf{X} - C_3\mathbf{Y}) &= \sum_{x \in \mathrm{Supp}(T)} \frac{1}{\binom{n}{|x|}}(X^x - C_3\mathbb{E}\mathbf{Y}^x) \\
&\geqslant \sum_{x \in \mathrm{Supp}(T)} \frac{1}{\binom{n}{n/2}}(X^x - C_3\mathbb{E}\mathbf{Y}^x) \geqslant \frac{\omega(\overline{\beta},T)}{\binom{n}{n/2}} - \frac{|\mathrm{Supp}(T)|}{\binom{n}{n/2}} C_3C_1\alpha \\
&\geqslant \frac{\omega(\overline{\beta},T)}{\binom{n}{n/2}}(1 - C_4^{-1}C_3C_1\alpha) \geqslant (\omega_{crit}(\overline{\beta}) + \alpha)(1 - \frac{1}{2\omega_{crit}(\overline{\beta})}\alpha) \\
&= \omega_{crit}(\overline{\beta}) + \frac{1}{2}\alpha(1 - \frac{\alpha}{\omega_{crit}(\overline{\beta})}) \\
&> \omega_{crit}(\overline{\beta}).
\end{aligned}$$

□

The following lemma, inspired by a corresponding lemma from [45] (adapted from an argument of [47]), gives us very good control over the number of coloured chains below a given vertex. It is surprising that given our non-transitive family $\mathscr{G}$ of forbidden chains that we still retain such excellent control.

Lemma 9.3.4. There is a constant $Q \geqslant 0$ independent of n such that the following is true. For any $x \in X$, $i \leqslant k$, and C a chain of maximal length whose top element is x, let $\mathbf{Z}^x_{c_1 \succ \cdots \succ c_i}$ be the random variable such that $\mathbf{Z}^x_{c_1 \succ \cdots \succ c_i}(C)$ is the number of coloured subchains of C of length i contained in T, whose top element is x and is coloured $c_1 \succ \cdots \succ c_i$. Then

$$\mathbf{Z}^x_{c_1 \succ \cdots \succ c_i}(C) \leqslant Q + \mathbf{Y}^x(C).$$

Proof. Recall that $\mathscr{G}_k$ contains all chains of length k. Therefore, for $i = k$ any $Q \geqslant 0$ works. If $i < k$, it is enough to ensure that Q satisfies

$$\mathbf{Z}^x_{c_1 \succ \cdots \succ c_i}(C) \leqslant Q + \sum_{j=1}^{m} \mathbf{Z}^x_{c_1 \succ \underbrace{j \succ \cdots \succ j}_{k-1}}(C).$$

Indeed, if we can show this then by using the trivial bound

$$Q + \sum_{j=1}^{m} \mathbf{Z}^x_{c_1 \succ \underbrace{j \succ \cdots \succ j}_{k-1}}(C) \leqslant Q + \mathbf{Y}^x(C),$$

the conclusion follows immediately.

If $c_1 \notin T(x)$ then the result is trivially true for any choice of $Q \geqslant 0$, so we assume that c_1 is a valid choice at x. Denoting $s = |C \cap \mathrm{Supp}(T)|$, we have the trival bounds

$$\mathbf{Z}^x_{c_1 \succ \cdots \succ c_i}(C) \leqslant (s-1)^{i-1} \text{ and } \sum_{j=1}^{m} \mathbf{Z}^x_{c_1 \succ \underbrace{j \succ \cdots \succ j}_{k-1}}(C) \geqslant \binom{\frac{s-1}{m}}{k-1},$$

where the second bound follows by observing that the most frequent colour on C appears at least $\frac{s-1}{m}$ times.

Thus it suffices to take Q such that

$$(s-1)^{i-1} \leqslant Q + \binom{\frac{s-1}{m}}{k-1}$$

for every $1 \leqslant i < k$, and every $s \geqslant 1$. □

Proof of Lemma 9.2.7. We build an auxiliary sub-hypergraph $\mathscr{H}$ of T one edge at a time, ensuring with each new edge that $\Delta_j(\mathscr{H}_\ell) \leqslant (\delta n)^{\ell-j}$ holds for all ℓ and $1 \leqslant j \leqslant \ell$, where $\mathscr{H}_\ell$ is the set of ℓ-edges of $\mathscr{H}$, until for some ℓ we have $e(\mathscr{H}_\ell) \geqslant \delta^{\ell-1} n^\ell \binom{n}{n/2}$, and then we output $\mathscr{H}_\ell$.[1] In particular we assume that at the current stage $e(\mathscr{H}_\ell) < \delta^{\ell-1} n^\ell \binom{n}{n/2}$ for all ℓ. Note that given a coloured chain B contained in T that also appears in $\mathscr{G}$, if it cannot be added to $\mathscr{H}$, then there exists a coloured subchain B' satisfying $d_{\mathscr{H}_\ell}(B') = (\delta n)^{\ell-|B'|}$ for some ℓ, so adding B to $\mathscr{H}$ would violate the codegree condition. We will implicitly find such a coloured chain B which we can add to $\mathscr{H}$ by constructing it one vertex at a time from the top down, ensuring that no codegree condition among the subsets of B is violated at each step. The following claim shows that there are very few ways of extending a "good" B to a "bad" B with the addition of a vertex.

Claim 9.3.5. Given an i-chain $x_1 \supsetneq \ldots \supsetneq x_i$ with x_j coloured by $c_j \in T(x_j)$, then there are at most $O(1)\delta n$ choices for (x_{i+1}, c_{i+1}) with $x_i \supsetneq x_{i+1}$ and $c_{i+1} \in T(x_{i+1})$ such that there exists some $2 \leqslant \ell \leqslant k$, and *nonempty* $A \subset \{(x_1,c_1),\ldots,(x_i,c_i)\}$ with $|A| \leqslant \ell - 1$ such that $d_{\mathscr{H}_\ell}(A \cup \{(x_{i+1},c_{i+1})\}) = (\delta n)^{\ell-(|A|+1)}$.

Proof. Fix some $A \subset \{(x_1,c_1),\ldots,(x_i,c_i)\}$ and $2 \leqslant \ell \leqslant k$, and let $\mathscr{B}$ be the set of all (x_{i+1},c_{i+1}) with $x_{i+1} \subsetneq x_i$ and $c_{i+1} \in T(x_{i+1})$ and $d_{\mathscr{H}_\ell}(A \cup \{(x_{i+1},c_{i+1})\}) = (\delta n)^{\ell-(|A|+1)}$. The multiset union

$$\bigsqcup_{(x_{i+1},c_{i+1}) \in \mathscr{B}} \text{hyperedges of } \mathscr{H}_\ell \text{ containing } A \cup \{(x_{i+1},c_{i+1})\}$$

has size $|\mathscr{B}|(\delta n)^{\ell-(|A|+1)}$, and each edge appears at most k times. Ignoring repeats, this is a collection of hyperedges of $\mathscr{H}_\ell$ containing A, so has at most $\Delta_{|A|}(\mathscr{H}_\ell) \leqslant (\delta n)^{\ell-|A|}$ distinct elements. Therefore, we conclude that $|\mathscr{B}| \leqslant k\delta n$. Summing over all choices of A, ℓ we obtain the desired result. □

[1] This is analogous to [45], except since they worked in a single uniformity they could directly construct their final hypergraph without using an auxiliary hypergraph.

We remark that the claim (via the nonemptiness condition on A) does not take into account the possibility that the new coloured vertex v added to B violates the $\Delta_1(\mathscr{H}_\ell)$ condition for the singleton $\{v\}$. However, we will explicitly deal with this possibility by disregarding such coloured vertices.

We continue the proof of Lemma 9.2.7 along the lines of [45], which we include for completeness (and rephrase in terms of random variables for convenience). By a double counting argument, the number of coloured vertices v with $d_{\mathscr{H}_\ell}(\{v\}) = (\delta n)^{\ell-1}$ is at most $\ell e(\mathscr{H}_\ell)/(\delta n)^{\ell-1} \leqslant \ell\delta\binom{n}{n/2}$. Omitting these coloured vertices, and assuming we take $\delta < \alpha/(2k\log(1+\sum_{i=1}^m \beta_i))$, we obtain a template T' with $\omega(\overline{\beta}, T') \geqslant (\omega_{crit}(\overline{\beta}) + \frac{\alpha}{2})\binom{n}{n/2}$. We will now show there is a coloured chain contained in T' which we can add to $\mathscr{H}$. For the remainder of the proof we take all random variables with respect to T' rather than T.

By Lemma 9.3.3 applied to T', we can take $x \in \mathrm{Supp}(T')$ minimal such that $\mathbb{E}\mathbf{Y}^x \geqslant C_1\frac{\alpha}{2}$. Then for $x' \subsetneq x$, and $c_1, \ldots, c_j$ with $j \leqslant k$ by Lemma 9.3.4 and the minimality of x we have

$$\mathbb{E}\mathbf{Z}^{x'}_{c_1 \succ \cdots \succ c_j} \leqslant Q + \mathbb{E}\mathbf{Y}^{x'} \leqslant Q + C_1\frac{\alpha}{2}$$

and

$$\mathbb{E}\mathbf{Z}^{x}_{c_1 \succ \cdots \succ c_j} \leqslant Q + \mathbb{E}\mathbf{Y}^{x}.$$

Our goal is to find a coloured chain with x as its top element contained in T' that we can add to $\mathscr{H}$ without violating any of the codegree conditions. To do this, we write the random variable $\mathbf{Y}^x = \mathbf{Y}^x_{bad} + \mathbf{Y}^x_{good}$ where $\mathbf{Y}^x_{bad}$ only counts coloured chains of $\mathscr{G}$ contained in T' we are not allowed to add to $\mathscr{H}$. It suffices to prove the upper bound

$$\mathbb{E}\mathbf{Y}^x_{bad} \leqslant \sum_\ell \sum_{c_1,\ldots,c_l} \sum_{1\leqslant i\leqslant \ell} \mathbb{E}\mathbf{Z}^x_{c_1\succ\cdots\succ c_i} \cdot \left(O(1)\delta n \frac{1}{n/3}\right) \cdot \max_{x'\subsetneq x} \mathbb{E}\mathbf{Z}^{x'}_{c_{i+1}\succ\cdots\succ c_\ell}.$$

Indeed, by the above the right hand side is bounded above by $(Q+\mathbb{E}\mathbf{Y}^x)(O(1)\delta)(Q+C_1\frac{\alpha}{2})$, and by choosing δ sufficiently small in terms of α and the absolute constant Q (independent of $\mathbb{E}\mathbf{Y}^x$), we can guarantee this is strictly less than $\mathbb{E}\mathbf{Y}^x$ (using the fact that $\mathbb{E}\mathbf{Y}^x \geqslant C_1\frac{\alpha}{2}$). Therefore $\mathbf{Y}^x_{good}$ is not identically zero and we can find a new hyperedge to add to $\mathscr{H}$.

To do this we first similarly split $\mathbf{Z}^x_{c_1\succ\cdots\succ c_\ell} = \mathbf{Z}^x_{c_1\succ\cdots\succ c_\ell,bad} + \mathbf{Z}^x_{c_1\succ\cdots\succ c_\ell,good}$, and write

$$\mathbf{Y}^x_{bad}(C) = \sum_\ell \sum_{(c_1\succ\cdots\succ c_\ell)\in\mathscr{G}_\ell} \mathbf{Z}^x_{c_1\succ\cdots\succ c_\ell,bad}(C).$$

Next, we upper bound

$$\mathbf{Z}^x_{c_1\succ\cdots\succ c_\ell,bad}(C) \leqslant \sum_i \sum_{x_i\subsetneq x} \sum_{x_{i+1}\subsetneq x_i} \mathbf{Z}^x_{c_1\succ\cdots\succ c_\ell,bad,i,x_i,x_{i+1}}(C)$$

where $\mathbf{Z}^x_{c_1\succ\cdots\succ c_\ell,bad,i,x_i,x_{i+1}}$ counts those coloured subchains of C contained in T', whose top element is x, coloured $c_1 \succ \cdots \succ c_\ell$, such that the i'th and $i+1$'st elements from the top are precisely at

the locations x_i, x_{i+1} respectively, and furthermore that x_{i+1} along with some subset of the coloured elements of the chain above it violate some codegree condition.

To bound the expectation of the right hand side of this triple sum, we first note that

$$\begin{aligned}\mathbb{E}\mathbf{Z}^x_{c_1\succ\cdots\succ c_\ell,bad,i,x_i,x_{i+1}} &= (\mathbb{E}\mathbf{Z}^{x_{i+1}}_{c_{i+1}\succ\cdots\succ c_\ell})\mathbb{E}\mathbf{Z}^x_{c_1\succ\cdots\succ c_{i+1},bad,i,x_i,x_{i+1}}\\ &\leqslant (\max_{x'\subsetneq x}\mathbb{E}\mathbf{Z}^{x'}_{c_{i+1}\succ\cdots\succ c_\ell})\cdot\mathbb{E}\mathbf{Z}^x_{c_1\succ\cdots\succ c_{i+1},bad,i,x_i,x_{i+1}}\end{aligned}$$

By Claim 9.3.5, we now have

$$\sum_{x_{i+1}\subsetneq x_i}\mathbb{E}\mathbf{Z}^x_{c_1\succ\cdots\succ c_{i+1},bad,i,x_i,x_{i+1}} \leqslant O(1)\delta n\frac{1}{n/3}\mathbb{E}\mathbf{Z}^x_{c_1\succ\cdots\succ c_i,x_i}$$

where $\mathbf{Z}^x_{c_1\succ\cdots\succ c_i,x_i}(C)$ counts those coloured subchains of C contained in T', whose top element is x, coloured $c_1\succ\cdots\succ c_i$, such that the bottom element is x_i.

Finally, note that

$$\sum_{x_i\subsetneq x}\mathbf{Z}^x_{c_1\succ\cdots\succ c_i,x_i}(C) = \mathbf{Z}^x_{c_1\succ\cdots\succ c_i}(C).$$

Putting this all together now yields the desired inequality.

□

9.4 Crude upper bound

Assume $\mathscr{G}$ contains a monochromatic chain of every colour (of the m colours). Then there exists $L = L(\mathscr{G})$ such that no validly coloured set (S,c) contains a chain of length L. We can easily show $|\Lambda(\mathscr{G},n)| \leqslant (m+1)^{(L-1)\binom{n}{n/2}(1+o(1))}$ using the L-chain containers constructed in [10].

Indeed, in [10] it was shown that there exists a family $\mathscr{C}$ of subsets of $\mathscr{P}(n)$ such that $|\mathscr{C}| \leqslant 2^{o(1)L\binom{n}{n/2}}$, $|C| \leqslant (1+o(1))(L-1)\binom{n}{n/2}$ for every $C\in\mathscr{C}$ and every set $S\subset\mathscr{P}(n)$ that is L-chain free is contained in some set $C\in\mathscr{C}$. To conclude, note that there are at most $(m+1)^{(L-1+o(1))\binom{n}{n/2}}$ coloured subsets of C for each $C\in\mathscr{C}$ giving a total of at most $2^{o(1)L\binom{n}{n/2}}\times(m+1)^{(1+o(1))(L-1)\binom{n}{n/2}} = (m+1)^{(1+o(1))(L-1)\binom{n}{n/2}}$ coloured sets that are L-chain free.

Chapter 10

Strong Ramsey Games in Unbounded Time

The work in this section is joint with Stefan David and Ivailo Hartarsky and is published in the European Journal of Combinatorics [50].

10.1 Introduction

10.1.1 Background

Positional games were first studied by Hales and Jewett [88] and Erdős and Selfridge [62]. The general setting was given by Berge [22], but the field was shaped by the numerous works of Beck since the early 80s including [13–15, 17–20]. Though the most natural positional games are the strong ones, in which both players compete to achieve the same objective, the theory has largely deviated from this direction due to the prohibitive difficulty of strong games. Many weak variants have been developed and have proved more suitable for study, like Maker-Breaker games, for which much is known (see e.g., [19]). However, the strong games remain very poorly understood.

In the present chapter we study instances of a particular type of strong game – the strong Ramsey game. This was first introduced by Harary [89] for cliques and later for arbitrary graphs [90]. In the strong Ramsey game, or simply the *Ramsey game*, $\mathscr{R}(B,H)$ on a finite or infinite graph B, called the *board*, with *target graph* H, two players, **P1** and **P2**, alternate to take previously unclaimed edges of B, starting with **P1**. The first player to claim an isomorphic copy of H *wins* and if none does so in finite time, the game is declared a *draw*. More generally, one might take B and H to be r-uniform hypergraphs. This setting was already mentioned by Beck and Csirmaz and Beck in [13, 20].

For fixed graphs B,H, in $\mathscr{R}(B,H)$, a very general strategy stealing argument due to Nash shows that **P2** cannot have a winning strategy. Moreover, for any fixed target graph H it follows from Ramsey's theorem [126] that $\mathscr{R}(K_n,H)$ cannot end in a draw for n sufficiently large. Therefore, in $\mathscr{R}(K_n,H)$ **P1** has a winning strategy for n sufficiently large. The strategy given by this argument is

not explicit, and indeed almost no examples of explicit strategies are known. In particular, no explicit strategy has been exhibited for $\mathscr{R}(K_n, K_k)$ with $k \geqslant 5$ and n large.

This fact makes it difficult to attack most natural questions in the field. For instance a notorious open question popularised by Beck [17–19] asks if, for fixed k, in the game $\mathscr{R}(K_n, K_k)$, **P1** has a winning strategy in bounded time as $n \to \infty$. An easy compactness argument shows that this is equivalent to the game $\mathscr{R}(K_\omega, K_k)$ being a **P1**-win. The answer is conjectured to be in the affirmative [17, 18, 125] but no progress has been made on the problem for $k \geqslant 5$. Beck emphasised the importance of the question when he listed the question among his "7 most humiliating open problems" [19]. Another notorious open question asks if for every fixed target graph H in the game $\mathscr{R}(K_n, H)$, **P1** has a winning strategy in bounded time as $n \to \infty$ or, equivalently, $\mathscr{R}(K_\omega, H)$ is a **P1** win. In a recent paper [94], Hefetz, Kusch, Narins, Pokrovskiy, Requilé and Sarid addressed the natural generalisation to hypergraphs, and changed the intuition about this phenomenon completely. Surprisingly, in [94] they exhibited a target 5-uniform hypergraph $\mathscr{H}$ for which they constructed an explicit drawing strategy for **P2** in the game $\mathscr{R}\left(K_\omega^{(5)}, \mathscr{H}\right)$. This result provides strong evidence that just strategy stealing and Ramsey-type arguments are insufficient to attack Beck's conjecture.

However, the corresponding question for graphs, as asked in [94], still remains open. As we explain in Section 10.1.3, as the rank decreases this question becomes much harder. In this chapter we make further progress in decreasing it. In our first result, Theorem 10.1.2 (see Corollary 10.1.3), we exhibit a graph G (in fact $G = K_6 \setminus K_4$, see Figure 1), for which we prove that in $\mathscr{R}(K_n \sqcup K_n, G)$ **P1** does not have a winning strategy in bounded time. In order to do so we build on the work of [94]. However, there is a serious obstacle in adapting the strategy developed in [94] from 5-uniform hypergraphs to graphs. As we further explain in Section 10.1.3, although the strategy developed in [94] is a strong game drawing strategy, the core of the argument is a weak game fast winning strategy that is inapplicable to graphs. Therefore, most of our effort is spent developing a much more intricate and elaborate strategy for graphs reflecting characteristic difficulties of strong games, which we discuss further in Section 10.5.

Turning to hypergraphs, in our second result, Theorem 10.1.7 (see Corollary 10.1.8), we exhibit a 4-uniform hypergraph G', obtained by adding 2 new vertices and including them in all edges of G, for which we prove that in $\mathscr{R}\left(K_n^{(4)}, G'\right)$ **P1** does not have a winning strategy in bounded time. In order to do so we cover the board $K_n^{(4)}$ with copies of the board $K_{n-2}^{(2)}$, noting that $K_n^{(4)} = \bigcup_{X \neq Y \in V(K_n^{(4)})} K_{n-2}^{X,Y}$, where $K_{n-2}^{X,Y}$ is the set of hyperedges containing X and Y, naturally identified with K_{n-2}. While in the graph setting the strategy is considerably more involved, in the hypergraph setting the strategy is much more difficult to analyse.

It is not clear whether one should expect $\mathscr{R}(K_n, G)$ to admit a bounded time winning strategy for **P1**, but it is known (see [33] for a simple proof) that for having a winning strategy on K_n is equivalent to having one on $K_n \sqcup K_n$. Thus, by compactness, our result refutes one of two natural conjectures extending simple finite board facts to infinite boards. Namely, it is not the case that on the one hand

for every graph H $\mathscr{R}(K_\omega, H)$ is a **P1**-win and on the other hand for every graph H if $\mathscr{R}(K_\omega, H)$ is a **P1**-win, then $\mathscr{R}(K_\omega \sqcup K_\omega, H)$ is also a **P1**-win.

10.1.2 Results

In this section we state our results.

Definition 10.1.1. Let $G = K_6 \setminus K_4$ be the graph with edge set $E(K_6) \setminus E(K_4)$ (see Figure 1).

In the graph setting the central result is as follows.

Theorem 10.1.2. The game $\mathscr{R}(K_\omega \sqcup K_\omega, G)$ is a draw.

As mentioned in the previous section, one can also formulate the result for finite boards. A standard compactness argument gives the following result.

Corollary 10.1.3. In the game $\mathscr{R}(K_n \sqcup K_n, G)$ **P1** does not have a strategy which guarantees him a win in a bounded number of moves, as $n \to \infty$. □

Remark 10.1.4. Note that we obtain an explicit drawing strategy for the first player in $\mathscr{R}(K_\omega, G)$.

An amusing corollary of Theorem 10.1.2 is the following.

Corollary 10.1.5. It is not the case that on the one hand for every graph H $\mathscr{R}(K_\omega, H)$ is a **P1**-win and on the other hand for every graph H if $\mathscr{R}(K_\omega, H)$ is a **P1**-win, then $\mathscr{R}(K_\omega \sqcup K_\omega, H)$ is also a **P1**-win. □

In the hypergraph setting we work with the following hypergraph.

Definition 10.1.6. Let G' be the 4-uniform hypergraph obtained from our graph G as $V(G') = V(G) \sqcup \{X, Y\}$ and $E(G') = \{e \sqcup \{X, Y\}, e \in E(G)\}$.

Below is the main result on hypergraphs.

Theorem 10.1.7. The game $\mathscr{R}\left(K_\omega^{(4)}, G'\right)$ is a draw.

As before, we can deduce a version of the result for finite boards.

Corollary 10.1.8. In the game $\mathscr{R}\left(K_n^{(4)}, G'\right)$ **P1** does not have a strategy which guarantees him a win in a bounded number of moves, as $n \to \infty$. □

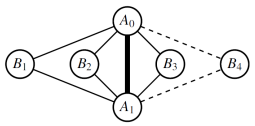


Figure 1: The graph G. The dashed lines form the pair, the solid lines form the core and the base is thickened.

10.1.3 Overview

Graph setting Throughout the chapter, unless otherwise stated, all configurations are considered after **P1**'s move and before **P2**'s move. The graph G is formed by a *base* and four *pairs* of edges connected to it (see Figure 1). We decompose the graph G into a pair and a *core* which is formed by the base and the other three pairs.

We construct the drawing strategy for **P2** in the game $\mathscr{R}(K_\omega \sqcup K_\omega, G)$ in three stages. We denote the copy of K_ω in which **P1** takes his first edge by K^1 and the other copy by K^2.

In the first stage **P2** builds a core in K^2. While doing this **P2** ensures that the following two statements hold. On the one hand, during the entire stage **P2** remains ahead of **P1** in building G in K^2, which is intuitively possible, since **P2** is the first player in K^2. In particular, at the end of the first stage **P1** does not have a threat in K^2. On the other hand, at the end of the stage **P1** has at most $|E(G)|-1$ edges in K^1. At the beginning of the second stage **P2** checks if **P1** has a threat in K^1. If this is the case, **P2** blocks **P1**'s (possibly infinite) threats as long as **P1** keeps making new ones in K^1. The graph G is such that **P1** cannot force a win by making such consecutive threats. If at some point **P1** does not have a threat in K^1, then, in the third stage, **P2** aims to build the pair in K^2 and complete G. He does so by making a (possibly infinite) series of threats from a well-chosen endpoint of the base to new vertices. The choice is such that **P1**'s responses to **P2**'s threats cannot be part of a threat of **P1**.

Roughly speaking, in [94] the strategy is divided into the same three stages: building a core, blocking threats and then making threats. However, there is a severe obstruction in transferring the first stage of the strategy from 5-uniform hypergraphs to graphs. Indeed, in [94] the target hypergraph has a non-trivial and identifiable core that admits a weak game fast building strategy (with number of moves equal to the size of the core). The construction of such a core relies heavily on the high rank. Unfortunately, for graphs this is not feasible anymore because the construction of any non-trivial identifiable core can be delayed by **P1**. Therefore, in the graph setting one should abandon such weak game strategy and face the essential difficulty of strong game strategies, by allowing delay and making sure to block the other player while building one's own target graph.

Hypergraph setting As it was discussed in the introduction, the board $K_\omega^{(4)}$ can be viewed as $\bigcup_{X,Y\in V\left(K_\omega^{(4)}\right)} K_\omega^{X,Y}$, where $K_\omega^{X,Y} = \{e \in E(K_\omega^{(4)}), X, Y \in e\}$ identifies naturally with the board K_ω. Note that a copy of G' is contained in exactly one of the boards $K_\omega^{X,Y}$ and identifies with a graph G in this board.

The drawing strategy of **P2** in the game $\mathscr{R}(K_\omega^{(4)}, G')$ still follows the same three stages. In the first stage **P2** builds a core in a board $K_\omega^{X,Y}$ (corresponding to K^2 from the graph setting) disjoint from **P1**'s first hyperedge by using a very simplified version of the first stage of the strategy for $\mathscr{R}(K_\omega \sqcup K_\omega, G)$, which exploits the larger number of symmetries in higher rank. **P2** ensures that at the end of the first stage **P1** does not have a threat in any board $K_\omega^{X,Y}$, except possibly exactly one of the six boards containing **P1**'s first hyperedge. The second and third stages remain unchanged. However, in the hypergraph setting we need a more subtle analysis than in the graph setting, as the different boards

intersect. The key observation is that for $\{X_1,Y_1\}\cap\{X_2,Y_2\}=\varnothing$ the boards $K_\omega^{X_1,Y_1}$ and $K_\omega^{X_2,Y_2}$ share only one hyperedge. In this respect it is clear that higher rank simplifies the problem.

10.1.4 Organisation of the chapter

The rest of the chapter is organised as follows. In Section 10.2 we establish the key lemmas which are used repeatedly in the proof. In Section 10.3 we construct a drawing strategy for **P2** in $\mathscr{R}(K_\omega\sqcup K_\omega,G)$. In Section 10.4 we derive a drawing strategy for **P2** in $\mathscr{R}\left(K_\omega^{(4)},G\right)$. In Section 10.5 we make some concluding remarks and state some open questions.

10.2 Preliminaries

10.2.1 Setup and notation

For this section and the next one we consider the game $\mathscr{R}(K_\omega\sqcup K_\omega,G)$, where the graph G is defined below. Denote the two disjoint copies of K_ω constituting the board by K^1 and K^2 and assume without loss of generality that **P1**'s first edge is taken in K^1.

For technical reasons, we consider the version of the game in which **P1** stops playing after a finite number of moves, and the game continues until **P2** builds a copy of G. In order to show that **P2** has a drawing strategy, it is enough to show that at the end **P1** does not have a copy of G.

Definition 10.2.1. Let $G=(V,E)$ be the graph with vertex set $V=\{A_0,A_1,B_1,B_2,B_3,B_4\}$ and edge set $E=\{A_0A_1,A_0B_i,A_1B_i\mid i=1,\dots,4\}$ (see Figure 1). With a slight abuse of notation we shall refer to any isomorphic copy of G as G.

We call the edge A_0A_1 the *base* of G. Note that any automorphism of G fixes the base, so it is well defined.

For the following set of definitions we consider the game at a certain stage. Recall from Section 10.1.3 that all configurations are considered after **P1**'s move and before **P2**'s move.

Definition 10.2.2. We say that a given graph G is **P1**-*free* if **P1** has no edges in G and we say it is **P2**-*free* if **P2** has no edges in G.

Definition 10.2.3. For a given graph G, let $e_{\mathbf{P1}}(G)$ be the number of edges in G taken by **P1** if G is **P2**-free; otherwise, define it to be 0. Define similarly $e_{\mathbf{P2}}(G)$. For a given vertex A let $\deg_{\mathbf{P1}}(A)$ and $\deg_{\mathbf{P2}}(A)$ be the number of edges that contain A taken by **P1** and **P2**, respectively.

Note that in the course of the game for a given G the quantities $e_{\mathbf{P1}}(G)$ and $e_{\mathbf{P2}}(G)$ may decrease, becoming 0.

Definition 10.2.4. We say that a vertex F is **P1**-*free* if $\deg_{\mathbf{P1}}(F)=0$, we say it is **P2**-*free* if $\deg_{\mathbf{P2}}(F)=0$ and we say it is *free* if $\deg_{\mathbf{P1}}(F)=\deg_{\mathbf{P2}}(F)=0$.

Definition 10.2.5. We call an edge A_0A_1 taken by **P2** in K^2 a *potential base* if there exist two vertices B_1, B_2 in K^2 such that **P2** has the edges A_0B_1, A_0B_2, A_1B_1 and A_1B_2 and there exists a special vertex $X \in \{A_0, A_1\}$ such that:

(i) **P1** does not have a triangle X, T_1, T_2;

(ii) **P1** does not have a 4-cycle X, C_1, C_2, C_3 with the edge XC_2 not taken by **P2**.

Remark 10.2.6. If the edge A_0A_1 is a potential base with special vertex X, then if later in the game **P1** constructs an additional star $XF_1, XF_2, \ldots$ from X to **P1**-free vertices $F_1, F_2, \ldots$ together with exactly r extra edges, then he has at most r triangles sharing a common edge that contain X.

10.2.2 Lemmas

In this section we present the main technical result, Lemma 10.2.7, which states that if the game has reached certain configurations, then **P2** has a drawing strategy. This result is used recurrently in conjunction with the strategy constructed in Section 10.3. In order to check one of the hypotheses of Lemma 10.2.7 we also establish Lemma 10.2.11.

Lemma 10.2.7. Assume that before **P2**'s turn, the game has the following properties:

(a) **P1** has at most six edges in K^1;

(b) for any G in K^2, $e_{\mathbf{P1}}(G) \leqslant 5$;

(c) **P2** has a potential base A_0A_1.

Then, **P2** has a drawing strategy.

Proof of Lemma 10.2.7. Let $\mathscr{E}_0^1$ and $\mathscr{E}_0^2$ be the set of initial edges of **P1** in K^1 and K^2 respectively. Without loss of generality let A_0 be the special vertex of the potential base A_0A_1. The line of play of **P2** is divided in three stages as follows. In the first stage, **P2** takes edges from A_1 to free vertices F_i for $i = 1, \ldots, k$ until at some point **P1** does not take the edge A_0F_k (this necessarily happens, since **P1** makes only a finite number of moves). Then **P2** takes A_0F_k and **P1** plays somewhere. Let $\mathscr{E}_1^2 = \{A_0F_i, 1 \leqslant i < k\}$ be the star taken by **P1** and $\mathscr{E}_2$ be the set consisting of the last two edges taken by **P1** (if he did not stop playing).

Claim 10.2.8. One can guarantee that after a finite number of moves, called second stage, the configuration (before **P2**'s move) satisfies the following two properties.

- For each G in K^1 we have that $e_{\mathbf{P1}}(G) \leqslant 7$.
- The set of edges **P1** has taken in K^2 is $\mathscr{E}_0^2 \sqcup \mathscr{E}_1^2 \sqcup \mathscr{E}^2$ for some $\mathscr{E}^2$ with $|\mathscr{E}^2| \leqslant 2$.

Proof of Claim 10.2.8. We consider three cases for the configuration at the end of the first stage (before **P2**'s move).

Case I. P1 has exactly 8 edges in K^1 forming a copy G^1 of G with its base C_0C_1 present but the edge C_0D_1 absent.

As $|\mathscr{E}_0^1| \leqslant 6$, we have that $\mathscr{E}_2$ is contained in G^1. **P2** takes the edge C_0D_1 and, while **P1** keeps taking edges $C_{\varepsilon_n}D_n$ to **P1**-free vertices D_n, **P2** keeps taking the edges $C_{1-\varepsilon_n}D_n$ for $\varepsilon_n \in \{0,1\}$. Eventually **P1** either stops or takes a different kind of edge ε. Let $\mathscr{E} = \{\varepsilon\}$. Note that $e_{\mathbf{P1}}(G) \leqslant 7$ for all G in K^1. Indeed, this immediately follows from the fact that for all vertices $C \in V(K^1) \setminus \{C_0, C_1\}$ we have that $\deg_{\mathbf{P1}}(C) \leqslant 3$.

Case II. P1 has exactly 8 edges in K^1 forming a copy G^1 of G without its base C_0C_1.

As $|\mathscr{E}_0^1| \leqslant 6$, we have that $\mathscr{E}_2$ is contained in G^1. **P2** takes C_0C_1 and then **P1** takes an edge ε. Let $\mathscr{E} = \{\varepsilon\}$. Note that $e_{\mathbf{P1}}(G) \leqslant 7$ for all G in K^1.

Case III. For any G in K^1, $e_{\mathbf{P1}}(G) \leqslant 7$.

In this case we directly skip to the third stage and set $\mathscr{E} = \mathscr{E}_2$.

In all cases, let $\mathscr{E}^2 = \mathscr{E} \cap K^2$ and note that the set of edges **P1** has taken in K^2 is $\mathscr{E}_0^2 \sqcup \mathscr{E}_1^2 \sqcup \mathscr{E}^2$, with $|\mathscr{E}^2| \leqslant 2$, as claimed. □

In the third stage **P2** takes edges from A_1 to free vertices F_i for $k < i \leqslant l$ until at some point **P1** does not take the edge A_0F_l. Then **P2** takes A_0F_l and constructs a G. The game stops after **P2**'s move. Let $\mathscr{E}_3^2 = \{A_0F_i, k < i < l\}$ and $\mathscr{E}_4$ be the set consisting of the last edge taken by **P1** (if he did not stop playing).

Claim 10.2.9. At the end of the third stage **P1** does not have a G.

Proof of Claim 10.2.9. Notice that by Claim 10.2.8 for any G in K^1 we have $e_{\mathbf{P1}}(G) \leqslant 7+1$, since $|\mathscr{E}_4| \leqslant 1$. We claim that the same holds in K^2. Indeed, **P1** cannot have a G in K^2 that does not intersect $\mathscr{E}_1^2 \cup \mathscr{E}_3^2$, as $e_{\mathbf{P1}}(G) \leqslant 5+3$ by b and $|\mathscr{E}_4| + |\mathscr{E}^2| \leqslant 3$. On the other hand, **P1** cannot have a G in K^2 that contains some A_0F_i, as neither of the vertices A_0 and F_i can be in the base. This is because **P1** has at most 3 triangles sharing an edge that contain vertex A_0 by Remark 10.2.6, and $\deg_{\mathbf{P1}}(F_i) \leqslant 1+3$. □

Clearly, Claim 10.2.9 implies that **P2** has a drawing strategy. □

As a tangential point, note **P2** could adopt a slightly different strategy in the first stage. Say that **P2** has the triangle $A_0A_1B_1$ and assume that **P2** takes the edges from A_1 to free vertices F_i for $i = 1, \ldots, 5$, while **P1** takes the edges A_0F_i. Then **P2** can win in three moves by taking three of the five edges B_1F_i. Thus, in the original version of the game $\mathscr{R}(K_\omega \sqcup K_\omega, G)$ (in which **P1** does not stop

playing after a finite time) **P2** can always build the core of G in his drawing strategy, as claimed in Section 10.1.3. As this point is not used in the rest of proof we shall not redo the analysis for this strategy.

We next provide a quick way to check that condition c of Lemma 10.2.7 holds, i.e., **P2** has a potential base A_0A_1. By Definition 10.2.5, the only structures taken by **P1** that could prevent A_0A_1 from being a potential base are 4-cycles A_i,C_1,C_2,C_3 where A_iC_2 is not taken by $P2$ and traingles A_i,C_1,C_2. Therefore, it is natural to track the edges of **P1** that could be part of such configurations.

Definition 10.2.10. At a certain stage of the game, and for a given edge A_0A_1 taken by **P2** in K^2, we call an edge taken by **P1** *good for* A_0A_1 if every 4-cycle A_i,C_1,C_2,C_3 with A_iC_2 not taken by **P2** and every triangle A_i,C_1,C_2 that contain this edge also contains an edge taken by **P2**. In particular, all edges of **P1** in K^1 are good for any edge of **P2** in K^2. We call *bad for* A_0A_1 an edge of **P1** in K^2 that is not good for A_0A_1.

Equivalently, an edge X_1X_2 is good for A_0A_1 if and only if it is in K^1 or

$$\{X_1,X_2\}\cap\{A_0,A_1\}=\varnothing, \qquad \textbf{P2 has } A_0X_1 \text{ or } A_0X_2, \qquad \textbf{P2 has } A_1X_1 \text{ or } A_1X_2. \tag{10.1}$$

We call a 2Δ-configuration for an edge A_0A_1 the edges $A_0X_1,A_0X_2,A_1X_1,A_1X_2,X_1X_2$ (see Figure 2) for any two vertices X_1,X_2 disjoint from A_0,A_1.

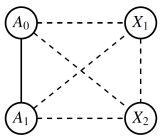


Figure 2: The 2Δ-configuration.

Lemma 10.2.11. Assume that before his turn **P2** has an edge A_0A_1 in K^2 such that:

(a) **P1** has at most 5 bad edges for A_0A_1;

(b) **P1** does not have a 2Δ-configuration for A_0A_1;

(c) **P2** has edges A_0B_1, A_0B_2, A_1B_1, A_1B_2 for some vertices B_1,B_2.

Then A_0A_1 is a potential base, i.e., condition c of Lemma 10.2.7 holds.

Proof of Lemma 10.2.11. Assume for a contradiction that A_0A_1 is not a potential-base. Let us call a triangle or 4-cycle as in Definition 10.2.5 an *obstruction* for X ($X\in\{A_0,A_1\}$). By condition c, **P1** has an obstruction for each of A_0 and A_1. Yet, since **P2** has A_0A_1, a single obstruction cannot contain both A_0 and A_1. So **P1** has two separate obstructions, one containing A_0 but not A_1, and one containing A_1 but not A_0. The obstruction containing A_0 has degree 2 at A_0, so the obstruction containing A_1 has at most $5-2$ edges by condition a, since all edges in an obstruction are bad. Thus, both obstructions are triangles, they contain A_0 and A_1 and they share a common edge by condition a, which contradicts condition b, as A_0A_1 is taken by **P2**. □

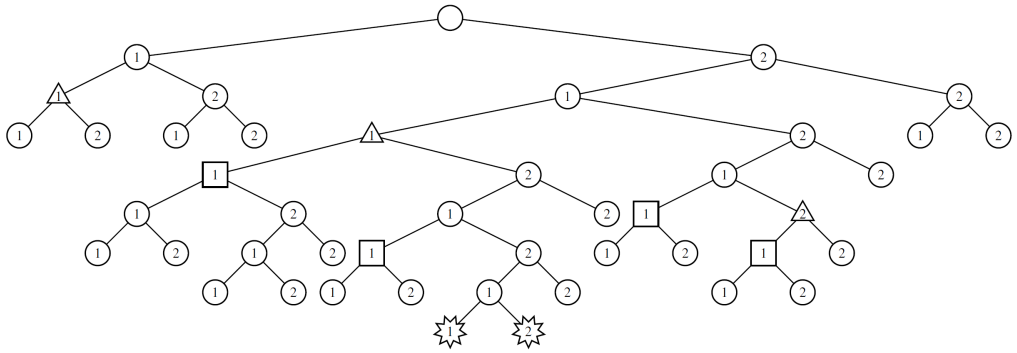


Figure 3: The case tree. Each vertex represents the case whose label is given by appending the labels of vertices along the path from the root to it. The two special end-cases are in stars. The marked split-cases are in triangles if **P1** lost 1 edge and in squares if he lost 2.

10.3 A drawing strategy

10.3.1 Setting

In this section we present a drawing strategy for **P2** in $\mathscr{R}(K_\omega \sqcup K_\omega, G)$. We construct it by considering various cases, according to the moves of **P1**. The following trick allows us to reduce significantly the number of cases considered: instead of considering all possible moves of **P1** up to isomorphism we often give him *additional edges*, i.e., edges taken by **P1** which are not specified immediately after **P1**'s turn, but are obviously taken into account. All figures depict (part of) the configuration in K^2 only (the first edge of **P1** being in K^1), not featuring the additional edges. At a certain stage of the game, the current number of additional edges is marked as "$+n$" on the corresponding figure and the rest of **P1**'s edges are specified. The first edge of **P1** is considered specified, although it does not appear drawn. In all the figures, **P1**'s edges are drawn as dashed lines while **P2**'s edges are drawn as continuous lines. Whenever two figures are drawn for the same case, the left one corresponds to the initial state of the game before **P2**'s move under the assumption defining the present case, while the right one corresponds to the final state of the game before **P2**'s subsequent move. If only one figure is drawn, it corresponds to the final state of the game, as the initial state of the game is the same as the parent case with an additional assumption defining the present case.

The cases considered in the strategy naturally form a strict binary tree (see Figure 3), whose leaves we call *end-cases* and whose internal nodes we call *split-cases*. When possible, we respect the convention that at each split-case we consider a particular edge **P2** is interested in, the left child (labeled by 1) corresponds to the edge being already taken by **P1**, while the right child (labeled by 2) corresponds to the edge being available for **P2** to take.

Finally, if, at a certain stage of the game, **P1** has $k+1$ edges in total (including his first edge in K^1 and at most k other edges in K^2), we say that ***P1** lost l edges*, if $e_{\mathbf{P1}}(G) \leqslant k-l$ for any G in K^2. It is clear that the number of edges lost by **P2** is non-decreasing in the course of the game. For that reason we shall indicate the loss of edges as soon as it arises. More precisely, a split-case is *marked* if at the

final stage of the case **P1** is certain to have lost more edges than in the parent case. These cases are denoted by triangles (resp. squares) in Figure 3 if **P1** lost 1 (resp. 2) edges.

For the reader's convenience, in Section 10.3.2 we make a detailed analysis of **Case 1**. This analysis contains on one hand the strategy of **P2** in **Case 1** and on the other hand the detailed verification of the hypotheses of Lemma 10.2.7 in each of the end-cases. This allows us to conclude that **P2** has a drawing strategy in **Case 1**. In Section 10.3.3 we present the automatic procedure to check that in all end-cases of **Case 2**, with the exception of the two special end-cases, the hypotheses of Lemma 10.2.7 are satisfied. In Section 10.3.4 we give the strategy of **P2** in **Case 2**, leaving the mechanical checks explained in Section 10.3.3 to the reader. Finally, the two *special end-cases*, namely **Case 2.1.1.2.1.2.1.1** and **Case 2.1.1.2.1.2.1.2**, are treated separately in Section 10.3.5, as they do not allow a direct application of Lemma 10.2.7. Altogether, this allows us to conclude that **P2** has a drawing strategy.

For the purpose of understanding the strategy of **P2**, the reader is expected to go through the figures in order, thus exploring the case tree by a depth-first search. General split-case (Figure 4) corresponds to the root of the case tree in Figure 3.

10.3.2 Split-case 1.

In the first part, the reader is expected first to go though Figures 4-8 to familiarize with the strategy of **P2** in **Marked split-case 1.1** and then to read the analysis below which explains how Lemma 10.2.7 can be applied to **End-case 1.1.1** and **End-case 1.1.2** to conclude that in these cases **P2** has a drawing strategy. Similarly, in the second part, the reader is expected to go though Figures 9-11 to familiarize with the strategy of **P2** in **Marked split-case 1.2** and then to read the analysis below.

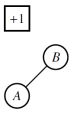


Figure 4: **General split-case.** Without loss of generality assume that the first edge **P1** takes is XY in K^1. In response, **P2** takes the edge AB in K^2. We mention that, excluding XY, **P1** has 1 unspecified additional edge (not appearing in the figure) by "+1".

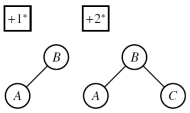


Figure 5: **Split-case 1.** Assume that the second edge taken by **P1** is either in K^1 or in K^2 and is incident with either A or B. In the latter case, we may assume that it is incident with B. We indicate that either one of these is the case by a "*" in the figures. Then **P2** takes the edge BC, where C is any free vertex in K^2.

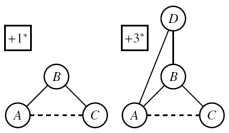


Figure 6: **Marked split-case 1.1.** Assume that **P1** has AC. Then **P2** takes BD, where D is any free vertex in K^2. Note that at least one of the edges DA, DC is not taken by any player, so assume without loss of generality that DA is not taken. Then **P2** takes DA. **P1** lost an edge.

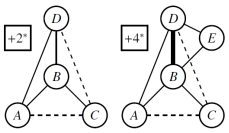


Figure 7: **End-case 1.1.1.** Assume that **P1** has DC. Then **P2** takes BE, where E is any free vertex in K^2. Note that at least one of the edges ED, EA is not taken by any player, so assume without loss of generality that ED is not taken. Then **P2** takes ED. The potential base is BD.

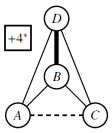


Figure 8: **End-case 1.1.2.** Assume that **P1** does not have DC. Then **P2** takes this edge. The potential base is BD.

Marked split-case 1.1. Recall that "*" (see Figure 5) means we assume that the second edge of **P1** is either in K^2 containing B or it is in K^1. Let us begin by checking that in **Marked split-case 1.1**, Figure 6, **P1** lost an edge. At this stage **P1** has a total of 5 edges including the first edge XY, which is in K^1. We need to check that $e_{\mathbf{P1}}(G) \leqslant 3$ for any G in K^2. Assume for the sake of contradiction that there exist a G in K^2 such that $e_{\mathbf{P1}}(G) \geqslant 4$. Then G contains all edges of **P1** except XY and contains no edges of **P2**. This forces the second edge of **P1** to be in K^2, and moreover, condition "*" forces this edge to be incident to B, say BF. In order for G to contain both AC and BF, the base X_0X_1 of G needs to satisfy without loss of generality $X_0 \in \{A,C\}$, as any edge in G is incident with the base. Then G contains the edge X_0B, as every vertex of G is connected to X_0, so X_0B needs not to be taken by **P2**. This yields the desired contradiction (see Figure 6).

In order to apply Lemma 10.2.7 to **End-case 1.1.1** and **End-case 1.1.2** and conclude that in these cases **P2** has a drawing strategy, we are now ready to check that the hypotheses of Lemma 10.2.7 are satisfied in these cases.

Firstly, let us check that condition b of Lemma 10.2.7 holds. Since **P1** lost an edge in **Marked split-case 1.1**, the same holds for **End-case 1.1.1** and **End-case 1.1.2**. Note that the total number of edges of **P1** including the first edge XY which is in K^1 is 7 and 6, respectively, for the two end-cases. Therefore, in **End-case 1.1.1** and **End-case 1.1.2** each G in K^2 satisfies $e_{\mathbf{P1}}(G) \leqslant 5$ and $e_{\mathbf{P1}}(G) \leqslant 4$, respectively, as desired.

Secondly, let us check that condition a of Lemma 10.2.7 holds. This follows immediately from the fact that in **End-case 1.1.1** and **End-case 1.1.2 P1** has 4 additional edges, so at most 5 edges in K^1.

Thirdly, let us check that condition c of Lemma 10.2.7 holds. We do so by applying Lemma 10.2.11 to the thickened edge BD. Thus, we are left with checking that the hypotheses of Lemma 10.2.11 are satisfied.

Condition c of Lemma 10.2.11 is checked directly on the figures. Note that by (10.1) we have that AC is good for BD, as both AB and AD are taken by **P2**. Recall that in **End-case 1.1.1** and in **End-case 1.1.2 P1** has at most 6 and 5 edges in K^2, respectively. Therefore, in **End-case 1.1.1** and in **End-case 1.1.2 P1** has at most 5 and 4 bad edges, respectively. It follows that condition a of Lemma 10.2.11 holds. Furthermore, in **End-case 1.1.2** condition b of Lemma 10.2.11 also holds. Finally, we are left with checking condition b in **End-case 1.1.1**, i.e., that CD, together with the other 4 possible bad edges, cannot form a 2Δ-configuration. Indeed, CD cannot appear in a 2Δ-configuration for BD, since BC is taken by **P2**. This concludes the proof.

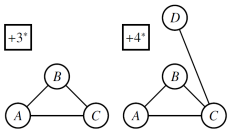


Figure 9: **Split-case 1.2.** Assume that **P1** does not have AC. Then **P2** takes this edge and then plays the edge CD, where D is a free vertex.

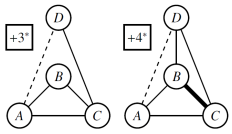


Figure 10: **End-case 1.2.1.** Assume that **P1** has AD. Then he does not have BD, as D was a **P1**-free vertex before his move. Then **P2** takes BD.
The potential base is BC.

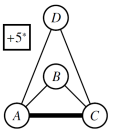


Figure 11: **End-case 1.2.2.** Assume that **P2** does not have AD. Then **P2** takes this edge. The potential base is AC.

Split-case 1.2. Note that **Split-case 1.2** is not marked. In order to apply Lemma 10.2.7 to **End-case 1.2.1** and **End-case 1.2.2** and conclude that in these cases **P2** has a drawing strategy, we shall check that the hypotheses of Lemma 10.2.7 are satisfied.

In both **End-case 1.2.1** and **End-case 1.2.2** condition b of Lemma 10.2.7 trivially holds as **P1** has at most 5 edges in K^2. Condition a of Lemma 10.2.7 is also automatic, as **P1** has 4 and 5 additional edges, respectively.

In order to check that condition c of Lemma 10.2.7 holds, we shall check that the hypotheses of Lemma 10.2.11 are satisfied. Condition c of Lemma 10.2.11 is checked directly on Figures 10 and 11. Note that in **End-case 1.2.1** the edge AD is good for BC, as AB and AC are both taken by **P2**. It follows that there are at most 4 bad edges, implying conditions a and b of Lemma 10.2.11. Also note that in **End-case 1.2.2** the second edge of **P1** is either in K^1 or is in K^2 incident to B by condition "*". In either case this edge is good for AC, as both AB and BC are taken by **P2**. This allows us to conclude as in the previous case.

10.3.3 Applying Lemma 10.2.7

Let us now explain how the conditions of Lemma 10.2.7 are verified in the remaining end-cases with the exception of the special ones which we shall discuss further in Section 10.3.5. The procedure generalises the one in the example cases discussed in detail in Section 10.3.2.

Condition a of Lemma 10.2.7 In each of the end-cases (including the special ones), we mechanically check that **P1** has at most 5 additional edges, so he has at most 6 edges in K^1.

Condition b of Lemma 10.2.7 It is just a little harder to check that in all non-special end-cases for all G in K^2, $e_{\mathbf{P1}}(G) \leqslant 5$. For each marked split-case (see Figure 3) we see that **P1** loses (at least) l edges for a certain number $l \in \{1,2\}$ which depends on the case. If **P1** loses l edges in a given split-case, it follows that he also loses l edges in all of its descendant cases. For each non-special end-case we refer to the closest ancestor marked split-case (if any) to see how many edges **P1** lost (see Figure 3) and check that condition b holds.

In order to establish that **P1** loses l edges in a marked split-case, we proceed as follows. Let $k+1$ be the total number of edges taken by **P1**. We consider all non-isomorphic edges X_0X_1 in K^2, such that X_0X_1 is not taken by **P2** (so X_0X_1 could potentially become the base of a G that **P1** constructs). We count the number of edges in K^2 taken by **P1** which are of the form X_iY with YX_{1-i} not taken by **P2**, and add the edge X_0X_1 if it is taken by **P1**. This number bounds $e_{\mathbf{P1}}(G)$ for copies of G with base X_0X_1. We check that for each choice of X_0X_1 there are at most $k-l$ such edges. To exclude a large number of edges X_0X_1 from the very beginning, we first investigate $\deg_{\mathbf{P1}}(v)$ for all vertices v.

Condition c of Lemma 10.2.7 In each of the non-special end-cases, we mechanically check that **P2** has a potential base A_0A_1 – which is declared and marked in all pictures by a thickened edge. To do

so it is enough to check the conditions of Lemma 10.2.11. Firstly, we inspect that **P1** has at most 5 bad edges for A_0A_1 (condition a of Lemma 10.2.11), using (10.1) to establish that some edges are good for A_0A_1 and further that the bad edges do not form a 2Δ-configuration if there are 5 of them (condition b of Lemma 10.2.11). Finally, we inspect that there exist two vertices B_1, B_2 such that **P2** has the edges $A_0B_1, A_0B_2, A_1B_1, A_1B_2$ (condition c of Lemma 10.2.11), so we can conclude.

Remark 10.3.1. While the verification of the conditions of Lemma 10.2.7 is very easy given the strategy for **P2**, it is a central part of the chapter to determine that right strategy. The idea of using additional edges and reducing drastically the number of cases helped us achieve this goal.

10.3.4 Split-case 2.

We now describe the rest of the strategy. Recall the **General split-case**, Figure 4.

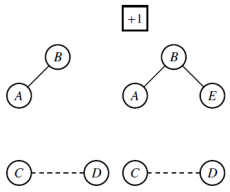


Figure 12: **Split-case 2.** Assume that the second edge taken by **P1** is CD with C and D free vertices in K^2. Then **P2** takes BE, where E is a free vertex in K^2.

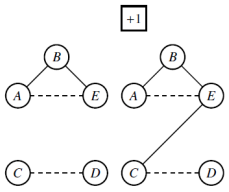


Figure 13: **Split-case 2.1.** Assume that **P1** has AE. Then **P2** takes CE.

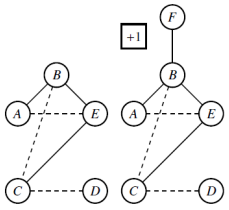


Figure 14: **Marked split-case 2.1.1.** Assume that **P1** has BC. Then **P2** takes BF, where F is a free vertex in K^2.
P1 lost an edge.

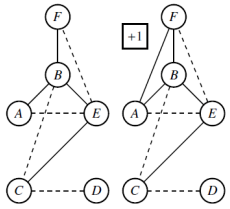


Figure 15: **Marked split-case 2.1.1.1.** Assume that **P1** has EF. Then **P2** takes AF.
P1 lost two edges.

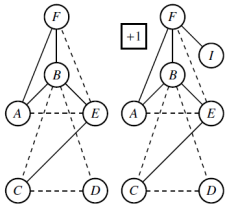


Figure 16: **Split-case 2.1.1.1.1.** Assume that **P1** has BD. Then **P2** takes FI, where I is a free vertex in K^2.

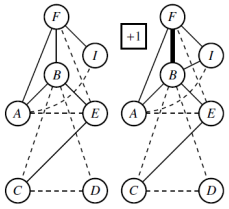


Figure 17: **End-case 2.1.1.1.1.1.** Assume that **P1** has AI. Then **P2** takes BI.
The potential base is BF.

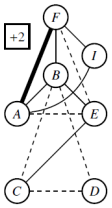


Figure 18: **End-case 2.1.1.1.1.2.** Assume that **P1** does not have AI. Then **P2** takes this edge.
The potential base is AF.

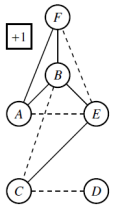


Figure 19: **Split-case 2.1.1.1.2.** Assume that **P1** does not have BD.

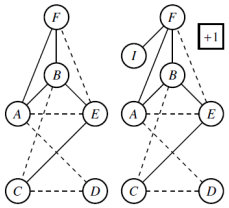


Figure 20: **Split-case 2.1.1.1.2.1.** Assume that **P1** has AD or DF and without loss of generality let this edge be AD. Then **P2** takes FI, where I is a free vertex in K^2.

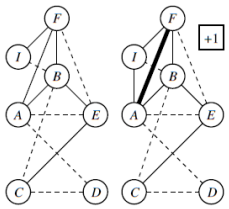


Figure 21: **End-case 2.1.1.1.2.1.1.** Assume that **P1** has BI. Then **P2** takes AI.
The potential base is AF.

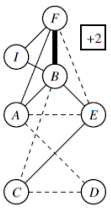


Figure 22: **End-case 2.1.1.1.2.1.2.** Assume that **P1** does not have BI. Then **P2** takes this edge.
The potential base is BF.

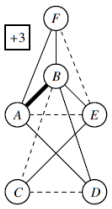


Figure 23: **End-case 2.1.1.1.2.2.** Assume that **P1** has none of DA, DB or DF. Then **P1** takes BD and then one of AD and DF, which is not taken. Without loss of generality, let this edge be AD.
The potential base is AB.

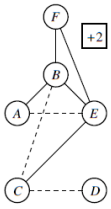


Figure 24: **Split-case 2.1.1.2.** Assume that **P1** does not have EF. Then **P2** takes this edge.

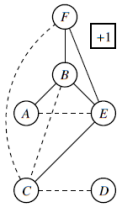


Figure 25: **Split-case 2.1.1.2.1.** Assume that **P1** has CF.

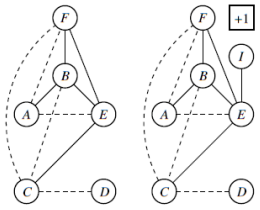


Figure 26: **Marked split-case 2.1.1.2.1.1.** Assume that **P1** has AF. Then **P2** takes EI, where I is a free vertex in K^2.
P1 lost 2 edges.

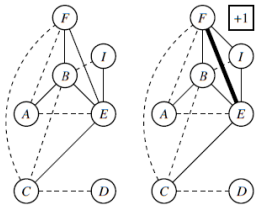


Figure 27: **End-case 2.1.1.2.1.1.1.** Assume that **P1** has BI. Then **P2** takes FI.
The potential base is EF.

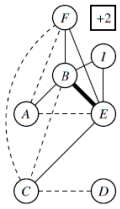


Figure 28: **End-case 2.1.1.2.1.1.2.** Assume that **P1** does not have BI. Then **P2** takes this edge.
The potential base is BE.

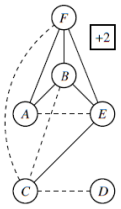


Figure 29: **Split-case 2.1.1.2.1.2.** Assume that **P1** does not have AF. Then **P2** takes this edge.

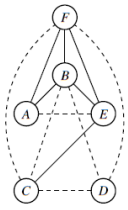


Figure 30: **Split-case 2.1.1.2.1.2.1.** Assume that **P1** has both BD and DF. Then **P2** takes FI, where I is a free vertex in K^2. If **P1** does not take BI, **P2** takes it and stops. Otherwise, **P1** takes BI, then **P2** takes FJ, where J is a free vertex in K^2. If **P1** does not take BJ, **P2** takes it and stops. Otherwise, **P1** takes BJ, then **P2** takes FK, where K is a free vertex in K^2. If **P1** does not take BK, **P2** takes it and stops.

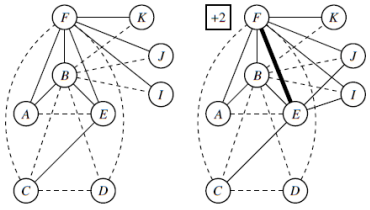


Figure 31: **Special end-case 2.1.1.2.1.2.1.1.** Assume that in the strategy described in the previous case **P1** did take BI, BJ and BK successively. Then **P2** takes EI and one of EJ and EK which remains free after **P1**'s turn. Without loss of generality, let this edge be EJ.
This case is treated separately in Section 3.5.

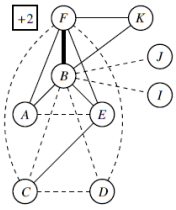


Figure 32: **Special end-case 2.1.1.2.1.2.1.2.** Assume that in the strategy described in split-case **2.1.1.2.1.2.1.1. P1** did not take BI, then BJ and then BK. Without loss of generality we can assume that BK and FK were taken by **P2**, provided that **P2** concede both FI and FJ if they were taken by **P2** and that we give **P1** the edges BI and BJ regardless if **P1** took them (this corresponds to the case where **P1** responded the first two times, but not the third, and **P2** promises not to use the first two edges he acquired).
This case is treated separately in Section 3.5.

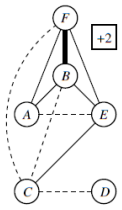


Figure 33: **End-case 2.1.1.2.1.2.2.** Assume that **P1** does not have both BD and DF, so that **P1** does not have a 2Δ-configuration for BF.
The potential base is BF.

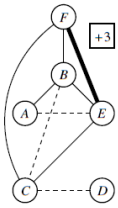


Figure 34: **End-case 2.1.1.2.2.** Assume that **P1** does not have CF. Then **P2** takes this edge.
The potential base is EF.

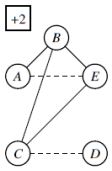


Figure 35: **Split-case 2.1.2.** Assume that **P1** does not have BC. Then **P2** takes this edge.

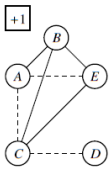


Figure 36: **Split-case 2.1.2.1.** Assume that **P1** has AC.

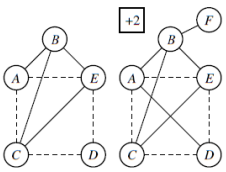


Figure 37: **Marked split-case 2.1.2.1.1.** Assume that **P1** has DE. Then **P2** takes AD and then BF, where F is a free vertex in K^2.
P1 lost 2 edges.

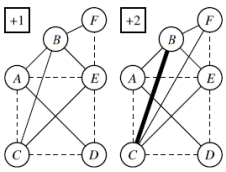


Figure 38: **End-case 2.1.2.1.1.1.** Assume that **P1** has EF. Then he does not have CF, as F was a **P1**-free vertex before his move. Then **P2** takes CF.
The potential base is BC.

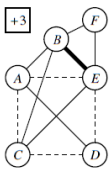


Figure 39: **End-case 2.1.2.1.1.2.** Assume that **P1** does not have EF. Then **P2** takes this edge.
The potential base is BE.

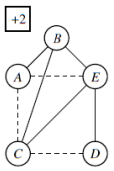


Figure 40: **Marked split-case 2.1.2.1.2.** Assume that **P1** does not have DE. Then **P2** takes this edge.
P1 lost an edge.

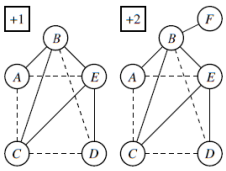


Figure 41: **Marked split-case 2.1.2.1.2.1.** Assume that **P1** has BD. Then **P2** takes BF, where F is a free vertex in K^2.
P1 lost 2 edges.

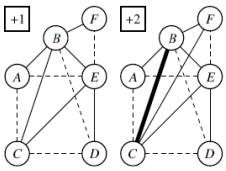


Figure 42: **End-case 2.1.2.1.2.1.1.** Assume that **P1** has EF. Then he does not have CF, as F was a **P1**-free vertex before his move. Then **P2** takes CF.
The potential base is BC.

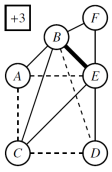


Figure 43: **End-case 2.1.2.1.2.1.2.** Assume that **P1** does not have EF. Then **P2** takes this edge.
The potential base is BE.

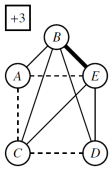


Figure 44: **End-case 2.1.2.1.2.2.** Assume that **P1** does not have BD. Then **P2** takes this edge.
The potential base is BE.

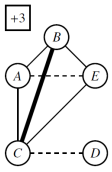


Figure 45: **End-case 2.1.2.2.** Assume **P1** does not have AC. Then **P2** takes this edge.
The potential base is BC.

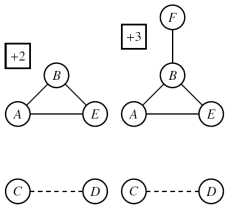


Figure 46: **Split-case 2.2.** Assume that **P1** does not have AE. Then **P2** takes this edge and plays from a **P1**-free vertex among A, B and E to a free vertex in K^2, say F. Without loss of generality assume **P2** played from B. Hence, after **P1**'s turn there is a vertex among A and E of **P1**-degree at most 1.

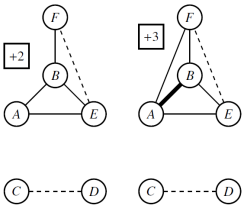


Figure 47: **End-case 2.2.1.** Assume **P1** has AF or EF. Without loss of generality assume **P1** has EF. Then he does not have AF, as F was a **P1**-free vertex before his last turn. Then **P2** takes AF.
The potential base is AB.

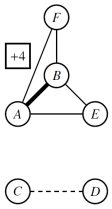


Figure 48: **End-case 2.2.2.** Assume **P1** does not have any of AF and EF. Without loss of generality assume $\deg_{\mathbf{P1}}(A) \leq 1$. Then **P2** takes AF.
The potential base is AB, as $\deg_{\mathbf{P1}}(A) + \deg_{\mathbf{P1}}(B) \leq 1 + 2$.

10.3.5 Special end-cases

In this section we present a drawing strategy for **P2** in the two special end-cases, which cannot be dealt with by directly applying Lemma 10.2.7.

End-case 2.1.1.2.1.2.1.1. Figure 31-right Note that **P1** lost 3 edges (i.e., he has not won yet). If **P1** does not have EK, **P2** takes it and creates a G before him. Otherwise, one of **P1**'s additional edges is EK and hence **P1** lost 4 edges. Moreover, by inspection we observe that **P1** has no 4-cycles E, C_1, C_2, C_3 with the edge EC_2 not taken by **P2** and at most one triangle ET_1T_2. Let $\mathscr{E}_0$ be the initial set of edges taken by **P1** in K^2, so that $|\mathscr{E}_0| \leqslant 11$. Then **P2** takes edges from F to free vertices L_i for $i = 1, \ldots, k$ until at some point **P1** does not take the edge EL_k. Then **P2** takes EL_k and completes a G. The game stops after **P2**'s move. Let $\mathscr{E}_1 = \{EL_i, 1 \leqslant i < k\}$ be the star taken by **P1** and $\mathscr{E}_2$ be his last edge. We claim that **P1** does not have a G. Indeed, **P1** cannot have a G that does not intersect $\mathscr{E}_1$, as $e_{\mathbf{P1}}(G) \leqslant 12 - 4$ since **P1** lost 4 edges and $|\mathscr{E}_0| + |\mathscr{E}_2| \leqslant 12$. On the other hand, **P1** cannot have a G that contains some EL_i, as neither of the vertices E and L_i can be the base. This is because **P1** has at most 2 triangles containing E by the above observation and the fact that $\deg_{\mathbf{P1}}(L_i) \leqslant 2$.

End-case 2.1.1.2.1.2.1.2. Figure 32 Note that **P1** lost 3 edges. (i.e., he has not won yet). Moreover, by inspection we observe that **P1** has either at most one 4-cycle F, C_1, C_2, C_3 with the edge FC_2 not taken by anyone and at most one triangle FT_1T_2, or no such 4-cycle and at most 2 such triangles. Let $\mathscr{E}_0$ be the initial set of edges taken by **P1** in K^2, so that $|\mathscr{E}_0| = 10$. Then **P2** takes edges from B to

free vertices L_i for $i = 1,\dots,k$ until at some point **P1** does not take the edge FL_k. Then **P2** takes FL_k and completes a G. The game stops after **P2**'s move. Let $\mathscr{E}_1 = \{FL_i, 1 \leqslant i < k\}$ be the star taken by **P1** and $\mathscr{E}_2$ be his last edge. We claim that **P1** does not have a G. Indeed, **P1** cannot have a G that does not intersect $\mathscr{E}_1$, as $e_{\mathbf{P1}}(G) \leqslant 11-3$ since **P1** lost 3 edges and $|\mathscr{E}_0| + |\mathscr{E}_2| \leqslant 11$. On the other hand, **P1** cannot have a G that contains some FL_i, as neither of the vertices F and L_i can be the base. This is because **P1** has at most 3 triangles containing F by the above observation and the fact that $\deg_{\mathbf{P1}}(L_i) \leqslant 2$.

This analysis concludes the proof of Theorem 10.1.2. □

10.4 From graphs on two copies to hypergaphs

In this section we show that $\mathscr{R}(K_\omega^{(4)}, G')$ is a draw. Recall from Definition 10.1.6 that G' is the 4-uniform hypergraph obtained from the graph G by adding 2 new vertices X,Y and including them in all the edges. We call X,Y the two *centres*. Note that every isomorphism of G' fixes $\{X,Y\}$ and by abuse of notation refer to any copy of G' in $K_\omega^{(4)}$ by G'. Given the two points $\{X,Y\}$ the set of all hyperedges containing these points naturally identifies with the set of all edges of $K_\omega^{(2)}$. We call this set of hyperedges the *XY board*, which corresponds to the $K_\omega^{X,Y}$ notation from Section 10.1.3. We use the vocabulary introduced for G on the XY board and we denote hyperedges on the XY board by their other two vertices when X,Y are clear from the context. Note that the two centres have individual and joint degrees $9 = |E(G)| = |E(G')|$. The two points in the base have (individual) degree 5. The other vertices have degree 2. For the sake of conciseness, we use the immediate extensions of definitions for graphs to hypergraphs.

Proof of Theorem 10.1.7. We give the following explicit drawing strategy for **P2** divided in four stages.

In the first stage **P2** plays as follows. Without loss of generality **P1** takes the hyperedge $TUVW$. **P2** takes the hyperedge $XYAB$, where all vertices are free. **P1** takes a hyperedge which without loss of generality does not contain X. **P2** takes the hyperedge $XYBC$ where C is a free vertex and then **P1** takes a hyperedge. Then **P1** does not have both $XAYC$ and $XABC$, as C was a **P1**-free vertex before his move, say he does not have $XAYC$. Then **P2** takes $XAYC$ and **P1** takes another hyperedge. At this point on the XY board **P2** has the hyperedges AB, BC, AC and **P1** has at most 2 hyperedges. **P2** takes the hyperedge CD, where D is a free vertex and **P1** takes another hyperedge. Note that **P1** cannot have both DA and DB, as D was a **P1**-free vertex before his move, say he does not have DA. Then **P2** takes DA and **P1** takes another hyperedge. At the end of the first stage **P1** has 6 hyperedges $\mathscr{E}_0$ in total including $TUVW$ and at most 4 of them are on the XY board.

Claim 10.4.1. At the end of the first stage AC is a potential base $AC = A_0A_1$ (see Figure 11) with special vertex A_0 such that on the entire 4-uniform board **P1** has at most 2 hyperedges that contain A_0 but do not contain A_1.

Proof of Claim 10.4.1. If both vertices A and C can be the special vertex of AC, it suffices to choose A_0 to be one of them with at most 2 hyperedges taken by **P1** containing it but not the other by the pigeonhole principle (as $TUVW$ is disjoint from AC). Otherwise, assume C cannot be the special vertex of AC. By Definition 10.2.5, **P1** has claimed on the XY board either a 4-cycle C, C_1, C_2, C_3 where the hyperedge $XYCC_2$ is not taken by **P2** or a triangle C, C_1, C_2. Note that none of the hyperedges claimed by **P1** in the triangle or cycle can contain vertex A. In addition, the first hyperedge $TUVW$ claimed by **P1** does not contain A and, moreover, is not in the XY board. Therefore, from the total of 6 hyperedges claimed by **P1**, at most 2 can contain A. Moreover, from the total of at most 4 hyperedges on the XY board claimed by **P1**, at most 1 can contain A. Therefore, A is the special vertex of the potential base AC. □

In the second stage we consider the XY board and proceed as in the proof of Lemma 10.2.7, though the proof requires more attention. **P2** takes edges from A_1 to free vertices F_i for $i = 1, \dots, k$ until at some point **P1** does not take the edge A_0F_k. Then **P2** takes A_0F_k and **P1** takes two extra hyperedges. Let $\mathscr{E}_1 = \{A_0F_i, 1 \leqslant i < k\}$ be the star taken by **P1** and $\mathscr{E}_2$ be the set consisting of the last two hyperedges taken by **P1**.

At the end of the second stage **P1** has $TUVW$, 7 other hyperedges and the star $\mathscr{E}_1$ (which does not have vertices in common with $TUVW$). For the third stage, consider three cases.

Case I. P1 has at least 8 hyperedges in a copy G^1 of G' containing $TUVW$ with its base present.

Assume without loss of generality that G^1 has centres TU and base C_0C_1. Note that $G^1 \cap \mathscr{E}_1 = \varnothing$ and so **P1** has exactly 8 hyperedges in G^1 and those are precisely $\mathscr{E}_0 \cup \mathscr{E}_2$. Let $C_{\varepsilon_1}D_1$ be the edge absent in G^1. Then **P2** takes the edge $C_{1-\varepsilon_1}D_1$ and, while **P1** keeps taking edges $C_{\varepsilon_i}D_i$ to some vertices D_i, **P2** keeps taking edges $C_{1-\varepsilon_i}D_i$ for $\varepsilon_i \in \{0,1\}$ and $2 \leqslant i \leqslant m$ until at some point **P1** either stops or takes a different kind of edge. Let $\mathscr{E}_3 = \{C_{\varepsilon_i}D_i, 1 \leqslant i \leqslant m\}$ and let $\mathscr{E}_4$ be the set consisting of the last edge.

In the fourth stage **P2** moves back to the XY board. He takes edges from A_1 to free vertices F_i for $k < i \leqslant l$ until at some point **P1** does not take the edge A_0F_l. Then **P2** takes A_0F_l and constructs a G. The game stops after **P2**'s move. Let $\mathscr{E}_5 = \{A_0F_i, k < i < l\}$ and $\mathscr{E}_6$ be the set consisting of the last edge taken by **P1**. We claim that **P1** does not have a G'.

Assume for a contradiction that **P1** has a G' with centres HI. Note that the XY and TU boards intersect only in the hyperedge $XYTU$, which is not of the form XYA_0F_j. Then on the HI board, **P1** has either at most 1 hyperedge of the TU board or no hyperedges of the form XYA_0F_j. Indeed if, on the contrary, the HI board contains TUJ_1K_1, TUJ_2K_2 and XYA_0F_j, then $HI = J_1K_1 = J_2K_2$, as $TU \cap XYA_0F_j = \varnothing$, a contradiction.

Assume that the HI board contains a hyperedge of the star $\mathscr{E}_1 \cup \mathscr{E}_5$ taken by **P1**. Then the HI board contains at most one hyperedge from the TU board and hence it contains at most one hyperedge from $\mathscr{E}_0 \cup \mathscr{E}_2 \cup \mathscr{E}_3$, since $\mathscr{E}_0 \cup \mathscr{E}_2 \subset G^1$ is in the TU board and so is $\mathscr{E}_3$ by definition. Notice that a star in one 2-uniform board is a star in any 2-uniform board, and hence the HI board contains a star

from $\mathscr{E}_1 \cup \mathscr{E}_5$. Finally, the HI board contains at most two hyperedges from $\mathscr{E}_4 \cup \mathscr{E}_6$. Altogether, on the HI board, **P1** has at most a star, and three extra hyperedges. However, by Remark 10.2.6, this is not enough to complete a G' – contradiction.

Otherwise, assume that G' has no hyperedge of the star $\mathscr{E}_1 \cup \mathscr{E}_5$. Further, by the proof of Lemma 10.2.7 we can assume that $\{H,I\} \neq \{T,U\}$. However, disregarding the star $\mathscr{E}_1 \cup \mathscr{E}_5$, the only vertices with **P1**-degree at least 9 are T, U, C_0 and C_1 and the last two have joint **P1**-degree at most $1+2$, since $\mathscr{E}_2 \subset G^1$ is in the TU board, which has only one hyperedge C_0C_1. So, without loss of generality, we can assume that $\{H,I\} = \{T,C_0\}$. Notice that the intersection of the TU and $HI = TC_0$ boards is a star from U when viewed in the TC_0 board and $\mathscr{E}_0 \cup \mathscr{E}_2 \subset G^1$ is in the TU board and so is $\mathscr{E}_3$ by definition. Therefore, on the TC_0 board, **P1** has only a star (from U) from $\mathscr{E}_0 \cup \mathscr{E}_2 \cup \mathscr{E}_3$ and at most 2 extra hyperedges from $\mathscr{E}_4 \cup \mathscr{E}_6$, which is not enough to complete a G' by Remark 10.2.6 – contradiction.

Case II. P1 has at least 8 hyperedges in a copy of G' containing $TUVW$ with its base C_0C_1 absent.

P2 takes C_0C_1 and then, in the fourth stage, proceeds as in Case I with the same proof.

Case III. For any G' containing $TUVW$, $e_{\mathbf{P1}}(G') \leqslant 7$.

In this case we directly skip to the fourth stage. **P2** takes hyperedges in the XY board from A_1 to free vertices F_i for $k < i \leqslant l$ until at some point **P1** does not take the hyperedge A_0F_l. Then **P2** takes A_0F_l and constructs a G'. The game stops after **P2**'s move. Let $\mathscr{E}_3 = \{A_0F_i, k < i < l\}$ and $\mathscr{E}_4$ be the set consisting of the last hyperedge taken by **P1**. Assume for a contradiction that **P1** has a G' with centres HI and base C_0C_1.

Note that G' cannot contain $TUVW$, since then $e_{\mathbf{P1}}(G') \leqslant 8$. Also, G' cannot be disjoint from both $TUVW$ and the star $\mathscr{E}_1 \cup \mathscr{E}_3$, as $e_{\mathbf{P1}}(G') \leqslant |(\mathscr{E}_0 \setminus \{TUVW\}) \cup \mathscr{E}_2 \cup \mathscr{E}_4| \leqslant 8$. Thus, G' intersects the star $\mathscr{E}_1 \cup \mathscr{E}_3$, but does not contain $TUVW$. Recall that $\deg_{\mathbf{P1}}(F_j) \leqslant 1 + |\mathscr{E}_2 \cup \mathscr{E}_4| \leqslant 4$ implies that F_j is not a centre nor is it in a base. Therefore, without loss of generality $\{H,I,C_0\} = \{X,Y,A_0\}$. In Lemma 10.2.7 we already proved that G' cannot be on the XY board, so without loss of generality $(H,I,C_0) = (X,A_0,Y)$. Moreover $C_1 \neq A_1$ as XYA_0A_1 is taken by **P2**. Since without $\mathscr{E}_2 \cup \mathscr{E}_4$ we have $\deg_{\mathbf{P1}}(F_i) = 1$ for all i, G' has at most $|\mathscr{E}_2 \cup \mathscr{E}_4| \leqslant 3$ hyperedges from the star $\mathscr{E}_1 \cup \mathscr{E}_3$. In addition, G' has at most 3 hyperedges from $\mathscr{E}_2 \cup \mathscr{E}_4$. This means that G' contains at least 3 hyperedges from $\mathscr{E}_0$. Recall that by construction, there are at most 2 hyperedges in $\mathscr{E}_0$ that contain A_0 but do not contain A_1. To arrive at the desired contradiction it is enough to show that G' does not contain any hyperedge of the form XA_0JA_1. Indeed, if such a hyperedge is in G', then necessarily $J = C_1$ or $J = C_0$, since it needs to intersect the base C_0C_1. However, the hyperedge $XA_0C_0A_1 = XA_0YA_1$ is taken by **P2** and the hyperedge $XA_0C_1A_1$ cannot be in G' because **P2** has $XA_0C_0A_1 = XA_0YA_1$. □

Remark 10.4.2. One can easily extend this proof to obtain r-uniform graphs with the same property for all $r \geqslant 4$.

10.5 Concluding remarks

10.5.1 Key aspects

As the strategy is quite cumbersome and does not have a clear structure, it is not hard to miss the forest for the trees. Let us point out a few aspects which are important for the understanding of the dynamic of the Ramsey game in general.

The strategy reflects the significant differences between weak and strong games. The most important feature which arises in this context is the notion of delay: unlike the strategy in [94], in ours **P2** does not focus solely on building a core fast, but rather needs to delay **P1** at least as much as **P1** delays him. We now point out some parts of the strategy which best highlight this feature.

Firstly, as it is clear from Figure 3, **Case 1** is much easier than **Case 2**, so the second move of **P1** is crucially important. Unexpectedly, the harder **Case 2** corresponds to two seemingly contradictory behaviours of **P1**. On the one hand, **P1** goes after **P2** in K^2, which can only have the purpose of blocking **P2**, since **P2** is the first player there. On the other hand, **P1** plays disjointly from **P2**'s edge, which is not the most natural of "blocking" moves. From another perspective, this is less surprising. Given that we prove exactly that **P2** can force a draw by moving away from **P1** in the other copy of K_ω, it is reasonable to expect that the most efficient move of **P1** would be to also start playing "away" from **P2**.

Another striking fact is that the "right" move of **P2** in **Case 2.1** is to go after **P1**, who is the "second player" in K^2. The purpose of this move is to constrain **P1**'s possibilities and later force **P1** to lose edges. This philosophy is even more visible in **Case 2.1.2.1.1**. There **P2** takes an edge which is completely useless to him from a "Maker" perspective, but makes **P1** lose 2 edges, so it is as though by playing this "Breaker" move **P2** gets a net advantage of 1 edge.

These examples show that a good strong game strategy should by all means seek to interact with the other player. However, the main approach to strong games so far relies on finding fast "Maker" strategies for the corresponding weak game and transforming them into strategies for the strong games [65, 66, 94]. Thus, our strategy suggests that the Ramsey game requires more sophisticated arguments.

On a more technical note, the idea of leaving additional edges unspecified and disregarding the stage of the game when an edge was played decreases the number of cases tremendously. This approach allows us to only have the *tiny* tree in Figure 3, which can readily be analysed without the help of a computer in short time. The driving force of the drawing strategy is Lemma 10.2.7, which provides us with intermediate desired configurations together with a drawing strategy from these configurations.

10.5.2 Open problems

The present work raises various natural closely related questions which, in view of our results, do not seem as widely open and intractable as previously.

The prime question to answer is the following.

Question 10.5.1. Is the game $\mathscr{R}(K_\omega, G)$ a draw?

As we already mentioned if the answer is positive, then this would be the first such known graph. Otherwise, G would be the answer to the next question.

Question 10.5.2. Does there exist a target graph H such that $\mathscr{R}(K_\omega, H)$ is a **P1**-win, but $\mathscr{R}(K_\omega \sqcup K_\omega, H)$ is not?

Such a counterexample would prove that yet another easy and folklore fact for finite boards breaks down for infinite ones.

Another line of thought, which might not lead directly towards $\mathscr{R}(K_n, K_k)$, was suggested by Bowler [33]. He proved the natural characterisation of positional games which are a draw on infinite disjoint copies of a finite one. One can wonder if the game on G gives a draw on an infinite number of copies of K_n. We believe that this is not the case.

Conjecture 10.5.3. For n sufficiently large $\mathscr{R}(\bigsqcup_\omega K_n, G)$ is a **P1**-win.

Finally, this last conjecture seems closely related to determining if the initiative in the sense of [33] of $\mathscr{R}(K_n, K_4)$ is 6, as asked in the next question.

Question 10.5.4. Let n be a fixed large integer. What is the minimal k such that **P1** has a winning strategy for $\mathscr{R}(K_n, K_4)$ such that if the second player is allowed to pass, he would not be able to do so more than k times before losing?

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